# THE JAVASCRIPT ALGORITHM WORKBOOK

A topic-first study edition of Josh Crozier's LeetCode JavaScript collection

## 12 guided chapters / 2,786 exercises

Learn the concept. Study the diagram. Practice the JavaScript pattern. Then attempt the grouped problems without looking at the solution companion.

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Problem IDs match the original LeetCode numbers and the separate numbered solutions PDF. Exercises are grouped by likely primary topic, not numeric sequence. Difficulty is inherited from the source repository.

CHAPTER 01 / 988 EXERCISES

# Arrays & Hashing

## 01 THE IDEA

Arrays store ordered values; maps and sets make repeated lookup efficient.

## 02 SEE IT

```
INPUT   [4] [7] [4] [2]
INDEX    0   1   2   3
COUNTS   4:2   7:1   2:1
```

## 03 JAVASCRIPT PATTERN

```
const counts = new Map();
for (const x of nums) counts.set(x, (counts.get(x) ?? 0) + 1);
```

## 04 HOW TO THINK

Learn to distinguish an index from a value. Trace a loop using a small array, then decide whether a Map or Set can replace repeated searches.

## 05 PERFORMANCE

Typical: linear scan O(n); hash lookup average O(1); sorting O(n log n).

## EXERCISE COLLECTION

Try each question before opening the matching problem number in the separate Solutions PDF. The topic classification is a study aid; some exercises combine multiple patterns.

### 0001 Two Sum

EASY

Given an array of integers `nums` and an integer `target`, return indices of the two numbers such that they add up to `target`.

You may assume that each input would have exactly one solution, and you may not use the same element twice.

You can return the answer in any order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0009 Palindrome Number

EASY

Given an integer `x`, return `true` if `x` is palindrome integer.

An integer is a palindrome when it reads the same backward as forward.
- For example, `121` is palindrome while `123` is not.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0027 Remove Element

EASY

Given an integer array nums and an integer val, remove all occurrences of val in nums in-place. The relative order of the elements may be changed.

Since it is impossible to change the length of the array in some languages, you must instead have the result be placed in the first part of the array nums. More formally, if there are k elements after removing the duplicates, then the first k elements of nums should hold the final result. It does not matter what you leave beyond the first k elements.

Return k after placing the final result in the first k slots of nums.

Do not allocate extra space for another array. You must do this by modifying the input array in-place with O(1) extra memory.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0038 Count And Say

MEDIUM

The count-and-say sequence is a sequence of digit strings defined by the recursive formula:
- countAndSay(1) = "1"
- countAndSay(n) is the way you would "say" the digit string from countAndSay(n-1), which is then converted into a different digit string.

To determine how you "say" a digit string, split it into the minimal number of substrings such that each substring contains exactly one unique digit. Then for each substring, say the number of digits, then say the digit. Finally, concatenate every said digit.

For example, the saying and conversion for digit string "3322251":
Given a positive integer n, return the nth term of the count-and-say sequence.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0054 Spiral Matrix

MEDIUM

Given an `m x n` `matrix`, return all elements of the `matrix` in spiral order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0059 Spiral Matrix Ii

MEDIUM

Given a positive integer n, generate an n x n matrix filled with elements from 1 to n2 in spiral order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0064 Minimum Path Sum

MEDIUM

Given a m x n grid filled with non-negative numbers, find a path from top left to bottom right, which minimizes the sum of all numbers along its path.

Note: You can only move either down or right at any point in time.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0065 Valid Number

HARD

Given a string s, return whether s is a valid number.

For example, all the following are valid numbers: "2", "0089", "-0.1", "+3.14", "4.", "-.9", "2e10", "-90E3", "3e+7", "+6e-1", "53.5e93", "-123.456e789", while the following are not valid numbers: "abc", "1a", "1e", "e3", "99e2.5", "--6", "-+3", "95a54e53".

Formally, a valid number is defined using one of the following definitions:
- An integer number followed by an optional exponent.
- A decimal number followed by an optional exponent.

An integer number is defined with an optional sign '-' or '+' followed by digits.
A decimal number is defined with an optional sign '-' or '+' followed by one of the following definitions:
- Digits followed by a dot '.'.
- Digits followed by a dot '.' followed by digits.
- A dot '.' followed by digits.
- An exponent is defined with an exponent notation 'e' or 'E' followed by an integer number.

The digits are defined as one or more digits.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0073 Set Matrix Zeroes

MEDIUM

Given an `m x n` integer `matrix` matrix, if an element is `0`, set its entire row and column to `0`'s, and return the matrix.

You must do it in place.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0074 Search A 2D Matrix

MEDIUM

Write an efficient algorithm that searches for a value in an m x n matrix.
This matrix has the following properties:

Integers in each row are sorted from left to right.
The first integer of each row is greater than the last integer of the previous row.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0112 Path Sum

EASY

Given the root of a binary tree and an integer targetSum, return true if the tree has a root-to-leaf path such that adding up all the values along the path equals targetSum.

A leaf is a node with no children.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0113 Path Sum Ii

MEDIUM

Given the root of a binary tree and an integer targetSum, return all root-to-leaf paths where the sum of the node values in the path equals targetSum. Each path should be returned as a list of the node values, not node references.

A root-to-leaf path is a path starting from the root and ending at any leaf node. A leaf is a node with no children.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0128 Longest Consecutive Sequence

MEDIUM

Given an unsorted array of integers nums, return the length of the longest consecutive elements sequence.

You must write an algorithm that runs in O(n) time.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0129 Sum Root To Leaf Numbers

MEDIUM

You are given the root of a binary tree containing digits from 0 to 9 only.

Each root-to-leaf path in the tree represents a number.

- For example, the root-to-leaf path 1 -> 2 -> 3 represents the number 123.

Return the total sum of all root-to-leaf numbers. Test cases are generated so that the answer will fit in a 32-bit integer.

A leaf node is a node with no children.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0136 Single Number

EASY

Given a non-empty array of integers nums, every element appears twice except for one. Find that single one.

You must implement a solution with a linear runtime complexity and use only constant extra space.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0137 Single Number Ii

MEDIUM

Given an integer array nums where every element appears three times except for one, which appears exactly once. Find the single element and return it.

You must implement a solution with a linear runtime complexity and use only constant extra space.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0138 Copy List With Random Pointer

MEDIUM

A linked list of length n is given such that each node contains an additional random pointer, which could point to any node in the list, or null.

Construct a deep copy of the list. The deep copy should consist of exactly n brand new nodes, where each new node has its value set to the value of its corresponding original node. Both the next and random pointer of the new nodes should point to new nodes in the copied list such that the pointers in the original list and copied list represent the same list state. None of the pointers in the new list should point to nodes in the original list.

For example, if there are two nodes X and Y in the original list, where X.random --> Y, then for the corresponding two nodes x and y in the copied list, x.random --> y.

Return the head of the copied linked list.

The linked list is represented in the input/output as a list of n nodes. Each node is represented as a pair of [val, random_index] where:
- val: an integer representing Node.val
- random_index: the index of the node (range from 0 to n-1) that the random pointer points to, or null if it does not point to any node.

Your code will only be given the head of the original linked list.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0146 Lru Cache

MEDIUM

Design a data structure that follows the constraints of a Least Recently Used (LRU) cache.

Implement the LRUCache class:
- LRUCache(int capacity) Initialize the LRU cache with positive size capacity.
- int get(int key) Return the value of the key if the key exists, otherwise return -1.
- void put(int key, int value) Update the value of the key if the key exists. Otherwise, add the key-value pair to the cache. If the number of keys exceeds the capacity from this operation, evict the least recently used key.

The functions get and put must each run in O(1) average time complexity.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0149 Max Points On A Line

HARD

Given an array of points where points[i] = [xi, yi] represents a point on the X-Y plane, return the maximum number of points that lie on the same straight line.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0162 Find Peak Element

MEDIUM

A peak element is an element that is strictly greater than its neighbors.

Given a 0-indexed integer array nums, find a peak element, and return its index. If the array contains multiple peaks, return the index to any of the peaks.

You may imagine that nums[-1] = nums[n] = -∞. In other words, an element is always considered to be strictly greater than a neighbor that is outside the array.

You must write an algorithm that runs in O(log n) time.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0165 Compare Version Numbers

MEDIUM

Given two version strings, version1 and version2, compare them. A version string consists of revisions separated by dots '.'. The value of the revision is its integer conversion ignoring leading zeros.

To compare version strings, compare their revision values in left-to-right order. If one of the version strings has fewer revisions, treat the missing revision values as 0.

Return the following:
- If version1 < version2, return -1.
- If version1 > version2, return 1.
- Otherwise, return 0.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0166 Fraction To Recurring Decimal

MEDIUM

Given two integers representing the numerator and denominator of a fraction, return the fraction in string format.

If the fractional part is repeating, enclose the repeating part in parentheses.

If multiple answers are possible, return any of them.

It is guaranteed that the length of the answer string is less than 104 for all the given inputs.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0169 Majority Element

EASY

Given an array nums of size n, return the majority element.

The majority element is the element that appears more than ⌊n / 2⌋ times. You may assume that the majority element always exists in the array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0171 Excel Sheet Column Number

EASY

Given a string columnTitle that represents the column title as appears in an Excel sheet, return its corresponding column number.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0179 Largest Number

MEDIUM

Given a list of non-negative integers nums, arrange them such that they form the largest number.

Note: The result may be very large, so you need to return a string instead of an integer.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0187 Repeated Dna Sequences

MEDIUM

The DNA sequence is composed of a series of nucleotides abbreviated as 'A', 'C', 'G', and 'T':
- For example, "ACGAATTCCG" is a DNA sequence.

When studying DNA, it is useful to identify repeated sequences within the DNA.

Given a string s that represents a DNA sequence, return all the 10-letter-long sequences (substrings) that occur more than once in a DNA molecule. You may return the answer in any order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0189 Rotate Array

MEDIUM

Given an array, rotate the array to the right by k steps, where k is non-negative.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0191 Number Of 1 Bits

EASY

Write a function that takes an unsigned integer and returns the number of '1' bits it has (also known as the Hamming weight).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0201 Bitwise And Of Numbers Range

MEDIUM

Given two integers left and right that represent the range [left, right], return the bitwise AND of all numbers in this range, inclusive.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0202 Happy Number

EASY

Write an algorithm to determine if a number n is happy.

A happy number is a number defined by the following process:
- Starting with any positive integer, replace the number by the sum of the squares of its digits.
- Repeat the process until the number equals 1 (where it will stay), or it loops endlessly in a cycle which does not include 1.
- Those numbers for which this process ends in 1 are happy.

Return true if n is a happy number, and false if not.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0204 Count Primes

MEDIUM

Given an integer n, return the number of prime numbers that are strictly less than n.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0217 Contains Duplicate

EASY

Given an integer array `nums`, return `true` if any value appears at least twice in the array, and return `false` if every element is distinct.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0219 Contains Duplicate Ii

EASY

Given an integer array `nums` and an integer `k`, return `true` if there are two distinct indices `i` and `j` in the array such that `nums[i] == nums[j]` and `abs(i - j) <= k`.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0220 Contains Duplicate Iii

HARD

You are given an integer array nums and two integers indexDiff and valueDiff.

Find a pair of indices (i, j) such that:
- i != j,
- abs(i - j) <= indexDiff.
- abs(nums[i] - nums[j]) <= valueDiff, and

Return true if such pair exists or false otherwise.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0228 Summary Ranges

EASY

You are given a sorted unique integer array nums.

A range [a,b] is the set of all integers from a to b (inclusive).

Return the smallest sorted list of ranges that cover all the numbers in the array exactly. That is, each element of nums is covered by exactly one of the ranges, and there is no integer x such that x is in one of the ranges but not in nums.

Each range [a,b] in the list should be output as:
- "a->b" if a != b
- "a" if a == b

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0229 Majority Element Ii

**MEDIUM**

Given an integer array of size n, find all elements that appear more than ■ n/3 ■ times.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0238 Product Of Array Except Self

**MEDIUM**

Given an integer array nums, return an array answer such that answer[i] is equal to the product of all the elements of nums except nums[i].

The product of any prefix or suffix of nums is guaranteed to fit in a 32-bit integer.

You must write an algorithm that runs in O(n) time and without using the division operation.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0240 Search A 2D Matrix Ii

**MEDIUM**

Write an efficient algorithm that searches for a value target in an m x n integer matrix matrix. This matrix has the following properties:
- Integers in each row are sorted in ascending from left to right.
- Integers in each column are sorted in ascending from top to bottom.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0246 Strobogrammatic Number

**EASY**

Given a string num which represents an integer, return true if num is a strobogrammatic number.

A strobogrammatic number is a number that looks the same when rotated 180 degrees (looked at upside down).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0247 Strobogrammatic Number Ii

**MEDIUM**

Given an integer n, return all the strobogrammatic numbers that are of length n. You may return the answer in any order.

A strobogrammatic number is a number that looks the same when rotated 180 degrees (looked at upside down).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0248 Strobogrammatic Number Iii

HARD

Given two strings low and high that represent two integers low and high where low <= high, return the number of strobogrammatic numbers in the range [low, high].

A strobogrammatic number is a number that looks the same when rotated 180 degrees (looked at upside down).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0260 Single Number Iii

MEDIUM

Given an integer array nums, in which exactly two elements appear only once and all the other elements appear exactly twice. Find the two elements that appear only once. You can return the answer in any order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0263 Ugly Number

EASY

Write a program to check whether a given number is an ugly number.

Ugly numbers are positive numbers whose prime factors only include 2, 3, 5.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0264 Ugly Number Ii

MEDIUM

Write a program to find the n-th ugly number.

Ugly numbers are positive numbers whose prime factors only include 2, 3, 5.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0268 Missing Number

EASY

Given an array `nums` containing `n` distinct numbers in the range `[0, n]`, return the only number in the range that is missing from the array.

Follow up: Could you implement a solution using only `O(1)` extra space complexity and `O(n)` runtime complexity?

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0269 Alien Dictionary

HARD

There is a new alien language that uses the English alphabet. However, the order of the letters is unknown to you.

You are given a list of strings words from the alien language's dictionary. Now it is claimed that the strings in words are sorted lexicographically by the rules of this new language.

If this claim is incorrect, and the given arrangement of string in words cannot correspond to any order of letters, return "".

Otherwise, return a string of the unique letters in the new alien language sorted in lexicographically increasing order by the new language's rules. If there are multiple solutions, return any of them.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0287 Find The Duplicate Number

**MEDIUM**

Given an array of integers nums containing n + 1 integers where each integer is in the range [1, n] inclusive.

There is only one repeated number in nums, return this repeated number.

You must solve the problem without modifying the array nums and using only constant extra space.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0294 Flip Game Ii

**MEDIUM**

You are playing a Flip Game with your friend.

You are given a string currentState that contains only '+' and '-'. You and your friend take turns to flip two consecutive "++" into "--". The game ends when a person can no longer make a move, and therefore the other person will be the winner.

Return true if the starting player can guarantee a win, and false otherwise.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0303 Range Sum Query Immutable

**EASY**

Given an integer array nums, handle multiple queries of the following type:
- Calculate the sum of the elements of nums between indices left and right inclusive where left <= right.

Implement the NumArray class:
- NumArray(int[] nums) Initializes the object with the integer array nums.
- int sumRange(int left, int right) Returns the sum of the elements of nums between indices left and right inclusive (i.e. nums[left] + nums[left + 1] + ... + nums[right]).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0304 Range Sum Query 2D Immutable

**MEDIUM**

Given a 2D matrix matrix, handle multiple queries of the following type:
- Calculate the sum of the elements of matrix inside the rectangle defined by its upper left corner (row1, col1) and lower right corner (row2, col2).

Implement the NumMatrix class:
- NumMatrix(int[][] matrix) Initializes the object with the integer matrix matrix.
- int sumRegion(int row1, int col1, int row2, int col2) Returns the sum of the elements of matrix inside the rectangle defined by its upper left corner (row1, col1) and lower right corner (row2, col2).

You must design an algorithm where sumRegion works on O(1) time complexity.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0306 Additive Number

**MEDIUM**

An additive number is a string whose digits can form an additive sequence.

A valid additive sequence should contain at least three numbers. Except for the first two numbers, each subsequent number in the sequence must be the sum of the preceding two.

Given a string containing only digits, return true if it is an additive number or false otherwise.

Note: Numbers in the additive sequence cannot have leading zeros, so sequence 1, 2, 03 or 1, 02, 3 is invalid.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0307 Range Sum Query Mutable

**MEDIUM**

Given an integer array nums, handle multiple queries of the following types:
where left <= right.

Implement the NumArray class:
- NumArray(int[] nums) Initializes the object with the integer array nums.
- void update(int index, int val) Updates the value of nums[index] to be val.
- int sumRange(int left, int right) Returns the sum of the elements of nums between indices left and right inclusive (i.e. nums[left] + nums[left + 1] + ... + nums[right]).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0308 Range Sum Query 2D Mutable

**MEDIUM**

Given a 2D matrix matrix, handle multiple queries of the following types:
corner (row1, col1) and lower right corner (row2, col2).

Implement the NumMatrix class:
- NumMatrix(int[][] matrix) Initializes the object with the integer matrix matrix.
- void update(int row, int col, int val) Updates the value of matrix[row][col] to be val.
- int sumRegion(int row1, int col1, int row2, int col2) Returns the sum of the elements of matrix inside the rectangle defined by its upper left corner (row1, col1) and lower right corner (row2, col2).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0311 Sparse Matrix Multiplication

**MEDIUM**

Given two sparse matrices mat1 of size m x k and mat2 of size k x n, return the result of mat1 x mat2. You may assume that multiplication is always possible.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0313 Super Ugly Number

**MEDIUM**

A super ugly number is a positive integer whose prime factors are in the array primes.

Given an integer n and an array of integers primes, return the nth super ugly number.

The nth super ugly number is guaranteed to fit in a 32-bit signed integer.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0315 Count Of Smaller Numbers After Self

**HARD**

Given an integer array nums, return an integer array counts where counts[i] is the number of smaller elements to the right of nums[i].

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0321 Create Maximum Number

**HARD**

You are given two integer arrays nums1 and nums2 of lengths m and n respectively. nums1 and nums2 represent the digits of two numbers. You are also given an integer k.

Create the maximum number of length k <= m + n from digits of the two numbers. The relative order of the digits from the same array must be preserved.

Return an array of the k digits representing the answer.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0327 Count Of Range Sum

**HARD**

Given an integer array nums and two integers lower and upper, return the number of range sums that lie in [lower, upper] inclusive.

Range sum S(i, j) is defined as the sum of the elements in nums between indices i and j inclusive, where i <= j.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0329 Longest Increasing Path In A Matrix

**HARD**

Given an m x n integers matrix, return the length of the longest increasing path in matrix.

From each cell, you can either move in four directions: left, right, up, or down. You may not move diagonally or move outside the boundary (i.e., wrap-around is not allowed).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0330 Patching Array

**HARD**

Given a sorted integer array nums and an integer n, add/patch elements to the array such that any number in the range [1, n] inclusive can be formed by the sum of some elements in the array.

Return the minimum number of patches required.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0334 Increasing Triplet Subsequence

MEDIUM

Given an integer array nums, return true if there exists a triple of indices (i, j, k) such that i < j < k and nums[i] < nums[j] < nums[k]. If no such indices exists, return false.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0336 Palindrome Pairs

HARD

You are given a 0-indexed array of unique strings words.

A palindrome pair is a pair of integers (i, j) such that:
- 0 <= i, j < words.length,
- i != j, and
- words[i] + words[j] (the concatenation of the two strings) is a palindrome.

Return an array of all the palindrome pairs of words.

You must write an algorithm with O(sum of words[i].length) runtime complexity.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0338 Counting Bits

EASY

Given an integer n, return an array ans of length n + 1 such that for each i (0 <= i <= n), ans[i] is the number of 1's in the binary representation of i.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0339 Nested List Weight Sum

MEDIUM

You are given a nested list of integers nestedList. Each element is either an integer or a list whose elements may also be integers or other lists.

The depth of an integer is the number of lists that it is inside of. For example, the nested list [1,[2,2],[[3],2],1] has each integer's value set to its depth.

Return the sum of each integer in nestedList multiplied by its depth.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0347 Top K Frequent Elements

MEDIUM

Given an integer array `nums` and an integer `k`, return the `k` most frequent elements. You may return the answer in any order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0349 Intersection Of Two Arrays

EASY

Given two integer arrays `nums1` and `nums2`, return an array of their intersection.

Each element in the result must be unique and you may return the result in any order.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0350 Intersection Of Two Arrays Ii

**EASY**

Given two integer arrays nums1 and nums2, return an array of their intersection.

Each element in the result must appear as many times as it shows in both arrays and you may return the result in any order.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0351 Android Unlock Patterns

**MEDIUM**

Android devices have a special lock screen with a 3 x 3 grid of dots. Users can set an "unlock pattern" by connecting the dots in a specific sequence, forming a series of joined line segments where each segment's endpoints are two consecutive dots in the sequence. A sequence of k dots is a valid unlock pattern if both of the following are true:
- All the dots in the sequence are distinct.
- If the line segment connecting two consecutive dots in the sequence passes through the center of any other dot, the other dot must have previously appeared in the sequence. No jumps through the center non-selected dots are allowed.
- For example, connecting dots 2 and 9 without dots 5 or 6 appearing beforehand is valid because the line from dot 2 to dot 9 does not pass through the center of either dot 5 or 6.
- However, connecting dots 1 and 3 without dot 2 appearing beforehand is invalid because the line from dot 1 to dot 3 passes through the center of dot 2.

Here are some example valid and invalid unlock patterns:
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0353 Design Snake Game

**MEDIUM**

Design a Snake game that is played on a device with screen size height x width. Play the game online if you are not familiar with the game.

The snake is initially positioned at the top left corner (0, 0) with a length of 1 unit.

You are given an array food where food[i] = (ri, ci) is the row and column position of a piece of food that the snake can eat. When a snake eats a piece of food, its length and the game's score both increase by 1.

Each piece of food appears one by one on the screen, meaning the second piece of food will not appear until the snake eats the first piece of food.

When a piece of food appears on the screen, it is guaranteed that it will not appear on a block occupied by the snake.

The game is over if the snake goes out of bounds (hits a wall) or if its head occupies a space that its body occupies after moving (i.e. a snake of length 4 cannot run into itself).

Implement the SnakeGame class:
- SnakeGame(int width, int height, int[][] food) Initializes the object with a screen of size height x width and the positions of the food.
- int move(String direction) Returns the score of the game after applying one direction move by the snake. If the game is over, return -1.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0355 Design Twitter

MEDIUM

Design a simplified version of Twitter where users can post tweets, follow/unfollow another user, and is able to see the 10 most recent tweets in the user's news feed.

Implement the Twitter class:
- Twitter() Initializes your twitter object.
- void postTweet(int userId, int tweetId) Composes a new tweet with ID tweetId by the user userId. Each call to this function will be made with a unique tweetId.
- List<Integer> getNewsFeed(int userId) Retrieves the 10 most recent tweet IDs in the user's news feed. Each item in the news feed must be posted by users who the user followed or by the user themself. Tweets must be ordered from most recent to least recent.
- void follow(int followerId, int followeeId) The user with ID followerId started following the user with ID followeeId.
- void unfollow(int followerId, int followeeId) The user with ID followerId started unfollowing the user with ID followeeId.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0356 Line Reflection

MEDIUM

Given n points on a 2D plane, find if there is such a line parallel to the y-axis that reflects the given points symmetrically.

In other words, answer whether or not if there exists a line that after reflecting all points over the given line, the original points' set is the same as the reflected ones.

Note that there can be repeated points.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0359 Logger Rate Limiter

EASY

Design a logger system that receives a stream of messages along with their timestamps. Each unique message should only be printed at most every 10 seconds (i.e. a message printed at timestamp t will prevent other identical messages from being printed until timestamp t + 10).

All messages will come in chronological order. Several messages may arrive at the same timestamp.

Implement the Logger class:
- Logger() Initializes the logger object.
- bool shouldPrintMessage(int timestamp, string message) Returns true if the message should be printed in the given timestamp, otherwise returns false.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0362 Design Hit Counter

MEDIUM

Design a hit counter which counts the number of hits received in the past 5 minutes (i.e., the past 300 seconds).

Your system should accept a timestamp parameter (in seconds granularity), and you may assume that calls are being made to the system in chronological order (i.e., timestamp is monotonically increasing). Several hits may arrive roughly at the same time.

Implement the HitCounter class:
- HitCounter() Initializes the object of the hit counter system.
- void hit(int timestamp) Records a hit that happened at timestamp (in seconds). Several hits may happen at the same timestamp.
- int getHits(int timestamp) Returns the number of hits in the past 5 minutes from timestamp (i.e., the past 300 seconds).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0363 Max Sum Of Rectangle No Larger Than K

**HARD**

Given an m x n matrix matrix and an integer k, return the max sum of a rectangle in the matrix such that its sum is no larger than k.

It is guaranteed that there will be a rectangle with a sum no larger than k.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0364 Nested List Weight Sum Ii

**MEDIUM**

You are given a nested list of integers nestedList. Each element is either an integer or a list whose elements may also be integers or other lists.

The depth of an integer is the number of lists that it is inside of. For example, the nested list [1,[2,2],[[3],2],1] has each integer's value set to its depth. Let maxDepth be the maximum depth of any integer.

The weight of an integer is maxDepth - (the depth of the integer) + 1.

Return the sum of each integer in nestedList multiplied by its weight.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0371 Sum Of Two Integers

**MEDIUM**

Given two integers a and b, return the sum of the two integers without using the operators + and -.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0373 Find K Pairs With Smallest Sums

**MEDIUM**

You are given two integer arrays nums1 and nums2 sorted in non-decreasing order and an integer k.

Define a pair (u, v) which consists of one element from the first array and one element from the second array.

Return the k pairs (u1, v1), (u2, v2), ..., (uk, vk) with the smallest sums.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0374 Guess Number Higher Or Lower

**MEDIUM**

We are playing the Guess Game. The game is as follows:

I pick a number from 1 to n. You have to guess which number I picked.

Every time you guess wrong, I will tell you whether the number I picked is higher or lower than your guess.

You call a pre-defined API int guess(int num), which returns 3 possible results:
- -1: The number I picked is lower than your guess (i.e. pick < num).
- 1: The number I picked is higher than your guess (i.e. pick > num).
- 0: The number I picked is equal to your guess (i.e. pick == num).

Return the number that I picked.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0375 Guess Number Higher Or Lower Ii

**MEDIUM**

We are playing the Guessing Game. The game will work as follows:
or lower, and you will continue guessing.
you lose the game.

Given a particular n, return the minimum amount of money you need to guarantee a win regardless of what number I pick.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0379 Design Phone Directory

**MEDIUM**

Design a phone directory that initially has maxNumbers empty slots that can store numbers. The directory should store numbers, check if a certain slot is empty or not, and empty a given slot.

Implement the PhoneDirectory class:
- PhoneDirectory(int maxNumbers) Initializes the phone directory with the number of available slots maxNumbers.
- int get() Provides a number that is not assigned to anyone. Returns -1 if no number is available.
- bool check(int number) Returns true if the slot number is available and false otherwise.
- void release(int number) Recycles or releases the slot number.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0380 Insert Delete Getrandom O1

**MEDIUM**

Implement the RandomizedSet class:
- RandomizedSet() Initializes the RandomizedSet object.
- bool insert(int val) Inserts an item val into the set if not present. Returns true if the item was not present, false otherwise.
- bool remove(int val) Removes an item val from the set if present. Returns true if the item was present, false otherwise.
- int getRandom() Returns a random element from the current set of elements (it's guaranteed that at least one element exists when this method is called). Each element must have the same probability of being returned.

You must implement the functions of the class such that each function works in average O(1) time complexity.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0381 Insert Delete Getrandom O1 Duplicates Allowed

HARD

RandomizedCollection is a data structure that contains a collection of numbers, possibly duplicates (i.e., a multiset). It should support inserting and removing specific elements and also reporting a random element.

Implement the RandomizedCollection class:
- RandomizedCollection() Initializes the empty RandomizedCollection object.
- bool insert(int val) Inserts an item val into the multiset, even if the item is already present. Returns true if the item is not present, false otherwise.
- bool remove(int val) Removes an item val from the multiset if present. Returns true if the item is present, false otherwise. Note that if val has multiple occurrences in the multiset, we only remove one of them.
- int getRandom() Returns a random element from the current multiset of elements. The probability of each element being returned is linearly related to the number of the same values the multiset contains.

You must implement the functions of the class such that each function works on average O(1) time complexity.

Note: The test cases are generated such that getRandom will only be called if there is at least one item in the RandomizedCollection.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0383 Ransom Note

EASY

Given two stings ransomNote and magazine, return true if ransomNote can be constructed from magazine and false otherwise.

Each letter in magazine can only be used once in ransomNote.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0384 Shuffle An Array

MEDIUM

Given an integer array nums, design an algorithm to randomly shuffle the array. All permutations of the array should be equally likely as a result of the shuffling.

Implement the Solution class:
- Solution(int[] nums) Initializes the object with the integer array nums.
- int[] reset() Resets the array to its original configuration and returns it.
- int[] shuffle() Returns a random shuffling of the array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0389 Find The Difference

EASY

You are given two strings s and t.

String t is generated by random shuffling string s and then add one more letter at a random position.

Return the letter that was added to t.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0391 Perfect Rectangle

HARD

Given an array rectangles where rectangles[i] = [xi, yi, ai, bi] represents an axis-aligned rectangle. The bottom-left point of the rectangle is (xi, yi) and the top-right point of it is (ai, bi).

Return true if all the rectangles together form an exact cover of a rectangular region.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0398 Random Pick Index

MEDIUM

Given an integer array nums with possible duplicates, randomly output the index of a given target number. You can assume that the given target number must exist in the array.

Implement the Solution class:
- Solution(int[] nums) Initializes the object with the array nums.
- int pick(int target) Picks a random index i from nums where nums[i] == target. If there are multiple valid i's, then each index should have an equal probability of returning.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0399 Evaluate Division

MEDIUM

You are given an array of variable pairs equations and an array of real numbers values, where equations[i] = [Ai, Bi] and values[i] represent the equation Ai / Bi = values[i]. Each Ai or Bi is a string that represents a single variable.

You are also given some queries, where queries[j] = [Cj, Dj] represents the jth query where you must find the answer for Cj / Dj = ?.

Return the answers to all queries. If a single answer cannot be determined, return -1.0.

Note: The input is always valid. You may assume that evaluating the queries will not result in division by zero and that there is no contradiction.

Note: The variables that do not occur in the list of equations are undefined, so the answer cannot be determined for them.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0403 Frog Jump

HARD

A frog is crossing a river. The river is divided into some number of units, and at each unit, there may or may not exist a stone. The frog can jump on a stone, but it must not jump into the water.

Given a list of stones positions (in units) in sorted ascending order, determine if the frog can cross the river by landing on the last stone. Initially, the frog is on the first stone and assumes the first jump must be 1 unit.

If the frog's last jump was k units, its next jump must be either k - 1, k, or k + 1 units. The frog can only jump in the forward direction.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0404 Sum Of Left Leaves

EASY

Given the root of a binary tree, return the sum of all left leaves.

A leaf is a node with no children. A left leaf is a leaf that is the left child of another node.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0405 Convert A Number To Hexadecimal

EASY

Given an integer num, return a string representing its hexadecimal representation. For negative integers, two's complement method is used.

All the letters in the answer string should be lowercase characters, and there should not be any leading zeros in the answer except for the zero itself.

Note: You are not allowed to use any built-in library method to directly solve this problem.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0409 Longest Palindrome

EASY

Given a string s which consists of lowercase or uppercase letters, return the length of the longest palindrome that can be built with those letters.

Letters are case sensitive, for example, "Aa" is not considered a palindrome.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0410 Split Array Largest Sum

HARD

Given an integer array nums and an integer k, split nums into k non-empty subarrays such that the largest sum of any subarray is minimized.

Return the minimized largest sum of the split.

A subarray is a contiguous part of the array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0414 Third Maximum Number

EASY

Given an integer array nums, return the third distinct maximum number in this array. If the third maximum does not exist, return the maximum number.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0421 Maximum Xor Of Two Numbers In An Array

MEDIUM

Given an integer array nums, return the maximum result of nums[i] XOR nums[j], where 0 <= i <= j < n.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0432 All Oone Data Structure

**HARD**

Design a data structure to store the strings' count with the ability to return the strings with minimum and maximum counts.

Implement the AllOne class:
- AllOne() Initializes the object of the data structure.
- inc(String key) Increments the count of the string key by 1. If key does not exist in the data structure, insert it with count 1.
- dec(String key) Decrements the count of the string key by 1. If the count of key is 0 after the decrement, remove it from the data structure. It is guaranteed that key exists in the data structure before the decrement.
- getMaxKey() Returns one of the keys with the maximal count. If no element exists, return an empty string "".
- getMinKey() Returns one of the keys with the minimum count. If no element exists, return an empty string "".

Note that each function must run in O(1) average time complexity.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0433 Minimum Genetic Mutation

**MEDIUM**

A gene string can be represented by an 8-character long string, with choices from 'A', 'C', 'G', and 'T'.

Suppose we need to investigate a mutation from a gene string startGene to a gene string endGene where one mutation is defined as one single character changed in the gene string.

For example, "AACCGGTT" --> "AACCGGTA" is one mutation.

There is also a gene bank bank that records all the valid gene mutations. A gene must be in bank to make it a valid gene string.

Given the two gene strings startGene and endGene and the gene bank bank, return the minimum number of mutations needed to mutate from startGene to endGene. If there is no such a mutation, return -1.

Note that the starting point is assumed to be valid, so it might not be included in the bank.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0437 Path Sum Iii

**MEDIUM**

Given the root of a binary tree and an integer targetSum, return the number of paths where the sum of the values along the path equals targetSum.

The path does not need to start or end at the root or a leaf, but it must go downwards (i.e., traveling only from parent nodes to child nodes).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0442 Find All Duplicates In An Array

MEDIUM

Given an integer array nums of length n where all the integers of nums are in the range [1, n] and each integer appears once or twice, return an array of all the integers that appears twice.

You must write an algorithm that runs in O(n) time and uses only constant extra space.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0444 Sequence Reconstruction

MEDIUM

You are given an integer array nums of length n where nums is a permutation of the integers in the range [1, n]. You are also given a 2D integer array sequences where sequences[i] is a subsequence of nums.

Check if nums is the shortest possible and the only supersequence. The shortest supersequence is a sequence with the shortest length and has all sequences[i] as subsequences. There could be multiple valid supersequences for the given array sequences.
- For example, for sequences = [[1,2],[1,3]], there are two shortest supersequences, [1,2,3] and [1,3,2].
- While for sequences = [[1,2],[1,3],[1,2,3]], the only shortest supersequence possible is [1,2,3]. [1,2,3,4] is a possible supersequence but not the shortest.

Return true if nums is the only shortest supersequence for sequences, or false otherwise.

A subsequence is a sequence that can be derived from another sequence by deleting some or no elements without changing the order of the remaining elements.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0446 Arithmetic Slices Ii Subsequence

HARD

Given an integer array nums, return the number of all the arithmetic subsequences of nums.

A sequence of numbers is called arithmetic if it consists of at least three elements and if the difference between any two consecutive elements is the same.

- For example, [1, 3, 5, 7, 9], [7, 7, 7, 7], and [3, -1, -5, -9] are arithmetic sequences.
- For example, [1, 1, 2, 5, 7] is not an arithmetic sequence.

A subsequence of an array is a sequence that can be formed by removing some elements (possibly none) of the array.

- For example, [2,5,10] is a subsequence of [1,2,1,2,4,1,5,10].

The test cases are generated so that the answer fits in 32-bit integer.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0447 Number Of Boomerangs

MEDIUM

You are given n points in the plane that are all distinct, where points[i] = [xi, yi]. A boomerang is a tuple of points (i, j, k) such that the distance between i and j equals the distance between i and k (the order of the tuple matters).

Return the number of boomerangs.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 0448 Find All Numbers Disappeared In An Array

EASY

Given an array `nums` of `n` integers where `nums[i]` is in the range `[1, n]`, return an array of all the integers in the range `[1, n]` that do not appear in nums.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 0453 Minimum Moves To Equal Array Elements

MEDIUM

Given an integer array nums of size n, return the minimum number of moves required to make all array elements equal.

In one move, you can increment n - 1 elements of the array by 1.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 0457 Circular Array Loop

MEDIUM

You are playing a game involving a circular array of non-zero integers nums. Each nums[i] denotes the number of indices forward/backward you must move if you are located at index i:
- If nums[i] is positive, move nums[i] steps forward, and
- If nums[i] is negative, move nums[i] steps backward.

Since the array is circular, you may assume that moving forward from the last element puts you on the first element, and moving backwards from the first element puts you on the last element.

A cycle in the array consists of a sequence of indices seq of length k where:
- Following the movement rules above results in the repeating index sequence seq[0] -> seq[1] -> ... -> seq[k - 1] -> seq[0] -> ...
- Every nums[seq[j]] is either all positive or all negative.
- k > 1

Return true if there is a cycle in nums, or false otherwise.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 0460 Lfu Cache

HARD

Design and implement a data structure for a Least Frequently Used (LFU) cache.

Implement the LFUCache class:
- LFUCache(int capacity) Initializes the object with the capacity of the data structure.
- int get(int key) Gets the value of the key if the key exists in the cache. Otherwise, returns -1.
- void put(int key, int value) Update the value of the key if present, or inserts the key if not already present. When the cache reaches its capacity, it should invalidate and remove the least frequently used key before inserting a new item. For this problem, when there is a tie (i.e., two or more keys with the same frequency), the least recently used key would be invalidated.

To determine the least frequently used key, a use counter is maintained for each key in the cache. The key with the smallest use counter is the least frequently used key.

When a key is first inserted into the cache, its use counter is set to 1 (due to the put operation). The use counter for a key in the cache is incremented either a get or put operation is called on it.

The functions get and put must each run in O(1) average time complexity.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0462 Minimum Moves To Equal Array Elements Ii

**MEDIUM**

Given an integer array nums of size n, return the minimum number of moves required to make all array elements equal.

In one move, you can increment or decrement an element of the array by 1.

Test cases are designed so that the answer will fit in a 32-bit integer.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0464 Can I Win

**MEDIUM**

In the "100 game" two players take turns adding, to a running total, any integer from 1 to 10. The player who first causes the running total to reach or exceed 100 wins.

What if we change the game so that players cannot re-use integers?

For example, two players might take turns drawing from a common pool of numbers from 1 to 15 without replacement until they reach a total >= 100.

Given two integers maxChoosableInteger and desiredTotal, return true if the first player to move can force a win, otherwise, return false. Assume both players play optimally.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0465 Optimal Account Balancing

**HARD**

You are given an array of transactions transactions where transactions[i] = [fromi, toi, amounti] indicates that the person with ID = fromi gave amounti $ to the person with ID = toi.

Return the minimum number of transactions required to settle the debt.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0466 Count The Repetitions

**HARD**

We define str = [s, n] as the string str which consists of the string s concatenated n times.

- For example, str == ["abc", 3] =="abcabcabc".

We define that string s1 can be obtained from string s2 if we can remove some characters from s2 such that it becomes s1.

- For example, s1 = "abc" can be obtained from s2 = "abdbec" based on our definition by removing the bolded underlined characters.

You are given two strings s1 and s2 and two integers n1 and n2. You have the two strings str1 = [s1, n1] and str2 = [s2, n2].

Return the maximum integer m such that str = [str2, m] can be obtained from str1.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 0476 Number Complement

EASY

The complement of an integer is the integer you get when you flip all the 0's to 1's and all the 1's to 0's in its binary representation.

For example, The integer 5 is "101" in binary and its complement is "010" which is the integer 2.

Given an integer num, return its complement.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 0479 Largest Palindrome Product

HARD

Given an integer n, return the largest palindromic integer that can be represented as the product of two n-digits integers. Since the answer can be very large, return it modulo 1337.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 0489 Robot Room Cleaner

HARD

You are controlling a robot that is located somewhere in a room. The room is modeled as an m x n binary grid where 0 represents a wall and 1 represents an empty slot.

The robot starts at an unknown location in the room that is guaranteed to be empty, and you do not have access to the grid, but you can move the robot using the given API Robot.

You are tasked to use the robot to clean the entire room (i.e., clean every empty cell in the room). The robot with the four given APIs can move forward, turn left, or turn right. Each turn is 90 degrees.

When the robot tries to move into a wall cell, its bumper sensor detects the obstacle, and it stays on the current cell.

Design an algorithm to clean the entire room using the following APIs:

```
interface Robot {
// returns true if next cell is open and robot moves into the cell.
// returns false if next cell is obstacle and robot stays on the current cell.
boolean move();

// Robot will stay on the same cell after calling turnLeft/turnRight.
// Each turn will be 90 degrees.
void turnLeft();
void turnRight();

// Clean the current cell.
void clean();
}
```

Note that the initial direction of the robot will be facing up. You can assume all four edges of the grid are all surrounded by a wall.

Custom testing:
The input is only given to initialize the room and the robot's position internally. You must solve this problem "blindfolded". In other words, you must control the robot using only the four mentioned APIs without knowing the room layout and the initial robot's position.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 0490 The Maze

MEDIUM

There is a ball in a maze with empty spaces (represented as 0) and walls (represented as 1). The ball can go through the empty spaces by rolling up, down, left or right, but it won't stop rolling until hitting a wall. When the ball stops, it could choose the next direction.

Given the m x n maze, the ball's start position and the destination, where start = [startrow, startcol] and destination = [destinationrow, destinationcol], return true if the ball can stop at the destination, otherwise return false.

You may assume that the borders of the maze are all walls (see examples).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0491 Non Decreasing Subsequences

MEDIUM

Given an integer array nums, return all the different possible non-decreasing subsequences of the given array with at least two elements. You may return the answer in any order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0493 Reverse Pairs

HARD

Given an integer array nums, return the number of reverse pairs in the array.

A reverse pair is a pair (i, j) where:
- 0 <= i < j < nums.length and
- nums[i] > 2 * nums[j].

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0494 Target Sum

MEDIUM

You are given an integer array nums and an integer target.

You want to build an expression out of nums by adding one of the symbols '+' and '-' before each integer in nums and then concatenate all the integers.

For example, if nums = [2, 1], you can add a '+' before 2 and a '-' before 1 and concatenate them to build the expression "+2-1".

Return the number of different expressions that you can build, which evaluates to target.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0506 Relative Ranks

EASY

You are given an integer array score of size n, where score[i] is the score of the ith athlete in a competition. All the scores are guaranteed to be unique.

The athletes are placed based on their scores, where the 1st place athlete has the highest score, the 2nd place athlete has the 2nd highest score, and so on. The placement of each athlete determines their rank:
- The 1st place athlete's rank is "Gold Medal".

- The 2nd place athlete's rank is "Silver Medal".
- The 3rd place athlete's rank is "Bronze Medal".
- For the 4th place to the nth place athlete, their rank is their placement number (i.e., the xth place athlete's rank is "x").

Return an array answer of size n where answer[i] is the rank of the ith athlete.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0507 Perfect Number

**EASY**

A perfect number is a positive integer that is equal to the sum of its positive divisors, excluding the number itself. A divisor of an integer x is an integer that can divide x evenly.

Given an integer n, return true if n is a perfect number, otherwise return false.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0514 Freedom Trail

**HARD**

In the video game Fallout 4, the quest "Road to Freedom" requires players to reach a metal dial called the "Freedom Trail Ring" and use the dial to spell a specific keyword to open the door.

Given a string ring that represents the code engraved on the outer ring and another string key that represents the keyword that needs to be spelled, return the minimum number of steps to spell all the characters in the keyword.

Initially, the first character of the ring is aligned at the "12:00" direction. You should spell all the characters in key one by one by rotating ring clockwise or anticlockwise to make each character of the string key aligned at the "12:00" direction and then by pressing the center button.

At the stage of rotating the ring to spell the key character key[i]:
- You can rotate the ring clockwise or anticlockwise by one place, which counts as one step. The final purpose of the rotation is to align one of ring's characters at the "12:00" direction, where this character must equal key[i].
- If the character key[i] has been aligned at the "12:00" direction, press the center button to spell, which also counts as one step. After the pressing, you could begin to spell the next character in the key (next stage). Otherwise, you have finished all the spelling.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0519 Random Flip Matrix

**MEDIUM**

There is an m x n binary grid matrix with all the values set 0 initially. Design an algorithm to randomly pick an index (i, j) where matrix[i][j] == 0 and flips it to 1. All the indices (i, j) where matrix[i][j] == 0 should be equally likely to be returned.

Optimize your algorithm to minimize the number of calls made to the built-in random function of your language and optimize the time and space complexity.

Implement the Solution class:
- Solution(int m, int n) Initializes the object with the size of the binary matrix m and n.
- int[] flip() Returns a random index [i, j] of the matrix where matrix[i][j] == 0 and flips it to 1.
- void reset() Resets all the values of the matrix to be 0.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0525 Contiguous Array

MEDIUM

Given a binary array nums, return the maximum length of a contiguous subarray with an equal number of 0 and 1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0532 K Diff Pairs In An Array

MEDIUM

Given an array of integers nums and an integer k, return the number of unique k-diff pairs in the array.

A k-diff pair is an integer pair (nums[i], nums[j]), where the following are true:
- 0 <= i, j < nums.length
- i != j
- |nums[i] - nums[j]| == k

Notice that |val| denotes the absolute value of val.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0533 Lonely Pixel Ii

MEDIUM

Given an m x n picture consisting of black 'B' and white 'W' pixels and an integer target, return the number of black lonely pixels.

A black lonely pixel is a character 'B' that located at a specific position (r, c) where:
- Row r and column c both contain exactly target black pixels.
- For all rows that have a black pixel at column c, they should be exactly the same as row r.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0535 Encode And Decode Tinyurl

MEDIUM

TinyURL is a URL shortening service where you enter a URL such as

There is no restriction on how your encode/decode algorithm should work. You just need to ensure that a URL can be encoded to a tiny URL and the tiny URL can be decoded to the original URL.

Implement the Solution class:
- Solution() Initializes the object of the system.
- String encode(String longUrl) Returns a tiny URL for the given longUrl.
- String decode(String shortUrl) Returns the original long URL for the given shortUrl.

It is guaranteed that the given shortUrl was encoded by the same object.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0537 Complex Number Multiplication

MEDIUM

A complex number can be represented as a string on the form "real+imaginaryi" where:
- real is the real part and is an integer in the range [-100, 100].
- imaginary is the imaginary part and is an integer in the range [-100, 100].
- i2 == -1.

Given two complex numbers num1 and num2 as strings, return a string of the complex number that represents their multiplications.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0539 Minimum Time Difference

**MEDIUM**

Given a list of 24-hour clock time points in "HH:MM" format, return the minimum minutes difference between any two time-points in the list.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0542 01 Matrix

**MEDIUM**

Given an m x n binary matrix mat, return the distance of the nearest 0 for each cell.

The distance between two adjacent cells is 1.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0548 Split Array With Equal Sum

**HARD**

Given an integer array nums of length n, return true if there is a triplet (i, j, k) which satisfies the following conditions:
- 0 < i, i + 1 < j, j + 1 < k < n - 1
- The sum of subarrays (0, i - 1), (i + 1, j - 1), (j + 1, k - 1) and (k + 1, n - 1) is equal.

A subarray (l, r) represents a slice of the original array starting from the element indexed l to the element indexed r.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0554 Brick Wall

**MEDIUM**

There is a rectangular brick wall in front of you with n rows of bricks. The ith row has some number of bricks each of the same height (i.e., one unit) but they can be of different widths. The total width of each row is the same.

Draw a vertical line from the top to the bottom and cross the least bricks. If your line goes through the edge of a brick, then the brick is not considered as crossed. You cannot draw a line just along one of the two vertical edges of the wall, in which case the line will obviously cross no bricks.

Given the 2D array wall that contains the information about the wall, return the minimum number of crossed bricks after drawing such a vertical line.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0561 Array Partition

**EASY**

Given an integer array nums of 2n integers, group these integers into n pairs (a1, b1), (a2, b2), ..., (an, bn) such that the sum of min(ai, bi) for all i is maximized.
Return the maximized sum.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0562 Longest Line Of Consecutive One In Matrix

MEDIUM

Given an m x n binary matrix mat, return the length of the longest line of consecutive one in the matrix.

The line could be horizontal, vertical, diagonal, or anti-diagonal.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0565 Array Nesting

MEDIUM

A zero-indexed array A of length N contains all integers from 0 to N-1. Find and return the longest length of set S, where S[i] = {A[i], A[A[i]], A[A[A[i]]], ... } subjected to the rule below.

Suppose the first element in S starts with the selection of element A[i] of index = i, the next element in S should be A[A[i]], and then A[A[A[i]]]… By that analogy, we stop adding right before a duplicate element occurs in S.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0566 Reshape The Matrix

EASY

In MATLAB, there is a handy function called reshape which can reshape an m x n matrix into a new one with a different size r x c keeping its original data.

You are given an m x n matrix mat and two integers r and c representing the number of rows and the number of columns of the wanted reshaped matrix.

The reshaped matrix should be filled with all the elements of the original matrix in the same row-traversing order as they were.

If the reshape operation with given parameters is possible and legal, output the new reshaped matrix; Otherwise, output the original matrix.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0575 Distribute Candies

EASY

Alice has n candies, where the ith candy is of type candyType[i]. Alice noticed that she started to gain weight, so she visited a doctor.

The doctor advised Alice to only eat n / 2 of the candies she has (n is always even). Alice likes her candies very much, and she wants to eat the maximum number of different types of candies while still following the doctor's advice.

Given the integer array candyType of length n, return the maximum number of different types of candies she can eat if she only eats n / 2 of them.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0582 Kill Process

MEDIUM

You have n processes forming a rooted tree structure. You are given two integer arrays pid

and ppid, where pid[i] is the ID of the ith process and ppid[i] is the ID of the ith process's parent process.

Each process has only one parent process but may have multiple children processes. Only one process has ppid[i] = 0, which means this process has no parent process (the root of the tree).

When a process is killed, all of its children processes will also be killed.

Given an integer kill representing the ID of a process you want to kill, return a list of the IDs of the processes that will be killed. You may return the answer in any order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0587 Erect The Fence

**HARD**

You are given an array trees where trees[i] = [xi, yi] represents the location of a tree in the garden.

Fence the entire garden using the minimum length of rope, as it is expensive. The garden is well-fenced only if all the trees are enclosed.

Return the coordinates of trees that are exactly located on the fence perimeter. You may return the answer in any order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0588 Design In Memory File System

**HARD**

Design a data structure that simulates an in-memory file system.

Implement the FileSystem class:
- FileSystem() Initializes the object of the system.
- List<String> ls(String path)
- If path is a file path, returns a list that only contains this file's name.
- If path is a directory path, returns the list of file and directory names in this directory.
- The answer should in lexicographic order.
- void mkdir(String path) Makes a new directory according to the given path. The given directory path does not exist. If the middle directories in the path do not exist, you should create them as well.
- void addContentToFile(String filePath, String content)
- If filePath does not exist, creates that file containing given content.
- If filePath already exists, appends the given content to original content.
- String readContentFromFile(String filePath) Returns the content in the file at filePath.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0593 Valid Square

**MEDIUM**

Given the coordinates of four points in 2D space p1, p2, p3 and p4, return true if the four points construct a square.

The coordinate of a point pi is represented as [xi, yi]. The input is not given in any order.

A valid square has four equal sides with positive length and four equal angles (90-degree angles).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0594 Longest Harmonious Subsequence

EASY

We define a harmonious array as an array where the difference between its maximum value and its minimum value is exactly 1.

Given an integer array nums, return the length of its longest harmonious subsequence among all its possible subsequences.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0599 Minimum Index Sum Of Two Lists

EASY

Suppose Andy and Doris want to choose a restaurant for dinner, and they both have a list of favorite restaurants represented by strings.

You need to help them find out their common interest with the least list index sum. If there is a choice tie between answers, output all of them with no order requirement. You could assume there always exists an answer.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0609 Find Duplicate File In System

MEDIUM

Given a list paths of directory info, including the directory path, and all the files with contents in this directory, return all the duplicate files in the file system in terms of their paths. You may return the answer in any order.

A group of duplicate files consists of at least two files that have the same content.

A single directory info string in the input list has the following format:
- "root/d1/d2/.../dm f1.txt(f1_content) f2.txt(f2_content) ... fn.txt(fn_content)"

It means there are n files (f1.txt, f2.txt ... fn.txt) with content (f1_content, f2_content ... fn_content) respectively in the directory "root/d1/d2/.../dm". Note that n >= 1 and m >= 0. If m = 0, it means the directory is just the root directory.

The output is a list of groups of duplicate file paths. For each group, it contains all the file paths of the files that have the same content. A file path is a string that has the following format:
- "directory_path/file_name.txt"

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0611 Valid Triangle Number

MEDIUM

Given an integer array nums, return the number of triplets chosen from the array that can make triangles if we take them as side lengths of a triangle.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0624 Maximum Distance In Arrays

MEDIUM

You are given m arrays, where each array is sorted in ascending order.

You can pick up two integers from two different arrays (each array picks one) and calculate

the distance. We define the distance between two integers a and b to be their absolute difference |a - b|.

Return the maximum distance.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0628 Maximum Product Of Three Numbers

**EASY**

Given an integer array, find three numbers whose
product is maximum and output the maximum product.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0629 K Inverse Pairs Array

**HARD**

For an integer array nums, an inverse pair is a pair of integers [i, j] where 0 <= i < j < nums.length and nums[i] > nums[j].

Given two integers n and k, return the number of different arrays consisting of numbers from 1 to n such that there are exactly k inverse pairs. Since the answer can be huge, return it modulo 109 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0631 Design Excel Sum Formula

**HARD**

Design the basic function of Excel and implement the function of the sum formula.

Implement the Excel class:
- Excel(int height, char width) Initializes the object with the height and the width of the sheet. The sheet is an integer matrix mat of size height x width with the row index in the range [1, height] and the column index in the range ['A', width]. All the values should be zero initially.
- void set(int row, char column, int val) Changes the value at mat[row][column] to be val.
- int get(int row, char column) Returns the value at mat[row][column].
- int sum(int row, char column, List<String> numbers) Sets the value at mat[row][column] to be the sum of cells represented by numbers and returns the value at mat[row][column]. This sum formula should exist until this cell is overlapped by another value or another sum formula. numbers[i] could be on the format:
- "ColRow" that represents a single cell.
- For example, "F7" represents the cell mat[7]['F'].
- "ColRow1:ColRow2" that represents a range of cells. The range will always be a rectangle where "ColRow1" represent the position of the top-left cell, and "ColRow2" represents the position of the bottom-right cell.
- For example, "B3:F7" represents the cells mat[i][j] for 3 <= i <= 7 and 'B' <= j <= 'F'.

Note: You could assume that there will not be any circular sum reference.

For example, mat[1]['A'] == sum(1, "B") and mat[1]['B'] == sum(1, "A").

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0632 Smallest Range Covering Elements From K Lists

**HARD**

You have k lists of sorted integers in non-decreasing order. Find the smallest range that

includes at least one number from each of the k lists.

We define the range [a, b] is smaller than range [c, d] if b - a < d - c or a < c if b - a == d - c.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0633 Sum Of Square Numbers

**MEDIUM**

Given a non-negative integer c, decide whether there're two integers a and b such that a2 + b2 = c.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0634 Find The Derangement Of An Array

**MEDIUM**

In combinatorial mathematics, a derangement is a permutation of the elements of a set, such that no element appears in its original position.

You are given an integer n. There is originally an array consisting of n integers from 1 to n in ascending order, return the number of derangements it can generate. Since the answer may be huge, return it modulo 109 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0638 Shopping Offers

**MEDIUM**

In LeetCode Store, there are n items to sell. Each item has a price. However, there are some special offers, and a special offer consists of one or more different kinds of items with a sale price.

You are given an integer array price where price[i] is the price of the ith item, and an integer array needs where needs[i] is the number of pieces of the ith item you want to buy.

You are also given an array special where special[i] is of size n + 1 where special[i][j] is the number of pieces of the jth item in the ith offer and special[i][n] (i.e., the last integer in the array) is the price of the ith offer.

Return the lowest price you have to pay for exactly certain items as given, where you could make optimal use of the special offers. You are not allowed to buy more items than you want, even if that would lower the overall price. You could use any of the special offers as many times as you want.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0645 Set Mismatch

**MEDIUM**

You have a set of integers s, which originally contains all the numbers from 1 to n. Unfortunately, due to some error, one of the numbers in s got duplicated to another number in the set, which results in repetition of one number and loss of another number.

You are given an integer array nums representing the data status of this set after the error.

Find the number that occurs twice and the number that is missing and return them in the form of an array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0646 Maximum Length Of Pair Chain

MEDIUM

You are given an array of n pairs pairs where pairs[i] = [lefti, righti] and lefti < righti.

A pair p2 = [c, d] follows a pair p1 = [a, b] if b < c. A chain of pairs can be formed in this fashion.

Return the length longest chain which can be formed.

You do not need to use up all the given intervals. You can select pairs in any order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0658 Find K Closest Elements

MEDIUM

Given a sorted integer array arr, two integers k and x, return the k closest integers to x in the array. The result should also be sorted in ascending order.

An integer a is closer to x than an integer b if:
- |a - x| < |b - x|, or
- |a - x| == |b - x| and a < b

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0659 Split Array Into Consecutive Subsequences

MEDIUM

You are given an integer array nums that is sorted in non-decreasing order.

Determine if it is possible to split nums into one or more subsequences such that both of the following conditions are true:
- Each subsequence is a consecutive increasing sequence (i.e. each integer is exactly one more than the previous integer).
- All subsequences have a length of 3 or more.

Return true if you can split nums according to the above conditions, or false otherwise.

A subsequence of an array is a new array that is formed from the original array by deleting some (can be none) of the elements without disturbing the relative positions of the remaining elements. (i.e., [1,3,5] is a subsequence of [1,2,3,4,5] while [1,3,2] is not).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0665 Non Decreasing Array

MEDIUM

Given an array nums with n integers, your task is to check if it could become non-decreasing by modifying at most one element.

We define an array is non-decreasing if nums[i] <= nums[i + 1] holds for every i (0-based) such that (0 <= i <= n - 2).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0666 Path Sum Iv

MEDIUM

If the depth of a tree is smaller than 5, then this tree can be represented by an array of three-digit integers. You are given an ascending array nums consisting of three-digit

integers representing a binary tree with a depth smaller than 5, where for each integer:
- The hundreds digit represents the depth d of this node, where 1 <= d <= 4.
- The tens digit represents the position p of this node within its level, where 1 <= p <= 8, corresponding to its position in a full binary tree.
- The units digit represents the value v of this node, where 0 <= v <= 9.

Return the sum of all paths from the root towards the leaves.

It is guaranteed that the given array represents a valid connected binary tree.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0668 Kth Smallest Number In Multiplication Table

**HARD**

Nearly everyone has used the Multiplication Table. The multiplication table of size m x n is an integer matrix mat where mat[i][j] == i * j (1-indexed).

Given three integers m, n, and k, return the kth smallest element in the m x n multiplication table.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0673 Number Of Longest Increasing Subsequence

**MEDIUM**

Given an integer array nums, return the number of longest increasing subsequences.

Notice that the sequence has to be strictly increasing.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0676 Implement Magic Dictionary

**MEDIUM**

Design a data structure that is initialized with a list of different words. Provided a string, you should determine if you can change exactly one character in this string to match any word in the data structure.

Implement the MagicDictionary class:
- MagicDictionary() Initializes the object.
- void buildDict(String[] dictionary) Sets the data structure with an array of distinct strings dictionary.
- bool search(String searchWord) Returns true if you can change exactly one character in searchWord to match any string in the data structure, otherwise returns false.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0677 Map Sum Pairs

**MEDIUM**

Design a map that allows you to do the following:
- Maps a string key to a given value.
- Returns the sum of the values that have a key with a prefix equal to a given string.

Implement the MapSum class:
- MapSum() Initializes the MapSum object.
- void insert(String key, int val) Inserts the key-val pair into the map. If the key already existed, the original key-value pair will be overridden to the new one.
- int sum(string prefix) Returns the sum of all the pairs' value whose key starts with the prefix.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 0681 Next Closest Time

MEDIUM

Given a time represented in the format "HH:MM", form the next closest time by reusing the current digits. There is no limit on how many times a digit can be reused.

You may assume the given input string is always valid. For example, "01:34", "12:09" are all valid. "1:34", "12:9" are all invalid.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 0684 Redundant Connection

MEDIUM

In this problem, a tree is an undirected graph that is connected and has no cycles.

You are given a graph that started as a tree with n nodes labeled from 1 to n, with one additional edge added. The added edge has two different vertices chosen from 1 to n, and was not an edge that already existed. The graph is represented as an array edges of length n where edges[i] = [ai, bi] indicates that there is an edge between nodes ai and bi in the graph.

Return an edge that can be removed so that the resulting graph is a tree of n nodes. If there are multiple answers, return the answer that occurs last in the input.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 0688 Knight Probability In Chessboard

MEDIUM

On an n x n chessboard, a knight starts at the cell (row, column) and attempts to make exactly k moves. The rows and columns are 0-indexed, so the top-left cell is (0, 0), and the bottom-right cell is (n - 1, n - 1).

A chess knight has eight possible moves it can make, as illustrated below. Each move is two cells in a cardinal direction, then one cell in an orthogonal direction.

Each time the knight is to move, it chooses one of eight possible moves uniformly at random (even if the piece would go off the chessboard) and moves there.

The knight continues moving until it has made exactly k moves or has moved off the chessboard.

Return the probability that the knight remains on the board after it has stopped moving.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 0693 Binary Number With Alternating Bits

EASY

Given a positive integer, check whether it has alternating bits: namely, if two adjacent bits will always have different values.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 0697 Degree Of An Array

EASY

Given a non-empty array of non-negative integers nums, the degree of this array is

defined as the maximum frequency of any one of its elements.

Your task is to find the smallest possible length of a (contiguous) subarray of nums, that has the same degree as nums.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0699 Falling Squares

HARD

There are several squares being dropped onto the X-axis of a 2D plane.

You are given a 2D integer array positions where positions[i] = [lefti, sideLengthi] represents the ith square with a side length of sideLengthi that is dropped with its left edge aligned with X-coordinate lefti.

Each square is dropped one at a time from a height above any landed squares. It then falls downward (negative Y direction) until it either lands on the top side of another square or on the X-axis. A square brushing the left/right side of another square does not count as landing on it. Once it lands, it freezes in place and cannot be moved.

After each square is dropped, you must record the height of the current tallest stack of squares.

Return an integer array ans where ans[i] represents the height described above after dropping the ith square.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0705 Design Hashset

EASY

Design a HashSet without using any built-in hash table libraries.

Implement MyHashSet class:
- void add(key) Inserts the value key into the HashSet.
- bool contains(key) Returns whether the value key exists in the HashSet or not.
- void remove(key) Removes the value key in the HashSet. If key does not exist in the HashSet, do nothing.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0706 Design Hashmap

EASY

Design a HashMap without using any built-in hash table libraries.
Implement the MyHashMap class:
- MyHashMap() initializes the object with an empty map.
- void put(int key, int value) inserts a (key, value) pair into the HashMap. If the key already exists in the map, update the corresponding value.
- int get(int key) returns the value to which the specified key is mapped, or -1 if this map contains no mapping for the key.
- void remove(key) removes the key and its corresponding value if the map contains the mapping for the key.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0710 Random Pick With Blacklist

HARD

You are given an integer n and an array of unique integers blacklist. Design an algorithm to pick a random integer in the range [0, n - 1] that is not in blacklist. Any integer that is in the mentioned range and not in blacklist should be equally likely to be returned.

Optimize your algorithm such that it minimizes the number of calls to the built-in random function of your language.

Implement the Solution class:
- Solution(int n, int[] blacklist) Initializes the object with the integer n and the blacklisted integers blacklist.
- int pick() Returns a random integer in the range [0, n - 1] and not in blacklist.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0719 Find K Th Smallest Pair Distance

HARD

The distance of a pair of integers a and b is defined as the absolute difference between a and b.

Given an integer array nums and an integer k, return the kth smallest distance among all the pairs nums[i] and nums[j] where 0 <= i < j < nums.length.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0721 Accounts Merge

MEDIUM

Given a list of accounts where each element accounts[i] is a list of strings, where the first element accounts[i][0] is a name, and the rest of the elements are emails representing emails of the account.

Now, we would like to merge these accounts. Two accounts definitely belong to the same person if there is some common email to both accounts. Note that even if two accounts have the same name, they may belong to different people as people could have the same name. A person can have any number of accounts initially, but all of their accounts definitely have the same name.

After merging the accounts, return the accounts in the following format: the first element of each account is the name, and the rest of the elements are emails in sorted order. The accounts themselves can be returned in any order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0726 Number Of Atoms

HARD

Given a string formula representing a chemical formula, return the count of each atom.

The atomic element always starts with an uppercase character, then zero or more lowercase letters, representing the name.

One or more digits representing that element's count may follow if the count is greater than 1. If the count is 1, no digits will follow.
- For example, "H2O" and "H2O2" are possible, but "H1O2" is impossible.

Two formulas are concatenated together to produce another formula.
- For example, "H2O2He3Mg4" is also a formula.

A formula placed in parentheses, and a count (optionally added) is also a formula.

- For example, "(H2O2)" and "(H2O2)3" are formulas.

Return the count of all elements as a string in the following form: the first name (in sorted order), followed by its count (if that count is more than 1), followed by the second name (in sorted order), followed by its count (if that count is more than 1), and so on.

The test cases are generated so that all the values in the output fit in a 32-bit integer.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0728 Self Dividing Numbers

**EASY**

A self-dividing number is a number that is divisible by every digit it contains.
- For example, 128 is a self-dividing number because 128 % 1 == 0, 128 % 2 == 0, and 128 % 8 == 0.

A self-dividing number is not allowed to contain the digit zero.

Given two integers left and right, return a list of all the self-dividing numbers in the range [left, right] (both inclusive).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0730 Count Different Palindromic Subsequences

**HARD**

Given a string s, return the number of different non-empty palindromic subsequences in s. Since the answer may be very large, return it modulo 109 + 7.

A subsequence of a string is obtained by deleting zero or more characters from the string.

A sequence is palindromic if it is equal to the sequence reversed.

Two sequences a1, a2, ... and b1, b2, ... are different if there is some i for which ai != bi.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0736 Parse Lisp Expression

**HARD**

You are given a string expression representing a Lisp-like expression to return the integer value of.

The syntax for these expressions is given as follows.
- An expression is either an integer, let expression, add expression, mult expression, or an assigned variable. Expressions always evaluate to a single integer.
- (An integer could be positive or negative.)
- A let expression takes the form "(let v1 e1 v2 e2 ... vn en expr)", where let is always the string "let", then there are one or more pairs of alternating variables and expressions, meaning that the first variable v1 is assigned the value of the expression e1, the second variable v2 is assigned the value of the expression e2, and so on sequentially; and then the value of this let expression is the value of the expression expr.
- An add expression takes the form "(add e1 e2)" where add is always the string "add", there are always two expressions e1, e2 and the result is the addition of the evaluation of e1 and the evaluation of e2.
- A mult expression takes the form "(mult e1 e2)" where mult is always the string "mult", there are always two expressions e1, e2 and the result is the multiplication of the evaluation of e1 and the evaluation of e2.
- For this question, we will use a smaller subset of variable names. A variable starts with a lowercase letter, then zero or more lowercase letters or digits. Additionally, for your convenience, the names "add", "let", and "mult" are protected and will never be used as

variable names.
- Finally, there is the concept of scope. When an expression of a variable name is evaluated, within the context of that evaluation, the innermost scope (in terms of parentheses) is checked first for the value of that variable, and then outer scopes are checked sequentially. It is guaranteed that every expression is legal. Please see the examples for more details on the scope.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0747 Largest Number At Least Twice Of Others

**EASY**

You are given an integer array nums where the largest integer is unique.

Determine whether the largest element in the array is at least twice as much as every other number in the array. If it is, return the index of the largest element, or return -1 otherwise.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0749 Contain Virus

**HARD**

A virus is spreading rapidly, and your task is to quarantine the infected area by installing walls.

The world is modeled as an m x n binary grid isInfected, where isInfected[i][j] == 0 represents uninfected cells, and isInfected[i][j] == 1 represents cells contaminated with the virus. A wall (and only one wall) can be installed between any two 4-directionally adjacent cells, on the shared boundary.

Every night, the virus spreads to all neighboring cells in all four directions unless blocked by a wall. Resources are limited. Each day, you can install walls around only one region (i.e., the affected area (continuous block of infected cells) that threatens the most uninfected cells the following night). There will never be a tie.

Return the number of walls used to quarantine all the infected regions. If the world will become fully infected, return the number of walls used.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0750 Number Of Corner Rectangles

**MEDIUM**

Given an m x n integer matrix grid where each entry is only 0 or 1, return the number of corner rectangles.

A corner rectangle is four distinct 1's on the grid that forms an axis-aligned rectangle. Note that only the corners need to have the value 1. Also, all four 1's used must be distinct.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0752 Open The Lock

**MEDIUM**

You have a lock in front of you with 4 circular wheels. Each wheel has 10 slots: '0', '1', '2', '3', '4', '5', '6', '7', '8', '9'. The wheels can rotate freely and wrap around: for example we can turn '9' to be '0', or '0' to be '9'. Each move consists of turning one wheel one slot.

The lock initially starts at '0000', a string representing the state of the 4 wheels.

You are given a list of deadends dead ends, meaning if the lock displays any of these codes, the wheels of the lock will stop turning and you will be unable to open it.

Given a target representing the value of the wheels that will unlock the lock, return the minimum total number of turns required to open the lock, or -1 if it is impossible.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0753 Cracking The Safe

**HARD**

There is a safe protected by a password. The password is a sequence of n digits where each digit can be in the range [0, k - 1].

The safe has a peculiar way of checking the password. When you enter in a sequence, it checks the most recent n digits that were entered each time you type a digit.

For example, the correct password is "345" and you enter in "012345":
- After typing 0, the most recent 3 digits is "0", which is incorrect.
- After typing 1, the most recent 3 digits is "01", which is incorrect.
- After typing 2, the most recent 3 digits is "012", which is incorrect.
- After typing 3, the most recent 3 digits is "123", which is incorrect.
- After typing 4, the most recent 3 digits is "234", which is incorrect.
- After typing 5, the most recent 3 digits is "345", which is correct and the safe unlocks.

Return any string of minimum length that will unlock the safe at some point of entering it.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0754 Reach A Number

**MEDIUM**

You are standing at position 0 on an infinite number line. There is a destination at position target.

You can make some number of moves numMoves so that:
- On each move, you can either go left or right.
- During the ith move (starting from i == 1 to i == numMoves), you take i steps in the chosen direction.

Given the integer target, return the minimum number of moves required (i.e., the minimum numMoves) to reach the destination.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0756 Pyramid Transition Matrix

**MEDIUM**

You are stacking blocks to form a pyramid. Each block has a color, which is represented by a single letter. Each row of blocks contains one less block than the row beneath it and is centered on top.

To make the pyramid aesthetically pleasing, there are only specific triangular patterns that are allowed. A triangular pattern consists of a single block stacked on top of two blocks. The patterns are given as a list of three-letter strings allowed, where the first two characters of a pattern represent the left and right bottom blocks respectively, and the third character is the top block.

For example, "ABC" represents a triangular pattern with a 'C' block stacked on top of an 'A' (left) and 'B' (right) block. Note that this is different from "BAC" where 'B' is on the left bottom and 'A' is on the right bottom.

You start with a bottom row of blocks bottom, given as a single string, that you must use as

the base of the pyramid.

Given bottom and allowed, return true if you can build the pyramid all the way to the top such that every triangular pattern in the pyramid is in allowed, or false otherwise.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0757 Set Intersection Size At Least Two

**HARD**

You are given a 2D integer array intervals where intervals[i] = [starti, endi] represents all the integers from starti to endi inclusively.

A containing set is an array nums where each interval from intervals has at least two integers in nums.

For example, if intervals = [[1,3], [3,7], [8,9]], then [1,2,4,7,8,9] and [2,3,4,8,9] are containing sets.

Return the minimum possible size of a containing set.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0762 Prime Number Of Set Bits In Binary Representation

**EASY**

Given two integers left and right, return the count of numbers in the inclusive range [left, right] having a prime number of set bits in their binary representation.

Recall that the number of set bits an integer has is the number of 1's present when written in binary.

For example, 21 written in binary is 10101, which has 3 set bits.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0763 Partition Labels

**MEDIUM**

You are given a string s. We want to partition the string into as many parts as possible so that each letter appears in at most one part. For example, the string "ababcc" can be partitioned into ["abab", "cc"], but partitions such as ["aba", "bcc"] or ["ab", "ab", "cc"] are invalid.

Note that the partition is done so that after concatenating all the parts in order, the resultant string should be s.

Return a list of integers representing the size of these parts.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0766 Toeplitz Matrix

**EASY**

Given an m x n matrix, return true if the matrix is Toeplitz. Otherwise, return false.

A matrix is Toeplitz if every diagonal from top-left to bottom-right has the same elements.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0771 Jewels And Stones

EASY

You're given strings jewels representing the types of stones that are jewels, and stones representing the stones you have. Each character in stones is a type of stone you have. You want to know how many of the stones you have are also jewels.

Letters are case sensitive, so "a" is considered a different type of stone from "A".

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0773 Sliding Puzzle

HARD

On an 2 x 3 board, there are five tiles labeled from 1 to 5, and an empty square represented by

The state of the board is solved if and only if the board is [[1,2,3],[4,5,0]].

Given the puzzle board board, return the least number of moves required so that the state of the board is solved. If it is impossible for the state of the board to be solved, return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0792 Number Of Matching Subsequences

MEDIUM

Given a string s and an array of strings words, return the number of words[i] that is a subsequence of s.

A subsequence of a string is a new string generated from the original string with some characters (can be none) deleted without changing the relative order of the remaining characters.

For example, "ace" is a subsequence of "abcde".

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0805 Split Array With Same Average

HARD

You are given an integer array nums.

You should move each element of nums into one of the two arrays A and B such that A and B are non-empty, and average(A) == average(B).

Return true if it is possible to achieve that and false otherwise.

Note that for an array arr, average(arr) is the sum of all the elements of arr over the length of arr.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0808 Soup Servings

MEDIUM

There are two types of soup: type A and type B. Initially, we have n ml of each type of soup. There are four kinds of operations:

When we serve some soup, we give it to someone, and we no longer have it. Each turn, we will choose from the four operations with an equal probability 0.25. If the remaining volume of soup is not enough to complete the operation, we will serve as much as possible. We stop once

we no longer have some quantity of both types of soup.

Note that we do not have an operation where all 100 ml's of soup B are used first.

Return the probability that soup A will be empty first, plus half the probability that A and B become empty at the same time. Answers within 10-5 of the actual answer will be accepted.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0811 Subdomain Visit Count

**MEDIUM**

A website domain "discuss.leetcode.com" consists of various subdomains. At the top level, we have "com", at the next level, we have "leetcode.com" and at the lowest level, "discuss.leetcode.com". When we visit a domain like "discuss.leetcode.com", we will also visit the parent domains "leetcode.com" and "com" implicitly.

A count-paired domain is a domain that has one of the two formats "rep d1.d2.d3" or "rep d1.d2" where rep is the number of visits to the domain and d1.d2.d3 is the domain itself.

For example, "9001 discuss.leetcode.com" is a count-paired domain that indicates that discuss.leetcode.com was visited 9001 times.

Given an array of count-paired domains cpdomains, return an array of the count-paired domains of each subdomain in the input. You may return the answer in any order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0813 Largest Sum Of Averages

**MEDIUM**

You are given an integer array nums and an integer k. You can partition the array into at most k non-empty adjacent subarrays. The score of a partition is the sum of the averages of each subarray.

Note that the partition must use every integer in nums, and that the score is not necessarily an integer.

Return the maximum score you can achieve of all the possible partitions. Answers within 10-6 of the actual answer will be accepted.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0815 Bus Routes

**HARD**

You are given an array routes representing bus routes where routes[i] is a bus route that the ith bus repeats forever.

For example, if routes[0] = [1, 5, 7], this means that the 0th bus travels in the sequence 1 -> 5 -> 7 -> 1 -> 5 -> 7 -> 1 -> ... forever.

You will start at the bus stop source (You are not on any bus initially), and you want to go to the bus stop target. You can travel between bus stops by buses only.

Return the least number of buses you must take to travel from source to target.

Return -1 if it is not possible.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0822 Card Flipping Game

MEDIUM

You are given two 0-indexed integer arrays fronts and backs of length n, where the ith card has the positive integer fronts[i] printed on the front and backs[i] printed on the back. Initially, each card is placed on a table such that the front number is facing up and the other is facing down. You may flip over any number of cards (possibly zero).

After flipping the cards, an integer is considered good if it is facing down on some card and not facing up on any card.

Return the minimum possible good integer after flipping the cards. If there are no good integers, return 0.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0829 Consecutive Numbers Sum

HARD

Given an integer n, return the number of ways you can write n as the sum of consecutive positive integers.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0835 Image Overlap

MEDIUM

You are given two images, img1 and img2, represented as binary, square matrices of size n x n. A binary matrix has only 0s and 1s as values.

We translate one image however we choose by sliding all the 1 bits left, right, up, and/or down any number of units. We then place it on top of the other image. We can then calculate the overlap by counting the number of positions that have a 1 in both images.

Note also that a translation does not include any kind of rotation. Any 1 bits that are translated outside of the matrix borders are erased.

Return the largest possible overlap.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0840 Magic Squares In Grid

MEDIUM

A 3 x 3 magic square is a 3 x 3 grid filled with distinct numbers from 1 to 9 such that each row, column, and both diagonals all have the same sum.

Given a row x col grid of integers, how many 3 x 3 magic square subgrids are there?

Note: while a magic square can only contain numbers from 1 to 9, grid may contain numbers up to 15.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0841 Keys And Rooms

MEDIUM

There are n rooms labeled from 0 to n - 1 and all the rooms are locked except for room 0. Your goal is to visit all the rooms. However, you cannot enter a locked room without having its key.

When you visit a room, you may find a set of distinct keys in it. Each key has a number on it, denoting which room it unlocks, and you can take all of them with you to unlock the other rooms.

Given an array rooms where rooms[i] is the set of keys that you can obtain if you visited room i, return true if you can visit all the rooms, or false otherwise.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0845 Longest Mountain In Array

**MEDIUM**

You may recall that an array arr is a mountain array if and only if:
- arr.length >= 3
- There exists some index i (0-indexed) with 0 < i < arr.length - 1 such that:
- arr[0] < arr[1] < ... < arr[i - 1] < arr[i]
- arr[i] > arr[i + 1] > ... > arr[arr.length - 1]

Given an integer array arr, return the length of the longest subarray, which is a mountain. Return 0 if there is no mountain subarray.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0846 Hand Of Straights

**MEDIUM**

Alice has some number of cards and she wants to rearrange the cards into groups so that each group is of size groupSize, and consists of groupSize consecutive cards.

Given an integer array hand where hand[i] is the value written on the ith card and an integer groupSize, return true if she can rearrange the cards, or false otherwise.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0850 Rectangle Area Ii

**HARD**

You are given a 2D array of axis-aligned rectangles. Each rectangle[i] = [xi1, yi1, xi2, yi2] denotes the ith rectangle where (xi1, yi1) are the coordinates of the bottom-left corner, and (xi2, yi2) are the coordinates of the top-right corner.

Calculate the total area covered by all rectangles in the plane. Any area covered by two or more rectangles should only be counted once.

Return the total area. Since the answer may be too large, return it modulo 109 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0852 Peak Index In A Mountain Array

**MEDIUM**

You are given an integer mountain array arr of length n where the values increase to a peak element and then decrease.

Return the index of the peak element.

Your task is to solve it in O(log(n)) time complexity.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0861 Score After Flipping Matrix

MEDIUM

You are given an m x n binary matrix grid.

A move consists of choosing any row or column and toggling each value in that row or column (i.e., changing all 0's to 1's, and all 1's to 0's).

Every row of the matrix is interpreted as a binary number, and the score of the matrix is the sum of these numbers.

Return the highest possible score after making any number of moves (including zero moves).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0867 Transpose Matrix

EASY

Given a matrix A, return the transpose of A.

The transpose of a matrix is the matrix flipped over it's main diagonal, switching the row and column indices of the matrix.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0871 Minimum Number Of Refueling Stops

HARD

A car travels from a starting position to a destination which is target miles east of the starting position.

There are gas stations along the way. The gas stations are represented as an array stations where stations[i] = [positioni, fueli] indicates that the ith gas station is positioni miles east of the starting position and has fueli liters of gas.

The car starts with an infinite tank of gas, which initially has startFuel liters of fuel in it. It uses one liter of gas per one mile that it drives. When the car reaches a gas station, it may stop and refuel, transferring all the gas from the station into the car.

Return the minimum number of refueling stops the car must make in order to reach its destination. If it cannot reach the destination, return -1.

Note that if the car reaches a gas station with 0 fuel left, the car can still refuel there. If the car reaches the destination with 0 fuel left, it is still considered to have arrived.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0874 Walking Robot Simulation

MEDIUM

A robot on an infinite XY-plane starts at point (0, 0) facing north. The robot receives an array of integers commands, which represents a sequence of moves that it needs to execute. There are only three possible types of instructions the robot can receive:
- -2: Turn left 90 degrees.
- -1: Turn right 90 degrees.
- 1 <= k <= 9: Move forward k units, one unit at a time.

Some of the grid squares are obstacles. The ith obstacle is at grid point obstacles[i] = (xi, yi). If the robot runs into an obstacle, it will stay in its current location (on the block adjacent to the obstacle) and move onto the next command.

Return the maximum squared Euclidean distance that the robot reaches at any point in its path

(i.e. if the distance is 5, return 25).

Note:
- There can be an obstacle at (0, 0). If this happens, the robot will ignore the obstacle until it has moved off the origin. However, it will be unable to return to (0, 0) due to the obstacle.
- North means +Y direction.
- East means +X direction.
- South means -Y direction.
- West means -X direction.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0878 Nth Magical Number

**HARD**

A positive integer is magical if it is divisible by either a or b.

Given the three integers n, a, and b, return the nth magical number. Since the answer may be very large, return it modulo 109 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0885 Spiral Matrix Iii

**MEDIUM**

You start at the cell (rStart, cStart) of an rows x cols grid facing east. The northwest corner is at the first row and column in the grid, and the southeast corner is at the last row and column.

You will walk in a clockwise spiral shape to visit every position in this grid. Whenever you move outside the grid's boundary, we continue our walk outside the grid (but may return to the grid boundary later.). Eventually, we reach all rows * cols spaces of the grid.

Return an array of coordinates representing the positions of the grid in the order you visited them.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0888 Fair Candy Swap

**EASY**

Alice and Bob have a different total number of candies. You are given two integer arrays aliceSizes and bobSizes where aliceSizes[i] is the number of candies of the ith box of candy that Alice has and bobSizes[j] is the number of candies of the jth box of candy that Bob has.

Since they are friends, they would like to exchange one candy box each so that after the exchange, they both have the same total amount of candy. The total amount of candy a person has is the sum of the number of candies in each box they have.

Return an integer array answer where answer[0] is the number of candies in the box that Alice must exchange, and answer[1] is the number of candies in the box that Bob must exchange. If there are multiple answers, you may return any one of them. It is guaranteed that at least one answer exists.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0891 Sum Of Subsequence Widths

HARD

The width of a sequence is the difference between the maximum and minimum elements in the sequence.

Given an array of integers nums, return the sum of the widths of all the non-empty subsequences of nums. Since the answer may be very large, return it modulo 109 + 7.

A subsequence is a sequence that can be derived from an array by deleting some or no elements without changing the order of the remaining elements. For example, [3,6,2,7] is a subsequence of the array [0,3,1,6,2,2,7].

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0896 Monotonic Array

EASY

An array is monotonic if it is either monotone increasing or monotone decreasing.

An array nums is monotone increasing if for all i <= j, nums[i] <= nums[j]. An array nums is monotone decreasing if for all i <= j, nums[i] >= nums[j].

Given an integer array nums, return true if the given array is monotonic, or false otherwise.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0904 Fruit Into Baskets

MEDIUM

You are visiting a farm that has a single row of fruit trees arranged from left to right. The trees are represented by an integer array fruits where fruits[i] is the type of fruit the ith tree produces.

You want to collect as much fruit as possible. However, the owner has some strict rules that you must follow:
- You only have two baskets, and each basket can only hold a single type of fruit. There is no limit on the amount of fruit each basket can hold.
- Starting from any tree of your choice, you must pick exactly one fruit from every tree (including the start tree) while moving to the right. The picked fruits must fit in one of your baskets.
- Once you reach a tree with fruit that cannot fit in your baskets, you must stop.

Given the integer array fruits, return the maximum number of fruits you can pick.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0909 Snakes And Ladders

MEDIUM

You are given an n x n integer matrix board where the cells are labeled from 1 to n2 in a Boustrophedon style starting from the bottom left of the board (i.e. board[n - 1][0]) and alternating direction each row.

You start on square 1 of the board. In each move, starting from square curr, do the following:
- Choose a destination square next with a label in the range [curr + 1, min(curr + 6, n2)].
- This choice simulates the result of a standard 6-sided die roll: i.e., there are always at most 6 destinations, regardless of the size of the board.
- If next has a snake or ladder, you must move to the destination of that snake or ladder. Otherwise, you move to next.

- The game ends when you reach the square n2.

A board square on row r and column c has a snake or ladder if board[r][c] != -1. The destination of that snake or ladder is board[r][c]. Squares 1 and n2 are not the starting points of any snake or ladder.

Note that you only take a snake or ladder at most once per dice roll. If the destination to a snake or ladder is the start of another snake or ladder, you do not follow the subsequent snake or ladder.

- For example, suppose the board is [[-1,4],[-1,3]], and on the first move, your destination square is 2. You follow the ladder to square 3, but do not follow the subsequent ladder to 4.

Return the least number of dice rolls required to reach the square n2. If it is not possible to reach the square, return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0911 Online Election

**MEDIUM**

You are given two integer arrays persons and times. In an election, the ith vote was cast for persons[i] at time times[i].

For each query at a time t, find the person that was leading the election at time t. Votes cast at time t will count towards our query. In the case of a tie, the most recent vote (among tied candidates) wins.

Implement the TopVotedCandidate class:
- TopVotedCandidate(int[] persons, int[] times) Initializes the object with the persons and times arrays.
- int q(int t) Returns the number of the person that was leading the election at time t according to the mentioned rules.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0914 X Of A Kind In A Deck Of Cards

**MEDIUM**

You are given an integer array deck where deck[i] represents the number written on the ith card.

Partition the cards into one or more groups such that:
- Each group has exactly x cards where x > 1, and
- All the cards in one group have the same integer written on them.

Return true if such partition is possible, or false otherwise.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0920 Number Of Music Playlists

**HARD**

Your music player contains n different songs. You want to listen to goal songs (not necessarily different) during your trip. To avoid boredom, you will create a playlist so that:
- Every song is played at least once.
- A song can only be played again only if k other songs have been played.

Given n, goal, and k, return the number of possible playlists that you can create. Since the answer can be very large, return it modulo 109 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0928 Minimize Malware Spread Ii

HARD

You are given a network of n nodes represented as an n x n adjacency matrix graph, where the ith node is directly connected to the jth node if graph[i][j] == 1.

Some nodes initial are initially infected by malware. Whenever two nodes are directly connected, and at least one of those two nodes is infected by malware, both nodes will be infected by malware. This spread of malware will continue until no more nodes can be infected in this manner.

Suppose M(initial) is the final number of nodes infected with malware in the entire network after the spread of malware stops.

We will remove exactly one node from initial, completely removing it and any connections from this node to any other node.

Return the node that, if removed, would minimize M(initial). If multiple nodes could be removed to minimize M(initial), return such a node with the smallest index.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0929 Unique Email Addresses

EASY

Every email consists of a local name and a domain name, separated by the @ sign.

For example, in alice@leetcode.com, alice is the local name, and leetcode.com is the domain name.

Besides lowercase letters, these emails may contain '.'s or '+'s.

If you add periods ('.') between some characters in the local name part of an email address, mail sent there will be forwarded to the same address without dots in the local name.
For example, "alice.z@leetcode.com" and "alicez@leetcode.com" forward to the same email address.
(Note that this rule does not apply for domain names.)

If you add a plus ('+') in the local name, everything after the first plus sign will be ignored.
This allows certain emails to be filtered, for example m.y+name@email.com will be forwarded to my@email.com. (Again, this rule does not apply for domain names.)

It is possible to use both of these rules at the same time.

Given a list of emails, we send one email to each address in the list.
How many different addresses actually receive mails?

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0931 Minimum Falling Path Sum

MEDIUM

Given an n x n array of integers matrix, return the minimum sum of any falling path through matrix.

A falling path starts at any element in the first row and chooses the element in the next row that is either directly below or diagonally left/right. Specifically, the next element from position (row, col) will be (row + 1, col - 1), (row + 1, col), or (row + 1, col + 1).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0932 Beautiful Array

MEDIUM

An array nums of length n is beautiful if:
- nums is a permutation of the integers in the range [1, n].
- For every 0 <= i < j < n, there is no index k with i < k < j where 2 * nums[k] == nums[i] + nums[j].

Given the integer n, return any beautiful array nums of length n. There will be at least one valid answer for the given n.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0933 Number Of Recent Calls

EASY

You have a RecentCounter class which counts the number of recent requests within a certain time frame.

Implement the RecentCounter class:
- RecentCounter() Initializes the counter with zero recent requests.
- int ping(int t) Adds a new request at time t, where t represents some time in milliseconds, and returns the number of requests that has happened in the past 3000 milliseconds (including the new request). Specifically, return the number of requests that have happened in the inclusive range [t - 3000, t].

It is guaranteed that every call to ping uses a strictly larger value of t than the previous call.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0939 Minimum Area Rectangle

MEDIUM

You are given an array of points in the X-Y plane points where points[i] = [xi, yi].

Return the minimum area of a rectangle formed from these points, with sides parallel to the X and Y axes. If there is not any such rectangle, return 0.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0941 Valid Mountain Array

EASY

Given an array of integers arr, return true if and only if it is a valid mountain array.

Recall that arr is a mountain array if and only if:
- arr.length >= 3
- There exists some i with 0 < i < arr.length - 1 such that:
- arr[0] < arr[1] < ... < arr[i - 1] < arr[i]
- arr[i] > arr[i + 1] > ... > arr[arr.length - 1]

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0945 Minimum Increment To Make Array Unique

MEDIUM

You are given an integer array nums. In one move, you can pick an index i where 0 <= i < nums.length and increment nums[i] by 1.

Return the minimum number of moves to make every value in nums unique.

The test cases are generated so that the answer fits in a 32-bit integer.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0947 Most Stones Removed With Same Row Or Column

**MEDIUM**

On a 2D plane, we place n stones at some integer coordinate points. Each coordinate point may have at most one stone.

A stone can be removed if it shares either the same row or the same column as another stone that has not been removed.

Given an array stones of length n where stones[i] = [xi, yi] represents the location of the ith stone, return the largest possible number of stones that can be removed.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0952 Largest Component Size By Common Factor

**HARD**

You are given an integer array of unique positive integers nums. Consider the following graph:
- There are nums.length nodes, labeled nums[0] to nums[nums.length - 1],
- There is an undirected edge between nums[i] and nums[j] if nums[i] and nums[j] share a common factor greater than 1.

Return the size of the largest connected component in the graph.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0953 Verifying An Alien Dictionary

**EASY**

In an alien language, surprisingly, they also use English lowercase letters, but possibly in a different order. The order of the alphabet is some permutation of lowercase letters.

Given a sequence of words written in the alien language, and the order of the alphabet, return true if and only if the given words are sorted lexicographically in this alien language.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0954 Array Of Doubled Pairs

**MEDIUM**

Given an integer array of even length arr, return true if it is possible to reorder arr such that arr[2 * i + 1] = 2 * arr[2 * i] for every 0 <= i < len(arr) / 2, or false otherwise.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0956 Tallest Billboard

**HARD**

You are installing a billboard and want it to have the largest height. The billboard will have two steel supports, one on each side. Each steel support must be an equal height.

You are given a collection of rods that can be welded together. For example, if you have rods of lengths 1, 2, and 3, you can weld them together to make a support of length 6.

Return the largest possible height of your billboard installation. If you cannot support the

billboard, return 0.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0957 Prison Cells After N Days

**MEDIUM**

There are 8 prison cells in a row and each cell is either occupied or vacant.

Each day, whether the cell is occupied or vacant changes according to the following rules:
- If a cell has two adjacent neighbors that are both occupied or both vacant, then the cell becomes occupied.
- Otherwise, it becomes vacant.

Note that because the prison is a row, the first and the last cells in the row can't have two adjacent neighbors.

You are given an integer array cells where cells[i] == 1 if the ith cell is occupied and cells[i] == 0 if the ith cell is vacant, and you are given an integer n.

Return the state of the prison after n days (i.e., n such changes described above).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0961 N Repeated Element In Size 2N Array

**EASY**

You are given an integer array nums with the following properties:
- nums.length == 2 * n.
- nums contains n + 1 unique elements.
- Exactly one element of nums is repeated n times.

Return the element that is repeated n times.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0963 Minimum Area Rectangle Ii

**MEDIUM**

You are given an array of points in the X-Y plane points where points[i] = [xi, yi].

Return the minimum area of any rectangle formed from these points, with sides not necessarily parallel to the X and Y axes. If there is not any such rectangle, return 0.

Answers within 10^-5 of the actual answer will be accepted.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0964 Least Operators To Express Number

**HARD**

Given a single positive integer x, we will write an expression of the form x (op1) x (op2) x (op3) x ... where each operator op1, op2, etc. is either addition, subtraction, multiplication, or division (+, -, *, or /). For example, with x = 3, we might write 3 * 3 / 3 + 3 - 3 which is a value of 3.

When writing such an expression, we adhere to the following conventions:
- The division operator (/) returns rational numbers.
- There are no parentheses placed anywhere.
- We use the usual order of operations: multiplication and division happen before addition and subtraction.
- It is not allowed to use the unary negation operator (-). For example, "x - x" is

a valid expression as it only uses subtraction, but "-x + x" is not because it uses negation.

We would like to write an expression with the least number of operators such that the expression equals the given target. Return the least number of operators used.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0967 Numbers With Same Consecutive Differences

**MEDIUM**

Given two integers n and k, return an array of all the integers of length n where the difference between every two consecutive digits is k. You may return the answer in any order.

Note that the integers should not have leading zeros. Integers as 02 and 043 are not allowed.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0970 Powerful Integers

**EASY**

Given two positive integers x and y, an integer is powerful if it is equal to x^i + y^j for some integers i >= 0 and j >= 0.

Return a list of all powerful integers that have value less than or equal to bound.

You may return the answer in any order. In your answer, each value should occur at most once.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0972 Equal Rational Numbers

**HARD**

Given two strings s and t, each of which represents a non-negative rational number, return true if and only if they represent the same number. The strings may use parentheses to denote the repeating part of the rational number.

A rational number can be represented using up to three parts: <IntegerPart>, <NonRepeatingPart>, and a <RepeatingPart>. The number will be represented in one of the following three ways:
- <IntegerPart>
- For example, 12, 0, and 123.
- <IntegerPart><.><NonRepeatingPart>
- For example, 0.5, 1., 2.12, and 123.0001.
- <IntegerPart><.><NonRepeatingPart><(><RepeatingPart><)>
- For example, 0.1(6), 1.(9), 123.00(1212).

The repeating portion of a decimal expansion is conventionally denoted within a pair of round brackets. For example:
- 1/6 = 0.16666666... = 0.1(6) = 0.1666(6) = 0.166(66).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0981 Time Based Key Value Store

**MEDIUM**

Design a time-based key-value data structure that can store multiple values for the same key at different time stamps and retrieve the key's value at a certain timestamp.

Implement the TimeMap class:
- TimeMap() Initializes the object of the data structure.
- void set(String key, String value, int timestamp) Stores the key key with the value value at the given time timestamp.
- String get(String key, int timestamp) Returns a value such that set was called previously, with timestamp_prev <= timestamp. If there are multiple such values, it returns the value associated with the largest timestamp_prev. If there are no values, it returns "".

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0982 Triples With Bitwise And Equal To Zero

**HARD**

Given an integer array nums, return the number of AND triples.

An AND triple is a triple of indices (i, j, k) such that:
- 0 <= i < nums.length
- 0 <= j < nums.length
- 0 <= k < nums.length
- nums[i] & nums[j] & nums[k] == 0, where & represents the bitwise-AND operator.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0983 Minimum Cost For Tickets

**MEDIUM**

You have planned some train traveling one year in advance. The days of the year in which you will travel are given as an integer array days. Each day is an integer from 1 to 365.

Train tickets are sold in three different ways:
- a 1-day pass is sold for costs[0] dollars,
- a 7-day pass is sold for costs[1] dollars, and
- a 30-day pass is sold for costs[2] dollars.

The passes allow that many days of consecutive travel.

For example, if we get a 7-day pass on day 2, then we can travel for 7 days: 2, 3, 4, 5, 6, 7, and 8.

Return the minimum number of dollars you need to travel every day in the given list of days.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0985 Sum Of Even Numbers After Queries

**EASY**

We have an array A of integers, and an array queries of queries.

For the i-th query val = queries[i][0], index = queries[i][1], we add val to A[index]. Then, the answer to the i-th query is the sum of the even values of A.

(Here, the given index = queries[i][1] is a 0-based index, and each query permanently modifies the array A.)

Return the answer to all queries. Your answer array should have answer[i] as the answer to the i-th query.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0989 Add To Array Form Of Integer

EASY

For a non-negative integer X, the array-form of X is
an array of its digits in left to right order.
For example, if X = 1231, then the array form is [1,2,3,1].

Given the array-form A of a non-negative integer X,
return the array-form of the integer X+K.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0995 Minimum Number Of K Consecutive Bit Flips

HARD

You are given a binary array nums and an integer k.

A k-bit flip is choosing a subarray of length k from nums and simultaneously changing every 0 in the subarray to 1, and every 1 in the subarray to 0.

Return the minimum number of k-bit flips required so that there is no 0 in the array. If it is not possible, return -1.

A subarray is a contiguous part of an array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0996 Number Of Squareful Arrays

HARD

An array is squareful if the sum of every pair of adjacent elements is a perfect square.

Given an integer array nums, return the number of permutations of nums that are squareful.

Two permutations perm1 and perm2 are different if there is some index i such that perm1[i] != perm2[i].

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0997 Find The Town Judge

EASY

In a town, there are n people labeled from 1 to n. There is a rumor that one of these people is secretly the town judge.

If the town judge exists, then:
- The town judge trusts nobody.
- Everybody (except for the town judge) trusts the town judge.
- There is exactly one person that satisfies properties 1 and 2.

You are given an array trust where trust[i] = [ai, bi] representing that the person labeled ai trusts the person labeled bi.

Return the label of the town judge if the town judge exists and can be identified, or return -1 otherwise.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1001 Grid Illumination

HARD

There is a 2D grid of size n x n where each cell of this grid has a lamp that is initially turned off.

You are given a 2D array of lamp positions lamps, where lamps[i] = [rowi, coli] indicates that the lamp at grid[rowi][coli] is turned on. Even if the same lamp is listed more than once, it is turned on.

When a lamp is turned on, it illuminates its cell and all other cells in the same row, column, or diagonal.

You are also given another 2D array queries, where queries[j] = [rowj, colj]. For the jth query, determine whether grid[rowj][colj] is illuminated or not. After answering the jth query, turn off the lamp at grid[rowj][colj] and its 8 adjacent lamps if they exist. A lamp is adjacent if its cell shares either a side or corner with grid[rowj][colj].

Return an array of integers ans, where ans[j] should be 1 if the cell in the jth query was illuminated, or 0 if the lamp was not.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1005 Maximize Sum Of Array After K Negations

EASY

Given an integer array nums and an integer k, modify the array in the following way:
- choose an index i and replace nums[i] with -nums[i].

You should apply this process exactly k times. You may choose the same index i multiple times.

Return the largest possible sum of the array after modifying it in this way.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1010 Pairs Of Songs With Total Durations Divisible By 60

MEDIUM

You are given a list of songs where the ith song has a duration of time[i] seconds.

Return the number of pairs of songs for which their total duration in seconds is divisible by 60. Formally, we want the number of indices i, j such that i < j with (time[i] + time[j]) % 60 == 0.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1013 Partition Array Into Three Parts With Equal Sum

EASY

Given an array of integers arr, return true if we can partition the array into three non-empty parts with equal sums.

Formally, we can partition the array if we can find indexes i + 1 < j with (arr[0] + arr[1] + ... + arr[i] == arr[i + 1] + arr[i + 2] + ... + arr[j - 1] == arr[j] + arr[j + 1] + ... + arr[arr.length - 1])

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1014 Best Sightseeing Pair

MEDIUM

You are given an integer array values where values[i] represents the value of the ith sightseeing spot. Two sightseeing spots i and j have a distance j - i between them.

The score of a pair (i < j) of sightseeing spots is values[i] + values[j] + i - j: the sum of the values of the sightseeing spots, minus the distance between them.

Return the maximum score of a pair of sightseeing spots.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1020 Number Of Enclaves

MEDIUM

You are given an m x n binary matrix grid, where 0 represents a sea cell and 1 represents a land cell.

A move consists of walking from one land cell to another adjacent (4-directionally) land cell or walking off the boundary of the grid.

Return the number of land cells in grid for which we cannot walk off the boundary of the grid in any number of moves.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1022 Sum Of Root To Leaf Binary Numbers

EASY

You are given the root of a binary tree where each node has a value 0 or 1. Each root-to-leaf path represents a binary number starting with the most significant bit.

For example, if the path is 0 -> 1 -> 1 -> 0 -> 1, then this could represent 01101 in binary, which is 13.

For all leaves in the tree, consider the numbers represented by the path from the root to that leaf. Return the sum of these numbers.

The test cases are generated so that the answer fits in a 32-bits integer.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1026 Maximum Difference Between Node And Ancestor

MEDIUM

Given the root of a binary tree, find the maximum value v for which there exist different nodes a and b where v = |a.val - b.val| and a is an ancestor of b.

A node a is an ancestor of b if either: any child of a is equal to b or any child of a is an ancestor of b.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1027 Longest Arithmetic Subsequence

MEDIUM

Given an array nums of integers, return the length of the longest arithmetic subsequence in nums.

Note that:

- A subsequence is an array that can be derived from another array by deleting some or no elements without changing the order of the remaining elements.
- A sequence seq is arithmetic if seq[i + 1] - seq[i] are all the same value (for 0 <= i < seq.length - 1).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1030 Matrix Cells In Distance Order

**EASY**

You are given four integers row, cols, rCenter, and cCenter. There is a rows x cols matrix and you are on the cell with the coordinates (rCenter, cCenter).

Return the coordinates of all cells in the matrix, sorted by their distance from (rCenter, cCenter) from the smallest distance to the largest distance. You may return the answer in any order that satisfies this condition.

The distance between two cells (r1, c1) and (r2, c2) is |r1 - r2| + |c1 - c2|.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1034 Coloring A Border

**MEDIUM**

You are given an m x n integer matrix grid, and three integers row, col, and color. Each value in the grid represents the color of the grid square at that location.

Two squares are called adjacent if they are next to each other in any of the 4 directions.

Two squares belong to the same connected component if they have the same color and they are adjacent.

The border of a connected component is all the squares in the connected component that are either adjacent to (at least) a square not in the component, or on the boundary of the grid (the first or last row or column).

You should color the border of the connected component that contains the square grid[row][col] with color.

Return the final grid.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1036 Escape A Large Maze

**HARD**

There is a 1 million by 1 million grid on an XY-plane, and the coordinates of each grid square are (x, y).

We start at the source = [sx, sy] square and want to reach the target = [tx, ty] square. There is also an array of blocked squares, where each blocked[i] = [xi, yi] represents a blocked square with coordinates (xi, yi).

Each move, we can walk one square north, east, south, or west if the square is not in the array of blocked squares. We are also not allowed to walk outside of the grid.

Return true if and only if it is possible to reach the target square from the source square through a sequence of valid moves.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1042 Flower Planting With No Adjacent

MEDIUM

You have n gardens, labeled from 1 to n, and an array paths where paths[i] = [xi, yi] describes a bidirectional path between garden xi to garden yi. In each garden, you want to plant one of 4 types of flowers.

All gardens have at most 3 paths coming into or leaving it.

Your task is to choose a flower type for each garden such that, for any two gardens connected by a path, they have different types of flowers.

Return any such a choice as an array answer, where answer[i] is the type of flower planted in the (i+1)th garden. The flower types are denoted 1, 2, 3, or 4. It is guaranteed an answer exists.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1043 Partition Array For Maximum Sum

MEDIUM

Given an integer array arr, partition the array into (contiguous) subarrays of length at most k. After partitioning, each subarray has their values changed to become the maximum value of that subarray.

Return the largest sum of the given array after partitioning. Test cases are generated so that the answer fits in a 32-bit integer.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1049 Last Stone Weight Ii

MEDIUM

You are given an array of integers stones where stones[i] is the weight of the ith stone.

We are playing a game with the stones. On each turn, we choose any two stones and smash them together. Suppose the stones have weights x and y with x <= y. The result of this smash is:
- If x == y, both stones are destroyed, and
- If x != y, the stone of weight x is destroyed, and the stone of weight y has new weight y - x.

At the end of the game, there is at most one stone left.

Return the smallest possible weight of the left stone. If there are no stones left, return 0.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1054 Distant Barcodes

MEDIUM

In a warehouse, there is a row of barcodes, where the ith barcode is barcodes[i].

Rearrange the barcodes so that no two adjacent barcodes are equal. You may return any answer, and it is guaranteed an answer exists.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1056 Confusing Number

EASY

A confusing number is a number that when rotated 180 degrees becomes a different number with each digit valid.

We can rotate digits of a number by 180 degrees to form new digits.
- When 0, 1, 6, 8, and 9 are rotated 180 degrees, they become 0, 1, 9, 8, and 6 respectively.
- When 2, 3, 4, 5, and 7 are rotated 180 degrees, they become invalid.

Note that after rotating a number, we can ignore leading zeros.
- For example, after rotating 8000, we have 0008 which is considered as just 8.

Given an integer n, return true if it is a confusing number, or false otherwise.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1057 Campus Bikes

**MEDIUM**

On a campus represented on the X-Y plane, there are n workers and m bikes, with n <= m.

You are given an array workers of length n where workers[i] = [xi, yi] is the position of the ith worker. You are also given an array bikes of length m where bikes[j] = [xj, yj] is the position of the jth bike. All the given positions are unique.

Assign a bike to each worker. Among the available bikes and workers, we choose the (workeri, bikej) pair with the shortest Manhattan distance between each other and assign the bike to that worker.

If there are multiple (workeri, bikej) pairs with the same shortest Manhattan distance, we choose the pair with the smallest worker index. If there are multiple ways to do that, we choose the pair with the smallest bike index. Repeat this process until there are no available workers.

Return an array answer of length n, where answer[i] is the index (0-indexed) of the bike that the ith worker is assigned to.

The Manhattan distance between two points p1 and p2 is
Manhattan(p1, p2) = |p1.x - p2.x| + |p1.y - p2.y|.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1072 Flip Columns For Maximum Number Of Equal Rows

**MEDIUM**

You are given an m x n binary matrix matrix.

You can choose any number of columns in the matrix and flip every cell in that column (i.e., Change the value of the cell from 0 to 1 or vice versa).

Return the maximum number of rows that have all values equal after some number of flips.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1073 Adding Two Negabinary Numbers

**MEDIUM**

Given two numbers arr1 and arr2 in base -2, return the result of adding them together.

Each number is given in array format: as an array of 0s and 1s, from most significant bit to least significant bit. For example, arr = [1,1,0,1] represents the number (-2)^3 + (-2)^2 + (-2)^0 = -3. A number arr in array, format is also guaranteed to have no leading zeros: either arr == [0] or arr[0] == 1.

Return the result of adding arr1 and arr2 in the same format: as an array of 0s and 1s with no leading zeros.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1074 Number Of Submatrices That Sum To Target

HARD

Given a matrix and a target, return the number of non-empty submatrices that sum to target.

A submatrix x1, y1, x2, y2 is the set of all cells matrix[x][y] with x1 <= x <= x2 and y1 <= y <= y2.

Two submatrices (x1, y1, x2, y2) and (x1', y1', x2', y2') are different if they have some coordinate that is different: for example, if x1 != x1'.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1086 High Five

EASY

Given a list of the scores of different students, items, where items[i] = [IDi, scorei] represents one score from a student with IDi, calculate each student's top five average.

Return the answer as an array of pairs result, where result[j] = [IDj, topFiveAveragej] represents the student with IDj and their top five average. Sort result by IDj in increasing order.

A student's top five average is calculated by taking the sum of their top five scores and dividing it by 5 using integer division.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1088 Confusing Number Ii

HARD

A confusing number is a number that when rotated 180 degrees becomes a different number with each digit valid.

We can rotate digits of a number by 180 degrees to form new digits.
- When 0, 1, 6, 8, and 9 are rotated 180 degrees, they become 0, 1, 9, 8, and 6 respectively.
- When 2, 3, 4, 5, and 7 are rotated 180 degrees, they become invalid.

Note that after rotating a number, we can ignore leading zeros.
- For example, after rotating 8000, we have 0008 which is considered as just 8.

Given an integer n, return the number of confusing numbers in the inclusive range [1, n].

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1089 Duplicate Zeros

EASY

Given a fixed-length integer array arr, duplicate each occurrence of zero, shifting the remaining elements to the right.

Note that elements beyond the length of the original array are not written. Do the above modifications to the input array in place and do not return anything.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1090 Largest Values From Labels

MEDIUM

You are given n item's value and label as two integer arrays values and labels. You are also given two integers numWanted and useLimit.

Your task is to find a subset of items with the maximum sum of their values such that:
- The number of items is at most numWanted.
- The number of items with the same label is at most useLimit.

Return the maximum sum.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1095 Find In Mountain Array

**HARD**

You may recall that an array arr is a mountain array if and only if:
- arr.length >= 3
- There exists some i with 0 < i < arr.length - 1 such that:
- arr[0] < arr[1] < ... < arr[i - 1] < arr[i]
- arr[i] > arr[i + 1] > ... > arr[arr.length - 1]

Given a mountain array mountainArr, return the minimum index such that mountainArr.get(index) == target. If such an index does not exist, return -1.

You cannot access the mountain array directly. You may only access the array using a MountainArray interface:
- MountainArray.get(k) returns the element of the array at index k (0-indexed).
- MountainArray.length() returns the length of the array.

Submissions making more than 100 calls to MountainArray.get will be judged Wrong Answer. Also, any solutions that attempt to circumvent the judge will result in disqualification.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1096 Brace Expansion Ii

**HARD**

Under the grammar given below, strings can represent a set of lowercase words. Let R(expr) denote the set of words the expression represents.

The grammar can best be understood through simple examples:
- Single letters represent a singleton set containing that word.
- R("a") = {"a"}
- R("w") = {"w"}
- When we take a comma-delimited list of two or more expressions, we take the union of possibilities.
- R("{a,b,c}") = {"a","b","c"}
- R("{{a,b},{b,c}}") = {"a","b","c"} (notice the final set only contains each word at most once)
- When we concatenate two expressions, we take the set of possible concatenations between two words where the first word comes from the first expression and the second word comes from the second expression.
- R("{a,b}{c,d}") = {"ac","ad","bc","bd"}
- R("a{b,c}{d,e}f{g,h}") = {"abdfg", "abdfh", "abefg", "abefh", "acdfg", "acdfh", "acefg", "acefh"}

Formally, the three rules for our grammar:
- For every lowercase letter x, we have R(x) = {x}.
- For expressions e1, e2, ... , ek with k >= 2, we have R({e1, e2, ...}) = R(e1) ∪ R(e2) ∪ ...
- For expressions e1 and e2, we have R(e1 + e2) = {a + b for (a, b) in R(e1) × R(e2)}, where + denotes concatenation, and × denotes the cartesian product.

Given an expression representing a set of words under the given grammar, return the sorted list of words that the expression represents.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1099 Two Sum Less Than K

EASY

Given an array nums of integers and integer k, return the maximum sum such that there exists i < j with nums[i] + nums[j] = sum and sum < k. If no i, j exist satisfying this equation, return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1118 Number Of Days In A Month

EASY

Given a year year and a month month, return the number of days of that month.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1121 Divide Array Into Increasing Sequences

HARD

Given an integer array nums sorted in non-decreasing order and an integer k, return true if this array can be divided into one or more disjoint increasing subsequences of length at least k, or false otherwise.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1125 Smallest Sufficient Team

HARD

In a project, you have a list of required skills reqSkills, and a list of people. The ith person people[i] contains a list of skills that the person has.

Consider a sufficient team: a set of people such that for every required skill in reqSkills, there is at least one person in the team who has that skill. We can represent these teams by the index of each person.

For example, team = [0, 1, 3] represents the people with skills people[0], people[1], and people[3].

Return any sufficient team of the smallest possible size, represented by the index of each person. You may return the answer in any order.

It is guaranteed an answer exists.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1128 Number Of Equivalent Domino Pairs

EASY

Given a list of dominoes, dominoes[i] = [a, b] is equivalent to dominoes[j] = [c, d] if and only if either (a == c and b == d), or (a == d and b == c) - that is, one domino can be rotated to be equal to another domino.

Return the number of pairs (i, j) for which 0 <= i < j < dominoes.length, and dominoes[i] is equivalent to dominoes[j].

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1133 Largest Unique Number

EASY

Given an integer array nums, return the largest integer that only occurs once.
If no integer occurs once, return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1134 Armstrong Number

EASY

Given an integer n, return true if and only if it is an Armstrong number.

The k-digit number n is an Armstrong number if and only if the kth power of each digit sums to n.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1136 Parallel Courses

MEDIUM

You are given an integer n, which indicates that there are n courses labeled from 1 to n. You are also given an array relations where relations[i] = [prevCoursei, nextCoursei], representing a prerequisite relationship between course prevCoursei and course nextCoursei: course prevCoursei has to be taken before course nextCoursei.

In one semester, you can take any number of courses as long as you have taken all the prerequisites in the previous semester for the courses you are taking.

Return the minimum number of semesters needed to take all courses. If there is no way to take all the courses, return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1137 N Th Tribonacci Number

EASY

The Tribonacci sequence Tn is defined as follows:

T0 = 0, T1 = 1, T2 = 1, and Tn+3 = Tn + Tn+1 + Tn+2 for n >= 0.

Given n, return the value of Tn.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1140 Stone Game Ii

MEDIUM

Alice and Bob continue their games with piles of stones. There are a number of piles arranged in a row, and each pile has a positive integer number of stones piles[i]. The objective of the game is to end with the most stones.

Alice and Bob take turns, with Alice starting first.

On each player's turn, that player can take all the stones in the first X remaining piles, where 1 <= X <= 2M. Then, we set M = max(M, X). Initially, M = 1.

The game continues until all the stones have been taken.

Assuming Alice and Bob play optimally, return the maximum number of stones Alice can get.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1146 Snapshot Array

MEDIUM

Implement a SnapshotArray that supports the following interface:
- SnapshotArray(int length) initializes an array-like data structure with the given length. Initially, each element equals 0.
- void set(index, val) sets the element at the given index to be equal to val.
- int snap() takes a snapshot of the array and returns the snapId: the total number of times we called snap() minus 1.
- int get(index, snapId) returns the value at the given index, at the time we took the snapshot with the given snapId

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1152 Analyze User Website Visit Pattern

MEDIUM

You are given two string arrays username and website and an integer array timestamp. All the given arrays are of the same length and the tuple [username[i], website[i], timestamp[i]] indicates that the user username[i] visited the website website[i] at time timestamp[i].

A pattern is a list of three websites (not necessarily distinct).
- For example, ["home", "away", "love"], ["leetcode", "love", "leetcode"], and ["luffy", "luffy", "luffy"] are all patterns.

The score of a pattern is the number of users that visited all the websites in the pattern in the same order they appeared in the pattern.
- For example, if the pattern is ["home", "away", "love"], the score is the number of users x such that x visited "home" then visited "away" and visited "love" after that.
- Similarly, if the pattern is ["leetcode", "love", "leetcode"], the score is the number of users x such that x visited "leetcode" then visited "love" and visited "leetcode" one more time after that.
- Also, if the pattern is ["luffy", "luffy", "luffy"], the score is the number of users x such that x visited "luffy" three different times at different timestamps.

Return the pattern with the largest score. If there is more than one pattern with the same largest score, return the lexicographically smallest such pattern.

Note that the websites in a pattern do not need to be visited contiguously, they only need to be visited in the order they appeared in the pattern.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1155 Number Of Dice Rolls With Target Sum

MEDIUM

You have n dice, and each dice has k faces numbered from 1 to k.

Given three integers n, k, and target, return the number of possible ways (out of the kn total ways) to roll the dice, so the sum of the face-up numbers equals target. Since the answer may be too large, return it modulo 109 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1165 Single Row Keyboard

EASY

There is a special keyboard with all keys in a single row.

Given a string keyboard of length 26 indicating the layout of the keyboard (indexed from 0 to 25). Initially, your finger is at index 0. To type a character, you have to move your finger to the index of the desired character. The time taken to move your finger from index i to index j is |i - j|.

You want to type a string word. Write a function to calculate how much time it takes to type it with one finger.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1166 Design File System

**MEDIUM**

You are asked to design a file system that allows you to create new paths and associate them with different values.

The format of a path is one or more concatenated strings of the form: / followed by one or more lowercase English letters. For example, "/leetcode" and "/leetcode/problems" are valid paths while an empty string "" and "/" are not.

Implement the FileSystem class:
- bool createPath(string path, int value) Creates a new path and associates a value to it if possible and returns true. Returns false if the path already exists or its parent path doesn't exist.
- int get(string path) Returns the value associated with path or returns -1 if the path doesn't exist.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1169 Invalid Transactions

**MEDIUM**

A transaction is possibly invalid if:
- the amount exceeds $1000, or;
- if it occurs within (and including) 60 minutes of another transaction with the same name in a different city.

You are given an array of strings transaction where transactions[i] consists of comma-separated values representing the name, time (in minutes), amount, and city of the transaction.

Return a list of transactions that are possibly invalid. You may return the answer in any order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1181 Before And After Puzzle

**MEDIUM**

Given a list of phrases, generate a list of Before and After puzzles.

A phrase is a string that consists of lowercase English letters and spaces only. No space appears in the start or the end of a phrase. There are no consecutive spaces in a phrase.

Before and After puzzles are phrases that are formed by merging two phrases where the last word of the first phrase is the same as the first word of the second phrase.

Return the Before and After puzzles that can be formed by every two phrases phrases[i] and phrases[j] where i != j. Note that the order of matching two phrases matters, we want to consider both orders.

You should return a list of distinct strings sorted lexicographically.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1183 Maximum Number Of Ones

HARD

Consider a matrix M with dimensions width * height, such that every cell has value 0 or 1, and any square sub-matrix of M of size sideLength * sideLength has at most maxOnes ones.

Return the maximum possible number of ones that the matrix M can have.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1187 Make Array Strictly Increasing

HARD

Given two integer arrays arr1 and arr2, return the minimum number of operations (possibly zero) needed to make arr1 strictly increasing.

In one operation, you can choose two indices 0 <= i < arr1.length and 0 <= j < arr2.length and do the assignment arr1[i] = arr2[j].

If there is no way to make arr1 strictly increasing, return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1189 Maximum Number Of Balloons

EASY

Given a string text, you want to use the characters of text to form as many instances of the word "balloon" as possible.

You can use each character in text at most once. Return the maximum number of instances that can be formed.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1191 K Concatenation Maximum Sum

MEDIUM

Given an integer array arr and an integer k, modify the array by repeating it k times.

For example, if arr = [1, 2] and k = 3 then the modified array will be [1, 2, 1, 2, 1, 2].

Return the maximum sub-array sum in the modified array. Note that the length of the sub-array can be 0 and its sum in that case is 0.

As the answer can be very large, return the answer modulo 109 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1198 Find Smallest Common Element In All Rows

MEDIUM

Given an m x n matrix mat where every row is sorted in strictly increasing order, return the smallest common element in all rows.

If there is no common element, return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1200 Minimum Absolute Difference

EASY

Given an array of distinct integers arr, find all pairs of elements with the minimum absolute difference of any two elements.

Return a list of pairs in ascending order(with respect to pairs), each pair [a, b] follows:
- a, b are from arr
- a < b
- b - a equals to the minimum absolute difference of any two elements in arr

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1201 Ugly Number Iii

MEDIUM

An ugly number is a positive integer that is divisible by a, b, or c.

Given four integers n, a, b, and c, return the nth ugly number.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1207 Unique Number Of Occurrences

EASY

Given an array of integers arr, return true if the number of occurrences of each value in the array is unique or false otherwise.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1210 Minimum Moves To Reach Target With Rotations

HARD

In an n*n grid, there is a snake that spans 2 cells and starts moving from the top left corner at (0, 0) and (0, 1). The grid has empty cells represented by zeros and blocked cells represented by ones. The snake wants to reach the lower right corner at (n-1, n-2) and (n-1, n-1).

In one move the snake can:
- Move one cell to the right if there are no blocked cells there. This move keeps the horizontal/vertical position of the snake as it is.
- Move down one cell if there are no blocked cells there. This move keeps the horizontal/vertical position of the snake as it is.
- Rotate clockwise if it's in a horizontal position and the two cells under it are both empty. In that case the snake moves from (r, c) and (r, c+1) to (r, c) and (r+1, c).
- Rotate counterclockwise if it's in a vertical position and the two cells to its right are both empty. In that case the snake moves from (r, c) and (r+1, c) to (r, c) and (r, c+1).

Return the minimum number of moves to reach the target.

If there is no way to reach the target, return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1215 Stepping Numbers

MEDIUM

A stepping number is an integer such that all of its adjacent digits have an absolute difference of exactly 1.
- For example, 321 is a stepping number while 421 is not.

Given two integers low and high, return a sorted list of all the stepping numbers in the inclusive range [low, high].

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1218 Longest Arithmetic Subsequence Of Given Difference

**MEDIUM**

Given an integer array arr and an integer difference, return the length of the longest subsequence in arr which is an arithmetic sequence such that the difference between adjacent elements in the subsequence equals difference.

A subsequence is a sequence that can be derived from arr by deleting some or no elements without changing the order of the remaining elements.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1222 Queens That Can Attack The King

**MEDIUM**

On a 0-indexed 8 x 8 chessboard, there can be multiple black queens and one white king.

You are given a 2D integer array queens where queens[i] = [xQueeni, yQueeni] represents the position of the ith black queen on the chessboard. You are also given an integer array king of length 2 where king = [xKing, yKing] represents the position of the white king.

Return the coordinates of the black queens that can directly attack the king. You may return the answer in any order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1224 Maximum Equal Frequency

**HARD**

Given an array nums of positive integers, return the longest possible length of an array prefix of nums, such that it is possible to remove exactly one element from this prefix so that every number that has appeared in it will have the same number of occurrences.

If after removing one element there are no remaining elements, it's still considered that every appeared number has the same number of ocurrences (0).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1228 Missing Number In Arithmetic Progression

**EASY**

In some array arr, the values were in arithmetic progression: the values arr[i + 1] - arr[i] are all equal for every 0 <= i < arr.length - 1.

A value from arr was removed that was not the first or last value in the array.

Given arr, return the removed value.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1233 Remove Sub Folders From The Filesystem

**MEDIUM**

Given a list of folders, remove all sub-folders in those folders and return in any order the folders after removing.

If a folder[i] is located within another folder[j], it is called a sub-folder of it. The format of a path is one or more concatenated strings of the form: / followed by one or more lowercase English letters. For example, /leetcode and /leetcode/problems are valid paths while an empty string and / are not.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1236 Web Crawler

**MEDIUM**

Given a url startUrl and an interface HtmlParser, implement a web crawler to crawl all links that are under the same hostname as startUrl.

Return all urls obtained by your web crawler in any order.

Your crawler should:
- Start from the page: startUrl
- Call HtmlParser.getUrls(url) to get all urls from a webpage of given url.
- Do not crawl the same link twice.
- Explore only the links that are under the same hostname as startUrl.

As shown in the example url above, the hostname is example.org. For simplicity sake, you may assume all urls use http protocol without any port specified. For example, the urls while urls http://example.org/test and http://example.com/abc are not under the same hostname.

The HtmlParser interface is defined as such:
interface HtmlParser {
// Return a list of all urls from a webpage of given url.
public List<String> getUrls(String url);
}

Below are two examples explaining the functionality of the problem, for custom testing purposes you'll have three variables urls, edges and startUrl. Notice that you will only have access to startUrl in your code, while urls and edges are not directly accessible to you in code.

Note: Consider the same URL with the trailing slash "/" as a different URL. For example, "http://news.yahoo.com", and "http://news.yahoo.com/" are different urls.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1243 Array Transformation

**EASY**

Given an initial array arr, every day you produce a new array using the array of the previous day.

On the i-th day, you do the following operations on the array of day i-1 to produce the array of day i:
- If an element is smaller than both its left neighbor and its right neighbor, then this element is incremented.
- If an element is bigger than both its left neighbor and its right neighbor, then this element is decremented.

The first and last elements never change.

After some days, the array does not change. Return that final array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1244 Design A Leaderboard

MEDIUM

Design a Leaderboard class, which has 3 functions:
- addScore(playerId, score): Update the leaderboard by adding score to the given player's score. If there is no player with such id in the leaderboard, add him to the leaderboard with the given score.
- top(K): Return the score sum of the top K players.
- reset(playerId): Reset the score of the player with the given id to 0 (in other words erase it from the leaderboard). It is guaranteed that the player was added to the leaderboard before calling this function.

Initially, the leaderboard is empty.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1250 Check If It Is A Good Array

HARD

Given an array nums of positive integers. Your task is to select some subset of nums, multiply each element by an integer and add all these numbers. The array is said to be good if you can obtain a sum of 1 from the array by any possible subset and multiplicand.

Return True if the array is good otherwise return False.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1252 Cells With Odd Values In A Matrix

EASY

Given `n` and `m` which are the dimensions of a matrix initialized by zeros and given an array `indices` where `indices[i] = [ri, ci]`.
For each pair of `[ri, ci]` you have to increment all cells in row `ri` and column `ci` by 1.

Return the number of cells with odd values in the matrix after applying the increment to all `indices`.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1253 Reconstruct A 2 Row Binary Matrix

MEDIUM

Given the following details of a matrix with n columns and 2 rows:
- The matrix is a binary matrix, which means each element in the matrix can be 0 or 1.
- The sum of elements of the 0-th(upper) row is given as upper.
- The sum of elements of the 1-st(lower) row is given as lower.
- The sum of elements in the i-th column(0-indexed) is colsum[i], where colsum is given as an integer array with length n.

Your task is to reconstruct the matrix with upper, lower and colsum.
Return it as a 2-D integer array.
If there are more than one valid solution, any of them will be accepted.
If no valid solution exists, return an empty 2-D array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1256 Encode Number

MEDIUM

Given a non-negative integer num, Return its encoding string.

The encoding is done by converting the integer to a string using a secret function that you should deduce from the following table:

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1257 Smallest Common Region

MEDIUM

You are given some lists of regions where the first region of each list directly contains all other regions in that list.

If a region x contains a region y directly, and region y contains region z directly, then region x is said to contain region z indirectly. Note that region x also indirectly contains all regions indirectly containd in y.

Naturally, if a region x contains (either directly or indirectly) another region y, then x is bigger than or equal to y in size. Also, by definition, a region x contains itself.

Given two regions: region1 and region2, return the smallest region that contains both of them.

It is guaranteed the smallest region exists.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1260 Shift 2D Grid

EASY

Given a 2D grid of size m x n and an integer k. You need to shift the grid k times.

In one shift operation:
- Element at grid[i][j] moves to grid[i][j + 1].
- Element at grid[i][n - 1] moves to grid[i + 1][0].
- Element at grid[m - 1][n - 1] moves to grid[0][0].

Return the 2D grid after applying shift operation k times.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1262 Greatest Sum Divisible By Three

MEDIUM

Given an integer array nums, return the maximum possible sum of elements of the array such that it is divisible by three.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1263 Minimum Moves To Move A Box To Their Target Location

HARD

A storekeeper is a game in which the player pushes boxes around in a warehouse trying to get them to target locations.

The game is represented by an m x n grid of characters grid where each element is a wall, floor, or box.

Your task is to move the box 'B' to the target position 'T' under the following rules:

- The character 'S' represents the player. The player can move up, down, left, right in grid if it is a floor (empty cell).
- The character '.' represents the floor which means a free cell to walk.
- The character '#' represents the wall which means an obstacle (impossible to walk there).
- There is only one box 'B' and one target cell 'T' in the grid.
- The box can be moved to an adjacent free cell by standing next to the box and then moving in the direction of the box. This is a push.
- The player cannot walk through the box.

Return the minimum number of pushes to move the box to the target. If there is no way to reach the target, return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1267 Count Servers That Communicate

**MEDIUM**

You are given a map of a server center, represented as a m * n integer matrix grid, where 1 means that on that cell there is a server and 0 means that it is no server. Two servers are said to communicate if they are on the same row or on the same column.

Return the number of servers that communicate with any other server.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1269 Number Of Ways To Stay In The Same Place After Some Steps

**HARD**

You have a pointer at index 0 in an array of size arrLen. At each step, you can move 1 position to the left, 1 position to the right in the array, or stay in the same place (The pointer should not be placed outside the array at any time).

Given two integers steps and arrLen, return the number of ways such that your pointer is still at index 0 after exactly steps steps. Since the answer may be too large, return it modulo 109 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1271 Hexspeak

**EASY**

A decimal number can be converted to its Hexspeak representation by first converting it to an uppercase hexadecimal string, then replacing all occurrences of the digit '0' with the letter 'O', and the digit '1' with the letter 'I'. Such a representation is valid if and only if it consists only of the letters in the set {'A', 'B', 'C', 'D', 'E', 'F', 'I', 'O'}.

Given a string num representing a decimal integer n, return the Hexspeak representation of n if it is valid, otherwise return "ERROR".

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1274 Number Of Ships In A Rectangle

**HARD**

Each ship is located at an integer point on the sea represented by a cartesian plane, and each integer point may contain at most 1 ship.

You have a function Sea.hasShips(topRight, bottomLeft) which takes two points as arguments and returns true If there is at least one ship in the rectangle represented by the two points, including on the boundary.

Given two points: the top right and bottom left corners of a rectangle, return the number of ships present in that rectangle. It is guaranteed that there are at most 10 ships in that rectangle.

Submissions making more than 400 calls to hasShips will be judged Wrong Answer. Also, any solutions that attempt to circumvent the judge will result in disqualification.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1276 Number Of Burgers With No Waste Of Ingredients

**MEDIUM**

Given two integers tomatoSlices and cheeseSlices. The ingredients of different burgers are as follows:
- Jumbo Burger: 4 tomato slices and 1 cheese slice.
- Small Burger: 2 Tomato slices and 1 cheese slice.

Return [total_jumbo, total_small] so that the number of remaining tomatoSlices equal to 0 and the number of remaining cheeseSlices equal to 0. If it is not possible to make the remaining tomatoSlices and cheeseSlices equal to 0 return [].

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1277 Count Square Submatrices With All Ones

**MEDIUM**

Given a m * n matrix of ones and zeros, return how many square submatrices have all ones.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1282 Group The People Given The Group Size They Belong To

**MEDIUM**

There are n people that are split into some unknown number of groups. Each person is labeled with a unique ID from 0 to n - 1.

You are given an integer array groupSizes, where groupSizes[i] is the size of the group that person i is in. For example, if groupSizes[1] = 3, then person 1 must be in a group of size 3.

Return a list of groups such that each person i is in a group of size groupSizes[i].

Each person should appear in exactly one group, and every person must be in a group. If there are multiple answers, return any of them. It is guaranteed that there will be at least one valid solution for the given input.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1284 Minimum Number Of Flips To Convert Binary Matrix To Zero Matrix

**HARD**

Given a m x n binary matrix mat. In one step, you can choose one cell and flip it and all the four neighbors of it if they exist (Flip is changing 1 to 0 and 0 to 1). A pair of cells are called neighbors if they share one edge.

Return the minimum number of steps required to convert mat to a zero matrix or -1 if you cannot.

A binary matrix is a matrix with all cells equal to 0 or 1 only.

A zero matrix is a matrix with all cells equal to 0.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1289 Minimum Falling Path Sum Ii

HARD

Given an n x n integer matrix grid, return the minimum sum of a falling path with non-zero shifts.

A falling path with non-zero shifts is a choice of exactly one element from each row of grid such that no two elements chosen in adjacent rows are in the same column.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1292 Maximum Side Length Of A Square With Sum Less Than Or Equal To Thr

MEDIUM

Given a `m x n` matrix `mat` and an integer `threshold`.
Return the maximum side-length of a square with a sum less than or equal to `threshold` or return 0 if there is no such square.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1296 Divide Array In Sets Of K Consecutive Numbers

MEDIUM

Given an array of integers nums and a positive integer k, find whether it is possible to divide this array into sets of k consecutive numbers.

Return true if it is possible. Otherwise, return false.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1298 Maximum Candies You Can Get From Boxes

HARD

You have n boxes labeled from 0 to n - 1. You are given four arrays: status, candies, keys, and containedBoxes where:
- status[i] is 1 if the ith box is open and 0 if the ith box is closed,
- candies[i] is the number of candies in the ith box,
- keys[i] is a list of the labels of the boxes you can open after opening the ith box.
- containedBoxes[i] is a list of the boxes you found inside the ith box.

You are given an integer array initialBoxes that contains the labels of the boxes you initially have. You can take all the candies in any open box and you can use the keys in it to open new boxes and you also can use the boxes you find in it.

Return the maximum number of candies you can get following the rules above.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1299 Replace Elements With Greatest Element On Right Side

EASY

Given an array arr, replace every element in that array with the greatest element among the elements to its right, and replace the last element with -1.

After doing so, return the array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1300 Sum Of Mutated Array Closest To Target

MEDIUM

Given an integer array arr and a target value target, return the integer value such that when we change all the integers larger than value in the given array to be equal to value, the sum of the array gets as close as possible (in absolute difference) to target.

In case of a tie, return the minimum such integer.

Notice that the answer is not neccesarilly a number from arr.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1301 Number Of Paths With Max Score

HARD

You are given a square board of characters. You can move on the board starting at the bottom right square marked with the character 'S'.

You need to reach the top left square marked with the character 'E'. The rest of the squares are labeled either with a numeric character 1, 2, ..., 9 or with an obstacle 'X'. In one move you can go up, left or up-left (diagonally) only if there is no obstacle there.

Return a list of two integers: the first integer is the maximum sum of numeric characters you can collect, and the second is the number of such paths that you can take to get that maximum sum, taken modulo 10^9 + 7.

In case there is no path, return [0, 0].

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1302 Deepest Leaves Sum

MEDIUM

Given the root of a binary tree, return the sum of values of its deepest leaves.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1304 Find N Unique Integers Sum Up To Zero

EASY

Given an integer n, return any array containing
n unique integers such that they add up to 0.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1307 Verbal Arithmetic Puzzle

HARD

Given an equation, represented by words on the left side and the result on the right side.

You need to check if the equation is solvable under the following rules:
- Each character is decoded as one digit (0 - 9).
- No two characters can map to the same digit.
- Each words[i] and result are decoded as one number without leading zeros.
- Sum of numbers on the left side (words) will equal to the number on the right side (result).

Return true if the equation is solvable, otherwise return false.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1311 Get Watched Videos By Your Friends

MEDIUM

There are n people, each person has a unique id between 0 and n-1. Given the arrays watchedVideos and friends, where watchedVideos[i] and friends[i] contain the list of watched videos and the list of friends respectively for the person with id = i.

Level 1 of videos are all watched videos by your friends, level 2 of videos are all watched videos by the friends of your friends and so on. In general, the level k of videos are all watched videos by people with the shortest path exactly equal to k with you. Given your id and the level of videos, return the list of videos ordered by their frequencies (increasing). For videos with the same frequency order them alphabetically from least to greatest.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1314 Matrix Block Sum

MEDIUM

Given a m x n matrix mat and an integer k, return a matrix answer where each answer[i][j] is the sum of all elements mat[r][c] for:
- i - k <= r <= i + k,
- j - k <= c <= j + k, and
- (r, c) is a valid position in the matrix.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1315 Sum Of Nodes With Even Valued Grandparent

MEDIUM

Given the root of a binary tree, return the sum of values of nodes with an even-valued grandparent. If there are no nodes with an even-valued grandparent, return 0.

A grandparent of a node is the parent of its parent if it exists.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1317 Convert Integer To The Sum Of Two No Zero Integers

EASY

Given an integer n. No-Zero integer is a positive integer which doesn't contain any 0 in its decimal representation.

Return a list of two integers [A, B] where:

- A and B are No-Zero integers.
- A + B = n

It's guarateed that there is at least one valid solution.
If there are many valid solutions you can return any of them.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1319 Number Of Operations To Make Network Connected

MEDIUM

There are n computers numbered from 0 to n - 1 connected by ethernet cables connections forming a network where connections[i] = [ai, bi] represents a connection between computers ai and bi. Any computer can reach any other computer directly or indirectly through the network.

You are given an initial computer network connections. You can extract certain cables between

two directly connected computers, and place them between any pair of disconnected computers to make them directly connected.

Return the minimum number of times you need to do this in order to make all the computers connected. If it is not possible, return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1323 Maximum 69 Number

**EASY**

Given a positive integer num consisting only of digits 6 and 9.

Return the maximum number you can get by changing at most one digit (6 becomes 9, and 9 becomes 6).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1326 Minimum Number Of Taps To Open To Water A Garden

**HARD**

There is a one-dimensional garden on the x-axis. The garden starts at the point 0 and ends at the point n. (i.e., the length of the garden is n).

There are n + 1 taps located at points [0, 1, ..., n] in the garden.

Given an integer n and an integer array ranges of length n + 1 where ranges[i] (0-indexed) means the i-th tap can water the area [i - ranges[i], i + ranges[i]] if it was open.

Return the minimum number of taps that should be open to water the whole garden, If the garden cannot be watered return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1331 Rank Transform Of An Array

**EASY**

Given an array of integers arr, replace each element with its rank.

The rank represents how large the element is. The rank has the following rules:
- Rank is an integer starting from 1.
- The larger the element, the larger the rank. If two elements are equal, their rank must be the same.
- Rank should be as small as possible.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1334 Find The City With The Smallest Number Of Neighbors At A Threshold

**MEDIUM**

There are n cities numbered from 0 to n-1. Given the array edges where edges[i] = [fromi, toi, weighti] represents a bidirectional and weighted edge between cities fromi and toi, and given the integer distanceThreshold.

Return the city with the smallest number of cities that are reachable through some path and whose distance is at most distanceThreshold, If there are multiple such cities, return the city with the greatest number.

Notice that the distance of a path connecting cities i and j is equal to the sum of the edges' weights along that path.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1337 The K Weakest Rows In A Matrix

EASY

You are given an m x n binary matrix mat of 1's (representing soldiers) and 0's (representing civilians). The soldiers are positioned in front of the civilians. That is, all the 1's will appear to the left of all the 0's in each row.

A row i is weaker than a row j if one of the following is true:
- The number of soldiers in row i is less than the number of soldiers in row j.
- Both rows have the same number of soldiers and i < j.

Return the indices of the k weakest rows in the matrix ordered from weakest to strongest.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1338 Reduce Array Size To The Half

MEDIUM

You are given an integer array arr. You can choose a set of integers and remove all the occurrences of these integers in the array.

Return the minimum size of the set so that at least half of the integers of the array are removed.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1342 Number Of Steps To Reduce A Number To Zero

EASY

Given an integer num, return the number of steps to reduce it to zero.

In one step, if the current number is even, you have to divide it by 2, otherwise, you have to subtract 1 from it.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1343 Number Of Sub Arrays Of Size K And Average Greater Than Or Equal T

MEDIUM

Given an array of integers arr and two integers k and threshold, return the number of sub-arrays of size k and average greater than or equal to threshold.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1346 Check If N And Its Double Exist

EASY

Given an array arr of integers, check if there exist two indices i and j such that:
- i != j
- 0 <= i, j < arr.length
- arr[i] == 2 * arr[j]

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1348 Tweet Counts Per Frequency

MEDIUM

A social media company is trying to monitor activity on their site by analyzing the number

of tweets that occur in select periods of time. These periods can be partitioned into smaller time chunks based on a certain frequency (every minute, hour, or day).

For example, the period [10, 10000] (in seconds) would be partitioned into the following time chunks with these frequencies:
- Every minute (60-second chunks): [10,69], [70,129], [130,189], ..., [9970,10000]
- Every hour (3600-second chunks): [10,3609], [3610,7209], [7210,10000]
- Every day (86400-second chunks): [10,10000]

Notice that the last chunk may be shorter than the specified frequency's chunk size and will always end with the end time of the period (10000 in the above example).

Design and implement an API to help the company with their analysis.

Implement the TweetCounts class:
- TweetCounts() Initializes the TweetCounts object.
- void recordTweet(String tweetName, int time) Stores the tweetName at the recorded time (in seconds).
- List<Integer> getTweetCountsPerFrequency(String freq, String tweetName, int startTime, int endTime) Returns a list of integers representing the number of tweets with tweetName in each time chunk for the given period of time [startTime, endTime] (in seconds) and frequency freq.
- freq is one of "minute", "hour", or "day" representing a frequency of every minute, hour, or day respectively.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1349 Maximum Students Taking Exam

**HARD**

Given a m * n matrix seats that represent seats distributions in a classroom. If a seat is broken, it is denoted by '#' character otherwise it is denoted by a '.' character.

Students can see the answers of those sitting next to the left, right, upper left and upper right, but he cannot see the answers of the student sitting directly in front or behind him. Return the maximum number of students that can take the exam together without any cheating being possible.

Students must be placed in seats in good condition.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1352 Product Of The Last K Numbers

**MEDIUM**

Design an algorithm that accepts a stream of integers and retrieves the product of the last k integers of the stream.

Implement the ProductOfNumbers class:
- ProductOfNumbers() Initializes the object with an empty stream.
- void add(int num) Appends the integer num to the stream.
- int getProduct(int k) Returns the product of the last k numbers in the current list.
You can assume that always the current list has at least k numbers.
- The test cases are generated so that, at any time, the product of any contiguous sequence of numbers will fit into a single 32-bit integer without overflowing.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1353 Maximum Number Of Events That Can Be Attended

MEDIUM

You are given an array of events where events[i] = [startDayi, endDayi]. Every event i starts at startDayi and ends at endDayi.

You can attend an event i at any day d where startTimei <= d <= endTimei. You can only attend one event at any time d.

Return the maximum number of events you can attend.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1354 Construct Target Array With Multiple Sums

HARD

You are given an array target of n integers. From a starting array arr consisting of n 1's, you may perform the following procedure:
- let x be the sum of all elements currently in your array.
- choose index i, such that 0 <= i < n and set the value of arr at index i to x.
- You may repeat this procedure as many times as needed.

Return true if it is possible to construct the target array from arr, otherwise, return false.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1357 Apply Discount Every N Orders

MEDIUM

There is a supermarket that is frequented by many customers. The products sold at the supermarket are represented as two parallel integer arrays products and prices, where the ith product has an ID of products[i] and a price of prices[i].

When a customer is paying, their bill is represented as two parallel integer arrays product and amount, where the jth product they purchased has an ID of product[j], and amount[j] is how much of the product they bought. Their subtotal is calculated as the sum of each amount[j] * (price of the jth product).

The supermarket decided to have a sale. Every nth customer paying for their groceries will be given a percentage discount. The discount amount is given by discount, where they will be given discount percent off their subtotal. More formally, if their subtotal is bill, then they would actually pay bill * ((100 - discount) / 100).

Implement the Cashier class:
- Cashier(int n, int discount, int[] products, int[] prices) Initializes the object with n, the discount, and the products and their prices.
- double getBill(int[] product, int[] amount) Returns the final total of the bill with the discount applied (if any). Answers within 10-5 of the actual value will be accepted.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1359 Count All Valid Pickup And Delivery Options

HARD

Given n orders, each order consists of a pickup and a delivery service.

Count all valid pickup/delivery possible sequences such that delivery(i) is always after of pickup(i).

Since the answer may be too large, return it modulo 10^9 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1360 Number Of Days Between Two Dates

EASY

Write a program to count the number of days between two dates.

The two dates are given as strings, their format is `YYYY-MM-DD`
as shown in the examples.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1365 How Many Numbers Are Smaller Than The Current Number

EASY

Given the array nums, for each nums[i] find out how many numbers in the array are smaller than it. That is, for each nums[i] you have to count the number of valid j's such that j != i and nums[j] < nums[i].

Return the answer in an array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1366 Rank Teams By Votes

MEDIUM

In a special ranking system, each voter gives a rank from highest to lowest to all teams participating in the competition.

The ordering of teams is decided by who received the most position-one votes. If two or more teams tie in the first position, we consider the second position to resolve the conflict, if they tie again, we continue this process until the ties are resolved. If two or more teams are still tied after considering all positions, we rank them alphabetically based on their team letter.

You are given an array of strings votes which is the votes of all voters in the ranking systems. Sort all teams according to the ranking system described above.

Return a string of all teams sorted by the ranking system.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1368 Minimum Cost To Make At Least One Valid Path In A Grid

HARD

Given an m x n grid. Each cell of the grid has a sign pointing to the next cell you should visit if you are currently in this cell. The sign of grid[i][j] can be:
- 1 which means go to the cell to the right. (i.e go from grid[i][j] to grid[i][j + 1])
- 2 which means go to the cell to the left. (i.e go from grid[i][j] to grid[i][j - 1])
- 3 which means go to the lower cell. (i.e go from grid[i][j] to grid[i + 1][j])
- 4 which means go to the upper cell. (i.e go from grid[i][j] to grid[i - 1][j])

Notice that there could be some signs on the cells of the grid that point outside the grid.

You will initially start at the upper left cell (0, 0). A valid path in the grid is a path that starts from the upper left cell (0, 0) and ends at the bottom-right cell (m - 1, n - 1) following the signs on the grid. The valid path does not have to be the shortest.

You can modify the sign on a cell with cost = 1. You can modify the sign on a cell one time only.

Return the minimum cost to make the grid have at least one valid path.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1377 Frog Position After T Seconds

HARD

Given an undirected tree consisting of n vertices numbered from 1 to n. A frog starts jumping from vertex 1. In one second, the frog jumps from its current vertex to another unvisited vertex if they are directly connected. The frog can not jump back to a visited vertex. In case the frog can jump to several vertices, it jumps randomly to one of them with the same probability. Otherwise, when the frog can not jump to any unvisited vertex, it jumps forever on the same vertex.

The edges of the undirected tree are given in the array edges, where edges[i] = [ai, bi] means that exists an edge connecting the vertices ai and bi.

Return the probability that after t seconds the frog is on the vertex target. Answers within 10-5 of the actual answer will be accepted.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1380 Lucky Numbers In A Matrix

EASY

Given an m x n matrix of distinct numbers, return all lucky numbers in the matrix in any order.

A lucky number is an element of the matrix such that it is the minimum element in its row and maximum in its column.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1385 Find The Distance Value Between Two Arrays

EASY

Given two integer arrays arr1 and arr2, and the integer d, return the distance value between the two arrays.

The distance value is defined as the number of elements arr1[i] such that there is not any element arr2[j] where |arr1[i]-arr2[j]| <= d.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1386 Cinema Seat Allocation

MEDIUM

A cinema has n rows of seats, numbered from 1 to n and there are ten seats in each row, labelled from 1 to 10 as shown in the figure above.

Given the array reservedSeats containing the numbers of seats already reserved, for example, reservedSeats[i] = [3,8] means the seat located in row 3 and labelled with 8 is already reserved.

Return the maximum number of four-person groups you can assign on the cinema seats. A four-person group occupies four adjacent seats in one single row. Seats across an aisle (such as [3,3] and [3,4]) are not considered to be adjacent, but there is an exceptional case on which an aisle split a four-person group, in that case, the aisle split a four-person group in the middle, which means to have two people on each side.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1388 Pizza With 3N Slices

HARD

There is a pizza with 3n slices of varying size, you and your friends will take slices of pizza as follows:
- You will pick any pizza slice.
- Your friend Alice will pick the next slice in the anti-clockwise direction of your pick.
- Your friend Bob will pick the next slice in the clockwise direction of your pick.
- Repeat until there are no more slices of pizzas.

Given an integer array slices that represent the sizes of the pizza slices in a clockwise direction, return the maximum possible sum of slice sizes that you can pick.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1389 Create Target Array In The Given Order

EASY

Given two arrays of integers nums and index. Your task is to create target array under the following rules:

- Initially target array is empty.
- From left to right read nums[i] and index[i], insert at index index[i] the value nums[i] in target array.
- Repeat the previous step until there are no elements to read in nums and index.
- Return the target array.

It is guaranteed that the insertion operations will be valid.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1390 Four Divisors

MEDIUM

Given an integer array nums, return the sum of divisors of the integers in that array that have exactly four divisors. If there is no such integer in the array, return 0.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1391 Check If There Is A Valid Path In A Grid

MEDIUM

You are given an m x n grid. Each cell of grid represents a street. The street of grid[i][j] can be:
- 1 which means a street connecting the left cell and the right cell.
- 2 which means a street connecting the upper cell and the lower cell.
- 3 which means a street connecting the left cell and the lower cell.
- 4 which means a street connecting the right cell and the lower cell.
- 5 which means a street connecting the left cell and the upper cell.
- 6 which means a street connecting the right cell and the upper cell.

You will initially start at the street of the upper-left cell (0, 0). A valid path in the grid is a path that starts from the upper left cell (0, 0) and ends at the bottom-right cell (m - 1, n - 1). The path should only follow the streets.

Notice that you are not allowed to change any street.

Return true if there is a valid path in the grid or false otherwise.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1394 Find Lucky Integer In An Array

EASY

Given an array of integers arr, a lucky integer is an integer that has a frequency in the array equal to its value.

Return the largest lucky integer in the array. If there is no lucky integer return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1395 Count Number Of Teams

MEDIUM

There are n soldiers standing in a line. Each soldier is assigned a unique rating value.

You have to form a team of 3 soldiers amongst them under the following rules:
- Choose 3 soldiers with index (i, j, k) with rating (rating[i], rating[j], rating[k]).
- A team is valid if: (rating[i] < rating[j] < rating[k]) or (rating[i] > rating[j] > rating[k]) where (0 <= i < j < k < n).

Return the number of teams you can form given the conditions. (soldiers can be part of multiple teams).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1396 Design Underground System

MEDIUM

An underground railway system is keeping track of customer travel times between different stations. They are using this data to calculate the average time it takes to travel from one station to another.

Implement the UndergroundSystem class:
- void checkIn(int id, string stationName, int t)
- A customer with a card ID equal to id, checks in at the station stationName at time t.
- A customer can only be checked into one place at a time.
- void checkOut(int id, string stationName, int t)
- A customer with a card ID equal to id, checks out from the station stationName at time t.
- double getAverageTime(string startStation, string endStation)
- Returns the average time it takes to travel from startStation to endStation.
- The average time is computed from all the previous traveling times from startStation to endStation that happened directly, meaning a check in at startStation followed by a check out from endStation.
- The time it takes to travel from startStation to endStation may be different from the time it takes to travel from endStation to startStation.

There will be at least one customer that has traveled from startStation to endStation before getAverageTime is called.
You may assume all calls to the checkIn and checkOut methods are consistent. If a customer checks in at time t1 then checks out at time t2, then t1 < t2. All events happen in chronological order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1399 Count Largest Group

EASY

You are given an integer n.

Each number from 1 to n is grouped according to the sum of its digits.

Return the number of groups that have the largest size.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1404 Number Of Steps To Reduce A Number In Binary Representation To One

**MEDIUM**

Given the binary representation of an integer as a string s, return the number of steps to reduce it to 1 under the following rules:
- If the current number is even, you have to divide it by 2.
- If the current number is odd, you have to add 1 to it.

It is guaranteed that you can always reach one for all test cases.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1411 Number Of Ways To Paint N 3 Grid

**HARD**

You have a grid of size n x 3 and you want to paint each cell of the grid with exactly one of the three colors: Red, Yellow, or Green while making sure that no two adjacent cells have the same color (i.e., no two cells that share vertical or horizontal sides have the same color).

Given n the number of rows of the grid, return the number of ways you can paint this grid. As the answer may grow large, the answer must be computed modulo 109 + 7.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1413 Minimum Value To Get Positive Step By Step Sum

**EASY**

Given an array of integers nums, you start with an initial positive value startValue.

In each iteration, you calculate the step by step sum of startValue plus elements in nums (from left to right).

Return the minimum positive value of startValue such that the step by step sum is never less than 1.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1416 Restore The Array

**HARD**

A program was supposed to print an array of integers. The program forgot to print whitespaces and the array is printed as a string of digits s and all we know is that all integers in the array were in the range [1, k] and there are no leading zeros in the array.

Given the string s and the integer k, return the number of the possible arrays that can be printed as s using the mentioned program. Since the answer may be very large, return it modulo 109 + 7.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1418 Display Table Of Food Orders In A Restaurant

**MEDIUM**

Given the array orders, which represents the orders that customers have done in a restaurant. More specifically orders[i]=[customerNamei,tableNumberi,foodItemi] where customerNamei is the name of the customer, tableNumberi is the table customer sit at, and foodItemi is the item

customer orders.

Return the restaurant's “display table”. The “display table” is a table whose row entries denote how many of each food item each table ordered. The first column is the table number and the remaining columns correspond to each food item in alphabetical order. The first row should be a header whose first column is “Table”, followed by the names of the food items. Note that the customer names are not part of the table. Additionally, the rows should be sorted in numerically increasing order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1419 Minimum Number Of Frogs Croaking

**MEDIUM**

You are given the string croakOfFrogs, which represents a combination of the string "croak" from different frogs, that is, multiple frogs can croak at the same time, so multiple "croak" are mixed.

Return the minimum number of different frogs to finish all the croaks in the given string.

A valid "croak" means a frog is printing five letters 'c', 'r', 'o', 'a', and 'k' sequentially. The frogs have to print all five letters to finish a croak. If the given string is not a combination of a valid "croak" return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1420 Build Array Where You Can Find The Maximum Exactly K Comparisons

**HARD**

You are given three integers n, m and k. Consider the following algorithm to find the maximum element of an array of positive integers.

You should build the array arr which has the following properties:
- arr has exactly n integers.
- 1 <= arr[i] <= m where (0 <= i < n).
- After applying the mentioned algorithm to arr, the value search_cost is equal to k.

Return the number of ways to build the array arr under the mentioned conditions. As the answer may grow large, the answer must be computed modulo 109 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1425 Constrained Subsequence Sum

**HARD**

Given an integer array nums and an integer k, return the maximum sum of a non-empty subsequence of that array such that for every two consecutive integers in the subsequence, nums[i] and nums[j], where i < j, the condition j - i <= k is satisfied.

A subsequence of an array is obtained by deleting some number of elements (can be zero) from the array, leaving the remaining elements in their original order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1426 Counting Elements

**EASY**

Given an integer array arr, count how many elements x there are, such that x + 1 is also in arr. If there are duplicates in arr, count them separately.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1429 First Unique Number

MEDIUM

You have a queue of integers, you need to retrieve the first unique integer in the queue.

Implement the FirstUnique class:
- FirstUnique(int[] nums) Initializes the object with the numbers in the queue.
- int showFirstUnique() returns the value of the first unique integer of the queue, and returns -1 if there is no such integer.
- void add(int value) insert value to the queue.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1431 Kids With The Greatest Number Of Candies

EASY

There are n kids with candies. You are given an integer array candies, where each candies[i] represents the number of candies the ith kid has, and an integer extraCandies, denoting the number of extra candies that you have.

Return a boolean array result of length n, where result[i] is true if, after giving the ith kid all the extraCandies, they will have the greatest number of candies among all the kids, or false otherwise.

Note that multiple kids can have the greatest number of candies.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1432 Max Difference You Can Get From Changing An Integer

MEDIUM

You are given an integer num. You will apply the following steps exactly two times:
- Pick a digit x (0 <= x <= 9).
- Pick another digit y (0 <= y <= 9). The digit y can be equal to x.
- Replace all the occurrences of x in the decimal representation of num by y.
- The new integer cannot have any leading zeros, also the new integer cannot be 0.

Let a and b be the results of applying the operations to num the first and second times, respectively.

Return the max difference between a and b.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1434 Number Of Ways To Wear Different Hats To Each Other

HARD

There are n people and 40 types of hats labeled from 1 to 40.

Given a 2D integer array hats, where hats[i] is a list of all hats preferred by the ith person.

Return the number of ways that n people can wear different hats from each other.

Since the answer may be too large, return it modulo 109 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1436 Destination City

EASY

You are given the array paths, where paths[i] = [cityAi, cityBi] means there exists a direct path going from cityAi to cityBi. Return the destination city, that is, the city without any path outgoing to another city.

It is guaranteed that the graph of paths forms a line without any loop, therefore, there will be exactly one destination city.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1442 Count Triplets That Can Form Two Arrays Of Equal Xor

MEDIUM

Given an array of integers arr.

We want to select three indices i, j and k where (0 <= i < j <= k < arr.length).

Let's define a and b as follows:
- a = arr[i] ^ arr[i + 1] ^ ... ^ arr[j - 1]
- b = arr[j] ^ arr[j + 1] ^ ... ^ arr[k]

Note that ^ denotes the bitwise-xor operation.

Return the number of triplets (i, j and k) Where a == b.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1444 Number Of Ways Of Cutting A Pizza

HARD

Given a rectangular pizza represented as a rows x cols matrix containing the following characters: 'A' (an apple) and '.' (empty cell) and given the integer k. You have to cut the pizza into k pieces using k-1 cuts.

For each cut you choose the direction: vertical or horizontal, then you choose a cut position at the cell boundary and cut the pizza into two pieces. If you cut the pizza vertically, give the left part of the pizza to a person. If you cut the pizza horizontally, give the upper part of the pizza to a person. Give the last piece of pizza to the last person.

Return the number of ways of cutting the pizza such that each piece contains at least one apple. Since the answer can be a huge number, return this modulo 10^9 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1450 Number Of Students Doing Homework At A Given Time

EASY

Given two integer arrays startTime and endTime and given an integer queryTime.

The ith student started doing their homework at the time startTime[i] and finished it at time endTime[i].

Return the number of students doing their homework at time queryTime. More formally, return the number of students where queryTime lays in the interval [startTime[i], endTime[i]] inclusive.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1453 Maximum Number Of Darts Inside Of A Circular Dartboard

HARD

Alice is throwing n darts on a very large wall. You are given an array darts where darts[i] = [xi, yi] is the position of the ith dart that Alice threw on the wall.

Bob knows the positions of the n darts on the wall. He wants to place a dartboard of radius r on the wall so that the maximum number of darts that Alice throws lie on the dartboard.

Given the integer r, return the maximum number of darts that can lie on the dartboard.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1458 Max Dot Product Of Two Subsequences

HARD

Given two arrays nums1 and nums2.

Return the maximum dot product between non-empty subsequences of nums1 and nums2 with the same length.

A subsequence of a array is a new array which is formed from the original array by deleting some (can be none) of the characters without disturbing the relative positions of the remaining characters. (ie, [2,3,5] is a subsequence of [1,2,3,4,5] while [1,5,3] is not).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1460 Make Two Arrays Equal By Reversing Sub Arrays

EASY

Given two integer arrays of equal length target and arr.

In one step, you can select any non-empty sub-array of arr and reverse it. You are allowed to make any number of steps.

Return True if you can make arr equal to target, or False otherwise.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1464 Maximum Product Of Two Elements In An Array

EASY

Given the array of integers nums, you will choose two different indices i and j of that array. Return the maximum value of (nums[i]-1)*(nums[j]-1).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1467 Probability Of A Two Boxes Having The Same Number Of Distinct Ball

HARD

Given 2n balls of k distinct colors. You will be given an integer array balls of size k where balls[i] is the number of balls of color i.

All the balls will be shuffled uniformly at random, then we will distribute the first n balls to the first box and the remaining n balls to the other box (Please read the explanation of the second example carefully).

Please note that the two boxes are considered different. For example, if we have two balls of colors a and b, and two boxes [] and (), then the distribution [a] (b) is considered different than the distribution [b] (a) (Please read the explanation of the first example carefully).

Return the probability that the two boxes have the same number of distinct balls. Answers within 10-5 of the actual value will be accepted as correct.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1470 Shuffle The Array

**EASY**

Given the array nums consisting of 2n elements in the form [x1,x2,...,xn,y1,y2,...,yn].

Return the array in the form [x1,y1,x2,y2,...,xn,yn].

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1471 The K Strongest Values In An Array

**MEDIUM**

Given an array of integers arr and an integer k.

A value arr[i] is said to be stronger than a value arr[j] if |arr[i] - m| > |arr[j] - m| where m is the centre of the array.

If |arr[i] - m| == |arr[j] - m|, then arr[i] is said to be stronger than arr[j] if arr[i] > arr[j].

Return a list of the strongest k values in the array. return the answer in any arbitrary order.

The centre is the middle value in an ordered integer list. More formally, if the length of the list is n, the centre is the element in position ((n - 1) / 2) in the sorted list (0-indexed).
- For arr = [6, -3, 7, 2, 11], n = 5 and the centre is obtained by sorting the array arr = [-3, 2, 6, 7, 11] and the centre is arr[m] where m = ((5 - 1) / 2) = 2. The centre is 6.
- For arr = [-7, 22, 17, 3], n = 4 and the centre is obtained by sorting the array arr = [-7, 3, 17, 22] and the centre is arr[m] where m = ((4 - 1) / 2) = 1. The centre is 3.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1473 Paint House Iii

**HARD**

There is a row of m houses in a small city, each house must be painted with one of the n colors (labeled from 1 to n), some houses that have been painted last summer should not be painted again.

A neighborhood is a maximal group of continuous houses that are painted with the same color.

For example: houses = [1,2,2,3,3,2,1,1] contains 5 neighborhoods [{1}, {2,2}, {3,3}, {2}, {1,1}].

Given an array houses, an m x n matrix cost and an integer target where:
- houses[i]: is the color of the house i, and 0 if the house is not painted yet.
- cost[i][j]: is the cost of paint the house i with the color j + 1.

Return the minimum cost of painting all the remaining houses in such a way that there are exactly target neighborhoods. If it is not possible, return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1475 Final Prices With A Special Discount In A Shop

**EASY**

Given the array prices where prices[i] is the price of the ith item in a shop.
There is a special discount for items in the shop, if you buy the ith item,
then you will receive a discount equivalent to prices[j] where j is the minimum

index such that j > i and prices[j] <= prices[i], otherwise, you will not receive any discount at all.

Return an array where the ith element is the final price you will pay for the ith item of the shop considering the special discount.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1477 Find Two Non Overlapping Sub Arrays Each With Target Sum

**MEDIUM**

You are given an array of integers arr and an integer target.

You have to find two non-overlapping sub-arrays of arr each with a sum equal target. There can be multiple answers so you have to find an answer where the sum of the lengths of the two sub-arrays is minimum.

Return the minimum sum of the lengths of the two required sub-arrays, or return -1 if you cannot find such two sub-arrays.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1478 Allocate Mailboxes

**HARD**

Given the array houses where houses[i] is the location of the ith house along a street and an integer k, allocate k mailboxes in the street.

Return the minimum total distance between each house and its nearest mailbox.

The test cases are generated so that the answer fits in a 32-bit integer.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1480 Running Sum Of 1D Array

**EASY**

Given an array nums. We define a running sum of an array as runningSum[i] = sum(nums[0]…nums[i]).

Return the running sum of nums.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1481 Least Number Of Unique Integers After K Removals

**MEDIUM**

Given an array of integers arr and an integer k. Find the least number of unique integers after removing exactly k elements.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1482 Minimum Number Of Days To Make M Bouquets

**MEDIUM**

You are given an integer array bloomDay, an integer m and an integer k.

You want to make m bouquets. To make a bouquet, you need to use k adjacent flowers from the garden.

The garden consists of n flowers, the ith flower will bloom in the bloomDay[i] and then can be used in exactly one bouquet.

Return the minimum number of days you need to wait to be able to make m bouquets from the garden.
If it is impossible to make m bouquets return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1486 Xor Operation In An Array

**EASY**

Given an integer n and an integer start.

Define an array nums where nums[i] = start + 2*i (0-indexed) and n == nums.length.

Return the bitwise XOR of all elements of nums.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1487 Making File Names Unique

**MEDIUM**

Given an array of strings names of size n. You will create n folders in your file system such that, at the ith minute, you will create a folder with the name names[i].

Since two files cannot have the same name, if you enter a folder name that was previously used, the system will have a suffix addition to its name in the form of (k), where, k is the smallest positive integer such that the obtained name remains unique.

Return an array of strings of length n where ans[i] is the actual name the system will assign to the ith folder when you create it.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1488 Avoid Flood In The City

**MEDIUM**

Your country has an infinite number of lakes. Initially, all the lakes are empty, but when it rains over the nth lake, the nth lake becomes full of water. If it rains over a lake that is full of water, there will be a flood. Your goal is to avoid floods in any lake.

Given an integer array rains where:
- rains[i] > 0 means there will be rains over the rains[i] lake.
- rains[i] == 0 means there are no rains this day and you can choose one lake this day and dry it.

Return an array ans where:
- ans.length == rains.length
- ans[i] == -1 if rains[i] > 0.
- ans[i] is the lake you choose to dry in the ith day if rains[i] == 0.

If there are multiple valid answers return any of them. If it is impossible to avoid flood return an empty array.

Notice that if you chose to dry a full lake, it becomes empty, but if you chose to dry an empty lake, nothing changes.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1496 Path Crossing

**EASY**

Given a string path, where path[i] = 'N', 'S', 'E' or 'W', each representing moving one unit north, south, east, or west, respectively. You start at the origin (0, 0) on a 2D plane and walk on the path specified by path.

Return true if the path crosses itself at any point, that is, if at any time
you are on a location you have previously visited. Return false otherwise.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1497 Check If Array Pairs Are Divisible By K

**MEDIUM**

Given an array of integers arr of even length n and an integer k.

We want to divide the array into exactly n / 2 pairs such that the sum of each pair is
divisible by k.

Return true If you can find a way to do that or false otherwise.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1498 Number Of Subsequences That Satisfy The Given Sum Condition

**MEDIUM**

You are given an array of integers nums and an integer target.

Return the number of non-empty subsequences of nums such that the sum of the minimum and
maximum element on it is less or equal to target. Since the answer may be too large,
return it modulo 109 + 7.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1500 Design A File Sharing System

**MEDIUM**

We will use a file-sharing system to share a very large file which consists of m small
chunks with IDs from 1 to m.

When users join the system, the system should assign a unique ID to them. The unique ID
should be used once for each user, but when a user leaves the system, the ID can be reused again.

Users can request a certain chunk of the file, the system should return a list of IDs of
all the users who own this chunk. If the user receives a non-empty list of IDs, they receive
the requested chunk successfully.

Implement the FileSharing class:
- FileSharing(int m) Initializes the object with a file of m chunks.
- int join(int[] ownedChunks): A new user joined the system owning some chunks of the file,
the system should assign an id to the user which is the smallest positive integer not
taken by any other user. Return the assigned id.
- void leave(int userID): The user with userID will leave the system, you cannot take file
chunks from them anymore.
- int[] request(int userID, int chunkID): The user userID requested the file chunk with chunkID.
Return a list of the IDs of all users that own this chunk sorted in ascending order.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1504 Count Submatrices With All Ones

**MEDIUM**

Given an m x n binary matrix mat, return the number of submatrices that have all ones.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1509 Minimum Difference Between Largest And Smallest Value In Three Mov

MEDIUM

You are given an integer array nums.

In one move, you can choose one element of nums and change it to any value.

Return the minimum difference between the largest and smallest value of nums after performing at most three moves.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1512 Number Of Good Pairs

EASY

Given an array of integers nums, return the number of good pairs.

A pair (i, j) is called good if nums[i] == nums[j] and i < j.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1521 Find A Value Of A Mysterious Function Closest To Target

HARD

Winston was given the above mysterious function func. He has an integer array arr and an integer target and he wants to find the values l and r that make the value |func(arr, l, r) - target| minimum possible.

Return the minimum possible value of |func(arr, l, r) - target|.

Notice that func should be called with the values l and r where 0 <= l, r < arr.length.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1524 Number Of Sub Arrays With Odd Sum

MEDIUM

Given an array of integers arr, return the number of subarrays with an odd sum.

Since the answer can be very large, return it modulo 109 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1534 Count Good Triplets

EASY

Given an array of integers arr, and three integers a, b and c. You need to find the number of good triplets.

A triplet (arr[i], arr[j], arr[k]) is good if the following conditions are true:
- 0 <= i < j < k < arr.length
- |arr[i] - arr[j]| <= a
- |arr[j] - arr[k]| <= b
- |arr[i] - arr[k]| <= c

Where |x| denotes the absolute value of x.

Return the number of good triplets.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1535 Find The Winner Of An Array Game

MEDIUM

Given an integer array arr of distinct integers and an integer k.

A game will be played between the first two elements of the array (i.e. arr[0] and arr[1]). In each round of the game, we compare arr[0] with arr[1], the larger integer wins and remains at position 0, and the smaller integer moves to the end of the array. The game ends when an integer wins k consecutive rounds.

Return the integer which will win the game.

It is guaranteed that there will be a winner of the game.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1536 Minimum Swaps To Arrange A Binary Grid

MEDIUM

Given an n x n binary grid, in one step you can choose two adjacent rows of the grid and swap them.

A grid is said to be valid if all the cells above the main diagonal are zeros.

Return the minimum number of steps needed to make the grid valid, or -1 if the grid cannot be valid.

The main diagonal of a grid is the diagonal that starts at cell (1, 1) and ends at cell (n, n).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1538 Guess The Majority In A Hidden Array

MEDIUM

We have an integer array nums, where all the integers in nums are 0 or 1. You will not be given direct access to the array, instead, you will have an API ArrayReader which have the following functions:
- int query(int a, int b, int c, int d): where 0 <= a < b < c < d < ArrayReader.length(). The function returns the distribution of the value of the 4 elements and returns:
- 4 : if the values of the 4 elements are the same (0 or 1).
- 2 : if three elements have a value equal to 0 and one element has value equal to 1 or vice versa.
- 0 : if two element have a value equal to 0 and two elements have a value equal to 1.
- int length(): Returns the size of the array.

You are allowed to call query() 2 * n times at most where n is equal to ArrayReader.length().

Return any index of the most frequent value in nums, in case of tie, return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1539 Kth Missing Positive Number

EASY

Given an array arr of positive integers sorted in a strictly increasing order, and an integer k.

Return the kth positive integer that is missing from this array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1547 Minimum Cost To Cut A Stick

HARD

Given a wooden stick of length n units. The stick is labelled from 0 to n. For example, a stick of length 6 is labelled as follows.

Given an integer array cuts where cuts[i] denotes a position you should perform a cut at.

You should perform the cuts in order, you can change the order of the cuts as you wish.

The cost of one cut is the length of the stick to be cut, the total cost is the sum of costs of all cuts. When you cut a stick, it will be split into two smaller sticks (i.e. the sum of their lengths is the length of the stick before the cut). Please refer to the first example for a better explanation.

Return the minimum total cost of the cuts.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1551 Minimum Operations To Make Array Equal

MEDIUM

You have an array arr of length n where arr[i] = (2 * i) + 1 for all valid values of i (i.e., 0 <= i < n).

In one operation, you can select two indices x and y where 0 <= x, y < n and subtract 1 from arr[x] and add 1 to arr[y] (i.e., perform arr[x] -=1 and arr[y] += 1). The goal is to make all the elements of the array equal. It is guaranteed that all the elements of the array can be made equal using some operations.

Given an integer n, the length of the array, return the minimum number of operations needed to make all the elements of arr equal.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1553 Minimum Number Of Days To Eat N Oranges

HARD

There are n oranges in the kitchen and you decided to eat some of these oranges every day as follows:
- Eat one orange.
- If the number of remaining oranges n is divisible by 2 then you can eat n / 2 oranges.
- If the number of remaining oranges n is divisible by 3 then you can eat 2 * (n / 3) oranges.

You can only choose one of the actions per day.

Given the integer n, return the minimum number of days to eat n oranges.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1557 Minimum Number Of Vertices To Reach All Nodes

MEDIUM

Given a directed acyclic graph, with n vertices numbered from 0 to n-1, and an array edges where edges[i] = [fromi, toi] represents a directed edge from node fromi to node toi.

Find the smallest set of vertices from which all nodes in the graph are reachable. It's guaranteed that a unique solution exists.

Notice that you can return the vertices in any order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1558 Minimum Numbers Of Function Calls To Make Target Array

MEDIUM

You are given an integer array nums. You have an integer array arr of the same length with all values set to 0 initially. You also have the following modify function.

You want to use the modify function to convert arr to nums using the minimum number of calls.

Return the minimum number of function calls to make nums from arr.

The test cases are generated so that the answer fits in a 32-bit signed integer.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1559 Detect Cycles In 2D Grid

MEDIUM

Given a 2D array of characters grid of size m x n, you need to find if there exists any cycle consisting of the same value in grid.

A cycle is a path of length 4 or more in the grid that starts and ends at the same cell. From a given cell, you can move to one of the cells adjacent to it - in one of the four directions (up, down, left, or right), if it has the same value of the current cell.

Also, you cannot move to the cell that you visited in your last move. For example, the cycle (1, 1) -> (1, 2) -> (1, 1) is invalid because from (1, 2) we visited (1, 1) which was the last visited cell.

Return true if any cycle of the same value exists in grid, otherwise, return false.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1561 Maximum Number Of Coins You Can Get

MEDIUM

There are 3n piles of coins of varying size, you and your friends will take piles of coins as follows:
- In each step, you will choose any 3 piles of coins (not necessarily consecutive).
- Of your choice, Alice will pick the pile with the maximum number of coins.
- You will pick the next pile with the maximum number of coins.
- Your friend Bob will pick the last pile.
- Repeat until there are no more piles of coins.

Given an array of integers piles where piles[i] is the number of coins in the ith pile.

Return the maximum number of coins that you can have.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1562 Find Latest Group Of Size M

MEDIUM

Given an array arr that represents a permutation of numbers from 1 to n.

You have a binary string of size n that initially has all its bits set to zero. At each step i (assuming both the binary string and arr are 1-indexed) from 1 to n, the bit at position arr[i] is set to 1.

You are also given an integer m. Find the latest step at which there exists a group of ones of length m. A group of ones is a contiguous substring of 1's such that it cannot be extended in either direction.

Return the latest step at which there exists a group of ones of length exactly m. If no such group exists, return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1570 Dot Product Of Two Sparse Vectors

MEDIUM

Given two sparse vectors, compute their dot product.

Implement class SparseVector:
- SparseVector(nums) Initializes the object with the vector nums
- dotProduct(vec) Compute the dot product between the instance of SparseVector and vec

A sparse vector is a vector that has mostly zero values, you should store the sparse vector efficiently and compute the dot product between two SparseVector.

Follow up: What if only one of the vectors is sparse?

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1572 Matrix Diagonal Sum

EASY

Given a square matrix mat, return the sum of the matrix diagonals.

Only include the sum of all the elements on the primary diagonal and all the elements on the secondary diagonal that are not part of the primary diagonal.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1575 Count All Possible Routes

HARD

You are given an array of distinct positive integers locations where locations[i] represents the position of city i. You are also given integers start, finish and fuel representing the starting city, ending city, and the initial amount of fuel you have, respectively.

At each step, if you are at city i, you can pick any city j such that j != i and 0 <= j < locations.length and move to city j. Moving from city i to city j reduces the amount of fuel you have by |locations[i] - locations[j]|. Please notice that |x| denotes the absolute value of x.

Notice that fuel cannot become negative at any point in time, and that you are allowed to visit any city more than once (including start and finish).

Return the count of all possible routes from start to finish. Since the answer may be too large, return it modulo 109 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1577 Number Of Ways Where Square Of Number Is Equal To Product Of Two N

MEDIUM

Given two arrays of integers nums1 and nums2, return the number of triplets formed (type 1 and type 2) under the following rules:
- Type 1: Triplet (i, j, k) if nums1[i]2 == nums2[j] * nums2[k] where 0 <= i < nums1.length and 0 <= j < k < nums2.length.
- Type 2: Triplet (i, j, k) if nums2[i]2 == nums1[j] * nums1[k] where 0 <= i < nums2.length and 0 <= j < k < nums1.length.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1582 Special Positions In A Binary Matrix

EASY

Given an m x n binary matrix mat, return the number of special positions in mat.

A position (i, j) is called special if mat[i][j] == 1 and all other elements in row i and column j are 0 (rows and columns are 0-indexed).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1583 Count Unhappy Friends

MEDIUM

You are given a list of preferences for n friends, where n is always even.

For each person i, preferences[i] contains a list of friends sorted in the order of preference. In other words, a friend earlier in the list is more preferred than a friend later in the list. Friends in each list are denoted by integers from 0 to n-1.

All the friends are divided into pairs. The pairings are given in a list pairs, where pairs[i] = [xi, yi] denotes xi is paired with yi and yi is paired with xi.

However, this pairing may cause some of the friends to be unhappy. A friend x is unhappy if x is paired with y and there exists a friend u who is paired with v but:
- x prefers u over y, and
- u prefers x over v.

Return the number of unhappy friends.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1584 Min Cost To Connect All Points

MEDIUM

You are given an array points representing integer coordinates of some points on a 2D-plane, where points[i] = [xi, yi].

The cost of connecting two points [xi, yi] and [xj, yj] is the manhattan distance between them: |xi - xj| + |yi - yj|, where |val| denotes the absolute value of val.

Return the minimum cost to make all points connected. All points are connected if there is exactly one simple path between any two points.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1590 Make Sum Divisible By P

MEDIUM

Given an array of positive integers nums, remove the smallest subarray (possibly empty) such that the sum of the remaining elements is divisible by p. It is not allowed to remove the whole array.

Return the length of the smallest subarray that you need to remove, or -1 if it's impossible.

A subarray is defined as a contiguous block of elements in the array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1591 Strange Printer Ii

HARD

There is a strange printer with the following two special requirements:

- On each turn, the printer will print a solid rectangular pattern of a single color on the grid. This will cover up the existing colors in the rectangle.
- Once the printer has used a color for the above operation, the same color cannot be used again.

You are given a m x n matrix targetGrid, where targetGrid[row][col] is the color in the position (row, col) of the grid.

Return true if it is possible to print the matrix targetGrid, otherwise, return false.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1594 Maximum Non Negative Product In A Matrix

**MEDIUM**

You are given a m x n matrix grid. Initially, you are located at the top-left corner (0, 0), and in each step, you can only move right or down in the matrix.

Among all possible paths starting from the top-left corner (0, 0) and ending in the bottom-right corner (m - 1, n - 1), find the path with the maximum non-negative product. The product of a path is the product of all integers in the grid cells visited along the path.

Return the maximum non-negative product modulo 109 + 7. If the maximum product is negative, return -1.

Notice that the modulo is performed after getting the maximum product.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1600 Throne Inheritance

**MEDIUM**

A kingdom consists of a king, his children, his grandchildren, and so on. Every once in a while, someone in the family dies or a child is born.

The kingdom has a well-defined order of inheritance that consists of the king as the first member. Let's define the recursive function Successor(x, curOrder), which given a person x and the inheritance order so far, returns who should be the next person after x in the order of inheritance.

For example, assume we have a kingdom that consists of the king, his children Alice and Bob (Alice is older than Bob), and finally Alice's son Jack.

get ["king", "Alice"].
get ["king", "Alice", "Jack"].
get ["king", "Alice", "Jack", "Bob"].
will be ["king", "Alice", "Jack", "Bob"].

Using the above function, we can always obtain a unique order of inheritance.

Implement the ThroneInheritance class:
- ThroneInheritance(string kingName) Initializes an object of the ThroneInheritance class. The name of the king is given as part of the constructor.
- void birth(string parentName, string childName) Indicates that parentName gave birth to childName.
- void death(string name) Indicates the death of name. The death of the person doesn't affect the Successor function nor the current inheritance order. You can treat it as just marking the person as dead.
- string[] getInheritanceOrder() Returns a list representing the current order of inheritance excluding dead people.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1601 Maximum Number Of Achievable Transfer Requests

HARD

We have n buildings numbered from 0 to n - 1. Each building has a number of employees. It's transfer season, and some employees want to change the building they reside in.

You are given an array requests where requests[i] = [fromi, toi] represents an employee's request to transfer from building fromi to building toi.

All buildings are full, so a list of requests is achievable only if for each building, the net change in employee transfers is zero. This means the number of employees leaving is equal to the number of employees moving in. For example if n = 3 and two employees are leaving building 0, one is leaving building 1, and one is leaving building 2, there should be two employees moving to building 0, one employee moving to building 1, and one employee moving to building 2.

Return the maximum number of achievable requests.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1604 Alert Using Same Key Card Three Or More Times In A One Hour Period

MEDIUM

LeetCode company workers use key-cards to unlock office doors. Each time a worker uses their key-card, the security system saves the worker's name and the time when it was used. The system emits an alert if any worker uses the key-card three or more times in a one-hour period.

You are given a list of strings keyName and keyTime where [keyName[i], keyTime[i]] corresponds to a person's name and the time when their key-card was used in a single day.

Access times are given in the 24-hour time format "HH:MM", such as "23:51" and "09:49".

Return a list of unique worker names who received an alert for frequent keycard use. Sort the names in ascending order alphabetically.

Notice that "10:00" - "11:00" is considered to be within a one-hour period, while "22:51" - "23:52" is not considered to be within a one-hour period.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1605 Find Valid Matrix Given Row And Column Sums

MEDIUM

You are given two arrays rowSum and colSum of non-negative integers where rowSum[i] is the sum of the elements in the ith row and colSum[j] is the sum of the elements of the jth column of a 2D matrix. In other words, you do not know the elements of the matrix, but you do know the sums of each row and column.

Find any matrix of non-negative integers of size rowSum.length x colSum.length that satisfies the rowSum and colSum requirements.

Return a 2D array representing any matrix that fulfills the requirements. It's guaranteed that at least one matrix that fulfills the requirements exists.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1606 Find Servers That Handled Most Number Of Requests

HARD

You have k servers numbered from 0 to k-1 that are being used to handle multiple requests simultaneously. Each server has infinite computational capacity but cannot handle more than

one request at a time. The requests are assigned to servers according to a specific algorithm:
- The ith (0-indexed) request arrives.
- If all servers are busy, the request is dropped (not handled at all).
- If the (i % k)th server is available, assign the request to that server.
- Otherwise, assign the request to the next available server (wrapping around the list of servers and starting from 0 if necessary). For example, if the ith server is busy, try to assign the request to the (i+1)th server, then the (i+2)th server, and so on.

You are given a strictly increasing array arrival of positive integers, where arrival[i] represents the arrival time of the ith request, and another array load, where load[i] represents the load of the ith request (the time it takes to complete). Your goal is to find the busiest server(s). A server is considered busiest if it handled the most number of requests successfully among all the servers.

Return a list containing the IDs (0-indexed) of the busiest server(s). You may return the IDs in any order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1608 Special Array With X Elements Greater Than Or Equal X

**EASY**

You are given an array nums of non-negative integers. nums is considered special if there exists a number x such that there are exactly x numbers in nums that are greater than or equal to x.

Notice that x does not have to be an element in nums.

Return x if the array is special, otherwise, return -1. It can be proven that if nums is special, the value for x is unique.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1610 Maximum Number Of Visible Points

**HARD**

You are given an array points, an integer angle, and your location, where location = [posx, posy] and points[i] = [xi, yi] both denote integral coordinates on the X-Y plane.

Initially, you are facing directly east from your position. You cannot move from your position, but you can rotate. In other words, posx and posy cannot be changed. Your field of view in degrees is represented by angle, determining how wide you can see from any given view direction. Let d be the amount in degrees that you rotate counterclockwise. Then, your field of view is the inclusive range of angles [d - angle/2, d + angle/2].

You can see some set of points if, for each point, the angle formed by the point, your position, and the immediate east direction from your position is in your field of view.

There can be multiple points at one coordinate. There may be points at your location, and you can always see these points regardless of your rotation. Points do not obstruct your vision to other points.

Return the maximum number of points you can see.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1615 Maximal Network Rank

**MEDIUM**

There is an infrastructure of n cities with some number of roads connecting these cities. Each roads[i] = [ai, bi] indicates that there is a bidirectional road between cities ai and bi.

The network rank of two different cities is defined as the total number of directly connected roads to either city. If a road is directly connected to both cities, it is only counted once.

The maximal network rank of the infrastructure is the maximum network rank of all pairs of different cities.

Given the integer n and the array roads, return the maximal network rank of the entire infrastructure.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1619 Mean Of Array After Removing Some Elements

**EASY**

Given an integer array arr, return the mean of the remaining integers after removing the smallest 5% and the largest 5% of the elements.

Answers within 10-5 of the actual answer will be considered accepted.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1621 Number Of Sets Of K Non Overlapping Line Segments

**MEDIUM**

Given n points on a 1-D plane, where the ith point (from 0 to n-1) is at x = i, find the number of ways we can draw exactly k non-overlapping line segments such that each segment covers two or more points. The endpoints of each segment must have integral coordinates. The k line segments do not have to cover all n points, and they are allowed to share endpoints.

Return the number of ways we can draw k non-overlapping line segments. Since this number can be huge, return it modulo 109 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1632 Rank Transform Of A Matrix

**HARD**

Given an m x n matrix, return a new matrix answer where answer[row][col] is the rank of matrix[row][col].

The rank is an integer that represents how large an element is compared to other elements. It is calculated using the following rules:
- The rank is an integer starting from 1.
- If two elements p and q are in the same row or column, then:
- If p < q then rank(p) < rank(q)
- If p == q then rank(p) == rank(q)
- If p > q then rank(p) > rank(q)
- The rank should be as small as possible.

The test cases are generated so that answer is unique under the given rules.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1640 Check Array Formation Through Concatenation

**EASY**

You are given an array of distinct integers arr and an array of integer arrays pieces, where the integers in pieces are distinct. Your goal is to form arr by concatenating the arrays in pieces in any order. However, you are not allowed to reorder the integers in each array pieces[i].

Return true if it is possible to form the array arr from pieces. Otherwise, return false.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1646 Get Maximum In Generated Array

EASY

You are given an integer n. A 0-indexed integer array nums of length n + 1 is generated in the following way:
- nums[0] = 0
- nums[1] = 1
- nums[2 * i] = nums[i] when 2 <= 2 * i <= n
- nums[2 * i + 1] = nums[i] + nums[i + 1] when 2 <= 2 * i + 1 <= n

Return the maximum integer in the array nums.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1654 Minimum Jumps To Reach Home

MEDIUM

A certain bug's home is on the x-axis at position x. Help them get there from position 0.

The bug jumps according to the following rules:
- It can jump exactly a positions forward (to the right).
- It can jump exactly b positions backward (to the left).
- It cannot jump backward twice in a row.
- It cannot jump to any forbidden positions.

The bug may jump forward beyond its home, but it cannot jump to positions numbered with negative integers.

Given an array of integers forbidden, where forbidden[i] means that the bug cannot jump to the position forbidden[i], and integers a, b, and x, return the minimum number of jumps needed for the bug to reach its home. If there is no possible sequence of jumps that lands the bug on position x, return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1655 Distribute Repeating Integers

HARD

You are given an array of n integers, nums, where there are at most 50 unique values in the array. You are also given an array of m customer order quantities, quantity, where quantity[i] is the amount of integers the ith customer ordered. Determine if it is possible to distribute nums such that:
- The ith customer gets exactly quantity[i] integers,
- The integers the ith customer gets are all equal, and
- Every customer is satisfied.

Return true if it is possible to distribute nums according to the above conditions.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1659 Maximize Grid Happiness

HARD

You are given four integers, m, n, introvertsCount, and extrovertsCount. You have an m x n grid, and there are two types of people: introverts and extroverts. There are introvertsCount introverts and extrovertsCount extroverts.

You should decide how many people you want to live in the grid and assign each of them one grid cell. Note that you do not have to have all the people living in the grid.

The happiness of each person is calculated as follows:
- Introverts start with 120 happiness and lose 30 happiness for each neighbor (introvert or extrovert).
- Extroverts start with 40 happiness and gain 20 happiness for each neighbor (introvert or extrovert).

Neighbors live in the directly adjacent cells north, east, south, and west of a person's cell.

The grid happiness is the sum of each person's happiness. Return the maximum possible grid happiness.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1664 Ways To Make A Fair Array

**MEDIUM**

You are given an integer array nums. You can choose exactly one index (0-indexed) and remove the element. Notice that the index of the elements may change after the removal.

For example, if nums = [6,1,7,4,1]:
- Choosing to remove index 1 results in nums = [6,7,4,1].
- Choosing to remove index 2 results in nums = [6,1,4,1].
- Choosing to remove index 4 results in nums = [6,1,7,4].

An array is fair if the sum of the odd-indexed values equals the sum of the even-indexed values.

Return the number of indices that you could choose such that after the removal, nums is fair.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1671 Minimum Number Of Removals To Make Mountain Array

**HARD**

You may recall that an array arr is a mountain array if and only if:
- arr.length >= 3
- There exists some index i (0-indexed) with 0 < i < arr.length - 1 such that:
- arr[0] < arr[1] < ... < arr[i - 1] < arr[i]
- arr[i] > arr[i + 1] > ... > arr[arr.length - 1]

Given an integer array nums, return the minimum number of elements to remove to make nums a mountain array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1674 Minimum Moves To Make Array Complementary

**MEDIUM**

You are given an integer array nums of even length n and an integer limit. In one move, you can replace any integer from nums with another integer between 1 and limit, inclusive.

The array nums is complementary if for all indices i (0-indexed), nums[i] + nums[n - 1 - i] equals the same number. For example, the array [1,2,3,4] is complementary because for all indices i, nums[i] + nums[n - 1 - i] = 5.

Return the minimum number of moves required to make nums complementary.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1675 Minimize Deviation In Array

**HARD**

You are given an array nums of n positive integers.

You can perform two types of operations on any element of the array any number of times:
- If the element is even, divide it by 2.
- For example, if the array is [1,2,3,4], then you can do this operation on the last element, and the array will be [1,2,3,2].
- If the element is odd, multiply it by 2.
- For example, if the array is [1,2,3,4], then you can do this operation on the first element, and the array will be [2,2,3,4].

The deviation of the array is the maximum difference between any two elements in the array.

Return the minimum deviation the array can have after performing some number of operations.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1679 Max Number Of K Sum Pairs

**MEDIUM**

You are given an integer array nums and an integer k.

In one operation, you can pick two numbers from the array whose sum equals k and remove them from the array.

Return the maximum number of operations you can perform on the array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1680 Concatenation Of Consecutive Binary Numbers

**MEDIUM**

Given an integer n, return the decimal value of the binary string formed by concatenating the binary representations of 1 to n in order, modulo 109 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1681 Minimum Incompatibility

**HARD**

You are given an integer array nums and an integer k. You are asked to distribute this array into k subsets of equal size such that there are no two equal elements in the same subset.

A subset's incompatibility is the difference between the maximum and minimum elements in that array.

Return the minimum possible sum of incompatibilities of the k subsets after distributing the array optimally, or return -1 if it is not possible.

A subset is a group integers that appear in the array with no particular order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1688 Count Of Matches In Tournament

**EASY**

You are given an integer n, the number of teams in a tournament that has strange rules:
- If the current number of teams is even, each team gets paired with another team. A total of n / 2 matches are played, and n / 2 teams advance to the next round.
- If the current number of teams is odd, one team randomly advances in the tournament, and the rest gets paired. A total of (n - 1) / 2 matches are played, and (n - 1) / 2 + 1 teams advance to the next round.

Return the number of matches played in the tournament until a winner is decided.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1689 Partitioning Into Minimum Number Of Deci Binary Numbers

MEDIUM

A decimal number is called deci-binary if each of its digits is either 0 or 1 without any leading zeros. For example, 101 and 1100 are deci-binary, while 112 and 3001 are not.

Given a string n that represents a positive decimal integer, return the minimum number of positive deci-binary numbers needed so that they sum up to n.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1692 Count Ways To Distribute Candies

HARD

There are n unique candies (labeled 1 through n) and k bags. You are asked to distribute all the candies into the bags such that every bag has at least one candy.

There can be multiple ways to distribute the candies. Two ways are considered different if the candies in one bag in the first way are not all in the same bag in the second way. The order of the bags and the order of the candies within each bag do not matter.

For example, (1), (2,3) and (2), (1,3) are considered different because candies 2 and 3 in the bag (2,3) in the first way are not in the same bag in the second way (they are split between the bags (2) and (1,3)). However, (1), (2,3) and (3,2), (1) are considered the same because the candies in each bag are all in the same bags in both ways.

Given two integers, n and k, return the number of different ways to distribute the candies. As the answer may be too large, return it modulo 109 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1694 Reformat Phone Number

EASY

You are given a phone number as a string number. number consists of digits, spaces ' ', and/or dashes '-'.

You would like to reformat the phone number in a certain manner. Firstly, remove all spaces and dashes. Then, group the digits from left to right into blocks of length 3 until there are 4 or fewer digits. The final digits are then grouped as follows:
- 2 digits: A single block of length 2.
- 3 digits: A single block of length 3.
- 4 digits: Two blocks of length 2 each.

The blocks are then joined by dashes. Notice that the reformatting process should never produce any blocks of length 1 and produce at most two blocks of length 2.

Return the phone number after formatting.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1695 Maximum Erasure Value

MEDIUM

You are given an array of positive integers nums and want to erase a subarray containing unique elements. The score you get by erasing the subarray is equal to the sum of its elements.

Return the maximum score you can get by erasing exactly one subarray.

An array b is called to be a subarray of a if it forms a contiguous subsequence of a, that is, if it is equal to a[l],a[l+1],...,a[r] for some (l,r).

TRY IT / Write examples, edge cases, and your own algorithm first.

## 1700 Number Of Students Unable To Eat Lunch

EASY

The school cafeteria offers circular and square sandwiches at lunch break, referred to by numbers 0 and 1 respectively. All students stand in a queue. Each student either prefers square or circular sandwiches.

The number of sandwiches in the cafeteria is equal to the number of students. The sandwiches are placed in a stack. At each step:
- If the student at the front of the queue prefers the sandwich on the top of the stack, they will take it and leave the queue.
- Otherwise, they will leave it and go to the queue's end.

This continues until none of the queue students want to take the top sandwich and are thus unable to eat.

You are given two integer arrays students and sandwiches where sandwiches[i] is the type of the ith sandwich in the stack (i = 0 is the top of the stack) and students[j] is the preference of the jth student in the initial queue (j = 0 is the front of the queue).
Return the number of students that are unable to eat.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 1705 Maximum Number Of Eaten Apples

MEDIUM

There is a special kind of apple tree that grows apples every day for n days. On the ith day, the tree grows apples[i] apples that will rot after days[i] days, that is on day i + days[i] the apples will be rotten and cannot be eaten. On some days, the apple tree does not grow any apples, which are denoted by apples[i] == 0 and days[i] == 0.

You decided to eat at most one apple a day (to keep the doctors away). Note that you can keep eating after the first n days.

Given two integer arrays days and apples of length n, return the maximum number of apples you can eat.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 1707 Maximum Xor With An Element From Array

HARD

You are given an array nums consisting of non-negative integers. You are also given a queries array, where queries[i] = [xi, mi].

The answer to the ith query is the maximum bitwise XOR value of xi and any element of nums that does not exceed mi. In other words, the answer is max(nums[j] XOR xi) for all j such that nums[j] <= mi. If all elements in nums are larger than mi, then the answer is -1.

Return an integer array answer where answer.length == queries.length and answer[i] is the answer to the ith query.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 1711 Count Good Meals

MEDIUM

A good meal is a meal that contains exactly two different food items with a sum of deliciousness

equal to a power of two.

You can pick any two different foods to make a good meal.

Given an array of integers deliciousness where deliciousness[i] is the deliciousness of the ith item of food, return the number of different good meals you can make from this list modulo 109 + 7.

Note that items with different indices are considered different even if they have the same deliciousness value.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1713 Minimum Operations To Make A Subsequence

**HARD**

You are given an array target that consists of distinct integers and another integer array arr that can have duplicates.

In one operation, you can insert any integer at any position in arr. For example, if arr = [1,4,1,2], you can add 3 in the middle and make it [1,4,3,1,2]. Note that you can insert the integer at the very beginning or end of the array.

Return the minimum number of operations needed to make target a subsequence of arr.

A subsequence of an array is a new array generated from the original array by deleting some elements (possibly none) without changing the remaining elements' relative order. For example, [2,7,4] is a subsequence of [4,2,3,7,2,1,4] (the underlined elements), while [2,4,2] is not.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1714 Sum Of Special Evenly Spaced Elements In Array

**HARD**

You are given a 0-indexed integer array nums consisting of n non-negative integers.

You are also given an array queries, where queries[i] = [xi, yi]. The answer to the ith query is the sum of all nums[j] where xi <= j < n and (j - xi) is divisible by yi.

Return an array answer where answer.length == queries.length and answer[i] is the answer to the ith query modulo 109 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1720 Decode Xored Array

**EASY**

There is a hidden integer array arr that consists of n non-negative integers.

It was encoded into another integer array encoded of length n - 1, such that encoded[i] = arr[i] XOR arr[i + 1]. For example, if arr = [1,0,2,1], then encoded = [1,2,3].

You are given the encoded array. You are also given an integer first, that is the first element of arr, i.e. arr[0].

Return the original array arr. It can be proved that the answer exists and is unique.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1722 Minimize Hamming Distance After Swap Operations

MEDIUM

You are given two integer arrays, source and target, both of length n. You are also given an array allowedSwaps where each allowedSwaps[i] = [ai, bi] indicates that you are allowed to swap the elements at index ai and index bi (0-indexed) of array source. Note that you can swap elements at a specific pair of indices multiple times and in any order.

The Hamming distance of two arrays of the same length, source and target, is the number of positions where the elements are different. Formally, it is the number of indices i for 0 <= i <= n-1 where source[i] != target[i] (0-indexed).

Return the minimum Hamming distance of source and target after performing any amount of swap operations on array source.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1724 Checking Existence Of Edge Length Limited Paths Ii

HARD

An undirected graph of n nodes is defined by edgeList, where edgeList[i] = [ui, vi, disi] denotes an edge between nodes ui and vi with distance disi. Note that there may be multiple edges between two nodes, and the graph may not be connected.

Implement the DistanceLimitedPathsExist class:
- DistanceLimitedPathsExist(int n, int[][] edgeList) Initializes the class with an undirected graph.
- boolean query(int p, int q, int limit) Returns true if there exists a path from p to q such that each edge on the path has a distance strictly less than limit, and otherwise false.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1725 Number Of Rectangles That Can Form The Largest Square

EASY

You are given an array rectangles where rectangles[i] = [li, wi] represents the ith rectangle of length li and width wi.

You can cut the ith rectangle to form a square with a side length of k if both k <= li and k <= wi. For example, if you have a rectangle [4,6], you can cut it to get a square with a side length of at most 4.

Let maxLen be the side length of the largest square you can obtain from any of the given rectangles.

Return the number of rectangles that can make a square with a side length of maxLen.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1726 Tuple With Same Product

MEDIUM

Given an array nums of distinct positive integers, return the number of tuples (a, b, c, d) such that a * b = c * d where a, b, c, and d are elements of nums, and a != b != c != d.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1727 Largest Submatrix With Rearrangements

MEDIUM

You are given a binary matrix matrix of size m x n, and you are allowed to rearrange the columns of the matrix in any order.

Return the area of the largest submatrix within matrix where every element of the submatrix is 1 after reordering the columns optimally.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1728 Cat And Mouse Ii

HARD

A game is played by a cat and a mouse named Cat and Mouse.

The environment is represented by a grid of size rows x cols, where each element is a wall, floor, player (Cat, Mouse), or food.
- Players are represented by the characters 'C'(Cat),'M'(Mouse).
- Floors are represented by the character '.' and can be walked on.
- Walls are represented by the character '#' and cannot be walked on.
- Food is represented by the character 'F' and can be walked on.
- There is only one of each character 'C', 'M', and 'F' in grid.

Mouse and Cat play according to the following rules:
- Mouse moves first, then they take turns to move.
- During each turn, Cat and Mouse can jump in one of the four directions (left, right, up, down). They cannot jump over the wall nor outside of the grid.
- catJump, mouseJump are the maximum lengths Cat and Mouse can jump at a time, respectively. Cat and Mouse can jump less than the maximum length.
- Staying in the same position is allowed.
- Mouse can jump over Cat.

The game can end in 4 ways:
- If Cat occupies the same position as Mouse, Cat wins.
- If Cat reaches the food first, Cat wins.
- If Mouse reaches the food first, Mouse wins.
- If Mouse cannot get to the food within 1000 turns, Cat wins.

Given a rows x cols matrix grid and two integers catJump and mouseJump, return true if Mouse can win the game if both Cat and Mouse play optimally, otherwise return false.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1733 Minimum Number Of People To Teach

MEDIUM

On a social network consisting of m users and some friendships between users, two users can communicate with each other if they know a common language.

You are given an integer n, an array languages, and an array friendships where:
- There are n languages numbered 1 through n,
- languages[i] is the set of languages the ith user knows, and
- friendships[i] = [ui, vi] denotes a friendship between the users ui and vi.

You can choose one language and teach it to some users so that all friends can communicate with each other. Return the minimum number of users you need to teach.

Note that friendships are not transitive, meaning if x is a friend of y and y is a friend of z, this doesn't guarantee that x is a friend of z.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1735 Count Ways To Make Array With Product

HARD

You are given a 2D integer array, queries. For each queries[i], where queries[i] = [ni, ki], find the number of different ways you can place positive integers into an array of size ni such that the product of the integers is ki. As the number of ways may be too large, the answer to the ith query is the number of ways modulo 109 + 7.

Return an integer array answer where answer.length == queries.length, and answer[i] is the answer to the ith query.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1742 Maximum Number Of Balls In A Box

EASY

You are working in a ball factory where you have n balls numbered from lowLimit up to highLimit inclusive (i.e., n == highLimit - lowLimit + 1), and an infinite number of boxes numbered from 1 to infinity.

Your job at this factory is to put each ball in the box with a number equal to the sum of digits of the ball's number. For example, the ball number 321 will be put in the box number 3 + 2 + 1 = 6 and the ball number 10 will be put in the box number 1 + 0 = 1.

Given two integers lowLimit and highLimit, return the number of balls in the box with the most balls.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1743 Restore The Array From Adjacent Pairs

MEDIUM

There is an integer array nums that consists of n unique elements, but you have forgotten it. However, you do remember every pair of adjacent elements in nums.

You are given a 2D integer array adjacentPairs of size n - 1 where each adjacentPairs[i] = [ui, vi] indicates that the elements ui and vi are adjacent in nums.

It is guaranteed that every adjacent pair of elements nums[i] and nums[i+1] will exist in adjacentPairs, either as [nums[i], nums[i+1]] or [nums[i+1], nums[i]]. The pairs can appear in any order.

Return the original array nums. If there are multiple solutions, return any of them.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1748 Sum Of Unique Elements

EASY

You are given an integer array nums. The unique elements of an array are the elements that appear exactly once in the array.

Return the sum of all the unique elements of nums.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1751 Maximum Number Of Events That Can Be Attended Ii

HARD

You are given an array of events where events[i] = [startDayi, endDayi, valuei]. The ith event

starts at startDayi and ends at endDayi, and if you attend this event, you will receive a value of valuei. You are also given an integer k which represents the maximum number of events you can attend.

You can only attend one event at a time. If you choose to attend an event, you must attend the entire event. Note that the end day is inclusive: that is, you cannot attend two events where one of them starts and the other ends on the same day.

Return the maximum sum of values that you can receive by attending events.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1755 Closest Subsequence Sum

**HARD**

You are given an integer array nums and an integer goal.

You want to choose a subsequence of nums such that the sum of its elements is the closest possible to goal. That is, if the sum of the subsequence's elements is sum, then you want to minimize the absolute difference abs(sum - goal).

Return the minimum possible value of abs(sum - goal).

Note that a subsequence of an array is an array formed by removing some elements (possibly all or none) of the original array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1769 Minimum Number Of Operations To Move All Balls To Each Box

**MEDIUM**

You have n boxes. You are given a binary string boxes of length n, where boxes[i] is '0' if the ith box is empty, and '1' if it contains one ball.

In one operation, you can move one ball from a box to an adjacent box. Box i is adjacent to box j if abs(i - j) == 1. Note that after doing so, there may be more than one ball in some boxes.

Return an array answer of size n, where answer[i] is the minimum number of operations needed to move all the balls to the ith box.

Each answer[i] is calculated considering the initial state of the boxes.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1773 Count Items Matching A Rule

**EASY**

You are given an array items, where each items[i] = [typei, colori, namei] describes the type, color, and name of the ith item. You are also given a rule represented by two strings, ruleKey and ruleValue.

The ith item is said to match the rule if one of the following is true:
- ruleKey == "type" and ruleValue == typei.
- ruleKey == "color" and ruleValue == colori.
- ruleKey == "name" and ruleValue == namei.

Return the number of items that match the given rule.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1775 Equal Sum Arrays With Minimum Number Of Operations

MEDIUM

You are given two arrays of integers nums1 and nums2, possibly of different lengths. The values in the arrays are between 1 and 6, inclusive.

In one operation, you can change any integer's value in any of the arrays to any value between 1 and 6, inclusive.

Return the minimum number of operations required to make the sum of values in nums1 equal to the sum of values in nums2. Return -1 if it is not possible to make the sum of the two arrays equal.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1780 Check If Number Is A Sum Of Powers Of Three

MEDIUM

Given an integer n, return true if it is possible to represent n as the sum of distinct powers of three. Otherwise, return false.

An integer y is a power of three if there exists an integer x such that y == 3x.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1782 Count Pairs Of Nodes

HARD

You are given an undirected graph defined by an integer n, the number of nodes, and a 2D integer array edges, the edges in the graph, where edges[i] = [ui, vi] indicates that there is an undirected edge between ui and vi. You are also given an integer array queries.

Let incident(a, b) be defined as the number of edges that are connected to either node a or b.

The answer to the jth query is the number of pairs of nodes (a, b) that satisfy both of the following conditions:
- a < b
- incident(a, b) > queries[j]

Return an array answers such that answers.length == queries.length and answers[j] is the answer of the jth query.

Note that there can be multiple edges between the same two nodes.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1785 Minimum Elements To Add To Form A Given Sum

MEDIUM

You are given an integer array nums and two integers limit and goal. The array nums has an interesting property that abs(nums[i]) <= limit.

Return the minimum number of elements you need to add to make the sum of the array equal to goal. The array must maintain its property that abs(nums[i]) <= limit.

Note that abs(x) equals x if x >= 0, and -x otherwise.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1786 Number Of Restricted Paths From First To Last Node

MEDIUM

There is an undirected weighted connected graph. You are given a positive integer n which denotes that the graph has n nodes labeled from 1 to n, and an array edges where each edges[i] = [ui, vi, weighti] denotes that there is an edge between nodes ui and vi with weight equal to weighti.

A path from node start to node end is a sequence of nodes [z0, z1, z2, ..., zk] such that z0 = start and zk = end and there is an edge between zi and zi+1 where 0 <= i <= k-1.

The distance of a path is the sum of the weights on the edges of the path. Let distanceToLastNode(x) denote the shortest distance of a path between node n and node x. A restricted path is a path that also satisfies that distanceToLastNode(zi) > distanceToLastNode(zi+1) where 0 <= i <= k-1.

Return the number of restricted paths from node 1 to node n. Since that number may be too large, return it modulo 109 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1787 Make The Xor Of All Segments Equal To Zero

HARD

You are given an array nums and an integer k. The XOR of a segment [left, right] where left <= right is the XOR of all the elements with indices between left and right, inclusive: nums[left] XOR nums[left+1] XOR ... XOR nums[right].

Return the minimum number of elements to change in the array such that the XOR of all segments of size k is equal to zero.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1788 Maximize The Beauty Of The Garden

HARD

There is a garden of n flowers, and each flower has an integer beauty value. The flowers are arranged in a line. You are given an integer array flowers of size n and each flowers[i] represents the beauty of the ith flower.

A garden is valid if it meets these conditions:
- The garden has at least two flowers.
- The first and the last flower of the garden have the same beauty value.

As the appointed gardener, you have the ability to remove any (possibly none) flowers from the garden. You want to remove flowers in a way that makes the remaining garden valid. The beauty of the garden is the sum of the beauty of all the remaining flowers.

Return the maximum possible beauty of some valid garden after you have removed any (possibly none) flowers.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1797 Design Authentication Manager

MEDIUM

There is an authentication system that works with authentication tokens. For each session, the user will receive a new authentication token that will expire timeToLive seconds after the currentTime. If the token is renewed, the expiry time will be extended to expire timeToLive seconds after the (potentially different) currentTime.

Implement the AuthenticationManager class:
- AuthenticationManager(int timeToLive) constructs the AuthenticationManager and sets the timeToLive.
- generate(string tokenId, int currentTime) generates a new token with the given tokenId at the given currentTime in seconds.
- renew(string tokenId, int currentTime) renews the unexpired token with the given tokenId at the given currentTime in seconds. If there are no unexpired tokens with the given tokenId, the request is ignored, and nothing happens.
- countUnexpiredTokens(int currentTime) returns the number of unexpired tokens at the given currentTime.

Note that if a token expires at time t, and another action happens on time t (renew or countUnexpiredTokens), the expiration takes place before the other actions.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1798 Maximum Number Of Consecutive Values You Can Make

**MEDIUM**

You are given an integer array coins of length n which represents the n coins that you own. The value of the ith coin is coins[i]. You can make some value x if you can choose some of your n coins such that their values sum up to x.

Return the maximum number of consecutive integer values that you can make with your coins starting from and including 0.

Note that you may have multiple coins of the same value.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1801 Number Of Orders In The Backlog

**MEDIUM**

You are given a 2D integer array orders, where each orders[i] = [pricei, amounti, orderTypei] denotes that amounti orders have been placed of type orderTypei at the price pricei.
The orderTypei is:
- 0 if it is a batch of buy orders, or
- 1 if it is a batch of sell orders.

Note that orders[i] represents a batch of amounti independent orders with the same price and order type. All orders represented by orders[i] will be placed before all orders represented by orders[i+1] for all valid i.

There is a backlog that consists of orders that have not been executed. The backlog is initially empty. When an order is placed, the following happens:
- If the order is a buy order, you look at the sell order with the smallest price in the backlog. If that sell order's price is smaller than or equal to the current buy order's price, they will match and be executed, and that sell order will be removed from the backlog. Else, the buy order is added to the backlog.
- Vice versa, if the order is a sell order, you look at the buy order with the largest price in the backlog. If that buy order's price is larger than or equal to the current sell order's price, they will match and be executed, and that buy order will be removed from the backlog. Else, the sell order is added to the backlog.

Return the total amount of orders in the backlog after placing all the orders from the input. Since this number can be large, return it modulo 109 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1802 Maximum Value At A Given Index In A Bounded Array

MEDIUM

You are given three positive integers: n, index, and maxSum. You want to construct an array nums (0-indexed) that satisfies the following conditions:
- nums.length == n
- nums[i] is a positive integer where 0 <= i < n.
- abs(nums[i] - nums[i+1]) <= 1 where 0 <= i < n-1.
- The sum of all the elements of nums does not exceed maxSum.
- nums[index] is maximized.

Return nums[index] of the constructed array.

Note that abs(x) equals x if x >= 0, and -x otherwise.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1803 Count Pairs With Xor In A Range

HARD

Given a (0-indexed) integer array nums and two integers low and high, return the number of nice pairs.

A nice pair is a pair (i, j) where 0 <= i < j < nums.length and low <= (nums[i] XOR nums[j]) <= high.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1808 Maximize Number Of Nice Divisors

HARD

You are given a positive integer primeFactors. You are asked to construct a positive integer n that satisfies the following conditions:
- The number of prime factors of n (not necessarily distinct) is at most primeFactors.
- The number of nice divisors of n is maximized. Note that a divisor of n is nice if it is divisible by every prime factor of n. For example, if n = 12, then its prime factors are [2,2,3], then 6 and 12 are nice divisors, while 3 and 4 are not.

Return the number of nice divisors of n. Since that number can be too large, return it modulo 109 + 7.

Note that a prime number is a natural number greater than 1 that is not a product of two smaller natural numbers. The prime factors of a number n is a list of prime numbers such that their product equals n.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1810 Minimum Path Cost In A Hidden Grid

MEDIUM

This is an interactive problem.

There is a robot in a hidden grid, and you are trying to get it from its starting cell to the target cell in this grid. The grid is of size m x n, and each cell in the grid is either empty or blocked. It is guaranteed that the starting cell and the target cell are different, and neither of them is blocked.

Each cell has a cost that you need to pay each time you move to the cell. The starting cell's cost is not applied before the robot moves.

You want to find the minimum total cost to move the robot to the target cell. However, you do

not know the grid's dimensions, the starting cell, nor the target cell. You are only allowed to ask queries to the GridMaster object.

The GridMaster class has the following functions:
- boolean canMove(char direction) Returns true if the robot can move in that direction. Otherwise, it returns false.
- int move(char direction) Moves the robot in that direction and returns the cost of moving to that cell. If this move would move the robot to a blocked cell or off the grid, the move will be ignored, the robot will remain in the same position, and the function will return -1.
- boolean isTarget() Returns true if the robot is currently on the target cell. Otherwise, it returns false.

Note that direction in the above functions should be a character from {'U','D','L','R'}, representing the directions up, down, left, and right, respectively.

Return the minimum total cost to get the robot from its initial starting cell to the target cell. If there is no valid path between the cells, return -1.

Custom testing:

The test input is read as a 2D matrix grid of size m x n and four integers r1, c1, r2, and c2 where:
- grid[i][j] == 0 indicates that the cell (i, j) is blocked.
- grid[i][j] >= 1 indicates that the cell (i, j) is empty and grid[i][j] is the cost to move to that cell.
- (r1, c1) is the starting cell of the robot.
- (r2, c2) is the target cell of the robot.

Remember that you will not have this information in your code.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1814 Count Nice Pairs In An Array

**MEDIUM**

You are given an array nums that consists of non-negative integers. Let us define rev(x) as the reverse of the non-negative integer x. For example, rev(123) = 321, and rev(120) = 21. A pair of indices (i, j) is nice if it satisfies all of the following conditions:
- 0 <= i < j < nums.length
- nums[i] + rev(nums[j]) == nums[j] + rev(nums[i])

Return the number of nice pairs of indices. Since that number can be too large, return it modulo 109 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1815 Maximum Number Of Groups Getting Fresh Donuts

**HARD**

There is a donuts shop that bakes donuts in batches of batchSize. They have a rule where they must serve all of the donuts of a batch before serving any donuts of the next batch. You are given an integer batchSize and an integer array groups, where groups[i] denotes that there is a group of groups[i] customers that will visit the shop. Each customer will get exactly one donut.

When a group visits the shop, all customers of the group must be served before serving any of the following groups. A group will be happy if they all get fresh donuts. That is, the first customer of the group does not receive a donut that was left over from the previous group.

You can freely rearrange the ordering of the groups. Return the maximum possible number of happy groups after rearranging the groups.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1817 Finding The Users Active Minutes

MEDIUM

You are given the logs for users' actions on LeetCode, and an integer k. The logs are represented by a 2D integer array logs where each logs[i] = [IDi, timei] indicates that the user with IDi performed an action at the minute timei.

Multiple users can perform actions simultaneously, and a single user can perform multiple actions in the same minute.

The user active minutes (UAM) for a given user is defined as the number of unique minutes in which the user performed an action on LeetCode. A minute can only be counted once, even if multiple actions occur during it.

You are to calculate a 1-indexed array answer of size k such that, for each j (1 <= j <= k), answer[j] is the number of users whose UAM equals j.

Return the array answer as described above.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1818 Minimum Absolute Sum Difference

MEDIUM

You are given two positive integer arrays nums1 and nums2, both of length n.

The absolute sum difference of arrays nums1 and nums2 is defined as the sum of |nums1[i] - nums2[i]| for each 0 <= i < n (0-indexed).

You can replace at most one element of nums1 with any other element in nums1 to minimize the absolute sum difference.

Return the minimum absolute sum difference after replacing at most one element in the array nums1. Since the answer may be large, return it modulo 109 + 7.

|x| is defined as:
- x if x >= 0, or
- -x if x < 0.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1819 Number Of Different Subsequences Gcds

HARD

You are given an array nums that consists of positive integers.

The GCD of a sequence of numbers is defined as the greatest integer that divides all the numbers in the sequence evenly.
- For example, the GCD of the sequence [4,6,16] is 2.

A subsequence of an array is a sequence that can be formed by removing some elements (possibly none) of the array.
- For example, [2,5,10] is a subsequence of [1,2,1,2,4,1,5,10].

Return the number of different GCDs among all non-empty subsequences of nums.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1820 Maximum Number Of Accepted Invitations

MEDIUM

There are m boys and n girls in a class attending an upcoming party.

You are given an m x n integer matrix grid, where grid[i][j] equals 0 or 1. If grid[i][j] == 1, then that means the ith boy can invite the jth girl to the party. A boy can invite at most one girl, and a girl can accept at most one invitation from a boy.

Return the maximum possible number of accepted invitations.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1822 Sign Of The Product Of An Array

**EASY**

Implement a function signFunc(x) that returns:
- 1 if x is positive.
- -1 if x is negative.
- 0 if x is equal to 0.

You are given an integer array nums. Let product be the product of all values in the array nums.

Return signFunc(product).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1827 Minimum Operations To Make The Array Increasing

**EASY**

You are given an integer array nums (0-indexed). In one operation, you can choose an element of the array and increment it by 1.
- For example, if nums = [1,2,3], you can choose to increment nums[1] to make nums = [1,3,3].

Return the minimum number of operations needed to make nums strictly increasing.

An array nums is strictly increasing if nums[i] < nums[i+1] for all 0 <= i < nums.length - 1. An array of length 1 is trivially strictly increasing.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1828 Queries On Number Of Points Inside A Circle

**MEDIUM**

You are given an array points where points[i] = [xi, yi] is the coordinates of the ith point on a 2D plane. Multiple points can have the same coordinates.

You are also given an array queries where queries[j] = [xj, yj, rj] describes a circle centered at (xj, yj) with a radius of rj.

For each query queries[j], compute the number of points inside the jth circle. Points on the border of the circle are considered inside.

Return an array answer, where answer[j] is the answer to the jth query.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1835 Find Xor Sum Of All Pairs Bitwise And

**HARD**

The XOR sum of a list is the bitwise XOR of all its elements. If the list only contains one element, then its XOR sum will be equal to this element.
- For example, the XOR sum of [1,2,3,4] is equal to 1 XOR 2 XOR 3 XOR 4 = 4, and the XOR sum of [3] is equal to 3.

You are given two 0-indexed arrays arr1 and arr2 that consist only of non-negative integers.

Consider the list containing the result of arr1[i] AND arr2[j] (bitwise AND) for every (i, j)

pair where 0 <= i < arr1.length and 0 <= j < arr2.length.

Return the XOR sum of the aforementioned list.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1838 Frequency Of The Most Frequent Element

**MEDIUM**

The frequency of an element is the number of times it occurs in an array.

You are given an integer array nums and an integer k. In one operation, you can choose an index of nums and increment the element at that index by 1.

Return the maximum possible frequency of an element after performing at most k operations.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1845 Seat Reservation Manager

**MEDIUM**

Design a system that manages the reservation state of n seats that are numbered from 1 to n.

Implement the SeatManager class:
- SeatManager(int n) Initializes a SeatManager object that will manage n seats numbered from 1 to n. All seats are initially available.
- int reserve() Fetches the smallest-numbered unreserved seat, reserves it, and returns its number.
- void unreserve(int seatNumber) Unreserves the seat with the given seatNumber.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1846 Maximum Element After Decreasing And Rearranging

**MEDIUM**

You are given an array of positive integers arr. Perform some operations (possibly none) on arr so that it satisfies these conditions:
- The value of the first element in arr must be 1.
- The absolute difference between any 2 adjacent elements must be less than or equal to 1. In other words, abs(arr[i] - arr[i - 1]) <= 1 for each i where 1 <= i < arr.length (0-indexed). abs(x) is the absolute value of x.

There are 2 types of operations that you can perform any number of times:
- Decrease the value of any element of arr to a smaller positive integer.
- Rearrange the elements of arr to be in any order.

Return the maximum possible value of an element in arr after performing the operations to satisfy the conditions.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1848 Minimum Distance To The Target Element

**EASY**

Given an integer array nums (0-indexed) and two integers target and start, find an index i such that nums[i] == target and abs(i - start) is minimized. Note that abs(x) is the absolute value of x.

Return abs(i - start).

It is guaranteed that target exists in nums.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1850 Minimum Adjacent Swaps To Reach The Kth Smallest Number

MEDIUM

You are given a string num, representing a large integer, and an integer k.

We call some integer wonderful if it is a permutation of the digits in num and is greater in value than num. There can be many wonderful integers. However, we only care about the smallest-valued ones.
- For example, when num = "5489355142":
- The 1st smallest wonderful integer is "5489355214".
- The 2nd smallest wonderful integer is "5489355241".
- The 3rd smallest wonderful integer is "5489355412".
- The 4th smallest wonderful integer is "5489355421".

Return the minimum number of adjacent digit swaps that needs to be applied to num to reach the kth smallest wonderful integer.

The tests are generated in such a way that kth smallest wonderful integer exists.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1855 Maximum Distance Between A Pair Of Values

MEDIUM

You are given two non-increasing 0-indexed integer arrays nums2 and nums2.

A pair of indices (i, j), where 0 <= i < nums1.length and 0 <= j < nums2.length, is valid if both i <= j and nums1[i] <= nums2[j]. The distance of the pair is j - i.

Return the maximum distance of any valid pair (i, j). If there are no valid pairs, return 0.

An array arr is non-increasing if arr[i-1] >= arr[i] for every 1 <= i < arr.length.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1862 Sum Of Floored Pairs

HARD

Given an integer array nums, return the sum of floor(nums[i] / nums[j]) for all pairs of indices 0 <= i, j < nums.length in the array. Since the answer may be too large, return it modulo 109 + 7.

The floor() function returns the integer part of the division.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1865 Finding Pairs With A Certain Sum

MEDIUM

You are given two integer arrays nums1 and nums2. You are tasked to implement a data structure that supports queries of two types:
value (0 <= i < nums1.length and 0 <= j < nums2.length).

Implement the FindSumPairs class:
- FindSumPairs(int[] nums1, int[] nums2) Initializes the FindSumPairs object with two integer arrays nums1 and nums2.
- void add(int index, int val) Adds val to nums2[index], i.e., apply nums2[index] += val.
- int count(int tot) Returns the number of pairs (i, j) such that nums1[i] + nums2[j] == tot.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1866 Number Of Ways To Rearrange Sticks With K Sticks Visible

HARD

There are n uniquely-sized sticks whose lengths are integers from 1 to n. You want to arrange the sticks such that exactly k sticks are visible from the left. A stick is visible from the left if there are no longer sticks to the left of it.
- For example, if the sticks are arranged [1,3,2,5,4], then the sticks with lengths 1, 3, and 5 are visible from the left.

Given n and k, return the number of such arrangements. Since the answer may be large, return it modulo 109 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1868 Product Of Two Run Length Encoded Arrays

MEDIUM

Run-length encoding is a compression algorithm that allows for an integer array nums with many segments of consecutive repeated numbers to be represented by a (generally smaller) 2D array encoded. Each encoded[i] = [vali, freqi] describes the ith segment of repeated numbers in nums where vali is the value that is repeated freqi times.
- For example, nums = [1,1,1,2,2,2,2,2] is represented by the run-length encoded array encoded = [[1,3],[2,5]]. Another way to read this is "three 1's followed by five 2's".

The product of two run-length encoded arrays encoded1 and encoded2 can be calculated using the following steps:

You are given two run-length encoded arrays encoded1 and encoded2 representing full arrays nums1 and nums2 respectively. Both nums1 and nums2 have the same length. Each encoded1[i] = [vali, freqi] describes the ith segment of nums1, and each encoded2[j] = [valj, freqj] describes the jth segment of nums2.

Return the product of encoded1 and encoded2.

Note: Compression should be done such that the run-length encoded array has the minimum possible length.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1874 Minimize Product Sum Of Two Arrays

MEDIUM

The product sum of two equal-length arrays a and b is equal to the sum of a[i] * b[i] for all 0 <= i < a.length (0-indexed).
- For example, if a = [1,2,3,4] and b = [5,2,3,1], the product sum would be 1*5 + 2*2 + 3*3 + 4*1 = 22.

Given two arrays nums1 and nums2 of length n, return the minimum product sum if you are allowed to rearrange the order of the elements in nums1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1877 Minimize Maximum Pair Sum In Array

MEDIUM

The pair sum of a pair (a,b) is equal to a + b. The maximum pair sum is the largest pair sum in a list of pairs.
- For example, if we have pairs (1,5), (2,3), and (4,4), the maximum pair sum would be max(1+5, 2+3, 4+4) = max(6, 5, 8) = 8.

Given an array nums of even length n, pair up the elements of nums into n / 2 pairs such that:
- Each element of nums is in exactly one pair, and
- The maximum pair sum is minimized.

Return the minimized maximum pair sum after optimally pairing up the elements.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1878 Get Biggest Three Rhombus Sums In A Grid

**MEDIUM**

You are given an m x n integer matrix grid.

A rhombus sum is the sum of the elements that form the border of a regular rhombus shape in grid. The rhombus must have the shape of a square rotated 45 degrees with each of the corners centered in a grid cell. Below is an image of four valid rhombus shapes with the corresponding colored cells that should be included in each rhombus sum.

Note that the rhombus can have an area of 0, which is depicted by the purple rhombus in the bottom right corner.

Return the biggest three distinct rhombus sums in the grid in descending order. If there are less than three distinct values, return all of them.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1879 Minimum Xor Sum Of Two Arrays

**HARD**

You are given two integer arrays nums1 and nums2 of length n.

The XOR sum of the two integer arrays is (nums1[0] XOR nums2[0]) + (nums1[1] XOR nums2[1]) + ... + (nums1[n - 1] XOR nums2[n - 1]) (0-indexed).
- For example, the XOR sum of [1,2,3] and [3,2,1] is equal to (1 XOR 3) + (2 XOR 2) + (3 XOR 1) = 2 + 0 + 2 = 4.

Rearrange the elements of nums2 such that the resulting XOR sum is minimized.

Return the XOR sum after the rearrangement.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1885 Count Pairs In Two Arrays

**MEDIUM**

Given two integer arrays nums1 and nums2 of length n, count the pairs of indices (i, j) such that i < j and nums1[i] + nums1[j] > nums2[i] + nums2[j].

Return the number of pairs satisfying the condition.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1886 Determine Whether Matrix Can Be Obtained By Rotation

**EASY**

Given two n x n binary matrices mat and target, return true if it is possible to make mat equal to target by rotating mat in 90-degree increments, or false otherwise.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1887 Reduction Operations To Make The Array Elements Equal

MEDIUM

Given an integer array nums, your goal is to make all elements in nums equal. To complete one operation, follow these steps:
If there are multiple elements with the largest value, pick the smallest i.
nextLargest.

Return the number of operations to make all elements in nums equal.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1899 Merge Triplets To Form Target Triplet

MEDIUM

A triplet is an array of three integers. You are given a 2D integer array triplets, where triplets[i] = [ai, bi, ci] describes the ith triplet. You are also given an integer array target = [x, y, z] that describes the triplet you want to obtain.

To obtain target, you may apply the following operation on triplets any number of times (possibly zero):
- Choose two indices (0-indexed) i and j (i != j) and update triplets[j] to become [max(ai, aj), max(bi, bj), max(ci, cj)].
- For example, if triplets[i] = [2, 5, 3] and triplets[j] = [1, 7, 5], triplets[j] will be updated to [max(2, 1), max(5, 7), max(3, 5)] = [2, 7, 5].

Return true if it is possible to obtain the target triplet [x, y, z] as an element of triplets, or false otherwise.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1901 Find A Peak Element Ii

MEDIUM

A peak element in a 2D grid is an element that is strictly greater than all of its adjacent neighbors to the left, right, top, and bottom.

Given a 0-indexed m x n matrix mat where no two adjacent cells are equal, find any peak element mat[i][j] and return the length 2 array [i,j].

You may assume that the entire matrix is surrounded by an outer perimeter with the value -1 in each cell.

You must write an algorithm that runs in O(m log(n)) or O(n log(m)) time.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1904 The Number Of Full Rounds You Have Played

MEDIUM

You are participating in an online chess tournament. There is a chess round that starts every 15 minutes. The first round of the day starts at 00:00, and after every 15 minutes, a new round starts.
- For example, the second round starts at 00:15, the fourth round starts at 00:45, and the seventh round starts at 01:30.

You are given two strings loginTime and logoutTime where:
- loginTime is the time you will login to the game, and
- logoutTime is the time you will logout from the game.

If logoutTime is earlier than loginTime, this means you have played from loginTime to midnight

and from midnight to logoutTime.

Return the number of full chess rounds you have played in the tournament.

Note: All the given times follow the 24-hour clock. That means the first round of the day starts at 00:00 and the last round of the day starts at 23:45.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1906 Minimum Absolute Difference Queries

**MEDIUM**

The minimum absolute difference of an array a is defined as the minimum value of |a[i] - a[j]|, where 0 <= i < j < a.length and a[i] != a[j]. If all elements of a are the same, the minimum absolute difference is -1.
- For example, the minimum absolute difference of the array [5,2,3,7,2] is |2 - 3| = 1. Note that it is not 0 because a[i] and a[j] must be different.

You are given an integer array nums and the array queries where queries[i] = [li, ri]. For each query i, compute the minimum absolute difference of the subarray nums[li...ri] containing the elements of nums between the 0-based indices li and ri (inclusive).

Return an array ans where ans[i] is the answer to the ith query.

A subarray is a contiguous sequence of elements in an array.

The value of |x| is defined as:
- x if x >= 0.
- -x if x < 0.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1908 Game Of Nim

**MEDIUM**

Alice and Bob take turns playing a game with Alice starting first.

In this game, there are n piles of stones. On each player's turn, the player should remove any positive number of stones from a non-empty pile of his or her choice. The first player who cannot make a move loses, and the other player wins.

Given an integer array piles, where piles[i] is the number of stones in the ith pile, return true if Alice wins, or false if Bob wins.

Both Alice and Bob play optimally.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1909 Remove One Element To Make The Array Strictly Increasing

**EASY**

Given a 0-indexed integer array nums, return true if it can be made strictly increasing after removing exactly one element, or false otherwise. If the array is already strictly increasing, return true.

The array nums is strictly increasing if nums[i - 1] < nums[i] for each index (1 <= i < nums.length).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1911 Maximum Alternating Subsequence Sum

MEDIUM

The alternating sum of a 0-indexed array is defined as the sum of the elements at even indices minus the sum of the elements at odd indices.
- For example, the alternating sum of [4,2,5,3] is (4 + 5) - (2 + 3) = 4.

Given an array nums, return the maximum alternating sum of any subsequence of nums (after reindexing the elements of the subsequence).

A subsequence of an array is a new array generated from the original array by deleting some elements (possibly none) without changing the remaining elements' relative order. For example, [2,7,4] is a subsequence of [4,2,3,7,2,1,4] (the underlined elements), while [2,4,2] is not.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1912 Design Movie Rental System

HARD

You have a movie renting company consisting of n shops. You want to implement a renting system that supports searching for, booking, and returning movies. The system should also support generating a report of the currently rented movies.

Each movie is given as a 2D integer array entries where entries[i] = [shopi, moviei, pricei] indicates that there is a copy of movie moviei at shop shopi with a rental price of pricei. Each shop carries at most one copy of a movie moviei.

The system should support the following functions:
- Search: Finds the cheapest 5 shops that have an unrented copy of a given movie. The shops should be sorted by price in ascending order, and in case of a tie, the one with the smaller shopi should appear first. If there are less than 5 matching shops, then all of them should be returned. If no shop has an unrented copy, then an empty list should be returned.
- Rent: Rents an unrented copy of a given movie from a given shop.
- Drop: Drops off a previously rented copy of a given movie at a given shop.
- Report: Returns the cheapest 5 rented movies (possibly of the same movie ID) as a 2D list res where res[j] = [shopj, moviej] describes that the jth cheapest rented movie moviej was rented from the shop shopj. The movies in res should be sorted by price in ascending order, and in case of a tie, the one with the smaller shopj should appear first, and if there is still tie, the one with the smaller moviej should appear first. If there are fewer than 5 rented movies, then all of them should be returned. If no movies are currently being rented, then an empty list should be returned.

Implement the MovieRentingSystem class:
- MovieRentingSystem(int n, int[][] entries) Initializes the MovieRentingSystem object with n shops and the movies in entries.
- List<Integer> search(int movie) Returns a list of shops that have an unrented copy of the given movie as described above.
- void rent(int shop, int movie) Rents the given movie from the given shop.
- void drop(int shop, int movie) Drops off a previously rented movie at the given shop.
- List<List<Integer>> report() Returns a list of cheapest rented movies as described above.

Note: The test cases will be generated such that rent will only be called if the shop has an unrented copy of the movie, and drop will only be called if the shop had previously rented out the movie.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1913 Maximum Product Difference Between Two Pairs

EASY

The product difference between two pairs (a, b) and (c, d) is defined as (a * b) - (c * d).
- For example, the product difference between (5, 6) and (2, 7) is (5 * 6) - (2 * 7) = 16.

Given an integer array nums, choose four distinct indices w, x, y, and z such that the product difference between pairs (nums[w], nums[x]) and (nums[y], nums[z]) is maximized.

Return the maximum such product difference.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1914 Cyclically Rotating A Grid

MEDIUM

You are given an m x n integer matrix grid, where m and n are both even integers, and an integer k.

The matrix is composed of several layers, which is shown in the below image, where each color is its own layer.

A cyclic rotation of the matrix is done by cyclically rotating each layer in the matrix. To cyclically rotate a layer once, each element in the layer will take the place of the adjacent element in the counter-clockwise direction. An example rotation is shown below.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1916 Count Ways To Build Rooms In An Ant Colony

HARD

You are an ant tasked with adding n new rooms numbered 0 to n-1 to your colony. You are given the expansion plan as a 0-indexed integer array of length n, prevRoom, where prevRoom[i] indicates that you must build room prevRoom[i] before building room i, and these two rooms must be connected directly. Room 0 is already built, so prevRoom[0] = -1. The expansion plan is given such that once all the rooms are built, every room will be reachable from room 0.

You can only build one room at a time, and you can travel freely between rooms you have already built only if they are connected. You can choose to build any room as long as its previous room is already built.

Return the number of different orders you can build all the rooms in. Since the answer may be large, return it modulo 109 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1921 Eliminate Maximum Number Of Monsters

MEDIUM

You are playing a video game where you are defending your city from a group of n monsters. You are given a 0-indexed integer array dist of size n, where dist[i] is the initial distance in kilometers of the ith monster from the city.

The monsters walk toward the city at a constant speed. The speed of each monster is given to you in an integer array speed of size n, where speed[i] is the speed of the ith monster in kilometers per minute.

You have a weapon that, once fully charged, can eliminate a single monster. However, the weapon takes one minute to charge. The weapon is fully charged at the very start.

You lose when any monster reaches your city. If a monster reaches the city at the exact moment

the weapon is fully charged, it counts as a loss, and the game ends before you can use your weapon.

Return the maximum number of monsters that you can eliminate before you lose, or n if you can eliminate all the monsters before they reach the city.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1922 Count Good Numbers

**MEDIUM**

A digit string is good if the digits (0-indexed) at even indices are even and the digits at odd indices are prime (2, 3, 5, or 7).

For example, "2582" is good because the digits (2 and 8) at even positions are even and the digits (5 and 2) at odd positions are prime. However, "3245" is not good because 3 is at an even index but is not even.

Given an integer n, return the total number of good digit strings of length n. Since the answer may be large, return it modulo 109 + 7.

A digit string is a string consisting of digits 0 through 9 that may contain leading zeros.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1923 Longest Common Subpath

**HARD**

There is a country of n cities numbered from 0 to n - 1. In this country, there is a road connecting every pair of cities.

There are m friends numbered from 0 to m - 1 who are traveling through the country. Each one of them will take a path consisting of some cities. Each path is represented by an integer array that contains the visited cities in order. The path may contain a city more than once, but the same city will not be listed consecutively.

Given an integer n and a 2D integer array paths where paths[i] is an integer array representing the path of the ith friend, return the length of the longest common subpath that is shared by every friend's path, or 0 if there is no common subpath at all.

A subpath of a path is a contiguous sequence of cities within that path.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1925 Count Square Sum Triples

**EASY**

A square triple (a,b,c) is a triple where a, b, and c are integers and a2 + b2 = c2.

Given an integer n, return the number of square triples such that 1 <= a, b, c <= n.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1927 Sum Game

**MEDIUM**

Alice and Bob take turns playing a game, with Alice starting first.

You are given a string num of even length consisting of digits and '?' characters. On each turn, a player will do the following if there is still at least one '?' in num:

The game ends when there are no more '?' characters in num.

For Bob to win, the sum of the digits in the first half of num must be equal to the sum of the digits in the second half. For Alice to win, the sums must not be equal.
- For example, if the game ended with num = "243801", then Bob wins because 2+4+3 = 8+0+1. If the game ended with num = "243803", then Alice wins because 2+4+3 != 8+0+3.

Assuming Alice and Bob play optimally, return true if Alice will win and false if Bob will win.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1929 Concatenation Of Array

**EASY**

Given an integer array `nums` of length `n`, you want to create an array `ans` of length `2n` where `ans[i] == nums[i`] and `ans[i + n] == nums[i]` for `0 <= i < n` (0-indexed).

Specifically, `ans` is the concatenation of two `nums` arrays.

Return the array `ans`.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1930 Unique Length 3 Palindromic Subsequences

**MEDIUM**

Given a string s, return the number of unique palindromes of length three that are a subsequence of s.

Note that even if there are multiple ways to obtain the same subsequence, it is still only counted once.

A palindrome is a string that reads the same forwards and backwards.

A subsequence of a string is a new string generated from the original string with some characters (can be none) deleted without changing the relative order of the remaining characters.

For example, "ace" is a subsequence of "abcde".

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1931 Painting A Grid With Three Different Colors

**HARD**

You are given two integers m and n. Consider an m x n grid where each cell is initially white. You can paint each cell red, green, or blue. All cells must be painted.

Return the number of ways to color the grid with no two adjacent cells having the same color. Since the answer can be very large, return it modulo 109 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1936 Add Minimum Number Of Rungs

**MEDIUM**

You are given a strictly increasing integer array rungs that represents the height of rungs on a ladder. You are currently on the floor at height 0, and you want to reach the last rung.

You are also given an integer dist. You can only climb to the next highest rung if the distance between where you are currently at (the floor or on a rung) and the next rung is at most dist. You are able to insert rungs at any positive integer height if a rung is not already there.

Return the minimum number of rungs that must be added to the ladder in order for you to climb to the last rung.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1937 Maximum Number Of Points With Cost

**MEDIUM**

You are given an m x n integer matrix points (0-indexed). Starting with 0 points, you want to maximize the number of points you can get from the matrix.

To gain points, you must pick one cell in each row. Picking the cell at coordinates (r, c) will add points[r][c] to your score.

However, you will lose points if you pick a cell too far from the cell that you picked in the previous row. For every two adjacent rows r and r + 1 (where 0 <= r < m - 1), picking cells at coordinates (r, c1) and (r + 1, c2) will subtract abs(c1 - c2) from your score.

Return the maximum number of points you can achieve.

abs(x) is defined as:
- x for x >= 0.
- -x for x < 0.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1938 Maximum Genetic Difference Query

**HARD**

There is a rooted tree consisting of n nodes numbered 0 to n - 1. Each node's number denotes its unique genetic value (i.e. the genetic value of node x is x). The genetic difference between two genetic values is defined as the bitwise-XOR of their values. You are given the integer array parents, where parents[i] is the parent for node i. If node x is the root of the tree, then parents[x] == -1.

You are also given the array queries where queries[i] = [nodei, vali]. For each query i, find the maximum genetic difference between vali and pi, where pi is the genetic value of any node that is on the path between nodei and the root (including nodei and the root). More formally, you want to maximize vali XOR pi.

Return an array ans where ans[i] is the answer to the ith query.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1942 The Number Of The Smallest Unoccupied Chair

**MEDIUM**

There is a party where n friends numbered from 0 to n - 1 are attending. There is an infinite number of chairs in this party that are numbered from 0 to infinity. When a friend arrives at the party, they sit on the unoccupied chair with the smallest number.
- For example, if chairs 0, 1, and 5 are occupied when a friend comes, they will sit on chair number 2.

When a friend leaves the party, their chair becomes unoccupied at the moment they leave. If another friend arrives at that same moment, they can sit in that chair.

You are given a 0-indexed 2D integer array times where times[i] = [arrivali, leavingi], indicating the arrival and leaving times of the ith friend respectively, and an integer targetFriend. All arrival times are distinct.

Return the chair number that the friend numbered targetFriend will sit on.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1947 Maximum Compatibility Score Sum

MEDIUM

There is a survey that consists of n questions where each question's answer is either 0 (no) or 1 (yes).

The survey was given to m students numbered from 0 to m - 1 and m mentors numbered from 0 to m - 1. The answers of the students are represented by a 2D integer array students where students[i] is an integer array that contains the answers of the ith student (0-indexed). The answers of the mentors are represented by a 2D integer array mentors where mentors[j] is an integer array that contains the answers of the jth mentor (0-indexed).

Each student will be assigned to one mentor, and each mentor will have one student assigned to them. The compatibility score of a student-mentor pair is the number of answers that are the same for both the student and the mentor.
- For example, if the student's answers were [1, 0, 1] and the mentor's answers were [0, 0, 1], then their compatibility score is 2 because only the second and the third answers are the same.

You are tasked with finding the optimal student-mentor pairings to maximize the sum of the compatibility scores.

Given students and mentors, return the maximum compatibility score sum that can be achieved.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1948 Delete Duplicate Folders In System

HARD

Due to a bug, there are many duplicate folders in a file system. You are given a 2D array paths, where paths[i] is an array representing an absolute path to the ith folder in the file system.
- For example, ["one", "two", "three"] represents the path "/one/two/three".

Two folders (not necessarily on the same level) are identical if they contain the same non-empty set of identical subfolders and underlying subfolder structure. The folders do not need to be at the root level to be identical. If two or more folders are identical, then mark the folders as well as all their subfolders.
- For example, folders "/a" and "/b" in the file structure below are identical. They (as well as their subfolders) should all be marked:
- /a
- /a/x
- /a/x/y
- /a/z
- /b
- /b/x
- /b/x/y
- /b/z
- However, if the file structure also included the path "/b/w", then the folders "/a" and "/b" would not be identical. Note that "/a/x" and "/b/x" would still be considered identical even with the added folder.

Once all the identical folders and their subfolders have been marked, the file system will delete all of them. The file system only runs the deletion once, so any folders that become identical after the initial deletion are not deleted.

Return the 2D array ans containing the paths of the remaining folders after deleting all the marked folders. The paths may be returned in any order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1953 Maximum Number Of Weeks For Which You Can Work

MEDIUM

There are n projects numbered from 0 to n - 1. You are given an integer array milestones where each milestones[i] denotes the number of milestones the ith project has.

You can work on the projects following these two rules:
- Every week, you will finish exactly one milestone of one project. You must work every week.
- You cannot work on two milestones from the same project for two consecutive weeks.
- Once all the milestones of all the projects are finished, or if the only milestones that you can work on will cause you to violate the above rules, you will stop working. Note that you may not be able to finish every project's milestones due to these constraints.

Return the maximum number of weeks you would be able to work on the projects without violating the rules mentioned above.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1955 Count Number Of Special Subsequences

HARD

A sequence is special if it consists of a positive number of 0s, followed by a positive number of 1s, then a positive number of 2s.
- For example, [0,1,2] and [0,0,1,1,1,2] are special.
- In contrast, [2,1,0], [1], and [0,1,2,0] are not special.

Given an array nums (consisting of only integers 0, 1, and 2), return the number of different subsequences that are special. Since the answer may be very large, return it modulo 109 + 7.

A subsequence of an array is a sequence that can be derived from the array by deleting some or no elements without changing the order of the remaining elements. Two subsequences are different if the set of indices chosen are different.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1968 Array With Elements Not Equal To Average Of Neighbors

MEDIUM

You are given a 0-indexed array nums of distinct integers. You want to rearrange the elements in the array such that every element in the rearranged array is not equal to the average of its neighbors.

More formally, the rearranged array should have the property such that for every i in the range 1 <= i < nums.length - 1, (nums[i-1] + nums[i+1]) / 2 is not equal to nums[i].

Return any rearrangement of nums that meets the requirements.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1969 Minimum Non Zero Product Of The Array Elements

MEDIUM

You are given a positive integer p. Consider an array nums (1-indexed) that consists of the integers in the inclusive range [1, 2p - 1] in their binary representations. You are allowed to do the following operation any number of times:
- Choose two elements x and y from nums.
- Choose a bit in x and swap it with its corresponding bit in y. Corresponding bit refers to the bit that is in the same position in the other integer.

For example, if x = 1101 and y = 0011, after swapping the 2nd bit from the right, we have

x = 1111 and y = 0001.

Find the minimum non-zero product of nums after performing the above operation any number of times. Return this product modulo 109 + 7.

Note: The answer should be the minimum product before the modulo operation is done.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1973 Count Nodes Equal To Sum Of Descendants

**MEDIUM**

Given the root of a binary tree, return the number of nodes where the value of the node is equal to the sum of the values of its descendants.

A descendant of a node x is any node that is on the path from node x to some leaf node. The sum is considered to be 0 if the node has no descendants.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1975 Maximum Matrix Sum

**MEDIUM**

You are given an n x n integer matrix. You can do the following operation any number of times:
- Choose any two adjacent elements of matrix and multiply each of them by -1.

Two elements are considered adjacent if and only if they share a border.

Your goal is to maximize the summation of the matrix's elements. Return the maximum sum of the matrix's elements using the operation mentioned above.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1976 Number Of Ways To Arrive At Destination

**MEDIUM**

You are in a city that consists of n intersections numbered from 0 to n - 1 with bi-directional roads between some intersections. The inputs are generated such that you can reach any intersection from any other intersection and that there is at most one road between any two intersections.

You are given an integer n and a 2D integer array roads where roads[i] = [ui, vi, timei] means that there is a road between intersections ui and vi that takes timei minutes to travel. You want to know in how many ways you can travel from intersection 0 to intersection n - 1 in the shortest amount of time.

Return the number of ways you can arrive at your destination in the shortest amount of time. Since the answer may be large, return it modulo 109 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1977 Number Of Ways To Separate Numbers

**HARD**

You wrote down many positive integers in a string called num. However, you realized that you forgot to add commas to seperate the different numbers. You remember that the list of integers was non-decreasing and that no integer had leading zeros.

Return the number of possible lists of integers that you could have written down to get the string num. Since the answer may be large, return it modulo 109 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1979 Find Greatest Common Divisor Of Array

EASY

Given an integer array nums, return the greatest common divisor of the smallest number and largest number in nums.

The greatest common divisor of two numbers is the largest positive integer that evenly divides both numbers.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1981 Minimize The Difference Between Target And Chosen Elements

MEDIUM

You are given an m x n integer matrix mat and an integer target.

Choose one integer from each row in the matrix such that the absolute difference between target and the sum of the chosen elements is minimized.

Return the minimum absolute difference.

The absolute difference between two numbers a and b is the absolute value of a - b.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1983 Widest Pair Of Indices With Equal Range Sum

MEDIUM

You are given two 0-indexed binary arrays nums1 and nums2. Find the widest pair of indices (i, j) such that i <= j and nums1[i] + nums1[i+1] + ... + nums1[j] == nums2[i] + nums2[i+1] + ... + nums2[j].

The widest pair of indices is the pair with the largest distance between i and j. The distance between a pair of indices is defined as j - i + 1.

Return the distance of the widest pair of indices. If no pair of indices meets the conditions, return 0.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1984 Minimum Difference Between Highest And Lowest Of K Scores

EASY

You are given a 0-indexed integer array nums, where nums[i] represents the score of the ith student. You are also given an integer k.

Pick the scores of any k students from the array so that the difference between the highest and the lowest of the k scores is minimized.

Return the minimum possible difference.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1986 Minimum Number Of Work Sessions To Finish The Tasks

MEDIUM

There are n tasks assigned to you. The task times are represented as an integer array tasks of length n, where the ith task takes tasks[i] hours to finish. A work session is when you work for at most sessionTime consecutive hours and then take a break.

You should finish the given tasks in a way that satisfies the following conditions:

- If you start a task in a work session, you must complete it in the same work session.
- You can start a new task immediately after finishing the previous one.
- You may complete the tasks in any order.

Given tasks and sessionTime, return the minimum number of work sessions needed to finish all the tasks following the conditions above.

The tests are generated such that sessionTime is greater than or equal to the maximum element in tasks[i].

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1987 Number Of Unique Good Subsequences

**HARD**

You are given a binary string binary. A subsequence of binary is considered good if it is not empty and has no leading zeros (with the exception of "0").

Find the number of unique good subsequences of binary.
- For example, if binary = "001", then all the good subsequences are ["0", "0", "1"], so the unique good subsequences are "0" and "1". Note that subsequences "00", "01", and "001" are not good because they have leading zeros.

Return the number of unique good subsequences of binary. Since the answer may be very large, return it modulo 109 + 7.

A subsequence is a sequence that can be derived from another sequence by deleting some or no elements without changing the order of the remaining elements.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1989 Maximum Number Of People That Can Be Caught In Tag

**MEDIUM**

You are playing a game of tag with your friends. In tag, people are divided into two teams: people who are "it", and people who are not "it". The people who are "it" want to catch as many people as possible who are not "it".

You are given a 0-indexed integer array team containing only zeros (denoting people who are not "it") and ones (denoting people who are "it"), and an integer dist. A person who is "it" at index i can catch any one person whose index is in the range [i - dist, i + dist] (inclusive) and is not "it".

Return the maximum number of people that the people who are "it" can catch.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1991 Find The Middle Index In Array

**EASY**

Given a 0-indexed integer array nums, find the leftmost middleIndex (i.e., the smallest amongst all the possible ones).

A middleIndex is an index where nums[0] + nums[1] + ... + nums[middleIndex-1] == nums[middleIndex+1] + nums[middleIndex+2] + ... + nums[nums.length-1].

If middleIndex == 0, the left side sum is considered to be 0. Similarly, if middleIndex == nums.length - 1, the right side sum is considered to be 0.

Return the leftmost middleIndex that satisfies the condition, or -1 if there is no such index.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1995 Count Special Quadruplets

EASY

Given a 0-indexed integer array nums, return the number of distinct quadruplets (a, b, c, d) such that:
- nums[a] + nums[b] + nums[c] == nums[d], and
- a < b < c < d

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2001 Number Of Pairs Of Interchangeable Rectangles

MEDIUM

You are given n rectangles represented by a 0-indexed 2D integer array rectangles, where rectangles[i] = [widthi, heighti] denotes the width and height of the ith rectangle.

Two rectangles i and j (i < j) are considered interchangeable if they have the same width-to-height ratio. More formally, two rectangles are interchangeable if widthi/heighti == widthj/heightj (using decimal division, not integer division).

Return the number of pairs of interchangeable rectangles in rectangles.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2002 Maximum Product Of The Length Of Two Palindromic Subsequences

MEDIUM

Given a string s, find two disjoint palindromic subsequences of s such that the product of their lengths is maximized. The two subsequences are disjoint if they do not both pick a character at the same index.

Return the maximum possible product of the lengths of the two palindromic subsequences.

A subsequence is a string that can be derived from another string by deleting some or no characters without changing the order of the remaining characters. A string is palindromic if it reads the same forward and backward.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2006 Count Number Of Pairs With Absolute Difference K

EASY

Given an integer array nums and an integer k, return the number of pairs (i, j) where i < j such that |nums[i] - nums[j]| == k.

The value of |x| is defined as:
- x if x >= 0.
- -x if x < 0.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2007 Find Original Array From Doubled Array

MEDIUM

An integer array original is transformed into a doubled array changed by appending twice the value of every element in original, and then randomly shuffling the resulting array.

Given an array changed, return original if changed is a doubled array. If changed is not a doubled array, return an empty array. The elements in original may be returned in any order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2009 Minimum Number Of Operations To Make Array Continuous

HARD

You are given an integer array nums. In one operation, you can replace any element in nums with any integer.

nums is considered continuous if both of the following conditions are fulfilled:
- All elements in nums are unique.
- The difference between the maximum element and the minimum element in nums equals nums.length - 1.

For example, nums = [4, 2, 5, 3] is continuous, but nums = [1, 2, 3, 5, 6] is not continuous.

Return the minimum number of operations to make nums continuous.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2012 Sum Of Beauty In The Array

MEDIUM

You are given a 0-indexed integer array nums. For each index i (1 <= i <= nums.length - 2) the beauty of nums[i] equals:
- 2, if nums[j] < nums[i] < nums[k], for all 0 <= j < i and for all i < k <= nums.length - 1.
- 1, if nums[i - 1] < nums[i] < nums[i + 1], and the previous condition is not satisfied.
- 0, if none of the previous conditions holds.

Return the sum of beauty of all nums[i] where 1 <= i <= nums.length - 2.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2013 Detect Squares

MEDIUM

You are given a stream of points on the X-Y plane. Design an algorithm that:
- Adds new points from the stream into a data structure. Duplicate points are allowed and should be treated as different points.
- Given a query point, counts the number of ways to choose three points from the data structure such that the three points and the query point form an axis-aligned square with positive area.

An axis-aligned square is a square whose edges are all the same length and are either parallel or perpendicular to the x-axis and y-axis.

Implement the DetectSquares class:
- DetectSquares() Initializes the object with an empty data structure.
- void add(int[] point) Adds a new point point = [x, y] to the data structure.
- int count(int[] point) Counts the number of ways to form axis-aligned squares with point point = [x, y] as described above.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2016 Maximum Difference Between Increasing Elements

EASY

Given a 0-indexed integer array nums of size n, find the maximum difference between nums[i] and nums[j] (i.e., nums[j] - nums[i]), such that 0 <= i < j < n and nums[i] < nums[j].

Return the maximum difference. If no such i and j exists, return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2017 Grid Game

MEDIUM

You are given a 0-indexed 2D array grid of size 2 x n, where grid[r][c] represents the number of points at position (r, c) on the matrix. Two robots are playing a game on this matrix.

Both robots initially start at (0, 0) and want to reach (1, n-1). Each robot may only move to the right ((r, c) to (r, c + 1)) or down ((r, c) to (r + 1, c)).

At the start of the game, the first robot moves from (0, 0) to (1, n-1), collecting all the points from the cells on its path. For all cells (r, c) traversed on the path, grid[r][c] is set to 0. Then, the second robot moves from (0, 0) to (1, n-1), collecting the points on its path. Note that their paths may intersect with one another.

The first robot wants to minimize the number of points collected by the second robot. In contrast, the second robot wants to maximize the number of points it collects. If both robots play optimally, return the number of points collected by the second robot.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2019 The Score Of Students Solving Math Expression

HARD

You are given a string s that contains digits 0-9, addition symbols '+', and multiplication symbols '*' only, representing a valid math expression of single digit numbers (e.g., 3+5*2). This expression was given to n elementary school students. The students were instructed to get the answer of the expression by following this order of operations:

You are given an integer array answers of length n, which are the submitted answers of the students in no particular order. You are asked to grade the answers, by following these rules:
- If an answer equals the correct answer of the expression, this student will be rewarded 5 points;
- Otherwise, if the answer could be interpreted as if the student applied the operators in the wrong order but had correct arithmetic, this student will be rewarded 2 points;
- Otherwise, this student will be rewarded 0 points.

Return the sum of the points of the students.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2022 Convert 1D Array Into 2D Array

EASY

You are given a 0-indexed 1-dimensional (1D) integer array original, and two integers, m and n. You are tasked with creating a 2-dimensional (2D) array with m rows and n columns using all the elements from original.

The elements from indices 0 to n - 1 (inclusive) of original should form the first row of the constructed 2D array, the elements from indices n to 2 * n - 1 (inclusive) should form the second row of the constructed 2D array, and so on.

Return an m x n 2D array constructed according to the above procedure, or an empty 2D array if it is impossible.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2025 Maximum Number Of Ways To Partition An Array

HARD

You are given a 0-indexed integer array nums of length n. The number of ways to partition

nums is the number of pivot indices that satisfy both conditions:
- 1 <= pivot < n
- nums[0] + nums[1] + ... + nums[pivot - 1] == nums[pivot] + nums[pivot + 1] + ... + nums[n - 1]

You are also given an integer k. You can choose to change the value of one element of nums to k, or to leave the array unchanged.

Return the maximum possible number of ways to partition nums to satisfy both conditions after changing at most one element.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2032 Two Out Of Three

**EASY**

Given three integer arrays nums1, nums2, and nums3, return a distinct array containing all the values that are present in at least two out of the three arrays. You may return the values in any order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2033 Minimum Operations To Make A Uni Value Grid

**MEDIUM**

You are given a 2D integer grid of size m x n and an integer x. In one operation, you can add x to or subtract x from any element in the grid.

A uni-value grid is a grid where all the elements of it are equal.

Return the minimum number of operations to make the grid uni-value. If it is not possible, return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2034 Stock Price Fluctuation

**MEDIUM**

You are given a stream of records about a particular stock. Each record contains a timestamp and the corresponding price of the stock at that timestamp.

Unfortunately due to the volatile nature of the stock market, the records do not come in order. Even worse, some records may be incorrect. Another record with the same timestamp may appear later in the stream correcting the price of the previous wrong record.

Design an algorithm that:
- Updates the price of the stock at a particular timestamp, correcting the price from any previous records at the timestamp.
- Finds the latest price of the stock based on the current records. The latest price is the price at the latest timestamp recorded.
- Finds the maximum price the stock has been based on the current records.
- Finds the minimum price the stock has been based on the current records.

Implement the StockPrice class:
- StockPrice() Initializes the object with no price records.
- void update(int timestamp, int price) Updates the price of the stock at the given timestamp.
- int current() Returns the latest price of the stock.
- int maximum() Returns the maximum price of the stock.
- int minimum() Returns the minimum price of the stock.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2035 Partition Array Into Two Arrays To Minimize Sum Difference

HARD

You are given an integer array nums of 2 * n integers. You need to partition nums into two arrays of length n to minimize the absolute difference of the sums of the arrays. To partition nums, put each element of nums into one of the two arrays.

Return the minimum possible absolute difference.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2037 Minimum Number Of Moves To Seat Everyone

EASY

There are n availabe seats and n students standing in a room. You are given an array seats of length n, where seats[i] is the position of the ith seat. You are also given the array students of length n, where students[j] is the position of the jth student.

You may perform the following move any number of times:
- Increase or decrease the position of the ith student by 1 (i.e., moving the ith student from position x to x + 1 or x - 1)

Return the minimum number of moves required to move each student to a seat such that no two students are in the same seat.

Note that there may be multiple seats or students in the same position at the beginning.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2049 Count Nodes With The Highest Score

MEDIUM

There is a binary tree rooted at 0 consisting of n nodes. The nodes are labeled from 0 to n - 1. You are given a 0-indexed integer array parents representing the tree, where parents[i] is the parent of node i. Since node 0 is the root, parents[0] == -1.

Each node has a score. To find the score of a node, consider if the node and the edges connected to it were removed. The tree would become one or more non-empty subtrees. The size of a subtree is the number of the nodes in it. The score of the node is the product of the sizes of all those subtrees.

Return the number of nodes that have the highest score.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2058 Find The Minimum And Maximum Number Of Nodes Between Critical Poin

MEDIUM

A critical point in a linked list is defined as either a local maxima or a local minima.

A node is a local maxima if the current node has a value strictly greater than the previous node and the next node.

A node is a local minima if the current node has a value strictly smaller than the previous node and the next node.

Note that a node can only be a local maxima/minima if there exists both a previous node and a next node.

Given a linked list head, return an array of length 2 containing [minDistance, maxDistance] where minDistance is the minimum distance between any two distinct critical points and maxDistance is the maximum distance between any two distinct critical points. If there are

fewer than two critical points, return [-1, -1].

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2059 Minimum Operations To Convert Number

**MEDIUM**

You are given a 0-indexed integer array nums containing distinct numbers, an integer start, and an integer goal. There is an integer x that is initially set to start, and you want to perform operations on x such that it is converted to goal. You can perform the following operation repeatedly on the number x:

If 0 <= x <= 1000, then for any index i in the array (0 <= i < nums.length), you can set x to any of the following:
- x + nums[i]
- x - nums[i]
- x ^ nums[i] (bitwise-XOR)

Note that you can use each nums[i] any number of times in any order. Operations that set x to be out of the range 0 <= x <= 1000 are valid, but no more operations can be done afterward.

Return the minimum number of operations needed to convert x = start into goal, and -1 if it is not possible.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2061 Number Of Spaces Cleaning Robot Cleaned

**MEDIUM**

A room is represented by a 0-indexed 2D binary matrix room where a 0 represents an empty space and a 1 represents a space with an object. The top left corner of the room will be empty in all test cases.

A cleaning robot starts at the top left corner of the room and is facing right. The robot will continue heading straight until it reaches the edge of the room or it hits an object, after which it will turn 90 degrees clockwise and repeat this process. The starting space and all spaces that the robot visits are cleaned by it.

Return the number of clean spaces in the room if the robot runs indefinitely.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2064 Minimized Maximum Of Products Distributed To Any Store

**MEDIUM**

You are given an integer n indicating there are n specialty retail stores. There are m product types of varying amounts, which are given as a 0-indexed integer array quantities, where quantities[i] represents the number of products of the ith product type.

You need to distribute all products to the retail stores following these rules:
- A store can only be given at most one product type but can be given any amount of it.
- After distribution, each store will have been given some number of products (possibly 0).

Let x represent the maximum number of products given to any store. You want x to be as small as possible, i.e., you want to minimize the maximum number of products that are given to any store.

Return the minimum possible x.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2071 Maximum Number Of Tasks You Can Assign

HARD

You have n tasks and m workers. Each task has a strength requirement stored in a 0-indexed integer array tasks, with the ith task requiring tasks[i] strength to complete. The strength of each worker is stored in a 0-indexed integer array workers, with the jth worker having workers[j] strength. Each worker can only be assigned to a single task and must have a strength greater than or equal to the task's strength requirement (i.e., workers[j] >= tasks[i]).

Additionally, you have pills magical pills that will increase a worker's strength by strength. You can decide which workers receive the magical pills, however, you may only give each worker at most one magical pill.

Given the 0-indexed integer arrays tasks and workers and the integers pills and strength, return the maximum number of tasks that can be completed.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2077 Paths In Maze That Lead To Same Room

MEDIUM

A maze consists of n rooms numbered from 1 to n, and some rooms are connected by corridors. You are given a 2D integer array corridors where corridors[i] = [room1i, room2i] indicates that there is a corridor connecting room1i and room2i, allowing a person in the maze to go from room1i to room2i and vice versa.

The designer of the maze wants to know how confusing the maze is. The confusion score of the maze is the number of different cycles of length 3.
- For example, 1 → 2 → 3 → 1 is a cycle of length 3, but 1 → 2 → 3 → 4 and 1 → 2 → 3 → 2 → 1 are not.

Two cycles are considered to be different if one or more of the rooms visited in the first cycle is not in the second cycle.

Return the confusion score of the maze.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2080 Range Frequency Queries

MEDIUM

Design a data structure to find the frequency of a given value in a given subarray.

The frequency of a value in a subarray is the number of occurrences of that value in the subarray.

Implement the RangeFreqQuery class:
- RangeFreqQuery(int[] arr) Constructs an instance of the class with the given 0-indexed integer array arr.
- int query(int left, int right, int value) Returns the frequency of value in the subarray arr[left...right].
- A subarray is a contiguous sequence of elements within an array. arr[left...right] denotes the subarray that contains the elements of nums between indices left and right (inclusive).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2081 Sum Of K Mirror Numbers

HARD

A k-mirror number is a positive integer without leading zeros that reads the same both forward

and backward in base-10 as well as in base-k.
- For example, 9 is a 2-mirror number. The representation of 9 in base-10 and base-2 are 9 and 1001 respectively, which read the same both forward and backward.
- On the contrary, 4 is not a 2-mirror number. The representation of 4 in base-2 is 100, which does not read the same both forward and backward.

Given the base k and the number n, return the sum of the n smallest k-mirror numbers.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2086 Minimum Number Of Food Buckets To Feed The Hamsters

**MEDIUM**

You are given a 0-indexed string hamsters where hamsters[i] is either:
- 'H' indicating that there is a hamster at index i, or
- '.' indicating that index i is empty.

You will add some number of food buckets at the empty indices in order to feed the hamsters. A hamster can be fed if there is at least one food bucket to its left or to its right. More formally, a hamster at index i can be fed if you place a food bucket at index i - 1 and/or at index i + 1.

Return the minimum number of food buckets you should place at empty indices to feed all the hamsters or -1 if it is impossible to feed all of them.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2087 Minimum Cost Homecoming Of A Robot In A Grid

**MEDIUM**

There is an m x n grid, where (0, 0) is the top-left cell and (m - 1, n - 1) is the bottom-right cell. You are given an integer array startPos where startPos = [startrow, startcol] indicates that initially, a robot is at the cell (startrow, startcol). You are also given an integer array homePos where homePos = [homerow, homecol] indicates that its home is at the cell (homerow, homecol).

The robot needs to go to its home. It can move one cell in four directions: left, right, up, or down, and it can not move outside the boundary. Every move incurs some cost. You are further given two 0-indexed integer arrays: rowCosts of length m and colCosts of length n.
- If the robot moves up or down into a cell whose row is r, then this move costs rowCosts[r].
- If the robot moves left or right into a cell whose column is c, then this move costs colCosts[c].

Return the minimum total cost for this robot to return home.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2088 Count Fertile Pyramids In A Land

**HARD**

A farmer has a rectangular grid of land with m rows and n columns that can be divided into unit cells. Each cell is either fertile (represented by a 1) or barren (represented by a 0). All cells outside the grid are considered barren.

A pyramidal plot of land can be defined as a set of cells with the following criteria: number of rows it covers. Let (r, c) be the apex of the pyramid, and its height be h. Then, the plot comprises of cells (i, j) where r <= i <= r + h - 1 and c - (i - r) <= j <= c + (i - r).

An inverse pyramidal plot of land can be defined as a set of cells with similar criteria: of an inverse pyramid is the number of rows it covers. Let (r, c) be the apex of the pyramid, and its height be h. Then, the plot comprises of cells (i, j) where

r - h + 1 <= i <= r and c - (r - i) <= j <= c + (r - i).

Some examples of valid and invalid pyramidal (and inverse pyramidal) plots are shown below. Black cells indicate fertile cells.

Given a 0-indexed m x n binary matrix grid representing the farmland, return the total number of pyramidal and inverse pyramidal plots that can be found in grid.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2091 Removing Minimum And Maximum From Array

**MEDIUM**

You are given a 0-indexed array of distinct integers nums.

There is an element in nums that has the lowest value and an element that has the highest value. We call them the minimum and maximum respectively. Your goal is to remove both these elements from the array.

A deletion is defined as either removing an element from the front of the array or removing an element from the back of the array.

Return the minimum number of deletions it would take to remove both the minimum and maximum element from the array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2093 Minimum Cost To Reach City With Discounts

**MEDIUM**

A series of highways connect n cities numbered from 0 to n - 1. You are given a 2D integer array highways where highways[i] = [city1i, city2i, tolli] indicates that there is a highway that connects city1i and city2i, allowing a car to go from city1i to city2i and vice versa for a cost of tolli.

You are also given an integer discounts which represents the number of discounts you have. You can use a discount to travel across the ith highway for a cost of tolli / 2 (integer division). Each discount may only be used once, and you can only use at most one discount per highway.

Return the minimum total cost to go from city 0 to city n - 1, or -1 if it is not possible to go from city 0 to city n - 1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2097 Valid Arrangement Of Pairs

**HARD**

You are given a 0-indexed 2D integer array pairs where pairs[i] = [starti, endi]. An arrangement of pairs is valid if for every index i where 1 <= i < pairs.length, we have endi-1 == starti.

Return any valid arrangement of pairs.

Note: The inputs will be generated such that there exists a valid arrangement of pairs.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2098 Subsequence Of Size K With The Largest Even Sum

**MEDIUM**

You are given an integer array nums and an integer k. Find the largest even sum of any subsequence of nums that has a length of k.

Return this sum, or -1 if such a sum does not exist.

A subsequence is an array that can be derived from another array by deleting some or no elements without changing the order of the remaining elements.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2099 Find Subsequence Of Length K With The Largest Sum

**MEDIUM**

You are given an integer array nums and an integer k. You want to find a subsequence of nums of length k that has the largest sum.

Return any such subsequence as an integer array of length k.

A subsequence is an array that can be derived from another array by deleting some or no elements without changing the order of the remaining elements.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2103 Rings And Rods

**EASY**

There are n rings and each ring is either red, green, or blue. The rings are distributed across ten rods labeled from 0 to 9.

You are given a string rings of length 2n that describes the n rings that are placed onto the rods. Every two characters in rings forms a color-position pair that is used to describe each ring where:
- The first character of the ith pair denotes the ith ring's color ('R', 'G', 'B').
- The second character of the ith pair denotes the rod that the ith ring is placed on ('0' to '9').

For example, "R3G2B1" describes n == 3 rings: a red ring placed onto the rod labeled 3, a green ring placed onto the rod labeled 2, and a blue ring placed onto the rod labeled 1.

Return the number of rods that have all three colors of rings on them.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2107 Number Of Unique Flavors After Sharing K Candies

**MEDIUM**

You are given a 0-indexed integer array candies, where candies[i] represents the flavor of the ith candy. Your mom wants you to share these candies with your little sister by giving her k consecutive candies, but you want to keep as many flavors of candies as possible.

Return the maximum number of unique flavors of candy you can keep after sharing with your sister.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2110 Number Of Smooth Descent Periods Of A Stock

**MEDIUM**

You are given an integer array prices representing the daily price history of a stock, where prices[i] is the stock price on the ith day.

A smooth descent period of a stock consists of one or more contiguous days such that the price on each day is lower than the price on the preceding day by exactly 1. The first day of the period is exempted from this rule.

Return the number of smooth descent periods.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 2111 Minimum Operations To Make The Array K Increasing

HARD

You are given a 0-indexed array arr consisting of n positive integers, and a positive integer k.

The array arr is called K-increasing if arr[i-k] <= arr[i] holds for every index i, where k <= i <= n-1.

- For example, arr = [4, 1, 5, 2, 6, 2] is K-increasing for k = 2 because:
- arr[0] <= arr[2] (4 <= 5)
- arr[1] <= arr[3] (1 <= 2)
- arr[2] <= arr[4] (5 <= 6)
- arr[3] <= arr[5] (2 <= 2)
- However, the same arr is not K-increasing for k = 1 (because arr[0] > arr[1]) or k = 3 (because arr[0] > arr[3]).

In one operation, you can choose an index i and change arr[i] into any positive integer.

Return the minimum number of operations required to make the array K-increasing for the given k.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 2113 Elements In Array After Removing And Replacing Elements

MEDIUM

You are given a 0-indexed integer array nums. Initially on minute 0, the array is unchanged. Every minute, the leftmost element in nums is removed until no elements remain. Then, every minute, one element is appended to the end of nums, in the order they were removed in, until the original array is restored. This process repeats indefinitely.
- For example, the array [0,1,2] would change as follows: [0,1,2] → [1,2] → [2] → [] → [0] → [0,1] → [0,1,2] → [1,2] → [2] → [] → [0] → [0,1] → [0,1,2] → ...

You are also given a 2D integer array queries of size n where queries[j] = [timej, indexj]. The answer to the jth query is:
- nums[indexj] if indexj < nums.length at minute timej
- -1 if indexj >= nums.length at minute timej

Return an integer array ans of size n where ans[j] is the answer to the jth query.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 2115 Find All Possible Recipes From Given Supplies

MEDIUM

You have information about n different recipes. You are given a string array recipes and a 2D string array ingredients. The ith recipe has the name recipes[i], and you can create it if you have all the needed ingredients from ingredients[i]. A recipe can also be an ingredient for other recipes, i.e., ingredients[i] may contain a string that is in recipes.

You are also given a string array supplies containing all the ingredients that you initially have, and you have an infinite supply of all of them.

Return a list of all the recipes that you can create. You may return the answer in any order.

Note that two recipes may contain each other in their ingredients.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 2117 Abbreviating The Product Of A Range

HARD

You are given two positive integers left and right with left <= right. Calculate the product of all integers in the inclusive range [left, right].

Since the product may be very large, you will abbreviate it following these steps:
- For example, there are 3 trailing zeros in 1000, and there are 0 trailing zeros in 546.
as <pre>...<suf> where <pre> denotes the first 5 digits of the product, and <suf> denotes the last 5 digits of the product after removing all trailing zeros. If d <= 10, we keep it unchanged.
- For example, we express 1234567654321 as 12345...54321, but 1234567 is represented as 1234567.
- For example, 12345678987600000 will be represented as "12345...89876e5".

Return a string denoting the abbreviated product of all integers in the inclusive range [left, right].

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2119 A Number After A Double Reversal

EASY

Reversing an integer means to reverse all its digits.
- For example, reversing 2021 gives 1202. Reversing 12300 gives 321 as the leading zeros are not retained.

Given an integer num, reverse num to get reversed1, then reverse reversed1 to get reversed2. Return true if reversed2 equals num. Otherwise return false.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2122 Recover The Original Array

HARD

Alice had a 0-indexed array arr consisting of n positive integers. She chose an arbitrary positive integer k and created two new 0-indexed integer arrays lower and higher in the following manner:

Unfortunately, Alice lost all three arrays. However, she remembers the integers that were present in the arrays lower and higher, but not the array each integer belonged to. Help Alice and recover the original array.

Given an array nums consisting of 2n integers, where exactly n of the integers were present in lower and the remaining in higher, return the original array arr. In case the answer is not unique, return any valid array.

Note: The test cases are generated such that there exists at least one valid array arr.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2123 Minimum Operations To Remove Adjacent Ones In Matrix

HARD

You are given a 0-indexed binary matrix grid. In one operation, you can flip any 1 in grid to be 0.

A binary matrix is well-isolated if there is no 1 in the matrix that is 4-directionally connected (i.e., horizontal and vertical) to another 1.

Return the minimum number of operations to make grid well-isolated.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 2125 Number Of Laser Beams In A Bank

MEDIUM

Anti-theft security devices are activated inside a bank. You are given a 0-indexed binary string array bank representing the floor plan of the bank, which is an m x n 2D matrix. bank[i] represents the ith row, consisting of '0's and '1's. '0' means the cell is empty, while'1' means the cell has a security device.

There is one laser beam between any two security devices if both conditions are met:
- The two devices are located on two different rows: r1 and r2, where r1 < r2.
- For each row i where r1 < i < r2, there are no security devices in the ith row.

Laser beams are independent, i.e., one beam does not interfere nor join with another.

Return the total number of laser beams in the bank.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 2132 Stamping The Grid

HARD

You are given an m x n binary matrix grid where each cell is either 0 (empty) or 1 (occupied).

You are then given stamps of size stampHeight x stampWidth. We want to fit the stamps such that they follow the given restrictions and requirements:

Return true if it is possible to fit the stamps while following the given restrictions and requirements. Otherwise, return false.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 2133 Check If Every Row And Column Contains All Numbers

EASY

An n x n matrix is valid if every row and every column contains all the integers from 1 to n (inclusive).

Given an n x n integer matrix matrix, return true if the matrix is valid. Otherwise, return false.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 2143 Choose Numbers From Two Arrays In Range

HARD

You are given two 0-indexed integer arrays nums1 and nums2 of length n.

A range [l, r] (inclusive) where 0 <= l <= r < n is balanced if:
- For every i in the range [l, r], you pick either nums1[i] or nums2[i].
- The sum of the numbers you pick from nums1 equals to the sum of the numbers you pick from nums2 (the sum is considered to be 0 if you pick no numbers from an array).

Two balanced ranges from [l1, r1] and [l2, r2] are considered to be different if at least one of the following is true:
- l1 != l2
- r1 != r2
- nums1[i] is picked in the first range, and nums2[i] is picked in the second range or vice versa for at least one i.

Return the number of different ranges that are balanced. Since the answer may be very large, return it modulo 109 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2144 Minimum Cost Of Buying Candies With Discount

**EASY**

A shop is selling candies at a discount. For every two candies sold, the shop gives a third candy for free.

The customer can choose any candy to take away for free as long as the cost of the chosen candy is less than or equal to the minimum cost of the two candies bought.

- For example, if there are 4 candies with costs 1, 2, 3, and 4, and the customer buys candies with costs 2 and 3, they can take the candy with cost 1 for free, but not the candy with cost 4.

Given a 0-indexed integer array cost, where cost[i] denotes the cost of the ith candy, return the minimum cost of buying all the candies.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2145 Count The Hidden Sequences

**MEDIUM**

You are given a 0-indexed array of n integers differences, which describes the differences between each pair of consecutive integers of a hidden sequence of length (n + 1). More formally, call the hidden sequence hidden, then we have that differences[i] = hidden[i + 1] - hidden[i].

You are further given two integers lower and upper that describe the inclusive range of values [lower, upper] that the hidden sequence can contain.

- For example, given differences = [1, -3, 4], lower = 1, upper = 6, the hidden sequence is a sequence of length 4 whose elements are in between 1 and 6 (inclusive).
- [3, 4, 1, 5] and [4, 5, 2, 6] are possible hidden sequences.
- [5, 6, 3, 7] is not possible since it contains an element greater than 6.
- [1, 2, 3, 4] is not possible since the differences are not correct.

Return the number of possible hidden sequences there are. If there are no possible sequences, return 0.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2146 K Highest Ranked Items Within A Price Range

**MEDIUM**

You are given a 0-indexed 2D integer array grid of size m x n that represents a map of the items in a shop. The integers in the grid represent the following:
- 0 represents a wall that you cannot pass through.
- 1 represents an empty cell that you can freely move to and from.

All other positive integers represent the price of an item in that cell. You may also freely move to and from these item cells.

It takes 1 step to travel between adjacent grid cells.

You are also given integer arrays pricing and start where pricing = [low, high] and start = [row, col] indicates that you start at the position (row, col) and are interested only in items with a price in the range of [low, high] (inclusive). You are further given an integer k.

You are interested in the positions of the k highest-ranked items whose prices are within

the given price range. The rank is determined by the first of these criteria that is different: a higher rank).

Return the k highest-ranked items within the price range sorted by their rank (highest to lowest). If there are fewer than k reachable items within the price range, return all of them.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2147 Number Of Ways To Divide A Long Corridor

**HARD**

Along a long library corridor, there is a line of seats and decorative plants. You are given a 0-indexed string corridor of length n consisting of letters 'S' and 'P' where each 'S' represents a seat and each 'P' represents a plant.

One room divider has already been installed to the left of index 0, and another to the right of index n - 1. Additional room dividers can be installed. For each position between indices i - 1 and i (1 <= i <= n - 1), at most one divider can be installed.

Divide the corridor into non-overlapping sections, where each section has exactly two seats with any number of plants. There may be multiple ways to perform the division. Two ways are different if there is a position with a room divider installed in the first way but not in the second way.

Return the number of ways to divide the corridor. Since the answer may be very large, return it modulo 109 + 7. If there is no way, return 0.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2148 Count Elements With Strictly Smaller And Greater Elements

**EASY**

Given an integer array nums, return the number of elements that have both a strictly smaller and a strictly greater element appear in nums.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2149 Rearrange Array Elements By Sign

**MEDIUM**

You are given a 0-indexed integer array nums of even length consisting of an equal number of positive and negative integers.

You should return the array of nums such that the the array follows the given conditions: preserved.

Return the modified array after rearranging the elements to satisfy the aforementioned conditions.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2150 Find All Lonely Numbers In The Array

**MEDIUM**

You are given an integer array nums. A number x is lonely when it appears only once, and no adjacent numbers (i.e. x + 1 and x - 1) appear in the array.

Return all lonely numbers in nums. You may return the answer in any order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2152 Minimum Number Of Lines To Cover Points

MEDIUM

You are given an array points where points[i] = [xi, yi] represents a point on an X-Y plane.

Straight lines are going to be added to the X-Y plane, such that every point is covered by at least one line.

Return the minimum number of straight lines needed to cover all the points.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2155 All Divisions With The Highest Score Of A Binary Array

MEDIUM

You are given a 0-indexed binary array nums of length n. nums can be divided at index i (where 0 <= i <= n) into two arrays (possibly empty) numsleft and numsright:
- numsleft has all the elements of nums between index 0 and i - 1 (inclusive), while numsright has all the elements of nums between index i and n - 1 (inclusive).
- If i == 0, numsleft is empty, while numsright has all the elements of nums.
- If i == n, numsleft has all the elements of nums, while numsright is empty.

The division score of an index i is the sum of the number of 0's in numsleft and the number of 1's in numsright.

Return all distinct indices that have the highest possible division score. You may return the answer in any order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2158 Amount Of New Area Painted Each Day

HARD

There is a long and thin painting that can be represented by a number line. You are given a 0-indexed 2D integer array paint of length n, where paint[i] = [starti, endi]. This means that on the ith day you need to paint the area between starti and endi.

Painting the same area multiple times will create an uneven painting so you only want to paint each area of the painting at most once.

Return an integer array worklog of length n, where worklog[i] is the amount of new area that you painted on the ith day.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2161 Partition Array According To Given Pivot

MEDIUM

You are given a 0-indexed integer array nums and an integer pivot. Rearrange nums such that the following conditions are satisfied:
- Every element less than pivot appears before every element greater than pivot.
- Every element equal to pivot appears in between the elements less than and greater than pivot.
- The relative order of the elements less than pivot and the elements greater than pivot is maintained.
- More formally, consider every pi, pj where pi is the new position of the ith element and pj is the new position of the jth element. If i < j and both elements are smaller (or larger) than pivot, then pi < pj.

Return nums after the rearrangement.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 2162 Minimum Cost To Set Cooking Time

MEDIUM

A generic microwave supports cooking times for:
- at least 1 second.
- at most 99 minutes and 99 seconds.

To set the cooking time, you push at most four digits. The microwave normalizes what you push as four digits by prepending zeroes. It interprets the first two digits as the minutes and the last two digits as the seconds. It then adds them up as the cooking time. For example,
- You push 9 5 4 (three digits). It is normalized as 0954 and interpreted as 9 minutes and 54 seconds.
- You push 0 0 0 8 (four digits). It is interpreted as 0 minutes and 8 seconds.
- You push 8 0 9 0. It is interpreted as 80 minutes and 90 seconds.
- You push 8 1 3 0. It is interpreted as 81 minutes and 30 seconds.

You are given integers startAt, moveCost, pushCost, and targetSeconds. Initially, your finger is on the digit startAt. Moving the finger above any specific digit costs moveCost units of fatigue. Pushing the digit below the finger once costs pushCost units of fatigue.

There can be multiple ways to set the microwave to cook for targetSeconds seconds but you are interested in the way with the minimum cost.

Return the minimum cost to set targetSeconds seconds of cooking time.

Remember that one minute consists of 60 seconds.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 2163 Minimum Difference In Sums After Removal Of Elements

HARD

You are given a 0-indexed integer array nums consisting of 3 * n elements.

You are allowed to remove any subsequence of elements of size exactly n from nums. The remaining 2 * n elements will be divided into two equal parts:
- The first n elements belonging to the first part and their sum is sumfirst.
- The next n elements belonging to the second part and their sum is sumsecond.

The difference in sums of the two parts is denoted as sumfirst - sumsecond.
- For example, if sumfirst = 3 and sumsecond = 2, their difference is 1.
- Similarly, if sumfirst = 2 and sumsecond = 3, their difference is -1.

Return the minimum difference possible between the sums of the two parts after the removal of n elements.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 2165 Smallest Value Of The Rearranged Number

MEDIUM

You are given an integer num. Rearrange the digits of num such that its value is minimized and it does not contain any leading zeros.

Return the rearranged number with minimal value.

Note that the sign of the number does not change after rearranging the digits.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 2166 Design Bitset

MEDIUM

A Bitset is a data structure that compactly stores bits.

Implement the Bitset class:
- Bitset(int size) Initializes the Bitset with size bits, all of which are 0.
- void fix(int idx) Updates the value of the bit at the index idx to 1. If the value was already 1, no change occurs.
- void unfix(int idx) Updates the value of the bit at the index idx to 0. If the value was already 0, no change occurs.
- void flip() Flips the values of each bit in the Bitset. In other words, all bits with value 0 will now have value 1 and vice versa.
- boolean all() Checks if the value of each bit in the Bitset is 1. Returns true if it satisfies the condition, false otherwise.
- boolean one() Checks if there is at least one bit in the Bitset with value 1. Returns true if it satisfies the condition, false otherwise.
- int count() Returns the total number of bits in the Bitset which have value 1.
- String toString() Returns the current composition of the Bitset. Note that in the resultant string, the character at the ith index should coincide with the value at the ith bit of the Bitset.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2169 Count Operations To Obtain Zero

EASY

You are given two non-negative integers num1 and num2.

In one operation, if num1 >= num2, you must subtract num2 from num1, otherwise subtract num1 from num2.
- For example, if num1 = 5 and num2 = 4, subtract num2 from num1, thus obtaining num1 = 1 and num2 = 4. However, if num1 = 4 and num2 = 5, after one operation, num1 = 4 and num2 = 1.

Return the number of operations required to make either num1 = 0 or num2 = 0.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2170 Minimum Operations To Make The Array Alternating

MEDIUM

You are given a 0-indexed array nums consisting of n positive integers.

The array nums is called alternating if:
- nums[i - 2] == nums[i], where 2 <= i <= n - 1.
- nums[i - 1] != nums[i], where 1 <= i <= n - 1.

In one operation, you can choose an index i and change nums[i] into any positive integer.

Return the minimum number of operations required to make the array alternating.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2171 Removing Minimum Number Of Magic Beans

MEDIUM

You are given an array of positive integers beans, where each integer represents the number of magic beans found in a particular magic bag.

Remove any number of beans (possibly none) from each bag such that the number of beans in

each remaining non-empty bag (still containing at least one bean) is equal. Once a bean has been removed from a bag, you are not allowed to return it to any of the bags.

Return the minimum number of magic beans that you have to remove.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2172 Maximum And Sum Of Array

**HARD**

You are given an integer array nums of length n and an integer numSlots such that 2 * numSlots >= n. There are numSlots slots numbered from 1 to numSlots.

You have to place all n integers into the slots such that each slot contains at most two numbers. The AND sum of a given placement is the sum of the bitwise AND of every number with its respective slot number.
- For example, the AND sum of placing the numbers [1, 3] into slot 1 and [4, 6] into slot 2 is equal to (1 AND 1) + (3 AND 1) + (4 AND 2) + (6 AND 2) = 1 + 1 + 0 + 2 = 4.

Return the maximum possible AND sum of nums given numSlots slots.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2174 Remove All Ones With Row And Column Flips Ii

**MEDIUM**

You are given a 0-indexed m x n binary matrix grid.

In one operation, you can choose any i and j that meet the following conditions:
- 0 <= i < m
- 0 <= j < n
- grid[i][j] == 1

and change the values of all cells in row i and column j to zero.

Return the minimum number of operations needed to remove all 1's from grid.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2176 Count Equal And Divisible Pairs In An Array

**EASY**

Given a 0-indexed integer array nums of length n and an integer k, return the number of pairs (i, j) where 0 <= i < j < n, such that nums[i] == nums[j] and (i * j) is divisible by k.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2177 Find Three Consecutive Integers That Sum To A Given Number

**MEDIUM**

Given an integer num, return three consecutive integers (as a sorted array) that sum to num. If num cannot be expressed as the sum of three consecutive integers, return an empty array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2179 Count Good Triplets In An Array

**HARD**

You are given two 0-indexed arrays nums1 and nums2 of length n, both of which are permutations of [0, 1, ..., n - 1].

A good triplet is a set of 3 distinct values which are present in increasing order by position both in nums1 and nums2. In other words, if we consider pos1v as the index of the value v in nums1 and pos2v as the index of the value v in nums2, then a good triplet will be a set (x, y, z) where 0 <= x, y, z <= n - 1, such that pos1x < pos1y < pos1z and pos2x < pos2y < pos2z.

Return the total number of good triplets.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2183 Count Array Pairs Divisible By K

**HARD**

Given a 0-indexed integer array nums of length n and an integer k, return the number of pairs (i, j) such that:
- 0 <= i < j <= n - 1 and
- nums[i] * nums[j] is divisible by k.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2184 Number Of Ways To Build Sturdy Brick Wall

**MEDIUM**

You are given integers height and width which specify the dimensions of a brick wall you are building. You are also given a 0-indexed array of unique integers bricks, where the ith brick has a height of 1 and a width of bricks[i]. You have an infinite supply of each type of brick and bricks may not be rotated.

Each row in the wall must be exactly width units long. For the wall to be sturdy, adjacent rows in the wall should not join bricks at the same location, except at the ends of the wall.

Return the number of ways to build a sturdy wall. Since the answer may be very large, return it modulo 109 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2189 Number Of Ways To Build House Of Cards

**MEDIUM**

You are given an integer n representing the number of playing cards you have. A house of cards meets the following conditions:
- A house of cards consists of one or more rows of triangles and horizontal cards.
- Triangles are created by leaning two cards against each other.
- One card must be placed horizontally between all adjacent triangles in a row.
- Any triangle on a row higher than the first must be placed on a horizontal card from the previous row.
- Each triangle is placed in the leftmost available spot in the row.

Return the number of distinct house of cards you can build using all n cards. Two houses of cards are considered distinct if there exists a row where the two houses contain a different number of cards.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2190 Most Frequent Number Following Key In An Array

**EASY**

You are given a 0-indexed integer array nums. You are also given an integer key, which is present in nums.

For every unique integer target in nums, count the number of times target immediately follows

an occurrence of key in nums. In other words, count the number of indices i such that:
- 0 <= i <= nums.length - 2,
- nums[i] == key and,
- nums[i + 1] == target.

Return the target with the maximum count. The test cases will be generated such that the target with maximum count is unique.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2193 Minimum Number Of Moves To Make Palindrome

**HARD**

You are given a string s consisting only of lowercase English letters.

In one move, you can select any two adjacent characters of s and swap them.

Return the minimum number of moves needed to make s a palindrome.

Note that the input will be generated such that s can always be converted to a palindrome.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2195 Append K Integers With Minimal Sum

**MEDIUM**

You are given an integer array nums and an integer k. Append k unique positive integers that do not appear in nums to nums such that the resulting total sum is minimum.

Return the sum of the k integers appended to nums.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2197 Replace Non Coprime Numbers In Array

**HARD**

You are given an array of integers nums. Perform the following steps:

Return the final modified array. It can be shown that replacing adjacent non-coprime numbers in any arbitrary order will lead to the same result.

The test cases are generated such that the values in the final array are less than or equal to 108.

Two values x and y are non-coprime if GCD(x, y) > 1 where GCD(x, y) is the Greatest Common Divisor of x and y.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2198 Number Of Single Divisor Triplets

**MEDIUM**

You are given a 0-indexed array of positive integers nums. A triplet of three distinct indices (i, j, k) is called a single divisor triplet of nums if nums[i] + nums[j] + nums[k] is divisible by exactly one of nums[i], nums[j], or nums[k].

Return the number of single divisor triplets of nums.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2200 Find All K Distant Indices In An Array

EASY

You are given a 0-indexed integer array nums and two integers key and k. A k-distant index is an index i of nums for which there exists at least one index j such that |i - j| <= k and nums[j] == key.

Return a list of all k-distant indices sorted in increasing order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2201 Count Artifacts That Can Be Extracted

MEDIUM

There is an n x n 0-indexed grid with some artifacts buried in it. You are given the integer n and a 0-indexed 2D integer array artifacts describing the positions of the rectangular artifacts where artifacts[i] = [r1i, c1i, r2i, c2i] denotes that the ith artifact is buried in the subgrid where:
- (r1i, c1i) is the coordinate of the top-left cell of the ith artifact and
- (r2i, c2i) is the coordinate of the bottom-right cell of the ith artifact.

You will excavate some cells of the grid and remove all the mud from them. If the cell has a part of an artifact buried underneath, it will be uncovered. If all the parts of an artifact are uncovered, you can extract it.

Given a 0-indexed 2D integer array dig where dig[i] = [ri, ci] indicates that you will excavate the cell (ri, ci), return the number of artifacts that you can extract.

The test cases are generated such that:
- No two artifacts overlap.
- Each artifact only covers at most 4 cells.
- The entries of dig are unique.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2202 Maximize The Topmost Element After K Moves

MEDIUM

You are given a 0-indexed integer array nums representing the contents of a pile, where nums[0] is the topmost element of the pile.

In one move, you can perform either of the following:
- If the pile is not empty, remove the topmost element of the pile.
- If there are one or more removed elements, add any one of them back onto the pile. This element becomes the new topmost element.

You are also given an integer k, which denotes the total number of moves to be made.

Return the maximum value of the topmost element of the pile possible after exactly k moves. In case it is not possible to obtain a non-empty pile after k moves, return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2206 Divide Array Into Equal Pairs

EASY

You are given an integer array nums consisting of 2 * n integers.

You need to divide nums into n pairs such that:
- Each element belongs to exactly one pair.
- The elements present in a pair are equal.

Return true if nums can be divided into n pairs, otherwise return false.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2208 Minimum Operations To Halve Array Sum

**MEDIUM**

You are given an array nums of positive integers. In one operation, you can choose any number from nums and reduce it to exactly half the number. (Note that you may choose this reduced number in future operations.)

Return the minimum number of operations to reduce the sum of nums by at least half.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2210 Count Hills And Valleys In An Array

**EASY**

You are given a 0-indexed integer array nums. An index i is part of a hill in nums if the closest non-equal neighbors of i are smaller than nums[i]. Similarly, an index i is part of a valley in nums if the closest non-equal neighbors of i are larger than nums[i]. Adjacent indices i and j are part of the same hill or valley if nums[i] == nums[j].

Note that for an index to be part of a hill or valley, it must have a non-equal neighbor on both the left and right of the index.

Return the number of hills and valleys in nums.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2211 Count Collisions On A Road

**MEDIUM**

There are n cars on an infinitely long road. The cars are numbered from 0 to n - 1 from left to right and each car is present at a unique point.

You are given a 0-indexed string directions of length n. directions[i] can be either 'L', 'R', or 'S' denoting whether the ith car is moving towards the left, towards the right, or staying at its current point respectively. Each moving car has the same speed.

The number of collisions can be calculated as follows:
- When two cars moving in opposite directions collide with each other, the number of collisions increases by 2.
- When a moving car collides with a stationary car, the number of collisions increases by 1.

After a collision, the cars involved can no longer move and will stay at the point where they collided. Other than that, cars cannot change their state or direction of motion.

Return the total number of collisions that will happen on the road.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2215 Find The Difference Of Two Arrays

**EASY**

Given two 0-indexed integer arrays nums1 and nums2, return a list answer of size 2 where:
- answer[0] is a list of all distinct integers in nums1 which are not present in nums2.
- answer[1] is a list of all distinct integers in nums2 which are not present in nums1.

Note that the integers in the lists may be returned in any order.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2216 Minimum Deletions To Make Array Beautiful

MEDIUM

You are given a 0-indexed integer array nums. The array nums is beautiful if:
- nums.length is even.
- nums[i] != nums[i + 1] for all i % 2 == 0.

Note that an empty array is considered beautiful.

You can delete any number of elements from nums. When you delete an element, all the elements to the right of the deleted element will be shifted one unit to the left to fill the gap created and all the elements to the left of the deleted element will remain unchanged.

Return the minimum number of elements to delete from nums to make it beautiful.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2219 Maximum Sum Score Of Array

MEDIUM

You are given a 0-indexed integer array nums of length n.

The sum score of nums at an index i where 0 <= i < n is the maximum of:
- The sum of the first i + 1 elements of nums.
- The sum of the last n - i elements of nums.

Return the maximum sum score of nums at any index.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2220 Minimum Bit Flips To Convert Number

EASY

A bit flip of a number x is choosing a bit in the binary representation of x and flipping it from either 0 to 1 or 1 to 0.
- For example, for x = 7, the binary representation is 111 and we may choose any bit (including any leading zeros not shown) and flip it. We can flip the first bit from the right to get 110, flip the second bit from the right to get 101, flip the fifth bit from the right (a leading zero) to get 10111, etc.

Given two integers start and goal, return the minimum number of bit flips to convert start to goal.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2221 Find Triangular Sum Of An Array

MEDIUM

You are given a 0-indexed integer array nums, where nums[i] is a digit between 0 and 9 (inclusive).

The triangular sum of nums is the value of the only element present in nums after the following process terminates:
0-indexed integer array newNums of length n - 1.
(nums[i] + nums[i+1]) % 10, where % denotes modulo operator.

Return the triangular sum of nums.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2222 Number Of Ways To Select Buildings

MEDIUM

You are given a 0-indexed binary string s which represents the types of buildings along a street where:
- s[i] = '0' denotes that the ith building is an office and
- s[i] = '1' denotes that the ith building is a restaurant.

As a city official, you would like to select 3 buildings for random inspection. However, to ensure variety, no two consecutive buildings out of the selected buildings can be of the same type.
- For example, given s = "001101", we cannot select the 1st, 3rd, and 5th buildings as that would form "011" which is not allowed due to having two consecutive buildings of the same type.

Return the number of valid ways to select 3 buildings.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2224 Minimum Number Of Operations To Convert Time

EASY

You are given two strings current and correct representing two 24-hour times.

24-hour times are formatted as "HH:MM", where HH is between 00 and 23, and MM is between 00 and 59. The earliest 24-hour time is 00:00, and the latest is 23:59.

In one operation you can increase the time current by 1, 5, 15, or 60 minutes. You can perform this operation any number of times.

Return the minimum number of operations needed to convert current to correct.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2225 Find Players With Zero Or One Losses

MEDIUM

You are given an integer array matches where matches[i] = [winneri, loseri] indicates that the player winneri defeated player loseri in a match.

Return a list answer of size 2 where:
- answer[0] is a list of all players that have not lost any matches.
- answer[1] is a list of all players that have lost exactly one match.

The values in the two lists should be returned in increasing order.

Note:
- You should only consider the players that have played at least one match.
- The testcases will be generated such that no two matches will have the same outcome.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2229 Check If An Array Is Consecutive

EASY

Given an integer array nums, return true if nums is consecutive, otherwise return false.

An array is consecutive if it contains every number in the range [x, x + n - 1] (inclusive), where x is the minimum number in the array and n is the length of the array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2233 Maximum Product After K Increments

MEDIUM

You are given an array of non-negative integers nums and an integer k. In one operation, you may choose any element from nums and increment it by 1.

Return the maximum product of nums after at most k operations. Since the answer may be very large, return it modulo 109 + 7. Note that you should maximize the product before taking the modulo.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2236 Root Equals Sum Of Children

EASY

You are given the root of a binary tree that consists of exactly 3 nodes: the root, its left child, and its right child.

Return true if the value of the root is equal to the sum of the values of its two children, or false otherwise.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2239 Find Closest Number To Zero

EASY

Given an integer array nums of size n, return the number with the value closest to 0 in nums. If there are multiple answers, return the number with the largest value.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2240 Number Of Ways To Buy Pens And Pencils

MEDIUM

You are given an integer total indicating the amount of money you have. You are also given two integers cost1 and cost2 indicating the price of a pen and pencil respectively. You can spend part or all of your money to buy multiple quantities (or none) of each kind of writing utensil.

Return the number of distinct ways you can buy some number of pens and pencils.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2244 Minimum Rounds To Complete All Tasks

MEDIUM

You are given a 0-indexed integer array tasks, where tasks[i] represents the difficulty level of a task. In each round, you can complete either 2 or 3 tasks of the same difficulty level.

Return the minimum rounds required to complete all the tasks, or -1 if it is not possible to complete all the tasks.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2247 Maximum Cost Of Trip With K Highways

HARD

A series of highways connect n cities numbered from 0 to n - 1. You are given a 2D integer array

highways where highways[i] = [city1i, city2i, tolli] indicates that there is a highway that connects city1i and city2i, allowing a car to go from city1i to city2i and vice versa for a cost of tolli.

You are also given an integer k. You are going on a trip that crosses exactly k highways. You may start at any city, but you may only visit each city at most once during your trip.

Return the maximum cost of your trip. If there is no trip that meets the requirements, return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2248 Intersection Of Multiple Arrays

**EASY**

Given a 2D integer array nums where nums[i] is a non-empty array of distinct positive integers, return the list of integers that are present in each array of nums sorted in ascending order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2249 Count Lattice Points Inside A Circle

**MEDIUM**

Given a 2D integer array circles where circles[i] = [xi, yi, ri] represents the center (xi, yi) and radius ri of the ith circle drawn on a grid, return the number of lattice points that are present inside at least one circle.

Note:
- A lattice point is a point with integer coordinates.
- Points that lie on the circumference of a circle are also considered to be inside it.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2250 Count Number Of Rectangles Containing Each Point

**MEDIUM**

You are given a 2D integer array rectangles where rectangles[i] = [li, hi] indicates that ith rectangle has a length of li and a height of hi. You are also given a 2D integer array points where points[j] = [xj, yj] is a point with coordinates (xj, yj).

The ith rectangle has its bottom-left corner point at the coordinates (0, 0) and its top-right corner point at (li, hi).

Return an integer array count of length points.length where count[j] is the number of rectangles that contain the jth point.

The ith rectangle contains the jth point if 0 <= xj <= li and 0 <= yj <= hi. Note that points that lie on the edges of a rectangle are also considered to be contained by that rectangle.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2251 Number Of Flowers In Full Bloom

**HARD**

You are given a 0-indexed 2D integer array flowers, where flowers[i] = [starti, endi] means the ith flower will be in full bloom from starti to endi (inclusive). You are also given a 0-indexed integer array people of size n, where people[i] is the time that the ith person will arrive to see the flowers.

Return an integer array answer of size n, where answer[i] is the number of flowers that are in full bloom when the ith person arrives.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2254 Design Video Sharing Platform

HARD

You have a video sharing platform where users can upload and delete videos. Each video is a string of digits, where the ith digit of the string represents the content of the video at minute i. For example, the first digit represents the content at minute 0 in the video, the second digit represents the content at minute 1 in the video, and so on. Viewers of videos can also like and dislike videos. Internally, the platform keeps track of the number of views, likes, and dislikes on each video.

When a video is uploaded, it is associated with the smallest available integer videoId starting from 0. Once a video is deleted, the videoId associated with that video can be reused for another video.

Implement the VideoSharingPlatform class:
- VideoSharingPlatform() Initializes the object.
- int upload(String video) The user uploads a video. Return the videoId associated with the video.
- void remove(int videoId) If there is a video associated with videoId, remove the video.
- String watch(int videoId, int startMinute, int endMinute) If there is a video associated with videoId, increase the number of views on the video by 1 and return the substring of the video string starting at startMinute and ending at min(endMinute, video.length - 1) (inclusive). Otherwise, return "-1".
- void like(int videoId) Increases the number of likes on the video associated with videoId by 1 if there is a video associated with videoId.
- void dislike(int videoId) Increases the number of dislikes on the video associated with videoId by 1 if there is a video associated with videoId.
- int[] getLikesAndDislikes(int videoId) Return a 0-indexed integer array values of length 2 where values[0] is the number of likes and values[1] is the number of dislikes on the video associated with videoId. If there is no video associated with videoId, return [-1].
- int getViews(int videoId) Return the number of views on the video associated with videoId, if there is no video associated with videoId, return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2256 Minimum Average Difference

MEDIUM

You are given a 0-indexed integer array nums of length n.

The average difference of the index i is the absolute difference between the average of the first i + 1 elements of nums and the average of the last n - i - 1 elements. Both averages should be rounded down to the nearest integer.

Return the index with the minimum average difference. If there are multiple such indices, return the smallest one.

Note:
- The absolute difference of two numbers is the absolute value of their difference.
- The average of n elements is the sum of the n elements divided (integer division) by n.
- The average of 0 elements is considered to be 0.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2257 Count Unguarded Cells In The Grid

MEDIUM

You are given two integers m and n representing a 0-indexed m x n grid. You are also given two 2D integer arrays guards and walls where guards[i] = [rowi, coli] and walls[j] = [rowj, colj]

represent the positions of the ith guard and jth wall respectively.

A guard can see every cell in the four cardinal directions (north, east, south, or west) starting from their position unless obstructed by a wall or another guard. A cell is guarded if there is at least one guard that can see it.

Return the number of unoccupied cells that are not guarded.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2260 Minimum Consecutive Cards To Pick Up

**MEDIUM**

You are given an integer array cards where cards[i] represents the value of the ith card. A pair of cards are matching if the cards have the same value.

Return the minimum number of consecutive cards you have to pick up to have a pair of matching cards among the picked cards. If it is impossible to have matching cards, return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2263 Make Array Non Decreasing Or Non Increasing

**HARD**

You are given a 0-indexed integer array nums. In one operation, you can:
- Choose an index i in the range 0 <= i < nums.length
- Set nums[i] to nums[i] + 1 or nums[i] - 1

Return the minimum number of operations to make nums non-decreasing or non-increasing.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2268 Minimum Number Of Keypresses

**MEDIUM**

You have a keypad with 9 buttons, numbered from 1 to 9, each mapped to lowercase English letters. You can choose which characters each button is matched to as long as:
- All 26 lowercase English letters are mapped to.
- Each character is mapped to by exactly 1 button.
- Each button maps to at most 3 characters.

To type the first character matched to a button, you press the button once. To type the second character, you press the button twice, and so on.

Given a string s, return the minimum number of keypresses needed to type s using your keypad.

Note that the characters mapped to by each button, and the order they are mapped in cannot be changed.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2269 Find The K Beauty Of A Number

**EASY**

The k-beauty of an integer num is defined as the number of substrings of num when it is read as a string that meet the following conditions:
- It has a length of k.
- It is a divisor of num.

Given integers num and k, return the k-beauty of num.

Note:

- Leading zeros are allowed.
- 0 is not a divisor of any value.

A substring is a contiguous sequence of characters in a string.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2270 Number Of Ways To Split Array

**MEDIUM**

You are given a 0-indexed integer array nums of length n.

nums contains a valid split at index i if the following are true:
- The sum of the first i + 1 elements is greater than or equal to the sum of the last n - i - 1 elements.
- There is at least one element to the right of i. That is, 0 <= i < n - 1.

Return the number of valid splits in nums.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2281 Sum Of Total Strength Of Wizards

**HARD**

As the ruler of a kingdom, you have an army of wizards at your command.

You are given a 0-indexed integer array strength, where strength[i] denotes the strength of the ith wizard. For a contiguous group of wizards (i.e. the wizards' strengths form a subarray of strength), the total strength is defined as the product of the following two values:
- The strength of the weakest wizard in the group.
- The total of all the individual strengths of the wizards in the group.

Return the sum of the total strengths of all contiguous groups of wizards. Since the answer may be very large, return it modulo 109 + 7.

A subarray is a contiguous non-empty sequence of elements within an array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2282 Number Of People That Can Be Seen In A Grid

**MEDIUM**

You are given an m x n 0-indexed 2D array of positive integers heights where heights[i][j] is the height of the person standing at position (i, j).

A person standing at position (row1, col1) can see a person standing at position (row2, col2) if:
- The person at (row2, col2) is to the right or below the person at (row1, col1). More formally, this means that either row1 == row2 and col1 < col2 or row1 < row2 and col1 == col2.
- Everyone in between them is shorter than both of them.

Return an m x n 2D array of integers answer where answer[i][j] is the number of people that the person at position (i, j) can see.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2288 Apply Discount To Prices

**MEDIUM**

A sentence is a string of single-space separated words where each word can contain digits, lowercase letters, and the dollar sign '$'. A word represents a price if it is a sequence of digits preceded by a dollar sign.

- For example, "$100", "$23", and "$6" represent prices while "100", "$", and "$1e5" do not.

You are given a string sentence representing a sentence and an integer discount. For each word representing a price, apply a discount of discount% on the price and update the word in the sentence. All updated prices should be represented with exactly two decimal places.

Return a string representing the modified sentence.

Note that all prices will contain at most 10 digits.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2289 Steps To Make Array Non Decreasing

**MEDIUM**

You are given a 0-indexed integer array nums. In one step, remove all elements nums[i] where nums[i - 1] > nums[i] for all 0 < i < nums.length.

Return the number of steps performed until nums becomes a non-decreasing array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2294 Partition Array Such That Maximum Difference Is K

**MEDIUM**

You are given an integer array nums and an integer k. You may partition nums into one or more subsequences such that each element in nums appears in exactly one of the subsequences.

Return the minimum number of subsequences needed such that the difference between the maximum and minimum values in each subsequence is at most k.

A subsequence is a sequence that can be derived from another sequence by deleting some or no elements without changing the order of the remaining elements.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2295 Replace Elements In An Array

**MEDIUM**

You are given a 0-indexed array nums that consists of n distinct positive integers. Apply m operations to this array, where in the ith operation you replace the number operations[i][0] with operations[i][1].

It is guaranteed that in the ith operation:
- operations[i][0] exists in nums.
- operations[i][1] does not exist in nums.

Return the array obtained after applying all the operations.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2300 Successful Pairs Of Spells And Potions

**MEDIUM**

You are given two positive integer arrays spells and potions, of length n and m respectively, where spells[i] represents the strength of the ith spell and potions[j] represents the strength of the jth potion.

You are also given an integer success. A spell and potion pair is considered successful if the product of their strengths is at least success.

Return an integer array pairs of length n where pairs[i] is the number of potions that will form a successful pair with the ith spell.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 2304 Minimum Path Cost In A Grid

MEDIUM

You are given a 0-indexed m x n integer matrix grid consisting of distinct integers from 0 to m * n - 1. You can move in this matrix from a cell to any other cell in the next row. That is, if you are in cell (x, y) such that x < m - 1, you can move to any of the cells (x + 1, 0), (x + 1, 1), ..., (x + 1, n - 1). Note that it is not possible to move from cells in the last row.

Each possible move has a cost given by a 0-indexed 2D array moveCost of size (m * n) x n, where moveCost[i][j] is the cost of moving from a cell with value i to a cell in column j of the next row. The cost of moving from cells in the last row of grid can be ignored.

The cost of a path in grid is the sum of all values of cells visited plus the sum of costs of all the moves made. Return the minimum cost of a path that starts from any cell in the first row and ends at any cell in the last row.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 2306 Naming A Company

HARD

You are given an array of strings ideas that represents a list of names to be used in the process of naming a company. The process of naming a company is as follows: (the concatenation of ideaA and ideaB, separated by a space) is a valid company name.

Return the number of distinct valid names for the company.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 2315 Count Asterisks

EASY

You are given a string s, where every two consecutive vertical bars '|' are grouped into a pair. In other words, the 1st and 2nd '|' make a pair, the 3rd and 4th '|' make a pair, and so forth.

Return the number of '*' in s, excluding the '*' between each pair of '|'.

Note that each '|' will belong to exactly one pair.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 2318 Number Of Distinct Roll Sequences

HARD

You are given an integer n. You roll a fair 6-sided dice n times. Determine the total number of distinct sequences of rolls possible such that the following conditions are satisfied: value of the ith roll is equal to the value of the jth roll, then abs(i - j) > 2.

Return the total number of distinct sequences possible. Since the answer may be very large, return it modulo 109 + 7.

Two sequences are considered distinct if at least one element is different.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 2319 Check If Matrix Is X Matrix

EASY

A square matrix is said to be an X-Matrix if both of the following conditions hold:

Given a 2D integer array grid of size n x n representing a square matrix, return true if grid is an X-Matrix. Otherwise, return false.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2320 Count Number Of Ways To Place Houses

MEDIUM

There is a street with n * 2 plots, where there are n plots on each side of the street. The plots on each side are numbered from 1 to n. On each plot, a house can be placed.

Return the number of ways houses can be placed such that no two houses are adjacent to each other on the same side of the street. Since the answer may be very large, return it modulo 109 + 7.

Note that if a house is placed on the ith plot on one side of the street, a house can also be placed on the ith plot on the other side of the street.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2321 Maximum Score Of Spliced Array

HARD

You are given two 0-indexed integer arrays nums1 and nums2, both of length n.

You can choose two integers left and right where 0 <= left <= right < n and swap the subarray nums1[left...right] with the subarray nums2[left...right].
- For example, if nums1 = [1,2,3,4,5] and nums2 = [11,12,13,14,15] and you choose left = 1 and right = 2, nums1 becomes [1,12,13,4,5] and nums2 becomes [11,2,3,14,15].

You may choose to apply the mentioned operation once or not do anything.

The score of the arrays is the maximum of sum(nums1) and sum(nums2), where sum(arr) is the sum of all the elements in the array arr.

Return the maximum possible score.

A subarray is a contiguous sequence of elements within an array. arr[left...right] denotes the subarray that contains the elements of nums between indices left and right (inclusive).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2325 Decode The Message

EASY

You are given the strings key and message, which represent a cipher key and a secret message, respectively. The steps to decode message are as follows:
substitution table.
letter in the alphabet), we have the partial substitution table of ('h' -> 'a', 'a' -> 'b', 'p' -> 'c', 'y' -> 'd', 'b' -> 'e', 'o' -> 'f').

Return the decoded message.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2326 Spiral Matrix Iv

MEDIUM

You are given two integers m and n, which represent the dimensions of a matrix.

You are also given the head of a linked list of integers.

Generate an m x n matrix that contains the integers in the linked list presented in spiral order (clockwise), starting from the top-left of the matrix. If there are remaining empty spaces, fill them with -1.

Return the generated matrix.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2327 Number Of People Aware Of A Secret

MEDIUM

On day 1, one person discovers a secret.

You are given an integer delay, which means that each person will share the secret with a new person every day, starting from delay days after discovering the secret. You are also given an integer forget, which means that each person will forget the secret forget days after discovering it. A person cannot share the secret on the same day they forgot it, or on any day afterwards.

Given an integer n, return the number of people who know the secret at the end of day n. Since the answer may be very large, return it modulo 109 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2328 Number Of Increasing Paths In A Grid

HARD

You are given an m x n integer matrix grid, where you can move from a cell to any adjacent cell in all 4 directions.

Return the number of strictly increasing paths in the grid such that you can start from any cell and end at any cell. Since the answer may be very large, return it modulo 109 + 7.

Two paths are considered different if they do not have exactly the same sequence of visited cells.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2332 The Latest Time To Catch A Bus

MEDIUM

You are given a 0-indexed integer array buses of length n, where buses[i] represents the departure time of the ith bus. You are also given a 0-indexed integer array passengers of length m, where passengers[j] represents the arrival time of the jth passenger. All bus departure times are unique. All passenger arrival times are unique.

You are given an integer capacity, which represents the maximum number of passengers that can get on each bus.

When a passenger arrives, they will wait in line for the next available bus. You can get on a bus that departs at x minutes if you arrive at y minutes where y <= x, and the bus is not full. Passengers with the earliest arrival times get on the bus first.

More formally when a bus arrives, either:
- If capacity or fewer passengers are waiting for a bus, they will all get on the bus, or

- The capacity passengers with the earliest arrival times will get on the bus.

Return the latest time you may arrive at the bus station to catch a bus. You cannot arrive at the same time as another passenger.

Note: The arrays buses and passengers are not necessarily sorted.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2333 Minimum Sum Of Squared Difference

**MEDIUM**

You are given two positive 0-indexed integer arrays nums1 and nums2, both of length n.

The sum of squared difference of arrays nums1 and nums2 is defined as the sum of (nums1[i] - nums2[i])2 for each 0 <= i < n.

You are also given two positive integers k1 and k2. You can modify any of the elements of nums1 by +1 or -1 at most k1 times. Similarly, you can modify any of the elements of nums2 by +1 or -1 at most k2 times.

Return the minimum sum of squared difference after modifying array nums1 at most k1 times and modifying array nums2 at most k2 times.

Note: You are allowed to modify the array elements to become negative integers.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2336 Smallest Number In Infinite Set

**MEDIUM**

You have a set which contains all positive integers [1, 2, 3, 4, 5, ...].

Implement the SmallestInfiniteSet class:
- SmallestInfiniteSet() Initializes the SmallestInfiniteSet object to contain all positive integers.
- int popSmallest() Removes and returns the smallest integer contained in the infinite set.
- void addBack(int num) Adds a positive integer num back into the infinite set, if it is not already in the infinite set.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2338 Count The Number Of Ideal Arrays

**HARD**

You are given two integers n and maxValue, which are used to describe an ideal array.

A 0-indexed integer array arr of length n is considered ideal if the following conditions hold:
- Every arr[i] is a value from 1 to maxValue, for 0 <= i < n.
- Every arr[i] is divisible by arr[i - 1], for 0 < i < n.

Return the number of distinct ideal arrays of length n. Since the answer may be very large, return it modulo 109 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2340 Minimum Adjacent Swaps To Make A Valid Array

**MEDIUM**

You are given a 0-indexed integer array nums.

Swaps of adjacent elements are able to be performed on nums.

A valid array meets the following conditions:
- The largest element (any of the largest elements if there are multiple) is at the rightmost position in the array.
- The smallest element (any of the smallest elements if there are multiple) is at the leftmost position in the array.

Return the minimum swaps required to make nums a valid array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2341 Maximum Number Of Pairs In Array

**EASY**

You are given a 0-indexed integer array nums. In one operation, you may do the following:
- Choose two integers in nums that are equal.
- Remove both integers from nums, forming a pair.

The operation is done on nums as many times as possible.

Return a 0-indexed integer array answer of size 2 where answer[0] is the number of pairs that are formed and answer[1] is the number of leftover integers in nums after doing the operation as many times as possible.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2343 Query Kth Smallest Trimmed Number

**MEDIUM**

You are given a 0-indexed array of strings nums, where each string is of equal length and consists of only digits.

You are also given a 0-indexed 2D integer array queries where queries[i] = [ki, trimi].
For each queries[i], you need to:
- Trim each number in nums to its rightmost trimi digits.
- Determine the index of the kith smallest trimmed number in nums. If two trimmed numbers are equal, the number with the lower index is considered to be smaller.
- Reset each number in nums to its original length.

Return an array answer of the same length as queries, where answer[i] is the answer to the ith query.

Note:
- To trim to the rightmost x digits means to keep removing the leftmost digit, until only x digits remain.
- Strings in nums may contain leading zeros.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2344 Minimum Deletions To Make Array Divisible

**HARD**

You are given two positive integer arrays nums and numsDivide. You can delete any number of elements from nums.

Return the minimum number of deletions such that the smallest element in nums divides all the elements of numsDivide. If this is not possible, return -1.

Note that an integer x divides y if y % x == 0.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2345 Finding The Number Of Visible Mountains

MEDIUM

You are given a 0-indexed 2D integer array peaks where peaks[i] = [xi, yi] states that mountain i has a peak at coordinates (xi, yi). A mountain can be described as a right-angled isosceles triangle, with its base along the x-axis and a right angle at its peak. More formally, the gradients of ascending and descending the mountain are 1 and -1 respectively.

A mountain is considered visible if its peak does not lie within another mountain (including the border of other mountains).

Return the number of visible mountains.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2347 Best Poker Hand

EASY

You are given an integer array ranks and a character array suits. You have 5 cards where the ith card has a rank of ranks[i] and a suit of suits[i].

The following are the types of poker hands you can make from best to worst:

Return a string representing the best type of poker hand you can make with the given cards.

Note that the return values are case-sensitive.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2349 Design A Number Container System

MEDIUM

Design a number container system that can do the following:
- Insert or Replace a number at the given index in the system.
- Return the smallest index for the given number in the system.

Implement the NumberContainers class:
- NumberContainers() Initializes the number container system.
- void change(int index, int number) Fills the container at index with the number. If there is already a number at that index, replace it.
- int find(int number) Returns the smallest index for the given number, or -1 if there is no index that is filled by number in the system.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2350 Shortest Impossible Sequence Of Rolls

HARD

You are given an integer array rolls of length n and an integer k. You roll a k sided dice numbered from 1 to k, n times, where the result of the ith roll is rolls[i].

Return the length of the shortest sequence of rolls so that there's no such subsequence in rolls.

A sequence of rolls of length len is the result of rolling a k sided dice len times.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2352 Equal Row And Column Pairs

MEDIUM

Given a 0-indexed n x n integer matrix grid, return the number of pairs (ri, cj) such that row ri and column cj are equal.

A row and column pair is considered equal if they contain the same elements in the same order (i.e., an equal array).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2353 Design A Food Rating System

MEDIUM

Design a food rating system that can do the following:
- Modify the rating of a food item listed in the system.
- Return the highest-rated food item for a type of cuisine in the system.

Implement the FoodRatings class:
- FoodRatings(String[] foods, String[] cuisines, int[] ratings) Initializes the system. The food items are described by foods, cuisines and ratings, all of which have a length of n.
- foods[i] is the name of the ith food,
- cuisines[i] is the type of cuisine of the ith food, and
- ratings[i] is the initial rating of the ith food.
- void changeRating(String food, int newRating) Changes the rating of the food item with the name food.
- String highestRated(String cuisine) Returns the name of the food item that has the highest rating for the given type of cuisine. If there is a tie, return the item with the lexicographically smaller name.

Note that a string x is lexicographically smaller than string y if x comes before y in dictionary order, that is, either x is a prefix of y, or if i is the first position such that x[i] != y[i], then x[i] comes before y[i] in alphabetic order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2354 Number Of Excellent Pairs

HARD

You are given a 0-indexed positive integer array nums and a positive integer k.

A pair of numbers (num1, num2) is called excellent if the following conditions are satisfied:
- Both the numbers num1 and num2 exist in the array nums.
- The sum of the number of set bits in num1 OR num2 and num1 AND num2 is greater than or equal to k, where OR is the bitwise OR operation and AND is the bitwise AND operation.

Return the number of distinct excellent pairs.

Two pairs (a, b) and (c, d) are considered distinct if either a != c or b != d. For example, (1, 2) and (2, 1) are distinct.

Note that a pair (num1, num2) such that num1 == num2 can also be excellent if you have at least one occurrence of num1 in the array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2355 Maximum Number Of Books You Can Take

HARD

You are given a 0-indexed integer array books of length n where books[i] denotes the number of books on the ith shelf of a bookshelf.

You are going to take books from a contiguous section of the bookshelf spanning from l to r where 0 <= l <= r < n. For each index i in the range l <= i < r, you must take strictly fewer books from shelf i than shelf i + 1.

Return the maximum number of books you can take from the bookshelf.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2357 Make Array Zero By Subtracting Equal Amounts

EASY

You are given a non-negative integer array nums. In one operation, you must:
- Choose a positive integer x such that x is less than or equal to the smallest non-zero element in nums.
- Subtract x from every positive element in nums.

Return the minimum number of operations to make every element in nums equal to 0.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2358 Maximum Number Of Groups Entering A Competition

MEDIUM

You are given a positive integer array grades which represents the grades of students in a university. You would like to enter all these students into a competition in ordered non-empty groups, such that the ordering meets the following conditions:
- The sum of the grades of students in the ith group is less than the sum of the grades of students in the (i + 1)th group, for all groups (except the last).
- The total number of students in the ith group is less than the total number of students in the (i + 1)th group, for all groups (except the last).

Return the maximum number of groups that can be formed.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2359 Find Closest Node To Given Two Nodes

MEDIUM

You are given a directed graph of n nodes numbered from 0 to n - 1, where each node has at most one outgoing edge.

The graph is represented with a given 0-indexed array edges of size n, indicating that there is a directed edge from node i to node edges[i]. If there is no outgoing edge from i, then edges[i] == -1.

You are also given two integers node1 and node2.

Return the index of the node that can be reached from both node1 and node2, such that the maximum between the distance from node1 to that node, and from node2 to that node is minimized. If there are multiple answers, return the node with the smallest index, and if no possible answer exists, return -1.

Note that edges may contain cycles.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2363 Merge Similar Items

EASY

You are given two 2D integer arrays, items1 and items2, representing two sets of items.
Each array items has the following properties:
- items[i] = [valuei, weighti] where valuei represents the value and weighti represents the weight of the ith item.
- The value of each item in items is unique.

Return a 2D integer array ret where ret[i] = [valuei, weighti], with weighti being the sum of weights of all items with value valuei.

Note: ret should be returned in ascending order by value.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2364 Count Number Of Bad Pairs

MEDIUM

You are given a 0-indexed integer array nums. A pair of indices (i, j) is a bad pair if i < j and j - i != nums[j] - nums[i].

Return the total number of bad pairs in nums.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2367 Number Of Arithmetic Triplets

EASY

You are given a 0-indexed, strictly increasing integer array nums and a positive integer diff.
A triplet (i, j, k) is an arithmetic triplet if the following conditions are met:
- i < j < k,
- nums[j] - nums[i] == diff, and
- nums[k] - nums[j] == diff.

Return the number of unique arithmetic triplets.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2368 Reachable Nodes With Restrictions

MEDIUM

There is an undirected tree with n nodes labeled from 0 to n - 1 and n - 1 edges.

You are given a 2D integer array edges of length n - 1 where edges[i] = [ai, bi] indicates that there is an edge between nodes ai and bi in the tree. You are also given an integer array restricted which represents restricted nodes.

Return the maximum number of nodes you can reach from node 0 without visiting a restricted node.

Note that node 0 will not be a restricted node.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2369 Check If There Is A Valid Partition For The Array

MEDIUM

You are given a 0-indexed integer array nums. You have to partition the array into one or more contiguous subarrays.

We call a partition of the array valid if each of the obtained subarrays satisfies one of

the following conditions:
between adjacent elements is 1. For example, the subarray [3,4,5] is good, but the subarray [1,3,5] is not.

Return true if the array has at least one valid partition. Otherwise, return false.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2371 Minimize Maximum Value In A Grid

HARD

You are given an m x n integer matrix grid containing distinct positive integers.

You have to replace each integer in the matrix with a positive integer satisfying the following conditions:
- The relative order of every two elements that are in the same row or column should stay the same after the replacements.
- The maximum number in the matrix after the replacements should be as small as possible.

The relative order stays the same if for all pairs of elements in the original matrix such that grid[r1][c1] > grid[r2][c2] where either r1 == r2 or c1 == c2, then it must be true that grid[r1][c1] > grid[r2][c2] after the replacements.

For example, if grid = [[2, 4, 5], [7, 3, 9]] then a good replacement could be either grid = [[1, 2, 3], [2, 1, 4]] or grid = [[1, 2, 3], [3, 1, 4]].

Return the resulting matrix. If there are multiple answers, return any of them.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2373 Largest Local Values In A Matrix

EASY

You are given an n x n integer matrix grid.

Generate an integer matrix maxLocal of size (n - 2) x (n - 2) such that:
- maxLocal[i][j] is equal to the largest value of the 3 x 3 matrix in grid centered around row i + 1 and column j + 1.

In other words, we want to find the largest value in every contiguous 3 x 3 matrix in grid.

Return the generated matrix.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2376 Count Special Integers

HARD

We call a positive integer special if all of its digits are distinct.

Given a positive integer n, return the number of special integers that belong to the interval [1, n].
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2382 Maximum Segment Sum After Removals

HARD

You are given two 0-indexed integer arrays nums and removeQueries, both of length n. For the ith query, the element in nums at the index removeQueries[i] is removed, splitting nums into different segments.

A segment is a contiguous sequence of positive integers in nums. A segment sum is the sum

of every element in a segment.

Return an integer array answer, of length n, where answer[i] is the maximum segment sum after applying the ith removal.

Note: The same index will not be removed more than once.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2384 Largest Palindromic Number

MEDIUM

You are given a string num consisting of digits only.

Return the largest palindromic integer (in the form of a string) that can be formed using digits taken from num. It should not contain leading zeroes.

Notes:
- You do not need to use all the digits of num, but you must use at least one digit.
- The digits can be reordered.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2386 Find The K Sum Of An Array

HARD

You are given an integer array nums and a positive integer k. You can choose any subsequence of the array and sum all of its elements together.

We define the K-Sum of the array as the kth largest subsequence sum that can be obtained (not necessarily distinct).

Return the K-Sum of the array.

A subsequence is an array that can be derived from another array by deleting some or no elements without changing the order of the remaining elements.

Note that the empty subsequence is considered to have a sum of 0.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2389 Longest Subsequence With Limited Sum

EASY

You are given an integer array nums of length n, and an integer array queries of length m.

Return an array answer of length m where answer[i] is the maximum size of a subsequence that you can take from nums such that the sum of its elements is less than or equal to queries[i].

A subsequence is an array that can be derived from another array by deleting some or no elements without changing the order of the remaining elements.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2392 Build A Matrix With Conditions

HARD

You are given a positive integer k. You are also given:
- a 2D integer array rowConditions of size n where rowConditions[i] = [abovei, belowi], and
- a 2D integer array colConditions of size m where colConditions[i] = [lefti, righti].

The two arrays contain integers from 1 to k.

You have to build a k x k matrix that contains each of the numbers from 1 to k exactly once.

The remaining cells should have the value 0.

The matrix should also satisfy the following conditions:
- The number abovei should appear in a row that is strictly above the row at which the number belowi appears for all i from 0 to n - 1.
- The number lefti should appear in a column that is strictly left of the column at which the number righti appears for all i from 0 to m - 1.

Return any matrix that satisfies the conditions. If no answer exists, return an empty matrix.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2396 Strictly Palindromic Number

**MEDIUM**

An integer n is strictly palindromic if, for every base b between 2 and n - 2 (inclusive), the string representation of the integer n in base b is palindromic.

Given an integer n, return true if n is strictly palindromic and false otherwise.

A string is palindromic if it reads the same forward and backward.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2398 Maximum Number Of Robots Within Budget

**HARD**

You have n robots. You are given two 0-indexed integer arrays, chargeTimes and runningCosts, both of length n. The ith robot costs chargeTimes[i] units to charge and costs runningCosts[i] units to run. You are also given an integer budget.

The total cost of running k chosen robots is equal to max(chargeTimes) + k * sum(runningCosts), where max(chargeTimes) is the largest charge cost among the k robots and sum(runningCosts) is the sum of running costs among the k robots.

Return the maximum number of consecutive robots you can run such that the total cost does not exceed budget.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2400 Number Of Ways To Reach A Position After Exactly K Steps

**MEDIUM**

You are given two positive integers startPos and endPos. Initially, you are standing at position startPos on an infinite number line. With one step, you can move either one position to the left, or one position to the right.

Given a positive integer k, return the number of different ways to reach the position endPos starting from startPos, such that you perform exactly k steps. Since the answer may be very large, return it modulo 109 + 7.

Two ways are considered different if the order of the steps made is not exactly the same.

Note that the number line includes negative integers.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2403 Minimum Time To Kill All Monsters

**HARD**

You are given an integer array power where power[i] is the power of the ith monster.

You start with 0 mana points, and each day you increase your mana points by gain where gain

initially is equal to 1.

Each day, after gaining gain mana, you can defeat a monster if your mana points are greater than or equal to the power of that monster. When you defeat a monster:
- your mana points will be reset to 0, and
- the value of gain increases by 1.

Return the minimum number of days needed to defeat all the monsters.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2404 Most Frequent Even Element

**EASY**

Given an integer array nums, return the most frequent even element.

If there is a tie, return the smallest one. If there is no such element, return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2408 Design Sql

**MEDIUM**

You are given two string arrays, names and columns, both of size n. The ith table is represented by the name names[i] and contains columns[i] number of columns.

You need to implement a class that supports the following operations:
- Insert a row in a specific table with an id assigned using an auto-increment method, where the id of the first inserted row is 1, and the id of each new row inserted into the same table is one greater than the id of the last inserted row, even if the last row was removed.
- Remove a row from a specific table. Removing a row does not affect the id of the next inserted row.
- Select a specific cell from any table and return its value.
- Export all rows from any table in csv format.

Implement the SQL class:
- SQL(String[] names, int[] columns)
- Creates the n tables.
- bool ins(String name, String[] row)
- Inserts row into the table name and returns true.
- If row.length does not match the expected number of columns, or name is not a valid table, returns false without any insertion.
- void rmv(String name, int rowId)
- Removes the row rowId from the table name.
- If name is not a valid table or there is no row with id rowId, no removal is performed.
- String sel(String name, int rowId, int columnId)
- Returns the value of the cell at the specified rowId and columnId in the table name.
- If name is not a valid table, or the cell (rowId, columnId) is invalid, returns "<null>".
- String[] exp(String name)
- Returns the rows present in the table name.
- If name is not a valid table, returns an empty array. Each row is represented as a string, with each cell value (including the row's id) separated by a ",".

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2409 Count Days Spent Together

**EASY**

Alice and Bob are traveling to Rome for separate business meetings.

You are given 4 strings arriveAlice, leaveAlice, arriveBob, and leaveBob. Alice will be in

the city from the dates arriveAlice to leaveAlice (inclusive), while Bob will be in the city from the dates arriveBob to leaveBob (inclusive). Each will be a 5-character string in the format "MM-DD", corresponding to the month and day of the date.

Return the total number of days that Alice and Bob are in Rome together.

You can assume that all dates occur in the same calendar year, which is not a leap year. Note that the number of days per month can be represented as: [31, 28, 31, 30, 31, 30, 31, 31, 30, 31, 30, 31].

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2421 Number Of Good Paths

**HARD**

There is a tree (i.e. a connected, undirected graph with no cycles) consisting of n nodes numbered from 0 to n - 1 and exactly n - 1 edges.

You are given a 0-indexed integer array vals of length n where vals[i] denotes the value of the ith node. You are also given a 2D integer array edges where edges[i] = [ai, bi] denotes that there exists an undirected edge connecting nodes ai and bi.

A good path is a simple path that satisfies the following conditions:
to the starting node (i.e. the starting node's value should be the maximum value along the path).

Return the number of distinct good paths.

Note that a path and its reverse are counted as the same path. For example, 0 -> 1 is considered to be the same as 1 -> 0. A single node is also considered as a valid path.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2422 Merge Operations To Turn Array Into A Palindrome

**MEDIUM**

You are given an array nums consisting of positive integers.

You can perform the following operation on the array any number of times:
- Choose any two adjacent elements and replace them with their sum.
- For example, if nums = [1,2,3,1], you can apply one operation to make it [1,5,1].

Return the minimum number of operations needed to turn the array into a palindrome.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2425 Bitwise Xor Of All Pairings

**MEDIUM**

You are given two 0-indexed arrays, nums1 and nums2, consisting of non-negative integers. There exists another array, nums3, which contains the bitwise XOR of all pairings of integers between nums1 and nums2 (every integer in nums1 is paired with every integer in nums2 exactly once).

Return the bitwise XOR of all integers in nums3.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2426 Number Of Pairs Satisfying Inequality

**HARD**

You are given two 0-indexed integer arrays nums1 and nums2, each of size n, and an

integer diff. Find the number of pairs (i, j) such that:
- 0 <= i < j <= n - 1 and
- nums1[i] - nums1[j] <= nums2[i] - nums2[j] + diff.

Return the number of pairs that satisfy the conditions.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2427 Number Of Common Factors

**EASY**

Given two positive integers a and b, return the number of common factors of a and b.

An integer x is a common factor of a and b if x divides both a and b.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2428 Maximum Sum Of An Hourglass

**MEDIUM**

You are given an m x n integer matrix grid.

We define an hourglass as a part of the matrix with the following form.

Return the maximum sum of the elements of an hourglass.

Note that an hourglass cannot be rotated and must be entirely contained within the matrix.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2431 Maximize Total Tastiness Of Purchased Fruits

**MEDIUM**

You are given two non-negative integer arrays price and tastiness, both arrays have the same length n. You are also given two non-negative integers maxAmount and maxCoupons.

For every integer i in range [0, n - 1]:
- price[i] describes the price of ith fruit.
- tastiness[i] describes the tastiness of ith fruit.

You want to purchase some fruits such that total tastiness is maximized and the total price does not exceed maxAmount.

Additionally, you can use a coupon to purchase fruit for half of its price (rounded down to the closest integer). You can use at most maxCoupons of such coupons.

Return the maximum total tastiness that can be purchased.

Note that:
- You can purchase each fruit at most once.
- You can use coupons on some fruit at most once.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2435 Paths In Matrix Whose Sum Is Divisible By K

**HARD**

You are given a 0-indexed m x n integer matrix grid and an integer k. You are currently at position (0, 0) and you want to reach position (m - 1, n - 1) moving only down or right.

Return the number of paths where the sum of the elements on the path is divisible by k. Since the answer may be very large, return it modulo 109 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2437 Number Of Valid Clock Times

EASY

You are given a string of length 5 called time, representing the current time on a digital clock in the format "hh:mm". The earliest possible time is "00:00" and the latest possible time is "23:59".

In the string time, the digits represented by the ? symbol are unknown, and must be replaced with a digit from 0 to 9.

Return an integer answer, the number of valid clock times that can be created by replacing every ? with a digit from 0 to 9.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2438 Range Product Queries Of Powers

MEDIUM

Given a positive integer n, there exists a 0-indexed array called powers, composed of the minimum number of powers of 2 that sum to n. The array is sorted in non-decreasing order, and there is only one way to form the array.

You are also given a 0-indexed 2D integer array queries, where queries[i] = [lefti, righti]. Each queries[i] represents a query where you have to find the product of all powers[j] with lefti <= j <= righti.

Return an array answers, equal in length to queries, where answers[i] is the answer to the ith query. Since the answer to the ith query may be too large, each answers[i] should be returned modulo 109 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2439 Minimize Maximum Of Array

MEDIUM

You are given a 0-indexed array nums comprising of n non-negative integers.

In one operation, you must:
- Choose an integer i such that 1 <= i < n and nums[i] > 0.
- Decrease nums[i] by 1.
- Increase nums[i - 1] by 1.

Return the minimum possible value of the maximum integer of nums after performing any number of operations.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2441 Largest Positive Integer That Exists With Its Negative

EASY

Given an integer array nums that does not contain any zeros, find the largest positive integer k such that -k also exists in the array.

Return the positive integer k. If there is no such integer, return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2442 Count Number Of Distinct Integers After Reverse Operations

MEDIUM

You are given an array nums consisting of positive integers.

You have to take each integer in the array, reverse its digits, and add it to the end of the array. You should apply this operation to the original integers in nums.

Return the number of distinct integers in the final array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2443 Sum Of Number And Its Reverse

MEDIUM

Given a non-negative integer num, return true if num can be expressed as the sum of any non-negative integer and its reverse, or false otherwise.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2445 Number Of Nodes With Value One

MEDIUM

There is an undirected connected tree with n nodes labeled from 1 to n and n - 1 edges. You are given the integer n. The parent node of a node with a label v is the node with the label floor (v / 2). The root of the tree is the node with the label 1.

- For example, if n = 7, then the node with the label 3 has the node with the label floor(3 / 2) = 1 as its parent, and the node with the label 7 has the node with the label floor(7 / 2) = 3 as its parent.

You are also given an integer array queries. Initially, every node has a value 0 on it. For each query queries[i], you should flip all values in the subtree of the node with the label queries[i].

Return the total number of nodes with the value 1 after processing all the queries.

Note that:
- Flipping the value of a node means that the node with the value 0 becomes 1 and vice versa.
- floor(x) is equivalent to rounding x down to the nearest integer.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2448 Minimum Cost To Make Array Equal

HARD

You are given two 0-indexed arrays nums and cost consisting each of n positive integers.

You can do the following operation any number of times:
- Increase or decrease any element of the array nums by 1.

The cost of doing one operation on the ith element is cost[i].

Return the minimum total cost such that all the elements of the array nums become equal.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2449 Minimum Number Of Operations To Make Arrays Similar

HARD

You are given two positive integer arrays nums and target, of the same length.

In one operation, you can choose any two distinct indices i and j where 0 <= i, j < nums.length and:
- set nums[i] = nums[i] + 2 and
- set nums[j] = nums[j] - 2.

Two arrays are considered to be similar if the frequency of each element is the same.

Return the minimum number of operations required to make nums similar to target. The test cases are generated such that nums can always be similar to target.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2453 Destroy Sequential Targets

**MEDIUM**

You are given a 0-indexed array nums consisting of positive integers, representing targets on a number line. You are also given an integer space.

You have a machine which can destroy targets. Seeding the machine with some nums[i] allows it to destroy all targets with values that can be represented as nums[i] + c * space, where c is any non-negative integer. You want to destroy the maximum number of targets in nums.

Return the minimum value of nums[i] you can seed the machine with to destroy the maximum number of targets.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2455 Average Value Of Even Numbers That Are Divisible By Three

**EASY**

Given an integer array nums of positive integers, return the average value of all even integers that are divisible by 3.

Note that the average of n elements is the sum of the n elements divided by n and rounded down to the nearest integer.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2456 Most Popular Video Creator

**MEDIUM**

You are given two string arrays creators and ids, and an integer array views, all of length n. The ith video on a platform was created by creators[i], has an id of ids[i], and has views[i] views.

The popularity of a creator is the sum of the number of views on all of the creator's videos. Find the creator with the highest popularity and the id of their most viewed video.
- If multiple creators have the highest popularity, find all of them.
- If multiple videos have the highest view count for a creator, find the lexicographically smallest id.

Note: It is possible for different videos to have the same id, meaning that ids do not uniquely identify a video. For example, two videos with the same ID are considered as distinct videos with their own viewcount.

Return a 2D array of strings answer where answer[i] = [creatorsi, idi] means that creatorsi has the highest popularity and idi is the id of their most popular video. The answer can be returned in any order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2460 Apply Operations To An Array

EASY

You are given a 0-indexed array nums of size n consisting of non-negative integers.

You need to apply n - 1 operations to this array where, in the ith operation (0-indexed), you will apply the following on the ith element of nums:
- If nums[i] == nums[i + 1], then multiply nums[i] by 2 and set nums[i + 1] to 0.
Otherwise, you skip this operation.

After performing all the operations, shift all the 0's to the end of the array.
- For example, the array [1,0,2,0,0,1] after shifting all its 0's to the end, is [1,2,1,0,0,0].

Return the resulting array.

Note that the operations are applied sequentially, not all at once.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2465 Number Of Distinct Averages

EASY

You are given a 0-indexed integer array nums of even length.

As long as nums is not empty, you must repetitively:
- Find the minimum number in nums and remove it.
- Find the maximum number in nums and remove it.
- Calculate the average of the two removed numbers.

The average of two numbers a and b is (a + b) / 2.
- For example, the average of 2 and 3 is (2 + 3) / 2 = 2.5.

Return the number of distinct averages calculated using the above process.

Note that when there is a tie for a minimum or maximum number, any can be removed.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2475 Number Of Unequal Triplets In Array

EASY

You are given a 0-indexed array of positive integers nums. Find the number of triplets (i, j, k) that meet the following conditions:
- 0 <= i < j < k < nums.length
- nums[i], nums[j], and nums[k] are pairwise distinct.
- In other words, nums[i] != nums[j], nums[i] != nums[k], and nums[j] != nums[k].

Return the number of triplets that meet the conditions.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2482 Difference Between Ones And Zeros In Row And Column

MEDIUM

You are given a 0-indexed m x n binary matrix grid.

A 0-indexed m x n difference matrix diff is created with the following procedure:
- Let the number of ones in the ith row be onesRowi.
- Let the number of ones in the jth column be onesColj.
- Let the number of zeros in the ith row be zerosRowi.
- Let the number of zeros in the jth column be zerosColj.
- diff[i][j] = onesRowi + onesColj - zerosRowi - zerosColj

Return the difference matrix diff.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2484 Count Palindromic Subsequences

**HARD**

Given a string of digits s, return the number of palindromic subsequences of s having length 5. Since the answer may be very large, return it modulo 109 + 7.

Note:
- A string is palindromic if it reads the same forward and backward.
- A subsequence is a string that can be derived from another string by deleting some or no characters without changing the order of the remaining characters.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2492 Minimum Score Of A Path Between Two Cities

**MEDIUM**

You are given a positive integer n representing n cities numbered from 1 to n. You are also given a 2D array roads where roads[i] = [ai, bi, distancei] indicates that there is a bidirectional road between cities ai and bi with a distance equal to distancei. The cities graph is not necessarily connected.

The score of a path between two cities is defined as the minimum distance of a road in this path.

Return the minimum possible score of a path between cities 1 and n.

Note:
- A path is a sequence of roads between two cities.
- It is allowed for a path to contain the same road multiple times, and you can visit cities 1 and n multiple times along the path.
- The test cases are generated such that there is at least one path between 1 and n.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2493 Divide Nodes Into The Maximum Number Of Groups

**HARD**

You are given a positive integer n representing the number of nodes in an undirected graph. The nodes are labeled from 1 to n.

You are also given a 2D integer array edges, where edges[i] = [ai, bi] indicates that there is a bidirectional edge between nodes ai and bi. Notice that the given graph may be disconnected.

Divide the nodes of the graph into m groups (1-indexed) such that:
- Each node in the graph belongs to exactly one group.
- For every pair of nodes in the graph that are connected by an edge [ai, bi], if ai belongs to the group with index x, and bi belongs to the group with index y, then |y - x| = 1.

Return the maximum number of groups (i.e., maximum m) into which you can divide the nodes. Return -1 if it is impossible to group the nodes with the given conditions.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2499 Minimum Total Cost To Make Arrays Unequal

**HARD**

You are given two 0-indexed integer arrays nums1 and nums2, of equal length n.

In one operation, you can swap the values of any two indices of nums1. The cost of this operation is the sum of the indices.

Find the minimum total cost of performing the given operation any number of times such that nums1[i] != nums2[i] for all 0 <= i <= n - 1 after performing all the operations.

Return the minimum total cost such that nums1 and nums2 satisfy the above condition. In case it is not possible, return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2501 Longest Square Streak In An Array

**MEDIUM**

You are given an integer array nums. A subsequence of nums is called a square streak if:
- The length of the subsequence is at least 2, and
- after sorting the subsequence, each element (except the first element) is the square of the previous number.

Return the length of the longest square streak in nums, or return -1 if there is no square streak.

A subsequence is an array that can be derived from another array by deleting some or no elements without changing the order of the remaining elements.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2503 Maximum Number Of Points From Grid Queries

**HARD**

You are given an m x n integer matrix grid and an array queries of size k.

Find an array answer of size k such that for each integer queries[i] you start in the top left cell of the matrix and repeat the following process:
- If queries[i] is strictly greater than the value of the current cell that you are in, then you get one point if it is your first time visiting this cell, and you can move to any adjacent cell in all 4 directions: up, down, left, and right.
- Otherwise, you do not get any points, and you end this process.

After the process, answer[i] is the maximum number of points you can get. Note that for each query you are allowed to visit the same cell multiple times.

Return the resulting array answer.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2505 Bitwise Or Of All Subsequence Sums

**MEDIUM**

Given an integer array nums, return the value of the bitwise OR of the sum of all possible subsequences in the array.

A subsequence is a sequence that can be derived from another sequence by removing zero or more elements without changing the order of the remaining elements.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2507 Smallest Value After Replacing With Sum Of Prime Factors

**MEDIUM**

You are given a positive integer n.

Continuously replace n with the sum of its prime factors.
- Note that if a prime factor divides n multiple times, it should be included in the sum as many times as it divides n.

Return the smallest value n will take on.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2510 Check If There Is A Path With Equal Number Of 0S And 1S

**MEDIUM**

You are given a 0-indexed m x n binary matrix grid. You can move from a cell (row, col) to any of the cells (row + 1, col) or (row, col + 1).

Return true if there is a path from (0, 0) to (m - 1, n - 1) that visits an equal number of 0's and 1's. Otherwise return false.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2512 Reward Top K Students

**MEDIUM**

You are given two string arrays positive_feedback and negative_feedback, containing the words denoting positive and negative feedback, respectively. Note that no word is both positive and negative.

Initially every student has 0 points. Each positive word in a feedback report increases the points of a student by 3, whereas each negative word decreases the points by 1.

You are given n feedback reports, represented by a 0-indexed string array report and a 0-indexed integer array student_id, where student_id[i] represents the ID of the student who has received the feedback report report[i]. The ID of each student is unique.

Given an integer k, return the top k students after ranking them in non-increasing order by their points. In case more than one student has the same points, the one with the lower ID ranks higher.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2519 Count The Number Of K Big Indices

**HARD**

You are given a 0-indexed integer array nums and a positive integer k.

We call an index i k-big if the following conditions are satisfied:
- There exist at least k different indices idx1 such that idx1 < i and nums[idx1] < nums[i].
- There exist at least k different indices idx2 such that idx2 > i and nums[idx2] < nums[i].

Return the number of k-big indices.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2521 Distinct Prime Factors Of Product Of Array

**MEDIUM**

Given an array of positive integers nums, return the number of distinct prime factors in the product of the elements of nums.

Note that:
- A number greater than 1 is called prime if it is divisible by only 1 and itself.
- An integer val1 is a factor of another integer val2 if val2 / val1 is an integer.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2523 Closest Prime Numbers In Range

MEDIUM

Given two positive integers left and right, find the two integers num1 and num2 such that:
- left <= num1 < num2 <= right
- Both num1 and num2 are prime numbers.
- num2 - num1 is the minimum amongst all other pairs satisfying the above conditions.

Return the positive integer array ans = [num1, num2]. If there are multiple pairs satisfying these conditions, return the one with the smallest num1 value. If no such numbers exist, return [-1, -1].

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2527 Find Xor Beauty Of Array

MEDIUM

You are given a 0-indexed integer array nums.

The effective value of three indices i, j, and k is defined as ((nums[i] | nums[j]) & nums[k]).

The xor-beauty of the array is the XORing of the effective values of all the possible triplets of indices (i, j, k) where 0 <= i, j, k < n.

Return the xor-beauty of nums.

Note that:
- val1 | val2 is bitwise OR of val1 and val2.
- val1 & val2 is bitwise AND of val1 and val2.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2529 Maximum Count Of Positive Integer And Negative Integer

EASY

Given an array nums sorted in non-decreasing order, return the maximum between the number of positive integers and the number of negative integers.

In other words, if the number of positive integers in nums is pos and the number of negative integers is neg, then return the maximum of pos and neg.

Note that 0 is neither positive nor negative.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2539 Count The Number Of Good Subsequences

MEDIUM

A subsequence of a string is good if it is not empty and the frequency of each one of its characters is the same.

Given a string s, return the number of good subsequences of s. Since the answer may be too large, return it modulo 109 + 7.

A subsequence is a string that can be derived from another string by deleting some or no characters without changing the order of the remaining characters.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2547 Minimum Cost To Split An Array

HARD

You are given an integer array nums and an integer k.

Split the array into some number of non-empty subarrays. The cost of a split is the sum of the importance value of each subarray in the split.

Let trimmed(subarray) be the version of the subarray where all numbers which appear only once are removed.
- For example, trimmed([3,1,2,4,3,4]) = [3,4,3,4].

The importance value of a subarray is k + trimmed(subarray).length.
- For example, if a subarray is [1,2,3,3,3,4,4], then trimmed([1,2,3,3,3,4,4]) = [3,3,3,4,4].
The importance value of this subarray will be k + 5.

Return the minimum possible cost of a split of nums.

A subarray is a contiguous non-empty sequence of elements within an array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2549 Count Distinct Numbers On Board

EASY

You are given a positive integer n, that is initially placed on a board. Every day, for 109 days, you perform the following procedure:
- For each number x present on the board, find all numbers 1 <= i <= n such that x % i == 1.
- Then, place those numbers on the board.

Return the number of distinct integers present on the board after 109 days have elapsed.

Note:
- Once a number is placed on the board, it will remain on it until the end.
- % stands for the modulo operation. For example, 14 % 3 is 2.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2554 Maximum Number Of Integers To Choose From A Range I

MEDIUM

You are given an integer array banned and two integers n and maxSum. You are choosing some number of integers following the below rules:
- The chosen integers have to be in the range [1, n].
- Each integer can be chosen at most once.
- The chosen integers should not be in the array banned.
- The sum of the chosen integers should not exceed maxSum.

Return the maximum number of integers you can choose following the mentioned rules.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2557 Maximum Number Of Integers To Choose From A Range Ii

MEDIUM

You are given an integer array banned and two integers n and maxSum. You are choosing some number of integers following the below rules:
- The chosen integers have to be in the range [1, n].
- Each integer can be chosen at most once.
- The chosen integers should not be in the array banned.
- The sum of the chosen integers should not exceed maxSum.

Return the maximum number of integers you can choose following the mentioned rules.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2561 Rearranging Fruits

**HARD**

You have two fruit baskets containing n fruits each. You are given two 0-indexed integer arrays basket1 and basket2 representing the cost of fruit in each basket. You want to make both baskets equal. To do so, you can use the following operation as many times as you want:
- Chose two indices i and j, and swap the ith fruit of basket1 with the jth fruit of basket2.
- The cost of the swap is min(basket1[i],basket2[j]).

Two baskets are considered equal if sorting them according to the fruit cost makes them exactly the same baskets.

Return the minimum cost to make both the baskets equal or -1 if impossible.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2562 Find The Array Concatenation Value

**EASY**

You are given a 0-indexed integer array nums.

The concatenation of two numbers is the number formed by concatenating their numerals.
- For example, the concatenation of 15, 49 is 1549.

The concatenation value of nums is initially equal to 0. Perform this operation until nums becomes empty:
- If nums has a size greater than one, add the value of the concatenation of the first and the last element to the concatenation value of nums, and remove those two elements from nums. For example, if the nums was [1, 2, 4, 5, 6], add 16 to the concatenation value.
- If only one element exists in nums, add its value to the concatenation value of nums, then remove it.

Return the concatenation value of nums.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2563 Count The Number Of Fair Pairs

**MEDIUM**

Given a 0-indexed integer array nums of size n and two integers lower and upper, return the number of fair pairs.

A pair (i, j) is fair if:
- 0 <= i < j < n, and
- lower <= nums[i] + nums[j] <= upper

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2567 Minimum Score By Changing Two Elements

**MEDIUM**

You are given an integer array nums.
- The low score of nums is the minimum absolute difference between any two integers.
- The high score of nums is the maximum absolute difference between any two integers.
- The score of nums is the sum of the high and low scores.

Return the minimum score after changing two elements of nums.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 2568 Minimum Impossible Or

MEDIUM

You are given a 0-indexed integer array nums.

We say that an integer x is expressible from nums if there exist some integers 0 <= index1 < index2 < ... < indexk < nums.length for which nums[index1] | nums[index2] | ... | nums[indexk] = x. In other words, an integer is expressible if it can be written as the bitwise OR of some subsequence of nums.

Return the minimum positive non-zero integer that is not expressible from nums.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 2570 Merge Two 2D Arrays By Summing Values

EASY

You are given two 2D integer arrays nums1 and nums2.
- nums1[i] = [idi, vali] indicate that the number with the id idi has a value equal to vali.
- nums2[i] = [idi, vali] indicate that the number with the id idi has a value equal to vali.

Each array contains unique ids and is sorted in ascending order by id.

Merge the two arrays into one array that is sorted in ascending order by id, respecting the following conditions:
- Only ids that appear in at least one of the two arrays should be included in the resulting array.
- Each id should be included only once and its value should be the sum of the values of this id in the two arrays. If the id does not exist in one of the two arrays, then assume its value in that array to be 0.

Return the resulting array. The returned array must be sorted in ascending order by id.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 2574 Left And Right Sum Differences

EASY

You are given a 0-indexed integer array nums of size n.

Define two arrays leftSum and rightSum where:
- leftSum[i] is the sum of elements to the left of the index i in the array nums. If there is no such element, leftSum[i] = 0.
- rightSum[i] is the sum of elements to the right of the index i in the array nums. If there is no such element, rightSum[i] = 0.

Return an integer array answer of size n where answer[i] = |leftSum[i] - rightSum[i]|.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 2576 Find The Maximum Number Of Marked Indices

MEDIUM

You are given a 0-indexed integer array nums.

Initially, all of the indices are unmarked. You are allowed to make this operation any number of times:
- Pick two different unmarked indices i and j such that 2 * nums[i] <= nums[j], then

mark i and j.

Return the maximum possible number of marked indices in nums using the above operation any number of times.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2578 Split With Minimum Sum

**EASY**

Given a positive integer num, split it into two non-negative integers num1 and num2 such that:
- The concatenation of num1 and num2 is a permutation of num.
- In other words, the sum of the number of occurrences of each digit in num1 and num2 is equal to the number of occurrences of that digit in num.
- num1 and num2 can contain leading zeros.

Return the minimum possible sum of num1 and num2.

Notes:
- It is guaranteed that num does not contain any leading zeros.
- The order of occurrence of the digits in num1 and num2 may differ from the order of occurrence of num.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2579 Count Total Number Of Colored Cells

**MEDIUM**

There exists an infinitely large two-dimensional grid of uncolored unit cells.
You are given a positive integer n, indicating that you must do the following routine for n minutes:
- At the first minute, color any arbitrary unit cell blue.
- Every minute thereafter, color blue every uncolored cell that touches a blue cell.

Below is a pictorial representation of the state of the grid after minutes 1, 2, and 3.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2581 Count Number Of Possible Root Nodes

**HARD**

Alice has an undirected tree with n nodes labeled from 0 to n - 1. The tree is represented as a 2D integer array edges of length n - 1 where edges[i] = [ai, bi] indicates that there is an edge between nodes ai and bi in the tree.

Alice wants Bob to find the root of the tree. She allows Bob to make several guesses about her tree. In one guess, he does the following:
- Chooses two distinct integers u and v such that there exists an edge [u, v] in the tree.
- He tells Alice that u is the parent of v in the tree.

Bob's guesses are represented by a 2D integer array guesses where guesses[j] = [uj, vj] indicates Bob guessed uj to be the parent of vj.

Alice being lazy, does not reply to each of Bob's guesses, but just says that at least k of his guesses are true.

Given the 2D integer arrays edges, guesses and the integer k, return the number of possible nodes that can be the root of Alice's tree. If there is no such tree, return 0.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2585 Number Of Ways To Earn Points

HARD

There is a test that has n types of questions. You are given an integer target and a 0-indexed 2D integer array types where types[i] = [counti, marksi] indicates that there are counti questions of the ith type, and each one of them is worth marksi points.

Return the number of ways you can earn exactly target points in the exam. Since the answer may be too large, return it modulo 109 + 7.

Note that questions of the same type are indistinguishable.
- For example, if there are 3 questions of the same type, then solving the 1st and 2nd questions is the same as solving the 1st and 3rd questions, or the 2nd and 3rd questions.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2590 Design A Todo List

MEDIUM

Design a Todo List Where users can add tasks, mark them as complete, or get a list of pending tasks. Users can also add tags to tasks and can filter the tasks by certain tags.

Implement the TodoList class:
- TodoList() Initializes the object.
- int addTask(int userId, String taskDescription, int dueDate, List<String> tags) Adds a task for the user with the ID userId with a due date equal to dueDate and a list of tags attached to the task. The return value is the ID of the task. This ID starts at 1 and is sequentially increasing. That is, the first task's id should be 1, the second task's id should be 2, and so on.
- List<String> getAllTasks(int userId) Returns a list of all the tasks not marked as complete for the user with ID userId, ordered by the due date. You should return an empty list if the user has no uncompleted tasks.
- List<String> getTasksForTag(int userId, String tag) Returns a list of all the tasks that are not marked as complete for the user with the ID userId and have tag as one of their tags, ordered by their due date. Return an empty list if no such task exists.
- void completeTask(int userId, int taskId) Marks the task with the ID taskId as completed only if the task exists and the user with the ID userId has this task, and it is uncompleted.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2592 Maximize Greatness Of An Array

MEDIUM

You are given a 0-indexed integer array nums. You are allowed to permute nums into a new array perm of your choosing.

We define the greatness of nums be the number of indices 0 <= i < nums.length for which perm[i] > nums[i].

Return the maximum possible greatness you can achieve after permuting nums.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2594 Minimum Time To Repair Cars

MEDIUM

You are given an integer array ranks representing the ranks of some mechanics. ranksi is the rank of the ith mechanic. A mechanic with a rank r can repair n cars in r * n2 minutes.

You are also given an integer cars representing the total number of cars waiting in the garage

to be repaired.

Return the minimum time taken to repair all the cars.

Note: All the mechanics can repair the cars simultaneously.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2595 Number Of Even And Odd Bits

**EASY**

You are given a positive integer n.

Let even denote the number of even indices in the binary representation of n with value 1.

Let odd denote the number of odd indices in the binary representation of n with value 1.

Note that bits are indexed from right to left in the binary representation of a number.

Return the array [even, odd].

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2598 Smallest Missing Non Negative Integer After Operations

**MEDIUM**

You are given a 0-indexed integer array nums and an integer value.

In one operation, you can add or subtract value from any element of nums.
- For example, if nums = [1,2,3] and value = 2, you can choose to subtract value from nums[0] to make nums = [-1,2,3].

The MEX (minimum excluded) of an array is the smallest missing non-negative integer in it.
- For example, the MEX of [-1,2,3] is 0 while the MEX of [1,0,3] is 2.

Return the maximum MEX of nums after applying the mentioned operation any number of times.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2600 K Items With The Maximum Sum

**EASY**

There is a bag that consists of items, each item has a number 1, 0, or -1 written on it.

You are given four non-negative integers numOnes, numZeros, numNegOnes, and k.

The bag initially contains:
- numOnes items with 1s written on them.
- numZeroes items with 0s written on them.
- numNegOnes items with -1s written on them.

We want to pick exactly k items among the available items. Return the maximum possible sum of numbers written on the items.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2610 Convert An Array Into A 2D Array With Conditions

**MEDIUM**

You are given an integer array nums. You need to create a 2D array from nums satisfying the following conditions:
- The 2D array should contain only the elements of the array nums.
- Each row in the 2D array contains distinct integers.
- The number of rows in the 2D array should be minimal.

Return the resulting array. If there are multiple answers, return any of them.

Note that the 2D array can have a different number of elements on each row.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2613 Beautiful Pairs

**HARD**

You are given two 0-indexed integer arrays nums1 and nums2 of the same length. A pair of indices (i,j) is called beautiful if|nums1[i] - nums1[j]| + |nums2[i] - nums2[j]| is the smallest amongst all possible indices pairs where i < j.

Return the beautiful pair. In the case that there are multiple beautiful pairs, return the lexicographically smallest pair.

Note that
- |x| denotes the absolute value of x.
- A pair of indices (i1, j1) is lexicographically smaller than (i2, j2) if i1 < i2 or i1 == i2 and j1 < j2.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2615 Sum Of Distances

**MEDIUM**

You are given a 0-indexed integer array nums. There exists an array arr of length nums.length, where arr[i] is the sum of |i - j| over all j such that nums[j] == nums[i] and j != i. If there is no such j, set arr[i] to be 0.

Return the array arr.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2616 Minimize The Maximum Difference Of Pairs

**MEDIUM**

You are given a 0-indexed integer array nums and an integer p. Find p pairs of indices of nums such that the maximum difference amongst all the pairs is minimized. Also, ensure no index appears more than once amongst the p pairs.

Note that for a pair of elements at the index i and j, the difference of this pair is |nums[i] - nums[j]|, where |x| represents the absolute value of x.

Return the minimum maximum difference among all p pairs. We define the maximum of an empty set to be zero.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2619 Array Prototype Last

**EASY**

Write code that enhances all arrays such that you can call the array.last() method on any array and it will return the last element. If there are no elements in the array, it should return -1.

You may assume the array is the output of JSON.parse.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2620 Counter

EASY

Given an integer n, return a counter function. This counter function initially returns n and then returns 1 more than the previous value every subsequent time it is called (n, n + 1, n + 2, etc).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2622 Cache With Time Limit

MEDIUM

Write a class that allows getting and setting key-value pairs, however a time until expiration is associated with each key.

The class has three public methods:
- set(key, value, duration): accepts an integer key, an integer value, and a duration in milliseconds. Once the duration has elapsed, the key should be inaccessible. The method should return true if the same un-expired key already exists and false otherwise. Both the value and duration should be overwritten if the key already exists.

- get(key): if an un-expired key exists, it should return the associated value. Otherwise it should return -1.

- count(): returns the count of un-expired keys.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2623 Memoize

MEDIUM

Given a function fn, return a memoized version of that function.

A memoized function is a function that will never be called twice with the same inputs. Instead it will return a cached value.

You can assume there are 3 possible input functions: sum, fib, and factorial.

- `sum` accepts two integers a and b and returns a + b. Assume that if a value has already been cached for the arguments (b, a) where a != b, it cannot be used for the arguments (a, b). For example, if the arguments are (3, 2) and (2, 3), two separate calls should be made.
- `fib` accepts a single integer n and returns 1 if n <= 1 or fib(n - 1) + fib(n - 2) otherwise.
- `factorial` accepts a single integer n and returns 1 if n <= 1 or factorial(n - 1) * n otherwise.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2625 Flatten Deeply Nested Array

MEDIUM

Given a multi-dimensional array arr and a depth n, return a flattened version of that array.

A multi-dimensional array is a recursive data structure that contains integers or other multi-dimensional arrays.

A flattened array is a version of that array with some or all of the sub-arrays removed and replaced with the actual elements in that sub-array. This flattening operation should only

be done if the current depth of nesting is less than n. The depth of the elements in the first array are considered to be 0.

Please solve it without the built-in Array.flat method.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2626 Array Reduce Transformation

**EASY**

Given an integer array nums, a reducer function fn, and an initial value init, return the final result obtained by executing the fn function on each element of the array, sequentially, passing in the return value from the calculation on the preceding element.

This result is achieved through the following operations: val = fn(init, nums[0]), val = fn(val, nums[1]), val = fn(val, nums[2]), ... until every element in the array has been processed. The ultimate value of val is then returned.

If the length of the array is 0, the function should return init.

Please solve it without using the built-in Array.reduce method.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2630 Memoize Ii

**HARD**

Given a function fn, return a memoized version of that function.

A memoized function is a function that will never be called twice with the same inputs. Instead it will return a cached value.

fn can be any function and there are no constraints on what type of values it accepts. Inputs are considered identical if they are === to each other.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2634 Filter Elements From Array

**EASY**

Given an integer array arr and a filtering function fn, return a filtered array filteredArr.

The fn function takes one or two arguments:
arr[i] - number from the arr
i - index of arr[i]

filteredArr should only contain the elements from the arr for which the expression fn(arr[i], i) evaluates to a truthy value. A truthy value is a value where Boolean(value) returns true.

Please solve it without the built-in Array.filter method.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2635 Apply Transform Over Each Element In Array

**EASY**

Given an integer array arr and a mapping function fn, return a new array with a transformation applied to each element.

The returned array should be created such that returnedArray[i] = fn(arr[i], i).

Please solve it without the built-in Array.map method.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2639 Find The Width Of Columns Of A Grid

**EASY**

You are given a 0-indexed m x n integer matrix grid. The width of a column is the maximum length of its integers.
- For example, if grid = [[-10], [3], [12]], the width of the only column is 3 since -10 is of length 3.

Return an integer array ans of size n where ans[i] is the width of the ith column.

The length of an integer x with len digits is equal to len if x is non-negative, and len + 1 otherwise.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2649 Nested Array Generator

**MEDIUM**

Given a multi-dimensional array of integers, return a generator object which yields integers in the same order as inorder traversal.

A multi-dimensional array is a recursive data structure that contains both integers and other multi-dimensional arrays.

inorder traversal iterates over each array from left to right, yielding any integers it encounters or applying inorder traversal to any arrays it encounters.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2652 Sum Multiples

**EASY**

Given a positive integer n, find the sum of all integers in the range [1, n] inclusive that are divisible by 3, 5, or 7.

Return an integer denoting the sum of all numbers in the given range satisfying the constraint.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2656 Maximum Sum With Exactly K Elements

**EASY**

You are given a 0-indexed integer array nums and an integer k. Your task is to perform the following operation exactly k times in order to maximize your score:

Return the maximum score you can achieve after performing the operation exactly k times.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2658 Maximum Number Of Fish In A Grid

**MEDIUM**

You are given a 0-indexed 2D matrix grid of size m x n, where (r, c) represents:
- A land cell if grid[r][c] = 0, or
- A water cell containing grid[r][c] fish, if grid[r][c] > 0.

A fisher can start at any water cell (r, c) and can do the following operations

any number of times:
- Catch all the fish at cell (r, c), or
- Move to any adjacent water cell.

Return the maximum number of fish the fisher can catch if he chooses his starting cell optimally, or 0 if no water cell exists.

An adjacent cell of the cell (r, c), is one of the cells (r, c + 1), (r, c - 1), (r + 1, c) or (r - 1, c) if it exists.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2661 First Completely Painted Row Or Column

**MEDIUM**

You are given a 0-indexed integer array arr, and an m x n integer matrix mat. arr and mat both contain all the integers in the range [1, m * n].

Go through each index i in arr starting from index 0 and paint the cell in mat containing the integer arr[i].

Return the smallest index i at which either a row or a column will be completely painted in mat.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2665 Counter Ii

**EASY**

Write a function createCounter. It should accept an initial integer init. It should return an object with three functions.

The three functions are:
- increment() increases the current value by 1 and then returns it.
- decrement() reduces the current value by 1 and then returns it.
- reset() sets the current value to init and then returns it.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2670 Find The Distinct Difference Array

**EASY**

You are given a 0-indexed array nums of length n.

The distinct difference array of nums is an array diff of length n such that diff[i] is equal to the number of distinct elements in the suffix nums[i + 1, ..., n - 1] subtracted from the number of distinct elements in the prefix nums[0, ..., i].

Return the distinct difference array of nums.

Note that nums[i, ..., j] denotes the subarray of nums starting at index i and ending at index j inclusive. Particularly, if i > j then nums[i, ..., j] denotes an empty subarray.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2672 Number Of Adjacent Elements With The Same Color

**MEDIUM**

You are given an integer n representing an array colors of length n where all elements are set to 0's meaning uncolored. You are also given a 2D integer array queries where queries[i] = [indexi, colori]. For the ith query:
- Set colors[indexi] to colori.

- Count the number of adjacent pairs in colors which have the same color (regardless of colori).

Return an array answer of the same length as queries where answer[i] is the answer to the ith query.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2675 Array Of Objects To Matrix

**HARD**

Write a function that converts an array of objects arr into a matrix m.

arr is an array of objects or arrays. Each item in the array can be deeply nested with child arrays and child objects. It can also contain numbers, strings, booleans, and null values.

The first row m should be the column names. If there is no nesting, the column names are the unique keys within the objects. If there is nesting, the column names are the respective paths in the object separated by ".".

Each of the remaining rows corresponds to an object in arr. Each value in the matrix corresponds to a value in an object. If a given object doesn't contain a value for a given column, the cell should contain an empty string "".

The columns in the matrix should be in lexographically ascending order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2677 Chunk Array

**EASY**

Given an array arr and a chunk size size, return a chunked array.

A chunked array contains the original elements in arr, but consists of subarrays each of length size. The length of the last subarray may be less than size if arr.length is not evenly divisible by size.

You may assume the array is the output of JSON.parse. In other words, it is valid JSON.

Please solve it without using lodash's _.chunk function.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2678 Number Of Senior Citizens

**EASY**

You are given a 0-indexed array of strings details. Each element of details provides information about a given passenger compressed into a string of length 15. The system is such that:
- The first ten characters consist of the phone number of passengers.
- The next character denotes the gender of the person.
- The following two characters are used to indicate the age of the person.
- The last two characters determine the seat allotted to that person.

Return the number of passengers who are strictly more than 60 years old.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2682 Find The Losers Of The Circular Game

**EASY**

There are n friends that are playing a game. The friends are sitting in a circle and are numbered from 1 to n in clockwise order. More formally, moving clockwise from the ith friend brings you to

the (i+1)th friend for 1 <= i < n, and moving clockwise from the nth friend brings you to the 1st friend.

The rules of the game are as follows:

1st friend receives the ball.
- After that, 1st friend passes it to the friend who is k steps away from them in the clockwise direction.
- After that, the friend who receives the ball should pass it to the friend who is 2 * k steps away from them in the clockwise direction.
- After that, the friend who receives the ball should pass it to the friend who is 3 * k steps away from them in the clockwise direction, and so on and so forth.

In other words, on the ith turn, the friend holding the ball should pass it to the friend who is i * k steps away from them in the clockwise direction.

The game is finished when some friend receives the ball for the second time.

The losers of the game are friends who did not receive the ball in the entire game.

Given the number of friends, n, and an integer k, return the array answer, which contains the losers of the game in the ascending order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2684 Maximum Number Of Moves In A Grid

**MEDIUM**

You are given a 0-indexed m x n matrix grid consisting of positive integers.

You can start at any cell in the first column of the matrix, and traverse the grid in the following way:
- From a cell (row, col), you can move to any of the cells: (row - 1, col + 1), (row, col + 1) and (row + 1, col + 1) such that the value of the cell you move to, should be strictly bigger than the value of the current cell.

Return the maximum number of moves that you can perform.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2685 Count The Number Of Complete Components

**MEDIUM**

You are given an integer n. There is an undirected graph with n vertices, numbered from 0 to n - 1. You are given a 2D integer array edges where edges[i] = [ai, bi] denotes that there exists an undirected edge connecting vertices ai and bi.

Return the number of complete connected components of the graph.

A connected component is a subgraph of a graph in which there exists a path between any two vertices, and no vertex of the subgraph shares an edge with a vertex outside of the subgraph.

A connected component is said to be complete if there exists an edge between every pair of its vertices.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2692 Make Object Immutable

**MEDIUM**

Write a function that takes an object obj and returns a new immutable version of this object.

An immutable object is an object that can't be altered and will throw an error if any attempt

is made to alter it.

There are three types of error messages that can be produced from this new object.
- Attempting to modify a key on the object will result in this error message:
- `Error Modifying: ${key}`.
- Attempting to modify an index on an array will result in this error message:
- `Error Modifying Index: ${index}`.
- Attempting to call a method that mutates an array will result in this error message:
- `Error Calling Method: ${methodName}`.
- You may assume the only methods that can mutate an array
are ['pop', 'push', 'shift', 'unshift', 'splice', 'sort', 'reverse'].

obj is a valid JSON object or array, meaning it is the output of JSON.parse().

Note that a string literal should be thrown, not an Error.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2694 Event Emitter

**MEDIUM**

Design an EventEmitter class. This interface is similar (but with some differences) to the one found in Node.js or the Event Target interface of the DOM. The EventEmitter should allow for subscribing to events and emitting them.

Your EventEmitter class should have the following two methods:
- subscribe - This method takes in two arguments: the name of an event as a string and a callback function. This callback function will later be called when the event is emitted.
An event should be able to have multiple listeners for the same event. When emitting an event with multiple callbacks, each should be called in the order in which they were subscribed.
An array of results should be returned. You can assume no callbacks passed to subscribe are referentially identical.
The subscribe method should also return an object with an unsubscribe method that enables the user to unsubscribe. When it is called, the callback should be removed from the list of subscriptions and undefined should be returned.
- emit - This method takes in two arguments: the name of an event as a string and an optional array of arguments that will be passed to the callback(s). If there are no callbacks subscribed to the given event, return an empty array. Otherwise, return an array of the results of all callback calls in the order they were subscribed.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2695 Array Wrapper

**EASY**

Create a class ArrayWrapper that accepts an array of integers in its constructor.

This class should have two features:
- When two instances of this class are added together with the + operator, the resulting value is the sum of all the elements in both arrays.
- When the String() function is called on the instance, it will return a comma separated string surrounded by brackets. For example, [1,2,3].

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2698 Find The Punishment Number Of An Integer

**MEDIUM**

Given a positive integer n, return the punishment number of n.

The punishment number of n is defined as the sum of the squares of all integers i such that:

- 1 <= i <= n
- The decimal representation of i * i can be partitioned into contiguous substrings such that the sum of the integer values of these substrings equals i.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2700 Differences Between Two Objects

**MEDIUM**

Write a function that accepts two deeply nested objects or arrays obj1 and obj2 and returns a new object representing their differences.

The function should compare the properties of the two objects and identify any changes. The returned object should only contains keys where the value is different from obj1 to obj2.

For each changed key, the value should be represented as an array [obj1 value, obj2 value]. Keys that exist in one object but not in the other should not be included in the returned object. The end result should be a deeply nested object where each leaf value is a difference array.

When comparing two arrays, the indices of the arrays are considered to be their keys.

You may assume that both objects are the output of JSON.parse.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2702 Minimum Operations To Make Numbers Non Positive

**HARD**

You are given a 0-indexed integer array nums and two integers x and y. In one operation, you must choose an index i such that 0 <= i < nums.length and perform the following:
- Decrement nums[i] by x.
- Decrement values by y at all indices except the ith one.

Return the minimum number of operations to make all the integers in nums less than or equal to zero.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2709 Greatest Common Divisor Traversal

**HARD**

You are given a 0-indexed integer array nums, and you are allowed to traverse between its indices. You can traverse between index i and index j, i != j, if and only if gcd(nums[i], nums[j]) > 1, where gcd is the greatest common divisor.

Your task is to determine if for every pair of indices i and j in nums, where i < j, there exists a sequence of traversals that can take us from i to j.

Return true if it is possible to traverse between all such pairs of indices, or false otherwise.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2711 Difference Of Number Of Distinct Values On Diagonals

**MEDIUM**

Given a 2D grid of size m x n, you should find the matrix answer of size m x n.

The cell answer[r][c] is calculated by looking at the diagonal values of the cell grid[r][c]:
- Let leftAbove[r][c] be the number of distinct values on the diagonal to the left and above the cell grid[r][c] not including the cell grid[r][c] itself.
- Let rightBelow[r][c] be the number of distinct values on the diagonal to the right and below

the cell grid[r][c], not including the cell grid[r][c] itself.
- Then answer[r][c] = |leftAbove[r][c] - rightBelow[r][c]|.

A matrix diagonal is a diagonal line of cells starting from some cell in either the topmost row or leftmost column and going in the bottom-right direction until the end of the matrix is reached.

- For example, in the below diagram the diagonal is highlighted using the cell with indices (2, 3) colored gray:
- Red-colored cells are left and above the cell.
- Blue-colored cells are right and below the cell.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2722 Join Two Arrays By Id

**MEDIUM**

Given two arrays arr1 and arr2, return a new array joinedArray. All the objects in each of the two inputs arrays will contain an id field that has an integer value.

joinedArray is an array formed by merging arr1 and arr2 based on their id key. The length of joinedArray should be the length of unique values of id. The returned array should be sorted in ascending order based on the id key.

If a given id exists in one array but not the other, the single object with that id should be included in the result array without modification.

If two objects share an id, their properties should be merged into a single object:
- If a key only exists in one object, that single key-value pair should be included in the object.
- If a key is included in both objects, the value in the object from arr2 should override the value from arr1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2729 Check If The Number Is Fascinating

**EASY**

You are given an integer n that consists of exactly 3 digits.

We call the number n fascinating if, after the following modification, the resulting number contains all the digits from 1 to 9 exactly once and does not contain any 0's:
- Concatenate n with the numbers 2 * n and 3 * n.

Return true if n is fascinating, or false otherwise.

Concatenating two numbers means joining them together. For example, the concatenation of 121 and 371 is 121371.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2737 Find The Closest Marked Node

**MEDIUM**

You are given a positive integer n which is the number of nodes of a 0-indexed directed weighted graph and a 0-indexed 2D array edges where edges[i] = [ui, vi, wi] indicates that there is an edge from node ui to node vi with weight wi.

You are also given a node s and a node array marked; your task is to find the minimum distance from s to any of the nodes in marked.

Return an integer denoting the minimum distance from s to any node in marked or -1 if there are no paths from s to any of the marked nodes.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 2748 Number Of Beautiful Pairs

EASY

You are given a 0-indexed integer array nums. A pair of indices i, j where 0 <= i < j < nums.length is called beautiful if the first digit of nums[i] and the last digit of nums[j] are coprime.

Return the total number of beautiful pairs in nums.

Two integers x and y are coprime if there is no integer greater than 1 that divides both of them. In other words, x and y are coprime if gcd(x, y) == 1, where gcd(x, y) is the greatest common divisor of x and y.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 2757 Generate Circular Array Values

MEDIUM

Given a circular array arr and an integer startIndex, return a generator object gen that yields values from arr.

The first time gen.next() is called on the generator, it should should yield arr[startIndex].

Each subsequent time gen.next() is called, an integer jump will be passed into the function (Ex: gen.next(-3)).
- If jump is positive, the index should increase by that value, however if the current index is the last index, it should instead jump to the first index.
- If jump is negative, the index should decrease by the magnitude of that value, however if the current index is the first index, it should instead jump to the last index.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 2766 Relocate Marbles

MEDIUM

You are given a 0-indexed integer array nums representing the initial positions of some marbles. You are also given two 0-indexed integer arrays moveFrom and moveTo of equal length.

Throughout moveFrom.length steps, you will change the positions of the marbles. On the ith step, you will move all marbles at position moveFrom[i] to position moveTo[i].

After completing all the steps, return the sorted list of occupied positions.

Notes:
- We call a position occupied if there is at least one marble in that position.
- There may be multiple marbles in a single position.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 2769 Find The Maximum Achievable Number

EASY

Given two integers, num and t. A number x is achievable if it can become equal to num after applying the following operation at most t times:
- Increase or decrease x by 1, and simultaneously increase or decrease num by 1.

Return the maximum possible value of x.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 2774 Array Upper Bound

EASY

Write code that enhances all arrays such that you can call the upperBound() method on any array and it will return the last index of a given target number. nums is a sorted ascending array of numbers that may contain duplicates. If the target number is not found in the array, return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2778 Sum Of Squares Of Special Elements

EASY

You are given a 1-indexed integer array nums of length n.

An element nums[i] of nums is called special if i divides n, i.e. n % i == 0.

Return the sum of the squares of all special elements of nums.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2779 Maximum Beauty Of An Array After Applying Operation

MEDIUM

You are given a 0-indexed array nums and a non-negative integer k.

In one operation, you can do the following:
- Choose an index i that hasn't been chosen before from the range [0, nums.length - 1].
- Replace nums[i] with any integer from the range [nums[i] - k, nums[i] + k].

The beauty of the array is the length of the longest subsequence consisting of equal elements.

Return the maximum possible beauty of the array nums after applying the operation any number of times.

Note that you can apply the operation to each index only once.

A subsequence of an array is a new array generated from the original array by deleting some elements (possibly none) without changing the order of the remaining elements.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2780 Minimum Index Of A Valid Split

MEDIUM

An element x of an integer array arr of length m is dominant if more than half the elements of arr have a value of x.

You are given a 0-indexed integer array nums of length n with one dominant element.

You can split nums at an index i into two arrays nums[0, ..., i] and nums[i + 1, ..., n - 1], but the split is only valid if:
- 0 <= i < n - 1
- nums[0, ..., i], and nums[i + 1, ..., n - 1] have the same dominant element.

Here, nums[i, ..., j] denotes the subarray of nums starting at index i and ending at index j, both ends being inclusive. Particularly, if j < i then nums[i, ..., j] denotes an empty subarray.

Return the minimum index of a valid split. If no valid split exists, return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2782 Number Of Unique Categories

MEDIUM

You are given an integer n and an object categoryHandler of class CategoryHandler.

There are n elements, numbered from 0 to n - 1. Each element has a category, and your task is to find the number of unique categories.

The class CategoryHandler contains the following function, which may help you:
- boolean haveSameCategory(integer a, integer b): Returns true if a and b are in the same category and false otherwise. Also, if either a or b is not a valid number (i.e. it's greater than or equal to nor less than 0), it returns false.

Return the number of unique categories.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2784 Check If Array Is Good

EASY

You are given an integer array nums. We consider an array good if it is a permutation of an array base[n].

base[n] = [1, 2, ..., n - 1, n, n] (in other words, it is an array of length n + 1 which contains 1 to n - 1 exactly once, plus two occurrences of n). For example, base[1] = [1, 1] and base[3] = [1, 2, 3, 3].

Return true if the given array is good, otherwise return false.

Note: A permutation of integers represents an arrangement of these numbers.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2787 Ways To Express An Integer As Sum Of Powers

MEDIUM

Given two positive integers n and x.

Return the number of ways n can be expressed as the sum of the xth power of unique positive integers, in other words, the number of sets of unique integers [n1, n2, ..., nk] where n = n1x + n2x + ... + nkx.

Since the result can be very large, return it modulo 109 + 7.

For example, if n = 160 and x = 3, one way to express n is n = 23 + 33 + 53.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2789 Largest Element In An Array After Merge Operations

MEDIUM

You are given a 0-indexed array nums consisting of positive integers.

You can do the following operation on the array any number of times:
- Choose an integer i such that 0 <= i < nums.length - 1 and nums[i] <= nums[i + 1]. Replace the element nums[i + 1] with nums[i] + nums[i + 1] and delete the element nums[i] from the array.

Return the value of the largest element that you can possibly obtain in the final array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2792 Count Nodes That Are Great Enough

HARD

You are given a root to a binary tree and an integer k. A node of this tree is called great enough if the followings hold:
- Its subtree has at least k nodes.
- Its value is greater than the value of at least k nodes in its subtree.

Return the number of nodes in this tree that are great enough.

The node u is in the subtree of the node v, if u == v or v is an ancestor of u.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2794 Create Object From Two Arrays

EASY

Given two arrays keysArr and valuesArr, return a new object obj. Each key-value pair in obj should come from keysArr[i] and valuesArr[i].

If a duplicate key exists at a previous index, that key-value should be excluded. In other words, only the first key should be added to the object.

If the key is not a string, it should be converted into a string by calling String() on it.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2798 Number Of Employees Who Met The Target

EASY

There are n employees in a company, numbered from 0 to n - 1. Each employee i has worked for hours[i] hours in the company.

The company requires each employee to work for at least target hours.

You are given a 0-indexed array of non-negative integers hours of length n and a non-negative integer target.

Return the integer denoting the number of employees who worked at least target hours.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2802 Find The K Th Lucky Number

MEDIUM

We know that 4 and 7 are lucky digits. Also, a number is called lucky if it contains only lucky digits.

You are given an integer k, return the kth lucky number represented as a string.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2804 Array Prototype Foreach

EASY

Write your version of method forEach that enhances all arrays such that you can call the array.forEach(callback, context) method on any array and it will execute callback on each element of the array. Method forEach should not return anything.

callback accepts the following arguments:
- currentValue - represents the current element being processed in the array. It is the

value of the element in the current iteration.
- index - represents the index of the current element being processed in the array.
- array - represents the array itself, allowing access to the entire array within the callback function.

The context is the object that should be passed as the function context parameter to the callback function, ensuring that the this keyword within the callback function refers to this context object.

Try to implement it without using the built-in array methods.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2806 Account Balance After Rounded Purchase

**EASY**

Initially, you have a bank account balance of 100 dollars.

You are given an integer purchaseAmount representing the amount you will spend on a purchase in dollars, in other words, its price.

When making the purchase, first the purchaseAmount is rounded to the nearest multiple of 10. Let us call this value roundedAmount. Then, roundedAmount dollars are removed from your bank account.

Return an integer denoting your final bank account balance after this purchase.

Notes:
- 0 is considered to be a multiple of 10 in this problem.
- When rounding, 5 is rounded upward (5 is rounded to 10, 15 is rounded to 20, 25 to 30, and so on).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2815 Max Pair Sum In An Array

**EASY**

You are given an integer array nums. You have to find the maximum sum of a pair of numbers from nums such that the largest digit in both numbers is equal.

For example, 2373 is made up of three distinct digits: 2, 3, and 7, where 7 is the largest among them.

Return the maximum sum or -1 if no such pair exists.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2824 Count Pairs Whose Sum Is Less Than Target

**EASY**

Given a 0-indexed integer array nums of length n and an integer target, return the number of pairs (i, j) where 0 <= i < j < n and nums[i] + nums[j] < target.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2829 Determine The Minimum Sum Of A K Avoiding Array

**MEDIUM**

You are given two integers, n and k.

An array of distinct positive integers is called a k-avoiding array if there does not exist any pair of distinct elements that sum to k.

Return the minimum possible sum of a k-avoiding array of length n.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2832 Maximal Range That Each Element Is Maximum In It

**MEDIUM**

You are given a 0-indexed array nums of distinct integers.

Let us define a 0-indexed array ans of the same length as nums in the following way:
- ans[i] is the maximum length of a subarray nums[l..r], such that the maximum element in that subarray is equal to nums[i].

Return the array ans.

Note that a subarray is a contiguous part of the array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2843 Count Symmetric Integers

**EASY**

You are given two positive integers low and high.

An integer x consisting of 2 * n digits is symmetric if the sum of the first n digits of x is equal to the sum of the last n digits of x. Numbers with an odd number of digits are never symmetric.

Return the number of symmetric integers in the range [low, high].

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2848 Points That Intersect With Cars

**EASY**

You are given a 0-indexed 2D integer array nums representing the coordinates of the cars parking on a number line. For any index i, nums[i] = [starti, endi] where starti is the starting point of the ith car and endi is the ending point of the ith car.

Return the number of integer points on the line that are covered with any part of a car.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2850 Minimum Moves To Spread Stones Over Grid

**MEDIUM**

You are given a 0-indexed 2D integer matrix grid of size 3 * 3, representing the number of stones in each cell. The grid contains exactly 9 stones, and there can be multiple stones in a single cell.

In one move, you can move a single stone from its current cell to any other cell if the two cells share a side.

Return the minimum number of moves required to place one stone in each cell.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2852 Sum Of Remoteness Of All Cells

**MEDIUM**

You are given a 0-indexed matrix grid of order n * n. Each cell in this matrix has a value grid[i][j], which is either a positive integer or -1 representing a blocked cell.

You can move from a non-blocked cell to any non-blocked cell that shares an edge.

For any cell (i, j), we represent its remoteness as R[i][j] which is defined as the following:
- If the cell (i, j) is a non-blocked cell, R[i][j] is the sum of the values grid[x][y] such that there is no path from the non-blocked cell (x, y) to the cell (i, j).
- For blocked cells, R[i][j] == 0.

Return the sum of R[i][j] over all cells.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2859 Sum Of Values At Indices With K Set Bits

**EASY**

You are given a 0-indexed integer array nums and an integer k.

Return an integer that denotes the sum of elements in nums whose corresponding indices have exactly k set bits in their binary representation.

The set bits in an integer are the 1's present when it is written in binary.

For example, the binary representation of 21 is 10101, which has 3 set bits.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2864 Maximum Odd Binary Number

**EASY**

You are given a binary string s that contains at least one '1'.

You have to rearrange the bits in such a way that the resulting binary number is the maximum odd binary number that can be created from this combination.

Return a string representing the maximum odd binary number that can be created from the given combination.

Note that the resulting string can have leading zeros.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2869 Minimum Operations To Collect Elements

**EASY**

You are given an array nums of positive integers and an integer k.

In one operation, you can remove the last element of the array and add it to your collection.

Return the minimum number of operations needed to collect elements 1, 2, ..., k.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2870 Minimum Number Of Operations To Make Array Empty

**MEDIUM**

You are given a 0-indexed array nums consisting of positive integers.

There are two types of operations that you can apply on the array any number of times:
- Choose two elements with equal values and delete them from the array.
- Choose three elements with equal values and delete them from the array.

Return the minimum number of operations required to make the array empty, or -1 if it is not possible.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2872 Maximum Number Of K Divisible Components

HARD

There is an undirected tree with n nodes labeled from 0 to n - 1. You are given the integer n and a 2D integer array edges of length n - 1, where edges[i] = [ai, bi] indicates that there is an edge between nodes ai and bi in the tree.

You are also given a 0-indexed integer array values of length n, where values[i] is the value associated with the ith node, and an integer k.

A valid split of the tree is obtained by removing any set of edges, possibly empty, from the tree such that the resulting components all have values that are divisible by k, where the value of a connected component is the sum of the values of its nodes.

Return the maximum number of components in any valid split.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2873 Maximum Value Of An Ordered Triplet I

EASY

You are given a 0-indexed integer array nums.

Return the maximum value over all triplets of indices (i, j, k) such that i < j < k. If all such triplets have a negative value, return 0.

The value of a triplet of indices (i, j, k) is equal to (nums[i] - nums[j]) * nums[k].

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2874 Maximum Value Of An Ordered Triplet Ii

MEDIUM

You are given a 0-indexed integer array nums.

Return the maximum value over all triplets of indices (i, j, k) such that i < j < k. If all such triplets have a negative value, return 0.

The value of a triplet of indices (i, j, k) is equal to (nums[i] - nums[j]) * nums[k].

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2892 Minimizing Array After Replacing Pairs With Their Product

MEDIUM

Given an integer array nums and an integer k, you can perform the following operation on the array any number of times:
- Select two adjacent elements of the array like x and y, such that x * y <= k, and replace both of them with a single element with value x * y (e.g. in one operation the array [1, 2, 2, 3] with k = 5 can become [1, 4, 3] or [2, 2, 3], but can't become [1, 2, 6]).

Return the minimum possible length of nums after any number of operations.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2894 Divisible And Non Divisible Sums Difference

EASY

You are given positive integers n and m.

Define two integers as follows:

- num1: The sum of all integers in the range [1, n] (both inclusive) that are not divisible by m.
- num2: The sum of all integers in the range [1, n] (both inclusive) that are divisible by m.

Return the integer num1 - num2.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2897 Apply Operations On Array To Maximize Sum Of Squares

HARD

You are given a 0-indexed integer array nums and a positive integer k.

You can do the following operation on the array any number of times:
- Choose any two distinct indices i and j and simultaneously update the values of nums[i] to (nums[i] AND nums[j]) and nums[j] to (nums[i] OR nums[j]). Here, OR denotes the bitwise OR operation, and AND denotes the bitwise AND operation.

You have to choose k elements from the final array and calculate the sum of their squares.

Return the maximum sum of squares you can achieve.

Since the answer can be very large, return it modulo 109 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2898 Maximum Linear Stock Score

MEDIUM

Given a 1-indexed integer array prices, where prices[i] is the price of a particular stock on the ith day, your task is to select some of the elements of prices such that your selection is linear.

A selection indexes, where indexes is a 1-indexed integer array of length k which is a subsequence of the array [1, 2, ..., n], is linear if:
- For every 1 < j <= k,
prices[indexes[j]] - prices[indexes[j - 1]] == indexes[j] - indexes[j - 1].

A subsequence is an array that can be derived from another array by deleting some or no elements without changing the order of the remaining elements.

The score of a selection indexes, is equal to the sum of the following array:
[prices[indexes[1]], prices[indexes[2]], ..., prices[indexes[k]].

Return the maximum score that a linear selection can have.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2903 Find Indices With Index And Value Difference I

EASY

You are given a 0-indexed integer array nums having length n, an integer indexDifference, and an integer valueDifference.

Your task is to find two indices i and j, both in the range [0, n - 1], that satisfy the following conditions:
- abs(i - j) >= indexDifference, and
- abs(nums[i] - nums[j]) >= valueDifference

Return an integer array answer, where answer = [i, j] if there are two such indices, and answer = [-1, -1] otherwise. If there are multiple choices for the two indices, return any of them.

Note: i and j may be equal.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2907 Maximum Profitable Triplets With Increasing Prices I

MEDIUM

Given the 0-indexed arrays prices and profits of length n. There are n items in an store where the ith item has a price of prices[i] and a profit of profits[i].

We have to pick three items with the following condition:
- prices[i] < prices[j] < prices[k] where i < j < k.

If we pick items with indices i, j and k satisfying the above condition, the profit would be profits[i] + profits[j] + profits[k].

Return the maximum profit we can get, and -1 if it's not possible to pick three items with the given condition.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2908 Minimum Sum Of Mountain Triplets I

EASY

You are given a 0-indexed array nums of integers.

A triplet of indices (i, j, k) is a mountain if:
- i < j < k
- nums[i] < nums[j] and nums[k] < nums[j]

Return the minimum possible sum of a mountain triplet of nums. If no such triplet exists, return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2909 Minimum Sum Of Mountain Triplets Ii

MEDIUM

You are given a 0-indexed array nums of integers.

A triplet of indices (i, j, k) is a mountain if:
- i < j < k
- nums[i] < nums[j] and nums[k] < nums[j]

Return the minimum possible sum of a mountain triplet of nums. If no such triplet exists, return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2912 Number Of Ways To Reach Destination In The Grid

HARD

You are given two integers n and m which represent the size of a 1-indexed grid. You are also given an integer k, a 1-indexed integer array source and a 1-indexed integer array dest, where source and dest are in the form [x, y] representing a cell on the given grid.

You can move through the grid in the following way:
- You can go from cell [x1, y1] to cell [x2, y2] if either x1 == x2 or y1 == y2.
- Note that you can't move to the cell you are already in e.g. x1 == x2 and y1 == y2.

Return the number of ways you can reach dest from source by moving through the grid exactly k times.

Since the answer may be very large, return it modulo 109 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2917 Find The K Or Of An Array

EASY

You are given an integer array nums, and an integer k. Let's introduce K-or operation by extending the standard bitwise OR. In K-or, a bit position in the result is set to 1 if at least k numbers in nums have a 1 in that position.

Return the K-or of nums.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2918 Minimum Equal Sum Of Two Arrays After Replacing Zeros

MEDIUM

You are given two arrays nums1 and nums2 consisting of positive integers.

You have to replace all the 0's in both arrays with strictly positive integers such that the sum of elements of both arrays becomes equal.

Return the minimum equal sum you can obtain, or -1 if it is impossible.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2921 Maximum Profitable Triplets With Increasing Prices Ii

HARD

Given the 0-indexed arrays prices and profits of length n. There are n items in an store where the ith item has a price of prices[i] and a profit of profits[i].

We have to pick three items with the following condition:
- prices[i] < prices[j] < prices[k] where i < j < k.

If we pick items with indices i, j and k satisfying the above condition, the profit would be profits[i] + profits[j] + profits[k].

Return the maximum profit we can get, and -1 if it's not possible to pick three items with the given condition.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2932 Maximum Strong Pair Xor I

EASY

You are given a 0-indexed integer array nums. A pair of integers x and y is called a strong pair if it satisfies the condition:
- |x - y| <= min(x, y)

You need to select two integers from nums such that they form a strong pair and their bitwise XOR is the maximum among all strong pairs in the array.

Return the maximum XOR value out of all possible strong pairs in the array nums.

Note that you can pick the same integer twice to form a pair.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2933 High Access Employees

MEDIUM

You are given a 2D 0-indexed array of strings, accessTimes, with size n. For each i where 0 <= i <= n - 1, accessTimes[i][0] represents the name of an employee, and accessTimes[i][1]

represents the access time of that employee. All entries in accessTimes are within the same day.

The access time is represented as four digits using a 24-hour time format, for example, "0800" or "2250".

An employee is said to be high-access if he has accessed the system three or more times within a one-hour period.

Times with exactly one hour of difference are not considered part of the same one-hour period. For example, "0815" and "0915" are not part of the same one-hour period.

Access times at the start and end of the day are not counted within the same one-hour period. For example, "0005" and "2350" are not part of the same one-hour period.

Return a list that contains the names of high-access employees with any order you want.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2934 Minimum Operations To Maximize Last Elements In Arrays

**MEDIUM**

You are given two 0-indexed integer arrays, nums1 and nums2, both having length n.

You are allowed to perform a series of operations (possibly none).

In an operation, you select an index i in the range [0, n - 1] and swap the values of nums1[i] and nums2[i].

Your task is to find the minimum number of operations required to satisfy the following conditions:
- nums1[n - 1] is equal to the maximum value among all elements of nums1, i.e., nums1[n - 1] = max(nums1[0], nums1[1], ..., nums1[n - 1]).
- nums2[n - 1] is equal to the maximum value among all elements of nums2, i.e., nums2[n - 1] = max(nums2[0], nums2[1], ..., nums2[n - 1]).

Return an integer denoting the minimum number of operations needed to meet both conditions, or -1 if it is impossible to satisfy both conditions.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2936 Number Of Equal Numbers Blocks

**MEDIUM**

You are given a 0-indexed array of integers, nums. The following property holds for nums:
- All occurrences of a value are adjacent. In other words, if there are two indices i < j such that nums[i] == nums[j], then for every index k that i < k < j, nums[k] == nums[i].

Since nums is a very large array, you are given an instance of the class BigArray which has the following functions:
- int at(long long index): Returns the value of nums[i].
- void size(): Returns nums.length.

Let's partition the array into maximal blocks such that each block contains equal values. Return the number of these blocks.

Note that if you want to test your solution using a custom test, behavior for tests with nums.length > 10 is undefined.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2946 Matrix Similarity After Cyclic Shifts

**EASY**

You are given an m x n integer matrix mat and an integer k. The matrix rows are 0-indexed.

The following proccess happens k times:
- Even-indexed rows (0, 2, 4, ...) are cyclically shifted to the left.
- Odd-indexed rows (1, 3, 5, ...) are cyclically shifted to the right.

Return true if the final modified matrix after k steps is identical to the original matrix, and false otherwise.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2952 Minimum Number Of Coins To Be Added

**MEDIUM**

You are given a 0-indexed integer array coins, representing the values of the coins available, and an integer target.

An integer x is obtainable if there exists a subsequence of coins that sums to x.

Return the minimum number of coins of any value that need to be added to the array so that every integer in the range [1, target] is obtainable.

A subsequence of an array is a new non-empty array that is formed from the original array by deleting some (possibly none) of the elements without disturbing the relative positions of the remaining elements.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2956 Find Common Elements Between Two Arrays

**EASY**

You are given two integer arrays nums1 and nums2 of sizes n and m, respectively.
Calculate the following values:
- answer1 : the number of indices i such that nums1[i] exists in nums2.
- answer2 : the number of indices i such that nums2[i] exists in nums1.

Return [answer1,answer2].

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2960 Count Tested Devices After Test Operations

**EASY**

You are given a 0-indexed integer array batteryPercentages having length n, denoting the battery percentages of n 0-indexed devices.

Your task is to test each device i in order from 0 to n - 1, by performing the following test operations:
- If batteryPercentages[i] is greater than 0:
- Increment the count of tested devices.
- Decrease the battery percentage of all devices with indices j in the range [i + 1, n - 1] by 1, ensuring their battery percentage never goes below 0, i.e, batteryPercentages[j] = max(0, batteryPercentages[j] - 1).
- Move to the next device.
- Otherwise, move to the next device without performing any test.

Return an integer denoting the number of devices that will be tested after performing the test operations in order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2963 Count The Number Of Good Partitions

HARD

You are given a 0-indexed array nums consisting of positive integers.

A partition of an array into one or more contiguous subarrays is called good if no two subarrays contain the same number.

Return the total number of good partitions of nums.

Since the answer may be large, return it modulo 109 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2964 Number Of Divisible Triplet Sums

MEDIUM

Given a 0-indexed integer array nums and an integer d, return the number of triplets (i, j, k) such that i < j < k and (nums[i] + nums[j] + nums[k]) % d == 0.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2965 Find Missing And Repeated Values

EASY

You are given a 0-indexed 2D integer matrix grid of size n * n with values in the range [1, n2]. Each integer appears exactly once except a which appears twice and b which is missing. The task is to find the repeating and missing numbers a and b.

Return a 0-indexed integer array ans of size 2 where ans[0] equals to a and ans[1] equals to b.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2966 Divide Array Into Arrays With Max Difference

MEDIUM

You are given an integer array nums of size n where n is a multiple of 3 and a positive integer k.

Divide the array nums into n / 3 arrays of size 3 satisfying the following condition:
- The difference between any two elements in one array is less than or equal to k.

Return a 2D array containing the arrays. If it is impossible to satisfy the conditions, return an empty array. And if there are multiple answers, return any of them.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2969 Minimum Number Of Coins For Fruits Ii

HARD

You are at a fruit market with different types of exotic fruits on display.

You are given a 1-indexed array prices, where prices[i] denotes the number of coins needed to purchase the ith fruit.

The fruit market has the following offer:
- If you purchase the ith fruit at prices[i] coins, you can get the next i fruits for free.

Note that even if you can take fruit j for free, you can still purchase it for prices[j] coins to receive a new offer.

Return the minimum number of coins needed to acquire all the fruits.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 2997 Minimum Number Of Operations To Make Array Xor Equal To K

MEDIUM

You are given a 0-indexed integer array nums and a positive integer k.

You can apply the following operation on the array any number of times:
- Choose any element of the array and flip a bit in its binary representation. Flipping a bit means changing a 0 to 1 or vice versa.

Return the minimum number of operations required to make the bitwise XOR of all elements of the final array equal to k.

Note that you can flip leading zero bits in the binary representation of elements. For example, for the number (101)2 you can flip the fourth bit and obtain (1101)2.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 2998 Minimum Number Of Operations To Make X And Y Equal

MEDIUM

You are given two positive integers x and y.

In one operation, you can do one of the four following operations:

Return the minimum number of operations required to make x and y equal.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 2999 Count The Number Of Powerful Integers

HARD

You are given three integers start, finish, and limit. You are also given a 0-indexed string s representing a positive integer.

A positive integer x is called powerful if it ends with s (in other words, s is a suffix of x) and each digit in x is at most limit.

Return the total number of powerful integers in the range [start..finish].

A string x is a suffix of a string y if and only if x is a substring of y that starts from some index (including 0) in y and extends to the index y.length - 1. For example, 25 is a suffix of 5125 whereas 512 is not.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 3002 Maximum Size Of A Set After Removals

MEDIUM

You are given two 0-indexed integer arrays nums1 and nums2 of even length n.

You must remove n / 2 elements from nums1 and n / 2 elements from nums2. After the removals, you insert the remaining elements of nums1 and nums2 into a set s.

Return the maximum possible size of the set s.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 3005 Count Elements With Maximum Frequency

EASY

You are given an array nums consisting of positive integers.

Return the total frequencies of elements in nums such that those elements all have the maximum frequency.

The frequency of an element is the number of occurrences of that element in the array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3009 Maximum Number Of Intersections On The Chart

HARD

There is a line chart consisting of n points connected by line segments. You are given a 1-indexed integer array y. The kth point has coordinates (k, y[k]). There are no horizontal lines; that is, no two consecutive points have the same y-coordinate.

We can draw an infinitely long horizontal line. Return the maximum number of points of intersection of the line with the chart.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3015 Count The Number Of Houses At A Certain Distance I

MEDIUM

You are given three positive integers n, x, and y.

In a city, there exist houses numbered 1 to n connected by n streets. There is a street connecting the house numbered i with the house numbered i + 1 for all 1 <= i <= n - 1. An additional street connects the house numbered x with the house numbered y.

For each k, such that 1 <= k <= n, you need to find the number of pairs of houses (house1, house2) such that the minimum number of streets that need to be traveled to reach house2 from house1 is k.

Return a 1-indexed array result of length n where result[k] represents the total number of pairs of houses such that the minimum streets required to reach one house from the other is k.

Note that x and y can be equal.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3018 Maximum Number Of Removal Queries That Can Be Processed I

HARD

You are given a 0-indexed array nums and a 0-indexed array queries.

You can do the following operation at the beginning at most once:
- Replace nums with a subsequence of nums.

We start processing queries in the given order; for each query, we do the following:
- If the first and the last element of nums is less than queries[i], the processing of queries ends.
- Otherwise, we choose either the first or the last element of nums if it is greater than or equal to queries[i], and we remove the chosen element from nums.

Return the maximum number of queries that can be processed by doing the operation optimally.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3019 Number Of Changing Keys

EASY

You are given a 0-indexed string s typed by a user. Changing a key is defined as using a key different from the last used key. For example, s = "ab" has a change of a key while s = "bBBb" does not have any.

Return the number of times the user had to change the key.

Note: Modifiers like shift or caps lock won't be counted in changing the key that is if a user typed the letter 'a' and then the letter 'A' then it will not be considered as a changing of key.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3025 Find The Number Of Ways To Place People I

MEDIUM

You are given a 2D array points of size n x 2 representing integer coordinates of some points on a 2D plane, where points[i] = [xi, yi].

Count the number of pairs of points (A, B), where
- A is on the upper left side of B, and
- there are no other points in the rectangle (or line) they make (including the border).

Return the count.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3027 Find The Number Of Ways To Place People Ii

HARD

You are given a 2D array points of size n x 2 representing integer coordinates of some points on a 2D-plane, where points[i] = [xi, yi].

We define the right direction as positive x-axis (increasing x-coordinate) and the left direction as negative x-axis (decreasing x-coordinate). Similarly, we define the up direction as positive y-axis (increasing y-coordinate) and the down direction as negative y-axis (decreasing y-coordinate)

You have to place n people, including Alice and Bob, at these points such that there is exactly one person at every point. Alice wants to be alone with Bob, so Alice will build a rectangular fence with Alice's position as the upper left corner and Bob's position as the lower right corner of the fence (Note that the fence might not enclose any area, i.e. it can be a line). If any person other than Alice and Bob is either inside the fence or on the fence, Alice will be sad.

Return the number of pairs of points where you can place Alice and Bob, such that Alice does not become sad on building the fence.

Note that Alice can only build a fence with Alice's position as the upper left corner, and Bob's position as the lower right corner. For example, Alice cannot build either of the fences in the picture below with four corners (1, 1), (1, 3), (3, 1), and (3, 3), because:
- With Alice at (3, 3) and Bob at (1, 1), Alice's position is not the upper left corner and Bob's position is not the lower right corner of the fence.
- With Alice at (1, 3) and Bob at (1, 1), Bob's position is not the lower right corner of the fence.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3030 Find The Grid Of Region Average

MEDIUM

You are given m x n grid image which represents a grayscale image, where image[i][j] represents a pixel with intensity in the range [0..255]. You are also given a non-negative integer threshold.

Two pixels are adjacent if they share an edge.

A region is a 3 x 3 subgrid where the absolute difference in intensity between any two adjacent pixels is less than or equal to threshold.

All pixels in a region belong to that region, note that a pixel can belong to multiple regions.

You need to calculate a m x n grid result, where result[i][j] is the average intensity of the regions to which image[i][j] belongs, rounded down to the nearest integer. If image[i][j] belongs to multiple regions, result[i][j] is the average of the rounded-down average intensities of these regions, rounded down to the nearest integer. If image[i][j] does not belong to any region, result[i][j] is equal to image[i][j].

Return the grid result.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3033 Modify The Matrix

EASY

Given a 0-indexed m x n integer matrix matrix, create a new 0-indexed matrix called answer. Make answer equal to matrix, then replace each element with the value -1 with the maximum element in its respective column.

Return the matrix answer.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3038 Maximum Number Of Operations With The Same Score I

EASY

You are given an array of integers nums. Consider the following operation:
- Delete the first two elements nums and define the score of the operation as the sum of these two elements.

You can perform this operation until nums contains fewer than two elements. Additionally, the same score must be achieved in all operations.

Return the maximum number of operations you can perform.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3044 Most Frequent Prime

MEDIUM

You are given a m x n 0-indexed 2D matrix mat. From every cell, you can create numbers in the following way:
- There could be at most 8 paths from the cells namely: east, south-east, south, south-west, west, north-west, north, and north-east.
- Select a path from them and append digits in this path to the number being formed by traveling in this direction.
- Note that numbers are generated at every step, for example, if the digits along the path are 1, 9, 1, then there will be three numbers generated along the way: 1, 19, 191.

Return the most frequent prime number greater than 10 out of all the numbers created by traversing the matrix or -1 if no such prime number exists. If there are multiple prime numbers with the highest frequency, then return the largest among them.

Note: It is invalid to change the direction during the move.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3046 Split The Array

**EASY**

You are given an integer array nums of even length. You have to split the array into two parts nums1 and nums2 such that:
- nums1.length == nums2.length == nums.length / 2.
- nums1 should contain distinct elements.
- nums2 should also contain distinct elements.

Return true if it is possible to split the array, and false otherwise.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3064 Guess The Number Using Bitwise Questions I

**MEDIUM**

There is a number n that you have to find.

There is also a pre-defined API int commonSetBits(int num), which returns the number of bits where both n and num are 1 in that position of their binary representation. In other words, it returns the number of set bits in n & num, where & is the bitwise AND operator.

Return the number n.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3068 Find The Maximum Sum Of Node Values

**HARD**

There exists an undirected tree with n nodes numbered 0 to n - 1. You are given a 0-indexed 2D integer array edges of length n - 1, where edges[i] = [ui, vi] indicates that there is an edge between nodes ui and vi in the tree. You are also given a positive integer k, and a 0-indexed array of non-negative integers nums of length n, where nums[i] represents the value of the node numbered i.

Alice wants the sum of values of tree nodes to be maximum, for which Alice can perform the following operation any number of times (including zero) on the tree:
- Choose any edge [u, v] connecting the nodes u and v, and update their values as follows:
- nums[u] = nums[u] XOR k
- nums[v] = nums[v] XOR k

Return the maximum possible sum of the values Alice can achieve by performing the operation any number of times.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3069 Distribute Elements Into Two Arrays I

**EASY**

You are given a 1-indexed array of distinct integers nums of length n.

You need to distribute all the elements of nums between two arrays arr1 and arr2 using n operations. In the first operation, append nums[1] to arr1. In the second operation,

append nums[2] to arr2. Afterwards, in the ith operation:
- If the last element of arr1 is greater than the last element of arr2, append nums[i] to arr1. Otherwise, append nums[i] to arr2.

The array result is formed by concatenating the arrays arr1 and arr2. For example, if arr1 == [1,2,3] and arr2 == [4,5,6], then result = [1,2,3,4,5,6].

Return the array result.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3070 Count Submatrices With Top Left Element And Sum Less Than K

**MEDIUM**

You are given a 0-indexed integer matrix grid and an integer k.

Return the number of submatrices that contain the top-left element of the grid, and have a sum less than or equal to k.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3073 Maximum Increasing Triplet Value

**MEDIUM**

Given an array nums, return the maximum value of a triplet (i, j, k) such that i < j < k and nums[i] < nums[j] < nums[k].

The value of a triplet (i, j, k) is nums[i] - nums[j] + nums[k].

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3078 Match Alphanumerical Pattern In Matrix I

**MEDIUM**

You are given a 2D integer matrix board and a 2D character matrix pattern. Where 0 <= board[r][c] <= 9 and each element of pattern is either a digit or a lowercase English letter.

Your task is to find a submatrix of board that matches pattern.

An integer matrix part matches pattern if we can replace cells containing letters in pattern with some digits (each distinct letter with a unique digit) in such a way that the resulting matrix becomes identical to the integer matrix part. In other words,
- The matrices have identical dimensions.
- If pattern[r][c] is a digit, then part[r][c] must be the same digit.
- If pattern[r][c] is a letter x:
- For every pattern[i][j] == x, part[i][j] must be the same as part[r][c].
- For every pattern[i][j] != x, part[i][j] must be different than part[r][c].

Return an array of length 2 containing the row number and column number of the upper-left corner of a submatrix of board which matches pattern. If there is more than one such submatrix, return the coordinates of the submatrix with the lowest row index, and in case there is still a tie, return the coordinates of the submatrix with the lowest column index. If there are no suitable answers, return [-1, -1].

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3079 Find The Sum Of Encrypted Integers

**EASY**

You are given an integer array nums containing positive integers. We define a function

encrypt such that encrypt(x) replaces every digit in x with the largest digit in x.
For example, encrypt(523) = 555 and encrypt(213) = 333.

Return the sum of encrypted elements.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3091 Apply Operations To Make Sum Of Array Greater Than Or Equal To K

**MEDIUM**

You are given a positive integer k. Initially, you have an array nums = [1].

You can perform any of the following operations on the array any number of times (possibly zero):
- Choose any element in the array and increase its value by 1.
- Duplicate any element in the array and add it to the end of the array.

Return the minimum number of operations required to make the sum of elements of the final array greater than or equal to k.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3094 Guess The Number Using Bitwise Questions Ii

**MEDIUM**

There is a number n between 0 and 230 - 1 (both inclusive) that you have to find.

There is a pre-defined API int commonBits(int num) that helps you with your mission.
But here is the challenge, every time you call this function, n changes in some way.
But keep in mind, that you have to find the initial value of n.

commonBits(int num) acts as follows:
- Calculate count which is the number of bits where both n and num have the same value in that position of their binary representation.
- n = n XOR num
- Return count.

Return the number n.

Note: In this world, all numbers are between 0 and 230 - 1 (both inclusive), thus for counting common bits, we see only the first 30 bits of those numbers.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3099 Harshad Number

**EASY**

An integer divisible by the sum of its digits is said to be a Harshad number. You are given an integer x. Return the sum of the digits of x if x is a Harshad number, otherwise, return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3119 Maximum Number Of Potholes That Can Be Fixed

**MEDIUM**

You are given a string road, consisting only of characters "x" and ".", where each "x" denotes a pothole and each "." denotes a smooth road, and an integer budget.

In one repair operation, you can repair n consecutive potholes for a price of n + 1.

Return the maximum number of potholes that can be fixed such that the sum of the prices of all of the fixes doesn't go over the given budget.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3125 Maximum Number That Makes Result Of Bitwise And Zero

MEDIUM

Given an integer n, return the maximum integer x such that x <= n, and the bitwise AND of all the numbers in the range [x, n] is 0.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3151 Special Array I

EASY

An array is considered special if every pair of its adjacent elements contains two numbers with different parity.

You are given an array of integers nums. Return true if nums is a special array, otherwise, return false.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3155 Maximum Number Of Upgradable Servers

MEDIUM

You have n data centers and need to upgrade their servers.

You are given four arrays count, upgrade, sell, and money of length n, which show:
- The number of servers
- The cost of upgrading a single server
- The money you get by selling a server
- The money you initially have
- for each data center respectively.

Return an array answer, where for each data center, the corresponding element in answer represents the maximum number of servers that can be upgraded.

Note that the money from one data center cannot be used for another data center.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3160 Find The Number Of Distinct Colors Among The Balls

MEDIUM

You are given an integer limit and a 2D array queries of size n x 2.

There are limit + 1 balls with distinct labels in the range [0, limit]. Initially, all balls are uncolored. For every query in queries that is of the form [x, y], you mark ball x with the color y. After each query, you need to find the number of distinct colors among the balls.

Return an array result of length n, where result[i] denotes the number of distinct colors after ith query.

Note that when answering a query, lack of a color will not be considered as a color.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3169 Count Days Without Meetings

MEDIUM

You are given a positive integer days representing the total number of days an employee is available for work (starting from day 1). You are also given a 2D array meetings of size n

where, meetings[i] = [start_i, end_i] represents the starting and ending days of meeting i (inclusive).

Return the count of days when the employee is available for work but no meetings are scheduled.

Note: The meetings may overlap.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3173 Bitwise Or Of Adjacent Elements

**EASY**

Given an array nums of length n, return an array answer of length n - 1 such that answer[i] = nums[i] | nums[i + 1] where | is the bitwise OR operation.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3183 The Number Of Ways To Make The Sum

**MEDIUM**

You have an infinite number of coins with values 1, 2, and 6, and only 2 coins with value 4.

Given an integer n, return the number of ways to make the sum of n with the coins you have.

Since the answer may be very large, return it modulo 109 + 7.

Note that the order of the coins doesn't matter and [2, 2, 3] is the same as [2, 3, 2].

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3186 Maximum Total Damage With Spell Casting

**MEDIUM**

A magician has various spells.

You are given an array power, where each element represents the damage of a spell. Multiple spells can have the same damage value.

It is a known fact that if a magician decides to cast a spell with a damage of power[i], they cannot cast any spell with a damage of power[i] - 2, power[i] - 1, power[i] + 1, or power[i] + 2.

Each spell can be cast only once.

Return the maximum possible total damage that a magician can cast.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3190 Find Minimum Operations To Make All Elements Divisible By Three

**EASY**

You are given an integer array nums. In one operation, you can add or subtract 1 from any element of nums.

Return the minimum number of operations to make all elements of nums divisible by 3.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3191 Minimum Operations To Make Binary Array Elements Equal To One I

**MEDIUM**

You are given a binary array nums.

You can do the following operation on the array any number of times (possibly zero):
- Choose any 3 consecutive elements from the array and flip all of them.

Flipping an element means changing its value from 0 to 1, and from 1 to 0.

Return the minimum number of operations required to make all elements in nums equal to 1.
If it is impossible, return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3199 Count Triplets With Even Xor Set Bits I

**EASY**

Given three integer arrays a, b, and c, return the number of triplets (a[i], b[j], c[k]),
such that the bitwise XOR of the elements of each triplet has an even number of set bits.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3205 Maximum Array Hopping Score I

**MEDIUM**

Given an array nums, you have to get the maximum score starting from index 0 and
hopping until you reach the last element of the array.

In each hop, you can jump from index i to an index j > i, and you get a score of
(j - i) * nums[j].

Return the maximum score you can get.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3215 Count Triplets With Even Xor Set Bits Ii

**MEDIUM**

Given three integer arrays a, b, and c, return the number of triplets (a[i], b[j], c[k]),
such that the bitwise XOR between the elements of each triplet has an even number of set bits.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3221 Maximum Array Hopping Score Ii

**MEDIUM**

Given an array nums, you have to get the maximum score starting from index 0 and
hopping until you reach the last element of the array.

In each hop, you can jump from index i to an index j > i, and you get a score of
(j - i) * nums[j].

Return the maximum score you can get.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3231 Minimum Number Of Increasing Subsequence To Be Removed

**HARD**

Given an array of integers nums, you are allowed to perform the following operation any
number of times:
- Remove a strictly increasing subsequence from the array.

Your task is to find the minimum number of operations required to make the array empty.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3237 Alt And Tab Simulation

MEDIUM

There are n windows open numbered from 1 to n, we want to simulate using alt + tab to navigate between the windows.

You are given an array windows which contains the initial order of the windows (the first element is at the top and the last one is at the bottom).

You are also given an array queries where for each query, the window queries[i] is brought to the top.

Return the final state of the array windows.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3247 Number Of Subsequences With Odd Sum

MEDIUM

Given an array nums, return the number of subsequences with an odd sum of elements.

Since the answer may be very large, return it modulo 109 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3269 Constructing Two Increasing Arrays

HARD

Given 2 integer arrays nums1 and nums2 consisting only of 0 and 1, your task is to calculate the minimum possible largest number in arrays nums1 and nums2, after doing the following.

Replace every 0 with an even positive integer and every 1 with an odd positive integer.
After replacement, both arrays should be increasing and each integer should be used at most once.

Return the minimum possible largest number after applying the changes.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3272 Find The Count Of Good Integers

HARD

You are given two positive integers n and k.

An integer x is called k-palindromic if:
- x is a palindrome.
- x is divisible by k.

An integer is called good if its digits can be rearranged to form a k-palindromic integer.
For example, for k = 2, 2020 can be rearranged to form the k-palindromic integer 2002, whereas 1010 cannot be rearranged to form a k-palindromic integer.

Return the count of good integers containing n digits.

Note that any integer must not have leading zeros, neither before nor after rearrangement.
For example, 1010 cannot be rearranged to form 101.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3279 Maximum Total Area Occupied By Pistons

HARD

There are several pistons in an old car engine, and we want to calculate the maximum possible area under the pistons.

You are given:
- An integer height, representing the maximum height a piston can reach.
- An integer array positions, where positions[i] is the current position of piston i, which is equal to the current area under it.
- A string directions, where directions[i] is the current moving direction of piston i, 'U' for up, and 'D' for down.

Each second:
- Every piston moves in its current direction 1 unit. e.g., if the direction is up, positions[i] is incremented by 1.
- If a piston has reached one of the ends, i.e., positions[i] == 0 or positions[i] == height, its direction will change.

Return the maximum possible area under all the pistons.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3299 Sum Of Consecutive Subsequences

HARD

We call an array arr of length n consecutive if one of the following holds:
- arr[i] - arr[i - 1] == 1 for all 1 <= i < n.
- arr[i] - arr[i - 1] == -1 for all 1 <= i < n.

The value of an array is the sum of its elements.

For example, [3, 4, 5] is a consecutive array of value 12 and [9, 8] is another of value 17. While [3, 4, 3] and [8, 6] are not consecutive.

Given an array of integers nums, return the sum of the values of all consecutive non-empty subsequences.

Since the answer may be very large, return it modulo 109 + 7.

Note that an array of length 1 is also considered consecutive.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3339 Find The Number Of K Even Arrays

MEDIUM

You are given three integers n, m, and k.

An array arr is called k-even if there are exactly k indices such that, for each of these indices i (0 <= i < n - 1):
- (arr[i] * arr[i + 1]) - arr[i] - arr[i + 1] is even.

Return the number of possible k-even arrays of size n where all elements are in the range [1, m].

Since the answer may be very large, return it modulo 109 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3344 Maximum Sized Array

MEDIUM

Given a positive integer s, let A be a 3D array of dimensions n × n × n, where each element A[i][j][k] is defined as:
- A[i][j][k] = i * (j OR k), where 0 <= i, j, k < n.

Return the maximum possible value of n such that the sum of all elements in array A does not exceed s.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3346 Maximum Frequency Of An Element After Performing Operations I

MEDIUM

You are given an integer array nums and two integers k and numOperations.

You must perform an operation numOperations times on nums, where in each operation you:
- Select an index i that was not selected in any previous operations.
- Add an integer in the range [-k, k] to nums[i].

Return the maximum possible frequency of any element in nums after performing the operations.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3347 Maximum Frequency Of An Element After Performing Operations Ii

HARD

You are given an integer array nums and two integers k and numOperations.

You must perform an operation numOperations times on nums, where in each operation you:
- Select an index i that was not selected in any previous operations.
- Add an integer in the range [-k, k] to nums[i].

Return the maximum possible frequency of any element in nums after performing the operations.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3354 Make Array Elements Equal To Zero

EASY

You are given an integer array nums.

Start by selecting a starting position curr such that nums[curr] == 0, and choose a movement direction of either left or right.

After that, you repeat the following process:
- If curr is out of the range [0, n - 1], this process ends.
- If nums[curr] == 0, move in the current direction by incrementing curr if you are moving right, or decrementing curr if you are moving left.
- Else if nums[curr] > 0:
- Decrement nums[curr] by 1.
- Reverse your movement direction (left becomes right and vice versa).
- Take a step in your new direction.

A selection of the initial position curr and movement direction is considered valid if every element in nums becomes 0 by the end of the process.

Return the number of possible valid selections.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3355 Zero Array Transformation I

MEDIUM

You are given an integer array nums of length n and a 2D array queries, where queries[i] = [li, ri].

For each queries[i]:
- Select a subset of indices within the range [li, ri] in nums.
- Decrement the values at the selected indices by 1.

A Zero Array is an array where all elements are equal to 0.

Return true if it is possible to transform nums into a Zero Array after processing all the queries sequentially, otherwise return false.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3356 Zero Array Transformation Ii

MEDIUM

You are given an integer array nums of length n and a 2D array queries where queries[i] = [li, ri, vali].

Each queries[i] represents the following action on nums:
- Decrement the value at each index in the range [li, ri] in nums by at most vali.
- The amount by which each value is decremented can be chosen independently for each index.

A Zero Array is an array with all its elements equal to 0.

Return the minimum possible non-negative value of k, such that after processing the first k queries in sequence, nums becomes a Zero Array. If no such k exists, return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3362 Zero Array Transformation Iii

MEDIUM

You are given an integer array nums of length n and a 2D array queries where queries[i] = [li, ri].

Each queries[i] represents the following action on nums:
- Decrement the value at each index in the range [li, ri] in nums by at most 1.
- The amount by which the value is decremented can be chosen independently for each index.

A Zero Array is an array with all its elements equal to 0.

Return the maximum number of elements that can be removed from queries, such that nums can still be converted to a zero array using the remaining queries. If it is not possible to convert nums to a zero array, return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3363 Find The Maximum Number Of Fruits Collected

HARD

There is a game dungeon comprised of n x n rooms arranged in a grid.

You are given a 2D array fruits of size n x n, where fruits[i][j] represents the number of fruits in the room (i, j). Three children will play in the game dungeon, with initial positions at the corner rooms (0, 0), (0, n - 1), and (n - 1, 0).

The children will make exactly n - 1 moves according to the following rules to reach the

room (n - 1, n - 1):
- The child starting from (0, 0) must move from their current room (i, j) to one of the rooms (i + 1, j + 1), (i + 1, j), and (i, j + 1) if the target room exists.
- The child starting from (0, n - 1) must move from their current room (i, j) to one of the rooms (i + 1, j - 1), (i + 1, j), and (i + 1, j + 1) if the target room exists.
- The child starting from (n - 1, 0) must move from their current room (i, j) to one of the rooms (i - 1, j + 1), (i, j + 1), and (i + 1, j + 1) if the target room exists.

When a child enters a room, they will collect all the fruits there. If two or more children enter the same room, only one child will collect the fruits, and the room will be emptied after they leave.

Return the maximum number of fruits the children can collect from the dungeon.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3369 Design An Array Statistics Tracker

**HARD**

Design a data structure that keeps track of the values in it and answers some queries regarding their mean, median, and mode.

Implement the StatisticsTracker class.
- StatisticsTracker(): Initialize the StatisticsTracker object with an empty array.
- void addNumber(int number): Add number to the data structure.
- void removeFirstAddedNumber(): Remove the earliest added number from the data structure.
- int getMean(): Return the floored mean of the numbers in the data structure.
- int getMedian(): Return the median of the numbers in the data structure.
- int getMode(): Return the mode of the numbers in the data structure. If there are multiple modes, return the smallest one.

Note:
- The mean of an array is the sum of all the values divided by the number of values in the array.
- The median of an array is the middle element of the array when it is sorted in non-decreasing order. If there are two choices for a median, the larger of the two values is taken.
- The mode of an array is the element that appears most often in the array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3370 Smallest Number With All Set Bits

**EASY**

You are given a positive number n.

Return the smallest number x greater than or equal to n, such that the binary representation of x contains only set bits

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3375 Minimum Operations To Make Array Values Equal To K

**EASY**

You are given an integer array nums and an integer k.

An integer h is called valid if all values in the array that are strictly greater than h are identical.

For example, if nums = [10, 8, 10, 8], a valid integer is h = 9 because all nums[i] > 9 are equal to 10, but 5 is not a valid integer.

You are allowed to perform the following operation on nums:
- Select an integer h that is valid for the current values in nums.

- For each index i where nums[i] > h, set nums[i] to h.

Return the minimum number of operations required to make every element in nums equal to k.
If it is impossible to make all elements equal to k, return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3391 Design A 3D Binary Matrix With Efficient Layer Tracking

**MEDIUM**

You are given a n x n x n binary 3D array matrix.

Implement the Matrix3D class:
- Matrix3D(int n) Initializes the object with the 3D binary array matrix, where all elements are initially set to 0.
- void setCell(int x, int y, int z) Sets the value at matrix[x][y][z] to 1.
- void unsetCell(int x, int y, int z) Sets the value at matrix[x][y][z] to 0.
- int largestMatrix() Returns the index x where matrix[x] contains the most number of 1's. If there are multiple such indices, return the largest x.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3394 Check If Grid Can Be Cut Into Sections

**MEDIUM**

You are given an integer n representing the dimensions of an n x n grid, with the origin at the bottom-left corner of the grid. You are also given a 2D array of coordinates rectangles, where rectangles[i] is in the form [startx, starty, endx, endy], representing a rectangle on the grid. Each rectangle is defined as follows:
- (startx, starty): The bottom-left corner of the rectangle.
- (endx, endy): The top-right corner of the rectangle.

Note that the rectangles do not overlap. Your task is to determine if it is possible to make either two horizontal or two vertical cuts on the grid such that:
- Each of the three resulting sections formed by the cuts contains at least one rectangle.
- Every rectangle belongs to exactly one section.

Return true if such cuts can be made; otherwise, return false.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3396 Minimum Number Of Operations To Make Elements In Array Distinct

**EASY**

You are given an integer array nums. You need to ensure that the elements in the array are distinct. To achieve this, you can perform the following operation any number of times:
- Remove 3 elements from the beginning of the array. If the array has fewer than 3 elements, remove all remaining elements.

Note that an empty array is considered to have distinct elements. Return the minimum number of operations needed to make the elements in the array distinct.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3397 Maximum Number Of Distinct Elements After Operations

**MEDIUM**

You are given an integer array nums and an integer k.

You are allowed to perform the following operation on each element of the array at most once:
- Add an integer in the range [-k, k] to the element.

Return the maximum possible number of distinct elements in nums after performing the operations.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3400 Maximum Number Of Matching Indices After Right Shifts

**MEDIUM**

You are given two integer arrays, nums1 and nums2, of the same length.

An index i is considered matching if nums1[i] == nums2[i].

Return the maximum number of matching indices after performing any number of right shifts on nums1.

A right shift is defined as shifting the element at index i to index (i + 1) % n, for all indices.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3405 Count The Number Of Arrays With K Matching Adjacent Elements

**HARD**

You are given three integers n, m, k. A good array arr of size n is defined as follows:
- Each element in arr is in the inclusive range [1, m].
- Exactly k indices i (where 1 <= i < n) satisfy the condition arr[i - 1] == arr[i].

Return the number of good arrays that can be formed.

Since the answer may be very large, return it modulo 109 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3416 Subsequences With A Unique Middle Mode Ii

**HARD**

Given an integer array nums, find the number of subsequences of size 5 of nums with a unique middle mode.

Since the answer may be very large, return it modulo 109 + 7.

A mode of a sequence of numbers is defined as the element that appears the maximum number of times in the sequence.

A sequence of numbers contains a unique mode if it has only one mode.

A sequence of numbers seq of size 5 contains a unique middle mode if the middle element (seq[2]) is a unique mode.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3423 Maximum Difference Between Adjacent Elements In A Circular Array

**EASY**

Given a circular array nums, find the maximum absolute difference between adjacent elements.

Note: In a circular array, the first and last elements are adjacent.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3432 Count Partitions With Even Sum Difference

EASY

You are given an integer array nums of length n.

A partition is defined as an index i where 0 <= i < n - 1, splitting the array into two non-empty subarrays such that:
- Left subarray contains indices [0, i].
- Right subarray contains indices [i + 1, n - 1].

Return the number of partitions where the difference between the sum of the left and right subarrays is even.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3433 Count Mentions Per User

MEDIUM

You are given an integer numberOfUsers representing the total number of users and an array events of size n x 3.

Each events[i] can be either of the following two types:
- This event indicates that a set of users was mentioned in a message at timestampi.
- The mentions_stringi string can contain one of the following tokens:
- id<number>: where <number> is an integer in range [0,numberOfUsers - 1]. There can be multiple ids separated by a single whitespace and may contain duplicates. This can mention even the offline users.
- ALL: mentions all users.
- HERE: mentions all online users.
- This event indicates that the user idi had become offline at timestampi for 60 time units. The user will automatically be online again at time timestampi + 60.

Return an array mentions where mentions[i] represents the number of mentions the user with id i has across all MESSAGE events.

All users are initially online, and if a user goes offline or comes back online, their status change is processed before handling any message event that occurs at the same timestamp.

Note that a user can be mentioned multiple times in a single message event, and each mention should be counted separately.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3442 Maximum Difference Between Even And Odd Frequency I

EASY

You are given a string s consisting of lowercase English letters.

Your task is to find the maximum difference diff = freq(a1) - freq(a2) between the frequency of characters a1 and a2 in the string such that:
- a1 has an odd frequency in the string.
- a2 has an even frequency in the string.

Return this maximum difference.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3445 Maximum Difference Between Even And Odd Frequency Ii

HARD

You are given a string s and an integer k. Your task is to find the maximum difference between the frequency of two characters, freq[a] - freq[b], in a substring subs of s, such that:
- subs has a size of at least k.
- Character a has an odd frequency in subs.
- Character b has an even frequency in subs.

Return the maximum difference.

Note that subs can contain more than 2 distinct characters.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3450 Maximum Students On A Single Bench

EASY

You are given a 2D integer array of student data students, where students[i] = [student_id, bench_id] represents that student student_id is sitting on the bench bench_id.

Return the maximum number of unique students sitting on any single bench. If no students are present, return 0.

Note: A student can appear multiple times on the same bench in the input, but they should be counted only once per bench.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3452 Sum Of Good Numbers

EASY

Given an array of integers nums and an integer k, an element nums[i] is considered good if it is strictly greater than the elements at indices i - k and i + k (if those indices exist). If neither of these indices exists, nums[i] is still considered good.

Return the sum of all the good elements in the array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3462 Maximum Sum With At Most K Elements

MEDIUM

You are given a 2D integer matrix grid of size n x m, an integer array limits of length n, and an integer k. The task is to find the maximum sum of at most k elements from the matrix grid such that:
- The number of elements taken from the ith row of grid does not exceed limits[i].

Return the maximum sum.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3476 Maximize Profit From Task Assignment

MEDIUM

You are given an integer array workers, where workers[i] represents the skill level of the ith worker. You are also given a 2D integer array tasks, where:
- tasks[i][0] represents the skill requirement needed to complete the task.
- tasks[i][1] represents the profit earned from completing the task.

Each worker can complete at most one task, and they can only take a task if their skill level is equal to the task's skill requirement. An additional worker joins today who can take up any task, regardless of the skill requirement.

Return the maximum total profit that can be earned by optimally assigning the tasks to the workers.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3484 Design Spreadsheet

**MEDIUM**

A spreadsheet is a grid with 26 columns (labeled from 'A' to 'Z') and a given number of rows. Each cell in the spreadsheet can hold an integer value between 0 and 105.

Implement the Spreadsheet class:
- Spreadsheet(int rows) Initializes a spreadsheet with 26 columns (labeled 'A' to 'Z') and the specified number of rows. All cells are initially set to 0.
- void setCell(String cell, int value) Sets the value of the specified cell. The cell reference is provided in the format "AX" (e.g., "A1", "B10"), where the letter represents the column (from 'A' to 'Z') and the number represents a 1-indexed row.
- void resetCell(String cell) Resets the specified cell to 0.
- int getValue(String formula) Evaluates a formula of the form "=X+Y", where X and Y are either cell references or non-negative integers, and returns the computed sum.

Note: If getValue references a cell that has not been explicitly set using setCell, its value is considered 0.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3495 Minimum Operations To Make Array Elements Zero

**HARD**

You are given a 2D array queries, where queries[i] is of the form [l, r]. Each queries[i] defines an array of integers nums consisting of elements ranging from l to r, both inclusive.

In one operation, you can:
- Select two integers a and b from the array.
- Replace them with floor(a / 4) and floor(b / 4).

Your task is to determine the minimum number of operations required to reduce all elements of the array to zero for each query. Return the sum of the results for all queries.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3496 Maximize Score After Pair Deletions

**MEDIUM**

You are given an array of integers nums. You must repeatedly perform one of the following operations while the array has more than two elements:
- Remove the first two elements.
- Remove the last two elements.
- Remove the first and last element.

For each operation, add the sum of the removed elements to your total score.

Return the maximum possible score you can achieve.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3508 Implement Router

MEDIUM

Design a data structure that can efficiently manage data packets in a network router.
Each data packet consists of the following attributes:
- source: A unique identifier for the machine that generated the packet.
- destination: A unique identifier for the target machine.
- timestamp: The time at which the packet arrived at the router.

Implement the Router class:
Router(int memoryLimit): Initializes the Router object with a fixed memory limit.
- memoryLimit is the maximum number of packets the router can store at any given time.
- If adding a new packet would exceed this limit, the oldest packet must be removed to free up space.

bool addPacket(int source, int destination, int timestamp): Adds a packet with the given attributes to the router.
- A packet is considered a duplicate if another packet with the same source, destination, and timestamp already exists in the router.
- Return true if the packet is successfully added (i.e., it is not a duplicate); otherwise return false.

int[] forwardPacket(): Forwards the next packet in FIFO (First In First Out) order.
- Remove the packet from storage.
- Return the packet as an array [source, destination, timestamp].
- If there are no packets to forward, return an empty array.

int getCount(int destination, int startTime, int endTime):
- Returns the number of packets currently stored in the router (i.e., not yet forwarded) that have the specified destination and have timestamps in the inclusive range [startTime, endTime].

Note that queries for addPacket will be made in increasing order of timestamp.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3511 Make A Positive Array

MEDIUM

You are given an array nums. An array is considered positive if the sum of all numbers in each subarray with more than two elements is positive.

You can perform the following operation any number of times:
- Replace one element in nums with any integer between -1018 and 1018.

Find the minimum number of operations needed to make nums positive.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3512 Minimum Operations To Make Array Sum Divisible By K

EASY

You are given an integer array nums and an integer k. You can perform the following operation any number of times:
- Select an index i and replace nums[i] with nums[i] - 1.

Return the minimum number of operations required to make the sum of the array divisible by k.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3520 Minimum Threshold For Inversion Pairs Count

MEDIUM

You are given an array of integers nums and an integer k.

An inversion pair with a threshold x is defined as a pair of indices (i, j) such that:
- i < j
- nums[i] > nums[j]
- The difference between the two numbers is at most x (i.e. nums[i] - nums[j] <= x).

Your task is to determine the minimum integer min_threshold such that there are at least k inversion pairs with threshold min_threshold.

If no such integer exists, return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3531 Count Covered Buildings

MEDIUM

You are given a positive integer n, representing an n x n city. You are also given a 2D grid buildings, where buildings[i] = [x, y] denotes a unique building located at coordinates [x, y].

A building is covered if there is at least one building in all four directions: left, right, above, and below.

Return the number of covered buildings.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3535 Unit Conversion Ii

MEDIUM

There are n types of units indexed from 0 to n - 1.

You are given a 2D integer array conversions of length n - 1, where conversions[i] = [sourceUniti, targetUniti, conversionFactori]. This indicates that a single unit of type sourceUniti is equivalent to conversionFactori units of type targetUniti.

You are also given a 2D integer array queries of length q, where queries[i] = [unitAi, unitBi].

Return an array answer of length q where answer[i] is the number of units of type unitBi equivalent to 1 unit of type unitAi, and can be represented as p/q where p and q are coprime. Return each answer[i] as pq-1 modulo 109 + 7, where q-1 represents the multiplicative inverse of q modulo 109 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3539 Find Sum Of Array Product Of Magical Sequences

HARD

You are given two integers, m and k, and an integer array nums.

A sequence of integers seq is called magical if:
- seq has a size of m.
- 0 <= seq[i] < nums.length
- The binary representation of 2seq[0] + 2seq[1] + ... + 2seq[m - 1] has k set bits.

The array product of this sequence is defined as prod(seq) = (nums[seq[0]] * nums[seq[1]] * ... * nums[seq[m - 1]]).

Return the sum of the array products for all valid magical sequences.

Since the answer may be large, return it modulo 109 + 7.

A set bit refers to a bit in the binary representation of a number that has a value of 1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3578 Count Partitions With Max Min Difference At Most K

**MEDIUM**

You are given an integer array nums and an integer k. Your task is to partition nums into one or more non-empty contiguous segments such that in each segment, the difference between its maximum and minimum elements is at most k.

Return the total number of ways to partition nums under this condition.

Since the answer may be too large, return it modulo 109 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3583 Count Special Triplets

**MEDIUM**

You are given an integer array nums.

A special triplet is defined as a triplet of indices (i, j, k) such that:
- 0 <= i < j < k < n, where n = nums.length
- nums[i] == nums[j] * 2
- nums[k] == nums[j] * 2

Return the total number of special triplets in the array.

Since the answer may be large, return it modulo 109 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3623 Count Number Of Trapezoids I

**MEDIUM**

You are given a 2D integer array points, where points[i] = [xi, yi] represents the coordinates of the ith point on the Cartesian plane.

A horizontal trapezoid is a convex quadrilateral with at least one pair of horizontal sides (i.e. parallel to the x-axis). Two lines are parallel if and only if they have the same slope.

Return the number of unique horizontal trapezoids that can be formed by choosing any four distinct points from points.

Since the answer may be very large, return it modulo 109 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

CHAPTER 02 / 355 EXERCISES

# Strings & Text Processing

## 01 THE IDEA

Strings are sequences of characters. Many tasks involve frequency counts, parsing, or finding contiguous substrings.

## 02 SEE IT

| | | | | |
|---|---|---|---|---|
| TEXT | c | o | d | e |
| INDEX | 0 | 1 | 2 | 3 |
| REVERSE | e | d | o | c |

## 03 JAVASCRIPT PATTERN

```
const chars = [...s];
const reversed = chars.reverse().join('');
```

## 04 HOW TO THINK

Check case, whitespace, Unicode and empty strings. A substring must be contiguous; a subsequence need not be.

## 05 PERFORMANCE

Typical: one scan O(n); nested comparisons O(n²).

## EXERCISE COLLECTION

Try each question before opening the matching problem number in the separate Solutions PDF. The topic classification is a study aid; some exercises combine multiple patterns.

### 0006 Zigzag Conversion

MEDIUM

The string `"PAYPALISHIRING"` is written in a zigzag pattern on a given number of rows like this: (you may want to display this pattern in a fixed font for better legibility)

```
> P   A   H   N
> A P L S I I G
> Y   I   R
```

And then read line by line: `"PAHNAPLSIIGYIR"`

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0008 String To Integer Atoi

MEDIUM

Implement the myAtoi(string s) function, which converts a string to a 32-bit signed integer (similar to C/C++'s atoi function).

The algorithm for myAtoi(string s) is as follows:

- Read in and ignore any leading whitespace.
- Check if the next character (if not already at the end of the string) is '-' or '+'. Read this character in if it is either. This determines if the final result is negative or positive respectively. Assume the result is positive if neither is present.
- Read in next the characters until the next non-digit charcter or the end of the input is reached. The rest of the string is ignored.
- Convert these digits into an integer (i.e. "123" -> 123, "0032" -> 32). If no digits were read, then the integer is 0. Change the sign as necessary (from step 2).
- If the integer is out of the 32-bit signed integer range [-231, 231 - 1], then clamp the integer so that it remains in the range. Specifically, integers less than -231 should be clamped to -231, and integers greater than 231 - 1 should be clamped to 231 - 1.
- Return the integer as the final result.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0012 Integer To Roman

MEDIUM

Roman numerals are represented by seven different symbols: `I`, `V`, `X`, `L`, `C`, `D` and `M`.

| Symbol | Value |
|---|---|
| I | 1 |
| V | 5 |
| X | 10 |
| L | 50 |
| C | 100 |
| D | 500 |
| M | 1000 |

For example, `2` is written as `II` in Roman numeral, just two one's added together. `12` is written as `XII`, which is simply `X + II`. The number `27` is written as `XXVII`, which is `XX + V + II`.

Roman numerals are usually written largest to smallest from left to right. However, the numeral for four is not `IIII`. Instead, the number four is written as `IV`. Because the one is before the five we subtract it making four. The same principle applies to the number nine, which is written as `IX`. There are six instances where subtraction is used:

- `I` can be placed before `V (5)` and `X (10)` to make 4 and 9.
- `X` can be placed before `L (50)` and `C (100)` to make 40 and 90.
- `C` can be placed before `D (500)` and `M (1000)` to make 400 and 900.

Given an integer, convert it to a roman numeral.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0013 Roman To Integer

EASY

Roman numerals are represented by seven different symbols: `I`, `V`, `X`, `L`, `C`, `D` and `M`.

| Symbol | Value |
|---|---|
| I | 1 |
| V | 5 |
| X | 10 |
| L | 50 |
| C | 100 |
| D | 500 |
| M | 1000 |

For example, `2` is written as `II` in Roman numeral, just two one's added together. `12` is written as `XII`, which is simply `X + II`. The number `27` is written as `XXVII`, which is `XX + V + II`.

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- `X` can be placed before `L (50)` and `C (100)` to make 40 and 90.
- `C` can be placed before `D (500)` and `M (1000)` to make 400 and 900.

Given a roman numeral, convert it to an integer.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0014 Longest Common Prefix

EASY

Write a function to find the longest common prefix string amongst an array of strings.

If there is no common prefix, return an empty string `""`.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0043 Multiply Strings

MEDIUM

Given two non-negative integers num1 and num2 represented as strings, return the product of num1 and num2, also represented as a string.

- The length of both num1 and num2 is < 110.
- Both num1 and num2 contain only digits 0-9.
- Both num1 and num2 do not contain any leading zero, except the number 0 itself.
- You must not use any built-in BigInteger library or convert the inputs to integer directly.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0049 Group Anagrams

MEDIUM

Given an array of strings `strs`, group the anagrams together. You can return the answer in any order.

An Anagram is a word or phrase formed by rearranging the letters of a different word or phrase, typically using all the original letters exactly once.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0058 Length Of Last Word

**EASY**

Given a string `s` consists of upper/lower-case alphabets and empty space characters ' ', return the length of last word (last word means the last appearing word if we loop from left to right) in the string.

If the last word does not exist, return 0.

Note: A word is defined as a maximal substring consisting of non-space characters only.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0068 Text Justification

**HARD**

Given an array of strings words and a width maxWidth, format the text such that each line has exactly maxWidth characters and is fully (left and right) justified.

You should pack your words in a greedy approach; that is, pack as many words as you can in each line. Pad extra spaces ' ' when necessary so that each line has exactly maxWidth characters.

Extra spaces between words should be distributed as evenly as possible. If the number of spaces on a line does not divide evenly between words, the empty slots on the left will be assigned more spaces than the slots on the right.

For the last line of text, it should be left-justified, and no extra space is inserted between words.

Note:
- A word is defined as a character sequence consisting of non-space characters only.
- Each word's length is guaranteed to be greater than 0 and not exceed maxWidth.
- The input array words contains at least one word.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0087 Scramble String

**HARD**

We can scramble a string s to get a string t using the following algorithm:

- Split the string into two non-empty substrings at a random index, i.e., if the string is s, divide it to x and y where s = x + y.
- Randomly decide to swap the two substrings or to keep them in the same order. i.e., after this step, s may become s = x + y or s = y + x.
- Apply step 1 recursively on each of the two substrings x and y.

Given two strings s1 and s2 of the same length, return true if s2 is a scrambled string of s1, otherwise, return false.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0097 Interleaving String

MEDIUM

Given strings s1, s2, and s3, find whether s3 is formed by an interleaving of s1 and s2.

An interleaving of two strings s and t is a configuration where s and t are divided into n and m substrings respectively, such that:
- s = s1 + s2 + ... + sn
- t = t1 + t2 + ... + tm
- |n - m| <= 1
- The interleaving is s1 + t1 + s2 + t2 + s3 + t3 + ... or t1 + s1 + t2 + s2 + t3 + s3 + ...

Note: a + b is the concatenation of strings a and b.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0139 Word Break

MEDIUM

Given a string s and a dictionary of strings wordDict, return true if s can be segmented into a space-separated sequence of one or more dictionary words.

Note that the same word in the dictionary may be reused multiple times in the segmentation.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0140 Word Break Ii

HARD

Given a string s and a dictionary of strings wordDict, add spaces in s to construct a sentence where each word is a valid dictionary word. Return all such possible sentences in any order.

Note that the same word in the dictionary may be reused multiple times in the segmentation.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0151 Reverse Words In A String

MEDIUM

Given an input string, reverse the string word by word.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0157 Read N Characters Given Read4

EASY

Given a file and assume that you can only read the file using a given method read4, implement a method to read n characters.

Method read4:
- The API read4 reads four consecutive characters from file, then writes those characters into the buffer array buf4.

The return value is the number of actual characters read.

Note that read4() has its own file pointer, much like FILE *fp in C.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0158 Read N Characters Given Read4 Ii Call Multiple Times

HARD

Given a file and assume that you can only read the file using a given method read4, implement a method read to read n characters. Your method read may be called multiple times.

Method read4:
- The API read4 reads four consecutive characters from file, then writes those characters into the buffer array buf4.

The return value is the number of actual characters read.

Note that read4() has its own file pointer, much like FILE *fp in C.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0186 Reverse Words In A String Ii

MEDIUM

Given a character array s, reverse the order of the words.

A word is defined as a sequence of non-space characters. The words in s will be separated by a single space.

Your code must solve the problem in-place, i.e. without allocating extra space.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0205 Isomorphic Strings

EASY

Given two strings s and t, determine if they are isomorphic.

Two strings s and t are isomorphic if the characters in s can be replaced to get t.

All occurrences of a character must be replaced with another character while preserving the order of characters. No two characters may map to the same character, but a character may map to itself.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0211 Design Add And Search Words Data Structure

MEDIUM

Design a data structure that supports adding new words and finding if a string matches any previously added string.

Implement the WordDictionary class:
- WordDictionary() Initializes the object.
- void addWord(word) Adds word to the data structure, it can be matched later.
- bool search(word) Returns true if there is any string in the data structure that matches word or false otherwise. word may contain dots '.' where dots can be matched with any letter.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0233 Number Of Digit One

HARD

Given an integer n, count the total number of digit 1 appearing in all non-negative integers less than or equal to n.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 0242 Valid Anagram

EASY

Given two strings s and t, return true if t is an anagram of s, and false otherwise.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 0243 Shortest Word Distance

EASY

Given an array of strings wordsDict and two different strings that already exist in the array word1 and word2, return the shortest distance between these two words in the list.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 0244 Shortest Word Distance Ii

MEDIUM

Design a data structure that will be initialized with a string array, and then it should answer queries of the shortest distance between two different strings from the array.

Implement the WordDistance class:
- WordDistance(String[] wordsDict) initializes the object with the strings array wordsDict.
- int shortest(String word1, String word2) returns the shortest distance between word1 and word2 in the array wordsDict.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 0245 Shortest Word Distance Iii

MEDIUM

Given an array of strings wordsDict and two strings that already exist in the array word1 and word2, return the shortest distance between the occurrence of these two words in the list.

Note that word1 and word2 may be the same. It is guaranteed that they represent two individual words in the list.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 0249 Group Shifted Strings

MEDIUM

Perform the following shift operations on a string:
- Right shift: Replace every letter with the successive letter of the English alphabet, where 'z' is replaced by 'a'. For example, "abc" can be right-shifted to "bcd" or "xyz" can be right-shifted to "yza".
- Left shift: Replace every letter with the preceding letter of the English alphabet, where 'a' is replaced by 'z'. For example, "bcd" can be left-shifted to "abc" or "yza" can be left-shifted to "xyz".

We can keep shifting the string in both directions to form an endless shifting sequence.
- For example, shift "abc" to form the sequence: ... <-> "abc" <-> "bcd" <-> ... <-> "xyz" <-> "yza" <-> .... <-> "zab" <-> "abc" <-> ...

You are given an array of strings strings, group together all strings[i] that belong to the same shifting sequence. You may return the answer in any order.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 0258 Add Digits

EASY

Given an integer num, repeatedly add all its digits until
the result has only one digit, and return it.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0271 Encode And Decode Strings

MEDIUM

Design an algorithm to encode a list of strings to a string. The encoded string is then sent over the network and is decoded back to the original list of strings.

Machine 1 (sender) has the function:
string encode(vector<string> strs) {
// ... your code
return encoded_string;
}
Machine 2 (receiver) has the function:
vector<string> decode(string s) {
//... your code
return strs;
}
So Machine 1 does:
string encoded_string = encode(strs);
and Machine 2 does:
vector<string> strs2 = decode(encoded_string);
strs2 in Machine 2 should be the same as strs in Machine 1.

Implement the encode and decode methods.

You are not allowed to solve the problem using any serialize methods (such as eval).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0273 Integer To English Words

HARD

Convert a non-negative integer `num` to its English words representation.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0281 Zigzag Iterator

MEDIUM

Given two vectors of integers v1 and v2, implement an iterator to return their elements alternately.

Implement the ZigzagIterator class:
- ZigzagIterator(List<int> v1, List<int> v2) initializes the object with the two vectors v1 and v2.
- boolean hasNext() returns true if the iterator still has elements, and false otherwise.
- int next() returns the current element of the iterator and moves the iterator to the next element.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0288 Unique Word Abbreviation

MEDIUM

The abbreviation of a word is a concatenation of its first letter, the number of characters between the first and last letter, and its last letter. If a word has only two characters, then it is an abbreviation of itself.

For example:
- dog --> d1g because there is one letter between the first letter 'd' and the last letter 'g'.
- internationalization --> i18n because there are 18 letters between the first letter 'i' and the last letter 'n'.
- it --> it because any word with only two characters is an abbreviation of itself.

Implement the ValidWordAbbr class:
- ValidWordAbbr(String[] dictionary) Initializes the object with a dictionary of words.
- boolean isUnique(string word) Returns true if either of the following conditions are met (otherwise returns false):
- There is no word in dictionary whose abbreviation is equal to word's abbreviation.
- For any word in dictionary whose abbreviation is equal to word's abbreviation, that word and word are the same.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0290 Word Pattern

EASY

Given a pattern and a string s, find if s follows the same pattern.

Here follow means a full match, such that there is a bijection between a letter in pattern and a non-empty word in s. Specifically:
- Each letter in pattern maps to exactly one unique word in s.
- Each unique word in s maps to exactly one letter in pattern.
- No two letters map to the same word, and no two words map to the same letter.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0291 Word Pattern Ii

MEDIUM

Given a pattern and a string s, return true if s matches the pattern.

A string s matches a pattern if there is some bijective mapping of single characters to non-empty strings such that if each character in pattern is replaced by the string it maps to, then the resulting string is s. A bijective mapping means that no two characters map to the same string, and no character maps to two different strings.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0316 Remove Duplicate Letters

MEDIUM

Given a string s, remove duplicate letters so that every letter appears once and only once. You must make sure your result is
the smallest in lexicographical order among all possible results.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0318 Maximum Product Of Word Lengths

MEDIUM

Given a string array words, return the maximum value of length(word[i]) * length(word[j]) where the two words do not share common letters. If no such two words exist, return 0.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0344 Reverse String

EASY

Write a function that reverses a string. The input string is given as an array of characters s.

You must do this by modifying the input array in-place with O(1) extra memory.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0345 Reverse Vowels Of A String

EASY

Write a function that takes a string as input and reverse only the vowels of a string.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0357 Count Numbers With Unique Digits

MEDIUM

Given an integer n, return the count of all numbers with unique digits, x, where 0 <= x < 10n.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0358 Rearrange String K Distance Apart

HARD

Given a string s and an integer k, rearrange s such that the same characters are at least distance k from each other. If it is not possible to rearrange the string, return an empty string "".

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0387 First Unique Character In A String

EASY

Given a string, find the first non-repeating character in it and return it's index. If it doesn't exist, return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0394 Decode String

MEDIUM

Given an encoded string, return its decoded string.

The encoding rule is: k[encoded_string], where the encoded_string inside the square brackets is being repeated exactly k times. Note that k is guaranteed to be a

positive integer.

You may assume that the input string is always valid; there are no extra white spaces, square brackets are well-formed, etc. Furthermore, you may assume that the original data does not contain any digits and that digits are only for those repeat numbers, k. For example, there will not be input like 3a or 2[4].

The test cases are generated so that the length of the output will never exceed 105.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0400 Nth Digit

**MEDIUM**

Given an integer n, return the nth digit of the infinite integer sequence [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, ...].

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0402 Remove K Digits

**MEDIUM**

Given string num representing a non-negative integer num, and an integer k, return the smallest possible integer after removing k digits from num.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0408 Valid Word Abbreviation

**EASY**

A string can be abbreviated by replacing any number of non-adjacent, non-empty substrings with their lengths. The lengths should not have leading zeros.

For example, a string such as "substitution" could be abbreviated as (but not limited to):
- "s10n" ("s ubstitutio n")
- "sub4u4" ("sub stit u tion")
- "12" ("substitution")
- "su3i1u2on" ("su bst i t u ti on")
- "substitution" (no substrings replaced)

The following are not valid abbreviations:
- "s55n" ("s ubsti tutio n", the replaced substrings are adjacent)
- "s010n" (has leading zeros)
- "s0ubstitution" (replaces an empty substring)

Given a string word and an abbreviation abbr, return whether the string matches the given abbreviation.

A substring is a contiguous non-empty sequence of characters within a string.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0411 Minimum Unique Word Abbreviation

**HARD**

A string can be abbreviated by replacing any number of non-adjacent substrings with their lengths. For example, a string such as "substitution" could be abbreviated as (but not limited to):
- "s10n" ("s ubstitutio n")
- "sub4u4" ("sub stit u tion")
- "12" ("substitution")

- "su3i1u2on" ("su bst i t u ti on")
- "substitution" (no substrings replaced)

Note that "s55n" ("s ubsti tutio n") is not a valid abbreviation of "substitution" because the replaced substrings are adjacent.

The length of an abbreviation is the number of letters that were not replaced plus the number of substrings that were replaced. For example, the abbreviation "s10n" has a length of 3 (2 letters + 1 substring) and "su3i1u2on" has a length of 9 (6 letters + 3 substrings).

Given a target string target and an array of strings dictionary, return an abbreviation of target with the shortest possible length such that it is not an abbreviation of any string in dictionary. If there are multiple shortest abbreviations, return any of them.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0415 Add Strings

**EASY**

Given two non-negative integers, num1 and num2 represented as string, return the sum of num1 and num2 as a string.

You must solve the problem without using any built-in library for handling large integers (such as BigInteger). You must also not convert the inputs to integers directly.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0418 Sentence Screen Fitting

**MEDIUM**

Given a rows x cols screen and a sentence represented as a list of strings, return the number of times the given sentence can be fitted on the screen.

The order of words in the sentence must remain unchanged, and a word cannot be split into two lines. A single space must separate two consecutive words in a line.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0420 Strong Password Checker

**HARD**

A password is considered strong if the below conditions are all met:
- It has at least 6 characters and at most 20 characters.
- It contains at least one lowercase letter, at least one uppercase letter, and at least one digit.
- It does not contain three repeating characters in a row (i.e., "Baaabb0" is weak, but "Baaba0" is strong).

Given a string password, return the minimum number of steps required to make password strong. if password is already strong, return 0.

In one step, you can:
- Insert one character to password,
- Delete one character from password, or
- Replace one character of password with another character.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0422 Valid Word Square

EASY

Given an array of strings words, return true if it forms a valid word square.

A sequence of strings forms a valid word square if the kth row and column read the same string, where 0 <= k < max(numRows, numColumns).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0423 Reconstruct Original Digits From English

MEDIUM

Given a string s containing an out-of-order English representation of digits 0-9, return the digits in ascending order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0424 Longest Repeating Character Replacement

MEDIUM

You are given a string s and an integer k. You can choose any character of the string and change it to any other uppercase English character. You can perform this operation at most k times.

Return the length of the longest substring containing the same letter you can get after performing the above operations.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0425 Word Squares

HARD

Given an array of unique strings words, return all the word squares you can build from words. The same word from words can be used multiple times. You can return the answer in any order.

A sequence of strings forms a valid word square if the kth row and column read the same string, where 0 <= k < max(numRows, numColumns).

- For example, the word sequence ["ball","area","lead","lady"] forms a word square because each word reads the same both horizontally and vertically.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0434 Number Of Segments In A String

EASY

Given a string s, return the number of segments in the string.

A segment is defined to be a contiguous sequence of non-space characters.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0438 Find All Anagrams In A String

MEDIUM

Given two strings s and p, return an array of all the start indices of p's anagrams in s.

You may return the answer in any order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0443 String Compression

**MEDIUM**

Given an array of characters chars, compress it using the following algorithm:

Begin with an empty string s. For each group of consecutive repeating characters in chars:
- If the group's length is 1, append the character to s.
- Otherwise, append the character followed by the group's length.

The compressed string s should not be returned separately, but instead, be stored in the input character array chars. Note that group lengths that are 10 or longer will be split into multiple characters in chars.

After you are done modifying the input array, return the new length of the array.

You must write an algorithm that uses only constant extra space.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0471 Encode String With Shortest Length

**HARD**

Given a string s, encode the string such that its encoded length is the shortest.

The encoding rule is: k[encoded_string], where the encoded_string inside the square brackets is being repeated exactly k times. k should be a positive integer.

If an encoding process does not make the string shorter, then do not encode it. If there are several solutions, return any of them.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0472 Concatenated Words

**HARD**

Given an array of strings words (without duplicates), return all the concatenated words in the given list of words.

A concatenated word is defined as a string that is comprised entirely of at least two shorter words (not necessarily distinct) in the given array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0481 Magical String

**MEDIUM**

A magical string s consists of only '1' and '2' and obeys the following rules:
- The string s is magical because concatenating the number of contiguous occurrences of characters '1' and '2' generates the string s itself.
- The first few elements of s is s = "1221121221221121122……". If we group the consecutive 1's and 2's in s, it will be "1 22 11 2 1 22 1 22 11 2 11 22 ....." and the occurrences of 1's or 2's in each group are "1 2 2 1 1 2 1 2 2 1 2 2 .....". You can see that the occurrence sequence is s itself.

Given an integer n, return the number of 1's in the first n number in the magical string s.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0524 Longest Word In Dictionary Through Deleting

MEDIUM

Given a string s and a string array dictionary, return the longest string in the dictionary that can be formed by deleting some of the given string characters. If there is more than one possible result, return the longest word with the smallest lexicographical order. If there is no possible result, return the empty string.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0527 Word Abbreviation

HARD

Given an array of distinct strings words, return the minimal possible abbreviations for every word.

The following are the rules for a string abbreviation:
characters in between, followed by the last character.
- Increase the prefix (characters in the first part) of each of their abbreviations by 1.
- For example, say you start with the words ["abcdef","abndef"] both initially abbreviated as "a4f". Then, a sequence of operations would be ["a4f","a4f"] -> ["ab3f","ab3f"] -> ["abc2f","abn2f"].
- This operation is repeated until every abbreviation is unique.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0541 Reverse String Ii

EASY

Given a string and an integer k, you need to reverse the first k characters for every 2k characters counting from the start of the string. If there are less than k characters left, reverse all of them. If there are less than 2k but greater than or equal to k characters, then reverse the first k characters and left the other as original.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0555 Split Concatenated Strings

MEDIUM

You are given an array of strings strs. You could concatenate these strings together into a loop, where for each string, you could choose to reverse it or not. Among all the possible loops

Return the lexicographically largest string after cutting the loop, which will make the looped string into a regular one.

Specifically, to find the lexicographically largest string, you need to experience two phases:
them in the same order as given.
a regular one starting from the character at the cutpoint.

And your job is to find the lexicographically largest one among all the possible regular strings.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0557 Reverse Words In A String Iii

EASY

Given a string s, reverse the order of characters in each word within a sentence while

still preserving whitespace and initial word order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0583 Delete Operation For Two Strings

**MEDIUM**

Given two strings word1 and word2, return the minimum number of steps required to make word1 and word2 the same.

In one step, you can delete exactly one character in either string.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0604 Design Compressed String Iterator

**EASY**

Design and implement a data structure for a compressed string iterator. The given compressed string will be in the form of each letter followed by a positive integer representing the number of this letter existing in the original uncompressed string.

Implement the StringIterator class:
- next() Returns the next character if the original string still has uncompressed characters, otherwise returns a white space.
- hasNext() Returns true if there is any letter needs to be uncompressed in the original string, otherwise returns false.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0616 Add Bold Tag In String

**MEDIUM**

You are given a string s and an array of strings words.

You should add a closed pair of bold tag  and  to wrap the substrings in s that exist in words.
- If two such substrings overlap, you should wrap them together with only one pair of closed bold-tag.
- If two substrings wrapped by bold tags are consecutive, you should combine them.

Return s after adding the bold tags.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0648 Replace Words

**MEDIUM**

In English, we have a concept called root, which can be followed by some other words to form another longer word - let's call this word successor. For example, the root an, followed by other, which can form another word another.

Now, given a dictionary consisting of many roots and a sentence. You need to replace all the successor in the sentence with the root forming it. If a successor has many roots can form it, replace it with the root with the shortest length.

You need to output the sentence after the replacement.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0678 Valid Parenthesis String

MEDIUM

Given a string s containing only three types of characters: '(', ')' and '*', return true if s is valid.

The following rules define a valid string:
- Any left parenthesis '(' must have a corresponding right parenthesis ')'.
- Any right parenthesis ')' must have a corresponding left parenthesis '('.
- Left parenthesis '(' must go before the corresponding right parenthesis ')'.
- '*' could be treated as a single right parenthesis ')' or a single left parenthesis '(' or an empty string "".

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0686 Repeated String Match

EASY

Given two strings A and B, find the minimum number of times A has to be repeated such that B is a substring of it. If no such solution, return -1.

For example, with A = "abcd" and B = "cdabcdab".

Return 3, because by repeating A three times (“abcdabcdabcd”), B is a substring of it; and B is not a substring of A repeated two times ("abcdabcd").

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0691 Stickers To Spell Word

HARD

We are given n different types of stickers. Each sticker has a lowercase English word on it.

You would like to spell out the given string target by cutting individual letters from your collection of stickers and rearranging them. You can use each sticker more than once if you want, and you have infinite quantities of each sticker.

Return the minimum number of stickers that you need to spell out target. If the task is impossible, return -1.

Note: In all test cases, all words were chosen randomly from the 1000 most common US English words, and target was chosen as a concatenation of two random words.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0692 Top K Frequent Words

MEDIUM

Given an array of strings words and an integer k, return the k most frequent strings.

Return the answer sorted by the frequency from highest to lowest. Sort the words with the same frequency by their lexicographical order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0712 Minimum Ascii Delete Sum For Two Strings

MEDIUM

Given two strings s1 and s2, return the lowest ASCII sum of deleted characters to make two strings equal.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0717 1 Bit And 2 Bit Characters

EASY

We have two special characters:
- The first character can be represented by one bit 0.
- The second character can be represented by two bits (10 or 11).

Given a binary array bits that ends with 0, return true if the last character must be a one-bit character.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0720 Longest Word In Dictionary

MEDIUM

Given an array of strings words representing an English Dictionary, return the longest word in words that can be built one character at a time by other words in words.

If there is more than one possible answer, return the longest word with the smallest lexicographical order. If there is no answer, return the empty string.

Note that the word should be built from left to right with each additional character being added to the end of a previous word.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0734 Sentence Similarity

EASY

We can represent a sentence as an array of words, for example, the sentence "I am happy with leetcode" can be represented as arr = ["I","am",happy","with","leetcode"].

Given two sentences sentence1 and sentence2 each represented as a string array and given an array of string pairs similarPairs where similarPairs[i] = [xi, yi] indicates that the two words xi and yi are similar.

Return true if sentence1 and sentence2 are similar, or false if they are not similar.

Two sentences are similar if:
- They have the same length (i.e., the same number of words)
- sentence1[i] and sentence2[i] are similar.

Notice that a word is always similar to itself, also notice that the similarity relation is not transitive. For example, if the words a and b are similar, and the words b and c are similar, a and c are not necessarily similar.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0737 Sentence Similarity Ii

MEDIUM

We can represent a sentence as an array of words, for example, the sentence "I am happy with leetcode" can be represented as arr = ["I","am",happy","with","leetcode"].

Given two sentences sentence1 and sentence2 each represented as a string array and given an array of string pairs similarPairs where similarPairs[i] = [xi, yi] indicates that the two words xi and yi are similar.

Return true if sentence1 and sentence2 are similar, or false if they are not similar.

Two sentences are similar if:
- They have the same length (i.e., the same number of words)
- sentence1[i] and sentence2[i] are similar.

Notice that a word is always similar to itself, also notice that the similarity relation is transitive. For example, if the words a and b are similar, and the words b and c are similar, then a and c are similar.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0738 Monotone Increasing Digits

**MEDIUM**

An integer has monotone increasing digits if and only if each pair of adjacent digits x and y satisfy x <= y.

Given an integer n, return the largest number that is less than or equal to n with monotone increasing digits.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0744 Find Smallest Letter Greater Than Target

**EASY**

You are given an array of characters letters that is sorted in non-decreasing order, and a character target. There are at least two different characters in letters.

Return the smallest character in letters that is lexicographically greater than target. If such a character does not exist, return the first character in letters.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0745 Prefix And Suffix Search

**HARD**

Design a special dictionary that searches the words in it by a prefix and a suffix.

Implement the WordFilter class:
- WordFilter(string[] words) Initializes the object with the words in the dictionary.
- f(string pref, string suff) Returns the index of the word in the dictionary, which has the prefix pref and the suffix suff. If there is more than one valid index, return the largest of them. If there is no such word in the dictionary, return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0748 Shortest Completing Word

**EASY**

Given a string licensePlate and an array of strings words, find the shortest completing word in words.

A completing word is a word that contains all the letters in licensePlate. Ignore numbers and spaces in licensePlate, and treat letters as case insensitive. If a letter appears more than once in licensePlate, then it must appear in the word the same number of times or more.

For example, if licensePlate = "aBc 12c", then it contains letters 'a', 'b' (ignoring case), and 'c' twice. Possible completing words are "abccdef", "caaacab", and "cbca".

Return the shortest completing word in words. It is guaranteed an answer exists. If there are multiple shortest completing words, return the first one that occurs in words.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0758 Bold Words In String

MEDIUM

Given an array of keywords words and a string s, make all appearances of all keywords words[i] in s bold. Any letters between  and  tags become bold.

Return s after adding the bold tags. The returned string should use the least number of tags possible, and the tags should form a valid combination.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0760 Find Anagram Mappings

EASY

You are given two integer arrays nums1 and nums2 where nums2 is an anagram of nums1. Both arrays may contain duplicates.

Return an index mapping array mapping from nums1 to nums2 where mapping[i] = j means the ith element in nums1 appears in nums2 at index j. If there are multiple answers, return any of them.

An array a is an anagram of an array b means b is made by randomizing the order of the elements in a.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0761 Special Binary String

HARD

Special binary strings are binary strings with the following two properties:
- The number of 0's is equal to the number of 1's.
- Every prefix of the binary string has at least as many 1's as 0's.

You are given a special binary string s.

A move consists of choosing two consecutive, non-empty, special substrings of s, and swapping them. Two strings are consecutive if the last character of the first string is exactly one index before the first character of the second string.

Return the lexicographically largest resulting string possible after applying the mentioned operations on the string.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0767 Reorganize String

MEDIUM

Given a string s, rearrange the characters of s so that any two adjacent characters are not the same.

Return any possible rearrangement of s or return "" if not possible.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0777 Swap Adjacent In Lr String

MEDIUM

In a string composed of 'L', 'R', and 'X' characters, like "RXXLRXRXL", a move consists of either replacing one occurrence of "XL" with "LX", or replacing one occurrence of "RX" with "XR". Given the starting string start and the ending string result, return True if and only if there exists

a sequence of moves to transform start to result.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0788 Rotated Digits

**MEDIUM**

An integer x is a good if after rotating each digit individually by 180 degrees, we get a valid number that is different from x. Each digit must be rotated - we cannot choose to leave it alone.

A number is valid if each digit remains a digit after rotation. For example:
- 0, 1, and 8 rotate to themselves,
- 2 and 5 rotate to each other (in this case they are rotated in a different direction, in other words, 2 or 5 gets mirrored),
- 6 and 9 rotate to each other, and
- the rest of the numbers do not rotate to any other number and become invalid.

Given an integer n, return the number of good integers in the range [1, n].

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0796 Rotate String

**EASY**

Given two strings s and goal, return true if and only if s can become goal after some number of shifts on s.

A shift on s consists of moving the leftmost character of s to the rightmost position.

For example, if s = "abcde", then it will be "bcdea" after one shift.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0804 Unique Morse Code Words

**EASY**

International Morse Code defines a standard encoding where each letter is mapped to a series of dots and dashes, as follows: "a" maps to ".-", "b" maps to "-...", "c" maps to "-.-.", and so on.

For convenience, the full table for the 26 letters of the English alphabet is given below:

[".-","-...","-.-.","-..",".","..-.","--.","....","..",
".---","-.-",".-..","--","-.","---",".--.","--.-",".-.",
"...","-","..-","...-",".--","-..-","-.--","--.."]

Now, given a list of words, each word can be written as a concatenation of the Morse code of each letter.
For example, "cba" can be written as "-.-..--...",
(which is the concatenation "-.-." + "-..." + ".-").
We'll call such a concatenation, the transformation of a word.

Return the number of different transformations among all words we have.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0806 Number Of Lines To Write String

EASY

You are given a string s of lowercase English letters and an array widths denoting how many pixels wide each lowercase English letter is. Specifically, widths[0] is the width of 'a', widths[1] is the width of 'b', and so on.

You are trying to write s across several lines, where each line is no longer than 100 pixels. Starting at the beginning of s, write as many letters on the first line such that the total width does not exceed 100 pixels. Then, from where you stopped in s, continue writing as many letters as you can on the second line. Continue this process until you have written all of s.

Return an array result of length 2 where:
- result[0] is the total number of lines.
- result[1] is the width of the last line in pixels.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0809 Expressive Words

MEDIUM

Sometimes people repeat letters to represent extra feeling. For example:
- "hello" -> "heeellooo"
- "hi" -> "hiiii"

In these strings like "heeellooo", we have groups of adjacent letters that are all the same: "h", "eee", "ll", "ooo".

You are given a string s and an array of query strings words. A query word is stretchy if it can be made to be equal to s by any number of applications of the following extension operation: choose a group consisting of characters c, and add some number of characters c to the group so that the size of the group is three or more.

For example, starting with "hello", we could do an extension on the group "o" to get "hellooo", but we cannot get "helloo" since the group "oo" has a size less than three. Also, we could do another extension like "ll" -> "llll" to get "hellllooo". If s = "hellllooo", then the query word "hello" would be stretchy because of these two extension operations: query = "hello" -> "hellooo" -> "hellllooo" = s.

Return the number of query strings that are stretchy.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0819 Most Common Word

EASY

Given a paragraph and a list of banned words, return the most frequent word that is not in the list of banned words. It is guaranteed there is at least one word that isn't banned, and that the answer is unique.

Words in the list of banned words are given in lowercase, and free of punctuation. Words in the paragraph are not case sensitive. The answer is in lowercase.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0820 Short Encoding Of Words

MEDIUM

A valid encoding of an array of words is any reference string s and array of indices such that:

- words.length == indices.length
- The reference string s ends with the '#' character.
- For each index indices[i], the substring of s starting from indices[i] and up to (but not including) the next '#' character is equal to words[i].

Given an array of words, return the length of the shortest reference string s possible of any valid encoding of words.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0821 Shortest Distance To A Character

**EASY**

Given a string s and a character c that occurs in s, return an array of integers answer where answer.length == s.length and answer[i] is the distance from index i to the closest occurrence of character c in s.

The distance between two indices i and j is abs(i - j), where abs is the absolute value function.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0833 Find And Replace In String

**MEDIUM**

You are given a 0-indexed string s that you must perform k replacement operations on. The replacement operations are given as three 0-indexed parallel arrays, indices, sources, and targets, all of length k.

To complete the ith replacement operation:

For example, if s = "abcd", indices[i] = 0, sources[i] = "ab", and targets[i] = "eee", then the result of this replacement will be "eeecd".

All replacement operations must occur simultaneously, meaning the replacement operations should not affect the indexing of each other. The testcases will be generated such that the replacements will not overlap.

- For example, a testcase with s = "abc", indices = [0, 1], and sources = ["ab","bc"] will not be generated because the "ab" and "bc" replacements overlap.

Return the resulting string after performing all replacement operations on s.

A substring is a contiguous sequence of characters in a string.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0839 Similar String Groups

**HARD**

Two strings, X and Y, are considered similar if either they are identical or we can make them equivalent by swapping at most two letters (in distinct positions) within the string X.

For example, "tars" and "rats" are similar (swapping at positions 0 and 2), and "rats" and "arts" are similar, but "star" is not similar to "tars", "rats", or "arts".

Together, these form two connected groups by similarity: {"tars", "rats", "arts"} and {"star"}. Notice that "tars" and "arts" are in the same group even though they are not similar. Formally, each group is such that a word is in the group if and only if it is similar to at least one other word in the group.

We are given a list strs of strings where every string in strs is an anagram of every other string in strs. How many groups are there?

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0843 Guess The Word

HARD

You are given an array of unique strings words where words[i] is six letters long. One word of words was chosen as a secret word.

You are also given the helper object Master. You may call Master.guess(word) where word is a six-letter-long string, and it must be from words. Master.guess(word) returns:
- -1 if word is not from words, or
- an integer representing the number of exact matches (value and position) of your guess to the secret word.

There is a parameter allowedGuesses for each test case where allowedGuesses is the maximum number of times you can call Master.guess(word).

For each test case, you should call Master.guess with the secret word without exceeding the maximum number of allowed guesses. You will get:
- "Either you took too many guesses, or you did not find the secret word." if you called Master.guess more than allowedGuesses times or if you did not call Master.guess with the secret word, or
- "You guessed the secret word correctly." if you called Master.guess with the secret word with the number of calls to Master.guess less than or equal to allowedGuesses.

The test cases are generated such that you can guess the secret word with a reasonable strategy (other than using the bruteforce method).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0844 Backspace String Compare

EASY

Given two strings `s` and `t`, return true if they are equal when both are typed into empty text editors. '#' means a backspace character.

Note that after backspacing an empty text, the text will continue empty.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0848 Shifting Letters

MEDIUM

You are given a string s of lowercase English letters and an integer array shifts of the same length.

Call the shift() of a letter, the next letter in the alphabet, (wrapping around so that 'z' becomes 'a').

For example, shift('a') = 'b', shift('t') = 'u', and shift('z') = 'a'.

Now for each shifts[i] = x, we want to shift the first i + 1 letters of s, x times.

Return the final string after all such shifts to s are applied.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0854 K Similar Strings

HARD

Strings s1 and s2 are k-similar (for some non-negative integer k) if we can swap the positions of two letters in s1 exactly k times so that the resulting string equals s2.

Given two anagrams s1 and s2, return the smallest k for which s1 and s2 are k-similar.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 0859 Buddy Strings

EASY

Given two strings s and goal, return true if you can swap two letters in s so the result is equal to goal, otherwise, return false.

Swapping letters is defined as taking two indices i and j (0-indexed) such that i != j and swapping the characters at s[i] and s[j].

For example, swapping at indices 0 and 2 in "abcd" results in "cbad".

TRY IT / Write examples, edge cases, and your own algorithm first.

## 0880 Decoded String At Index

MEDIUM

You are given an encoded string s. To decode the string to a tape, the encoded string is read one character at a time and the following steps are taken:
- If the character read is a letter, that letter is written onto the tape.
- If the character read is a digit d, the entire current tape is repeatedly written d - 1 more times in total.

Given an integer k, return the kth letter (1-indexed) in the decoded string.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 0884 Uncommon Words From Two Sentences

EASY

A sentence is a string of single-space separated words where each word consists only of lowercase letters.

A word is uncommon if it appears exactly once in one of the sentences, and does not appear in the other sentence.

Given two sentences s1 and s2, return a list of all the uncommon words. You may return the answer in any order.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 0893 Groups Of Special Equivalent Strings

MEDIUM

You are given an array of strings of the same length words.

In one move, you can swap any two even indexed characters or any two odd indexed characters of a string words[i].

Two strings words[i] and words[j] are special-equivalent if after any number of moves, words[i] == words[j].
- For example, words[i] = "zzxy" and words[j] = "xyzz" are special-equivalent because we may make the moves "zzxy" -> "xzzy" -> "xyzz".

A group of special-equivalent strings from words is a non-empty subset of words such that:
- Every pair of strings in the group are special equivalent, and
- The group is the largest size possible (i.e., there is not a string words[i] not in the group such that words[i] is special-equivalent to every string in the group).

Return the number of groups of special-equivalent strings from words.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 0902 Numbers At Most N Given Digit Set

HARD

Given an array of digits which is sorted in non-decreasing order. You can write numbers using each digits[i] as many times as we want. For example, if digits = ['1','3','5'], we may write numbers such as '13', '551', and '1351315'.

Return the number of positive integers that can be generated that are less than or equal to a given integer n.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 0917 Reverse Only Letters

EASY

Given a string s, reverse the string according to the following rules:
- All the characters that are not English letters remain in the same position.
- All the English letters (lowercase or uppercase) should be reversed.

Return s after reversing it.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 0926 Flip String To Monotone Increasing

MEDIUM

A binary string is monotone increasing if it consists of some number of 0's (possibly none), followed by some number of 1's (also possibly none).

You are given a binary string s. You can flip s[i] changing it from 0 to 1 or from 1 to 0.

Return the minimum number of flips to make s monotone increasing.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 0942 Di String Match

EASY

A permutation perm of n + 1 integers of all the integers in the range [0, n] can be represented as a string s of length n where:
- s[i] == 'I' if perm[i] < perm[i + 1], and
- s[i] == 'D' if perm[i] > perm[i + 1].

Given a string s, reconstruct the permutation perm and return it. If there are multiple valid permutations perm, return any of them.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 0943 Find The Shortest Superstring

HARD

Given an array of strings words, return the smallest string that contains each string in words as a substring. If there are multiple valid strings of the smallest length, return any of them.

You may assume that no string in words is a substring of another string in words.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 0949 Largest Time For Given Digits

MEDIUM

Given an array arr of 4 digits, find the latest 24-hour time that can be made using each digit exactly once.

24-hour times are formatted as "HH:MM", where HH is between 00 and 23, and MM is between 00 and 59. The earliest 24-hour time is 00:00, and the latest is 23:59.

Return the latest 24-hour time in "HH:MM" format. If no valid time can be made, return an empty string.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0966 Vowel Spellchecker

MEDIUM

Given a wordlist, we want to implement a spellchecker that converts a query word into a correct word.
For a given query word, the spell checker handles two categories of spelling mistakes:

- Capitalization: If the query matches a word in the wordlist (case-insensitive), then the query word is returned with the same case as the case in the wordlist.
Example: wordlist = ["yellow"], query = "YellOw": correct = "yellow"
Example: wordlist = ["Yellow"], query = "yellow": correct = "Yellow"
Example: wordlist = ["yellow"], query = "yellow": correct = "yellow"
- Vowel Errors: If after replacing the vowels ('a', 'e', 'i', 'o', 'u') of the query word with any vowel individually, it matches a word in the wordlist (case-insensitive), then the query word is returned with the same case as the match in the wordlist.
Example: wordlist = ["YellOw"], query = "yollow": correct = "YellOw"
Example: wordlist = ["YellOw"], query = "yeellow": correct = "" (no match)
Example: wordlist = ["YellOw"], query = "yllw": correct = "" (no match)

In addition, the spell checker operates under the following precedence rules:
- When the query exactly matches a word in the wordlist (case-sensitive), you should return the same word back.
- When the query matches a word up to capitlization, you should return the first such match in the wordlist.
- When the query matches a word up to vowel errors, you should return the first such match in the wordlist.
- If the query has no matches in the wordlist, you should return the empty string.

Given some queries, return a list of words answer, where answer[i] is the correct word for query = queries[i].

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0984 String Without Aaa Or Bbb

MEDIUM

Given two integers a and b, return any string s such that:
- s has length a + b and contains exactly a 'a' letters, and exactly b 'b' letters,
- The substring 'aaa' does not occur in s, and
- The substring 'bbb' does not occur in s.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0988 Smallest String Starting From Leaf

MEDIUM

You are given the root of a binary tree where each node has a value in the range [0, 25] representing the letters 'a' to 'z'.

Return the lexicographically smallest string that starts at a leaf of this tree and ends at the root.

As a reminder, any shorter prefix of a string is lexicographically smaller.

For example, "ab" is lexicographically smaller than "aba".

A leaf of a node is a node that has no children.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1002 Find Common Characters

EASY

Given a string array words, return an array of all characters that show up in all strings within the words (including duplicates). You may return the answer in any order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1012 Numbers With Repeated Digits

HARD

Given an integer n, return the number of positive integers in the range [1, n] that have at least one repeated digit.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1018 Binary Prefix Divisible By 5

EASY

You are given a binary array nums (0-indexed).

We define xi as the number whose binary representation is the subarray nums[0..i] (from most-significant-bit to least-significant-bit).

For example, if nums = [1,0,1], then x0 = 1, x1 = 2, and x2 = 5.

Return an array of booleans answer where answer[i] is true if xi is divisible by 5.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1032 Stream Of Characters

HARD

Design an algorithm that accepts a stream of characters and checks if a suffix of these characters is a string of a given array of strings words.

For example, if words = ["abc", "xyz"] and the stream added the four characters (one by one) 'a', 'x', 'y', and 'z', your algorithm should detect that the suffix "xyz" of the characters "axyz" matches "xyz" from words.

Implement the StreamChecker class:
- StreamChecker(String[] words) Initializes the object with the strings array words.
- boolean query(char letter) Accepts a new character from the stream and returns true if any non-empty suffix from the stream forms a word that is in words.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1047 Remove All Adjacent Duplicates In String

EASY

You are given a string s consisting of lowercase English letters. A duplicate removal consists of choosing two adjacent and equal letters and removing them.

We repeatedly make duplicate removals on s until we no longer can.

Return the final string after all such duplicate removals have been made. It can be proven that the answer is unique.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1048 Longest String Chain

**MEDIUM**

You are given an array of words where each word consists of lowercase English letters.

wordA is a predecessor of wordB if and only if we can insert exactly one letter anywhere in wordA without changing the order of the other characters to make it equal to wordB.

For example, "abc" is a predecessor of "abac", while "cba" is not a predecessor of "bcad".

A word chain is a sequence of words [word1, word2, ..., wordk] with k >= 1, where word1 is a predecessor of word2, word2 is a predecessor of word3, and so on. A single word is trivially a word chain with k == 1.

Return the length of the longest possible word chain with words chosen from the given list of words.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1055 Shortest Way To Form String

**MEDIUM**

A subsequence of a string is a new string that is formed from the original string by deleting some (can be none) of the characters without disturbing the relative positions of the remaining characters. (i.e., "ace" is a subsequence of "abcde" while "aec" is not).

Given two strings source and target, return the minimum number of subsequences of source such that their concatenation equals target. If the task is impossible, return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1065 Index Pairs Of A String

**EASY**

Given a string text and an array of strings words, return an array of all index pairs [i, j] so that the substring text[i...j] is in words.

Return the pairs [i, j] in sorted order (i.e., sort them by their first coordinate, and in case of ties sort them by their second coordinate).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1067 Digit Count In Range

**HARD**

Given a single-digit integer d and two integers low and high, return the number of times that d occurs as a digit in all integers in the inclusive range [low, high].

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1071 Greatest Common Divisor Of Strings

**EASY**

For two strings s and t, we say "t divides s" if and only if s = t + t + t + ... + t + t (i.e., t is concatenated with itself one or more times).

Given two strings str1 and str2, return the largest string x such that x divides both str1 and str2.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1079 Letter Tile Possibilities

**MEDIUM**

You have n tiles, where each tile has one letter tiles[i] printed on it.

Return the number of possible non-empty sequences of letters you can make using the letters printed on those tiles.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1081 Smallest Subsequence Of Distinct Characters

**MEDIUM**

Given a string s, return the lexicographically smallest subsequence of s that contains all the distinct characters of s exactly once.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1085 Sum Of Digits In The Minimum Number

**EASY**

Given an integer array nums, return 0 if the sum of the digits of the minimum integer in nums is odd, or 1 otherwise.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1119 Remove Vowels From A String

**EASY**

Given a string s, remove the vowels 'a', 'e', 'i', 'o', and 'u' from it, and return the new string.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1144 Decrease Elements To Make Array Zigzag

**MEDIUM**

Given an array nums of integers, a move consists of choosing any element and decreasing it by 1.

An array A is a zigzag array if either:
- Every even-indexed element is greater than adjacent elements, ie.
A[0] > A[1] < A[2] > A[3] < A[4] > ...
- OR, every odd-indexed element is greater than adjacent elements, ie.
A[0] < A[1] > A[2] < A[3] > A[4] < ...

Return the minimum number of moves to transform the given array nums into a zigzag array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1153 String Transforms Into Another String

**HARD**

Given two strings str1 and str2 of the same length, determine whether you can transform str1 into str2 by doing zero or more conversions.

In one conversion you can convert all occurrences of one character in str1 to any other lowercase English character.

Return true if and only if you can transform str1 into str2.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1160 Find Words That Can Be Formed By Characters

**EASY**

You are given an array of strings words and a string chars.

A string is good if it can be formed by characters from chars (each character can only be used once).

Return the sum of lengths of all good strings in words.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1170 Compare Strings By Frequency Of The Smallest Character

**MEDIUM**

Let the function f(s) be the frequency of the lexicographically smallest character in a non-empty string s. For example, if s = "dcce" then f(s) = 2 because the lexicographically smallest character is 'c', which has a frequency of 2.

You are given an array of strings words and another array of query strings queries. For each query queries[i], count the number of words in words such that f(queries[i]) < f(W) for each W in words.

Return an integer array answer, where each answer[i] is the answer to the ith query.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1178 Number Of Valid Words For Each Puzzle

**HARD**

With respect to a given puzzle string, a word is valid if both the following conditions are satisfied:
- word contains the first letter of puzzle.
- For each letter in word, that letter is in puzzle.
- For example, if the puzzle is "abcdefg", then valid words are "faced", "cabbage", and "baggage", while
- invalid words are "beefed" (does not include 'a') and "based" (includes 's' which is not in the puzzle).

Return an array answer, where answer[i] is the number of words in the given word list words that is valid with respect to the puzzle puzzles[i].

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1202 Smallest String With Swaps

**MEDIUM**

You are given a string s, and an array of pairs of indices in the string pairs where pairs[i] = [a, b] indicates 2 indices(0-indexed) of the string.

You can swap the characters at any pair of indices in the given pairs any number of times.

Return the lexicographically smallest string that s can be changed to after using the swaps.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1209 Remove All Adjacent Duplicates In String Ii

MEDIUM

You are given a string s and an integer k, a k duplicate removal consists of choosing k adjacent and equal letters from s and removing them, causing the left and the right side of the deleted substring to concatenate together.

We repeatedly make k duplicate removals on s until we no longer can.

Return the final string after all such duplicate removals have been made. It is guaranteed that the answer is unique.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1221 Split A String In Balanced Strings

EASY

Balanced strings are those that have an equal quantity of 'L' and 'R' characters.

Given a balanced string s, split it into some number of substrings such that:
- Each substring is balanced.

Return the maximum number of balanced strings you can obtain.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1239 Maximum Length Of A Concatenated String With Unique Characters

MEDIUM

You are given an array of strings arr. A string s is formed by the concatenation of a subsequence of arr that has unique characters.

Return the maximum possible length of s.

A subsequence is an array that can be derived from another array by deleting some or no elements without changing the order of the remaining elements.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1247 Minimum Swaps To Make Strings Equal

MEDIUM

You are given two strings s1 and s2 of equal length consisting of letters "x" and "y" only. Your task is to make these two strings equal to each other. You can swap any two characters that belong to different strings, which means: swap s1[i] and s2[j].

Return the minimum number of swaps required to make s1 and s2 equal, or return -1 if it is impossible to do so.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1255 Maximum Score Words Formed By Letters

HARD

Given a list of words, list of  single letters (might be repeating) and score of every character.

Return the maximum score of any valid set of words formed by using the given letters (words[i] cannot be used two or more times).

It is not necessary to use all characters in letters and each letter can only be used once. Score of letters 'a', 'b', 'c', ... ,'z' is given by score[0], score[1], ... , score[25]

respectively.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1258 Synonymous Sentences

**MEDIUM**

You are given a list of equivalent string pairs synonyms where synonyms[i] = [si, ti] indicates that si and ti are equivalent strings. You are also given a sentence text.

Return all possible synonymous sentences sorted lexicographically.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1281 Subtract The Product And Sum Of Digits Of An Integer

**EASY**

Given an integer number n, return the difference between the product of its digits and the sum of its digits.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1291 Sequential Digits

**MEDIUM**

An integer has sequential digits if and only if each digit in the number is one more than the previous digit.
Return a sorted list of all the integers in the range [low, high] inclusive that have sequential digits.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1295 Find Numbers With Even Number Of Digits

**EASY**

Given an array nums of integers, return how many of them contain an even number of digits.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1309 Decrypt String From Alphabet To Integer Mapping

**EASY**

Given a string s formed by digits ('0' - '9') and '#' .
We want to map s to English lowercase characters as follows:

- Characters ('a' to 'i') are represented by ('1' to '9') respectively.
- Characters ('j' to 'z') are represented by ('10#' to '26#') respectively.

Return the string formed after mapping.

It's guaranteed that a unique mapping will always exist.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1312 Minimum Insertion Steps To Make A String Palindrome

**HARD**

Given a string s. In one step you can insert any character at any index of the string.

Return the minimum number of steps to make s palindrome.

A Palindrome String is one that reads the same backward as well as forward.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1320 Minimum Distance To Type A Word Using Two Fingers

**HARD**

You have a keyboard layout as shown above in the X-Y plane, where each English uppercase letter is located at some coordinate.

For example, the letter 'A' is located at coordinate (0, 0), the letter 'B' is located at coordinate (0, 1), the letter 'P' is located at coordinate (2, 3) and the letter 'Z' is located at coordinate (4, 1).

Given the string word, return the minimum total distance to type such string using only two fingers.

The distance between coordinates (x1, y1) and (x2, y2) is |x1 - x2| + |y1 - y2|.

Note that the initial positions of your two fingers are considered free so do not count towards your total distance, also your two fingers do not have to start at the first letter or the first two letters.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1324 Print Words Vertically

**MEDIUM**

Given a string s. Return all the words vertically in the same order in which they appear in s.
Words are returned as a list of strings, complete with spaces when is necessary. (Trailing spaces are not allowed).
Each word would be put on only one column and that in one column there will be only one word.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1347 Minimum Number Of Steps To Make Two Strings Anagram

**MEDIUM**

You are given two strings of the same length s and t. In one step you can choose any character of t and replace it with another character.

Return the minimum number of steps to make t an anagram of s.

An Anagram of a string is a string that contains the same characters with a different (or the same) ordering.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1370 Increasing Decreasing String

**EASY**

You are given a string s. Reorder the string using the following algorithm:
and append it to the result.
and append it to the result.

If the smallest or largest character appears more than once, you may choose any occurrence to append to the result.

Return the resulting string after reordering s using this algorithm.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1374 Generate A String With Characters That Have Odd Counts

EASY

Given an integer n, return a string with n characters such that each character in such string occurs an odd number of times.

The returned string must contain only lowercase English letters. If there are multiples valid strings, return any of them.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1375 Number Of Times Binary String Is Prefix Aligned

MEDIUM

You have a 1-indexed binary string of length n where all the bits are 0 initially. We will flip all the bits of this binary string (i.e., change them from 0 to 1) one by one. You are given a 1-indexed integer array flips where flips[i] indicates that the bit at index i will be flipped in the ith step.

A binary string is prefix-aligned if, after the ith step, all the bits in the inclusive range [1, i] are ones and all the other bits are zeros.

Return the number of times the binary string is prefix-aligned during the flipping process.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1392 Longest Happy Prefix

HARD

A string is called a happy prefix if is a non-empty prefix which is also a suffix (excluding itself).

Given a string s, return the longest happy prefix of s. Return an empty string "" if no such prefix exists.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1397 Find All Good Strings

HARD

Given the strings s1 and s2 of size n and the string evil, return the number of good strings.

A good string has size n, it is alphabetically greater than or equal to s1, it is alphabetically smaller than or equal to s2, and it does not contain the string evil as a substring. Since the answer can be a huge number, return this modulo 109 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1400 Construct K Palindrome Strings

MEDIUM

Given a string s and an integer k, return true if you can use all the characters in s to construct k palindrome strings or false otherwise.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1405 Longest Happy String

MEDIUM

A string s is called happy if it satisfies the following conditions:
- s only contains the letters 'a', 'b', and 'c'.
- s does not contain any of "aaa", "bbb", or "ccc" as a substring.
- s contains at most a occurrences of the letter 'a'.
- s contains at most b occurrences of the letter 'b'.
- s contains at most c occurrences of the letter 'c'.

Given three integers a, b, and c, return the longest possible happy string. If there are multiple longest happy strings, return any of them. If there is no such string, return the empty string "".

A substring is a contiguous sequence of characters within a string.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1408 String Matching In An Array

EASY

Given an array of string words. Return all strings in words which is substring of another word in any order.

String words[i] is substring of words[j], if can be obtained removing some characters to left and/or right side of words[j].

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1417 Reformat The String

EASY

You are given an alphanumeric string s. (Alphanumeric string is a string consisting of lowercase English letters and digits).

You have to find a permutation of the string where no letter is followed by another letter and no digit is followed by another digit. That is, no two adjacent characters have the same type.

Return the reformatted string or return an empty string if it is impossible to reformat the string.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1422 Maximum Score After Splitting A String

EASY

Given a string s of zeros and ones, return the maximum score after splitting the string into two non-empty substrings (i.e. left substring and right substring).

The score after splitting a string is the number of zeros in the left substring plus the number of ones in the right substring.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1427 Perform String Shifts

EASY

You are given a string s containing lowercase English letters, and a matrix shift, where shift[i] = [directioni, amounti]:
- directioni can be 0 (for left shift) or 1 (for right shift).

- amounti is the amount by which string s is to be shifted.
- A left shift by 1 means remove the first character of s and append it to the end.
- Similarly, a right shift by 1 means remove the last character of s and add it to the beginning.

Return the final string after all operations.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1433 Check If A String Can Break Another String

**MEDIUM**

Given two strings: s1 and s2 with the same size, check if some permutation of string s1 can break some permutation of string s2 or vice-versa. In other words s2 can break s1 or vice-versa.

A string x can break string y (both of size n) if x[i] >= y[i] (in alphabetical order) for all i between 0 and n-1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1446 Consecutive Characters

**EASY**

The power of the string is the maximum length of a non-empty substring that contains only one unique character.

Given a string s, return the power of s.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1449 Form Largest Integer With Digits That Add Up To Target

**HARD**

Given an array of integers cost and an integer target, return the maximum integer you can paint under the following rules:
- The cost of painting a digit (i + 1) is given by cost[i] (0-indexed).
- The total cost used must be equal to target.
- The integer does not have 0 digits.

Since the answer may be very large, return it as a string. If there is no way to paint any integer given the condition, return "0".

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1451 Rearrange Words In A Sentence

**MEDIUM**

Given a sentence text (A sentence is a string of space-separated words) in the following format:
- First letter is in upper case.
- Each word in text are separated by a single space.

Your task is to rearrange the words in text such that all words are rearranged in an increasing order of their lengths. If two words have the same length, arrange them in their original order.

Return the new text following the format shown above.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1455 Check If A Word Occurs As A Prefix Of Any Word In A Sentence

EASY

Given a sentence that consists of some words separated by a single space, and a searchWord, check if searchWord is a prefix of any word in sentence.

Return the index of the word in sentence (1-indexed) where searchWord is a prefix of this word. If searchWord is a prefix of more than one word, return the index of the first word (minimum index). If there is no such word return -1.

A prefix of a string s is any leading contiguous substring of s.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1461 Check If A String Contains All Binary Codes Of Size K

MEDIUM

Given a binary string s and an integer k, return true if every binary code of length k is a substring of s. Otherwise, return false.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1505 Minimum Possible Integer After At Most K Adjacent Swaps On Digits

HARD

You are given a string num representing the digits of a very large integer and an integer k. You are allowed to swap any two adjacent digits of the integer at most k times.

Return the minimum integer you can obtain also as a string.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1525 Number Of Good Ways To Split A String

MEDIUM

You are given a string s.

A split is called good if you can split s into two non-empty strings sleft and sright where their concatenation is equal to s (i.e., sleft + sright = s) and the number of distinct letters in sleft and sright is the same.

Return the number of good splits you can make in s.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1528 Shuffle String

EASY

You are given a string s and an integer array indices of the same length. The string s will be shuffled such that the character at the ith position moves to indices[i] in the shuffled string.

Return the shuffled string.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1529 Minimum Suffix Flips

MEDIUM

You are given a 0-indexed binary string target of length n. You have another binary string s of length n that is initially set to all zeros. You want to make s equal to target.

In one operation, you can pick an index i where 0 <= i < n and flip all bits in the inclusive range [i, n - 1]. Flip means changing '0' to '1' and '1' to '0'.

Return the minimum number of operations needed to make s equal to target.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1531 String Compression Ii

HARD

Run-length encoding is a string compression method that works by replacing consecutive identical characters (repeated 2 or more times) with the concatenation of the character and the number marking the count of the characters (length of the run). For example, to compress the string "aabccc" we replace "aa" by "a2" and replace "ccc" by "c3". Thus the compressed string becomes "a2bc3".

Notice that in this problem, we are not adding '1' after single characters.

Given a string s and an integer k. You need to delete at most k characters from s such that the run-length encoded version of s has minimum length.

Find the minimum length of the run-length encoded version of s after deleting at most k characters.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1540 Can Convert String In K Moves

MEDIUM

Given two strings s and t, your goal is to convert s into t in k moves or less.

During the ith (1 <= i <= k) move you can:
- Choose any index j (1-indexed) from s, such that 1 <= j <= s.length and j has not been chosen in any previous move, and shift the character at that index i times.
- Do nothing.

Shifting a character means replacing it by the next letter in the alphabet (wrapping around so that 'z' becomes 'a'). Shifting a character by i means applying the shift operations i times.

Remember that any index j can be picked at most once.

Return true if it's possible to convert s into t in no more than k moves, otherwise return false.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1544 Make The String Great

EASY

Given a string s of lower and upper case English letters.

A good string is a string which doesn't have two adjacent characters s[i] and s[i + 1] where:
- 0 <= i <= s.length - 2
- s[i] is a lower-case letter and s[i + 1] is the same letter but in upper-case or vice-versa.

To make the string good, you can choose two adjacent characters that make the string bad and remove them. You can keep doing this until the string becomes good.

Return the string after making it good. The answer is guaranteed to be unique under the given constraints.

Notice that an empty string is also good.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1545 Find Kth Bit In Nth Binary String

**MEDIUM**

Given two positive integers n and k, the binary string Sn is formed as follows:
- S1 = "0"
- Si = Si - 1 + "1" + reverse(invert(Si - 1)) for i > 1

Where + denotes the concatenation operation, reverse(x) returns the reversed string x, and invert(x) inverts all the bits in x (0 changes to 1 and 1 changes to 0).

For example, the first four strings in the above sequence are:
- S1 = "0"
- S2 = "011"
- S3 = "0111001"
- S4 = "011100110110001"

Return the kth bit in Sn. It is guaranteed that k is valid for the given n.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1554 Strings Differ By One Character

**MEDIUM**

Given a list of strings dict where all the strings are of the same length.

Return true if there are 2 strings that only differ by 1 character in the same index, otherwise return false.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1573 Number Of Ways To Split A String

**MEDIUM**

Given a binary string s, you can split s into 3 non-empty strings s1, s2, and s3 where s1 + s2 + s3 = s.

Return the number of ways s can be split such that the number of ones is the same in s1, s2, and s3. Since the answer may be too large, return it modulo 109 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1576 Replace All S To Avoid Consecutive Repeating Characters

**MEDIUM**

Given a string s containing only lowercase English letters and the '?' character, convert all the '?' characters into lowercase letters such that the final string does not contain any consecutive repeating characters. You cannot modify the non '?' characters.

It is guaranteed that there are no consecutive repeating characters in the given string except for '?'.

Return the final string after all the conversions (possibly zero) have been made. If there is more than one solution, return any of them. It can be shown that an answer is always possible with the given constraints.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1592 Rearrange Spaces Between Words

EASY

You are given a string text of words that are placed among some number of spaces. Each word consists of one or more lowercase English letters and are separated by at least one space. It's guaranteed that text contains at least one word.

Rearrange the spaces so that there is an equal number of spaces between every pair of adjacent words and that number is maximized. If you cannot redistribute all the spaces equally, place the extra spaces at the end, meaning the returned string should be the same length as text.

Return the string after rearranging the spaces.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1616 Split Two Strings To Make Palindrome

MEDIUM

You are given two strings a and b of the same length. Choose an index and split both strings at the same index, splitting a into two strings: aprefix and asuffix where a = aprefix + asuffix, and splitting b into two strings: bprefix and bsuffix where b = bprefix + bsuffix. Check if aprefix + bsuffix or bprefix + asuffix forms a palindrome.

When you split a string s into sprefix and ssuffix, either ssuffix or sprefix is allowed to be empty. For example, if s = "abc", then "" + "abc", "a" + "bc", "ab" + "c" , and "abc" + "" are valid splits.

Return true if it is possible to form a palindrome string, otherwise return false.

Notice that x + y denotes the concatenation of strings x and y.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1618 Maximum Font To Fit A Sentence In A Screen

MEDIUM

You are given a string text. We want to display text on a screen of width w and height h. You can choose any font size from array fonts, which contains the available font sizes in ascending order.

You can use the FontInfo interface to get the width and height of any character at any available font size.

The FontInfo interface is defined as such:
interface FontInfo {
// Returns the width of character ch on the screen using font size fontSize.
// O(1) per call
public int getWidth(int fontSize, char ch);

// Returns the height of any character on the screen using font size fontSize.
// O(1) per call
public int getHeight(int fontSize);
}

The calculated width of text for some fontSize is the sum of every getWidth(fontSize, text[i]) call for each 0 <= i < text.length (0-indexed). The calculated height of text for some fontSize is getHeight(fontSize). Note that text is displayed on a single line.

It is guaranteed that FontInfo will return the same value if you call getHeight or getWidth with the same parameters.

It is also guaranteed that for any font size fontSize and any character ch:

- getHeight(fontSize) <= getHeight(fontSize+1)
- getWidth(fontSize, ch) <= getWidth(fontSize+1, ch)

Return the maximum font size you can use to display text on the screen. If text cannot fit on the display with any font size, return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1639 Number Of Ways To Form A Target String Given A Dictionary

**HARD**

You are given a list of strings of the same length words and a string target.

Your task is to form target using the given words under the following rules:
- target should be formed from left to right.
- To form the ith character (0-indexed) of target, you can choose the kth character of the jth string in words if target[i] = words[j][k].
- Once you use the kth character of the jth string of words, you can no longer use the xth character of any string in words where x <= k. In other words, all characters to the left of or at index k become unusable for every string.
- Repeat the process until you form the string target.

Notice that you can use multiple characters from the same string in words provided the conditions above are met.

Return the number of ways to form target from words. Since the answer may be too large, return it modulo 109 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1647 Minimum Deletions To Make Character Frequencies Unique

**MEDIUM**

A string s is called good if there are no two different characters in s that have the same frequency.

Given a string s, return the minimum number of characters you need to delete to make s good.

The frequency of a character in a string is the number of times it appears in the string. For example, in the string "aab", the frequency of 'a' is 2, while the frequency of 'b' is 1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1653 Minimum Deletions To Make String Balanced

**MEDIUM**

You are given a string s consisting only of characters 'a' and 'b'.

You can delete any number of characters in s to make s balanced. s is balanced if there is no pair of indices (i,j) such that i < j and s[i] = 'b' and s[j]= 'a'.

Return the minimum number of deletions needed to make s balanced.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1657 Determine If Two Strings Are Close

**MEDIUM**

Two strings are considered close if you can attain one from the other using the following operations:

- Operation 1: Swap any two existing characters.

For example, abcde -> aecdb
- Operation 2: Transform every occurrence of one existing character into another existing character, and do the same with the other character.
For example, aacabb -> bbcbaa (all a's turn into b's, and all b's turn into a's)

You can use the operations on either string as many times as necessary.

Given two strings, word1 and word2, return true if word1 and word2 are close, and false otherwise.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1662 Check If Two String Arrays Are Equivalent

**EASY**

Given two string arrays word1 and word2, return true if the two arrays represent the same string, and false otherwise.

A string is represented by an array if the array elements concatenated in order forms the string.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1663 Smallest String With A Given Numeric Value

**MEDIUM**

The numeric value of a lowercase character is defined as its position (1-indexed) in the alphabet, so the numeric value of a is 1, the numeric value of b is 2, the numeric value of c is 3, and so on.

The numeric value of a string consisting of lowercase characters is defined as the sum of its characters' numeric values. For example, the numeric value of the string "abe" is equal to 1 + 2 + 5 = 8.

You are given two integers n and k. Return the lexicographically smallest string with length equal to n and numeric value equal to k.

Note that a string x is lexicographically smaller than string y if x comes before y in dictionary order, that is, either x is a prefix of y, or if i is the first position such that x[i] != y[i], then x[i] comes before y[i] in alphabetic order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1684 Count The Number Of Consistent Strings

**EASY**

You are given a string allowed consisting of distinct characters and an array of strings words. A string is consistent if all characters in the string appear in the string allowed.

Return the number of consistent strings in the array words.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1702 Maximum Binary String After Change

**MEDIUM**

You are given a binary string binary consisting of only 0's or 1's. You can apply each of the following operations any number of times:
- Operation 1: If the number contains the substring "00", you can replace it with "10".
- For example, "00010" -> "10010"
- Operation 2: If the number contains the substring "10", you can replace it with "01".
- For example, "00010" -> "00001"

Return the maximum binary string you can obtain after any number of operations. Binary string x is greater than binary string y if x's decimal representation is greater than y's decimal representation.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1704 Determine If String Halves Are Alike

**EASY**

You are given a string s of even length. Split this string into two halves of equal lengths, and let a be the first half and b be the second half.

Two strings are alike if they have the same number of vowels ('a', 'e', 'i', 'o', 'u', 'A', 'E', 'I', 'O', 'U'). Notice that s contains uppercase and lowercase letters.

Return true if a and b are alike. Otherwise, return false.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1736 Latest Time By Replacing Hidden Digits

**EASY**

You are given a string time in the form of hh:mm, where some of the digits in the string are hidden (represented by ?).

The valid times are those inclusively between 00:00 and 23:59.

Return the latest valid time you can get from time by replacing the hidden digits.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1737 Change Minimum Characters To Satisfy One Of Three Conditions

**MEDIUM**

You are given two strings a and b that consist of lowercase letters. In one operation, you can change any character in a or b to any lowercase letter.

Your goal is to satisfy one of the following three conditions:
- Every letter in a is strictly less than every letter in b in the alphabet.
- Every letter in b is strictly less than every letter in a in the alphabet.
- Both a and b consist of only one distinct letter.

Return the minimum number of operations needed to achieve your goal.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1750 Minimum Length Of String After Deleting Similar Ends

**MEDIUM**

Given a string s consisting only of characters 'a', 'b', and 'c'. You are asked to apply the following algorithm on the string any number of times:

Return the minimum length of s after performing the above operation any number of times (possibly zero times).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1754 Largest Merge Of Two Strings

**MEDIUM**

You are given two strings word1 and word2. You want to construct a string merge in the

following way: while either word1 or word2 are non-empty, choose one of the following options:
- If word1 is non-empty, append the first character in word1 to merge and delete it from word1.
- For example, if word1 = "abc" and merge = "dv", then after choosing this operation, word1 = "bc" and merge = "dva".
- If word2 is non-empty, append the first character in word2 to merge and delete it from word2.
- For example, if word2 = "abc" and merge = "", then after choosing this operation, word2 = "bc" and merge = "a".

Return the lexicographically largest merge you can construct.

A string a is lexicographically larger than a string b (of the same length) if in the first position where a and b differ, a has a character strictly larger than the corresponding character in b. For example, "abcd" is lexicographically larger than "abcc" because the first position they differ is at the fourth character, and d is greater than c.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1758 Minimum Changes To Make Alternating Binary String

**EASY**

You are given a string s consisting only of the characters '0' and '1'. In one operation, you can change any '0' to '1' or vice versa.

The string is called alternating if no two adjacent characters are equal. For example, the string "010" is alternating, while the string "0100" is not.

Return the minimum number of operations needed to make s alternating.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1768 Merge Strings Alternately

**EASY**

You are given two strings word1 and word2. Merge the strings by adding letters in alternating order, starting with word1. If a string is longer than the other, append the additional letters onto the end of the merged string.

Return the merged string.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1784 Check If Binary String Has At Most One Segment Of Ones

**EASY**

Given a binary string s without leading zeros, return true if s contains at most one contiguous segment of ones. Otherwise, return false.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1790 Check If One String Swap Can Make Strings Equal

**EASY**

You are given two strings s1 and s2 of equal length. A string swap is an operation where you choose two indices in a string (not necessarily different) and swap the characters at these indices.

Return true if it is possible to make both strings equal by performing at most one string swap on exactly one of the strings. Otherwise, return false.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1796 Second Largest Digit In A String

EASY

Given an alphanumeric string s, return the second largest numerical digit that appears in s, or -1 if it does not exist.

An alphanumeric string is a string consisting of lowercase English letters and digits.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1805 Number Of Different Integers In A String

EASY

You are given a string word that consists of digits and lowercase English letters.

You will replace every non-digit character with a space. For example, "a123bc34d8ef34" will become " 123  34 8  34". Notice that you are left with some integers that are separated by at least one space: "123", "34", "8", and "34".

Return the number of different integers after performing the replacement operations on word.

Two integers are considered different if their decimal representations without any leading zeros are different.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1813 Sentence Similarity Iii

MEDIUM

You are given two strings sentence1 and sentence2, each representing a sentence composed of words. A sentence is a list of words that are separated by a single space with no leading or trailing spaces. Each word consists of only uppercase and lowercase English characters.

Two sentences s1 and s2 are considered similar if it is possible to insert an arbitrary sentence (possibly empty) inside one of these sentences such that the two sentences become equal. Note that the inserted sentence must be separated from existing words by spaces.

For example:
- s1 = "Hello Jane" and s2 = "Hello my name is Jane" can be made equal by inserting "my name is" between "Hello" and "Jane" in s1.
- s1 = "Frog cool" and s2 = "Frogs are cool" are not similar, since although there is a sentence "s are" inserted into s1, it is not separated from "Frog" by a space.

Given two sentences sentence1 and sentence2, return true if sentence1 and sentence2 are similar. Otherwise, return false.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1816 Truncate Sentence

EASY

A sentence is a list of words that are separated by a single space with no leading or trailing spaces. Each of the words consists of only uppercase and lowercase English letters (no punctuation).
- For example, "Hello World", "HELLO", and "hello world hello world" are all sentences.

You are given a sentence s and an integer k. You want to truncate s such that it contains only the first k words. Return s after truncating it.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1832 Check If The Sentence Is Pangram

EASY

A pangram is a sentence where every letter of the English alphabet appears at least once.

Given a string sentence containing only lowercase English letters, return true if sentence is a pangram, or false otherwise.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1837 Sum Of Digits In Base K

EASY

Given an integer n (in base 10) and a base k, return the sum of the digits of n after converting n from base 10 to base k.

After converting, each digit should be interpreted as a base 10 number, and the sum should be returned in base 10.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1842 Next Palindrome Using Same Digits

HARD

You are given a numeric string num, representing a very large palindrome.

Return the smallest palindrome larger than num that can be created by rearranging its digits. If no such palindrome exists, return an empty string "".

A palindrome is a number that reads the same backward as forward.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1844 Replace All Digits With Characters

EASY

You are given a 0-indexed string s that has lowercase English letters in its even indices and digits in its odd indices.

You must perform an operation shift(c, x), where c is a character and x is a digit, that returns the xth character after c.
- For example, shift('a', 5) = 'f' and shift('x', 0) = 'x'.

For every odd index i, you want to replace the digit s[i] with the result of the shift(s[i-1], s[i]) operation.

Return s after replacing all digits. It is guaranteed that shift(s[i-1], s[i]) will never exceed 'z'.

Note that shift(c, x) is not a preloaded function, but an operation to be implemented as part of the solution.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1849 Splitting A String Into Descending Consecutive Values

MEDIUM

You are given a string s that consists of only digits.

Check if we can split s into two or more non-empty substrings such that the numerical values of

the substrings are in descending order and the difference between numerical values of every two adjacent substrings is equal to 1.
- For example, the string s = "0090089" can be split into ["0090", "089"] with numerical values [90,89]. The values are in descending order and adjacent values differ by 1, so this way is valid.
- Another example, the string s = "001" can be split into ["0", "01"], ["00", "1"], or ["0", "0", "1"]. However all the ways are invalid because they have numerical values [0,1], [0,1], and [0,0,1] respectively, all of which are not in descending order.

Return true if it is possible to split s as described above, or false otherwise.

A substring is a contiguous sequence of characters in a string.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1858 Longest Word With All Prefixes

**MEDIUM**

Given an array of strings words, find the longest string in words such that every prefix of it is also in words.
- For example, let words = ["a", "app", "ap"]. The string "app" has prefixes "ap" and "a", all of which are in words.

Return the string described above. If there is more than one string with the same length, return the lexicographically smallest one, and if no string exists, return "".
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1864 Minimum Number Of Swaps To Make The Binary String Alternating

**MEDIUM**

Given a binary string s, return the minimum number of character swaps to make it alternating, or -1 if it is impossible.

The string is called alternating if no two adjacent characters are equal. For example, the strings "010" and "1010" are alternating, while the string "0100" is not.

Any two characters may be swapped, even if they are not adjacent.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1880 Check If Word Equals Summation Of Two Words

**EASY**

The letter value of a letter is its position in the alphabet starting from 0 (i.e. 'a' -> 0, 'b' -> 1, 'c' -> 2, etc.).

The numerical value of some string of lowercase English letters s is the concatenation of the letter values of each letter in s, which is then converted into an integer.

For example, if s = "acb", we concatenate each letter's letter value, resulting in "021". After converting it, we get 21.

You are given three strings firstWord, secondWord, and targetWord, each consisting of lowercase English letters 'a' through 'j' inclusive.

Return true if the summation of the numerical values of firstWord and secondWord equals the numerical value of targetWord, or false otherwise.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1888 Minimum Number Of Flips To Make The Binary String Alternating

MEDIUM

You are given a binary string s. You are allowed to perform two types of operations on the string in any sequence:
- Type-1: Remove the character at the start of the string s and append it to the end of the string.
- Type-2: Pick any character in s and flip its value, i.e., if its value is '0' it becomes '1' and vice-versa.

Return the minimum number of type-2 operations you need to perform such that s becomes alternating.

The string is called alternating if no two adjacent characters are equal.

For example, the strings "010" and "1010" are alternating, while the string "0100" is not.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1897 Redistribute Characters To Make All Strings Equal

EASY

You are given an array of strings words (0-indexed).

In one operation, pick two distinct indices i and j, where words[i] is a non-empty string, and move any character from words[i] to any position in words[j].

Return true if you can make every string in words equal using any number of operations, and false otherwise.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1898 Maximum Number Of Removable Characters

MEDIUM

You are given two strings s and p where p is a subsequence of s. You are also given a distinct 0-indexed integer array removable containing a subset of indices of s (s is also 0-indexed).

You want to choose an integer k (0 <= k <= removable.length) such that, after removing k characters from s using the first k indices in removable, p is still a subsequence of s. More formally, you will mark the character at s[removable[i]] for each 0 <= i < k, then remove all marked characters and check if p is still a subsequence.

Return the maximum k you can choose such that p is still a subsequence of s after the removals.

A subsequence of a string is a new string generated from the original string with some characters (can be none) deleted without changing the relative order of the remaining characters.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1903 Largest Odd Number In String

EASY

You are given a string num, representing a large integer. Return the largest-valued odd integer (as a string) that is a non-empty substring of num, or an empty string "" if no odd integer exists.

A substring is a contiguous sequence of characters within a string.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1935 Maximum Number Of Words You Can Type

EASY

There is a malfunctioning keyboard where some letter keys do not work. All other keys on the keyboard work properly.

Given a string text of words separated by a single space (no leading or trailing spaces) and a string brokenLetters of all distinct letter keys that are broken, return the number of words in text you can fully type using this keyboard.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1941 Check If All Characters Have Equal Number Of Occurrences

EASY

Given a string s, return true if s is a good string, or false otherwise.

A string s is good if all the characters that appear in s have the same number of occurrences (i.e., the same frequency).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1945 Sum Of Digits Of String After Convert

EASY

You are given a string s consisting of lowercase English letters, and an integer k. Your task is to convert the string into an integer by a special process, and then transform it by summing its digits repeatedly k times. More specifically, perform the following steps: replace 'a' with 1, 'b' with 2, ..., 'z' with 26).

For example, if s = "zbax" and k = 2, then the resulting integer would be 8 by the following operations:

Return the resulting integer after performing the operations described above.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1957 Delete Characters To Make Fancy String

EASY

A fancy string is a string where no three consecutive characters are equal.

Given a string s, delete the minimum possible number of characters from s to make it fancy.

Return the final string after the deletion. It can be shown that the answer will always be unique.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1961 Check If String Is A Prefix Of Array

EASY

Given a string s and an array of strings words, determine whether s is a prefix string of words.

A string s is a prefix string of words if s can be made by concatenating the first k strings in words for some positive k no larger than words.length.

Return true if s is a prefix string of words, or false otherwise.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1963 Minimum Number Of Swaps To Make The String Balanced

MEDIUM

You are given a 0-indexed string s of even length n. The string consists of exactly n / 2 opening brackets '[' and n / 2 closing brackets ']'.

A string is called balanced if and only if:
- It is the empty string, or
- It can be written as AB, where both A and B are balanced strings, or
- It can be written as [C], where C is a balanced string.

You may swap the brackets at any two indices any number of times.

Return the minimum number of swaps to make s balanced.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1974 Minimum Time To Type Word Using Special Typewriter

EASY

There is a special typewriter with lowercase English letters 'a' to 'z' arranged in a circle with a pointer. A character can only be typed if the pointer is pointing to that character. The pointer is initially pointing to the character 'a'.

Each second, you may perform one of the following operations:
- Move the pointer one character counterclockwise or clockwise.
- Type the character the pointer is currently on.

Given a string word, return the minimum number of seconds to type out the characters in word.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1980 Find Unique Binary String

MEDIUM

Given an array of strings nums containing n unique binary strings each of length n, return a binary string of length n that does not appear in nums. If there are multiple answers, you may return any of them.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1996 The Number Of Weak Characters In The Game

MEDIUM

You are playing a game that contains multiple characters, and each of the characters has two main properties: attack and defense. You are given a 2D integer array properties where properties[i] = [attacki, defensei] represents the properties of the ith character in the game.

A character is said to be weak if any other character has both attack and defense levels strictly greater than this character's attack and defense levels. More formally, a character i is said to be weak if there exists another character j where attackj > attacki and defensej > defensei.

Return the number of weak characters.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1999 Smallest Greater Multiple Made Of Two Digits

MEDIUM

Given three integers, k, digit1, and digit2, you want to find the smallest integer that is:
- Larger than k,
- A multiple of k, and
- Comprised of only the digits digit1 and/or digit2.

Return the smallest such integer. If no such integer exists or the integer exceeds the limit of a signed 32-bit integer (231 - 1), return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2000 Reverse Prefix Of Word

EASY

Given a 0-indexed string word and a character ch, reverse the segment of word that starts at index 0 and ends at the index of the first occurrence of ch (inclusive). If the character ch does not exist in word, do nothing.

For example, if word = "abcdefd" and ch = "d", then you should reverse the segment that starts at 0 and ends at 3 (inclusive). The resulting string will be "dcbaefd".

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2018 Check If Word Can Be Placed In Crossword

MEDIUM

You are given an m x n matrix board, representing the current state of a crossword puzzle. The crossword contains lowercase English letters (from solved words), ' ' to represent any empty cells, and '#' to represent any blocked cells.

A word can be placed horizontally (left to right or right to left) or vertically (top to bottom or bottom to top) in the board if:
- It does not occupy a cell containing the character '#'.
- The cell each letter is placed in must either be ' ' (empty) or match the letter already on the board.
- There must not be any empty cells ' ' or other lowercase letters directly left or right of the word if the word was placed horizontally.
- There must not be any empty cells ' ' or other lowercase letters directly above or below the word if the word was placed vertically.

Given a string word, return true if word can be placed in board, or false otherwise.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2023 Number Of Pairs Of Strings With Concatenation Equal To Target

MEDIUM

Given an array of digit strings nums and a digit string target, return the number of pairs of indices (i, j) (where i != j) such that the concatenation of nums[i] + nums[j] equals target.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2027 Minimum Moves To Convert String

EASY

You are given a string s consisting of n characters which are either 'X' or 'O'.

A move is defined as selecting three consecutive characters of s and converting them to 'O'. Note that if a move is applied to the character 'O', it will stay the same.

Return the minimum number of moves required so that all the characters of s are converted ]to 'O'.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2030 Smallest K Length Subsequence With Occurrences Of A Letter

**HARD**

You are given a string s, an integer k, a letter letter, and an integer repetition.

Return the lexicographically smallest subsequence of s of length k that has the letter letter appear at least repetition times. The test cases are generated so that the letter appears in s at least repetition times.

A subsequence is a string that can be derived from another string by deleting some or no characters without changing the order of the remaining characters.

A string a is lexicographically smaller than a string b if in the first position where a and b differ, string a has a letter that appears earlier in the alphabet than the corresponding letter in b.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2042 Check If Numbers Are Ascending In A Sentence

**EASY**

A sentence is a list of tokens separated by a single space with no leading or trailing spaces. Every token is either a positive number consisting of digits 0-9 with no leading zeros, or a word consisting of lowercase English letters.
- For example, "a puppy has 2 eyes 4 legs" is a sentence with seven tokens: "2" and "4" are numbers and the other tokens such as "puppy" are words.

Given a string s representing a sentence, you need to check if all the numbers in s are strictly increasing from left to right (i.e., other than the last number, each number is strictly smaller than the number on its right in s).

Return true if so, or false otherwise.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2047 Number Of Valid Words In A Sentence

**EASY**

A sentence consists of lowercase letters ('a' to 'z'), digits ('0' to '9'), hyphens ('-'), punctuation marks ('!', '.', and ','), and spaces (' ') only. Each sentence can be broken down into one or more tokens separated by one or more spaces ' '.

A token is a valid word if all three of the following are true:
- It only contains lowercase letters, hyphens, and/or punctuation (no digits).
- There is at most one hyphen '-'. If present, it must be surrounded by lowercase characters ("a-b" is valid, but "-ab" and "ab-" are not valid).
- There is at most one punctuation mark. If present, it must be at the end of the token ("ab,", "cd!", and "." are valid, but "a!b" and "c.," are not valid).

Examples of valid words include "a-b.", "afad", "ba-c", "a!", and "!".

Given a string sentence, return the number of valid words in sentence.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2052 Minimum Cost To Separate Sentence Into Rows

MEDIUM

You are given a string sentence containing words separated by spaces, and an integer k. Your task is to separate sentence into rows where the number of characters in each row is at most k. You may assume that sentence does not begin or end with a space, and the words in sentence are separated by a single space.

You can split sentence into rows by inserting line breaks between words in sentence. A word cannot be split between two rows. Each word must be used exactly once, and the word order cannot be rearranged. Adjacent words in a row should be separated by a single space, and rows should not begin or end with spaces.

The cost of a row with length n is (k - n)2, and the total cost is the sum of the costs for all rows except the last one.
- For example if sentence = "i love leetcode" and k = 12:
- Separating sentence into "i", "love", and "leetcode" has a cost of (12 - 1)2 + (12 - 4)2 = 185.
- Separating sentence into "i love", and "leetcode" has a cost of (12 - 6)2 = 36.
- Separating sentence into "i", and "love leetcode" is not possible because the length of "love leetcode" is greater than k.

Return the minimum possible total cost of separating sentence into rows.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2053 Kth Distinct String In An Array

MEDIUM

A distinct string is a string that is present only once in an array.

Given an array of strings arr, and an integer k, return the kth distinct string present in arr. If there are fewer than k distinct strings, return an empty string "".

Note that the strings are considered in the order in which they appear in the array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2060 Check If An Original String Exists Given Two Encoded Strings

HARD

An original string, consisting of lowercase English letters, can be encoded by the following steps:
- Arbitrarily split it into a sequence of some number of non-empty substrings.
- Arbitrarily choose some elements (possibly none) of the sequence, and replace each with its length (as a numeric string).
- Concatenate the sequence as the encoded string.

For example, one way to encode an original string "abcdefghijklmnop" might be:
- Split it as a sequence: ["ab", "cdefghijklmn", "o", "p"].
- Choose the second and third elements to be replaced by their lengths, respectively. The sequence becomes ["ab", "12", "1", "p"].
- Concatenate the elements of the sequence to get the encoded string: "ab121p".

Given two encoded strings s1 and s2, consisting of lowercase English letters and digits 1-9 (inclusive), return true if there exists an original string that could be encoded as both s1 and s2. Otherwise, return false.

Note: The test cases are generated such that the number of consecutive digits in s1 and s2

does not exceed 3.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2068 Check Whether Two Strings Are Almost Equivalent

**EASY**

Two strings word1 and word2 are considered almost equivalent if the differences between the frequencies of each letter from 'a' to 'z' between word1 and word2 is at most 3.

Given two strings word1 and word2, each of length n, return true if word1 and word2 are almost equivalent, or false otherwise.

The frequency of a letter x is the number of times it occurs in the string.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2075 Decode The Slanted Ciphertext

**MEDIUM**

A string originalText is encoded using a slanted transposition cipher to a string encodedText with the help of a matrix having a fixed number of rows rows.

originalText is placed first in a top-left to bottom-right manner.

The blue cells are filled first, followed by the red cells, then the yellow cells, and so on, until we reach the end of originalText. The arrow indicates the order in which the cells are filled. All empty cells are filled with ' '. The number of columns is chosen such that the rightmost column will not be empty after filling in originalText.

encodedText is then formed by appending all characters of the matrix in a row-wise fashion.

The characters in the blue cells are appended first to encodedText, then the red cells, and so on, and finally the yellow cells. The arrow indicates the order in which the cells are accessed.

For example, if originalText = "cipher" and rows = 3, then we encode it in the following manner.

The blue arrows depict how originalText is placed in the matrix, and the red arrows denote the order in which encodedText is formed. In the above example, encodedText = "ch ie pr".

Given the encoded string encodedText and number of rows rows, return the original string originalText.

Note: originalText does not have any trailing spaces ' '. The test cases are generated such that there is only one possible originalText.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2085 Count Common Words With One Occurrence

**EASY**

Given two string arrays words1 and words2, return the number of strings that appear exactly once in each of the two arrays.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2094 Finding 3 Digit Even Numbers

**EASY**

You are given an integer array digits, where each element is a digit. The array may contain duplicates.

You need to find all the unique integers that follow the given requirements:

- The integer consists of the concatenation of three elements from digits in any arbitrary order.
- The integer does not have leading zeros.
- The integer is even.

For example, if the given digits were [1, 2, 3], integers 132 and 312 follow the requirements.

Return a sorted array of the unique integers.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2108 Find First Palindromic String In The Array

**EASY**

Given an array of strings words, return the first palindromic string in the array. If there is no such string, return an empty string "".

A string is palindromic if it reads the same forward and backward.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2109 Adding Spaces To A String

**MEDIUM**

You are given a 0-indexed string s and a 0-indexed integer array spaces that describes the indices in the original string where spaces will be added. Each space should be inserted before the character at the given index.
- For example, given s = "EnjoyYourCoffee" and spaces = [5, 9], we place spaces before 'Y' and 'C', which are at indices 5 and 9 respectively. Thus, we obtain "Enjoy Your Coffee".

Return the modified string after the spaces have been added.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2114 Maximum Number Of Words Found In Sentences

**EASY**

A sentence is a list of words that are separated by a single space with no leading or trailing spaces.

You are given an array of strings sentences, where each sentences[i] represents a single sentence.

Return the maximum number of words that appear in a single sentence.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2120 Execution Of All Suffix Instructions Staying In A Grid

**MEDIUM**

There is an n x n grid, with the top-left cell at (0, 0) and the bottom-right cell at (n - 1, n - 1). You are given the integer n and an integer array startPos where startPos = [startrow, startcol] indicates that a robot is initially at cell (startrow, startcol).

You are also given a 0-indexed string s of length m where s[i] is the ith instruction for the robot: 'L' (move left), 'R' (move right), 'U' (move up), and 'D' (move down).

The robot can begin executing from any ith instruction in s. It executes the instructions one by one towards the end of s but it stops if either of these conditions is met:
- The next instruction will move the robot off the grid.
- There are no more instructions left to execute.

Return an array answer of length m where answer[i] is the number of instructions the robot can execute if the robot begins executing from the ith instruction in s.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2131 Longest Palindrome By Concatenating Two Letter Words

MEDIUM

You are given an array of strings words. Each element of words consists of two lowercase English letters.

Create the longest possible palindrome by selecting some elements from words and concatenating them in any order. Each element can be selected at most once.

Return the length of the longest palindrome that you can create. If it is impossible to create any palindrome, return 0.

A palindrome is a string that reads the same forward and backward.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2135 Count Words Obtained After Adding A Letter

MEDIUM

You are given two 0-indexed arrays of strings startWords and targetWords. Each string consists of lowercase English letters only.

For each string in targetWords, check if it is possible to choose a string from startWords and perform a conversion operation on it to be equal to that from targetWords.

The conversion operation is described in the following two steps:
- For example, if the string is "abc", the letters 'd', 'e', or 'y' can be added to it, but not 'a'. If 'd' is added, the resulting string will be "abcd".
- For example, "abcd" can be rearranged to "acbd", "bacd", "cbda", and so on. Note that it can also be rearranged to "abcd" itself.

Return the number of strings in targetWords that can be obtained by performing the operations on any string of startWords.

Note that you will only be verifying if the string in targetWords can be obtained from a string in startWords by performing the operations. The strings in startWords do not actually change during this process.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2138 Divide A String Into Groups Of Size K

EASY

A string s can be partitioned into groups of size k using the following procedure:
- The first group consists of the first k characters of the string, the second group consists of the next k characters of the string, and so on. Each element can be a part of exactly one group.
- For the last group, if the string does not have k characters remaining, a character fill is used to complete the group.

Note that the partition is done so that after removing the fill character from the last group (if it exists) and concatenating all the groups in order, the resultant string should be s.

Given the string s, the size of each group k and the character fill, return a string array denoting the composition of every group s has been divided into, using the above procedure.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2157 Groups Of Strings

HARD

You are given a 0-indexed array of strings words. Each string consists of lowercase English letters only. No letter occurs more than once in any string of words.

Two strings s1 and s2 are said to be connected if the set of letters of s2 can be obtained from the set of letters of s1 by any one of the following operations:
- Adding exactly one letter to the set of the letters of s1.
- Deleting exactly one letter from the set of the letters of s1.
- Replacing exactly one letter from the set of the letters of s1 with any letter, including itself.

The array words can be divided into one or more non-intersecting groups. A string belongs to a group if any one of the following is true:
- It is connected to at least one other string of the group.
- It is the only string present in the group.

Note that the strings in words should be grouped in such a manner that a string belonging to a group cannot be connected to a string present in any other group. It can be proved that such an arrangement is always unique.

Return an array ans of size 2 where:
- ans[0] is the maximum number of groups words can be divided into, and
- ans[1] is the size of the largest group.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2160 Minimum Sum Of Four Digit Number After Splitting Digits

EASY

You are given a positive integer num consisting of exactly four digits. Split num into two new integers new1 and new2 by using the digits found in num. Leading zeros are allowed in new1 and new2, and all the digits found in num must be used.
- For example, given num = 2932, you have the following digits: two 2's, one 9 and one 3. Some of the possible pairs [new1, new2] are [22, 93], [23, 92], [223, 9] and [2, 329].

Return the minimum possible sum of new1 and new2.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2180 Count Integers With Even Digit Sum

EASY

Given a positive integer num, return the number of positive integers less than or equal to num whose digit sums are even.

The digit sum of a positive integer is the sum of all its digits.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2182 Construct String With Repeat Limit

MEDIUM

You are given a string s and an integer repeatLimit. Construct a new string repeatLimitedString using the characters of s such that no letter appears more than repeatLimit times in a row. You do not have to use all characters from s.

Return the lexicographically largest repeatLimitedString possible.

A string a is lexicographically larger than a string b if in the first position where

a and b differ, string a has a letter that appears later in the alphabet than the corresponding letter in b. If the first min(a.length, b.length) characters do not differ, then the longer string is the lexicographically larger one.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2185 Counting Words With A Given Prefix

**EASY**

You are given an array of strings words and a string pref.

Return the number of strings in words that contain pref as a prefix.

A prefix of a string s is any leading contiguous substring of s.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2186 Minimum Number Of Steps To Make Two Strings Anagram Ii

**MEDIUM**

You are given two strings s and t. In one step, you can append any character to either s or t.

Return the minimum number of steps to make s and t anagrams of each other.

An anagram of a string is a string that contains the same characters with a different (or the same) ordering.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2207 Maximize Number Of Subsequences In A String

**MEDIUM**

You are given a 0-indexed string text and another 0-indexed string pattern of length 2, both of which consist of only lowercase English letters.

You can add either pattern[0] or pattern[1] anywhere in text exactly once. Note that the character can be added even at the beginning or at the end of text.

Return the maximum number of times pattern can occur as a subsequence of the modified text.

A subsequence is a string that can be derived from another string by deleting some or no characters without changing the order of the remaining characters.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2223 Sum Of Scores Of Built Strings

**HARD**

You are building a string s of length n one character at a time, prepending each new character to the front of the string. The strings are labeled from 1 to n, where the string with length i is labeled si.
- For example, for s = "abaca", s1 == "a", s2 == "ca", s3 == "aca", etc.

The score of si is the length of the longest common prefix between si and sn (Note that s == sn).

Given the final string s, return the sum of the score of every si.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2227 Encrypt And Decrypt Strings

HARD

You are given a character array keys containing unique characters and a string array values containing strings of length 2. You are also given another string array dictionary that contains all permitted original strings after decryption. You should implement a data structure that can encrypt or decrypt a 0-indexed string.

A string is encrypted with the following process:

Note that in case a character of the string is not present in keys, the encryption process cannot be carried out, and an empty string "" is returned.

A string is decrypted with the following process:
that values[i] == s. If there are multiple valid i, we choose any one of them. This means a string could have multiple possible strings it can decrypt to.

Implement the Encrypter class:
- Encrypter(char[] keys, String[] values, String[] dictionary) Initializes the Encrypter class with keys, values, and dictionary.
- String encrypt(String word1) Encrypts word1 with the encryption process described above and returns the encrypted string.
- int decrypt(String word2) Returns the number of possible strings word2 could decrypt to that also appear in dictionary.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2231 Largest Number After Digit Swaps By Parity

EASY

You are given a positive integer num. You may swap any two digits of num that have the same parity (i.e. both odd digits or both even digits).

Return the largest possible value of num after any number of swaps.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2243 Calculate Digit Sum Of A String

EASY

You are given a string s consisting of digits and an integer k.

A round can be completed if the length of s is greater than k. In one round, do the following:
group, the next k characters are in the second group, and so on. Note that the size of the last group can be smaller than k.
"346" is replaced with "13" because 3 + 4 + 6 = 13.
than k, repeat from step 1.

Return s after all rounds have been completed.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2246 Longest Path With Different Adjacent Characters

HARD

You are given a tree (i.e. a connected, undirected graph that has no cycles) rooted at node 0 consisting of n nodes numbered from 0 to n - 1. The tree is represented by a 0-indexed array parent of size n, where parent[i] is the parent of node i. Since node 0 is the root, parent[0] == -1.

You are also given a string s of length n, where s[i] is the character assigned to node i.

Return the length of the longest path in the tree such that no pair of adjacent nodes on the path have the same character assigned to them.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2255 Count Prefixes Of A Given String

**EASY**

You are given a string array words and a string s, where words[i] and s comprise only of lowercase English letters.

Return the number of strings in words that are a prefix of s.

A prefix of a string is a substring that occurs at the beginning of the string. A substring is a contiguous sequence of characters within a string.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2259 Remove Digit From Number To Maximize Result

**EASY**

You are given a string number representing a positive integer and a character digit.

Return the resulting string after removing exactly one occurrence of digit from number such that the value of the resulting string in decimal form is maximized. The test cases are generated such that digit occurs at least once in number.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2262 Total Appeal Of A String

**HARD**

The appeal of a string is the number of distinct characters found in the string.
- For example, the appeal of "abbca" is 3 because it has 3 distinct characters: 'a', 'b', and 'c'.

Given a string s, return the total appeal of all of its substrings.

A substring is a contiguous sequence of characters within a string.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2264 Largest 3 Same Digit Number In String

**EASY**

You are given a string num representing a large integer. An integer is good if it meets the following conditions:
- It is a substring of num with length 3.
- It consists of only one unique digit.

Return the maximum good integer as a string or an empty string "" if no such integer exists.

Note:
- A substring is a contiguous sequence of characters within a string.
- There may be leading zeroes in num or a good integer.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2266 Count Number Of Texts

MEDIUM

Alice is texting Bob using her phone. The mapping of digits to letters is shown in the figure below.

In order to add a letter, Alice has to press the key of the corresponding digit i times, where i is the position of the letter in the key.
- For example, to add the letter 's', Alice has to press '7' four times. Similarly, to add the letter 'k', Alice has to press '5' twice.
- Note that the digits '0' and '1' do not map to any letters, so Alice does not use them.

However, due to an error in transmission, Bob did not receive Alice's text message but received a string of pressed keys instead.
- For example, when Alice sent the message "bob", Bob received the string "2266622".

Given a string pressedKeys representing the string received by Bob, return the total number of possible text messages Alice could have sent.

Since the answer may be very large, return it modulo 109 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2273 Find Resultant Array After Removing Anagrams

EASY

You are given a 0-indexed string array words, where words[i] consists of lowercase English letters.

In one operation, select any index i such that 0 < i < words.length and words[i - 1] and words[i] are anagrams, and delete words[i] from words. Keep performing this operation as long as you can select an index that satisfies the conditions.

Return words after performing all operations. It can be shown that selecting the indices for each operation in any arbitrary order will lead to the same result.

An Anagram is a word or phrase formed by rearranging the letters of a different word or phrase using all the original letters exactly once. For example, "dacb" is an anagram of "abdc".

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2278 Percentage Of Letter In String

EASY

Given a string s and a character letter, return the percentage of characters in s that equal letter rounded down to the nearest whole percent.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2283 Check If Number Has Equal Digit Count And Digit Value

EASY

You are given a 0-indexed string num of length n consisting of digits.

Return true if for every index i in the range 0 <= i < n, the digit i occurs num[i] times in num, otherwise return false.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2284 Sender With Largest Word Count

MEDIUM

You have a chat log of n messages. You are given two string arrays messages and senders where messages[i] is a message sent by senders[i].

A message is list of words that are separated by a single space with no leading or trailing spaces. The word count of a sender is the total number of words sent by the sender. Note that a sender may send more than one message.

Return the sender with the largest word count. If there is more than one sender with the largest word count, return the one with the lexicographically largest name.

Note:
- Uppercase letters come before lowercase letters in lexicographical order.
- "Alice" and "alice" are distinct.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2287 Rearrange Characters To Make Target String

EASY

You are given two 0-indexed strings s and target. You can take some letters from s and rearrange them to form new strings.

Return the maximum number of copies of target that can be formed by taking letters from s and rearranging them.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2296 Design A Text Editor

HARD

Design a text editor with a cursor that can do the following:
- Add text to where the cursor is.
- Delete text from where the cursor is (simulating the backspace key).
- Move the cursor either left or right.

When deleting text, only characters to the left of the cursor will be deleted. The cursor will also remain within the actual text and cannot be moved beyond it. More formally, we have that 0 <= cursor.position <= currentText.length always holds.

Implement the TextEditor class:
- TextEditor() Initializes the object with empty text.
- void addText(string text) Appends text to where the cursor is. The cursor ends to the right of text.
- int deleteText(int k) Deletes k characters to the left of the cursor. Returns the number of characters actually deleted.
- string cursorLeft(int k) Moves the cursor to the left k times. Returns the last min(10, len) characters to the left of the cursor, where len is the number of characters to the left of the cursor.
- string cursorRight(int k) Moves the cursor to the right k times. Returns the last min(10, len) characters to the left of the cursor, where len is the number of characters to the left of the cursor.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2299 Strong Password Checker Ii

EASY

A password is said to be strong if it satisfies all the following criteria:
- It has at least 8 characters.
- It contains at least one lowercase letter.
- It contains at least one uppercase letter.
- It contains at least one digit.
- It contains at least one special character. The special characters are the characters in the following string: "!@#$%^&*()-+".
- It does not contain 2 of the same character in adjacent positions (i.e., "aab" violates this condition, but "aba" does not).

Given a string password, return true if it is a strong password. Otherwise, return false.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2309 Greatest English Letter In Upper And Lower Case

EASY

Given a string of English letters s, return the greatest English letter which occurs as both a lowercase and uppercase letter in s. The returned letter should be in uppercase. If no such letter exists, return an empty string.

An English letter b is greater than another letter a if b appears after a in the English alphabet.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2310 Sum Of Numbers With Units Digit K

MEDIUM

Given two integers num and k, consider a set of positive integers with the following properties:
- The units digit of each integer is k.
- The sum of the integers is num.

Return the minimum possible size of such a set, or -1 if no such set exists.

Note:
- The set can contain multiple instances of the same integer, and the sum of an empty set is considered 0.
- The units digit of a number is the rightmost digit of the number.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2337 Move Pieces To Obtain A String

MEDIUM

You are given two strings start and target, both of length n. Each string consists only of the characters 'L', 'R', and '_' where:
- The characters 'L' and 'R' represent pieces, where a piece 'L' can move to the left only if there is a blank space directly to its left, and a piece 'R' can move to the right only if there is a blank space directly to its right.
- The character '_' represents a blank space that can be occupied by any of the 'L' or 'R' pieces.

Return true if it is possible to obtain the string target by moving the pieces of the string start any number of times. Otherwise, return false.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2342 Max Sum Of A Pair With Equal Sum Of Digits

MEDIUM

You are given a 0-indexed array nums consisting of positive integers. You can choose two indices i and j, such that i != j, and the sum of digits of the number nums[i] is equal to that of nums[j].

Return the maximum value of nums[i] + nums[j] that you can obtain over all possible indices i and j that satisfy the conditions.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2351 First Letter To Appear Twice

EASY

Given a string s consisting of lowercase English letters, return the first letter to appear twice.

Note:
- A letter a appears twice before another letter b if the second occurrence of a is before the second occurrence of b.
- s will contain at least one letter that appears twice.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2375 Construct Smallest Number From Di String

MEDIUM

You are given a 0-indexed string pattern of length n consisting of the characters 'I' meaning increasing and 'D' meaning decreasing.

A 0-indexed string num of length n + 1 is created using the following conditions:
- num consists of the digits '1' to '9', where each digit is used at most once.
- If pattern[i] == 'I', then num[i] < num[i + 1].
- If pattern[i] == 'D', then num[i] > num[i + 1].

Return the lexicographically smallest possible string num that meets the conditions.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2380 Time Needed To Rearrange A Binary String

MEDIUM

You are given a binary string s. In one second, all occurrences of "01" are simultaneously replaced with "10". This process repeats until no occurrences of "01" exist.

Return the number of seconds needed to complete this process.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2381 Shifting Letters Ii

MEDIUM

You are given a string s of lowercase English letters and a 2D integer array shifts where shifts[i] = [starti, endi, directioni]. For every i, shift the characters in s from the index starti to the index endi (inclusive) forward if directioni = 1, or shift the characters backward if directioni = 0.

Shifting a character forward means replacing it with the next letter in the alphabet (wrapping around so that 'z' becomes 'a'). Similarly, shifting a character backward means replacing it

with the previous letter in the alphabet (wrapping around so that 'a' becomes 'z').

Return the final string after all such shifts to s are applied.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2390 Removing Stars From A String

**MEDIUM**

You are given a string s, which contains stars *.

In one operation, you can:
- Choose a star in s.
- Remove the closest non-star character to its left, as well as remove the star itself.

Return the string after all stars have been removed.

Note:
- The input will be generated such that the operation is always possible.
- It can be shown that the resulting string will always be unique.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2399 Check Distances Between Same Letters

**EASY**

You are given a 0-indexed string s consisting of only lowercase English letters, where each letter in s appears exactly twice. You are also given a 0-indexed integer array distance of length 26.

Each letter in the alphabet is numbered from 0 to 25 (i.e. 'a' -> 0, 'b' -> 1, 'c' -> 2, ... , 'z' -> 25).

In a well-spaced string, the number of letters between the two occurrences of the ith letter is distance[i]. If the ith letter does not appear in s, then distance[i] can be ignored.

Return true if s is a well-spaced string, otherwise return false.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2405 Optimal Partition Of String

**MEDIUM**

Given a string s, partition the string into one or more substrings such that the characters in each substring are unique. That is, no letter appears in a single substring more than once.

Return the minimum number of substrings in such a partition.

Note that each character should belong to exactly one substring in a partition.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2416 Sum Of Prefix Scores Of Strings

**HARD**

You are given an array words of size n consisting of non-empty strings.

We define the score of a string term as the number of strings words[i] such that term is a prefix of words[i].
- For example, if words = ["a", "ab", "abc", "cab"], then the score of "ab" is 2, since "ab" is a prefix of both "ab" and "abc".

Return an array answer of size n where answer[i] is the sum of scores of every non-empty

prefix of words[i].

Note that a string is considered as a prefix of itself.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2433 Find The Original Array Of Prefix Xor

**MEDIUM**

You are given an integer array pref of size n. Find and return the array arr of size n that satisfies:
- pref[i] = arr[0] ^ arr[1] ^ ... ^ arr[i].

Note that ^ denotes the bitwise-xor operation.

It can be proven that the answer is unique.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2450 Number Of Distinct Binary Strings After Applying Operations

**MEDIUM**

You are given a binary string s and a positive integer k.

You can apply the following operation on the string any number of times:
- Choose any substring of size k from s and flip all its characters, that is, turn all 1's into 0's, and all 0's into 1's.
- Return the number of distinct strings you can obtain. Since the answer may be too large, return it modulo 109 + 7.

Note that:
- A binary string is a string that consists only of the characters 0 and 1.
- A substring is a contiguous part of a string.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2451 Odd String Difference

**EASY**

You are given an array of equal-length strings words. Assume that the length of each string is n.

Each string words[i] can be converted into a difference integer array difference[i] of length n - 1 where difference[i][j] = words[i][j+1] - words[i][j] where 0 <= j <= n - 2. Note that the difference between two letters is the difference between their positions in the alphabet i.e. the position of 'a' is 0, 'b' is 1, and 'z' is 25.
- For example, for the string "acb", the difference integer array is [2 - 0, 1 - 2] = [2, -1].

All the strings in words have the same difference integer array, except one. You should find that string.

Return the string in words that has different difference integer array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2452 Words Within Two Edits Of Dictionary

**MEDIUM**

You are given two string arrays, queries and dictionary. All words in each array comprise of lowercase English letters and have the same length.

In one edit you can take a word from queries, and change any letter in it to any other letter.

Find all words from queries that, after a maximum of two edits, equal some word from dictionary.

Return a list of all words from queries, that match with some word from dictionary after a maximum of two edits. Return the words in the same order they appear in queries.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2466 Count Ways To Build Good Strings

**MEDIUM**

Given the integers zero, one, low, and high, we can construct a string by starting with an empty string, and then at each step perform either of the following:
- Append the character '0' zero times.
- Append the character '1' one times.

This can be performed any number of times.

A good string is a string constructed by the above process having a length between low and high (inclusive).

Return the number of different good strings that can be constructed satisfying these properties. Since the answer can be large, return it modulo 109 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2486 Append Characters To String To Make Subsequence

**MEDIUM**

You are given two strings s and t consisting of only lowercase English letters.

Return the minimum number of characters that need to be appended to the end of s so that t becomes a subsequence of s.

A subsequence is a string that can be derived from another string by deleting some or no characters without changing the order of the remaining characters.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2490 Circular Sentence

**EASY**

A sentence is a list of words that are separated by a single space with no leading or trailing spaces.

For example, "Hello World", "HELLO", "hello world hello world" are all sentences.
Words consist of only uppercase and lowercase English letters. Uppercase and lowercase English letters are considered different.

A sentence is circular if:
- The last character of a word is equal to the first character of the next word.
- The last character of the last word is equal to the first character of the first word.

For example, "leetcode exercises sound delightful", "eetcode", "leetcode eats soul" are all circular sentences. However, "Leetcode is cool", "happy Leetcode", "Leetcode" and "I like Leetcode" are not circular sentences.

Given a string sentence, return true if it is circular. Otherwise, return false.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2496 Maximum Value Of A String In An Array

**EASY**

The value of an alphanumeric string can be defined as:

- The numeric representation of the string in base 10, if it comprises of digits only.
- The length of the string, otherwise.

Given an array strs of alphanumeric strings, return the maximum value of any string in strs.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2506 Count Pairs Of Similar Strings

**EASY**

You are given a 0-indexed string array words.

Two strings are similar if they consist of the same characters.
- For example, "abca" and "cba" are similar since both consist of characters 'a', 'b', and 'c'.
- However, "abacba" and "bcfd" are not similar since they do not consist of the same characters.

Return the number of pairs (i, j) such that 0 <= i < j <= word.length - 1 and the two strings words[i] and words[j] are similar.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2515 Shortest Distance To Target String In A Circular Array

**EASY**

You are given a 0-indexed circular string array words and a string target. A circular array means that the array's end connects to the array's beginning.
- Formally, the next element of words[i] is words[(i + 1) % n] and the previous element of words[i] is words[(i - 1 + n) % n], where n is the length of words.

Starting from startIndex, you can move to either the next word or the previous word with 1 step at a time.

Return the shortest distance needed to reach the string target. If the string target does not exist in words, return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2516 Take K Of Each Character From Left And Right

**MEDIUM**

You are given a string s consisting of the characters 'a', 'b', and 'c' and a non-negative integer k. Each minute, you may take either the leftmost character of s, or the rightmost character of s.

Return the minimum number of minutes needed for you to take at least k of each character, or return -1 if it is not possible to take k of each character.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2520 Count The Digits That Divide A Number

**EASY**

Given an integer num, return the number of digits in num that divide num.

An integer val divides nums if nums % val == 0.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2533 Number Of Good Binary Strings

MEDIUM

You are given four integers minLength, maxLength, oneGroup and zeroGroup.

A binary string is good if it satisfies the following conditions:
- The length of the string is in the range [minLength, maxLength].
- The size of each block of consecutive 1's is a multiple of oneGroup.
- For example in a binary string 00110111100 sizes of each block of consecutive ones are [2,4].
- The size of each block of consecutive 0's is a multiple of zeroGroup.
- For example, in a binary string 00110111100 sizes of each block of consecutive zeros are [2,1,2].

Return the number of good binary strings. Since the answer may be too large, return it modulo 109 + 7.

Note that 0 is considered a multiple of all the numbers.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2535 Difference Between Element Sum And Digit Sum Of An Array

EASY

You are given a positive integer array nums:
- The element sum is the sum of all the elements in nums.
- The digit sum is the sum of all the digits (not necessarily distinct) that appear in nums.

Return the absolute difference between the element sum and digit sum of nums. Note that the absolute difference between two integers x and y is defined as |x - y|.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2544 Alternating Digit Sum

EASY

You are given a positive integer n. Each digit of n has a sign according to the following rules:
- The most significant digit is assigned a positive sign.
- Each other digit has an opposite sign to its adjacent digits.

Return the sum of all digits with their corresponding sign.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2546 Apply Bitwise Operations To Make Strings Equal

MEDIUM

You are given two 0-indexed binary strings s and target of the same length n. You can do the following operation on s any number of times:
- Choose two different indices i and j where 0 <= i, j < n.
- Simultaneously, replace s[i] with (s[i] OR s[j]) and s[j] with (s[i] XOR s[j]).

For example, if s = "0110", you can choose i = 0 and j = 2, then simultaneously replace s[0] with (s[0] OR s[2] = 0 OR 1 = 1), and s[2] with (s[0] XOR s[2] = 0 XOR 1 = 1), so we will have s = "1110".

Return true if you can make the string s equal to target, or false otherwise.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2553 Separate The Digits In An Array

EASY

Given an array of positive integers nums, return an array answer that consists of the digits of each integer in nums after separating them in the same order they appear in nums.

To separate the digits of an integer is to get all the digits it has in the same order.
- For example, for the integer 10921, the separation of its digits is [1,0,9,2,1].

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2559 Count Vowel Strings In Ranges

MEDIUM

You are given a 0-indexed array of strings words and a 2D array of integers queries.

Each query queries[i] = [li, ri] asks us to find the number of strings present at the indices ranging from li to ri (both inclusive) of words that start and end with a vowel.

Return an array ans of size queries.length, where ans[i] is the answer to the ith query.

Note that the vowel letters are 'a', 'e', 'i', 'o', and 'u'.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2566 Maximum Difference By Remapping A Digit

EASY

You are given an integer num. You know that Bob will sneakily remap one of the 10 possible digits (0 to 9) to another digit.

Return the difference between the maximum and minimum values Bob can make by remapping exactly one digit in num.

Notes:
- When Bob remaps a digit d1 to another digit d2, Bob replaces all occurrences of d1 in num with d2.
- Bob can remap a digit to itself, in which case num does not change.
- Bob can remap different digits for obtaining minimum and maximum values respectively.
- The resulting number after remapping can contain leading zeroes.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2586 Count The Number Of Vowel Strings In Range

EASY

You are given a 0-indexed array of string words and two integers left and right.

A string is called a vowel string if it starts with a vowel character and ends with a vowel character where vowel characters are 'a', 'e', 'i', 'o', and 'u'.

Return the number of vowel strings words[i] where i belongs to the inclusive range [left, right].

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2587 Rearrange Array To Maximize Prefix Score

MEDIUM

You are given a 0-indexed integer array nums. You can rearrange the elements of nums to any order (including the given order).

Let prefix be the array containing the prefix sums of nums after rearranging it. In other words, prefix[i] is the sum of the elements from 0 to i in nums after rearranging it. The score of nums is the number of positive integers in the array prefix.

Return the maximum score you can achieve.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2599 Make The Prefix Sum Non Negative

**MEDIUM**

You are given a 0-indexed integer array nums. You can apply the following operation any number of times:
- Pick any element from nums and put it at the end of nums.

The prefix sum array of nums is an array prefix of the same length as nums such that prefix[i] is the sum of all the integers nums[j] where j is in the inclusive range [0, i].

Return the minimum number of operations such that the prefix sum array does not contain negative integers. The test cases are generated such that it is always possible to make the prefix sum array non-negative.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2605 Form Smallest Number From Two Digit Arrays

**EASY**

Given two arrays of unique digits nums1 and nums2, return the smallest number that contains at least one digit from each array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2633 Convert Object To Json String

**MEDIUM**

Given a value, return a valid JSON string of that value. The value can be a string, number, array, object, boolean, or null. The returned string should not include extra spaces. The order of keys should be the same as the order returned by Object.keys().

Please solve it without using the built-in JSON.stringify method.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2640 Find The Score Of All Prefixes Of An Array

**MEDIUM**

We define the conversion array conver of an array arr as follows:
- conver[i] = arr[i] + max(arr[0..i]) where max(arr[0..i]) is the maximum value of arr[j] over 0 <= j <= i.

We also define the score of an array arr as the sum of the values of the conversion array of arr.

Given a 0-indexed integer array nums of length n, return an array ans of length n where ans[i] is the score of the prefix nums[0..i].

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2645 Minimum Additions To Make Valid String

MEDIUM

Given a string word to which you can insert letters "a", "b" or "c" anywhere and any number of times, return the minimum number of letters that must be inserted so that word becomes valid.

A string is called valid if it can be formed by concatenating the string "abc" several times.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2657 Find The Prefix Common Array Of Two Arrays

MEDIUM

You are given two 0-indexed integer permutations A and B of length n.

A prefix common array of A and B is an array C such that C[i] is equal to the count of numbers that are present at or before the index i in both A and B.

Return the prefix common array of A and B.

A sequence of n integers is called a permutation if it contains all integers from 1 to n exactly once.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2693 Call Function With Custom Context

MEDIUM

Enhance all functions to have the callPolyfill method. The method accepts an object obj as its first parameter and any number of additional arguments. The obj becomes the this context for the function. The additional arguments are passed to the function (that the callPolyfill method belongs on).

For example if you had the function:

```
function tax(price, taxRate) {
const totalCost = price * (1 + taxRate);
console.log(`The cost of ${this.item} is ${totalCost}`);
}
```

Calling this function like tax(10, 0.1) will log "The cost of undefined is 11". This is because the this context was not defined.

However, calling the function like tax.callPolyfill({item: "salad"}, 10, 0.1) will log "The cost of salad is 11". The this context was appropriately set, and the function logged an appropriate output.

Please solve this without using the built-in Function.call method.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2707 Extra Characters In A String

MEDIUM

You are given a 0-indexed string s and a dictionary of words dictionary. You have to break s into one or more non-overlapping substrings such that each substring is present in dictionary. There may be some extra characters in s which are not present in any of the substrings.

Return the minimum number of extra characters left over if you break up s optimally.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2710 Remove Trailing Zeros From A String

EASY

Given a positive integer num represented as a string, return the integer num without trailing zeros as a string.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2712 Minimum Cost To Make All Characters Equal

MEDIUM

You are given a 0-indexed binary string s of length n on which you can apply two types of operations:
- Choose an index i and invert all characters from index 0 to index i (both inclusive), with a cost of i + 1
- Choose an index i and invert all characters from index i to index n - 1 (both inclusive), with a cost of n - i

Return the minimum cost to make all characters of the string equal.

Invert a character means if its value is '0' it becomes '1' and vice-versa.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2716 Minimize String Length

EASY

Given a string s, you have two types of operation:
closest occurrence of c to the left of i (if exists).
closest occurrence of c to the right of i (if exists).

Your task is to minimize the length of s by performing the above operations zero or more times.

Return an integer denoting the length of the minimized string.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2744 Find Maximum Number Of String Pairs

EASY

You are given a 0-indexed array words consisting of distinct strings.

The string words[i] can be paired with the string words[j] if:
- The string words[i] is equal to the reversed string of words[j].
- 0 <= i < j < words.length.

Return the maximum number of pairs that can be formed from the array words.

Note that each string can belong in at most one pair.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2745 Construct The Longest New String

MEDIUM

You are given three integers x, y, and z.

You have x strings equal to "AA", y strings equal to "BB", and z strings equal to "AB". You want to choose some (possibly all or none) of these strings and concatenate them in some order to form a new string. This new string must not contain "AAA" or "BBB" as a substring.

Return the maximum possible length of the new string.

A substring is a contiguous non-empty sequence of characters within a string.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2754 Bind Function To Context

**MEDIUM**

Enhance all functions to have the bindPolyfill method. When bindPolyfill is called with a passed object obj, that object becomes the this context for the function.

For example, if you had the code:

```
function f() {
console.log('My context is ' + this.ctx);
}
f();
```

The output would be "My context is undefined". However, if you bound the function:

```
function f() {
console.log('My context is ' + this.ctx);
}
const boundFunc = f.boundPolyfill({ "ctx": "My Object" })
boundFunc();
```

The output should be "My context is My Object".

You may assume that a single non-null object will be passed to the bindPolyfill method.

Please solve it without the built-in Function.bind method.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2759 Convert Json String To Object

**HARD**

Given a string str, return parsed JSON parsedStr. You may assume the str is a valid JSON string hence it only includes strings, numbers, arrays, objects, booleans, and null. str will not include invisible characters and escape characters.

Please solve it without using the built-in JSON.parse method.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2788 Split Strings By Separator

**EASY**

Given an array of strings words and a character separator, split each string in words by separator.

Return an array of strings containing the new strings formed after the splits, excluding empty strings.

Notes
- separator is used to determine where the split should occur, but it is not included as part of the resulting strings.
- A split may result in more than two strings.
- The resulting strings must maintain the same order as they were initially given.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2796 Repeat String

EASY

Write code that enhances all strings such that you can call the string.replicate(x) method on any string and it will return repeated string x times.

Try to implement it without using the built-in method string.repeat.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2825 Make String A Subsequence Using Cyclic Increments

MEDIUM

You are given two 0-indexed strings str1 and str2.

In an operation, you select a set of indices in str1, and for each index i in the set, increment str1[i] to the next character cyclically. That is 'a' becomes 'b', 'b' becomes 'c', and so on, and 'z' becomes 'a'.

Return true if it is possible to make str2 a subsequence of str1 by performing the operation at most once, and false otherwise.

Note: A subsequence of a string is a new string that is formed from the original string by deleting some (possibly none) of the characters without disturbing the relative positions of the remaining characters.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2828 Check If A String Is An Acronym Of Words

EASY

Given an array of strings words and a string s, determine if s is an acronym of words.

The string s is considered an acronym of words if it can be formed by concatenating the first character of each string in words in order. For example, "ab" can be formed from ["apple", "banana"], but it can't be formed from ["bear", "aardvark"].

Return true if s is an acronym of words, and false otherwise.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2839 Check If Strings Can Be Made Equal With Operations I

EASY

You are given two strings s1 and s2, both of length 4, consisting of lowercase English letters.

You can apply the following operation on any of the two strings any number of times:
- Choose any two indices i and j such that j - i = 2, then swap the two characters at those indices in the string.

Return true if you can make the strings s1 and s2 equal, and false otherwise.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2840 Check If Strings Can Be Made Equal With Operations Ii

MEDIUM

You are given two strings s1 and s2, both of length n, consisting of lowercase English letters.

You can apply the following operation on any of the two strings any number of times:

- Choose any two indices i and j such that i < j and the difference j - i is even, then swap the two characters at those indices in the string.

Return true if you can make the strings s1 and s2 equal, and false otherwise.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2847 Smallest Number With Given Digit Product

**MEDIUM**

Given a positive integer n, return a string representing the smallest positive integer such that the product of its digits is equal to n, or "-1" if no such number exists.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2868 The Wording Game

**HARD**

Alice and Bob each have a lexicographically sorted array of strings named a and b respectively.

They are playing a wording game with the following rules:
- On each turn, the current player should play a word from their list such that the new word is closely greater than the last played word; then it's the other player's turn.
- If a player can't play a word on their turn, they lose.

Alice starts the game by playing her lexicographically smallest word.

Given a and b, return true if Alice can win knowing that both players play their best, and false otherwise.

A word w is closely greater than a word z if the following conditions are met:
- w is lexicographically greater than z.
- If w1 is the first letter of w and z1 is the first letter of z, w1 should either be equal to z1 or be the letter after z1 in the alphabet.
- For example, the word "care" is closely greater than "book" and "car", but is not closely greater than "ant" or "cook".

A string s is lexicographically greater than a string t if in the first position where s and t differ, string s has a letter that appears later in the alphabet than the corresponding letter in t. If the first min(s.length, t.length) characters do not differ, then the longer string is the lexicographically greater one.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2914 Minimum Number Of Changes To Make Binary String Beautiful

**MEDIUM**

You are given a 0-indexed binary string s having an even length.

A string is beautiful if it's possible to partition it into one or more substrings such that:
- Each substring has an even length.
- Each substring contains only 1's or only 0's.

You can change any character in s to 0 or 1.

Return the minimum number of changes required to make the string s beautiful.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2937 Make Three Strings Equal

EASY

You are given three strings: s1, s2, and s3. In one operation you can choose one of these strings and delete its rightmost character. Note that you cannot completely empty a string.

Return the minimum number of operations required to make the strings equal. If it is impossible to make them equal, return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2942 Find Words Containing Character

EASY

You are given a 0-indexed array of strings words and a character x.

Return an array of indices representing the words that contain the character x.

Note that the returned array may be in any order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2957 Remove Adjacent Almost Equal Characters

MEDIUM

You are given a 0-indexed string word.

In one operation, you can pick any index i of word and change word[i] to any lowercase English letter.

Return the minimum number of operations needed to remove all adjacent almost-equal characters from word.

Two characters a and b are almost-equal if a == b or a and b are adjacent in the alphabet.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2976 Minimum Cost To Convert String I

MEDIUM

You are given two 0-indexed strings source and target, both of length n and consisting of lowercase English letters. You are also given two 0-indexed character arrays original and changed, and an integer array cost, where cost[i] represents the cost of changing the character original[i] to the character changed[i].

You start with the string source. In one operation, you can pick a character x from the string and change it to the character y at a cost of z if there exists any index j such that cost[j] == z, original[j] == x, and changed[j] == y.

Return the minimum cost to convert the string source to the string target using any number of operations. If it is impossible to convert source to target, return -1.

Note that there may exist indices i, j such that original[j] == original[i] and changed[j] == changed[i].

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3014 Minimum Number Of Pushes To Type Word I

EASY

You are given a string word containing distinct lowercase English letters.

Telephone keypads have keys mapped with distinct collections of lowercase English letters, which can be used to form words by pushing them. For example, the key 2 is mapped with ["a","b","c"], we need to push the key one time to type "a", two times to type "b", and three times to type "c".

It is allowed to remap the keys numbered 2 to 9 to distinct collections of letters. The keys can be remapped to any amount of letters, but each letter must be mapped to exactly one key. You need to find the minimum number of times the keys will be pushed to type the string word.

Return the minimum number of pushes needed to type word after remapping the keys.

An example mapping of letters to keys on a telephone keypad is given below. Note that 1, *, #, and 0 do not map to any letters.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3016 Minimum Number Of Pushes To Type Word Ii

**MEDIUM**

You are given a string word containing lowercase English letters.

Telephone keypads have keys mapped with distinct collections of lowercase English letters, which can be used to form words by pushing them. For example, the key 2 is mapped with ["a","b","c"], we need to push the key one time to type "a", two times to type "b", and three times to type "c".

It is allowed to remap the keys numbered 2 to 9 to distinct collections of letters. The keys can be remapped to any amount of letters, but each letter must be mapped to exactly one key. You need to find the minimum number of times the keys will be pushed to type the string word.

Return the minimum number of pushes needed to type word after remapping the keys.

An example mapping of letters to keys on a telephone keypad is given below. Note that 1, *, #, and 0 do not map to any letters.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3029 Minimum Time To Revert Word To Initial State I

**MEDIUM**

You are given a 0-indexed string word and an integer k.

At every second, you must perform the following operations:
- Remove the first k characters of word.
- Add any k characters to the end of word.

Note that you do not necessarily need to add the same characters that you removed. However, you must perform both operations at every second.

Return the minimum time greater than zero required for word to revert to its initial state.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3032 Count Numbers With Unique Digits Ii

**EASY**

Given two positive integers a and b, return the count of numbers having unique digits in the range [a, b] (inclusive).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3039 Apply Operations To Make String Empty

MEDIUM

You are given a string s.

Consider performing the following operation until s becomes empty:
- For every alphabet character from 'a' to 'z', remove the first occurrence of that character in s (if it exists).

For example, let initially s = "aabcbbca". We do the following operations:
- Remove the underlined characters s = "aabcbbca". The resulting string is s = "abbca".
- Remove the underlined characters s = "abbca". The resulting string is s = "ba".
- Remove the underlined characters s = "ba". The resulting string is s = "".

Return the value of the string s right before applying the last operation. In the example above, answer is "ba".

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3042 Count Prefix And Suffix Pairs I

EASY

You are given a 0-indexed string array words.

Let's define a boolean function isPrefixAndSuffix that takes two strings, str1 and str2:
- isPrefixAndSuffix(str1, str2) returns true if str1 is both a prefix and a suffix of str2, and false otherwise.

For example, isPrefixAndSuffix("aba", "ababa") is true because "aba" is a prefix of "ababa" and also a suffix, but isPrefixAndSuffix("abc", "abcd") is false.

Return an integer denoting the number of index pairs (i, j) such that i < j, and isPrefixAndSuffix(words[i], words[j]) is true.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3043 Find The Length Of The Longest Common Prefix

MEDIUM

You are given two arrays with positive integers arr1 and arr2.

A prefix of a positive integer is an integer formed by one or more of its digits, starting from its leftmost digit. For example, 123 is a prefix of the integer 12345, while 234 is not.

A common prefix of two integers a and b is an integer c, such that c is a prefix of both a and b. For example, 5655359 and 56554 have common prefixes 565 and 5655 while 1223 and 43456 do not have a common prefix.

You need to find the length of the longest common prefix between all pairs of integers (x, y) such that x belongs to arr1 and y belongs to arr2.

Return the length of the longest common prefix among all pairs. If no common prefix exists among them, return 0.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3071 Minimum Operations To Write The Letter Y On A Grid

MEDIUM

You are given a 0-indexed n x n grid where n is odd, and grid[r][c] is 0, 1, or 2.

We say that a cell belongs to the Letter Y if it belongs to one of the following:

- The diagonal starting at the top-left cell and ending at the center cell of the grid.
- The diagonal starting at the top-right cell and ending at the center cell of the grid.
- The vertical line starting at the center cell and ending at the bottom border of the grid.

The Letter Y is written on the grid if and only if:
- All values at cells belonging to the Y are equal.
- All values at cells not belonging to the Y are equal.
- The values at cells belonging to the Y are different from the values at cells not belonging to the Y.

Return the minimum number of operations needed to write the letter Y on the grid given that in one operation you can change the value at any cell to 0, 1, or 2.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3085 Minimum Deletions To Make String K Special

**MEDIUM**

You are given a string word and an integer k.

We consider word to be k-special if |freq(word[i]) - freq(word[j])| <= k for all indices i and j in the string.

Here, freq(x) denotes the frequency of the character x in word, and |y| denotes the absolute value of y.

Return the minimum number of characters you need to delete to make word k-special.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3088 Make String Anti Palindrome

**HARD**

We call a string s of even length n an anti-palindrome if for each index 0 <= i < n, s[i] != s[n - i - 1].

Given a string s, your task is to make s an anti-palindrome by doing any number of operations (including zero).

In one operation, you can select two characters from s and swap them.

Return the resulting string. If multiple strings meet the conditions, return the lexicographically smallest one. If it can't be made into an anti-palindrome, return "-1".

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3110 Score Of A String

**EASY**

You are given a string s. The score of a string is defined as the sum of the absolute difference between the ASCII values of adjacent characters.

Return the score of s.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3135 Equalize Strings By Adding Or Removing Characters At Ends

**MEDIUM**

Given two strings initial and target, your task is to modify initial by performing a series of operations to make it equal to target.

In one operation, you can add or remove one character only at the beginning or the end of

the string initial.

Return the minimum number of operations required to transform initial into target.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3136 Valid Word

**EASY**

A word is considered valid if:
- It contains a minimum of 3 characters.
- It contains only digits (0-9), and English letters (uppercase and lowercase).
- It includes at least one vowel.
- It includes at least one consonant.

You are given a string word.

Return true if word is valid, otherwise, return false.

Notes:
- 'a', 'e', 'i', 'o', 'u', and their uppercases are vowels.
- A consonant is an English letter that is not a vowel.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3167 Better Compression Of String

**MEDIUM**

You are given a string compressed representing a compressed version of a string. The format is a character followed by its frequency. For example, "a3b1a1c2" is a compressed version of the string "aaabacc".

We seek a better compression with the following conditions:

Return the better compression of compressed.

Note: In the better version of compression, the order of letters may change, which is acceptable.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3174 Clear Digits

**EASY**

You are given a string s.

Your task is to remove all digits by doing this operation repeatedly:
- Delete the first digit and the closest non-digit character to its left.

Return the resulting string after removing all digits.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3223 Minimum Length Of String After Operations

**MEDIUM**

You are given a string s.

You can perform the following process on s any number of times:
- Choose an index i in the string such that there is at least one character to the left of index i that is equal to s[i], and at least one character to the right that is also equal to s[i].
- Delete the closest character to the left of index i that is equal to s[i].
- Delete the closest character to the right of index i that is equal to s[i].

Return the minimum length of the final string s that you can achieve.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3227 Vowels Game In A String

**MEDIUM**

Alice and Bob are playing a game on a string.

You are given a string s, Alice and Bob will take turns playing the following game where Alice starts first:
- On Alice's turn, she has to remove any non-empty substring from s that contains an odd number of vowels.
- On Bob's turn, he has to remove any non-empty substring from s that contains an even number of vowels.

The first player who cannot make a move on their turn loses the game. We assume that both Alice and Bob play optimally.

Return true if Alice wins the game, and false otherwise.

The English vowels are: a, e, i, o, and u.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3253 Construct String With Minimum Cost Easy

**MEDIUM**

You are given a string target, an array of strings words, and an integer array costs, both arrays of the same length.

Imagine an empty string s.

You can perform the following operation any number of times (including zero):
- Choose an index i in the range [0, words.length - 1].
- Append words[i] to s.
- The cost of operation is costs[i].

Return the minimum cost to make s equal to target. If it's not possible, return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3289 The Two Sneaky Numbers Of Digitville

**EASY**

In the town of Digitville, there was a list of numbers called nums containing integers from 0 to n - 1. Each number was supposed to appear exactly once in the list, however, two mischievous numbers sneaked in an additional time, making the list longer than usual.

As the town detective, your task is to find these two sneaky numbers. Return an array of size two containing the two numbers (in any order), so peace can return to Digitville.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3304 Find The K Th Character In String Game I

**EASY**

Alice and Bob are playing a game. Initially, Alice has a string word = "a".

You are given a positive integer k.

Now Bob will ask Alice to perform the following operation forever:
- Generate a new string by changing each character in word to its next character in the

English alphabet, and append it to the original word.

For example, performing the operation on "c" generates "cd" and performing the operation on "zb" generates "zbac".

Return the value of the kth character in word, after enough operations have been done for word to have at least k characters.

Note that the character 'z' can be changed to 'a' in the operation.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3307 Find The K Th Character In String Game Ii

**HARD**

Alice and Bob are playing a game. Initially, Alice has a string word = "a".

You are given a positive integer k. You are also given an integer array operations, where operations[i] represents the type of the ith operation.

Now Bob will ask Alice to perform all operations in sequence:
- If operations[i] == 0, append a copy of word to itself.
- If operations[i] == 1, generate a new string by changing each character in word to its next character in the English alphabet, and append it to the original word.
For example, performing the operation on "c" generates "cd" and performing the operation on "zb" generates "zbac".

Return the value of the kth character in word after performing all the operations.

Note that the character 'z' can be changed to 'a' in the second type of operation.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3330 Find The Original Typed String I

**EASY**

Alice is attempting to type a specific string on her computer. However, she tends to be clumsy and may press a key for too long, resulting in a character being typed multiple times.

Although Alice tried to focus on her typing, she is aware that she may still have done this at most once.

You are given a string word, which represents the final output displayed on Alice's screen.

Return the total number of possible original strings that Alice might have intended to type.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3333 Find The Original Typed String Ii

**HARD**

Alice is attempting to type a specific string on her computer. However, she tends to be clumsy and may press a key for too long, resulting in a character being typed multiple times.

You are given a string word, which represents the final output displayed on Alice's screen. You are also given a positive integer k.

Return the total number of possible original strings that Alice might have intended to type, if she was trying to type a string of size at least k.

Since the answer may be very large, return it modulo 109 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3335 Total Characters In String After Transformations I

MEDIUM

You are given a string s and an integer t, representing the number of transformations to perform. In one transformation, every character in s is replaced according to the following rules:
- If the character is 'z', replace it with the string "ab".
- Otherwise, replace it with the next character in the alphabet. For example, 'a' is replaced with 'b', 'b' is replaced with 'c', and so on.

Return the length of the resulting string after exactly t transformations.

Since the answer may be very large, return it modulo 109 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3337 Total Characters In String After Transformations Ii

HARD

You are given a string s consisting of lowercase English letters, an integer t representing the number of transformations to perform, and an array nums of size 26. In one transformation, every character in s is replaced according to the following rules:
- Replace s[i] with the next nums[s[i] - 'a'] consecutive characters in the alphabet. For example, if s[i] = 'a' and nums[0] = 3, the character 'a' transforms into the next 3 consecutive characters ahead of it, which results in "bcd".
- The transformation wraps around the alphabet if it exceeds 'z'. For example, if s[i] = 'y' and nums[24] = 3, the character 'y' transforms into the next 3 consecutive characters ahead of it, which results in "zab".

Return the length of the resulting string after exactly t transformations.

Since the answer may be very large, return it modulo 109 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3460 Longest Common Prefix After At Most One Removal

MEDIUM

You are given two strings s and t.

Return the length of the longest common prefix between s and t after removing at most one character from s.

Note: s can be left without any removal.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3461 Check If Digits Are Equal In String After Operations I

EASY

You are given a string s consisting of digits. Perform the following operation repeatedly until the string has exactly two digits:
- For each pair of consecutive digits in s, starting from the first digit, calculate a new digit as the sum of the two digits modulo 10.
- Replace s with the sequence of newly calculated digits, maintaining the order in which they are computed.

Return true if the final two digits in s are the same; otherwise, return false.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3463 Check If Digits Are Equal In String After Operations Ii

HARD

You are given a string s consisting of digits. Perform the following operation repeatedly until the string has exactly two digits:
- For each pair of consecutive digits in s, starting from the first digit, calculate a new digit as the sum of the two digits modulo 10.
- Replace s with the sequence of newly calculated digits, maintaining the order in which they are computed.

Return true if the final two digits in s are the same; otherwise, return false.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3491 Phone Number Prefix

EASY

You are given a string array numbers that represents phone numbers. Return true if no phone number is a prefix of any other phone number; otherwise, return false.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3541 Find Most Frequent Vowel And Consonant

EASY

You are given a string s consisting of lowercase English letters ('a' to 'z').

Your task is to:
- Find the vowel (one of 'a', 'e', 'i', 'o', or 'u') with the maximum frequency.
- Find the consonant (all other letters excluding vowels) with the maximum frequency.

Return the sum of the two frequencies.

Note: If multiple vowels or consonants have the same maximum frequency, you may choose any one of them. If there are no vowels or no consonants in the string, consider their frequency as 0.

The frequency of a letter x is the number of times it occurs in the string.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

CHAPTER 03 / 189 EXERCISES

# Two Pointers & Sliding Windows

## 01 THE IDEA

Two pointers track positions while avoiding unnecessary repeated work. A window represents a contiguous interval.

## 02 SEE IT




## 03 JAVASCRIPT PATTERN

```
let left = 0;
for (let right = 0; right < nums.length; right++) {
  // expand, then shrink when invalid
}
```

## 04 HOW TO THINK

Ask what condition makes the current window valid. Identify which pointer moves and why no useful answer is skipped.

## 05 PERFORMANCE

Often O(n) when each pointer moves forward at most n times.

## EXERCISE COLLECTION

Try each question before opening the matching problem number in the separate Solutions PDF. The topic classification is a study aid; some exercises combine multiple patterns.

### 0003 Longest Substring Without Repeating Characters

MEDIUM

Given a string `s`, find the length of the longest substring without repeating characters.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0005 Longest Palindromic Substring

MEDIUM

Given a string `s`, return the longest palindromic substring in `s`.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0011 Container With Most Water

MEDIUM

Given n non-negative integers `a1, a2, ..., an ,` where each represents a point at coordinate `(i, ai)`. `n` vertical lines are drawn such that the two endpoints of the line `i` is at `(i, ai)` and `(i, 0)`. Find two lines, which, together with the x-axis forms a container, such that the container contains the most water.

Notice that you may not slant the container.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0015 3Sum

MEDIUM

Given an integer array nums, return all the triplets [nums[i], nums[j], nums[k]] such that i != j, i != k, and j != k, and nums[i] + nums[j] + nums[k] == 0.

Notice that the solution set must not contain duplicate triplets.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0016 3Sum Closest

MEDIUM

Given an integer array nums of length n and an integer target, find three integers in nums such that the sum is closest to target.

Return the sum of the three integers.

You may assume that each input would have exactly one solution.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0018 4Sum

MEDIUM

Given an array nums of n integers, return an array of all the unique quadruplets [nums[a], nums[b], nums[c], nums[d]] such that:
- 0 <= a, b, c, d < n
- a, b, c, and d are distinct.
- nums[a] + nums[b] + nums[c] + nums[d] == target

You may return the answer in any order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0030 Substring With Concatenation Of All Words

HARD

You are given a string s and an array of strings words. All the strings of words are of the same length.

A concatenated substring in s is a substring that contains all the strings of any permutation of words concatenated.

- For example, if words = ["ab","cd","ef"], then "abcdef", "abefcd", "cdabef", "cdefab", "efabcd", and "efcdab" are all concatenated strings. "acdbef" is not a concatenated substring because it is not the concatenation of any permutation of words.

Return the starting indices of all the concatenated substrings in s. You can return the answer in any order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0042 Trapping Rain Water

HARD

Given n non-negative integers representing an elevation map where the width of each bar is 1, compute how much water it can trap after raining.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0053 Maximum Subarray

EASY

Given an integer array `nums`, find the contiguous subarray (containing at least one number) which has the largest sum and return its sum.

A subarray is a contiguous part of an array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0076 Minimum Window Substring

HARD

Given two strings s and t of lengths m and n respectively, return the minimum window substring of s such that every character in t (including duplicates) is included in the window. If there is no such substring, return the empty string "".

The testcases will be generated such that the answer is unique.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0152 Maximum Product Subarray

MEDIUM

Given an integer array nums, find the contiguous subarray within an array (containing at least one number) which has the largest product.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0159 Longest Substring With At Most Two Distinct Characters

MEDIUM

Given a string s, return the length of the longest substring that contains at most two distinct characters.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0167 Two Sum Ii Input Array Is Sorted

EASY

Given a 1-indexed array of integers numbers that is already sorted in non-decreasing order, find two numbers such that they add up to a specific target number. Let these two numbers be numbers[index1] and numbers[index2] where 1 <= index1 < index2 <= numbers.length.

Return the indices of the two numbers, index1 and index2, added by one as an integer array [index1, index2] of length 2.

The tests are generated such that there is exactly one solution. You may not use the same element twice.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0170 Two Sum Iii Data Structure Design

EASY

Design a data structure that accepts a stream of integers and checks if it has a pair of integers that sum up to a particular value.

Implement the TwoSum class:
- TwoSum() Initializes the TwoSum object, with an empty array initially.
- void add(int number) Adds number to the data structure.
- boolean find(int value) Returns true if there exists any pair of numbers whose sum is equal to value, otherwise, it returns false.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0209 Minimum Size Subarray Sum

MEDIUM

Given an array of positive integers nums and a positive integer target, return the minimal length of a subarray whose sum is greater than or equal to target. If there is no such subarray, return 0 instead.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0239 Sliding Window Maximum

HARD

You are given an array of integers nums, there is a sliding window of size k which is moving from the very left of the array to the very right. You can only see the k numbers in the window. Each time the sliding window moves right by one position.

Return the max sliding window.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0259 3Sum Smaller

MEDIUM

Given an array of n integers nums and an integer target, find the number of index triplets i, j, k with 0 <= i < j < k < n that satisfy the condition nums[i] + nums[j] + nums[k] < target.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0325 Maximum Size Subarray Sum Equals K

MEDIUM

Given an integer array nums and an integer k, return the maximum length of a subarray that sums to k. If there is not one, return 0 instead.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0340 Longest Substring With At Most K Distinct Characters

MEDIUM

Given a string s and an integer k, return the length of the longest substring of s that contains at most k distinct characters.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0395 Longest Substring With At Least K Repeating Characters

MEDIUM

Given a string s and an integer k, return the length of the longest substring of s such that the frequency of each character in this substring is greater than or equal to k.

If no such substring exists, return 0.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0407 Trapping Rain Water Ii

HARD

Given an m x n integer matrix heightMap representing the height of each unit cell in a 2D elevation map, return the volume of water it can trap after raining.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0454 4Sum Ii

MEDIUM

Given four integer arrays nums1, nums2, nums3, and nums4 all of length n, return the number of tuples (i, j, k, l) such that:
- 0 <= i, j, k, l < n
- nums1[i] + nums2[j] + nums3[k] + nums4[l] == 0

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0459 Repeated Substring Pattern

EASY

Given a non-empty string check if it can be constructed by taking a substring of it and appending multiple copies of the substring together. You may assume the given string consists of lowercase English letters only and its length will not exceed 10000.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0467 Unique Substrings In Wraparound String

MEDIUM

We define the string base to be the infinite wraparound string of "abcdefghijklmnopqrstuvwxyz", so base will look like this:
- "...zabcdefghijklmnopqrstuvwxyzabcdefghijklmnopqrstuvwxyzabcd....".

Given a string s, return the number of unique non-empty substrings of s are present in base.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0480 Sliding Window Median

HARD

The median is the middle value in an ordered integer list. If the size of the list is even, there is no middle value. So the median is the mean of the two middle values.

- For examples, if arr = [2,3,4], the median is 3.
- For examples, if arr = [1,2,3,4], the median is (2 + 3) / 2 = 2.5.

You are given an integer array nums and an integer k. There is a sliding window of size k which is moving from the very left of the array to the very right. You can only see the k numbers in the window. Each time the sliding window moves right by one position.

Return the median array for each window in the original array. Answers within 10-5 of the actual value will be accepted.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0523 Continuous Subarray Sum

MEDIUM

Given an integer array nums and an integer k, return true if nums has a good subarray or false otherwise.

A good subarray is a subarray where:
- its length is at least two, and
- the sum of the elements of the subarray is a multiple of k.

Note that:
- A subarray is a contiguous part of the array.
- An integer x is a multiple of k if there exists an integer n such that x = n * k.

0 is always a multiple of k.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0560 Subarray Sum Equals K

MEDIUM

Given an array of integers nums and an integer k, return the total number of subarrays whose sum equals to k.

A subarray is a contiguous non-empty sequence of elements within an array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0581 Shortest Unsorted Continuous Subarray

MEDIUM

Given an integer array nums, you need to find one continuous subarray such that if you

only sort this subarray in non-decreasing order, then the whole array will be sorted in non-decreasing order.

Return the shortest such subarray and output its length.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0643 Maximum Average Subarray I

**EASY**

You are given an integer array nums consisting of n elements, and an integer k.

Find a contiguous subarray whose length is equal to k that has the maximum average value and return this value. Any answer with a calculation error less than 10-5 will be accepted.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0644 Maximum Average Subarray Ii

**HARD**

You are given an integer array nums consisting of n elements, and an integer k.

Find a contiguous subarray whose length is greater than or equal to k that has the maximum average value and return this value. Any answer with a calculation error less than 10-5 will be accepted.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0647 Palindromic Substrings

**MEDIUM**

Given a string s, return the number of palindromic substrings in it.

A string is a palindrome when it reads the same backward as forward.

A substring is a contiguous sequence of characters within the string.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0689 Maximum Sum Of 3 Non Overlapping Subarrays

**HARD**

Given an integer array nums and an integer k, find three non-overlapping subarrays of length k with maximum sum and return them.

Return the result as a list of indices representing the starting position of each interval (0-indexed). If there are multiple answers, return the lexicographically smallest one.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0696 Count Binary Substrings

**EASY**

Given a binary string s, return the number of non-empty substrings that have the same number of 0's and 1's, and all the 0's and all the 1's in these substrings are grouped consecutively.

Substrings that occur multiple times are counted the number of times they occur.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0713 Subarray Product Less Than K

MEDIUM

Your are given an array of positive integers nums.

Count and print the number of (contiguous) subarrays where the product of all the elements in the subarray is less than k.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0718 Maximum Length Of Repeated Subarray

MEDIUM

Given two integer arrays nums1 and nums2, return the maximum length of a subarray that appears in both arrays.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0727 Minimum Window Subsequence

HARD

Given strings s1 and s2, return the minimum contiguous substring part of s1, so that s2 is a subsequence of the part.

If there is no such window in s1 that covers all characters in s2, return the empty string "". If there are multiple such minimum-length windows, return the one with the left-most starting index.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0795 Number Of Subarrays With Bounded Maximum

MEDIUM

Given an integer array nums and two integers left and right, return the number of contiguous non-empty subarrays such that the value of the maximum array element in that subarray is in the range [left, right].

The test cases are generated so that the answer will fit in a 32-bit integer.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0828 Count Unique Characters Of All Substrings Of A Given String

HARD

Let's define a function countUniqueChars(s) that returns the number of unique characters in s. - For example, calling countUniqueChars(s) if s = "LEETCODE" then "L", "T", "C", "O", "D" are the unique characters since they appear only once in s, therefore countUniqueChars(s) = 5.

Given a string s, return the sum of countUniqueChars(t) where t is a substring of s. The test cases are generated such that the answer fits in a 32-bit integer.

Notice that some substrings can be repeated so in this case you have to count the repeated ones too.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0862 Shortest Subarray With Sum At Least K

HARD

Given an integer array nums and an integer k, return the length of the shortest non-empty subarray of nums with a sum of at least k. If there is no such subarray, return -1.

A subarray is a contiguous part of an array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0898 Bitwise Ors Of Subarrays

MEDIUM

Given an integer array arr, return the number of distinct bitwise ORs of all the non-empty subarrays of arr.

The bitwise OR of a subarray is the bitwise OR of each integer in the subarray. The bitwise OR of a subarray of one integer is that integer.

A subarray is a contiguous non-empty sequence of elements within an array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0907 Sum Of Subarray Minimums

MEDIUM

Given an array of integers arr, find the sum of min(b), where b ranges over every (contiguous) subarray of arr. Since the answer may be large, return the answer modulo 109 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0918 Maximum Sum Circular Subarray

MEDIUM

Given a circular integer array nums of length n, return the maximum possible sum of a non-empty subarray of nums.

A circular array means the end of the array connects to the beginning of the array. Formally, the next element of nums[i] is nums[(i + 1) % n] and the previous element of nums[i] is nums[(i - 1 + n) % n].

A subarray may only include each element of the fixed buffer nums at most once. Formally, for a subarray nums[i], nums[i + 1], ..., nums[j], there does not exist i <= k1, k2 <= j with k1 % n == k2 % n.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0923 3Sum With Multiplicity

MEDIUM

Given an integer array arr, and an integer target, return the number of tuples i, j, k such that i < j < k and arr[i] + arr[j] + arr[k] == target.

As the answer can be very large, return it modulo 109 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0930 Binary Subarrays With Sum

MEDIUM

Given a binary array nums and an integer goal, return the number of non-empty subarrays with a sum goal.

A subarray is a contiguous part of the array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0974 Subarray Sums Divisible By K

MEDIUM

Given an integer array nums and an integer k, return the number of non-empty subarrays that have a sum divisible by k.

A subarray is a contiguous part of an array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0978 Longest Turbulent Subarray

MEDIUM

Given an integer array arr, return the length of a maximum size turbulent subarray of arr.

A subarray is turbulent if the comparison sign flips between each adjacent pair of elements in the subarray.

More formally, a subarray [arr[i], arr[i + 1], ..., arr[j]] of arr is said to be turbulent if and only if:
- For i <= k < j:
- arr[k] > arr[k + 1] when k is odd, and
- arr[k] < arr[k + 1] when k is even.
- Or, for i <= k < j:
- arr[k] > arr[k + 1] when k is even, and
- arr[k] < arr[k + 1] when k is odd.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0992 Subarrays With K Different Integers

HARD

Given an integer array nums and an integer k, return the number of good subarrays of nums.

A good array is an array where the number of different integers in that array is exactly k.

For example, [1,2,3,1,2] has 3 different integers: 1, 2, and 3.

A subarray is a contiguous part of an array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1016 Binary String With Substrings Representing 1 To N

MEDIUM

Given a binary string s and a positive integer n, return true if the binary representation of all the integers in the range [1, n] are substrings of s, or false otherwise.

A substring is a contiguous sequence of characters within a string.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1031 Maximum Sum Of Two Non Overlapping Subarrays

MEDIUM

Given an integer array nums and two integers firstLen and secondLen, return the maximum sum of elements in two non-overlapping subarrays with lengths firstLen and secondLen.

The array with length firstLen could occur before or after the array with length secondLen, but they have to be non-overlapping.

A subarray is a contiguous part of an array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1044 Longest Duplicate Substring

HARD

Given a string s, consider all duplicated substrings: (contiguous) substrings of s that occur 2 or more times. The occurrences may overlap.

Return any duplicated substring that has the longest possible length. If s does not have a duplicated substring, the answer is "".

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1062 Longest Repeating Substring

MEDIUM

Given a string s, return the length of the longest repeating substrings. If no repeating substring exists, return 0.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1063 Number Of Valid Subarrays

HARD

Given an integer array nums, return the number of non-empty subarrays with the leftmost element of the subarray not larger than other elements in the subarray.

A subarray is a contiguous part of an array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1100 Find K Length Substrings With No Repeated Characters

MEDIUM

Given a string s and an integer k, return the number of substrings in s of length k with no repeated characters.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1156 Swap For Longest Repeated Character Substring

MEDIUM

You are given a string text. You can swap two of the characters in the text.

Return the length of the longest substring with repeated characters.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1157 Online Majority Element In Subarray

HARD

Design a data structure that efficiently finds the majority element of a given subarray.

The majority element of a subarray is an element that occurs threshold times or more in the subarray.

Implementing the MajorityChecker class:
- MajorityChecker(int[] arr) Initializes the instance of the class with the given array arr.
- int query(int left, int right, int threshold) returns the element in the subarray arr[left...right] that occurs at least threshold times, or -1 if no such element exists.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1163 Last Substring In Lexicographical Order

HARD

Given a string s, return the last substring of s in lexicographical order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1177 Can Make Palindrome From Substring

MEDIUM

You are given a string s and array queries where queries[i] = [lefti, righti, ki]. We may rearrange the substring s[lefti...righti] for each query and then choose up to ki of them to replace with any lowercase English letter.

If the substring is possible to be a palindrome string after the operations above, the result of the query is true. Otherwise, the result is false.

Return a boolean array answer where answer[i] is the result of the ith query queries[i].

Note that each letter is counted individually for replacement, so if, for example s[lefti...righti] = "aaa", and ki = 2, we can only replace two of the letters. Also, note that no query modifies the initial string s.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1180 Count Substrings With Only One Distinct Letter

EASY

Given a string s, return the number of substrings that have only one distinct letter.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1186 Maximum Subarray Sum With One Deletion

MEDIUM

Given an array of integers, return the maximum sum for a non-empty subarray (contiguous elements) with at most one element deletion. In other words, you want to choose a subarray and optionally delete one element from it so that there is still at least one element left and the sum of the remaining elements is maximum possible.

Note that the subarray needs to be non-empty after deleting one element.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1190 Reverse Substrings Between Each Pair Of Parentheses

MEDIUM

You are given a string s that consists of lower case English letters and brackets.

Reverse the strings in each pair of matching parentheses, starting from the innermost one.

Your result should not contain any brackets.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1208 Get Equal Substrings Within Budget

MEDIUM

You are given two strings s and t of the same length and an integer maxCost.

You want to change s to t. Changing the ith character of s to ith character of t costs |s[i] - t[i]| (i.e., the absolute difference between the ASCII values of the characters).

Return the maximum length of a substring of s that can be changed to be the same as the corresponding substring of t with a cost less than or equal to maxCost. If there is no substring from s that can be changed to its corresponding substring from t, return 0.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1234 Replace The Substring For Balanced String

MEDIUM

You are given a string s of length n containing only four kinds of characters: 'Q', 'W', 'E', and 'R'.

A string is said to be balanced if each of its characters appears n / 4 times where n is the length of the string.

Return the minimum length of the substring that can be replaced with any other string of the same length to make s balanced. If s is already balanced, return 0.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1248 Count Number Of Nice Subarrays

MEDIUM

Given an array of integers nums and an integer k. A continuous subarray is called nice if there are k odd numbers on it.

Return the number of nice sub-arrays.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1297 Maximum Number Of Occurrences Of A Substring

MEDIUM

Given a string s, return the maximum number of ocurrences of any substring under the following rules:

- The number of unique characters in the substring must be less than or equal to maxLetters.
- The substring size must be between minSize and maxSize inclusive.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1310 Xor Queries Of A Subarray

MEDIUM

You are given an array arr of positive integers. You are also given the array queries where queries[i] = [lefti, righti].

For each query i compute the XOR of elements from lefti to righti (that is, arr[lefti] XOR arr[lefti + 1] XOR ... XOR arr[righti]).

Return an array answer where answer[i] is the answer to the ith query.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1316 Distinct Echo Substrings

HARD

Return the number of distinct non-empty substrings of text that can be written as the concatenation of some string with itself (i.e. it can be written as a + a where a is some string).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1330 Reverse Subarray To Maximize Array Value

HARD

You are given an integer array nums. The value of this array is defined as the sum of |nums[i] - nums[i + 1]| for all 0 <= i < nums.length - 1.

You are allowed to select any subarray of the given array and reverse it. You can perform this operation only once.

Find maximum possible value of the final array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1358 Number Of Substrings Containing All Three Characters

MEDIUM

Given a string s consisting only of characters a, b and c.

Return the number of substrings containing at least one occurrence of all these characters a, b and c.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1371 Find The Longest Substring Containing Vowels In Even Counts

MEDIUM

Given the string s, return the size of the longest substring containing each vowel an even number of times. That is, 'a', 'e', 'i', 'o', and 'u' must appear an even number of times.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1438 Longest Continuous Subarray With Absolute Diff Less Than Or Equal

MEDIUM

Given an array of integers nums and an integer limit, return the size of the longest non-empty subarray such that the absolute difference between any two elements of this subarray is less than or equal to limit.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 1456 Maximum Number Of Vowels In A Substring Of Given Length

MEDIUM

Given a string s and an integer k.

Return the maximum number of vowel letters in any substring of s with length k.

Vowel letters in English are (a, e, i, o, u).

TRY IT / Write examples, edge cases, and your own algorithm first.

## 1493 Longest Subarray Of 1S After Deleting One Element

MEDIUM

Given a binary array nums, you should delete one element from it.

Return the size of the longest non-empty subarray containing only 1's in the resulting array. Return 0 if there is no such subarray.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 1508 Range Sum Of Sorted Subarray Sums

MEDIUM

You are given the array nums consisting of n positive integers. You computed the sum of all non-empty continuous subarrays from the array and then sorted them in non-decreasing order, creating a new array of n * (n + 1) / 2 numbers.

Return the sum of the numbers from index left to index right (indexed from 1), inclusive, in the new array. Since the answer can be a huge number return it modulo 109 + 7.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 1513 Number Of Substrings With Only 1S

MEDIUM

Given a binary string s, return the number of substrings with all characters 1's. Since the answer may be too large, return it modulo 109 + 7.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 1520 Maximum Number Of Non Overlapping Substrings

HARD

Given a string s of lowercase letters, you need to find the maximum number of non-empty substrings of s that meet the following conditions:
either j < x or i > y is true.

Find the maximum number of substrings that meet the above conditions. If there are multiple solutions with the same number of substrings, return the one with minimum total length. It can be shown that there exists a unique solution of minimum total length.

Notice that you can return the substrings in any order.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 1526 Minimum Number Of Increments On Subarrays To Form A Target Array

HARD

You are given an integer array target. You have an integer array initial of the same size as target with all elements initially zeros.

In one operation you can choose any subarray from initial and increment each value by one.

Return the minimum number of operations to form a target array from initial.

The test cases are generated so that the answer fits in a 32-bit integer.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1542 Find Longest Awesome Substring

HARD

You are given a string s. An awesome substring is a non-empty substring of s such that we can make any number of swaps in order to make it a palindrome.

Return the length of the maximum length awesome substring of s.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1546 Maximum Number Of Non Overlapping Subarrays With Sum Equals Target

MEDIUM

Given an array nums and an integer target, return the maximum number of non-empty non-overlapping subarrays such that the sum of values in each subarray is equal to target.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1567 Maximum Length Of Subarray With Positive Product

MEDIUM

Given an array of integers nums, find the maximum length of a subarray where the product of all its elements is positive.

A subarray of an array is a consecutive sequence of zero or more values taken out of that array.

Return the maximum length of a subarray with positive product.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1574 Shortest Subarray To Be Removed To Make Array Sorted

MEDIUM

Given an integer array arr, remove a subarray (can be empty) from arr such that the remaining elements in arr are non-decreasing.

Return the length of the shortest subarray to remove.

A subarray is a contiguous subsequence of the array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1585 Check If String Is Transformable With Substring Sort Operations

HARD

Given two strings s and t, transform string s into string t using the following operation any

number of times:
- Choose a non-empty substring in s and sort it in place so the characters are in ascending order.
- For example, applying the operation on the underlined substring in "14234" results in "12344".

Return true if it is possible to transform s into t. Otherwise, return false.

A substring is a contiguous sequence of characters within a string.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1588 Sum Of All Odd Length Subarrays

**EASY**

Given an array of positive integers arr, return the sum of all possible odd-length subarrays of arr.

A subarray is a contiguous subsequence of the array.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1593 Split A String Into The Max Number Of Unique Substrings

**MEDIUM**

Given a string s, return the maximum number of unique substrings that the given string can be split into.

You can split string s into any list of non-empty substrings, where the concatenation of the substrings forms the original string. However, you must split the substrings such that all of them are unique.

A substring is a contiguous sequence of characters within a string.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1624 Largest Substring Between Two Equal Characters

**EASY**

Given a string s, return the length of the longest substring between two equal characters, excluding the two characters. If there is no such substring return -1.

A substring is a contiguous sequence of characters within a string.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1630 Arithmetic Subarrays

**MEDIUM**

A sequence of numbers is called arithmetic if it consists of at least two elements, and the difference between every two consecutive elements is the same. More formally, a sequence s is arithmetic if and only if s[i+1] - s[i] == s[1] - s[0] for all valid i.

For example, these are arithmetic sequences:
- 1, 3, 5, 7, 9
- 7, 7, 7, 7
- 3, -1, -5, -9

The following sequence is not arithmetic:
- 1, 1, 2, 5, 7

You are given an array of n integers, nums, and two arrays of m integers each, l and r,

representing the m range queries, where the ith query is the range [l[i], r[i]]. All the arrays are 0-indexed.

Return a list of boolean elements answer, where answer[i] is true if the subarray nums[l[i]], nums[l[i]+1], ... , nums[r[i]] can be rearranged to form an arithmetic sequence, and false otherwise.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1638 Count Substrings That Differ By One Character

**MEDIUM**

Given two strings s and t, find the number of ways you can choose a non-empty substring of s and replace a single character by a different character such that the resulting substring is a substring of t. In other words, find the number of substrings in s that differ from some substring in t by exactly one character.

For example, the underlined substrings in "computer" and "computation" only differ by the 'e'/'a', so this is a valid way.

Return the number of substrings that satisfy the condition above.

A substring is a contiguous sequence of characters within a string.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1668 Maximum Repeating Substring

**EASY**

For a string sequence, a string word is k-repeating if word concatenated k times is a substring of sequence. The word's maximum k-repeating value is the highest value k where word is k-repeating in sequence. If word is not a substring of sequence, word's maximum k-repeating value is 0.

Given strings sequence and word, return the maximum k-repeating value of word in sequence.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1698 Number Of Distinct Substrings In A String

**MEDIUM**

Given a string s, return the number of distinct substrings of s.

A substring of a string is obtained by deleting any number of characters (possibly zero) from the front of the string and any number (possibly zero) from the back of the string.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1708 Largest Subarray Length K

**EASY**

An array A is larger than some array B if for the first index i where A[i] != B[i], A[i] > B[i].

For example, consider 0-indexing:
- [1,3,2,4] > [1,2,2,4], since at index 1, 3 > 2.
- [1,4,4,4] < [2,1,1,1], since at index 0, 1 < 2.

A subarray is a contiguous subsequence of the array.

Given an integer array nums of distinct integers, return the largest subarray of nums of length k.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1712 Ways To Split Array Into Three Subarrays

MEDIUM

A split of an integer array is good if:
- The array is split into three non-empty contiguous subarrays - named left, mid, right respectively from left to right.
- The sum of the elements in left is less than or equal to the sum of the elements in mid, and the sum of the elements in mid is less than or equal to the sum of the elements in right.

Given nums, an array of non-negative integers, return the number of good ways to split nums. As the number may be too large, return it modulo 109 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1717 Maximum Score From Removing Substrings

MEDIUM

You are given a string s and two integers x and y. You can perform two types of operations any number of times.

- Remove substring "ab" and gain x points.
- For example, when removing "ab" from "cabxbae" it becomes "cxbae".
- Remove substring "ba" and gain y points.
- For example, when removing "ba" from "cabxbae" it becomes "cabxe".

Return the maximum points you can gain after applying the above operations on s.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1746 Maximum Subarray Sum After One Operation

MEDIUM

You are given an integer array nums. You must perform exactly one operation where you can replace one element nums[i] with nums[i] * nums[i].

Return the maximum possible subarray sum after exactly one operation. The subarray must be non-empty.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1749 Maximum Absolute Sum Of Any Subarray

MEDIUM

You are given an integer array nums. The absolute sum of a subarray [numsl, numsl+1, ..., numsr-1, numsr] is abs(numsl + numsl+1 + ... + numsr-1 + numsr).

Return the maximum absolute sum of any (possibly empty) subarray of nums.

Note that abs(x) is defined as follows:
- If x is a negative integer, then abs(x) = -x.
- If x is a non-negative integer, then abs(x) = x.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1759 Count Number Of Homogenous Substrings

MEDIUM

Given a string s, return the number of homogenous substrings of s. Since the answer may be too large, return it modulo 109 + 7.

A string is homogenous if all the characters of the string are the same.

A substring is a contiguous sequence of characters within a string.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1763 Longest Nice Substring

**EASY**

A string s is nice if, for every letter of the alphabet that s contains, it appears both in uppercase and lowercase. For example, "abABB" is nice because 'A' and 'a' appear, and 'B' and 'b' appear. However, "abA" is not because 'b' appears, but 'B' does not.

Given a string s, return the longest substring of s that is nice. If there are multiple, return the substring of the earliest occurrence. If there are none, return an empty string.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1764 Form Array By Concatenating Subarrays Of Another Array

**MEDIUM**

You are given a 2D integer array groups of length n. You are also given an integer array nums.

You are asked if you can choose n disjoint subarrays from the array nums such that the ith subarray is equal to groups[i] (0-indexed), and if i > 0, the (i-1)th subarray appears before the ith subarray in nums (i.e. the subarrays must be in the same order as groups).

Return true if you can do this task, and false otherwise.

Note that the subarrays are disjoint if and only if there is no index k such that nums[k] belongs to more than one subarray. A subarray is a contiguous sequence of elements within an array.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1781 Sum Of Beauty Of All Substrings

**MEDIUM**

The beauty of a string is the difference in frequencies between the most frequent and least frequent characters.
- For example, the beauty of "abaacc" is 3 - 1 = 2.

Given a string s, return the sum of beauty of all of its substrings.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1793 Maximum Score Of A Good Subarray

**HARD**

You are given an array of integers nums (0-indexed) and an integer k.

The score of a subarray (i, j) is defined as min(nums[i], nums[i+1], ..., nums[j]) * (j - i + 1). A good subarray is a subarray where i <= k <= j.

Return the maximum possible score of a good subarray.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1794 Count Pairs Of Equal Substrings With Minimum Difference

MEDIUM

You are given two strings firstString and secondString that are 0-indexed and consist only of lowercase English letters. Count the number of index quadruples (i,j,a,b) that satisfy the following conditions:
- 0 <= i <= j < firstString.length
- 0 <= a <= b < secondString.length
- The substring of firstString that starts at the ith character and ends at the jth character (inclusive) is equal to the substring of secondString that starts at the ath character and ends at the bth character (inclusive).
- j - a is the minimum possible value among all quadruples that satisfy the previous conditions.

Return the number of such quadruples.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1800 Maximum Ascending Subarray Sum

EASY

Given an array of positive integers nums, return the maximum possible sum of an ascending subarray in nums.

A subarray is defined as a contiguous sequence of numbers in an array.

A subarray [numsl, numsl+1, ..., numsr-1, numsr] is ascending if for all i where l <= i < r, numsi < numsi+1. Note that a subarray of size 1 is ascending.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1839 Longest Substring Of All Vowels In Order

MEDIUM

A string is considered beautiful if it satisfies the following conditions:
- Each of the 5 English vowels ('a', 'e', 'i', 'o', 'u') must appear at least once in it.
- The letters must be sorted in alphabetical order (i.e. all 'a's before 'e's, all 'e's before 'i's, etc.).

For example, strings "aeiou" and "aaaaaaeiiiioou" are considered beautiful, but "uaeio", "aeoiu", and "aaaeeeooo" are not beautiful.

Given a string word consisting of English vowels, return the length of the longest beautiful substring of word. If no such substring exists, return 0.

A substring is a contiguous sequence of characters in a string.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1852 Distinct Numbers In Each Subarray

MEDIUM

You are given an integer array nums of length n and an integer k. Your task is to find the number of distinct elements in every subarray of size k within nums.

Return an array ans such that ans[i] is the count of distinct elements in nums[i..(i + k - 1)] for each index 0 <= i < n - k.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1856 Maximum Subarray Min Product

MEDIUM

The min-product of an array is equal to the minimum value in the array multiplied by the array's sum.
- For example, the array [3,2,5] (minimum value is 2) has a min-product of 2 * (3+2+5) = 2 * 10 = 20.

Given an array of integers nums, return the maximum min-product of any non-empty subarray of nums. Since the answer may be large, return it modulo 109 + 7.

Note that the min-product should be maximized before performing the modulo operation. Testcases are generated such that the maximum min-product without modulo will fit in a 64-bit signed integer.

A subarray is a contiguous part of an array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1876 Substrings Of Size Three With Distinct Characters

EASY

A string is good if there are no repeated characters.

Given a string s, return the number of good substrings of length three in s.

Note that if there are multiple occurrences of the same substring, every occurrence should be counted.

A substring is a contiguous sequence of characters in a string.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1910 Remove All Occurrences Of A Substring

MEDIUM

Given two strings s and part, perform the following operation on s until all occurrences of the substring part are removed:
- Find the leftmost occurrence of the substring part and remove it from s.

Return s after removing all occurrences of part.

A substring is a contiguous sequence of characters in a string.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1915 Number Of Wonderful Substrings

MEDIUM

A wonderful string is a string where at most one letter appears an odd number of times.
- For example, "ccjjc" and "abab" are wonderful, but "ab" is not.

Given a string word that consists of the first ten lowercase English letters ('a' through 'j'), return the number of wonderful non-empty substrings in word. If the same substring appears multiple times in word, then count each occurrence separately.

A substring is a contiguous sequence of characters in a string.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1918 Kth Smallest Subarray Sum

MEDIUM

Given an integer array nums of length n and an integer k, return the kth smallest subarray sum.

A subarray is defined as a non-empty contiguous sequence of elements in an array. A subarray sum is the sum of all elements in the subarray.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1933 Check If String Is Decomposable Into Value Equal Substrings

EASY

A value-equal string is a string where all characters are the same.
- For example, "1111" and "33" are value-equal strings.
- In contrast, "123" is not a value-equal string.

Given a digit string s, decompose the string into some number of consecutive value-equal substrings where exactly one substring has a length of 2 and the remaining substrings have a length of 3.

Return true if you can decompose s according to the above rules. Otherwise, return false.

A substring is a contiguous sequence of characters in a string.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1946 Largest Number After Mutating Substring

MEDIUM

You are given a string num, which represents a large integer. You are also given a 0-indexed integer array change of length 10 that maps each digit 0-9 to another digit. More formally, digit d maps to digit change[d].

You may choose to mutate a single substring of num. To mutate a substring, replace each digit num[i] with the digit it maps to in change (i.e. replace num[i] with change[num[i]]).

Return a string representing the largest possible integer after mutating (or choosing not to) a single substring of num.

A substring is a contiguous sequence of characters within the string.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1950 Maximum Of Minimum Values In All Subarrays

MEDIUM

You are given an integer array nums of size n. You are asked to solve n queries for each integer i in the range 0 <= i < n.

To solve the ith query:

Return a 0-indexed integer array ans of size n such that ans[i] is the answer to the ith query.

A subarray is a contiguous sequence of elements in an array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1960 Maximum Product Of The Length Of Two Palindromic Substrings

HARD

You are given a 0-indexed string s and are tasked with finding two non-intersecting palindromic substrings of odd length such that the product of their lengths is maximized.

More formally, you want to choose four integers i, j, k, l such that 0 <= i <= j < k <= l < s.length and both the substrings s[i...j] and s[k...l] are palindromes and have odd lengths. s[i...j] denotes a substring from index i to index j inclusive.

Return the maximum possible product of the lengths of the two non-intersecting palindromic substrings.

A palindrome is a string that is the same forward and backward. A substring is a contiguous sequence of characters in a string.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1967 Number Of Strings That Appear As Substrings In Word

EASY

Given an array of strings patterns and a string word, return the number of strings in patterns that exist as a substring in word.

A substring is a contiguous sequence of characters within a string.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2031 Count Subarrays With More Ones Than Zeros

MEDIUM

You are given a binary array nums containing only the integers 0 and 1. Return the number of subarrays in nums that have more 1's than 0's. Since the answer may be very large, return it modulo 109 + 7.

A subarray is a contiguous sequence of elements within an array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2036 Maximum Alternating Subarray Sum

MEDIUM

A subarray of a 0-indexed integer array is a contiguous non-empty sequence of elements within an array.

The alternating subarray sum of a subarray that ranges from index i to j (inclusive, 0 <= i <= j < nums.length) is nums[i] - nums[i+1] + nums[i+2] - ... +/- nums[j].

Given a 0-indexed integer array nums, return the maximum alternating subarray sum of any subarray of nums.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2062 Count Vowel Substrings Of A String

EASY

A substring is a contiguous (non-empty) sequence of characters within a string.

A vowel substring is a substring that only consists of vowels ('a', 'e', 'i', 'o', and 'u') and has all five vowels present in it.

Given a string word, return the number of vowel substrings in word.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2063 Vowels Of All Substrings

**MEDIUM**

Given a string word, return the sum of the number of vowels ('a', 'e', 'i', 'o', and 'u') in every substring of word.

A substring is a contiguous (non-empty) sequence of characters within a string.

Note: Due to the large constraints, the answer may not fit in a signed 32-bit integer. Please be careful during the calculations.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2067 Number Of Equal Count Substrings

**MEDIUM**

You are given a 0-indexed string s consisting of only lowercase English letters, and an integer count. A substring of s is said to be an equal count substring if, for each unique letter in the substring, it appears exactly count times in the substring.

Return the number of equal count substrings in s.

A substring is a contiguous non-empty sequence of characters within a string.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2083 Substrings That Begin And End With The Same Letter

**MEDIUM**

You are given a 0-indexed string s consisting of only lowercase English letters. Return the number of substrings in s that begin and end with the same character.

A substring is a contiguous non-empty sequence of characters within a string.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2090 K Radius Subarray Averages

**MEDIUM**

You are given a 0-indexed array nums of n integers, and an integer k.

The k-radius average for a subarray of nums centered at some index i with the radius k is the average of all elements in nums between the indices i - k and i + k (inclusive). If there are less than k elements before or after the index i, then the k-radius average is -1.

Build and return an array avgs of length n where avgs[i] is the k-radius average for the subarray centered at index i.

The average of x elements is the sum of the x elements divided by x, using integer division. The integer division truncates toward zero, which means losing its fractional part.
- For example, the average of four elements 2, 3, 1, and
5 is (2 + 3 + 1 + 5) / 4 = 11 / 4 = 2.75, which truncates to 2.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2104 Sum Of Subarray Ranges

MEDIUM

You are given an integer array nums. The range of a subarray of nums is the difference between the largest and smallest element in the subarray.

Return the sum of all subarray ranges of nums.

A subarray is a contiguous non-empty sequence of elements within an array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2156 Find Substring With Given Hash Value

HARD

The hash of a 0-indexed string s of length k, given integers p and m, is computed using the following function:
- hash(s, p, m) = (val(s[0]) * p0 + val(s[1]) * p1 + ... + val(s[k-1]) * pk-1) mod m.

Where val(s[i]) represents the index of s[i] in the alphabet from val('a') = 1 to val('z') = 26.

You are given a string s and the integers power, modulo, k, and hashValue. Return sub, the first substring of s of length k such that hash(sub, power, modulo) == hashValue.

The test cases will be generated such that an answer always exists.

A substring is a contiguous non-empty sequence of characters within a string.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2168 Unique Substrings With Equal Digit Frequency

MEDIUM

Given a digit string s, return the number of unique substrings of s where every digit appears the same number of times.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2213 Longest Substring Of One Repeating Character

HARD

You are given a 0-indexed string s. You are also given a 0-indexed string queryCharacters of length k and a 0-indexed array of integer indices queryIndices of length k, both of which are used to describe k queries.

The ith query updates the character in s at index queryIndices[i] to the character queryCharacters[i].

Return an array lengths of length k where lengths[i] is the length of the longest substring of s consisting of only one repeating character after the ith query is performed.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2261 K Divisible Elements Subarrays

MEDIUM

Given an integer array nums and two integers k and p, return the number of distinct subarrays, which have at most k elements that are divisible by p.

Two arrays nums1 and nums2 are said to be distinct if:
- They are of different lengths, or

- There exists at least one index i where nums1[i] != nums2[i].

A subarray is defined as a non-empty contiguous sequence of elements in an array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2272 Substring With Largest Variance

**HARD**

The variance of a string is defined as the largest difference between the number of occurrences of any 2 characters present in the string. Note the two characters may or may not be the same.

Given a string s consisting of lowercase English letters only, return the largest variance possible among all substrings of s.

A substring is a contiguous sequence of characters within a string.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2301 Match Substring After Replacement

**HARD**

You are given two strings s and sub. You are also given a 2D character array mappings where mappings[i] = [oldi, newi] indicates that you may perform the following operation any number of times:
- Replace a character oldi of sub with newi.

Each character in sub cannot be replaced more than once.

Return true if it is possible to make sub a substring of s by replacing zero or more characters according to mappings. Otherwise, return false.

A substring is a contiguous non-empty sequence of characters within a string.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2302 Count Subarrays With Score Less Than K

**HARD**

The score of an array is defined as the product of its sum and its length.
- For example, the score of [1, 2, 3, 4, 5] is (1 + 2 + 3 + 4 + 5) * 5 = 75.

Given a positive integer array nums and an integer k, return the number of non-empty subarrays of nums whose score is strictly less than k.

A subarray is a contiguous sequence of elements within an array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2334 Subarray With Elements Greater Than Varying Threshold

**HARD**

You are given an integer array nums and an integer threshold.

Find any subarray of nums of length k such that every element in the subarray is greater than threshold / k.

Return the size of any such subarray. If there is no such subarray, return -1.

A subarray is a contiguous non-empty sequence of elements within an array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2348 Number Of Zero Filled Subarrays

MEDIUM

Given an integer array nums, return the number of subarrays filled with 0.

A subarray is a contiguous non-empty sequence of elements within an array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2393 Count Strictly Increasing Subarrays

MEDIUM

You are given an array nums consisting of positive integers.

Return the number of subarrays of nums that are in strictly increasing order.

A subarray is a contiguous part of an array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2395 Find Subarrays With Equal Sum

EASY

Given a 0-indexed integer array nums, determine whether there exist two subarrays of length 2 with equal sum. Note that the two subarrays must begin at different indices.

Return true if these subarrays exist, and false otherwise.

A subarray is a contiguous non-empty sequence of elements within an array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2401 Longest Nice Subarray

MEDIUM

You are given an array nums consisting of positive integers.

We call a subarray of nums nice if the bitwise AND of every pair of elements that are in different positions in the subarray is equal to 0.

Return the length of the longest nice subarray.

A subarray is a contiguous part of an array.

Note that subarrays of length 1 are always considered nice.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2411 Smallest Subarrays With Maximum Bitwise Or

MEDIUM

You are given a 0-indexed array nums of length n, consisting of non-negative integers. For each index i from 0 to n - 1, you must determine the size of the minimum sized non-empty subarray of nums starting at i (inclusive) that has the maximum possible bitwise OR.
- In other words, let Bij be the bitwise OR of the subarray nums[i...j]. You need to find the smallest subarray starting at i, such that bitwise OR of this subarray is equal to max(Bik) where i <= k <= n - 1.

The bitwise OR of an array is the bitwise OR of all the numbers in it.

Return an integer array answer of size n where answer[i] is the length of the minimum sized subarray starting at i with maximum bitwise OR.

A subarray is a contiguous non-empty sequence of elements within an array.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2414 Length Of The Longest Alphabetical Continuous Substring

**MEDIUM**

An alphabetical continuous string is a string consisting of consecutive letters in the alphabet.
In other words, it is any substring of the string "abcdefghijklmnopqrstuvwxyz".
- For example, "abc" is an alphabetical continuous string, while "acb" and "za" are not.

Given a string s consisting of lowercase letters only, return the length of the longest alphabetical continuous substring.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2419 Longest Subarray With Maximum Bitwise And

**MEDIUM**

You are given an integer array nums of size n.

Consider a non-empty subarray from nums that has the maximum possible bitwise AND.
- In other words, let k be the maximum value of the bitwise AND of any subarray of nums.
Then, only subarrays with a bitwise AND equal to k should be considered.

Return the length of the longest such subarray.

The bitwise AND of an array is the bitwise AND of all the numbers in it.

A subarray is a contiguous sequence of elements within an array.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2436 Minimum Split Into Subarrays With Gcd Greater Than One

**MEDIUM**

You are given an array nums consisting of positive integers.

Split the array into one or more disjoint subarrays such that:
- Each element of the array belongs to exactly one subarray, and
- The GCD of the elements of each subarray is strictly greater than 1.

Return the minimum number of subarrays that can be obtained after the split.

Note that:
- The GCD of a subarray is the largest positive integer that evenly divides all the elements of the subarray.
- A subarray is a contiguous part of the array.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2444 Count Subarrays With Fixed Bounds

**HARD**

You are given an integer array nums and two integers minK and maxK.

A fixed-bound subarray of nums is a subarray that satisfies the following conditions:
- The minimum value in the subarray is equal to minK.
- The maximum value in the subarray is equal to maxK.
- Return the number of fixed-bound subarrays.

A subarray is a contiguous part of an array.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2447 Number Of Subarrays With Gcd Equal To K

MEDIUM

Given an integer array nums and an integer k, return the number of subarrays of nums where the greatest common divisor of the subarray's elements is k.

A subarray is a contiguous non-empty sequence of elements within an array.

The greatest common divisor of an array is the largest integer that evenly divides all the array elements.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2461 Maximum Sum Of Distinct Subarrays With Length K

MEDIUM

You are given an integer array nums and an integer k. Find the maximum subarray sum of all the subarrays of nums that meet the following conditions:
- The length of the subarray is k, and
- All the elements of the subarray are distinct.

Return the maximum subarray sum of all the subarrays that meet the conditions. If no subarray meets the conditions, return 0.

A subarray is a contiguous non-empty sequence of elements within an array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2464 Minimum Subarrays In A Valid Split

MEDIUM

You are given an integer array nums.

Splitting of an integer array nums into subarrays is valid if:
- the greatest common divisor of the first and last elements of each subarray is greater than 1, and
- each element of nums belongs to exactly one subarray.

Return the minimum number of subarrays in a valid subarray splitting of nums. If a valid subarray splitting is not possible, return -1.

Note that:
- The greatest common divisor of two numbers is the largest positive integer that evenly divides both numbers.
- A subarray is a contiguous non-empty part of an array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2470 Number Of Subarrays With Lcm Equal To K

MEDIUM

Given an integer array nums and an integer k, return the number of subarrays of nums where the least common multiple of the subarray's elements is k.

A subarray is a contiguous non-empty sequence of elements within an array.

The least common multiple of an array is the smallest positive integer that is divisible by all the array elements.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2472 Maximum Number Of Non Overlapping Palindrome Substrings

HARD

You are given a string s and a positive integer k.

Select a set of non-overlapping substrings from the string s that satisfy the following conditions:
- The length of each substring is at least k.
- Each substring is a palindrome.

Return the maximum number of substrings in an optimal selection.

A substring is a contiguous sequence of characters within a string.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2488 Count Subarrays With Median K

HARD

You are given an array nums of size n consisting of distinct integers from 1 to n and a positive integer k.

Return the number of non-empty subarrays in nums that have a median equal to k.

Note:
- The median of an array is the middle element after sorting the array in ascending order. If the array is of even length, the median is the left middle element.
- For example, the median of [2,3,1,4] is 2, and the median of [8,4,3,5,1] is 4.
- A subarray is a contiguous part of an array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2489 Number Of Substrings With Fixed Ratio

MEDIUM

You are given a binary string s, and two integers num1 and num2. num1 and num2 are coprime numbers.

A ratio substring is a substring of s where the ratio between the number of 0's and the number of 1's in the substring is exactly num1 : num2.
- For example, if num1 = 2 and num2 = 3, then "01011" and "1110000111" are ratio substrings, while "11000" is not.

Return the number of non-empty ratio substrings of s.

Note that:
- A substring is a contiguous sequence of characters within a string.
- Two values x and y are coprime if gcd(x, y) == 1 where gcd(x, y) is the greatest common divisor of x and y.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2495 Number Of Subarrays Having Even Product

MEDIUM

Given a 0-indexed integer array nums, return the number of subarrays of nums having an even product.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2522 Partition String Into Substrings With Values At Most K

MEDIUM

You are given a string s consisting of digits from 1 to 9 and an integer k.

A partition of a string s is called good if:
- Each digit of s is part of exactly one substring.
- The value of each substring is less than or equal to k.

Return the minimum number of substrings in a good partition of s. If no good partition of s exists, return -1.

Note that:
- The value of a string is its result when interpreted as an integer. For example, the value of "123" is 123 and the value of "1" is 1.
- A substring is a contiguous sequence of characters within a string.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2524 Maximum Frequency Score Of A Subarray

HARD

You are given an integer array nums and a positive integer k.

The frequency score of an array is the sum of the distinct values in the array raised to the power of their frequencies, taking the sum modulo 109 + 7.
- For example, the frequency score of the array [5,4,5,7,4,4] is (43 + 52 + 71) modulo (109 + 7) = 96.

Return the maximum frequency score of a subarray of size k in nums. You should maximize the value under the modulo and not the actual value.

A subarray is a contiguous part of an array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2537 Count The Number Of Good Subarrays

MEDIUM

Given an integer array nums and an integer k, return the number of good subarrays of nums.

A subarray arr is good if there are at least k pairs of indices (i, j) such that i < j and arr[i] == arr[j].

A subarray is a contiguous non-empty sequence of elements within an array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2588 Count The Number Of Beautiful Subarrays

MEDIUM

You are given a 0-indexed integer array nums. In one operation, you can:
- Choose two different indices i and j such that 0 <= i, j < nums.length.
- Choose a non-negative integer k such that the kth bit (0-indexed) in the binary representation of nums[i] and nums[j] is 1.
- Subtract 2k from nums[i] and nums[j].

A subarray is beautiful if it is possible to make all of its elements equal to 0 after applying the above operation any number of times.

Return the number of beautiful subarrays in the array nums.

A subarray is a contiguous non-empty sequence of elements within an array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2606 Find The Substring With Maximum Cost

**MEDIUM**

You are given a string s, a string chars of distinct characters and an integer array vals of the same length as chars.

The cost of the substring is the sum of the values of each character in the substring. The cost of an empty string is considered 0.

The value of the character is defined in the following way:
- If the character is not in the string chars, then its value is its corresponding position (1-indexed) in the alphabet.
- For example, the value of 'a' is 1, the value of 'b' is 2, and so on. The value of 'z' is 26.
- Otherwise, assuming i is the index where the character occurs in the string chars, then its value is vals[i].

Return the maximum cost among all substrings of the string s.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2609 Find The Longest Balanced Substring Of A Binary String

**EASY**

You are given a binary string s consisting only of zeroes and ones.

A substring of s is considered balanced if all zeroes are before ones and the number of zeroes is equal to the number of ones inside the substring. Notice that the empty substring is considered a balanced substring.

Return the length of the longest balanced substring of s.

A substring is a contiguous sequence of characters within a string.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2696 Minimum String Length After Removing Substrings

**EASY**

You are given a string s consisting only of uppercase English letters.

You can apply some operations to this string where, in one operation, you can remove any occurrence of one of the substrings "AB" or "CD" from s.

Return the minimum possible length of the resulting string that you can obtain.

Note that the string concatenates after removing the substring and could produce new "AB" or "CD" substrings.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2743 Count Substrings Without Repeating Character

**MEDIUM**

You are given a string s consisting only of lowercase English letters. We call a substring special if it contains no character which has occurred at least twice (in other words, it does not contain a repeating character). Your task is to count the number of special substrings. For example, in the string "pop", the substring "po" is a special substring, however, "pop" is not special (since 'p' has occurred twice).

Return the number of special substrings.

A substring is a contiguous sequence of characters within a string. For example, "abc" is a substring of "abcd", but "acd" is not.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2763 Sum Of Imbalance Numbers Of All Subarrays

**HARD**

The imbalance number of a 0-indexed integer array arr of length n is defined as the number of indices in sarr = sorted(arr) such that:
- 0 <= i < n - 1, and
- sarr[i+1] - sarr[i] > 1

Here, sorted(arr) is the function that returns the sorted version of arr.

Given a 0-indexed integer array nums, return the sum of imbalance numbers of all its subarrays.

A subarray is a contiguous non-empty sequence of elements within an array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2767 Partition String Into Minimum Beautiful Substrings

**MEDIUM**

Given a binary string s, partition the string into one or more substrings such that each substring is beautiful.

A string is beautiful if:
- It doesn't contain leading zeros.
- It's the binary representation of a number that is a power of 5.

Return the minimum number of substrings in such partition. If it is impossible to partition the string s into beautiful substrings, return -1.

A substring is a contiguous sequence of characters in a string.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2799 Count Complete Subarrays In An Array

**MEDIUM**

You are given an array nums consisting of positive integers.

We call a subarray of an array complete if the following condition is satisfied:
- The number of distinct elements in the subarray is equal to the number of distinct elements in the whole array.

Return the number of complete subarrays.

A subarray is a contiguous non-empty part of an array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2841 Maximum Sum Of Almost Unique Subarray

**MEDIUM**

You are given an integer array nums and two positive integers m and k.

Return the maximum sum out of all almost unique subarrays of length k of nums. If no such subarray exists, return 0.

A subarray of nums is almost unique if it contains at least m distinct elements.

A subarray is a contiguous non-empty sequence of elements within an array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2845 Count Of Interesting Subarrays

**MEDIUM**

You are given a 0-indexed integer array nums, an integer modulo, and an integer k.

Your task is to find the count of subarrays that are interesting.

A subarray nums[l..r] is interesting if the following condition holds:
- Let cnt be the number of indices i in the range [l, r] such that nums[i] % modulo == k.
Then, cnt % modulo == k.

Return an integer denoting the count of interesting subarrays.

Note: A subarray is a contiguous non-empty sequence of elements within an array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2863 Maximum Length Of Semi Decreasing Subarrays

**MEDIUM**

You are given an integer array nums.

Return the length of the longest semi-decreasing subarray of nums, and 0 if there are no such subarrays.
- A subarray is a contiguous non-empty sequence of elements within an array.
- A non-empty array is semi-decreasing if its first element is strictly greater than its last element.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2871 Split Array Into Maximum Number Of Subarrays

**MEDIUM**

You are given an array nums consisting of non-negative integers.

We define the score of subarray nums[l..r] such that l <= r as nums[l] AND nums[l + 1] AND ... AND nums[r] where AND is the bitwise AND operation.

Consider splitting the array into one or more subarrays such that the following conditions are satisfied:
- Each element of the array belongs to exactly one subarray.
- The sum of scores of the subarrays is the minimum possible.

Return the maximum number of subarrays in a split that satisfies the conditions above.

A subarray is a contiguous part of an array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2913 Subarrays Distinct Element Sum Of Squares I

**EASY**

You are given a 0-indexed integer array nums.

The distinct count of a subarray of nums is defined as:
- Let nums[i..j] be a subarray of nums consisting of all the indices from i to j such that 0 <= i <= j < nums.length. Then the number of distinct values in nums[i..j] is called the distinct count of nums[i..j].

Return the sum of the squares of distinct counts of all subarrays of nums.

A subarray is a contiguous non-empty sequence of elements within an array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2941 Maximum Gcd Sum Of A Subarray

**HARD**

You are given an array of integers nums and an integer k.

The gcd-sum of an array a is calculated as follows:
- Let s be the sum of all the elements of a.
- Let g be the greatest common divisor of all the elements of a.
- The gcd-sum of a is equal to s * g.

Return the maximum gcd-sum of a subarray of nums with at least k elements.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2947 Count Beautiful Substrings I

**MEDIUM**

You are given a string s and a positive integer k.

Let vowels and consonants be the number of vowels and consonants in a string.

A string is beautiful if:
- vowels == consonants.
- (vowels * consonants) % k == 0, in other terms the multiplication of vowels and consonants is divisible by k.

Return the number of non-empty beautiful substrings in the given string s.

A substring is a contiguous sequence of characters in a string.

Vowel letters in English are 'a', 'e', 'i', 'o', and 'u'.

Consonant letters in English are every letter except vowels.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2950 Number Of Divisible Substrings

**MEDIUM**

Each character of the English alphabet has been mapped to a digit as shown below.

A string is divisible if the sum of the mapped values of its characters is divisible by its length.

Given a string s, return the number of divisible substrings of s.

A substring is a contiguous non-empty sequence of characters within a string.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2955 Number Of Same End Substrings

**MEDIUM**

You are given a 0-indexed string s, and a 2D array of integers queries, where queries[i] = [li, ri] indicates a substring of s starting from the index li and ending at the index ri (both inclusive), i.e. s[li..ri].

Return an array ans where ans[i] is the number of same-end substrings of queries[i].

A 0-indexed string t of length n is called same-end if it has the same character at both of its ends, i.e., t[0] == t[n - 1].

A substring is a contiguous non-empty sequence of characters within a string.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2958 Length Of Longest Subarray With At Most K Frequency

**MEDIUM**

You are given an integer array nums and an integer k.

The frequency of an element x is the number of times it occurs in an array.

An array is called good if the frequency of each element in this array is less than or equal to k.

Return the length of the longest good subarray of nums.

A subarray is a contiguous non-empty sequence of elements within an array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2962 Count Subarrays Where Max Element Appears At Least K Times

**MEDIUM**

You are given an integer array nums and a positive integer k.

Return the number of subarrays where the maximum element of nums appears at least k times in that subarray.

A subarray is a contiguous sequence of elements within an array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2970 Count The Number Of Incremovable Subarrays I

**EASY**

You are given a 0-indexed array of positive integers nums.

A subarray of nums is called incremovable if nums becomes strictly increasing on removing the subarray. For example, the subarray [3, 4] is an incremovable subarray of [5, 3, 4, 6, 7] because removing this subarray changes the array [5, 3, 4, 6, 7] to [5, 6, 7] which is strictly increasing.

Return the total number of incremovable subarrays of nums.

Note that an empty array is considered strictly increasing.

A subarray is a contiguous non-empty sequence of elements within an array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2981 Find Longest Special Substring That Occurs Thrice I

**MEDIUM**

You are given a string s that consists of lowercase English letters.

A string is called special if it is made up of only a single character. For example, the string "abc" is not special, whereas the strings "ddd", "zz", and "f" are special.

Return the length of the longest special substring of s which occurs at least thrice, or -1 if no special substring occurs at least thrice.

A substring is a contiguous non-empty sequence of characters within a string.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3010 Divide An Array Into Subarrays With Minimum Cost I

**EASY**

You are given an array of integers nums of length n.

The cost of an array is the value of its first element. For example, the cost of [1,2,3] is 1 while the cost of [3,4,1] is 3.

You need to divide nums into 3 disjoint contiguous subarrays.

Return the minimum possible sum of the cost of these subarrays.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3034 Number Of Subarrays That Match A Pattern I

**MEDIUM**

You are given a 0-indexed integer array nums of size n, and a 0-indexed integer array pattern of size m consisting of integers -1, 0, and 1.

A subarray nums[i..j] of size m + 1 is said to match the pattern if the following conditions hold for each element pattern[k]:
- nums[i + k + 1] > nums[i + k] if pattern[k] == 1.
- nums[i + k + 1] == nums[i + k] if pattern[k] == 0.
- nums[i + k + 1] < nums[i + k] if pattern[k] == -1.

Return the count of subarrays in nums that match the pattern.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3076 Shortest Uncommon Substring In An Array

**MEDIUM**

You are given an array arr of size n consisting of non-empty strings.

Find a string array answer of size n such that:
- answer[i] is the shortest substring of arr[i] that does not occur as a substring in any other string in arr. If multiple such substrings exist, answer[i] should be the lexicographically smallest. And if no such substring exists, answer[i] should be an empty string.

Return the array answer.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3083 Existence Of A Substring In A String And Its Reverse

**EASY**

Given a string s, find any substring of length 2 which is also present in the reverse of s.

Return true if such a substring exists, and false otherwise.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3084 Count Substrings Starting And Ending With Given Character

**MEDIUM**

You are given a string s and a character c. Return the total number of substrings of s that start and end with c.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 3090 Maximum Length Substring With Two Occurrences

EASY

Given a string s, return the maximum length of a substring such that it contains at most two occurrences of each character.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 3095 Shortest Subarray With Or At Least K I

EASY

You are given an array nums of non-negative integers and an integer k.

An array is called special if the bitwise OR of all of its elements is at least k.

Return the length of the shortest special non-empty subarray of nums, or return -1 if no special subarray exists.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 3097 Shortest Subarray With Or At Least K Ii

MEDIUM

You are given an array nums of non-negative integers and an integer k.

An array is called special if the bitwise OR of all of its elements is at least k.

Return the length of the shortest special non-empty subarray of nums, or return -1 if no special subarray exists.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 3104 Find Longest Self Contained Substring

HARD

Given a string s, your task is to find the length of the longest self-contained substring of s.

A substring t of a string s is called self-contained if t != s and for every character in t, it doesn't exist in the rest of s.

Return the length of the longest self-contained substring of s if it exists, otherwise, return -1.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 3105 Longest Strictly Increasing Or Strictly Decreasing Subarray

EASY

You are given an array of integers nums. Return the length of the longest subarray of nums which is either strictly increasing or strictly decreasing.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 3284 Sum Of Consecutive Subarrays

MEDIUM

We call an array arr of length n consecutive if one of the following holds:
- arr[i] - arr[i - 1] == 1 for all 1 <= i < n.
- arr[i] - arr[i - 1] == -1 for all 1 <= i < n.

The value of an array is the sum of its elements.

For example, [3, 4, 5] is a consecutive array of value 12 and [9, 8] is another of value 17. While [3, 4, 3] and [8, 6] are not consecutive.

Given an array of integers nums, return the sum of the values of all consecutive subarrays.

Since the answer may be very large, return it modulo 109 + 7.

Note that an array of length 1 is also considered consecutive.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3306 Count Of Substrings Containing Every Vowel And K Consonants Ii

MEDIUM

You are given a string word and a non-negative integer k.

Return the total number of substrings of word that contain every vowel ('a', 'e', 'i', 'o', and 'u') at least once and exactly k consonants.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3318 Find X Sum Of All K Long Subarrays I

EASY

You are given an array nums of n integers and two integers k and x.

The x-sum of an array is calculated by the following procedure:
- Count the occurrences of all elements in the array.
- Keep only the occurrences of the top x most frequent elements. If two elements have the same number of occurrences, the element with the bigger value is considered more frequent.
- Calculate the sum of the resulting array.

Note that if an array has less than x distinct elements, its x-sum is the sum of the array.

Return an integer array answer of length n - k + 1 where answer[i] is the x-sum of the subarray nums[i..i + k - 1].

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3329 Count Substrings With K Frequency Characters Ii

HARD

Given a string s and an integer k, return the total number of substrings of s where at least one character appears at least k times.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3349 Adjacent Increasing Subarrays Detection I

EASY

Given an array nums of n integers and an integer k, determine whether there exist two

adjacent subarrays of length k such that both subarrays are strictly increasing. Specifically, check if there are two subarrays starting at indices a and b (a < b), where:
- Both subarrays nums[a..a + k - 1] and nums[b..b + k - 1] are strictly increasing.
- The subarrays must be adjacent, meaning b = a + k.

Return true if it is possible to find two such subarrays, and false otherwise.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3350 Adjacent Increasing Subarrays Detection Ii

**MEDIUM**

Given an array nums of n integers, your task is to find the maximum value of k for which there exist two adjacent subarrays of length k each, such that both subarrays are strictly increasing. Specifically, check if there are two subarrays of length k starting at indices a and b (a < b), where:
- Both subarrays nums[a..a + k - 1] and nums[b..b + k - 1] are strictly increasing.
- The subarrays must be adjacent, meaning b = a + k.

Return the maximum possible value of k.

A subarray is a contiguous non-empty sequence of elements within an array.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3392 Count Subarrays Of Length Three With A Condition

**EASY**

Given an integer array nums, return the number of subarrays of length 3 such that the sum of the first and third numbers equals exactly half of the second number.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3422 Minimum Operations To Make Subarray Elements Equal

**MEDIUM**

You are given an integer array nums and an integer k. You can perform the following operation any number of times:
- Increase or decrease any element of nums by 1.

Return the minimum number of operations required to ensure that at least one subarray of size k in nums has all elements equal.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3480 Maximize Subarrays After Removing One Conflicting Pair

**HARD**

You are given an integer n which represents an array nums containing the numbers from 1 to n in order. Additionally, you are given a 2D array conflictingPairs, where conflictingPairs[i] = [a, b] indicates that a and b form a conflicting pair.

Remove exactly one element from conflictingPairs. Afterward, count the number of non-empty subarrays of nums which do not contain both a and b for any remaining conflicting pair [a, b].

Return the maximum number of subarrays possible after removing exactly one conflicting pair.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3487 Maximum Unique Subarray Sum After Deletion

EASY

You are given an integer array nums.

You are allowed to delete any number of elements from nums without making it empty.
After performing the deletions, select a subarray of nums such that:

Return the maximum sum of such a subarray.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

CHAPTER 04 / 52 EXERCISES

# Linked Lists

## 01 THE IDEA

A linked list consists of nodes connected by next references, rather than contiguous indexes.

## 02 SEE IT

```
HEAD -> [3|next] -> [8|next] -> [1|null]
```

## 03 JAVASCRIPT PATTERN

```
let current = head;
while (current) { current = current.next; }
```

## 04 HOW TO THINK

Draw every node and arrow. Track current, previous and next before changing links; handle empty and one-node lists.

## 05 PERFORMANCE

Traversal O(n); extra pointers O(1).

## EXERCISE COLLECTION

Try each question before opening the matching problem number in the separate Solutions PDF. The topic classification is a study aid; some exercises combine multiple patterns.

### 0002 Add Two Numbers

MEDIUM

You are given two non-empty linked lists representing two non-negative integers.

The digits are stored in reverse order, and each of their nodes contains a single digit.
Add the two numbers and return the sum as a linked list.

You may assume the two numbers do not contain any leading zero, except the number 0 itself.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0019 Remove Nth Node From End Of List

MEDIUM

Given the head of a linked list, remove the nth node from the end of the list and return its head.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 0021 Merge Two Sorted Lists

EASY

Merge two sorted linked lists and return it as a sorted list. The list should be made by splicing together the nodes of the first two lists.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 0024 Swap Nodes In Pairs

MEDIUM

Given a linked list, swap every two adjacent nodes and return its head. You must solve the problem without modifying the values in the list's nodes (i.e., only nodes themselves may be changed.)

TRY IT / Write examples, edge cases, and your own algorithm first.

## 0025 Reverse Nodes In K Group

HARD

Given the head of a linked list, reverse the nodes of the list k at a time, and return the modified list.

k is a positive integer and is less than or equal to the length of the linked list. If the number of nodes is not a multiple of k then left-out nodes, in the end, should remain as it is.

You may not alter the values in the list's nodes, only nodes themselves may be changed.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 0061 Rotate List

MEDIUM

Given the head of a linked list, rotate the list to the right by k places.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 0086 Partition List

MEDIUM

Given the head of a linked list and a value x, partition it such that all nodes less than x come before nodes greater than or equal to x.

You should preserve the original relative order of the nodes in each of the two partitions.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 0092 Reverse Linked List Ii

MEDIUM

Given the head of a singly linked list and two integers left and right where left <= right, reverse the nodes of the list from position left to position right, and return the reversed list.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0114 Flatten Binary Tree To Linked List

MEDIUM

Given the root of a binary tree, flatten the tree into a "linked list":
- The "linked list" should use the same TreeNode class where the right child pointer points to the next node in the list and the left child pointer is always null.
- The "linked list" should be in the same order as a pre-order traversal of the binary tree.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0141 Linked List Cycle

EASY

Given head, the head of a linked list, determine if the linked list has a cycle in it.

There is a cycle in a linked list if there is some node in the list that can be reached again by continuously following the next pointer. Internally, pos is used to denote the index of the node that tail's next pointer is connected to. Note that pos is not passed as a parameter.

Return true if there is a cycle in the linked list. Otherwise, return false.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0142 Linked List Cycle Ii

MEDIUM

Given the head of a linked list, return the node where the cycle begins. If there is no cycle, return null.

There is a cycle in a linked list if there is some node in the list that can be reached again by continuously following the next pointer. Internally, pos is used to denote the index of the node that tail's next pointer is connected to (0-indexed). It is -1 if there is no cycle. Note that pos is not passed as a parameter.

Do not modify the linked list.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0143 Reorder List

MEDIUM

You are given the head of a singly linked-list. The list can be represented as:

L0 → L1 → … → Ln - 1 → Ln
Reorder the list to be on the following form:

L0 → Ln → L1 → Ln - 1 → L2 → Ln - 2 → …
You may not modify the values in the list's nodes. Only nodes themselves may be changed.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0160 Intersection Of Two Linked Lists

MEDIUM

Given the heads of two singly linked-lists headA and headB, return the node at which the two lists intersect. If the two linked lists have no intersection at all, return null.

For example, the following two linked lists begin to intersect at node c1:

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0203 Remove Linked List Elements

EASY

Given the head of a linked list and an integer val, remove all the nodes of the linked list that has Node.val == val, and return the new head.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0206 Reverse Linked List

EASY

Given the head of a singly linked list, reverse the list, and return the reversed list.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0234 Palindrome Linked List

EASY

Given the `head` of a singly linked list, return `true` if it is a palindrome.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0237 Delete Node In A Linked List

EASY

Write a function to delete a node in a singly-linked list. You will not be given access to the head of the list, instead you will be given access to the node to be deleted directly.

It is guaranteed that the node to be deleted is not a tail node in the list.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0328 Odd Even Linked List

MEDIUM

Given the head of a singly linked list, group all the nodes with odd indices together followed by the nodes with even indices, and return the reordered list.

The first node is considered odd, and the second node is even, and so on.

Note that the relative order inside both the even and odd groups should remain as it was in the input.

You must solve the problem in O(1) extra space complexity and O(n) time complexity.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0369 Plus One Linked List

MEDIUM

Given a non-negative integer represented as a linked list of digits, plus one to the integer.

The digits are stored such that the most significant digit is at the head of the list.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0382 Linked List Random Node

MEDIUM

Given a singly linked list, return a random node's value from the linked list. Each node must have the same probability of being chosen.

Implement the Solution class:
- Solution(ListNode head) Initializes the object with the head of the singly-linked list head.
- int getRandom() Chooses a node randomly from the list and returns its value. All the nodes of the list should be equally likely to be chosen.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0426 Convert Binary Search Tree To Sorted Doubly Linked List

MEDIUM

Convert a Binary Search Tree to a sorted Circular Doubly-Linked List in place.

You can think of the left and right pointers as synonymous to the predecessor and successor pointers in a doubly-linked list. For a circular doubly linked list, the predecessor of the first element is the last element, and the successor of the last element is the first element.

We want to do the transformation in place. After the transformation, the left pointer of the tree node should point to its predecessor, and the right pointer should point to its successor. You should return the pointer to the smallest element of the linked list.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0430 Flatten A Multilevel Doubly Linked List

MEDIUM

You are given a doubly linked list, which contains nodes that have a next pointer, a previous pointer, and an additional child pointer. This child pointer may or may not point to a separate doubly linked list, also containing these special nodes. These child lists may have one or more children of their own, and so on, to produce a multilevel data structure as shown in the example below.

Given the head of the first level of the list, flatten the list so that all the nodes appear in a single-level, doubly linked list. Let curr be a node with a child list. The nodes in the child list should appear after curr and before curr.next in the flattened list.

Return the head of the flattened list. The nodes in the list must have all of their child pointers set to null.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0445 Add Two Numbers Ii

MEDIUM

You are given two non-empty linked lists representing two non-negative integers. The most significant digit comes first and each of their nodes contains a single digit. Add the two

numbers and return the sum as a linked list.

You may assume the two numbers do not contain any leading zero, except the number 0 itself.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0707 Design Linked List

**MEDIUM**

Design your implementation of the linked list. You can choose to use a singly or doubly linked list.

A node in a singly linked list should have two attributes: val and next. val is the value of the current node, and next is a pointer/reference to the next node.

If you want to use the doubly linked list, you will need one more attribute prev to indicate the previous node in the linked list. Assume all nodes in the linked list are 0-indexed.

Implement the MyLinkedList class:
- MyLinkedList() Initializes the MyLinkedList object.
- int get(int index) Get the value of the indexth node in the linked list. If the index is invalid, return -1.
- void addAtHead(int val) Add a node of value val before the first element of the linked list. After the insertion, the new node will be the first node of the linked list.
- void addAtTail(int val) Append a node of value val as the last element of the linked list.
- void addAtIndex(int index, int val) Add a node of value val before the indexth node in the linked list. If index equals the length of the linked list, the node will be appended to the end of the linked list. If index is greater than the length, the node will not be inserted.
- void deleteAtIndex(int index) Delete the indexth node in the linked list, if the index is valid.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0708 Insert Into A Sorted Circular Linked List

**MEDIUM**

Given a Circular Linked List node, which is sorted in non-descending order, write a function to insert a value insertVal into the list such that it remains a sorted circular list. The given node can be a reference to any single node in the list and may not necessarily be the smallest value in the circular list.

If there are multiple suitable places for insertion, you may choose any place to insert the new value. After the insertion, the circular list should remain sorted.

If the list is empty (i.e., the given node is null), you should create a new single circular list and return the reference to that single node. Otherwise, you should return the originally given node.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0725 Split Linked List In Parts

**MEDIUM**

Given the head of a singly linked list and an integer k, split the linked list into k consecutive linked list parts.

The length of each part should be as equal as possible: no two parts should have a size differing by more than one. This may lead to some parts being null.

The parts should be in the order of occurrence in the input list, and parts occurring earlier should always have a size greater than or equal to parts occurring later.

Return an array of the k parts.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0817 Linked List Components

**MEDIUM**

You are given the head of a linked list containing unique integer values and an integer array nums that is a subset of the linked list values.

Return the number of connected components in nums where two values are connected if they appear consecutively in the linked list.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0876 Middle Of The Linked List

**EASY**

Given the head of a singly linked list, return the middle node of the linked list.

If there are two middle nodes, return the second middle node.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1019 Next Greater Node In Linked List

**MEDIUM**

You are given the head of a linked list with n nodes.

For each node in the list, find the value of the next greater node. That is, for each node, find the value of the first node that is next to it and has a strictly larger value than it.

Return an integer array answer where answer[i] is the value of the next greater node of the ith node (1-indexed). If the ith node does not have a next greater node, set answer[i] = 0.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1171 Remove Zero Sum Consecutive Nodes From Linked List

**MEDIUM**

Given the head of a linked list, we repeatedly delete consecutive sequences of nodes that sum to 0 until there are no such sequences.

After doing so, return the head of the final linked list. You may return any such answer.

(Note that in the examples below, all sequences are serializations of ListNode objects.)

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1265 Print Immutable Linked List In Reverse

**MEDIUM**

You are given an immutable linked list, print out all values of each node in reverse with the help of the following interface:
- ImmutableListNode: An interface of immutable linked list, you are given the head of the list.

You need to use the following functions to access the linked list (you can't access the ImmutableListNode directly):
- ImmutableListNode.printValue(): Print value of the current node.
- ImmutableListNode.getNext(): Return the next node.

The input is only given to initialize the linked list internally. You must solve this problem

without modifying the linked list. In other words, you must operate the linked list using only the mentioned APIs.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1290 Convert Binary Number In A Linked List To Integer

**EASY**

Given `head` which is a reference node to a singly-linked list.
The value of each node in the linked list is either 0 or 1.
The linked list holds the binary representation of a number.
Return the decimal value of the number in the linked list.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1367 Linked List In Binary Tree

**MEDIUM**

Given a binary tree root and a linked list with head as the first node.

Return True if all the elements in the linked list starting from the head correspond to some downward path connected in the binary tree otherwise return False.

In this context downward path means a path that starts at some node and goes downwards.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1474 Delete N Nodes After M Nodes Of A Linked List

**EASY**

You are given the head of a linked list and two integers m and n.

Traverse the linked list and remove some nodes in the following way:
- Start with the head as the current node.
- Keep the first m nodes starting with the current node.
- Remove the next n nodes
- Keep repeating steps 2 and 3 until you reach the end of the list.

Return the head of the modified list after removing the mentioned nodes.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1634 Add Two Polynomials Represented As Linked Lists

**MEDIUM**

A polynomial linked list is a special type of linked list where every node represents a term in a polynomial expression.

Each node has three attributes:
- coefficient: an integer representing the number multiplier of the term. The coefficient of the term 9x4 is 9.
- power: an integer representing the exponent. The power of the term 9x4 is 4.
- next: a pointer to the next node in the list, or null if it is the last node of the list.

For example, the polynomial 5x3 + 4x - 7 is represented by the polynomial linked list illustrated below:

The polynomial linked list must be in its standard form: the polynomial must be in strictly descending order by its power value. Also, terms with a coefficient of 0 are omitted.

Given two polynomial linked list heads, poly1 and poly2, add the polynomials together and return the head of the sum of the polynomials.

PolyNode format:
The input/output format is as a list of n nodes, where each node is represented as its [coefficient, power]. For example, the polynomial 5x3 + 4x - 7 would be represented as: [[5,3],[4,1],[-7,0]].

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1669 Merge In Between Linked Lists

**MEDIUM**

You are given two linked lists: list1 and list2 of sizes n and m respectively.

Remove list1's nodes from the ath node to the bth node, and put list2 in their place.

The blue edges and nodes in the following figure indicate the result:

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1721 Swapping Nodes In A Linked List

**MEDIUM**

You are given the head of a linked list, and an integer k.

Return the head of the linked list after swapping the values of the kth node from the beginning and the kth node from the end (the list is 1-indexed).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1836 Remove Duplicates From An Unsorted Linked List

**MEDIUM**

Given the head of a linked list, find all the values that appear more than once in the list and delete the nodes that have any of those values.

Return the linked list after the deletions.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2046 Sort Linked List Already Sorted Using Absolute Values

**MEDIUM**

Given the head of a singly linked list that is sorted in non-decreasing order using the absolute values of its nodes, return the list sorted in non-decreasing order using the actual values of its nodes.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2074 Reverse Nodes In Even Length Groups

**MEDIUM**

You are given the head of a linked list.

The nodes in the linked list are sequentially assigned to non-empty groups whose lengths form the sequence of the natural numbers (1, 2, 3, 4, ...). The length of a group is the number of nodes assigned to it. In other words:
- The 1st node is assigned to the first group.
- The 2nd and the 3rd nodes are assigned to the second group.
- The 4th, 5th, and 6th nodes are assigned to the third group, and so on.

Note that the length of the last group may be less than or equal to 1 + the length of the second to last group.

Reverse the nodes in each group with an even length, and return the head of the modified linked list.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2095 Delete The Middle Node Of A Linked List

**MEDIUM**

You are given the head of a linked list. Delete the middle node, and return the head of the modified linked list.

The middle node of a linked list of size n is the ■n / 2■th node from the start using 0-based indexing, where ■x■ denotes the largest integer less than or equal to x.

For n = 1, 2, 3, 4, and 5, the middle nodes are 0, 1, 1, 2, and 2, respectively.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2130 Maximum Twin Sum Of A Linked List

**MEDIUM**

In a linked list of size n, where n is even, the ith node (0-indexed) of the linked list is known as the twin of the (n-1-i)th node, if 0 <= i <= (n / 2) - 1.

For example, if n = 4, then node 0 is the twin of node 3, and node 1 is the twin of node 2. These are the only nodes with twins for n = 4.

The twin sum is defined as the sum of a node and its twin.

Given the head of a linked list with even length, return the maximum twin sum of the linked list.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2181 Merge Nodes In Between Zeros

**MEDIUM**

You are given the head of a linked list, which contains a series of integers separated by 0's. The beginning and end of the linked list will have Node.val == 0.

For every two consecutive 0's, merge all the nodes lying in between them into a single node whose value is the sum of all the merged nodes. The modified list should not contain any 0's.

Return the head of the modified linked list.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2487 Remove Nodes From Linked List

**MEDIUM**

You are given the head of a linked list.

Remove every node which has a node with a greater value anywhere to the right side of it.

Return the head of the modified linked list.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2674 Split A Circular Linked List

**MEDIUM**

Given a circular linked list list of positive integers, your task is to split it into 2 circular

linked lists so that the first one contains the first half of the nodes in list (exactly ceil(list.length / 2) nodes) in the same order they appeared in list, and the second one contains the rest of the nodes in list in the same order they appeared in list.

Return an array answer of length 2 in which the first element is a circular linked list representing the first half and the second element is a circular linked list representing the second half.

A circular linked list is a normal linked list with the only difference being that the last node's next node, is the first node.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2807 Insert Greatest Common Divisors In Linked List

**MEDIUM**

Given the head of a linked list head, in which each node contains an integer value.

Between every pair of adjacent nodes, insert a new node with a value equal to the greatest common divisor of them.

Return the linked list after insertion.

The greatest common divisor of two numbers is the largest positive integer that evenly divides both numbers.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2816 Double A Number Represented As A Linked List

**MEDIUM**

You are given the head of a non-empty linked list representing a non-negative integer without leading zeroes.

Return the head of the linked list after doubling it.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3062 Winner Of The Linked List Game

**EASY**

You are given the head of a linked list of even length containing integers.

Each odd-indexed node contains an odd integer and each even-indexed node contains an even integer.

We call each even-indexed node and its next node a pair, e.g., the nodes with indices 0 and 1 are a pair, the nodes with indices 2 and 3 are a pair, and so on.

For every pair, we compare the values of the nodes in the pair:
- If the odd-indexed node is higher, the "Odd" team gets a point.
- If the even-indexed node is higher, the "Even" team gets a point.

Return the name of the team with the higher points, if the points are equal, return "Tie".

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3063 Linked List Frequency

**EASY**

Given the head of a linked list containing k distinct elements, return the head to a linked list of length k containing the frequency of each distinct element in the given linked list in any order.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 3217 Delete Nodes From Linked List Present In Array

MEDIUM

You are given an array of integers nums and the head of a linked list. Return the head of the modified linked list after removing all nodes from the linked list that have a value that exists in nums.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 3263 Convert Doubly Linked List To Array I

EASY

You are given the head of a doubly linked list, which contains nodes that have a next pointer and a previous pointer.

Return an integer array which contains the elements of the linked list in order.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 3294 Convert Doubly Linked List To Array Ii

MEDIUM

You are given an arbitrary node from a doubly linked list, which contains nodes that have a next pointer and a previous pointer.

Return an integer array which contains the elements of the linked list in order.

TRY IT / Write examples, edge cases, and your own algorithm first.

CHAPTER 05 / 44 EXERCISES

# Stacks, Queues & Monotonic Structures

## 01 THE IDEA

A stack removes the most recent item (LIFO); a queue removes the oldest (FIFO). Monotonic stacks keep a useful ordering.

## 02 SEE IT




## 03 JAVASCRIPT PATTERN

```
const stack = [];
stack.push(3); stack.push(8); stack.pop();
```

## 04 HOW TO THINK

Sketch pushes and pops one operation at a time. For a monotonic stack, explain why removed entries can never help later.

## 05 PERFORMANCE

Push/pop usually O(1); many monotonic-stack scans O(n).

## EXERCISE COLLECTION

Try each question before opening the matching problem number in the separate Solutions PDF. The topic classification is a study aid; some exercises combine multiple patterns.

### 0020 Valid Parentheses

EASY

Given a string s containing just the characters '(', ')', '{', '}', '[' and ']', determine if the input string is valid.

An input string is valid if:
- Open brackets must be closed by the same type of brackets.
- Open brackets must be closed in the correct order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0022 Generate Parentheses

MEDIUM

Given n pairs of parentheses, write a function to generate all combinations of well-formed parentheses.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0032 Longest Valid Parentheses

HARD

Given a string containing just the characters '(' and ')', return the length of the longest valid (well-formed) parentheses substring.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0084 Largest Rectangle In Histogram

HARD

Given an array of integers heights representing the histogram's bar height where the width of each bar is 1, return the area of the largest rectangle in the histogram.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0155 Min Stack

MEDIUM

Design a stack that supports push, pop, top, and retrieving the minimum element in constant time.

Implement the MinStack class:
- MinStack() initializes the stack object.
- void push(int val) pushes the element val onto the stack.
- void pop() removes the element on the top of the stack.
- int top() gets the top element of the stack.
- int getMin() retrieves the minimum element in the stack.

You must implement a solution with O(1) time complexity for each function.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0224 Basic Calculator

HARD

Given a string s representing a valid expression, implement a basic calculator to evaluate it, and return the result of the evaluation.

Note: You are not allowed to use any built-in function which evaluates strings as mathematical expressions, such as eval().

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0225 Implement Stack Using Queues

EASY

Implement a last-in-first-out (LIFO) stack using only two queues. The implemented stack should support all the functions of a normal stack (push, top, pop, and empty).

Implement the MyStack class:

- void push(int x) Pushes element x to the top of the stack.
- int pop() Removes the element on the top of the stack and returns it.
- int top() Returns the element on the top of the stack.
- boolean empty() Returns true if the stack is empty, false otherwise.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0227 Basic Calculator Ii

**MEDIUM**

Given a string s which represents an expression, evaluate this expression and return its value.

The integer division should truncate toward zero.

You may assume that the given expression is always valid. All intermediate results will be in the range of [-231, 231 - 1].

Note: You are not allowed to use any built-in function which evaluates strings as mathematical expressions, such as eval().

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0232 Implement Queue Using Stacks

**EASY**

Implement a first in first out (FIFO) queue using only two stacks.
The implemented queue should support all the functions of a normal queue (push, peek, pop, and empty).

Implement the MyQueue class:
- void push(int x) Pushes element x to the back of the queue.
- int pop() Removes the element from the front of the queue and returns it.
- int peek() Returns the element at the front of the queue.
- boolean empty() Returns true if the queue is empty, false otherwise.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0241 Different Ways To Add Parentheses

**MEDIUM**

Given a string expression of numbers and operators, return all possible results from computing all the different possible ways to group numbers and operators. You may return the answer in any order.

The test cases are generated such that the output values fit in a 32-bit integer and the number of different results does not exceed 104.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0301 Remove Invalid Parentheses

**HARD**

Given a string s that contains parentheses and letters, remove the minimum number of invalid parentheses to make the input string valid.

Return a list of unique strings that are valid with the minimum number of removals. You may return the answer in any order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0406 Queue Reconstruction By Height

MEDIUM

You are given an array of people, people, which are the attributes of some people in a queue (not necessarily in order). Each people[i] = [hi, ki] represents the ith person of height hi with exactly ki other people in front who have a height greater than or equal to hi.

Reconstruct and return the queue that is represented by the input array people. The returned queue should be formatted as an array queue, where queue[j] = [hj, kj] is the attributes of the jth person in the queue (queue[0] is the person at the front of the queue).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0496 Next Greater Element I

EASY

The next greater element of some element x in an array is the first greater element that is to the right of x in the same array.

You are given two distinct 0-indexed integer arrays nums1 and nums2, where nums1 is a subset of nums2.

For each 0 <= i < nums1.length, find the index j such that nums1[i] == nums2[j] and determine the next greater element of nums2[j] in nums2. If there is no next greater element, then the answer for this query is -1.

Return an array ans of length nums1.length such that ans[i] is the next greater element as described above.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0503 Next Greater Element Ii

MEDIUM

Given a circular integer array nums (i.e., the next element of nums[nums.length - 1] is nums[0]), return the next greater number for every element in nums.

The next greater number of a number x is the first greater number to its traversing-order next in the array, which means you could search circularly to find its next greater number. If it doesn't exist, return -1 for this number.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0556 Next Greater Element Iii

MEDIUM

Given a positive integer n, find the smallest integer which has exactly the same digits existing in the integer n and is greater in value than n. If no such positive integer exists, return -1.

Note that the returned integer should fit in 32-bit integer, if there is a valid answer but it does not fit in 32-bit integer, return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0622 Design Circular Queue

MEDIUM

Design your implementation of the circular queue. The circular queue is a linear data structure in which the operations are performed based on FIFO (First In First Out) principle, and the last position is connected back to the first position to make a circle. It is also called "Ring Buffer".

One of the benefits of the circular queue is that we can make use of the spaces in front of the queue. In a normal queue, once the queue becomes full, we cannot insert the next element even if there is a space in front of the queue. But using the circular queue, we can use the space to store new values.

Implement the MyCircularQueue class:
- MyCircularQueue(k) Initializes the object with the size of the queue to be k.
- int Front() Gets the front item from the queue. If the queue is empty, return -1.
- int Rear() Gets the last item from the queue. If the queue is empty, return -1.
- boolean enQueue(int value) Inserts an element into the circular queue. Return true if the operation is successful.
- boolean deQueue() Deletes an element from the circular queue. Return true if the operation is successful.
- boolean isEmpty() Checks whether the circular queue is empty or not.
- boolean isFull() Checks whether the circular queue is full or not.

You must solve the problem without using the built-in queue data structure in your programming language.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0716 Max Stack

HARD

Design a max stack data structure that supports the stack operations and supports finding the stack's maximum element.

Implement the MaxStack class:
- MaxStack() Initializes the stack object.
- void push(int x) Pushes element x onto the stack.
- int pop() Removes the element on top of the stack and returns it.
- int top() Gets the element on the top of the stack without removing it.
- int peekMax() Retrieves the maximum element in the stack without removing it.
- int popMax() Retrieves the maximum element in the stack and removes it. If there is more than one maximum element, only remove the top-most one.

You must come up with a solution that supports O(1) for each top call and O(logn) for each other call.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0739 Daily Temperatures

MEDIUM

Given a list of daily temperatures T, return a list such that, for each day in the input, tells you how many days you would have to wait until a warmer temperature. If there is no future day for which this is possible, put 0 instead.

For example, given the list of temperatures T = [73, 74, 75, 71, 69, 72, 76, 73], your output should be [1, 1, 4, 2, 1, 1, 0, 0].

Note: The length of temperatures will be in the range [1, 30000].
Each temperature will be an integer in the range [30, 100].
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0770 Basic Calculator Iv

**HARD**

Given an expression such as expression = "e + 8 - a + 5" and an evaluation map such as {"e": 1} (given in terms of evalvars = ["e"] and evalints = [1]), return a list of tokens representing the simplified expression, such as ["-1*a","14"]
- An expression alternates chunks and symbols, with a space separating each chunk and symbol.
- A chunk is either an expression in parentheses, a variable, or a non-negative integer.
- A variable is a string of lowercase letters (not including digits.) Note that variables can be multiple letters, and note that variables never have a leading coefficient or unary operator like "2x" or "-x".

Expressions are evaluated in the usual order: brackets first, then multiplication, then addition and subtraction.

- For example, expression = "1 + 2 * 3" has an answer of ["7"].

The format of the output is as follows:
- For each term of free variables with a non-zero coefficient, we write the free variables within a term in sorted order lexicographically.
- For example, we would never write a term like "b*a*c", only "a*b*c".
- Terms have degrees equal to the number of free variables being multiplied, counting multiplicity. We write the largest degree terms of our answer first, breaking ties by lexicographic order ignoring the leading coefficient of the term.
- For example, "a*a*b*c" has degree 4.
- The leading coefficient of the term is placed directly to the left with an asterisk separating it from the variables (if they exist.) A leading coefficient of 1 is still printed.
- An example of a well-formatted answer is ["-2*a*a*a", "3*a*a*b", "3*b*b", "4*a", "5*c", "-6"].
- Terms (including constant terms) with coefficient 0 are not included.
- For example, an expression of "0" has an output of [].

Note: You may assume that the given expression is always valid. All intermediate results will be in the range of [-231, 231 - 1].
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0772 Basic Calculator Iii

**HARD**

Implement a basic calculator to evaluate a simple expression string.

The expression string contains only non-negative integers, '+', '-', '*', '/' operators, and open '(' and closing parentheses ')'. The integer division should truncate toward zero.

You may assume that the given expression is always valid. All intermediate results will be in the range of [-231, 231 - 1].

Note: You are not allowed to use any built-in function which evaluates strings as mathematical expressions, such as eval().
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0856 Score Of Parentheses

**MEDIUM**

Given a balanced parentheses string s, return the score of the string.

The score of a balanced parentheses string is based on the following rule:
- "()" has score 1.
- AB has score A + B, where A and B are balanced parentheses strings.
- (A) has score 2 * A, where A is a balanced parentheses string.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0895 Maximum Frequency Stack

**HARD**

Design a stack-like data structure to push elements to the stack and pop the most frequent element from the stack.

Implement the FreqStack class:
- FreqStack() constructs an empty frequency stack.
- void push(int val) pushes an integer val onto the top of the stack.
- int pop() removes and returns the most frequent element in the stack.
- If there is a tie for the most frequent element, the element closest to the stack's top is removed and returned.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0899 Orderly Queue

**HARD**

You are given a string s and an integer k. You can choose one of the first k letters of s and append it at the end of the string.

Return the lexicographically smallest string you could have after applying the mentioned step any number of moves.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0921 Minimum Add To Make Parentheses Valid

**MEDIUM**

A parentheses string is valid if and only if:
- It is the empty string,
- It can be written as AB (A concatenated with B), where A and B are valid strings, or
- It can be written as (A), where A is a valid string.

You are given a parentheses string s. In one move, you can insert a parenthesis at any position of the string.
- For example, if s = "())", you can insert an opening parenthesis to be "(())" or a closing parenthesis to be "()))".

Return the minimum number of moves required to make s valid.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0946 Validate Stack Sequences

**MEDIUM**

Given two integer arrays pushed and popped each with distinct values, return true if this could have been the result of a sequence of push and pop operations on an initially empty stack, or false otherwise.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0991 Broken Calculator

MEDIUM

There is a broken calculator that has the integer startValue on its display initially.
In one operation, you can:
- multiply the number on display by 2, or
- subtract 1 from the number on display.

Given two integers startValue and target, return the minimum number of operations needed to display target on the calculator.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1021 Remove Outermost Parentheses

EASY

A valid parentheses string is either empty "", "(" + A + ")", or A + B, where A and B are valid parentheses strings, and + represents string concatenation.
- For example, "", "()", "(())()", and "(()(()))" are all valid parentheses strings.

A valid parentheses string s is primitive if it is nonempty, and there does not exist a way to split it into s = A + B, with A and B nonempty valid parentheses strings.

Given a valid parentheses string s, consider its primitive decomposition: s = P1 + P2 + ... + Pk, where Pi are primitive valid parentheses strings.

Return s after removing the outermost parentheses of every primitive string in the primitive decomposition of s.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1111 Maximum Nesting Depth Of Two Valid Parentheses Strings

MEDIUM

A string is a valid parentheses string (denoted VPS) if and only if it consists of "(" and ")" characters only, and:
- It is the empty string, or
- It can be written as AB (A concatenated with B), where A and B are VPS's, or
- It can be written as (A), where A is a VPS.

We can similarly define the nesting depth depth(S) of any VPS S as follows:
- depth("") = 0
- depth(A + B) = max(depth(A), depth(B)), where A and B are VPS's
- depth("(" + A + ")") = 1 + depth(A), where A is a VPS.

For example, "", "()()", and "()(()())" are VPS's (with nesting depths 0, 1, and 2), and ")(" and "(()" are not VPS's.

Given a VPS seq, split it into two disjoint subsequences A and B, such that A and B are VPS's (and A.length + B.length = seq.length).

Now choose any such A and B such that max(depth(A), depth(B)) is the minimum possible value.

Return an answer array (of length seq.length) that encodes such a choice of A and B: answer[i] = 0 if seq[i] is part of A, else answer[i] = 1. Note that even though multiple answers may exist, you may return any of them.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1172 Dinner Plate Stacks

HARD

You have an infinite number of stacks arranged in a row and numbered (left to right) from 0, each of the stacks has the same maximum capacity.

Implement the DinnerPlates class:
- DinnerPlates(int capacity) Initializes the object with the maximum capacity of the stacks capacity.
- void push(int val) Pushes the given integer val into the leftmost stack with a size less than capacity.
- int pop() Returns the value at the top of the rightmost non-empty stack and removes it from that stack, and returns -1 if all the stacks are empty.
- int popAtStack(int index) Returns the value at the top of the stack with the given index index and removes it from that stack or returns -1 if the stack with that given index is empty.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1249 Minimum Remove To Make Valid Parentheses

MEDIUM

Given a string s of '(' , ')' and lowercase English characters.

Your task is to remove the minimum number of parentheses ( '(' or ')', in any positions ) so that the resulting parentheses string is valid and return any valid string.

Formally, a parentheses string is valid if and only if:
- It is the empty string, contains only lowercase characters, or
- It can be written as AB (A concatenated with B), where A and B are valid strings, or
- It can be written as (A), where A is a valid string.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1381 Design A Stack With Increment Operation

MEDIUM

Design a stack that supports increment operations on its elements.

Implement the CustomStack class:
- CustomStack(int maxSize) Initializes the object with maxSize which is the maximum number of elements in the stack.
- void push(int x) Adds x to the top of the stack if the stack has not reached the maxSize.
- int pop() Pops and returns the top of the stack or -1 if the stack is empty.
- void inc(int k, int val) Increments the bottom k elements of the stack by val. If there are less than k elements in the stack, increment all the elements in the stack.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1441 Build An Array With Stack Operations

MEDIUM

You are given an integer array target and an integer n.

You have an empty stack with the two following operations:
- "Push": pushes an integer to the top of the stack.
- "Pop": removes the integer on the top of the stack.

You also have a stream of the integers in the range [1, n].

Use the two stack operations to make the numbers in the stack (from the bottom to the top) equal to target. You should follow the following rules:
- If the stream of the integers is not empty, pick the next integer from the stream and push it to the top of the stack.
- If the stack is not empty, pop the integer at the top of the stack.
- If, at any moment, the elements in the stack (from the bottom to the top) are equal to target, do not read new integers from the stream and do not do more operations on the stack.

Return the stack operations needed to build target following the mentioned rules. If there are multiple valid answers, return any of them.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1541 Minimum Insertions To Balance A Parentheses String

**MEDIUM**

Given a parentheses string s containing only the characters '(' and ')'. A parentheses string is balanced if:
- Any left parenthesis '(' must have a corresponding two consecutive right parenthesis '))'.
- Left parenthesis '(' must go before the corresponding two consecutive right parenthesis '))'.

In other words, we treat '(' as an opening parenthesis and '))' as a closing parenthesis.
- For example, "())", "())(()))" and "(())()))" are balanced, ")()", "()))" and "(()))" are not balanced.

You can insert the characters '(' and ')' at any position of the string to balance it if needed.

Return the minimum number of insertions needed to make s balanced.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1614 Maximum Nesting Depth Of The Parentheses

**EASY**

Given a valid parentheses string s, return the nesting depth of s. The nesting depth is the maximum number of nested parentheses.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1670 Design Front Middle Back Queue

**MEDIUM**

Design a queue that supports push and pop operations in the front, middle, and back.

Implement the FrontMiddleBack class:
- FrontMiddleBack() Initializes the queue.
- void pushFront(int val) Adds val to the front of the queue.
- void pushMiddle(int val) Adds val to the middle of the queue.
- void pushBack(int val) Adds val to the back of the queue.
- int popFront() Removes the front element of the queue and returns it. If the queue is empty, return -1.
- int popMiddle() Removes the middle element of the queue and returns it. If the queue is empty, return -1.
- int popBack() Removes the back element of the queue and returns it. If the queue is empty, return -1.

Notice that when there are two middle position choices, the operation is performed on the frontmost middle position choice. For example:
- Pushing 6 into the middle of [1, 2, 3, 4, 5] results in [1, 2, 6, 3, 4, 5].
- Popping the middle from [1, 2, 3, 4, 5, 6] returns 3 and results in [1, 2, 4, 5, 6].

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1691 Maximum Height By Stacking Cuboids

HARD

Given n cuboids where the dimensions of the ith cuboid is cuboids[i] = [widthi, lengthi, heighti] (0-indexed). Choose a subset of cuboids and place them on each other.

You can place cuboid i on cuboid j if widthi <= widthj and lengthi <= lengthj and heighti <= heightj. You can rearrange any cuboid's dimensions by rotating it to put it on another cuboid.

Return the maximum height of the stacked cuboids.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1756 Design Most Recently Used Queue

MEDIUM

Design a queue-like data structure that moves the most recently used element to the end of the queue.

Implement the MRUQueue class:
- MRUQueue(int n) constructs the MRUQueue with n elements: [1,2,3,...,n].
- int fetch(int k) moves the kth element (1-indexed) to the end of the queue and returns it.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1807 Evaluate The Bracket Pairs Of A String

MEDIUM

You are given a string s that contains some bracket pairs, with each pair containing a non-empty key.

- For example, in the string "(name)is(age)yearsold", there are two bracket pairs that contain the keys "name" and "age".

You know the values of a wide range of keys. This is represented by a 2D string array knowledge where each knowledge[i] = [keyi, valuei] indicates that key keyi has a value of valuei.

You are tasked to evaluate all of the bracket pairs. When you evaluate a bracket pair that contains some key keyi, you will:
- Replace keyi and the bracket pair with the key's corresponding valuei.
- If you do not know the value of the key, you will replace keyi and the bracket pair with a question mark "?" (without the quotation marks).

Each key will appear at most once in your knowledge. There will not be any nested brackets in s.

Return the resulting string after evaluating all of the bracket pairs.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1944 Number Of Visible People In A Queue

HARD

There are n people standing in a queue, and they numbered from 0 to n - 1 in left to right order. You are given an array heights of distinct integers where heights[i] represents the height of the ith person.

A person can see another person to their right in the queue if everybody in between is shorter than both of them. More formally, the ith person can see the jth person if i < j and min(heights[i], heights[j]) > max(heights[i+1], heights[i+2], ..., heights[j-1]).

Return an array answer of length n where answer[i] is the number of people the ith person

can see to their right in the queue.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2048 Next Greater Numerically Balanced Number

**MEDIUM**

An integer x is numerically balanced if for every digit d in the number x, there are exactly d occurrences of that digit in x.

Given an integer n, return the smallest numerically balanced number strictly greater than n.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2116 Check If A Parentheses String Can Be Valid

**MEDIUM**

A parentheses string is a non-empty string consisting only of '(' and ')'. It is valid if any of the following conditions is true:
- It is ().
- It can be written as AB (A concatenated with B), where A and B are valid parentheses strings.
- It can be written as (A), where A is a valid parentheses string.

You are given a parentheses string s and a string locked, both of length n. locked is a binary string consisting only of '0's and '1's. For each index i of locked,
- If locked[i] is '1', you cannot change s[i].
- But if locked[i] is '0', you can change s[i] to either '(' or ')'.

Return true if you can make s a valid parentheses string. Otherwise, return false.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2232 Minimize Result By Adding Parentheses To Expression

**MEDIUM**

You are given a 0-indexed string expression of the form "<num1>+<num2>" where <num1> and <num2> represent positive integers.

Add a pair of parentheses to expression such that after the addition of parentheses, expression is a valid mathematical expression and evaluates to the smallest possible value. The left parenthesis must be added to the left of '+' and the right parenthesis must be added to the right of '+'.

Return expression after adding a pair of parentheses such that expression evaluates to the smallest possible value. If there are multiple answers that yield the same result, return any of them.

The input has been generated such that the original value of expression, and the value of expression after adding any pair of parentheses that meets the requirements fits within a signed 32-bit integer.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2267 Check If There Is A Valid Parentheses String Path

**HARD**

A parentheses string is a non-empty string consisting only of '(' and ')'. It is valid if any of the following conditions is true:
- It is ().
- It can be written as AB (A concatenated with B), where A and B are valid parentheses strings.

- It can be written as (A), where A is a valid parentheses string.

You are given an m x n matrix of parentheses grid. A valid parentheses string path in the grid is a path satisfying all of the following conditions:
- The path starts from the upper left cell (0, 0).
- The path ends at the bottom-right cell (m - 1, n - 1).
- The path only ever moves down or right.
- The resulting parentheses string formed by the path is valid.

Return true if there exists a valid parentheses string path in the grid. Otherwise, return false.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2726 Calculator With Method Chaining

**EASY**

Design a Calculator class. The class should provide the mathematical operations of addition, subtraction, multiplication, division, and exponentiation. It should also allow consecutive operations to be performed using method chaining. The Calculator class constructor should accept a number which serves as the initial value of result.

Your Calculator class should have the following methods:
- add - This method adds the given number value to the result and returns the updated Calculator.
- subtract - This method subtracts the given number value from the result and returns the updated Calculator.
- multiply - This method multiplies the result by the given number value and returns the updated Calculator.
- divide - This method divides the result by the given number value and returns the updated Calculator. If the passed value is 0, an error "Division by zero is not allowed" should be thrown.
- power - This method raises the result to the power of the given number value and returns the updated Calculator.
- getResult - This method returns the result.

Solutions within 10-5 of the actual result are considered correct.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

CHAPTER 06 / 97 EXERCISES

# Binary Search & Sorting

## 01 THE IDEA

Binary search discards half of an ordered search range per comparison. Sorting establishes useful order for later work.

## 02 SEE IT

```
SORTED  [1] [3] [5] [7] [9]
          LO      MID     HI
COMPARE           ^
```

## 03 JAVASCRIPT PATTERN

```
let lo = 0, hi = a.length - 1;
while (lo <= hi) {
 const mid = Math.floor((lo + hi) / 2);
}
```

## 04 HOW TO THINK

State the invariant: the answer remains in the surviving range. Check how boundaries change to avoid infinite loops.

## 05 PERFORMANCE

Binary search O(log n); comparison sort commonly O(n log n).

## EXERCISE COLLECTION

Try each question before opening the matching problem number in the separate Solutions PDF. The topic classification is a study aid; some exercises combine multiple patterns.

### 0004 Median Of Two Sorted Arrays

HARD

There are two sorted arrays nums1 and nums2 of size m and n respectively.

Find the median of the two sorted arrays.
The overall run time complexity should be O(log (m+n)).

You may assume nums1 and nums2 cannot be both empty.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0023 Merge K Sorted Lists

HARD

You are given an array of k linked-lists lists, each linked-list is sorted in ascending order.

Merge all the linked-lists into one sorted linked-list and return it.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0026 Remove Duplicates From Sorted Array

EASY

Given an integer array nums sorted in non-decreasing order, remove the duplicates in-place such that each unique element appears only once. The relative order of the elements should be kept the same.

Since it is impossible to change the length of the array in some languages, you must instead have the result be placed in the first part of the array nums. More formally, if there are k elements after removing the duplicates, then the first k elements of nums should hold the final result. It does not matter what you leave beyond the first k elements.

Return k after placing the final result in the first k slots of nums.

Do not allocate extra space for another array. You must do this by modifying the input array in-place with O(1) extra memory.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0033 Search In Rotated Sorted Array

MEDIUM

There is an integer array nums sorted in ascending order (with distinct values).

Prior to being passed to your function, nums is possibly rotated at an unknown pivot index k (1 <= k < nums.length) such that the resulting array is [nums[k], nums[k+1], ..., nums[n-1], nums[0], nums[1], ..., nums[k-1]] (0-indexed). For example, [0,1,2,4,5,6,7] might be rotated at pivot index 3 and become [4,5,6,7,0,1,2].

Given the array nums after the possible rotation and an integer target, return the index of target if it is in nums, or -1 if it is not in nums.

You must write an algorithm with O(log n) runtime complexity.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0034 Find First And Last Position Of Element In Sorted Array

MEDIUM

Given an array of integers nums sorted in non-decreasing order, find the starting and ending position of a given target value.

If target is not found in the array, return [-1, -1].

You must write an algorithm with O(log n) runtime complexity.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0035 Search Insert Position

EASY

Given a sorted array of distinct integers and a target value, return the index if the target is found. If not, return the index where it would be if it were inserted in order.

You must write an algorithm with O(log n) runtime complexity.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0056 Merge Intervals

MEDIUM

Given an array of intervals where intervals[i] = [starti, endi], merge all overlapping intervals, and return an array of the non-overlapping intervals that cover all the intervals in the input.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0075 Sort Colors

MEDIUM

Given an array nums with n objects colored red, white, or blue, sort them in-place so that objects of the same color are adjacent, with the colors in the order red, white, and blue.

We will use the integers 0, 1, and 2 to represent the color red, white, and blue, respectively.

You must solve this problem without using the library's sort function.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0080 Remove Duplicates From Sorted Array Ii

MEDIUM

Given an integer array nums sorted in non-decreasing order, remove some duplicates in-place such that each unique element appears at most twice. The relative order of the elements should be kept the same.

Since it is impossible to change the length of the array in some languages, you must instead have the result be placed in the first part of the array nums. More formally, if there are k elements after removing the duplicates, then the first k elements of nums should hold the final result. It does not matter what you leave beyond the first k elements.

Return k after placing the final result in the first k slots of nums.

Do not allocate extra space for another array. You must do this by modifying the input array in-place with O(1) extra memory.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0081 Search In Rotated Sorted Array Ii

MEDIUM

There is an integer array nums sorted in non-decreasing order (not necessarily with distinct values).

Before being passed to your function, nums is rotated at an unknown

pivot index k (0 <= k < nums.length) such that the resulting array is [nums[k], nums[k+1], ..., nums[n-1], nums[0], nums[1], ..., nums[k-1]] (0-indexed). For example, [0,1,2,4,4,4,5,6,6,7] might be rotated at pivot index 5 and become [4,5,6,6,7,0,1,2,4,4].

Given the array nums after the rotation and an integer target, return true if target is in nums, or false if it is not in nums.

You must decrease the overall operation steps as much as possible.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0082 Remove Duplicates From Sorted List Ii

**MEDIUM**

Given the head of a sorted linked list, delete all nodes that have duplicate numbers, leaving only distinct numbers from the original list. Return the linked list sorted as well.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0083 Remove Duplicates From Sorted List

**EASY**

Given the head of a sorted linked list, delete all duplicates such that each element appears only once. Return the linked list sorted as well.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0088 Merge Sorted Array

**EASY**

You are given two integer arrays `nums1` and `nums2`, sorted in non-decreasing order, and two integers `m` and `n`, representing the number of elements in `nums1` and `nums2` respectively.

Merge `nums1` and `nums2` into a single array sorted in non-decreasing order.

The final sorted array should not be returned by the function, but instead be stored inside the array `nums1`. To accommodate this, `nums1` has a length of `m + n`, where the first `m` elements denote the elements that should be merged, and the last `n` elements are set to `0` and should be ignored. `nums2` has a length of `n`.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0095 Unique Binary Search Trees Ii

**MEDIUM**

Given an integer n, return all the structurally unique BST's (binary search trees), which has exactly n nodes of unique values from 1 to n. Return the answer in any order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0096 Unique Binary Search Trees

**MEDIUM**

Given an integer n, return the number of structurally unique BST's (binary search trees) which has exactly n nodes of unique values from 1 to n.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0098 Validate Binary Search Tree

MEDIUM

Given the root of a binary tree, determine if it is a valid binary search tree (BST).

A valid BST is defined as follows:
- The left subtree of a node contains only nodes with keys less than the node's key.
- The right subtree of a node contains only nodes with keys greater than the node's key.
- Both the left and right subtrees must also be binary search trees.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0099 Recover Binary Search Tree

MEDIUM

You are given the root of a binary search tree (BST), where the values of exactly two nodes of the tree were swapped by mistake. Recover the tree without changing its structure.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0108 Convert Sorted Array To Binary Search Tree

EASY

Given an integer array nums where the elements are sorted in ascending order, convert it to a height-balanced binary search tree.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0109 Convert Sorted List To Binary Search Tree

MEDIUM

Given the head of a singly linked list where elements are sorted in ascending order, convert it to a height-balanced binary search tree.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0147 Insertion Sort List

MEDIUM

Given the head of a singly linked list, sort the list using insertion sort, and return the sorted list's head.

The steps of the insertion sort algorithm:
- Insertion sort iterates, consuming one input element each repetition and growing a sorted output list.
- At each iteration, insertion sort removes one element from the input data, finds the location it belongs within the sorted list and inserts it there.
- It repeats until no input elements remain.

The following is a graphical example of the insertion sort algorithm. The partially sorted list (black) initially contains only the first element in the list. One element (red) is removed from the input data and inserted in-place into the sorted list with each iteration.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0148 Sort List

MEDIUM

Given the head of a linked list, return the list after sorting it in ascending order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0153 Find Minimum In Rotated Sorted Array

MEDIUM

Suppose an array of length n sorted in ascending order is rotated between 1 and n times.
For example, the array nums = [0,1,2,4,5,6,7] might become:
- [4,5,6,7,0,1,2] if it was rotated 4 times.
- [0,1,2,4,5,6,7] if it was rotated 7 times.

Notice that rotating an array [a[0], a[1], a[2], ..., a[n-1]] 1 time results in the array [a[n-1], a[0], a[1], a[2], ..., a[n-2]].

Given the sorted rotated array nums of unique elements, return the minimum element of this array.

You must write an algorithm that runs in O(log n) time.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0154 Find Minimum In Rotated Sorted Array Ii

HARD

Suppose an array of length n sorted in ascending order is rotated between 1 and n times.
For example, the array nums = [0,1,4,4,5,6,7] might become:
- [4,5,6,7,0,1,4] if it was rotated 4 times.
- [0,1,4,4,5,6,7] if it was rotated 7 times.

Notice that rotating an array [a[0], a[1], a[2], ..., a[n-1]] 1 time results in the array [a[n-1], a[0], a[1], a[2], ..., a[n-2]].

Given the sorted rotated array nums that may contain duplicates, return the minimum element of this array.

You must decrease the overall operation steps as much as possible.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0173 Binary Search Tree Iterator

MEDIUM

Implement the BSTIterator class that represents an iterator over the in-order traversal of a binary search tree (BST):
- BSTIterator(TreeNode root) Initializes an object of the BSTIterator class. The root of the BST is given as part of the constructor. The pointer should be initialized to a non-existent number smaller than any element in the BST.
- boolean hasNext() Returns true if there exists a number in the traversal to the right of the pointer, otherwise returns false.
- int next() Moves the pointer to the right, then returns the number at the pointer.

Notice that by initializing the pointer to a non-existent smallest number, the first call to next() will return the smallest element in the BST.

You may assume that next() calls will always be valid. That is, there will be at least a next number in the in-order traversal when next() is called.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0235 Lowest Common Ancestor Of A Binary Search Tree

EASY

Given a binary search tree (BST), find the lowest common ancestor (LCA) of two given nodes in the BST.

According to the definition of LCA on Wikipedia: “The lowest common ancestor is defined between two nodes p and q as the lowest node in T that has both p and q as descendants (where we allow a node to be a descendant of itself).”

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0255 Verify Preorder Sequence In Binary Search Tree

MEDIUM

Given an array of unique integers preorder, return true if it is the correct preorder traversal sequence of a binary search tree.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0270 Closest Binary Search Tree Value

EASY

Given the root of a binary search tree and a target value, return the value in the BST that is closest to the target. If there are multiple answers, print the smallest.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0272 Closest Binary Search Tree Value Ii

HARD

Given the root of a binary search tree, a target value, and an integer k, return the k values in the BST that are closest to the target. You may return the answer in any order.

You are guaranteed to have only one unique set of k values in the BST that are closest to the target.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0280 Wiggle Sort

MEDIUM

Given an integer array nums, reorder it such that nums[0] <= nums[1] >= nums[2] <= nums[3]....

You may assume the input array always has a valid answer.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0324 Wiggle Sort Ii

MEDIUM

Given an integer array nums, reorder it such that nums[0] < nums[1] > nums[2] < nums[3]

You may assume the input array always has a valid answer.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0360 Sort Transformed Array

MEDIUM

Given a sorted integer array nums and three integers a, b and c, apply a quadratic function of the form f(x) = ax2 + bx + c to each element nums[i] in the array, and return the array in a sorted order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0378 Kth Smallest Element In A Sorted Matrix

MEDIUM

Given an n x n matrix where each of the rows and columns is sorted in ascending order, return the kth smallest element in the matrix.

Note that it is the kth smallest element in the sorted order, not the kth distinct element.

You must find a solution with a memory complexity better than O(n2).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0451 Sort Characters By Frequency

MEDIUM

Given a string, sort it in decreasing order based on the frequency of characters.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0501 Find Mode In Binary Search Tree

EASY

Given the root of a binary search tree (BST) with duplicates, return all the mode(s) (i.e., the most frequently occurred element) in it.

If the tree has more than one mode, return them in any order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0540 Single Element In A Sorted Array

MEDIUM

You are given a sorted array consisting of only integers where every element appears exactly twice, except for one element which appears exactly once.

Return the single element that appears only once.

Your solution must run in O(log n) time and O(1) space.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0669 Trim A Binary Search Tree

MEDIUM

Given the root of a binary search tree and the lowest and highest boundaries as low and high, trim the tree so that all its elements lies in [low, high]. Trimming the tree should not change the relative structure of the elements that will remain in the tree (i.e., any node's descendant should remain a descendant). It can be proven that there is a unique answer.

Return the root of the trimmed binary search tree. Note that the root may change depending on the given bounds.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0700 Search In A Binary Search Tree

**EASY**

You are given the root of a binary search tree (BST) and an integer val.

Find the node in the BST that the node's value equals val and return the subtree rooted with that node. If such a node does not exist, return null.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0701 Insert Into A Binary Search Tree

**MEDIUM**

You are given the root node of a binary search tree (BST) and a value to insert into the tree. Return the root node of the BST after the insertion. It is guaranteed that the new value does not exist in the original BST.

Notice that there may exist multiple valid ways for the insertion, as long as the tree remains a BST after insertion. You can return any of them.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0702 Search In A Sorted Array Of Unknown Size

**MEDIUM**

This is an interactive problem.

You have a sorted array of unique elements and an unknown size. You do not have an access to the array but you can use the ArrayReader interface to access it. You can call ArrayReader.get(i) that:
- returns the value at the ith index (0-indexed) of the secret array (i.e., secret[i]), or
- returns 231 - 1 if the i is out of the boundary of the array.

You are also given an integer target.

Return the index k of the hidden array where secret[k] == target or return -1 otherwise.

You must write an algorithm with O(log n) runtime complexity.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0704 Binary Search

**EASY**

Given an array of integers nums which is sorted in ascending order, and an integer target, write a function to search target in nums. If target exists, then return its index. Otherwise, return -1.

You must write an algorithm with O(log n) runtime complexity.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0768 Max Chunks To Make Sorted Ii

HARD

You are given an integer array arr.

We split arr into some number of chunks (i.e., partitions), and individually sort each chunk. After concatenating them, the result should equal the sorted array.

Return the largest number of chunks we can make to sort the array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0769 Max Chunks To Make Sorted

MEDIUM

You are given an integer array arr of length n that represents a permutation of the integers in the range [0, n - 1].

We split arr into some number of chunks (i.e., partitions), and individually sort each chunk. After concatenating them, the result should equal the sorted array.

Return the largest number of chunks we can make to sort the array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0791 Custom Sort String

MEDIUM

S and T are strings composed of lowercase letters. In S, no letter occurs more than once.

S was sorted in some custom order previously. We want to permute the characters of T so that they match the order that S was sorted. More specifically, if x occurs before y in S, then x should occur before y in the returned string.

Return any permutation of T (as a string) that satisfies this property.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0905 Sort Array By Parity

EASY

Given an integer array nums, move all the even integers at the beginning of the array followed by all the odd integers.

Return any array that satisfies this condition.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0912 Sort An Array

MEDIUM

Given an array of integers nums, sort the array in ascending order and return it.

You must solve the problem without using any built-in functions in O(nlog(n)) time complexity and with the smallest space complexity possible.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0922 Sort Array By Parity Ii

EASY

Given an array of integers nums, half of the integers in nums are odd, and the other half are even.

Sort the array so that whenever nums[i] is odd, i is odd, and whenever nums[i] is even, i is even.

Return any answer array that satisfies this condition.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0944 Delete Columns To Make Sorted

EASY

You are given an array of n strings strs, all of the same length.

The strings can be arranged such that there is one on each line, making a grid.

For example, strs = ["abc", "bce", "cae"] can be arranged as follows:
- abc
- bce
- cae

You want to delete the columns that are not sorted lexicographically. In the above example (0-indexed), columns 0 ('a', 'b', 'c') and 2 ('c', 'e', 'e') are sorted, while column 1 ('b', 'c', 'a') is not, so you would delete column 1.

Return the number of columns that you will delete.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0955 Delete Columns To Make Sorted Ii

MEDIUM

You are given an array of n strings strs, all of the same length.

We may choose any deletion indices, and we delete all the characters in those indices for each string.

For example, if we have strs = ["abcdef","uvwxyz"] and deletion indices {0, 2, 3}, then the final array after deletions is ["bef", "vyz"].

Suppose we chose a set of deletion indices answer such that after deletions, the final array has its elements in lexicographic order (i.e., strs[0] <= strs[1] <= strs[2] <= ... <= strs[n - 1]). Return the minimum possible value of answer.length.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0960 Delete Columns To Make Sorted Iii

HARD

You are given an array of n strings strs, all of the same length.

We may choose any deletion indices, and we delete all the characters in those indices for each string.

For example, if we have strs = ["abcdef","uvwxyz"] and deletion indices {0, 2, 3}, then the final array after deletions is ["bef", "vyz"].

Suppose we chose a set of deletion indices answer such that after deletions, the final array has every string (row) in lexicographic order. (i.e.,

(strs[0][0] <= strs[0][1] <= ... <= strs[0][strs[0].length - 1]),
and (strs[1][0] <= strs[1][1] <= ... <= strs[1][strs[1].length - 1]),
and so on). Return the minimum possible value of answer.length.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0969 Pancake Sorting

**MEDIUM**

Given an array of integers arr, sort the array by performing a series of pancake flips.

In one pancake flip we do the following steps:
- Choose an integer k where 1 <= k <= arr.length.
- Reverse the sub-array arr[0...k-1] (0-indexed).

For example, if arr = [3,2,1,4] and we performed a pancake flip choosing k = 3, we reverse the sub-array [3,2,1], so arr = [1,2,3,4] after the pancake flip at k = 3.

Return an array of the k-values corresponding to a sequence of pancake flips that sort arr. Any valid answer that sorts the array within 10 * arr.length flips will be judged as correct.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0977 Squares Of A Sorted Array

**EASY**

Given an array of integers A sorted in non-decreasing order,
return an array of the squares of each number,
also in sorted non-decreasing order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1008 Construct Binary Search Tree From Preorder Traversal

**MEDIUM**

Given an array of integers preorder, which represents the preorder traversal of a BST (i.e., binary search tree), construct the tree and return its root.

It is guaranteed that there is always possible to find a binary search tree with the given requirements for the given test cases.

A binary search tree is a binary tree where for every node, any descendant of Node.left has a value strictly less than Node.val, and any descendant of Node.right has a value strictly greater than Node.val.

A preorder traversal of a binary tree displays the value of the node first, then traverses Node.left, then traverses Node.right.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1038 Binary Search Tree To Greater Sum Tree

**MEDIUM**

Given the root of a Binary Search Tree (BST), convert it to a Greater Tree such that every key of the original BST is changed to the original key plus the sum of all keys greater than the original key in BST.

As a reminder, a binary search tree is a tree that satisfies these constraints:
- The left subtree of a node contains only nodes with keys less than the node's key.
- The right subtree of a node contains only nodes with keys greater than the node's key.
- Both the left and right subtrees must also be binary search trees.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 1060 Missing Element In Sorted Array

MEDIUM

Given an integer array nums which is sorted in ascending order and all of its elements are unique and given also an integer k, return the kth missing number starting from the leftmost number of the array.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 1122 Relative Sort Array

EASY

Given two arrays arr1 and arr2, the elements of arr2 are distinct, and all elements in arr2 are also in arr1.

Sort the elements of arr1 such that the relative ordering of items in arr1 are the same as in arr2. Elements that do not appear in arr2 should be placed at the end of arr1 in ascending order.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 1150 Check If A Number Is Majority Element In A Sorted Array

EASY

Given an integer array nums sorted in non-decreasing order and an integer target, return true if target is a majority element, or false otherwise.

A majority element in an array nums is an element that appears more than nums.length / 2 times in the array.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 1203 Sort Items By Groups Respecting Dependencies

HARD

There are n items each belonging to zero or one of m groups where group[i] is the group that the i-th item belongs to and it's equal to -1 if the i-th item belongs to no group. The items and the groups are zero indexed. A group can have no item belonging to it.

Return a sorted list of the items such that:
- The items that belong to the same group are next to each other in the sorted list.
- There are some relations between these items where beforeItems[i] is a list containing all the items that should come before the i-th item in the sorted array (to the left of the i-th item).

Return any solution if there is more than one solution and return an empty list if there is no solution.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 1213 Intersection Of Three Sorted Arrays

EASY

Given three integer arrays arr1, arr2 and arr3 sorted in strictly increasing order, return a sorted array of only the integers that appeared in all three arrays.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 1287 Element Appearing More Than 25 In Sorted Array

EASY

Given an integer array sorted in non-decreasing order,
there is exactly one integer in the array that occurs more than 25% of the time.

Return that integer.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1305 All Elements In Two Binary Search Trees

MEDIUM

Given two binary search trees root1 and root2, return a list containing all the integers from both trees sorted in ascending order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1329 Sort The Matrix Diagonally

MEDIUM

A matrix diagonal is a diagonal line of cells starting from some cell in either the topmost row or leftmost column and going in the bottom-right direction until reaching the matrix's end. For example, the matrix diagonal starting from mat[2][0], where mat is a 6 x 3 matrix, includes cells mat[2][0], mat[3][1], and mat[4][2].

Given an m x n matrix mat of integers, sort each matrix diagonal in ascending order and return the resulting matrix.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1351 Count Negative Numbers In A Sorted Matrix

EASY

Given a m x n matrix grid which is sorted in non-increasing order both row-wise and column-wise, return the number of negative numbers in grid.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1356 Sort Integers By The Number Of 1 Bits

EASY

Given an integer array arr. You have to sort the integers in the array in ascending order by the number of 1's in their binary representation and in case of two or more integers have the same number of 1's you have to sort them in ascending order.

Return the sorted array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1382 Balance A Binary Search Tree

MEDIUM

Given the root of a binary search tree, return a balanced binary search tree with the same node values. If there is more than one answer, return any of them.

A binary search tree is balanced if the depth of the two subtrees of every node never differs

by more than 1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1387 Sort Integers By The Power Value

**MEDIUM**

The power of an integer x is defined as the number of steps needed to transform x into 1 using the following steps:
- if x is even then x = x / 2
- if x is odd then x = 3 * x + 1

For example, the power of x = 3 is 7 because 3 needs 7 steps to become 1 (3 --> 10 --> 5 --> 16 --> 8 --> 4 --> 2 --> 1).

Given three integers lo, hi and k. The task is to sort all integers in the interval [lo, hi] by the power value in ascending order, if two or more integers have the same power value sort them by ascending order.

Return the kth integer in the range [lo, hi] sorted by the power value.

Notice that for any integer x (lo <= x <= hi) it is guaranteed that x will transform into 1 using these steps and that the power of x is will fit in a 32-bit signed integer.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1439 Find The Kth Smallest Sum Of A Matrix With Sorted Rows

**HARD**

You are given an m x n matrix mat that has its rows sorted in non-decreasing order and an integer k.

You are allowed to choose exactly one element from each row to form an array.

Return the kth smallest array sum among all possible arrays.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1586 Binary Search Tree Iterator Ii

**MEDIUM**

Implement the BSTIterator class that represents an iterator over the in-order traversal of a binary search tree (BST):
- BSTIterator(TreeNode root) Initializes an object of the BSTIterator class. The root of the BST is given as part of the constructor. The pointer should be initialized to a non-existent number smaller than any element in the BST.
- boolean hasNext() Returns true if there exists a number in the traversal to the right of the pointer, otherwise returns false.
- int next() Moves the pointer to the right, then returns the number at the pointer.
- boolean hasPrev() Returns true if there exists a number in the traversal to the left of the pointer, otherwise returns false.
- int prev() Moves the pointer to the left, then returns the number at the pointer.

Notice that by initializing the pointer to a non-existent smallest number, the first call to next() will return the smallest element in the BST.

You may assume that next() and prev() calls will always be valid. That is, there will be at least a next/previous number in the in-order traversal when next()/prev() is called.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1636 Sort Array By Increasing Frequency

EASY

Given an array of integers nums, sort the array in increasing order based on the frequency of the values. If multiple values have the same frequency, sort them in decreasing order.

Return the sorted array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1641 Count Sorted Vowel Strings

MEDIUM

Given an integer n, return the number of strings of length n that consist only of vowels (a, e, i, o, u) and are lexicographically sorted.

A string s is lexicographically sorted if for all valid i, s[i] is the same as or comes before s[i+1] in the alphabet.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1649 Create Sorted Array Through Instructions

HARD

Given an integer array instructions, you are asked to create a sorted array from the elements in instructions. You start with an empty container nums. For each element from left to right in instructions, insert it into nums. The cost of each insertion is the minimum of the following:
- The number of elements currently in nums that are strictly less than instructions[i].
- The number of elements currently in nums that are strictly greater than instructions[i].

For example, if inserting element 3 into nums = [1,2,3,5], the cost of insertion is min(2, 1) (elements 1 and 2 are less than 3, element 5 is greater than 3) and nums will become [1,2,3,3,5].

Return the total cost to insert all elements from instructions into nums. Since the answer may be large, return it modulo 109 + 7

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1685 Sum Of Absolute Differences In A Sorted Array

MEDIUM

You are given an integer array nums sorted in non-decreasing order.

Build and return an integer array result with the same length as nums such that result[i] is equal to the summation of absolute differences between nums[i] and all the other elements in the array.

In other words, result[i] is equal to sum(|nums[i]-nums[j]|) where 0 <= j < nums.length and j != i (0-indexed).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1752 Check If Array Is Sorted And Rotated

EASY

Given an array nums, return true if the array was originally sorted in non-decreasing order, then rotated some number of positions (including zero). Otherwise, return false.

There may be duplicates in the original array.

Note: An array A rotated by x positions results in an array B of the same length such

that A[i] == B[(i+x) % A.length], where % is the modulo operation.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1772 Sort Features By Popularity

**MEDIUM**

You are given a string array features where features[i] is a single word that represents the name of a feature of the latest product you are working on. You have made a survey where users have reported which features they like. You are given a string array responses, where each responses[i] is a string containing space-separated words.

The popularity of a feature is the number of responses[i] that contain the feature. You want to sort the features in non-increasing order by their popularity. If two features have the same popularity, order them by their original index in features. Notice that one response could contain the same feature multiple times; this feature is only counted once in its popularity.

Return the features in sorted order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1830 Minimum Number Of Operations To Make String Sorted

**HARD**

You are given a string s (0-indexed). You are asked to perform the following operation on s until you get a sorted string:
possible values of k in the range [i, j] inclusive.

Return the number of operations needed to make the string sorted. Since the answer can be too large, return it modulo 109 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1859 Sorting The Sentence

**EASY**

A sentence is a list of words that are separated by a single space with no leading or trailing spaces. Each word consists of lowercase and uppercase English letters.

A sentence can be shuffled by appending the 1-indexed word position to each word then rearranging the words in the sentence.
- For example, the sentence "This is a sentence" can be shuffled as "sentence4 a3 is2 This1" or "is2 sentence4 This1 a3".

Given a shuffled sentence s containing no more than 9 words, reconstruct and return the original sentence.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1940 Longest Common Subsequence Between Sorted Arrays

**MEDIUM**

Given an array of integer arrays arrays where each arrays[i] is sorted in strictly increasing order, return an integer array representing the longest common subsequence among all the arrays.

A subsequence is a sequence that can be derived from another sequence by deleting some elements (possibly none) without changing the order of the remaining elements.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1966 Binary Searchable Numbers In An Unsorted Array

MEDIUM

Consider a function that implements an algorithm similar to Binary Search. The function has two input parameters: sequence is a sequence of integers, and target is an integer value. The purpose of the function is to find if the target exists in the sequence.

The pseudocode of the function is as follows:

func(sequence, target)
while sequence is not empty
randomly choose an element from sequence as the pivot
if pivot = target, return true
else if pivot < target, remove pivot and all elements to its left from the sequence
else, remove pivot and all elements to its right from the sequence
end while
return false

When the sequence is sorted, the function works correctly for all values. When the sequence is not sorted, the function does not work for all values, but may still work for some values.

Given an integer array nums, representing the sequence, that contains unique numbers and may or may not be sorted, return the number of values that are guaranteed to be found using the function, for every possible pivot selection.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1998 Gcd Sort Of An Array

HARD

You are given an integer array nums, and you can perform the following operation any number of times on nums:
- Swap the positions of two elements nums[i] and nums[j] if gcd(nums[i], nums[j]) > 1 where gcd(nums[i], nums[j]) is the greatest common divisor of nums[i] and nums[j].

Return true if it is possible to sort nums in non-decreasing order using the above swap method, or false otherwise.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2040 Kth Smallest Product Of Two Sorted Arrays

HARD

Given two sorted 0-indexed integer arrays nums1 and nums2 as well as an integer k, return the kth (1-based) smallest product of nums1[i] * nums2[j] where 0 <= i < nums1.length and 0 <= j < nums2.length.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2089 Find Target Indices After Sorting Array

EASY

You are given a 0-indexed integer array nums and a target element target.

A target index is an index i such that nums[i] == target.

Return a list of the target indices of nums after sorting nums in non-decreasing order. If there are no target indices, return an empty list. The returned list must be sorted in increasing order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2164 Sort Even And Odd Indices Independently

EASY

You are given a 0-indexed integer array nums. Rearrange the values of nums according to the following rules:
- For example, if nums = [4,1,2,3] before this step, it becomes [4,3,2,1] after. The values at odd indices 1 and 3 are sorted in non-increasing order.
- For example, if nums = [4,1,2,3] before this step, it becomes [2,1,4,3] after. The values at even indices 0 and 2 are sorted in non-decreasing order.

Return the array formed after rearranging the values of nums.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2191 Sort The Jumbled Numbers

MEDIUM

You are given a 0-indexed integer array mapping which represents the mapping rule of a shuffled decimal system. mapping[i] = j means digit i should be mapped to digit j in this system.

The mapped value of an integer is the new integer obtained by replacing each occurrence of digit i in the integer with mapping[i] for all 0 <= i <= 9.

You are also given another integer array nums. Return the array nums sorted in non-decreasing order based on the mapped values of its elements.

Notes:
- Elements with the same mapped values should appear in the same relative order as in the input.
- The elements of nums should only be sorted based on their mapped values and not be replaced by them.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2366 Minimum Replacements To Sort The Array

HARD

You are given a 0-indexed integer array nums. In one operation you can replace any element of the array with any two elements that sum to it.
- For example, consider nums = [5,6,7]. In one operation, we can replace nums[1] with 2 and 4 and convert nums to [5,2,4,7].

Return the minimum number of operations to make an array that is sorted in non-decreasing order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2387 Median Of A Row Wise Sorted Matrix

MEDIUM

Given an m x n matrix grid containing an odd number of integers where each row is sorted in non-decreasing order, return the median of the matrix.

You must solve the problem in less than O(m * n) time complexity.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2418 Sort The People

EASY

You are given an array of strings names, and an array heights that consists of distinct positive integers. Both arrays are of length n.

For each index i, names[i] and heights[i] denote the name and height of the ith person.

Return names sorted in descending order by the people's heights.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2459 Sort Array By Moving Items To Empty Space

**HARD**

You are given an integer array nums of size n containing each element from 0 to n - 1 (inclusive). Each of the elements from 1 to n - 1 represents an item, and the element 0 represents an empty space.

In one operation, you can move any item to the empty space. nums is considered to be sorted if the numbers of all the items are in ascending order and the empty space is either at the beginning or at the end of the array.

For example, if n = 4, nums is sorted if:
- nums = [0,1,2,3] or
- nums = [1,2,3,0]

...and considered to be unsorted otherwise.

Return the minimum number of operations needed to sort nums.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2471 Minimum Number Of Operations To Sort A Binary Tree By Level

**MEDIUM**

You are given the root of a binary tree with unique values.

In one operation, you can choose any two nodes at the same level and swap their values.

Return the minimum number of operations needed to make the values at each level sorted in a strictly increasing order.

The level of a node is the number of edges along the path between it and the root node.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2476 Closest Nodes Queries In A Binary Search Tree

**MEDIUM**

You are given the root of a binary search tree and an array queries of size n consisting of positive integers.

Find a 2D array answer of size n where answer[i] = [mini, maxi]:
- mini is the largest value in the tree that is smaller than or equal to queries[i]. If a such value does not exist, add -1 instead.
- maxi is the smallest value in the tree that is greater than or equal to queries[i]. If a such value does not exist, add -1 instead.

Return the array answer.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2545 Sort The Students By Their Kth Score

**MEDIUM**

There is a class with m students and n exams. You are given a 0-indexed m x n integer matrix score, where each row represents one student and score[i][j] denotes the score the ith student got in the jth exam. The matrix score contains distinct integers only.

You are also given an integer k. Sort the students (i.e., the rows of the matrix) by their scores in the kth (0-indexed) exam from the highest to the lowest.

Return the matrix after sorting it.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2724 Sort By

**EASY**

Given an array arr and a function fn, return a sorted array sortedArr.
You can assume fn only returns numbers and those numbers determine the sort order of sortedArr. sortedArr must be sorted in ascending order by fn output.

You may assume that fn will never duplicate numbers for a given array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2785 Sort Vowels In A String

**MEDIUM**

Given a 0-indexed string s, permute s to get a new string t such that:
- All consonants remain in their original places. More formally, if there is an index i with 0 <= i < s.length such that s[i] is a consonant, then t[i] = s[i].
- The vowels must be sorted in the nondecreasing order of their ASCII values. More formally, for pairs of indices i, j with 0 <= i < j < s.length such that s[i] and s[j] are vowels, then t[i] must not have a higher ASCII value than t[j].

Return the resulting string.

The vowels are 'a', 'e', 'i', 'o', and 'u', and they can appear in lowercase or uppercase. Consonants comprise all letters that are not vowels.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2826 Sorting Three Groups

**MEDIUM**

You are given an integer array nums. Each element in nums is 1, 2 or 3. In each operation, you can remove an element from nums. Return the minimum number of operations to make nums non-decreasing.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2855 Minimum Right Shifts To Sort The Array

**EASY**

You are given a 0-indexed array nums of length n containing distinct positive integers. Return the minimum number of right shifts required to sort nums and -1 if this is not possible.

A right shift is defined as shifting the element at index i to index (i + 1) % n, for all indices.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3011 Find If Array Can Be Sorted

MEDIUM

You are given a 0-indexed array of positive integers nums.

In one operation, you can swap any two adjacent elements if they have the same number of set bits. You are allowed to do this operation any number of times (including zero).

Return true if you can sort the array in ascending order, else return false.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3359 Find Sorted Submatrices With Maximum Element At Most K

HARD

You are given a 2D matrix grid of size m x n. You are also given a non-negative integer k.

Return the number of submatrices of grid that satisfy the following conditions:
- The maximum element in the submatrix less than or equal to k.
- Each row in the submatrix is sorted in non-increasing order.

A submatrix (x1, y1, x2, y2) is a matrix that forms by choosing all cells grid[x][y] where x1 <= x <= x2 and y1 <= y <= y2.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3431 Minimum Unlocked Indices To Sort Nums

MEDIUM

You are given an array nums consisting of integers between 1 and 3, and a binary array locked of the same size.

We consider nums sortable if it can be sorted using adjacent swaps, where a swap between two indices i and i + 1 is allowed if nums[i] - nums[i + 1] == 1 and locked[i] == 0.

In one operation, you can unlock any index i by setting locked[i] to 0.

Return the minimum number of operations needed to make nums sortable. If it is not possible to make nums sortable, return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3446 Sort Matrix By Diagonals

MEDIUM

You are given an n x n square matrix of integers grid. Return the matrix such that:
- The diagonals in the bottom-left triangle (including the middle diagonal) are sorted in non-increasing order.
- The diagonals in the top-right triangle are sorted in non-decreasing order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

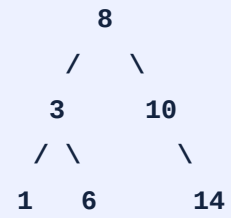
CHAPTER 07 / 183 EXERCISES

# Trees & Binary Search Trees

## 01 THE IDEA

Trees model hierarchical relationships. Recursive traversal visits a node and then its descendants.

## 02 SEE IT




## 03 JAVASCRIPT PATTERN

```
function dfs(node) {
 if (!node) return;
 dfs(node.left); dfs(node.right);
}
```

## 04 HOW TO THINK

Identify root, leaf, height, and traversal order. For BSTs, values to the left are smaller than the node.

## 05 PERFORMANCE

DFS O(n) time; recursion depth O(h) space.

## EXERCISE COLLECTION

Try each question before opening the matching problem number in the separate Solutions PDF. The topic classification is a study aid; some exercises combine multiple patterns.

### 0094 Binary Tree Inorder Traversal

EASY

Given the root of a binary tree, return the inorder traversal of its nodes' values.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0100 Same Tree

EASY

Given the roots of two binary trees p and q, write a function to check if they are the same or not.

Two binary trees are considered the same if they are structurally identical, and the nodes have the same value.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0101 Symmetric Tree

EASY

Given the root of a binary tree, check whether it is a mirror of itself (i.e., symmetric around its center).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0102 Binary Tree Level Order Traversal

MEDIUM

Given the root of a binary tree, return the level order traversal of its nodes' values. (i.e., from left to right, level by level).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0103 Binary Tree Zigzag Level Order Traversal

MEDIUM

Given the root of a binary tree, return the zigzag level order traversal of its nodes' values. (i.e., from left to right, then right to left for the next level and alternate between).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0104 Maximum Depth Of Binary Tree

EASY

Given the root of a binary tree, return its maximum depth.

A binary tree's maximum depth is the number of nodes along the longest path from the root node down to the farthest leaf node.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0105 Construct Binary Tree From Preorder And Inorder Traversal

MEDIUM

Given two integer arrays preorder and inorder where preorder is the preorder traversal of a binary tree and inorder is the inorder traversal of the same tree, construct and return the binary tree.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0106 Construct Binary Tree From Inorder And Postorder Traversal

MEDIUM

Given two integer arrays inorder and postorder where inorder is the inorder traversal of a binary tree and postorder is the postorder traversal of the same tree, construct and return the binary tree.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0107 Binary Tree Level Order Traversal Ii

MEDIUM

Given the root of a binary tree, return the bottom-up level order traversal of its nodes' values. (i.e., from left to right, level by level from leaf to root).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0110 Balanced Binary Tree

EASY

Given a binary tree, determine if it is height-balanced.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0111 Minimum Depth Of Binary Tree

EASY

Given a binary tree, find its minimum depth.

The minimum depth is the number of nodes along the shortest path from the root node down to the nearest leaf node.

Note: A leaf is a node with no children.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0124 Binary Tree Maximum Path Sum

HARD

A path in a binary tree is a sequence of nodes where each pair of adjacent nodes in the sequence has an edge connecting them. A node can only appear in the sequence at most once. Note that the path does not need to pass through the root.

The path sum of a path is the sum of the node's values in the path.

Given the root of a binary tree, return the maximum path sum of any non-empty path.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0144 Binary Tree Preorder Traversal

EASY

Given the root of a binary tree, return the preorder traversal of its nodes' values.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0145 Binary Tree Postorder Traversal

EASY

Given the root of a binary tree, return the postorder traversal
of its nodes' values.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0156 Binary Tree Upside Down

MEDIUM

Given the root of a binary tree, turn the tree upside down and return the new root.

You can turn a binary tree upside down with the following steps:
- The original left child becomes the new root.
- The original root becomes the new right child.
- The original right child becomes the new left child.

The mentioned steps are done level by level. It is guaranteed that every right node has
a sibling (a left node with the same parent) and has no children.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0199 Binary Tree Right Side View

MEDIUM

Given the root of a binary tree, imagine yourself standing on the right side of it, return the
values of the nodes you can see ordered from top to bottom.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0208 Implement Trie Prefix Tree

MEDIUM

A trie (pronounced as "try") or prefix tree is a tree data structure used to efficiently store
and retrieve keys in a dataset of strings. There are various applications of this data structure,
such as autocomplete and spellchecker.

Implement the Trie class:
- Trie() Initializes the trie object.
- void insert(String word) Inserts the string word into the trie.
- boolean search(String word) Returns true if the string word is in the trie (i.e., was inserted
before), and false otherwise.
- boolean startsWith(String prefix) Returns true if there is a previously inserted string word
that has the prefix prefix, and false otherwise.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0222 Count Complete Tree Nodes

EASY

Given the root of a complete binary tree, return the number of the nodes in the tree.

According to Wikipedia, every level, except possibly the last, is completely filled
in a complete binary tree, and all nodes in the last level are as far left as possible.
It can have between 1 and 2h nodes inclusive at the last level h.

Design an algorithm that runs in less than O(n) time complexity.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0226 Invert Binary Tree

EASY

Invert a binary tree.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0230 Kth Smallest Element In A Bst

MEDIUM

Given the root of a binary search tree, and an integer k, return the kth smallest value (1-indexed) of all the values of the nodes in the tree.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0236 Lowest Common Ancestor Of A Binary Tree

MEDIUM

Given a binary tree, find the lowest common ancestor (LCA) of two given nodes in the tree.

According to the definition of LCA on Wikipedia: “The lowest common ancestor is defined between two nodes p and q as the lowest node in T that has both p and q as descendants (where we allow a node to be a descendant of itself).”

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0250 Count Univalue Subtrees

MEDIUM

Given the root of a binary tree, return the number of uni-value subtrees.

A uni-value subtree means all nodes of the subtree have the same value.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0257 Binary Tree Paths

EASY

Given the root of a binary tree, return all root-to-leaf paths in any order.

A leaf is a node with no children.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0261 Graph Valid Tree

MEDIUM

You have a graph of n nodes labeled from 0 to n - 1. You are given an integer n and a list of edges where edges[i] = [ai, bi] indicates that there is an undirected edge between nodes ai and bi in the graph.

Return true if the edges of the given graph make up a valid tree, and false otherwise.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0285 Inorder Successor In Bst

MEDIUM

Given the root of a binary search tree and a node p in it, return the in-order successor of that node in the BST. If the given node has no in-order successor in the tree, return null.

The successor of a node p is the node with the smallest key greater than p.val.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0297 Serialize And Deserialize Binary Tree

HARD

Serialization is the process of converting a data structure or object into a sequence of bits so that it can be stored in a file or memory buffer, or transmitted across a network connection link to be reconstructed later in the same or another computer environment.

Design an algorithm to serialize and deserialize a binary tree. There is no restriction on how your serialization/deserialization algorithm should work. You just need to ensure that a binary tree can be serialized to a string and this string can be deserialized to the original tree structure.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0298 Binary Tree Longest Consecutive Sequence

MEDIUM

Given the root of a binary tree, return the length of the longest consecutive sequence path.

A consecutive sequence path is a path where the values increase by one along the path.

Note that the path can start at any node in the tree, and you cannot go from a node to its parent in the path.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0310 Minimum Height Trees

MEDIUM

A tree is an undirected graph in which any two vertices are connected by exactly one path. In other words, any connected graph without simple cycles is a tree.

Given a tree of n nodes labelled from 0 to n - 1, and an array of n - 1 edges where edges[i] = [ai, bi] indicates that there is an undirected edge between the two nodes ai and bi in the tree, you can choose any node of the tree as the root. When you select a node x as the root, the result tree has height h. Among all possible rooted trees, those with minimum height (i.e. min(h)) are called minimum height trees (MHTs).

Return a list of all MHTs' root labels. You can return the answer in any order.

The height of a rooted tree is the number of edges on the longest downward path between the root and a leaf.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0314 Binary Tree Vertical Order Traversal

MEDIUM

Given the root of a binary tree, return the vertical order traversal of its nodes' values. (i.e., from top to bottom, column by column).

If two nodes are in the same row and column, the order should be from left to right.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0331 Verify Preorder Serialization Of A Binary Tree

**MEDIUM**

One way to serialize a binary tree is to use preorder traversal. When we encounter a non-null node, we record the node's value. If it is a null node, we record using a sentinel value such as '#'.

For example, the above binary tree can be serialized to the string "9,3,4,#,#,1,#,#,2,#,6,#,#", where '#' represents a null node.

Given a string of comma-separated values preorder, return true if it is a correct preorder traversal serialization of a binary tree.

It is guaranteed that each comma-separated value in the string must be either an integer or a character '#' representing null pointer.

You may assume that the input format is always valid.
For example, it could never contain two consecutive commas, such as "1,,3".
Note: You are not allowed to reconstruct the tree.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0333 Largest Bst Subtree

**MEDIUM**

Given the root of a binary tree, find the largest subtree, which is also a Binary Search Tree (BST), where the largest means subtree has the largest number of nodes.

A Binary Search Tree (BST) is a tree in which all the nodes follow the below-mentioned properties:
- The left subtree values are less than the value of their parent (root) node's value.
- The right subtree values are greater than the value of their parent (root) node's value.

Note: A subtree must include all of its descendants.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0366 Find Leaves Of Binary Tree

**MEDIUM**

Given the root of a binary tree, collect a tree's nodes as if you were doing this:
- Collect all the leaf nodes.
- Remove all the leaf nodes.
- Repeat until the tree is empty.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0427 Construct Quad Tree

**MEDIUM**

Given a n * n matrix grid of 0's and 1's only. We want to represent grid with a Quad-Tree.

Return the root of the Quad-Tree representing grid.

A Quad-Tree is a tree data structure in which each internal node has exactly four children. Besides, each node has two attributes:
- val: True if the node represents a grid of 1's or False if the node represents a grid of 0's. Notice that you can assign the val to True or False when isLeaf is False, and both are

accepted in the answer.
- isLeaf: True if the node is a leaf node on the tree or False if the node has four children.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0428 Serialize And Deserialize N Ary Tree

HARD

Serialization is the process of converting a data structure or object into a sequence of bits so that it can be stored in a file or memory buffer, or transmitted across a network connection link to be reconstructed later in the same or another computer environment.

Design an algorithm to serialize and deserialize an N-ary tree. An N-ary tree is a rooted tree in which each node has no more than N children. There is no restriction on how your serialization/deserialization algorithm should work. You just need to ensure that an N-ary tree can be serialized to a string and this string can be deserialized to the original tree structure.

For example, you may serialize the following 3-ary tree as [1 [3[5 6] 2 4]]. Note that this is just an example, you do not necessarily need to follow this format.

Or you can follow LeetCode's level order traversal serialization format, where each group of children is separated by the null value.

For example, the above tree may be serialized as [1,null,2,3,4,5,null,null,6,7,null,8,null,9,10,null,null,11,null,12,null,13,null,null,14].

You do not necessarily need to follow the above-suggested formats, there are many more different formats that work so please be creative and come up with different approaches yourself.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0429 N Ary Tree Level Order Traversal

MEDIUM

Given an n-ary tree, return the level order traversal of its nodes' values.

Nary-Tree input serialization is represented in their level order traversal, each group of children is separated by the null value.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0431 Encode N Ary Tree To Binary Tree

HARD

Design an algorithm to encode an N-ary tree into a binary tree and decode the binary tree to get the original N-ary tree. An N-ary tree is a rooted tree in which each node has no more than N children. Similarly, a binary tree is a rooted tree in which each node has no more than 2 children. There is no restriction on how your encode/decode algorithm should work. You just need to ensure that an N-ary tree can be encoded to a binary tree and this binary tree can be decoded to the original N-nary tree structure.

Nary-Tree input serialization is represented in their level order traversal, each group of children is separated by the null value (See following example).

For example, you may encode the following 3-ary tree to a binary tree in this way:
- Input: root = [1,null,3,2,4,null,5,6]

Note that the above is just an example which might or might not work. You do not necessarily need to follow this format, so please be creative and come up with different approaches yourself.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0449 Serialize And Deserialize Bst

MEDIUM

Serialization is converting a data structure or object into a sequence of bits so that it can be stored in a file or memory buffer, or transmitted across a network connection link to be reconstructed later in the same or another computer environment.

Design an algorithm to serialize and deserialize a binary search tree. There is no restriction on how your serialization/deserialization algorithm should work. You need to ensure that a binary search tree can be serialized to a string, and this string can be deserialized to the original tree structure.

The encoded string should be as compact as possible.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0450 Delete Node In A Bst

MEDIUM

Given a root node reference of a BST and a key, delete the node with the given key in the BST. Return the root node reference (possibly updated) of the BST.

Basically, the deletion can be divided into two stages:

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0508 Most Frequent Subtree Sum

MEDIUM

Given the root of a binary tree, return the most frequent subtree sum. If there is a tie, return all the values with the highest frequency in any order.

The subtree sum of a node is defined as the sum of all the node values formed by the subtree rooted at that node (including the node itself).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0510 Inorder Successor In Bst Ii

MEDIUM

Given a node in a binary search tree, return the in-order successor of that node in the BST. If that node has no in-order successor, return null.

The successor of a node is the node with the smallest key greater than node.val.

You will have direct access to the node but not to the root of the tree. Each node will have a reference to its parent node. Below is the definition for Node:

```
class Node {
public int val;
public Node left;
public Node right;
public Node parent;
}
```

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0513 Find Bottom Left Tree Value

MEDIUM

Given the root of a binary tree, return the leftmost value in the last row of the tree.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0515 Find Largest Value In Each Tree Row

MEDIUM

Given the root of a binary tree, return an array of the largest value in each row of the tree (0-indexed).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0530 Minimum Absolute Difference In Bst

EASY

Given the root of a Binary Search Tree (BST), return the minimum absolute difference between the values of any two different nodes in the tree.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0536 Construct Binary Tree From String

MEDIUM

You need to construct a binary tree from a string consisting of parenthesis and integers.

The whole input represents a binary tree. It contains an integer followed by zero, one or two pairs of parenthesis. The integer represents the root's value and a pair of parenthesis contains a child binary tree with the same structure.

You always start to construct the left child node of the parent first if it exists.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0538 Convert Bst To Greater Tree

MEDIUM

Given the root of a Binary Search Tree (BST), convert it to a Greater Tree such that every key of the original BST is changed to the original key plus the sum of all keys greater than the original key in BST.

As a reminder, a binary search tree is a tree that satisfies these constraints:
- The left subtree of a node contains only nodes with keys less than the node's key.
- The right subtree of a node contains only nodes with keys greater than the node's key.
- Both the left and right subtrees must also be binary search trees.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0543 Diameter Of Binary Tree

EASY

Given the root of a binary tree, return the length of the diameter of the tree.

The diameter of a binary tree is the length of the longest path between any two nodes in a tree. This path may or may not pass through the root.

The length of a path between two nodes is represented by the number of edges between them.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 0545 Boundary Of Binary Tree

MEDIUM

The boundary of a binary tree is the concatenation of the root, the left boundary, the leaves ordered from left-to-right, and the reverse order of the right boundary.

The left boundary is the set of nodes defined by the following:
- The root node's left child is in the left boundary. If the root does not have a left child, then the left boundary is empty.
- If a node in the left boundary and has a left child, then the left child is in the left boundary.
- If a node is in the left boundary, has no left child, but has a right child, then the right child is in the left boundary.
- The leftmost leaf is not in the left boundary.

The right boundary is similar to the left boundary, except it is the right side of the root's right subtree. Again, the leaf is not part of the right boundary, and the right boundary is empty if the root does not have a right child.

The leaves are nodes that do not have any children. For this problem, the root is not a leaf.

Given the root of a binary tree, return the values of its boundary.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 0549 Binary Tree Longest Consecutive Sequence Ii

MEDIUM

Given the root of a binary tree, return the length of the longest consecutive path in the tree.

A consecutive path is a path where the values of the consecutive nodes in the path differ by one. This path can be either increasing or decreasing.
- For example, [1,2,3,4] and [4,3,2,1] are both considered valid, but the path [1,2,4,3] is not valid.

On the other hand, the path can be in the child-Parent-child order, where not necessarily be parent-child order.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 0558 Logical Or Of Two Binary Grids Represented As Quad Trees

MEDIUM

A Binary Matrix is a matrix in which all the elements are either 0 or 1.

Given quadTree1 and quadTree2. quadTree1 represents a n * n binary matrix and quadTree2 represents another n * n binary matrix.

Return a Quad-Tree representing the n * n binary matrix which is the result of logical bitwise OR of the two binary matrixes represented by quadTree1 and quadTree2.

Notice that you can assign the value of a node to True or False when isLeaf is False, and both are accepted in the answer.

A Quad-Tree is a tree data structure in which each internal node has exactly four children. Besides, each node has two attributes:
- val: True if the node represents a grid of 1's or False if the node represents a grid of 0's.
- isLeaf: True if the node is leaf node on the tree or False if the node has the four children.

```
class Node {
public boolean val;
```

public boolean isLeaf;
public Node topLeft;
public Node topRight;
public Node bottomLeft;
public Node bottomRight;
}.

We can construct a Quad-Tree from a two-dimensional area using the following steps:
the value of the grid and set the four children to Null and stop.
divide the current grid into four sub-grids as shown in the photo.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0559 Maximum Depth Of N Ary Tree

**EASY**

Given a n-ary tree, find its maximum depth.

The maximum depth is the number of nodes along the longest path from the root node down to the farthest leaf node.

Nary-Tree input serialization is represented in their level order traversal, each group of children is separated by the null value (See examples).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0563 Binary Tree Tilt

**EASY**

Given the root of a binary tree, return the sum of every tree node's tilt.

The tilt of a tree node is the absolute difference between the sum of all left subtree node values and all right subtree node values. If a node does not have a left child, then the sum of the left subtree node values is treated as 0. The rule is similar if the node does not have a right child.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0572 Subtree Of Another Tree

**EASY**

Given the roots of two binary trees root and subRoot, return true if there is a subtree of root with the same structure and node values of subRoot and false otherwise.

A subtree of a binary tree tree is a tree that consists of a node in tree and all of this node's descendants. The tree tree could also be considered as a subtree of itself.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0589 N Ary Tree Preorder Traversal

**EASY**

Given the root of an n-ary tree, return the preorder traversal of its nodes' values.

Nary-Tree input serialization is represented in their level order traversal. Each group of children is separated by the null value (See examples)

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0590 N Ary Tree Postorder Traversal

EASY

Given the root of an n-ary tree, return the postorder traversal of its nodes' values.

Nary-Tree input serialization is represented in their level order traversal. Each group of children is separated by the null value (See examples)

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0606 Construct String From Binary Tree

EASY

You need to construct a string consists of parenthesis and integers from a binary tree with the preorder traversing way.

The null node needs to be represented by empty parenthesis pair "()". And you need to omit all the empty parenthesis pairs that don't affect the one-to-one mapping relationship between the string and the original binary tree.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0617 Merge Two Binary Trees

EASY

Given two binary trees and imagine that when you put one of them to cover the other, some nodes of the two trees are overlapped while the others are not.

You need to merge them into a new binary tree. The merge rule is that if two nodes overlap, then sum node values up as the new value of the merged node. Otherwise, the NOT null node will be used as the node of new tree.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0623 Add One Row To Tree

MEDIUM

Given the root of a binary tree and two integers val and depth, add a row of nodes with value val at the given depth depth.

Note that the root node is at depth 1.

The adding rule is:
- Given the integer depth, for each not null tree node cur at the depth depth - 1, create two tree nodes with value val as cur's left subtree root and right subtree root.
- cur's original left subtree should be the left subtree of the new left subtree root.
- cur's original right subtree should be the right subtree of the new right subtree root.
- If depth == 1 that means there is no depth depth - 1 at all, then create a tree node with value val as the new root of the whole original tree, and the original tree is the new root's left subtree.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0637 Average Of Levels In Binary Tree

EASY

Given the root of a binary tree, return the average value of the nodes on each level in the form of an array. Answers within 10-5 of the actual answer will be accepted.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0652 Find Duplicate Subtrees

MEDIUM

Given the root of a binary tree, return all duplicate subtrees.

For each kind of duplicate subtrees, you only need to return the root node of any one of them.

Two trees are duplicate if they have the same structure with the same node values.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0653 Two Sum Iv Input Is A Bst

EASY

Given the root of a Binary Search Tree and a target number k, return true if there exist two elements in the BST such that their sum is equal to the given target.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0654 Maximum Binary Tree

MEDIUM

You are given an integer array nums with no duplicates. A maximum binary tree can be built recursively from nums using the following algorithm:

Return the maximum binary tree built from nums.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0655 Print Binary Tree

MEDIUM

Given the root of a binary tree, construct a 0-indexed m x n string matrix res that represents a formatted layout of the tree. The formatted layout matrix should be constructed using the following rules:
- The height of the tree is height and the number of rows m should be equal to height + 1.
- The number of columns n should be equal to 2height+1 - 1.
- Place the root node in the middle of the top row (more formally, at location res[0][(n-1)/2]).
- For each node that has been placed in the matrix at position res[r][c], place its left child at res[r+1][c-2height-r-1] and its right child at res[r+1][c+2height-r-1].
- Continue this process until all the nodes in the tree have been placed.
- Any empty cells should contain the empty string "".

Return the constructed matrix res.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0662 Maximum Width Of Binary Tree

MEDIUM

Given the root of a binary tree, return the maximum width of the given tree.

The maximum width of a tree is the maximum width among all levels.

The width of one level is defined as the length between the end-nodes (the leftmost and rightmost non-null nodes), where the null nodes between the end-nodes that would be present in a complete binary tree extending down to that level are also counted into the length calculation.

It is guaranteed that the answer will in the range of a 32-bit signed integer.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0663 Equal Tree Partition

MEDIUM

Given the root of a binary tree, return true if you can partition the tree into two trees with equal sums of values after removing exactly one edge on the original tree.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0671 Second Minimum Node In A Binary Tree

EASY

Given a non-empty special binary tree consisting of nodes with the non-negative value, where each node in this tree has exactly two or zero sub-node. If the node has two sub-nodes, then this node's value is the smaller value among its two sub-nodes. More formally, the property root.val = min(root.left.val, root.right.val) always holds.

Given such a binary tree, you need to output the second minimum value in the set made of all the nodes' value in the whole tree.

If no such second minimum value exists, output -1 instead.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0675 Cut Off Trees For Golf Event

HARD

You are asked to cut off all the trees in a forest for a golf event. The forest is represented as an m x n matrix. In this matrix:
- 0 means the cell cannot be walked through.
- 1 represents an empty cell that can be walked through.
- A number greater than 1 represents a tree in a cell that can be walked through, and this number is the tree's height.

In one step, you can walk in any of the four directions: north, east, south, and west. If you are standing in a cell with a tree, you can choose whether to cut it off.

You must cut off the trees in order from shortest to tallest. When you cut off a tree, the value at its cell becomes 1 (an empty cell).

Starting from the point (0, 0), return the minimum steps you need to walk to cut off all the trees. If you cannot cut off all the trees, return -1.

Note: The input is generated such that no two trees have the same height, and there is at least one tree needs to be cut off.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0687 Longest Univalue Path

MEDIUM

Given the root of a binary tree, return the length of the longest path, where each node in the path has the same value. This path may or may not pass through the root.

The length of the path between two nodes is represented by the number of edges between them.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0742 Closest Leaf In A Binary Tree

MEDIUM

Given the root of a binary tree where every node has a unique value and a target integer k, return the value of the nearest leaf node to the target k in the tree.

Nearest to a leaf means the least number of edges traveled on the binary tree to reach any leaf of the tree. Also, a node is called a leaf if it has no children.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0776 Split Bst

MEDIUM

Given the root of a binary search tree (BST) and an integer target, split the tree into two subtrees where the first subtree has nodes that are all smaller or equal to the target value, while the second subtree has all nodes that are greater than the target value. It is not necessarily the case that the tree contains a node with the value target.

Additionally, most of the structure of the original tree should remain. Formally, for any child c with parent p in the original tree, if they are both in the same subtree after the split, then node c should still have the parent p.

Return an array of the two roots of the two subtrees in order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0783 Minimum Distance Between Bst Nodes

EASY

Given the root of a Binary Search Tree (BST), return the minimum difference between the values of any two different nodes in the tree.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0814 Binary Tree Pruning

MEDIUM

Given the root of a binary tree, return the same tree where every subtree (of the given tree) not containing a 1 has been removed.

A subtree of a node node is node plus every node that is a descendant of node.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0823 Binary Trees With Factors

MEDIUM

Given an array of unique integers, arr, where each integer arr[i] is strictly greater than 1.

We make a binary tree using these integers, and each number may be used for any number of times. Each non-leaf node's value should be equal to the product of the values of its children.

Return the number of binary trees we can make. The answer may be too large so return the answer modulo 109 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0834 Sum Of Distances In Tree

**HARD**

There is an undirected connected tree with n nodes labeled from 0 to n - 1 and n - 1 edges.

You are given the integer n and the array edges where edges[i] = [ai, bi] indicates that there is an edge between nodes ai and bi in the tree.

Return an array answer of length n where answer[i] is the sum of the distances between the ith node in the tree and all other nodes.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0863 All Nodes Distance K In Binary Tree

**MEDIUM**

Given the root of a binary tree, the value of a target node target, and an integer k, return an array of the values of all nodes that have a distance k from the target node.

You can return the answer in any order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0865 Smallest Subtree With All The Deepest Nodes

**MEDIUM**

Given the root of a binary tree, the depth of each node is the shortest distance to the root.

Return the smallest subtree such that it contains all the deepest nodes in the original tree.

A node is called the deepest if it has the largest depth possible among any node in the entire tree.

The subtree of a node is a tree consisting of that node, plus the set of all descendants of that node.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0872 Leaf Similar Trees

**EASY**

Consider all the leaves of a binary tree, from left to right order, the values of those leaves form a leaf value sequence.

For example, in the given tree above, the leaf value sequence is (6, 7, 4, 9, 8).

Two binary trees are considered leaf-similar if their leaf value sequence is the same.

Return true if and only if the two given trees with head nodes root1 and root2 are leaf-similar.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0889 Construct Binary Tree From Preorder And Postorder Traversal

MEDIUM

Given two integer arrays, preorder and postorder where preorder is the preorder traversal of a binary tree of distinct values and postorder is the postorder traversal of the same tree, reconstruct and return the binary tree.

If there exist multiple answers, you can return any of them.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0894 All Possible Full Binary Trees

MEDIUM

Given an integer n, return a list of all possible full binary trees with n nodes. Each node of each tree in the answer must have Node.val == 0.

Each element of the answer is the root node of one possible tree. You may return the final list of trees in any order.

A full binary tree is a binary tree where each node has exactly 0 or 2 children.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0897 Increasing Order Search Tree

EASY

Given the root of a binary search tree, rearrange the tree in in-order so that the leftmost node in the tree is now the root of the tree, and every node has no left child and only one right child.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0919 Complete Binary Tree Inserter

MEDIUM

A complete binary tree is a binary tree in which every level, except possibly the last, is completely filled, and all nodes are as far left as possible.

Design an algorithm to insert a new node to a complete binary tree keeping it complete after the insertion.

Implement the CBTInserter class:
- CBTInserter(TreeNode root) Initializes the data structure with the root of the complete binary tree.
- int insert(int v) Inserts a TreeNode into the tree with value Node.val == val so that the tree remains complete, and returns the value of the parent of the inserted TreeNode.
- TreeNode get_root() Returns the root node of the tree.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0938 Range Sum Of Bst

EASY

Given the root node of a binary search tree and two integers low and high, return the sum of values of all nodes with a value in the inclusive range [low, high].

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0951 Flip Equivalent Binary Trees

MEDIUM

For a binary tree T, we can define a flip operation as follows: choose any node, and swap the left and right child subtrees.

A binary tree X is flip equivalent to a binary tree Y if and only if we can make X equal to Y after some number of flip operations.

Given the roots of two binary trees root1 and root2, return true if the two trees are flip equivalent or false otherwise.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0958 Check Completeness Of A Binary Tree

MEDIUM

Given the root of a binary tree, determine if it is a complete binary tree.

In a complete binary tree, every level, except possibly the last, is completely filled, and all nodes in the last level are as far left as possible. It can have between 1 and 2h nodes inclusive at the last level h.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0965 Univalued Binary Tree

EASY

A binary tree is uni-valued if every node in the tree has the same value.

Given the root of a binary tree, return true if the given tree is uni-valued, or false otherwise.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0968 Binary Tree Cameras

HARD

You are given the root of a binary tree. We install cameras on the tree nodes where each camera at a node can monitor its parent, itself, and its immediate children.

Return the minimum number of cameras needed to monitor all nodes of the tree.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0971 Flip Binary Tree To Match Preorder Traversal

MEDIUM

You are given the root of a binary tree with n nodes, where each node is uniquely assigned a value from 1 to n. You are also given a sequence of n values voyage, which is the desired pre-order traversal of the binary tree.

Any node in the binary tree can be flipped by swapping its left and right subtrees. For example, flipping node 1 will have the following effect.

Flip the smallest number of nodes so that the pre-order traversal of the tree matches voyage.

Return a list of the values of all flipped nodes. You may return the answer in any order. If it is impossible to flip the nodes in the tree to make the pre-order traversal match voyage, return the list [-1].

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0979 Distribute Coins In Binary Tree

MEDIUM

You are given the root of a binary tree with n nodes where each node in the tree has node.val coins. There are n coins in total throughout the whole tree.

In one move, we may choose two adjacent nodes and move one coin from one node to another. A move may be from parent to child, or from child to parent.

Return the minimum number of moves required to make every node have exactly one coin.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0987 Vertical Order Traversal Of A Binary Tree

HARD

Given the root of a binary tree, calculate the vertical order traversal of the binary tree.

For each node at position (row, col), its left and right children will be at positions (row + 1, col - 1) and (row + 1, col + 1) respectively. The root of the tree is at (0, 0).

The vertical order traversal of a binary tree is a list of top-to-bottom orderings for each column index starting from the leftmost column and ending on the rightmost column. There may be multiple nodes in the same row and same column. In such a case, sort these nodes by their values.

Return the vertical order traversal of the binary tree.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0993 Cousins In Binary Tree

EASY

Given the root of a binary tree with unique values and the values of two different nodes of the tree x and y, return true if the nodes corresponding to the values x and y in the tree are cousins, or false otherwise.

Two nodes of a binary tree are cousins if they have the same depth with different parents.

Note that in a binary tree, the root node is at the depth 0, and children of each depth k node are at the depth k + 1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0998 Maximum Binary Tree Ii

MEDIUM

A maximum tree is a tree where every node has a value greater than any other value in its subtree.

You are given the root of a maximum binary tree and an integer val.

Just as in the previous problem, the given tree was constructed from a list a (root = Construct(a)) recursively with the following Construct(a) routine:
- If a is empty, return null.
- Otherwise, let a[i] be the largest element of a. Create a root node with the value a[i].
- The left child of root will be Construct([a[0], a[1], ..., a[i - 1]]).
- The right child of root will be Construct([a[i + 1], a[i + 2], ..., a[a.length - 1]]).
- Return root.

Note that we were not given a directly, only a root node root = Construct(a).

Suppose b is a copy of a with the value val appended to it. It is guaranteed that b has unique values.

Return Construct(b).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1003 Check If Word Is Valid After Substitutions

**MEDIUM**

Given a string s, determine if it is valid.

A string s is valid if, starting with an empty string t = "", you can transform t into s after performing the following operation any number of times:
- Insert string "abc" into any position in t. More formally, t becomes tleft + "abc" + tright, where t == tleft + tright. Note that tleft and tright may be empty.

Return true if s is a valid string, otherwise, return false.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1028 Recover A Tree From Preorder Traversal

**HARD**

We run a preorder depth-first search (DFS) on the root of a binary tree.

At each node in this traversal, we output D dashes (where D is the depth of this node), then we output the value of this node. If the depth of a node is D, the depth of its immediate child is D + 1. The depth of the root node is 0.

If a node has only one child, that child is guaranteed to be the left child.

Given the output traversal of this traversal, recover the tree and return its root.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1080 Insufficient Nodes In Root To Leaf Paths

**MEDIUM**

Given the root of a binary tree and an integer limit, delete all insufficient nodes in the tree simultaneously, and return the root of the resulting binary tree.

A node is insufficient if every root to leaf path intersecting this node has a sum strictly less than limit.

A leaf is a node with no children.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1104 Path In Zigzag Labelled Binary Tree

**MEDIUM**

In an infinite binary tree where every node has two children, the nodes are labelled in row order.

In the odd numbered rows (ie., the first, third, fifth,...), the labelling is left to right, while in the even numbered rows (second, fourth, sixth,...), the labelling is right to left.

Given the label of a node in this tree, return the labels in the path from the root of the tree to the node with that label.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1110 Delete Nodes And Return Forest

MEDIUM

Given the root of a binary tree, each node in the tree has a distinct value.

After deleting all nodes with a value in toDelete, we are left with a forest (a disjoint union of trees).

Return the roots of the trees in the remaining forest. You may return the result in any order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1120 Maximum Average Subtree

MEDIUM

Given the root of a binary tree, return the maximum average value of a subtree of that tree. Answers within 10-5 of the actual answer will be accepted.

A subtree of a tree is any node of that tree plus all its descendants.

The average value of a tree is the sum of its values, divided by the number of nodes.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1123 Lowest Common Ancestor Of Deepest Leaves

MEDIUM

Given the root of a binary tree, return the lowest common ancestor of its deepest leaves.

Recall that:
- The node of a binary tree is a leaf if and only if it has no children
- The depth of the root of the tree is 0. if the depth of a node is d, the depth of each of its children is d + 1.
- The lowest common ancestor of a set S of nodes, is the node A with the largest depth such that every node in S is in the subtree with root A.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1130 Minimum Cost Tree From Leaf Values

MEDIUM

Given an array arr of positive integers, consider all binary trees such that:
- Each node has either 0 or 2 children;
- The values of arr correspond to the values of each leaf in an in-order traversal of the tree.
- The value of each non-leaf node is equal to the product of the largest leaf value in its left and right subtree, respectively.

Among all possible binary trees considered, return the smallest possible sum of the values of each non-leaf node. It is guaranteed this sum fits into a 32-bit integer.

A node is a leaf if and only if it has zero children.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1145 Binary Tree Coloring Game

MEDIUM

Two players play a turn based game on a binary tree. We are given the root of this binary tree, and the number of nodes n in the tree. n is odd, and each node has a distinct value from 1 to n.

Initially, the first player names a value x with 1 <= x <= n, and the second player names a value

y with 1 <= y <= n and y != x. The first player colors the node with value x red, and the second player colors the node with value y blue.

Then, the players take turns starting with the first player. In each turn, that player chooses a node of their color (red if player 1, blue if player 2) and colors an uncolored neighbor of the chosen node (either the left child, right child, or parent of the chosen node.)

If (and only if) a player cannot choose such a node in this way, they must pass their turn. If both players pass their turn, the game ends, and the winner is the player that colored more nodes.

You are the second player. If it is possible to choose such a y to ensure you win the game, return true. If it is not possible, return false.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1161 Maximum Level Sum Of A Binary Tree

**MEDIUM**

Given the root of a binary tree, the level of its root is 1, the level of its children is 2, and so on.

Return the smallest level x such that the sum of all the values of nodes at level x is maximal.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1214 Two Sum Bsts

**MEDIUM**

Given the roots of two binary search trees, root1 and root2, return true if and only if there is a node in the first tree and a node in the second tree whose values sum up to a given integer target.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1245 Tree Diameter

**MEDIUM**

The diameter of a tree is the number of edges in the longest path in that tree.

There is an undirected tree of n nodes labeled from 0 to n - 1. You are given a 2D array edges where edges.length == n - 1 and edges[i] = [ai, bi] indicates that there is an undirected edge between nodes ai and bi in the tree.

Return the diameter of the tree.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1261 Find Elements In A Contaminated Binary Tree

**MEDIUM**

Given a binary tree with the following rules:
then treeNode.left.val == 2 * x + 1
then treeNode.right.val == 2 * x + 2

Now the binary tree is contaminated, which means all treeNode.val have been changed to -1.

Implement the FindElements class:
- FindElements(TreeNode* root) Initializes the object with a contaminated binary tree and recovers it.

- bool find(int target) Returns true if the target value exists in the recovered binary tree.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1273 Delete Tree Nodes

**MEDIUM**

A tree rooted at node 0 is given as follows:
- The number of nodes is nodes;
- The value of the ith node is value[i];
- The parent of the ith node is parent[i].

Remove every subtree whose sum of values of nodes is zero.

Return the number of the remaining nodes in the tree.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1293 Shortest Path In A Grid With Obstacles Elimination

**HARD**

You are given an m x n integer matrix grid where each cell is either 0 (empty) or 1 (obstacle). You can move up, down, left, or right from and to an empty cell in one step.

Return the minimum number of steps to walk from the upper left corner (0, 0) to the lower right corner (m - 1, n - 1) given that you can eliminate at most k obstacles. If it is not possible to find such walk return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1325 Delete Leaves With A Given Value

**MEDIUM**

Given a binary tree root and an integer target, delete all the leaf nodes with value target.

Note that once you delete a leaf node with value target, if its parent node becomes a leaf node and has the value target, it should also be deleted (you need to continue doing that until you cannot).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1339 Maximum Product Of Splitted Binary Tree

**MEDIUM**

Given the root of a binary tree, split the binary tree into two subtrees by removing one edge such that the product of the sums of the subtrees is maximized.

Return the maximum product of the sums of the two subtrees. Since the answer may be too large, return it modulo 109 + 7.

Note that you need to maximize the answer before taking the mod and not after taking it.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1361 Validate Binary Tree Nodes

**MEDIUM**

You have n binary tree nodes numbered from 0 to n - 1 where node i has two children leftChild[i] and rightChild[i], return true if and only if all the given nodes form exactly one valid binary tree.

If node i has no left child then leftChild[i] will equal -1, similarly for the right child.

Note that the nodes have no values and that we only use the node numbers in this problem.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1372 Longest Zigzag Path In A Binary Tree

**MEDIUM**

You are given the root of a binary tree.

A ZigZag path for a binary tree is defined as follow:
- Choose any node in the binary tree and a direction (right or left).
- If the current direction is right, move to the right child of the current node; otherwise, move to the left child.
- Change the direction from right to left or from left to right.
- Repeat the second and third steps until you can't move in the tree.

Zigzag length is defined as the number of nodes visited - 1. (A single node has a length of 0).

Return the longest ZigZag path contained in that tree.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1373 Maximum Sum Bst In Binary Tree

**HARD**

Given a binary tree root, return the maximum sum of all keys of any sub-tree which is also a Binary Search Tree (BST).

Assume a BST is defined as follows:
- The left subtree of a node contains only nodes with keys less than the node's key.
- The right subtree of a node contains only nodes with keys greater than the node's key.
- Both the left and right subtrees must also be binary search trees.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1379 Find A Corresponding Node Of A Binary Tree In A Clone Of That Tree

**EASY**

Given two binary trees original and cloned and given a reference to a node target in the original tree.

The cloned tree is a copy of the original tree.

Return a reference to the same node in the cloned tree.

Note that you are not allowed to change any of the two trees or the target node and the answer must be a reference to a node in the cloned tree.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1430 Check If A String Is A Valid Sequence From Root To Leaves Path In

**MEDIUM**

Given a binary tree where each path going from the root to any leaf form a valid sequence, check if a given string is a valid sequence in such binary tree.

We get the given string from the concatenation of an array of integers arr and the concatenation of all values of the nodes along a path results in a sequence in the given binary tree.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 1443 Minimum Time To Collect All Apples In A Tree

MEDIUM

Given an undirected tree consisting of n vertices numbered from 0 to n-1, which has some apples in their vertices. You spend 1 second to walk over one edge of the tree. Return the minimum time in seconds you have to spend to collect all apples in the tree, starting at vertex 0 and coming back to this vertex.

The edges of the undirected tree are given in the array edges, where edges[i] = [ai, bi] means that exists an edge connecting the vertices ai and bi. Additionally, there is a boolean array hasApple, where hasApple[i] = true means that vertex i has an apple; otherwise, it does not have any apple.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 1448 Count Good Nodes In Binary Tree

MEDIUM

Given a binary tree root, a node X in the tree is named good if in the path from root to X there are no nodes with a value greater than X.

Return the number of good nodes in the binary tree.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 1457 Pseudo Palindromic Paths In A Binary Tree

MEDIUM

Given a binary tree where node values are digits from 1 to 9. A path in the binary tree is said to be pseudo-palindromic if at least one permutation of the node values in the path is a palindrome.

Return the number of pseudo-palindromic paths going from the root node to leaf nodes.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 1469 Find All The Lonely Nodes

EASY

In a binary tree, a lonely node is a node that is the only child of its parent node. The root of the tree is not lonely because it does not have a parent node.

Given the root of a binary tree, return an array containing the values of all lonely nodes in the tree. Return the list in any order.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 1483 Kth Ancestor Of A Tree Node

HARD

You are given a tree with n nodes numbered from 0 to n - 1 in the form of a parent array parent where parent[i] is the parent of ith node. The root of the tree is node 0. Find the kth ancestor of a given node.

The kth ancestor of a tree node is the kth node in the path from that node to the root node.

Implement the TreeAncestor class:
- TreeAncestor(int n, int[] parent) Initializes the object with the number of nodes in the tree

and the parent array.
- int getKthAncestor(int node, int k) return the kth ancestor of the given node node. If there is no such ancestor, return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1485 Clone Binary Tree With Random Pointer

**MEDIUM**

A binary tree is given such that each node contains an additional random pointer which could point to any node in the tree or null.

Return a deep copy of the tree.

The tree is represented in the same input/output way as normal binary trees where each node is represented as a pair of [val, random_index] where:
- val: an integer representing Node.val
- random_index: the index of the node (in the input) where the random pointer points to, or null if it does not point to any node.

You will be given the tree in class Node and you should return the cloned tree in class NodeCopy. NodeCopy class is just a clone of Node class with the same attributes and constructors.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1489 Find Critical And Pseudo Critical Edges In Minimum Spanning Tree

**HARD**

Given a weighted undirected connected graph with n vertices numbered from 0 to n - 1, and an array edges where edges[i] = [ai, bi, weighti] represents a bidirectional and weighted edge between nodes ai and bi. A minimum spanning tree (MST) is a subset of the graph's edges that connects all vertices without cycles and with the minimum possible total edge weight.

Find all the critical and pseudo-critical edges in the given graph's minimum spanning tree (MST). An MST edge whose deletion from the graph would cause the MST weight to increase is called a critical edge. On the other hand, a pseudo-critical edge is that which can appear in some MSTs but not all.

Note that you can return the indices of the edges in any order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1490 Clone N Ary Tree

**MEDIUM**

Given a root of an N-ary tree, return a deep copy (clone) of the tree.

Each node in the n-ary tree contains a val (int) and a list (List[Node]) of its children.

```
class Node {
public int val;
public List<Node> children;
}
```

Nary-Tree input serialization is represented in their level order traversal, each group of children is separated by the null value (See examples).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1506 Find Root Of N Ary Tree

MEDIUM

You are given all the nodes of an N-ary tree as an array of Node objects, where each node has a unique value.

Return the root of the N-ary tree.

Custom testing:
- An N-ary tree can be serialized as represented in its level order traversal where each group of children is separated by the null value (see examples).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1516 Move Sub Tree Of N Ary Tree

HARD

Given the root of an N-ary tree of unique values, and two nodes of the tree p and q.

You should move the subtree of the node p to become a direct child of node q. If p is already a direct child of q, do not change anything. Node p must be the last child in the children list of node q.

Return the root of the tree after adjusting it.

There are 3 cases for nodes p and q:

In cases 2 and 3, you just need to move p (with its sub-tree) to be a child of q, but in case 1 the tree may be disconnected, thus you need to reconnect the tree again. Please read the examples carefully before solving this problem.

Nary-Tree input serialization is represented in their level order traversal, each group of children is separated by the null value (See examples).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1519 Number Of Nodes In The Sub Tree With The Same Label

MEDIUM

You are given a tree (i.e. a connected, undirected graph that has no cycles) consisting of n nodes numbered from 0 to n - 1 and exactly n - 1 edges. The root of the tree is the node 0, and each node of the tree has a label which is a lower-case character given in the string labels (i.e. The node with the number i has the label labels[i]).

The edges array is given on the form edges[i] = [ai, bi], which means there is an edge between nodes ai and bi in the tree.

Return an array of size n where ans[i] is the number of nodes in the subtree of the ith node which have the same label as node i.

A subtree of a tree T is the tree consisting of a node in T and all of its descendant nodes.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1522 Diameter Of N Ary Tree

MEDIUM

Given a root of an N-ary tree, you need to compute the length of the diameter of the tree.

The diameter of an N-ary tree is the length of the longest path between any two nodes in

the tree. This path may or may not pass through the root.

(Nary-Tree input serialization is represented in their level order traversal, each group of children is separated by the null value.)

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1530 Number Of Good Leaf Nodes Pairs

**MEDIUM**

You are given the root of a binary tree and an integer distance. A pair of two different leaf nodes of a binary tree is said to be good if the length of the shortest path between them is less than or equal to distance.

Return the number of good leaf node pairs in the tree.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1569 Number Of Ways To Reorder Array To Get Same Bst

**HARD**

Given an array nums that represents a permutation of integers from 1 to n. We are going to construct a binary search tree (BST) by inserting the elements of nums in order into an initially empty BST. Find the number of different ways to reorder nums so that the constructed BST is identical to that formed from the original array nums.

For example, given nums = [2,1,3], we will have 2 as the root, 1 as a left child, and 3 as a right child. The array [2,3,1] also yields the same BST but [3,2,1] yields a different BST.

Return the number of ways to reorder nums such that the BST formed is identical to the original BST formed from nums.

Since the answer may be very large, return it modulo 109 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1597 Build Binary Expression Tree From Infix Expression

**HARD**

A binary expression tree is a kind of binary tree used to represent arithmetic expressions. Each node of a binary expression tree has either zero or two children. Leaf nodes (nodes with 0 children) correspond to operands (numbers), and internal nodes (nodes with 2 children) correspond to the operators '+' (addition), '-' (subtraction), '*' (multiplication), and '/' (division).

For each internal node with operator o, the infix expression it represents is (A o B), where A is the expression the left subtree represents and B is the expression the right subtree represents.

You are given a string s, an infix expression containing operands, the operators described above, and parentheses '(' and ')'.

Return any valid binary expression tree, whose in-order traversal reproduces s after omitting the parenthesis from it.

Please note that order of operations applies in s. That is, expressions in parentheses are evaluated first, and multiplication and division happen before addition and subtraction.

Operands must also appear in the same order in both s and the in-order traversal of the tree.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1602 Find Nearest Right Node In Binary Tree

MEDIUM

Given the root of a binary tree and a node u in the tree, return the nearest node on the same level that is to the right of u, or return null if u is the rightmost node in its level.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1609 Even Odd Tree

MEDIUM

A binary tree is named Even-Odd if it meets the following conditions:
- The root of the binary tree is at level index 0, its children are at level index 1, their children are at level index 2, etc.
- For every even-indexed level, all nodes at the level have odd integer values in strictly increasing order (from left to right).
- For every odd-indexed level, all nodes at the level have even integer values in strictly decreasing order (from left to right).

Given the root of a binary tree, return true if the binary tree is Even-Odd, otherwise return false.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1612 Check If Two Expression Trees Are Equivalent

MEDIUM

A binary expression tree is a kind of binary tree used to represent arithmetic expressions. Each node of a binary expression tree has either zero or two children. Leaf nodes (nodes with 0 children) correspond to operands (variables), and internal nodes (nodes with two children) correspond to the operators. In this problem, we only consider the '+' operator (i.e. addition).

You are given the roots of two binary expression trees, root1 and root2. Return true if the two binary expression trees are equivalent. Otherwise, return false.

Two binary expression trees are equivalent if they evaluate to the same value regardless of what the variables are set to.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1617 Count Subtrees With Max Distance Between Cities

HARD

There are n cities numbered from 1 to n. You are given an array edges of size n-1, where edges[i] = [ui, vi] represents a bidirectional edge between cities ui and vi. There exists a unique path between each pair of cities. In other words, the cities form a tree.

A subtree is a subset of cities where every city is reachable from every other city in the subset, where the path between each pair passes through only the cities from the subset. Two subtrees are different if there is a city in one subtree that is not present in the other.

For each d from 1 to n-1, find the number of subtrees in which the maximum distance between any two cities in the subtree is equal to d.

Return an array of size n-1 where the dth element (1-indexed) is the number of subtrees in which the maximum distance between any two cities is equal to d.

Notice that the distance between the two cities is the number of edges in the path between them.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1628 Design An Expression Tree With Evaluate Function

MEDIUM

Given the postfix tokens of an arithmetic expression, build and return the binary expression tree that represents this expression.

Postfix notation is a notation for writing arithmetic expressions in which the operands (numbers) appear before their operators. For example, the postfix tokens of the expression 4*(5-(7+2)) are represented in the array postfix = ["4","5","7","2","+","-","*"].

The class Node is an interface you should use to implement the binary expression tree. The returned tree will be tested using the evaluate function, which is supposed to evaluate the tree's value. You should not remove the Node class; however, you can modify it as you wish, and you can define other classes to implement it if needed.

A binary expression tree is a kind of binary tree used to represent arithmetic expressions. Each node of a binary expression tree has either zero or two children. Leaf nodes (nodes with 0 children) correspond to operands (numbers), and internal nodes (nodes with two children) correspond to the operators '+' (addition), '-' (subtraction), '*' (multiplication), and '/' (division).

It's guaranteed that no subtree will yield a value that exceeds 109 in absolute value, and all the operations are valid (i.e., no division by zero).

Follow up: Could you design the expression tree such that it is more modular? For example, is your design able to support additional operators without making changes to your existing evaluate implementation?

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1644 Lowest Common Ancestor Of A Binary Tree Ii

MEDIUM

Given the root of a binary tree, return the lowest common ancestor (LCA) of two given nodes, p and q. If either node p or q does not exist in the tree, return null. All values of the nodes in the tree are unique.

According to the definition of LCA on Wikipedia: "The lowest common ancestor of two nodes p and q in a binary tree T is the lowest node that has both p and q as descendants (where we allow a node to be a descendant of itself)". A descendant of a node x is a node y that is on the path from node x to some leaf node.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1650 Lowest Common Ancestor Of A Binary Tree Iii

MEDIUM

Given two nodes of a binary tree p and q, return their lowest common ancestor (LCA).

Each node will have a reference to its parent node. The definition for Node is below:

```
class Node {
public int val;
public Node left;
public Node right;
public Node parent;
}
```

According to the definition of LCA on Wikipedia: "The lowest common ancestor of two nodes p and q in a tree T is the lowest node that has both p and q as descendants (where we allow a node to be a descendant of itself)."

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1660 Correct A Binary Tree

MEDIUM

You have a binary tree with a small defect. There is exactly one invalid node where its right child incorrectly points to another node at the same depth but to the invalid node's right.

Given the root of the binary tree with this defect, root, return the root of the binary tree after removing this invalid node and every node underneath it (minus the node it incorrectly points to).

Custom testing:
The test input is read as 3 lines:
- TreeNode root
- int fromNode (not available to correctBinaryTree)
- int toNode (not available to correctBinaryTree)

After the binary tree rooted at root is parsed, the TreeNode with value of fromNode will have its right child pointer pointing to the TreeNode with a value of toNode. Then, root is passed to correctBinaryTree.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1666 Change The Root Of A Binary Tree

MEDIUM

Given the root of a binary tree and a leaf node, reroot the tree so that the leaf is the new root.

You can reroot the tree with the following steps for each node cur on the path starting from the leaf up to the root excluding the root:
parent's pointer to cur becomes null, making it have at most one child.

Return the new root of the rerooted tree.

Note: Ensure that your solution sets the Node.parent pointers correctly after rerooting or you will receive "Wrong Answer".

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1676 Lowest Common Ancestor Of A Binary Tree Iv

MEDIUM

Given the root of a binary tree and an array of TreeNode objects nodes, return the lowest common ancestor (LCA) of all the nodes in nodes. All the nodes will exist in the tree, and all values of the tree's nodes are unique.

Extending the definition of LCA on Wikipedia: "The lowest common ancestor of n nodes p1, p2, ..., pn in a binary tree T is the lowest node that has every pi as a descendant (where we allow a node to be a descendant of itself) for every valid i". A descendant of a node x is a node y that is on the path from node x to some leaf node.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1719 Number Of Ways To Reconstruct A Tree

HARD

You are given an array pairs, where pairs[i] = [xi, yi], and:
- There are no duplicates.
- xi < yi

Let ways be the number of rooted trees that satisfy the following conditions:

- The tree consists of nodes whose values appeared in pairs.
- A pair [xi, yi] exists in pairs if and only if xi is an ancestor of yi or yi is an ancestor of xi.
- Note: the tree does not have to be a binary tree.

Two ways are considered to be different if there is at least one node that has different parents in both ways.

Return:
- 0 if ways == 0
- 1 if ways == 1
- 2 if ways > 1

A rooted tree is a tree that has a single root node, and all edges are oriented to be outgoing from the root.

An ancestor of a node is any node on the path from the root to that node (excluding the node itself). The root has no ancestors.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1740 Find Distance In A Binary Tree

**MEDIUM**

Given the root of a binary tree and two integers p and q, return the distance between the nodes of value p and value q in the tree.

The distance between two nodes is the number of edges on the path from one to the other.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1766 Tree Of Coprimes

**HARD**

There is a tree (i.e., a connected, undirected graph that has no cycles) consisting of n nodes numbered from 0 to n - 1 and exactly n - 1 edges. Each node has a value associated with it, and the root of the tree is node 0.

To represent this tree, you are given an integer array nums and a 2D array edges. Each nums[i] represents the ith node's value, and each edges[j] = [uj, vj] represents an edge between nodes uj and vj in the tree.

Two values x and y are coprime if gcd(x, y) == 1 where gcd(x, y) is the greatest common divisor of x and y.

An ancestor of a node i is any other node on the shortest path from node i to the root. A node is not considered an ancestor of itself.

Return an array ans of size n, where ans[i] is the closest ancestor to node i such that nums[i] and nums[ans[i]] are coprime, or -1 if there is no such ancestor.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1804 Implement Trie Ii Prefix Tree

**MEDIUM**

A trie (pronounced as "try") or prefix tree is a tree data structure used to efficiently store and retrieve keys in a dataset of strings. There are various applications of this data structure, such as autocomplete and spellchecker.

Implement the Trie class:
- Trie() Initializes the trie object.
- void insert(String word) Inserts the string word into the trie.

- int countWordsEqualTo(String word) Returns the number of instances of the string word in the trie.
- int countWordsStartingWith(String prefix) Returns the number of strings in the trie that have the string prefix as a prefix.
- void erase(String word) Erases the string word from the trie.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1902 Depth Of Bst Given Insertion Order

**MEDIUM**

You are given a 0-indexed integer array order of length n, a permutation of integers from 1 to n representing the order of insertion into a binary search tree.

A binary search tree is defined as follows:
- The left subtree of a node contains only nodes with keys less than the node's key.
- The right subtree of a node contains only nodes with keys greater than the node's key.
- Both the left and right subtrees must also be binary search trees.

The binary search tree is constructed as follows:
- order[0] will be the root of the binary search tree.
- All subsequent elements are inserted as the child of any existing node such that the binary search tree properties hold.

Return the depth of the binary search tree.

A binary tree's depth is the number of nodes along the longest path from the root node down to the farthest leaf node.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1932 Merge Bsts To Create Single Bst

**HARD**

You are given n BST (binary search tree) root nodes for n separate BSTs stored in an array trees (0-indexed). Each BST in trees has at most 3 nodes, and no two roots have the same value. In one operation, you can:
- Select two distinct indices i and j such that the value stored at one of the leaves of trees[i] is equal to the root value of trees[j].
- Replace the leaf node in trees[i] with trees[j].
- Remove trees[j] from trees.

Return the root of the resulting BST if it is possible to form a valid BST after performing n - 1 operations, or null if it is impossible to create a valid BST.

A BST (binary search tree) is a binary tree where each node satisfies the following property:
- Every node in the node's left subtree has a value strictly less than the node's value.
- Every node in the node's right subtree has a value strictly greater than the node's value.

A leaf is a node that has no children.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1964 Find The Longest Valid Obstacle Course At Each Position

**HARD**

You want to build some obstacle courses. You are given a 0-indexed integer array obstacles of length n, where obstacles[i] describes the height of the ith obstacle.

For every index i between 0 and n - 1 (inclusive), find the length of the longest obstacle course in obstacles such that:
- You choose any number of obstacles between 0 and i inclusive.

- You must include the ith obstacle in the course.
- You must put the chosen obstacles in the same order as they appear in obstacles.
- Every obstacle (except the first) is taller than or the same height as the obstacle immediately before it.

Return an array ans of length n, where ans[i] is the length of the longest obstacle course for index i as described above.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1993 Operations On Tree

**MEDIUM**

You are given a tree with n nodes numbered from 0 to n - 1 in the form of a parent array parent where parent[i] is the parent of the ith node. The root of the tree is node 0, so parent[0] = -1 since it has no parent. You want to design a data structure that allows users to lock, unlock, and upgrade nodes in the tree.

The data structure should support the following functions:
- Lock: Locks the given node for the given user and prevents other users from locking the same node. You may only lock a node using this function if the node is unlocked.
- Unlock: Unlocks the given node for the given user. You may only unlock a node using this function if it is currently locked by the same user.
- Upgrade: Locks the given node for the given user and unlocks all of its descendants regardless of who locked it. You may only upgrade a node if all 3 conditions are true:
- The node is unlocked,
- It has at least one locked descendant (by any user), and
- It does not have any locked ancestors.

Implement the LockingTree class:
- LockingTree(int[] parent) initializes the data structure with the parent array.
- lock(int num, int user) returns true if it is possible for the user with id user to lock the node num, or false otherwise. If it is possible, the node num will become locked by the user with id user.
- unlock(int num, int user) returns true if it is possible for the user with id user to unlock the node num, or false otherwise. If it is possible, the node num will become unlocked.
- upgrade(int num, int user) returns true if it is possible for the user with id user to upgrade the node num, or false otherwise. If it is possible, the node num will be upgraded.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2003 Smallest Missing Genetic Value In Each Subtree

**HARD**

There is a family tree rooted at 0 consisting of n nodes numbered 0 to n - 1. You are given a 0-indexed integer array parents, where parents[i] is the parent for node i. Since node 0 is the root, parents[0] == -1.

There are 105 genetic values, each represented by an integer in the inclusive range [1, 105]. You are given a 0-indexed integer array nums, where nums[i] is a distinct genetic value for node i.

Return an array ans of length n where ans[i] is the smallest genetic value that is missing from the subtree rooted at node i.

The subtree rooted at a node x contains node x and all of its descendant nodes.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2005 Subtree Removal Game With Fibonacci Tree

HARD

A Fibonacci tree is a binary tree created using the order function order(n):
- order(0) is the empty tree.
- order(1) is a binary tree with only one node.
- order(n) is a binary tree that consists of a root node with the left subtree as order(n - 2) and the right subtree as order(n - 1).

Alice and Bob are playing a game with a Fibonacci tree with Alice staring first. On each turn, a player selects a node and removes that node and its subtree. The player that is forced to delete root loses.

Given the integer n, return true if Alice wins the game or false if Bob wins, assuming both players play optimally.

A subtree of a binary tree tree is a tree that consists of a node in tree and all of this node's descendants. The tree tree could also be considered as a subtree of itself.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2021 Brightest Position On Street

MEDIUM

A perfectly straight street is represented by a number line. The street has street lamp(s) on it and is represented by a 2D integer array lights. Each lights[i] = [positioni, rangei] indicates that there is a street lamp at position positioni that lights up the area from [positioni - rangei, positioni + rangei] (inclusive).

The brightness of a position p is defined as the number of street lamp that light up the position p.

Given lights, return the brightest position on the street. If there are multiple brightest positions, return the smallest one.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2096 Step By Step Directions From A Binary Tree Node To Another

MEDIUM

You are given the root of a binary tree with n nodes. Each node is uniquely assigned a value from 1 to n. You are also given an integer startValue representing the value of the start node s, and a different integer destValue representing the value of the destination node t.

Find the shortest path starting from node s and ending at node t. Generate step-by-step directions of such path as a string consisting of only the uppercase letters 'L', 'R', and 'U'. Each letter indicates a specific direction:
- 'L' means to go from a node to its left child node.
- 'R' means to go from a node to its right child node.
- 'U' means to go from a node to its parent node.

Return the step-by-step directions of the shortest path from node s to node t.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2196 Create Binary Tree From Descriptions

MEDIUM

You are given a 2D integer array descriptions where descriptions[i] = [parenti, childi, isLefti] indicates that parenti is the parent of childi in a binary tree of unique values. Furthermore,

- If isLefti == 1, then childi is the left child of parenti.
- If isLefti == 0, then childi is the right child of parenti.

Construct the binary tree described by descriptions and return its root.

The test cases will be generated such that the binary tree is valid.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2237 Count Positions On Street With Required Brightness

**MEDIUM**

You are given an integer n. A perfectly straight street is represented by a number line ranging from 0 to n - 1. You are given a 2D integer array lights representing the street lamp(s) on the street. Each lights[i] = [positioni, rangei] indicates that there is a street lamp at position positioni that lights up the area from [max(0, positioni - rangei), min(n - 1, positioni + rangei)] (inclusive).

The brightness of a position p is defined as the number of street lamps that light up the position p. You are given a 0-indexed integer array requirement of size n where requirement[i] is the minimum brightness of the ith position on the street.

Return the number of positions i on the street between 0 and n - 1 that have a brightness of at least requirement[i].

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2265 Count Nodes Equal To Average Of Subtree

**MEDIUM**

Given the root of a binary tree, return the number of nodes where the value of the node is equal to the average of the values in its subtree.

Note:
- The average of n elements is the sum of the n elements divided by n and rounded down to the nearest integer.
- A subtree of root is a tree consisting of root and all of its descendants.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2277 Closest Node To Path In Tree

**HARD**

You are given a positive integer n representing the number of nodes in a tree, numbered from 0 to n - 1 (inclusive). You are also given a 2D integer array edges of length n - 1, where edges[i] = [node1i, node2i] denotes that there is a bidirectional edge connecting node1i and node2i in the tree.

You are given a 0-indexed integer array query of length m where query[i] = [starti, endi, nodei] means that for the ith query, you are tasked with finding the node on the path from starti to endi that is closest to nodei.

Return an integer array answer of length m, where answer[i] is the answer to the ith query.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2290 Minimum Obstacle Removal To Reach Corner

**HARD**

You are given a 0-indexed 2D integer array grid of size m x n. Each cell has one of two values:
- 0 represents an empty cell,

- 1 represents an obstacle that may be removed.

You can move up, down, left, or right from and to an empty cell.

Return the minimum number of obstacles to remove so you can move from the upper left corner (0, 0) to the lower right corner (m - 1, n - 1).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2313 Minimum Flips In Binary Tree To Get Result

**HARD**

You are given the root of a binary tree with the following properties:
- Leaf nodes have either the value 0 or 1, representing false and true respectively.
- Non-leaf nodes have either the value 2, 3, 4, or 5, representing the boolean operations OR, AND, XOR, and NOT, respectively.

You are also given a boolean result, which is the desired result of the evaluation of the root node.

The evaluation of a node is as follows:
- If the node is a leaf node, the evaluation is the value of the node, i.e. true or false.
- Otherwise, evaluate the node's children and apply the boolean operation of its value with the children's evaluations.

In one operation, you can flip a leaf node, which causes a false node to become true, and a true node to become false.

Return the minimum number of operations that need to be performed such that the evaluation of root yields result. It can be shown that there is always a way to achieve result.

A leaf node is a node that has zero children.

Note: NOT nodes have either a left child or a right child, but other non-leaf nodes have both a left child and a right child.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2322 Minimum Score After Removals On A Tree

**HARD**

There is an undirected connected tree with n nodes labeled from 0 to n - 1 and n - 1 edges.

You are given a 0-indexed integer array nums of length n where nums[i] represents the value of the ith node. You are also given a 2D integer array edges of length n - 1 where edges[i] = [ai, bi] indicates that there is an edge between nodes ai and bi in the tree.

Remove two distinct edges of the tree to form three connected components. For a pair of removed edges, the following steps are defined:
the pair.
The three XOR values are 4 ^ 5 ^ 7 = 6, 1 ^ 9 = 8, and 3 ^ 3 ^ 3 = 3. The largest XOR value is 8 and the smallest XOR value is 3. The score is then 8 - 3 = 5.

Return the minimum score of any possible pair of edge removals on the given tree.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2331 Evaluate Boolean Binary Tree

**EASY**

You are given the root of a full binary tree with the following properties:
- Leaf nodes have either the value 0 or 1, where 0 represents False and 1 represents True.
- Non-leaf nodes have either the value 2 or 3, where 2 represents the boolean OR and 3 represents the boolean AND.

The evaluation of a node is as follows:
- If the node is a leaf node, the evaluation is the value of the node, i.e. True or False.
- Otherwise, evaluate the node's two children and apply the boolean operation of its value with the children's evaluations.

Return the boolean result of evaluating the root node.

A full binary tree is a binary tree where each node has either 0 or 2 children.

A leaf node is a node that has zero children.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2378 Choose Edges To Maximize Score In A Tree

**MEDIUM**

You are given a weighted tree consisting of n nodes numbered from 0 to n - 1.

The tree is rooted at node 0 and represented with a 2D array edges of size n where edges[i] = [pari, weighti] indicates that node pari is the parent of node i, and the edge between them has a weight equal to weighti. Since the root does not have a parent, you have edges[0] = [-1, -1].

Choose some edges from the tree such that no two chosen edges are adjacent and the sum of the weights of the chosen edges is maximized.

Return the maximum sum of the chosen edges.

Note:
- You are allowed to not choose any edges in the tree, the sum of weights in this case will be 0.
- Two edges Edge1 and Edge2 in the tree are adjacent if they have a common node.
- In other words, they are adjacent if Edge1 connects nodes a and b and Edge2 connects nodes b and c.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2385 Amount Of Time For Binary Tree To Be Infected

**MEDIUM**

You are given the root of a binary tree with unique values, and an integer start.
At minute 0, an infection starts from the node with value start.

Each minute, a node becomes infected if:
- The node is currently uninfected.
- The node is adjacent to an infected node.

Return the number of minutes needed for the entire tree to be infected.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2415 Reverse Odd Levels Of Binary Tree

**MEDIUM**

Given the root of a perfect binary tree, reverse the node values at each odd level of the tree.
- For example, suppose the node values at level 3 are [2,1,3,4,7,11,29,18], then it should become [18,29,11,7,4,3,1,2].

Return the root of the reversed tree.

A binary tree is perfect if all parent nodes have two children and all leaves are on the same level.

The level of a node is the number of edges along the path between it and the root node.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2458 Height Of Binary Tree After Subtree Removal Queries

HARD

You are given the root of a binary tree with n nodes. Each node is assigned a unique value from 1 to n. You are also given an array queries of size m.

You have to perform m independent queries on the tree where in the ith query you do the following:
- Remove the subtree rooted at the node with the value queries[i] from the tree. It is guaranteed that queries[i] will not be equal to the value of the root.

Return an array answer of size m where answer[i] is the height of the tree after performing the ith query.

Note:
- The queries are independent, so the tree returns to its initial state after each query.
- The height of a tree is the number of edges in the longest simple path from the root to some node in the tree.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2467 Most Profitable Path In A Tree

MEDIUM

There is an undirected tree with n nodes labeled from 0 to n - 1, rooted at node 0. You are given a 2D integer array edges of length n - 1 where edges[i] = [ai, bi] indicates that there is an edge between nodes ai and bi in the tree.

At every node i, there is a gate. You are also given an array of even integers amount, where amount[i] represents:
- the price needed to open the gate at node i, if amount[i] is negative, or,
- the cash reward obtained on opening the gate at node i, otherwise.

The game goes on as follows:
- Initially, Alice is at node 0 and Bob is at node bob.
- At every second, Alice and Bob each move to an adjacent node. Alice moves towards some leaf node, while Bob moves towards node 0.
- For every node along their path, Alice and Bob either spend money to open the gate at that node, or accept the reward. Note that:
- If the gate is already open, no price will be required, nor will there be any cash reward.
- If Alice and Bob reach the node simultaneously, they share the price/reward for opening the gate there. In other words, if the price to open the gate is c, then both Alice and Bob pay c / 2 each. Similarly, if the reward at the gate is c, both of them receive c / 2 each.
- If Alice reaches a leaf node, she stops moving. Similarly, if Bob reaches node 0, he stops moving. Note that these events are independent of each other.

Return the maximum net income Alice can have if she travels towards the optimal leaf node.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2479 Maximum Xor Of Two Non Overlapping Subtrees

HARD

There is an undirected tree with n nodes labeled from 0 to n - 1. You are given the integer n and a 2D integer array edges of length n - 1, where edges[i] = [ai, bi] indicates that there is an edge between nodes ai and bi in the tree. The root of the tree is the node labeled 0.

Each node has an associated value. You are given an array values of length n, where values[i] is the value of the ith node.

Select any two non-overlapping subtrees. Your score is the bitwise XOR of the sum of the values within those subtrees.

Return the maximum possible score you can achieve. If it is impossible to find two nonoverlapping subtrees, return 0.

Note that:
- The subtree of a node is the tree consisting of that node and all of its descendants.
- Two subtrees are non-overlapping if they do not share any common node.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2509 Cycle Length Queries In A Tree

**HARD**

You are given an integer n. There is a complete binary tree with 2n - 1 nodes. The root of that tree is the node with the value 1, and every node with a value val in the range [1, 2n - 1 - 1] has two children where:
- The left node has the value 2 * val, and
- The right node has the value 2 * val + 1.

You are also given a 2D integer array queries of length m, where queries[i] = [ai, bi]. For each query, solve the following problem:

Note that:
- A cycle is a path that starts and ends at the same node, and each edge in the path is visited only once.
- The length of a cycle is the number of edges visited in the cycle.
- There could be multiple edges between two nodes in the tree after adding the edge of the query.

Return an array answer of length m where answer[i] is the answer to the ith query.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2583 Kth Largest Sum In A Binary Tree

**MEDIUM**

You are given the root of a binary tree and a positive integer k.

The level sum in the tree is the sum of the values of the nodes that are on the same level.

Return the kth largest level sum in the tree (not necessarily distinct). If there are fewer than k levels in the tree, return -1.

Note that two nodes are on the same level if they have the same distance from the root.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2641 Cousins In Binary Tree Ii

**MEDIUM**

Given the root of a binary tree, replace the value of each node in the tree with the sum of all its cousins' values.

Two nodes of a binary tree are cousins if they have the same depth with different parents.

Return the root of the modified tree.

Note that the depth of a node is the number of edges in the path from the root node to it.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2673 Make Costs Of Paths Equal In A Binary Tree

MEDIUM

You are given an integer n representing the number of nodes in a perfect binary tree consisting of nodes numbered from 1 to n. The root of the tree is node 1 and each node i in the tree has two children where the left child is the node 2 * i and the right child is 2 * i + 1.

Each node in the tree also has a cost represented by a given 0-indexed integer array cost of size n where cost[i] is the cost of node i + 1. You are allowed to increment the cost of any node by 1 any number of times.

Return the minimum number of increments you need to make the cost of paths from the root to each leaf node equal.

Note:
- A perfect binary tree is a tree where each node, except the leaf nodes, has exactly 2 children.
- The cost of a path is the sum of costs of nodes in the path.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2689 Extract Kth Character From The Rope Tree

EASY

You are given the root of a binary tree and an integer k. Besides the left and right children, every node of this tree has two other properties, a string node.val containing only lowercase English letters (possibly empty) and a non-negative integer node.len. There are two types of nodes in this tree:
- Leaf: These nodes have no children, node.len = 0, and node.val is some non-empty string.
- Internal: These nodes have at least one child (also at most two children), node.len > 0, and node.val is an empty string.

The tree described above is called a Rope binary tree. Now we define S[node] recursively as follows:
- If node is some leaf node, S[node] = node.val,
- Otherwise if node is some internal node, S[node] = concat(S[node.left], S[node.right]) and S[node].length = node.len.

Return k-th character of the string S[root].

Note: If s and p are two strings, concat(s, p) is a string obtained by concatenating p to s. For example, concat("ab", "zz") = "abzz".

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2728 Count Houses In A Circular Street

EASY

You are given an object street of class Street that represents a circular street and a positive integer k which represents a maximum bound for the number of houses in that street (in other words, the number of houses is less than or equal to k). Houses' doors could be open or closed initially.

Initially, you are standing in front of a door to a house on this street. Your task is to count the number of houses in the street.

The class Street contains the following functions which may help you:
- void openDoor(): Open the door of the house you are in front of.
- void closeDoor(): Close the door of the house you are in front of.
- boolean isDoorOpen(): Returns true if the door of the current house is open and false otherwise.
- void moveRight(): Move to the right house.

- void moveLeft(): Move to the left house.

Return ans which represents the number of houses on this street.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2753 Count Houses In A Circular Street Ii

**HARD**

You are given an object street of class Street that represents a circular street and a positive integer k which represents a maximum bound for the number of houses in that street (in other words, the number of houses is less than or equal to k). Houses' doors could be open or closed initially (at least one is open).

Initially, you are standing in front of a door to a house on this street. Your task is to count the number of houses in the street.

The class Street contains the following functions which may help you:
- void closeDoor(): Close the door of the house you are in front of.
- boolean isDoorOpen(): Returns true if the door of the current house is open and false otherwise.
- void moveRight(): Move to the right house.

Note that by circular street, we mean if you number the houses from 1 to n, then the right house of housei is housei+1 for i < n, and the right house of housen is house1.

Return ans which represents the number of houses on this street.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2764 Is Array A Preorder Of Some Binary Tree

**MEDIUM**

Given a 0-indexed integer 2D array nodes, your task is to determine if the given array represents the preorder traversal of some binary tree.

For each index i, nodes[i] = [id, parentId], where id is the id of the node at the index i and parentId is the id of its parent in the tree (if the node has no parent, then parentId == -1).

Return true if the given array represents the preorder traversal of some tree, and false otherwise.

Note: the preorder traversal of a tree is a recursive way to traverse a tree in which we first visit the current node, then we do the preorder traversal for the left child, and finally, we do it for the right child.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2773 Height Of Special Binary Tree

**MEDIUM**

You are given a root, which is the root of a special binary tree with n nodes. The nodes of the special binary tree are numbered from 1 to n. Suppose the tree has k leaves in the following order: b1 < b2 < ... < bk.

The leaves of this tree have a special property! That is, for every leaf bi, the following conditions hold:
- The right child of bi is bi + 1 if i < k, and b1 otherwise.
- The left child of bi is bi - 1 if i > 1, and bk otherwise.

Return the height of the given tree.

Note: The height of a binary tree is the length of the longest path from the root to any other node.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 2791 Count Paths That Can Form A Palindrome In A Tree

HARD

You are given a tree (i.e. a connected, undirected graph that has no cycles) rooted at node 0 consisting of n nodes numbered from 0 to n - 1. The tree is represented by a 0-indexed array parent of size n, where parent[i] is the parent of node i. Since node 0 is the root, parent[0] == -1.

You are also given a string s of length n, where s[i] is the character assigned to the edge between i and parent[i]. s[0] can be ignored.

Return the number of pairs of nodes (u, v) such that u < v and the characters assigned to edges on the path from u to v can be rearranged to form a palindrome.

A string is a palindrome when it reads the same backwards as forwards.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 2795 Parallel Execution Of Promises For Individual Results Retrieval

MEDIUM

Given an array functions, return a promise promise. functions is an array of functions that return promises fnPromise. Each fnPromise can be resolved or rejected.

If fnPromise is resolved:

obj = { status: "fulfilled", value: resolved value}

If fnPromise is rejected:

obj = { status: "rejected", reason: reason of rejection (catched error message)}

The promise should resolve with an array of these objects obj. Each obj in the array should correspond to the promises in the original array function, maintaining the same order.

Try to implement it without using the built-in method Promise.allSettled().

TRY IT / Write examples, edge cases, and your own algorithm first.

## 2846 Minimum Edge Weight Equilibrium Queries In A Tree

HARD

There is an undirected tree with n nodes labeled from 0 to n - 1. You are given the integer n and a 2D integer array edges of length n - 1, where edges[i] = [ui, vi, wi] indicates that there is an edge between nodes ui and vi with weight wi in the tree.

You are also given a 2D integer array queries of length m, where queries[i] = [ai, bi]. For each query, find the minimum number of operations required to make the weight of every edge on the path from ai to bi equal. In one operation, you can choose any edge of the tree and change its weight to any value.

Note that:
- Queries are independent of each other, meaning that the tree returns to its initial state on each new query.
- The path from ai to bi is a sequence of distinct nodes starting with node ai and ending with node bi such that every two adjacent nodes in the sequence share an edge in the tree.

Return an array answer of length m where answer[i] is the answer to the ith query.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 2925 Maximum Score After Applying Operations On A Tree

MEDIUM

There is an undirected tree with n nodes labeled from 0 to n - 1, and rooted at node 0. You are given a 2D integer array edges of length n - 1, where edges[i] = [ai, bi] indicates that there is an edge between nodes ai and bi in the tree.

You are also given a 0-indexed integer array values of length n, where values[i] is the value associated with the ith node.

You start with a score of 0. In one operation, you can:
- Pick any node i.
- Add values[i] to your score.
- Set values[i] to 0.

A tree is healthy if the sum of values on the path from the root to any leaf node is different than zero.

Return the maximum score you can obtain after performing these operations on the tree any number of times so that it remains healthy.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3004 Maximum Subtree Of The Same Color

MEDIUM

You are given a 2D integer array edges representing a tree with n nodes, numbered from 0 to n - 1, rooted at node 0, where edges[i] = [ui, vi] means there is an edge between the nodes vi and ui.

You are also given a 0-indexed integer array colors of size n, where colors[i] is the color assigned to node i.

We want to find a node v such that every node in the subtree of v has the same color.

Return the size of such subtree with the maximum number of nodes possible.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3067 Count Pairs Of Connectable Servers In A Weighted Tree Network

MEDIUM

You are given an unrooted weighted tree with n vertices representing servers numbered from 0 to n - 1, an array edges where edges[i] = [ai, bi, weighti] represents a bidirectional edge between vertices ai and bi of weight weighti. You are also given an integer signalSpeed.

Two servers a and b are connectable through a server c if:
- a < b, a != c and b != c.
- The distance from c to a is divisible by signalSpeed.
- The distance from c to b is divisible by signalSpeed.
- The path from c to b and the path from c to a do not share any edges.

Return an integer array count of length n where count[i] is the number of server pairs that are connectable through the server i.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3157 Find The Level Of Tree With Minimum Sum

MEDIUM

Given the root of a binary tree root where each node has a value, return the level of the

tree that has the minimum sum of values among all the levels (in case of a tie, return the lowest level).

Note that the root of the tree is at level 1 and the level of any other node is its distance from the root + 1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3313 Find The Last Marked Nodes In Tree

**HARD**

There exists an undirected tree with n nodes numbered 0 to n - 1. You are given a 2D integer array edges of length n - 1, where edges[i] = [ui, vi] indicates that there is an edge between nodes ui and vi in the tree.

Initially, all nodes are unmarked. After every second, you mark all unmarked nodes which have at least one marked node adjacent to them.

Return an array nodes where nodes[i] is the last node to get marked in the tree, if you mark node i at time t = 0. If nodes[i] has multiple answers for any node i, you can choose any one answer.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3372 Maximize The Number Of Target Nodes After Connecting Trees I

**MEDIUM**

There exist two undirected trees with n and m nodes, with distinct labels in ranges [0, n - 1] and [0, m - 1], respectively.

You are given two 2D integer arrays edges1 and edges2 of lengths n - 1 and m - 1, respectively, where edges1[i] = [ai, bi] indicates that there is an edge between nodes ai and bi in the first tree and edges2[i] = [ui, vi] indicates that there is an edge between nodes ui and vi in the second tree. You are also given an integer k.

Node u is target to node v if the number of edges on the path from u to v is less than or equal to k. Note that a node is always target to itself.

Return an array of n integers answer, where answer[i] is the maximum possible number of nodes target to node i of the first tree if you have to connect one node from the first tree to another node in the second tree.

Note that queries are independent from each other. That is, for every query you will remove the added edge before proceeding to the next query.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3373 Maximize The Number Of Target Nodes After Connecting Trees Ii

**HARD**

There exist two undirected trees with n and m nodes, labeled from [0, n - 1] and [0, m - 1], respectively.

You are given two 2D integer arrays edges1 and edges2 of lengths n - 1 and m - 1, respectively, where edges1[i] = [ai, bi] indicates that there is an edge between nodes ai and bi in the first tree and edges2[i] = [ui, vi] indicates that there is an edge between nodes ui and vi in the second tree.

Node u is target to node v if the number of edges on the path from u to v is even. Note that a node is always target to itself.

Return an array of n integers answer, where answer[i] is the maximum possible number of nodes that are target to node i of the first tree if you had to connect one node from the first tree

to another node in the second tree.

Note that queries are independent from each other. That is, for every query you will remove the added edge before proceeding to the next query.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3481 Apply Substitutions

**MEDIUM**

You are given a replacements mapping and a text string that may contain placeholders formatted as %var%, where each var corresponds to a key in the replacements mapping. Each replacement value may itself contain one or more such placeholders. Each placeholder is replaced by the value associated with its corresponding replacement key.

Return the fully substituted text string which does not contain any placeholders.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

CHAPTER 08 / 80 EXERCISES

# Graphs & Traversal

## 01 THE IDEA

Graphs contain vertices and edges. BFS explores in layers; DFS follows paths deeply before backtracking.

## 02 SEE IT




## 03 JAVASCRIPT PATTERN

```
const seen = new Set([start]);
const queue = [start];
// visit neighbors once
```

## 04 HOW TO THINK

Draw edges and mark visited vertices. Ask whether the graph is directed, weighted or disconnected.

## 05 PERFORMANCE

BFS/DFS O(V + E) with adjacency lists.

## EXERCISE COLLECTION

Try each question before opening the matching problem number in the separate Solutions PDF. The topic classification is a study aid; some exercises combine multiple patterns.

### 0126 Word Ladder Ii

HARD

A transformation sequence from word beginWord to word endWord using a dictionary wordList is a sequence of words beginWord -> s1 -> s2 -> ... -> sk such that:
- Every adjacent pair of words differs by a single letter.
- Every si for 1 <= i <= k is in wordList. Note that beginWord does not need to be in wordList.
- sk == endWord

Given two words, beginWord and endWord, and a dictionary wordList, return all the shortest transformation sequences from beginWord to endWord, or an empty list if no such sequence exists. Each sequence should be returned as a list of the words [beginWord, s1, s2, ..., sk].

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0127 Word Ladder

HARD

A transformation sequence from word beginWord to word endWord using a dictionary wordList is a sequence of words beginWord -> s1 -> s2 -> ... -> sk such that:
- Every adjacent pair of words differs by a single letter.
- Every si for 1 <= i <= k is in wordList. Note that beginWord does not need to be in wordList.
- sk == endWord
Given two words, beginWord and endWord, and a dictionary wordList, return the number of words in the shortest transformation sequence from beginWord to endWord, or 0 if no such sequence exists.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0130 Surrounded Regions

MEDIUM

You are given an m x n matrix board containing letters 'X' and 'O', capture regions that are surrounded:
- Connect: A cell is connected to adjacent cells horizontally or vertically.
- Region: To form a region connect every 'O' cell.
- Surround: The region is surrounded with 'X' cells if you can connect the region with 'X' cells and none of the region cells are on the edge of the board.

To capture a surrounded region, replace all 'O's with 'X's in-place within the original board. You do not need to return anything.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0133 Clone Graph

MEDIUM

Given a reference of a node in a connected undirected graph.
Return a deep copy (clone) of the graph.
Each node in the graph contains a value (int) and a list (List[Node]) of its neighbors.

Test case format:
- For simplicity, each node's value is the same as the node's index (1-indexed).
For example, the first node with val == 1, the second node with val == 2, and so on.
The graph is represented in the test case using an adjacency list.

- An adjacency list is a collection of unordered lists used to represent a finite graph.
Each list describes the set of neighbors of a node in the graph.

- The given node will always be the first node with val = 1. You must return the copy of the given node as a reference to the cloned graph.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0200 Number Of Islands

MEDIUM

Given an m x n 2D binary grid grid which represents a map of '1's (land) and '0's (water), return the number of islands.

An island is surrounded by water and is formed by connecting adjacent lands horizontally or vertically. You may assume all four edges of the grid are all surrounded by water.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0207 Course Schedule

MEDIUM

There are a total of numCourses courses you have to take, labeled from 0 to numCourses - 1. You are given an array prerequisites where prerequisites[i] = [ai, bi] indicates that you must take course bi first if you want to take course ai.

For example, the pair [0, 1], indicates that to take course 0 you have to first take course 1.

Return true if you can finish all courses. Otherwise, return false.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0210 Course Schedule Ii

MEDIUM

There are a total of numCourses courses you have to take, labeled from 0 to numCourses - 1. You are given an array prerequisites where prerequisites[i] = [ai, bi] indicates that you must take course bi first if you want to take course ai.

For example, the pair [0, 1], indicates that to take course 0 you have to first take course 1.

Return the ordering of courses you should take to finish all courses. If there are many valid answers, return any of them. If it is impossible to finish all courses, return an empty array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0305 Number Of Islands Ii

HARD

You are given an empty 2D binary grid grid of size m x n. The grid represents a map where 0's represent water and 1's represent land. Initially, all the cells of grid are water cells (i.e., all the cells are 0's).

We may perform an add land operation which turns the water at position into a land. You are given an array positions where positions[i] = [ri, ci] is the position (ri, ci) at which we should operate the ith operation.

Return an array of integers answer where answer[i] is the number of islands after turning the cell (ri, ci) into a land.

An island is surrounded by water and is formed by connecting adjacent lands horizontally or vertically. You may assume all four edges of the grid are all surrounded by water.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0317 Shortest Distance From All Buildings

HARD

You are given an m x n grid grid of values 0, 1, or 2, where:
- each 0 marks an empty land that you can pass by freely,
- each 1 marks a building that you cannot pass through, and
- each 2 marks an obstacle that you cannot pass through.

You want to build a house on an empty land that reaches all buildings in the shortest total travel distance. You can only move up, down, left, and right.

Return the shortest travel distance for such a house. If it is not possible to build such a house according to the above rules, return -1.

The total travel distance is the sum of the distances between the houses of the friends and the meeting point.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 0323 Number Of Connected Components In An Undirected Graph

MEDIUM

You have a graph of n nodes. You are given an integer n and an array edges where edges[i] = [ai, bi] indicates that there is an edge between ai and bi in the graph.

Return the number of connected components in the graph.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 0332 Reconstruct Itinerary

HARD

You are given a list of airline tickets where tickets[i] = [fromi, toi] represent the departure and the arrival airports of one flight. Reconstruct the itinerary in order and return it.

All of the tickets belong to a man who departs from "JFK", thus, the itinerary must begin with "JFK". If there are multiple valid itineraries, you should return the itinerary that has the smallest lexical order when read as a single string.

For example, the itinerary ["JFK", "LGA"] has a smaller lexical order than ["JFK", "LGB"].

You may assume all tickets form at least one valid itinerary. You must use all the tickets once and only once.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 0386 Lexicographical Numbers

MEDIUM

Given an integer n, return all the numbers in the range [1, n] sorted in lexicographical order.

You must write an algorithm that runs in O(n) time.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 0417 Pacific Atlantic Water Flow

MEDIUM

There is an m x n rectangular island that borders both the Pacific Ocean and Atlantic Ocean. The Pacific Ocean touches the island's left and top edges, and the Atlantic Ocean touches the island's right and bottom edges.

The island is partitioned into a grid of square cells. You are given an m x n integer matrix heights where heights[r][c] represents the height above sea level of the cell at coordinate (r, c).

The island receives a lot of rain, and the rain water can flow to neighboring cells directly north, south, east, and west if the neighboring cell's height is less than or equal to the current cell's height. Water can flow from any cell adjacent to an ocean into the ocean.

Return a 2D list of grid coordinates result where result[i] = [ri, ci] denotes that rain water can flow from cell (ri, ci) to both the Pacific and Atlantic oceans.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 0440 K Th Smallest In Lexicographical Order

HARD

Given two integers n and k, return the kth lexicographically smallest integer in the range [1, n].

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0463 Island Perimeter

MEDIUM

You are given row x col grid representing a map where grid[i][j] = 1 represents land and grid[i][j] = 0 represents water.

Grid cells are connected horizontally/vertically (not diagonally). The grid is completely surrounded by water, and there is exactly one island (i.e., one or more connected land cells).

The island doesn't have "lakes", meaning the water inside isn't connected to the water around the island. One cell is a square with side length 1. The grid is rectangular, width and height don't exceed 100. Determine the perimeter of the island.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0488 Zuma Game

HARD

You are playing a variation of the game Zuma.

In this variation of Zuma, there is a single row of colored balls on a board, where each ball can be colored red 'R', yellow 'Y', blue 'B', green 'G', or white 'W'. You also have several colored balls in your hand.

Your goal is to clear all of the balls from the board. On each turn:
- Pick any ball from your hand and insert it in between two balls in the row or on either end of the row.
- If there is a group of three or more consecutive balls of the same color, remove the group of balls from the board.
- If this removal causes more groups of three or more of the same color to form, then continue removing each group until there are none left.
- If there are no more balls on the board, then you win the game.
- Repeat this process until you either win or do not have any more balls in your hand.

Given a string board, representing the row of balls on the board, and a string hand, representing the balls in your hand, return the minimum number of balls you have to insert to clear all the balls from the board. If you cannot clear all the balls from the board using the balls in your hand, return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0529 Minesweeper

MEDIUM

Let's play the minesweeper game (Wikipedia, online game)!

You are given an m x n char matrix board representing the game board where:
- 'M' represents an unrevealed mine,
- 'E' represents an unrevealed empty square,
- 'B' represents a revealed blank square that has no adjacent mines (i.e., above, below, left, right, and all 4 diagonals),
- digit ('1' to '8') represents how many mines are adjacent to this revealed square, and

- 'X' represents a revealed mine.

You are also given an integer array click where click = [clickr, clickc] represents the next click position among all the unrevealed squares ('M' or 'E').

Return the board after revealing this position according to the following rules:
blank 'B' and all of its adjacent unrevealed squares should be revealed recursively.
digit ('1' to '8') representing the number of adjacent mines.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0546 Remove Boxes

**HARD**

You are given several boxes with different colors represented by different positive numbers.

You may experience several rounds to remove boxes until there is no box left. Each time you can choose some continuous boxes with the same color (i.e., composed of k boxes, k >= 1), remove them and get k * k points.

Return the maximum points you can get.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0547 Number Of Provinces

**MEDIUM**

There are n cities. Some of them are connected, while some are not. If city a is connected directly with city b, and city b is connected directly with city c, then city a is connected indirectly with city c.

A province is a group of directly or indirectly connected cities and no other cities outside of the group.

You are given an n x n matrix isConnected where isConnected[i][j] = 1 if the ith city and the jth city are directly connected, and isConnected[i][j] = 0 otherwise.

Return the total number of provinces.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0630 Course Schedule Iii

**HARD**

There are n different online courses numbered from 1 to n. You are given an array courses where courses[i] = [durationi, lastDayi] indicate that the ith course should be taken continuously for durationi days and must be finished before or on lastDayi.

You will start on the 1st day and you cannot take two or more courses simultaneously.

Return the maximum number of courses that you can take.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0690 Employee Importance

**MEDIUM**

You have a data structure of employee information, including the employee's unique ID, importance value, and direct subordinates' IDs.

You are given an array of employees employees where:
- employees[i].id is the ID of the ith employee.
- employees[i].importance is the importance value of the ith employee.

- employees[i].subordinates is a list of the IDs of the direct subordinates of the ith employee.

Given an integer id that represents an employee's ID, return the total importance value of this employee and all their direct and indirect subordinates.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0694 Number Of Distinct Islands

**MEDIUM**

You are given an m x n binary matrix grid. An island is a group of 1's (representing land) connected 4-directionally (horizontal or vertical.) You may assume all four edges of the grid are surrounded by water.

An island is considered to be the same as another if and only if one island can be translated (and not rotated or reflected) to equal the other.

Return the number of distinct islands.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0695 Max Area Of Island

**MEDIUM**

You are given an m x n binary matrix grid. An island is a group of 1's (representing land) connected 4-directionally (horizontal or vertical.) You may assume all four edges of the grid are surrounded by water.

The area of an island is the number of cells with a value 1 in the island.

Return the maximum area of an island in grid. If there is no island, return 0.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0711 Number Of Distinct Islands Ii

**HARD**

You are given an m x n binary matrix grid. An island is a group of 1's (representing land) connected 4-directionally (horizontal or vertical.) You may assume all four edges of the grid are surrounded by water.

An island is considered to be the same as another if they have the same shape, or have the same shape after rotation (90, 180, or 270 degrees only) or reflection (left/right direction or up/down direction).

Return the number of distinct islands.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0733 Flood Fill

**EASY**

An image is represented by an m x n integer grid image where image[i][j] represents the pixel value of the image.

You are also given three integers sr, sc, and newColor. You should perform a flood fill on the image starting from the pixel image[sr][sc].

To perform a flood fill, consider the starting pixel, plus any pixels connected 4-directionally to the starting pixel of the same color as the starting pixel, plus any pixels connected 4-directionally to those pixels (also with the same color), and so on. Replace the color of all of the aforementioned pixels with newColor.

Return the modified image after performing the flood fill.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0743 Network Delay Time

**MEDIUM**

You are given a network of n nodes, labeled from 1 to n. You are also given times, a list of travel times as directed edges times[i] = (ui, vi, wi), where ui is the source node, vi is the target node, and wi is the time it takes for a signal to travel from source to target.

We will send a signal from a given node k. Return the minimum time it takes for all the n nodes to receive the signal. If it is impossible for all the n nodes to receive the signal, return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0785 Is Graph Bipartite

**MEDIUM**

There is an undirected graph with n nodes, where each node is numbered between 0 and n - 1. You are given a 2D array graph, where graph[u] is an array of nodes that node u is adjacent to. More formally, for each v in graph[u], there is an undirected edge between node u and node v. The graph has the following properties:
- There are no self-edges (graph[u] does not contain u).
- There are no parallel edges (graph[u] does not contain duplicate values).
- If v is in graph[u], then u is in graph[v] (the graph is undirected).
- The graph may not be connected, meaning there may be two nodes u and v such that there is no path between them.

A graph is bipartite if the nodes can be partitioned into two independent sets A and B such that every edge in the graph connects a node in set A and a node in set B.

Return true if and only if it is bipartite.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0787 Cheapest Flights Within K Stops

**MEDIUM**

There are n cities connected by some number of flights. You are given an array flights where flights[i] = [fromi, toi, pricei] indicates that there is a flight from city fromi to city toi with cost pricei.

You are also given three integers src, dst, and k, return the cheapest price from src to dst with at most k stops. If there is no such route, return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0797 All Paths From Source To Target

**MEDIUM**

Given a directed acyclic graph (DAG) of n nodes labeled from 0 to n - 1, find all possible paths from node 0 to node n - 1 and return them in any order.

The graph is given as follows: graph[i] is a list of all nodes you can visit from node i (i.e., there is a directed edge from node i to node graph[i][j]).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0827 Making A Large Island

HARD

You are given an n x n binary matrix grid. You are allowed to change at most one 0 to be 1.

Return the size of the largest island in grid after applying this operation.

An island is a 4-directionally connected group of 1s.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0847 Shortest Path Visiting All Nodes

HARD

You have an undirected, connected graph of n nodes labeled from 0 to n - 1. You are given an array graph where graph[i] is a list of all the nodes connected with node i by an edge.

Return the length of the shortest path that visits every node. You may start and stop at any node, you may revisit nodes multiple times, and you may reuse edges.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0851 Loud And Rich

MEDIUM

There is a group of n people labeled from 0 to n - 1 where each person has a different amount of money and a different level of quietness.

You are given an array richer where richer[i] = [ai, bi] indicates that ai has more money than bi and an integer array quiet where quiet[i] is the quietness of the ith person. All the given data in richer are logically correct (i.e., the data will not lead you to a situation where x is richer than y and y is richer than x at the same time).

Return an integer array answer where answer[x] = y if y is the least quiet person (that is, the person y with the smallest value of quiet[y]) among all people who definitely have equal to or more money than the person x.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0864 Shortest Path To Get All Keys

HARD

You are given an m x n grid grid where:
- '.' is an empty cell.
- '#' is a wall.
- '@' is the starting point.
- Lowercase letters represent keys.
- Uppercase letters represent locks.

You start at the starting point and one move consists of walking one space in one of the four cardinal directions. You cannot walk outside the grid, or walk into a wall.

If you walk over a key, you can pick it up and you cannot walk over a lock unless you have its corresponding key.

For some 1 <= k <= 6, there is exactly one lowercase and one uppercase letter of the first k letters of the English alphabet in the grid. This means that there is exactly one key for each lock, and one lock for each key; and also that the letters used to represent the keys and locks were chosen in the same order as the English alphabet.

Return the lowest number of moves to acquire all keys. If it is impossible, return -1.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 0882 Reachable Nodes In Subdivided Graph

HARD

You are given an undirected graph (the "original graph") with n nodes labeled from 0 to n - 1. You decide to subdivide each edge in the graph into a chain of nodes, with the number of new nodes varying between each edge.

The graph is given as a 2D array of edges where edges[i] = [ui, vi, cnti] indicates that there is an edge between nodes ui and vi in the original graph, and cnti is the total number of new nodes that you will subdivide the edge into. Note that cnti == 0 means you will not subdivide the edge.

To subdivide the edge [ui, vi], replace it with (cnti + 1) new edges and cnti new nodes. The new nodes are x1, x2, ..., xcnti, and the new edges are [ui, x1], [x1, x2], [x2, x3], ..., [xcnti-1, xcnti], [xcnti, vi].

In this new graph, you want to know how many nodes are reachable from the node 0, where a node is reachable if the distance is maxMoves or less.

Given the original graph and maxMoves, return the number of nodes that are reachable from node 0 in the new graph.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 1059 All Paths From Source Lead To Destination

MEDIUM

Given the edges of a directed graph where edges[i] = [ai, bi] indicates there is an edge between nodes ai and bi, and two nodes source and destination of this graph, determine whether or not all paths starting from source eventually, end at destination, that is:
- At least one path exists from the source node to the destination node
- If a path exists from the source node to a node with no outgoing edges, then that node is equal to destination.
- The number of possible paths from source to destination is a finite number.

Return true if and only if all roads from source lead to destination.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 1061 Lexicographically Smallest Equivalent String

MEDIUM

You are given two strings of the same length s1 and s2 and a string baseStr.

We say s1[i] and s2[i] are equivalent characters.
- For example, if s1 = "abc" and s2 = "cde", then we have 'a' == 'c', 'b' == 'd', and 'c' == 'e'.

Equivalent characters follow the usual rules of any equivalence relation:
- Reflexivity: 'a' == 'a'.
- Symmetry: 'a' == 'b' implies 'b' == 'a'.
- Transitivity: 'a' == 'b' and 'b' == 'c' implies 'a' == 'c'.

For example, given the equivalency information from s1 = "abc" and s2 = "cde", "acd" and "aab" are equivalent strings of baseStr = "eed", and "aab" is the lexicographically smallest equivalent string of baseStr.

Return the lexicographically smallest equivalent string of baseStr by using the equivalency information from s1 and s2.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 1066 Campus Bikes Ii

MEDIUM

On a campus represented as a 2D grid, there are n workers and m bikes, with n <= m. Each worker and bike is a 2D coordinate on this grid.

We assign one unique bike to each worker so that the sum of the Manhattan distances between each worker and their assigned bike is minimized.

Return the minimum possible sum of Manhattan distances between each worker and their assigned bike.

The Manhattan distance between two points p1 and p2 is Manhattan(p1, p2) = |p1.x - p2.x| + |p1.y - p2.y|.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1091 Shortest Path In Binary Matrix

MEDIUM

Given an n x n binary matrix grid, return the length of the shortest clear path in the matrix. If there is no clear path, return -1.

A clear path in a binary matrix is a path from the top-left cell (i.e., (0, 0)) to the bottom-right cell (i.e., (n - 1, n - 1)) such that:
- All the visited cells of the path are 0.
- All the adjacent cells of the path are 8-directionally connected (i.e., they are different and they share an edge or a corner).

The length of a clear path is the number of visited cells of this path.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1109 Corporate Flight Bookings

MEDIUM

There are n flights that are labeled from 1 to n.

You are given an array of flight bookings bookings, where bookings[i] = [firsti, lasti, seatsi] represents a booking for flights firsti through lasti (inclusive) with seatsi seats reserved for each flight in the range.

Return an array answer of length n, where answer[i] is the total number of seats reserved for flight i.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1129 Shortest Path With Alternating Colors

MEDIUM

You are given an integer n, the number of nodes in a directed graph where the nodes are labeled from 0 to n - 1. Each edge is red or blue in this graph, and there could be self-edges and parallel edges.

You are given two arrays redEdges and blueEdges where:
- redEdges[i] = [ai, bi] indicates that there is a directed red edge from node ai to node bi in the graph, and
- blueEdges[j] = [uj, vj] indicates that there is a directed blue edge from node uj to node vj in the graph.

Return an array answer of length n, where each answer[x] is the length of the shortest

path from node 0 to node x such that the edge colors alternate along the path, or -1 if such a path does not exist.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1197 Minimum Knight Moves

**MEDIUM**

In an infinite chess board with coordinates from -infinity to +infinity, you have a knight at square [0, 0].

A knight has 8 possible moves it can make, as illustrated below. Each move is two squares in a cardinal direction, then one square in an orthogonal direction.

Return the minimum number of steps needed to move the knight to the square [x, y]. It is guaranteed the answer exists.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1254 Number Of Closed Islands

**MEDIUM**

Given a 2D grid consists of 0s (land) and 1s (water). An island is a maximal 4-directionally connected group of 0s and a closed island is an island totally (all left, top, right, bottom) surrounded by 1s.

Return the number of closed islands.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1415 The K Th Lexicographical String Of All Happy Strings Of Length N

**MEDIUM**

A happy string is a string that:
- consists only of letters of the set ['a', 'b', 'c'].
- s[i] != s[i + 1] for all values of i from 1 to s.length - 1 (string is 1-indexed).

For example, strings "abc", "ac", "b" and "abcbabcbcb" are all happy strings and strings "aa", "baa" and "ababbc" are not happy strings.

Given two integers n and k, consider a list of all happy strings of length n sorted in lexicographical order.

Return the kth string of this list or return an empty string if there are less than k happy strings of length n.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1462 Course Schedule Iv

**MEDIUM**

There are a total of numCourses courses you have to take, labeled from 0 to numCourses - 1. You are given an array prerequisites where prerequisites[i] = [ai, bi] indicates that you must take course ai first if you want to take course bi.

- For example, the pair [0, 1] indicates that you have to take course 0 before you can take course 1.

Prerequisites can also be indirect. If course a is a prerequisite of course b, and course b is a prerequisite of course c, then course a is a prerequisite of course c.

You are also given an array queries where queries[j] = [uj, vj]. For the jth query, you

should answer whether course uj is a prerequisite of course vj or not.

Return a boolean array answer, where answer[j] is the answer to the jth query.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1466 Reorder Routes To Make All Paths Lead To The City Zero

**MEDIUM**

There are n cities numbered from 0 to n - 1 and n - 1 roads such that there is only one way to travel between two different cities (this network form a tree). Last year, The ministry of transport decided to orient the roads in one direction because they are too narrow.

Roads are represented by connections where connections[i] = [ai, bi] represents a road from city ai to city bi.

This year, there will be a big event in the capital (city 0), and many people want to travel to this city.

Your task consists of reorienting some roads such that each city can visit the city 0. Return the minimum number of edges changed.

It's guaranteed that each city can reach city 0 after reorder.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1548 The Most Similar Path In A Graph

**HARD**

We have n cities and m bi-directional roads where roads[i] = [ai, bi] connects city ai with city bi. Each city has a name consisting of exactly three upper-case English letters given in the string array names. Starting at any city x, you can reach any city y where y != x (i.e., the cities and the roads are forming an undirected connected graph).

You will be given a string array targetPath. You should find a path in the graph of the same length and with the minimum edit distance to targetPath.

You need to return the order of the nodes in the path with the minimum edit distance. The path should be of the same length of targetPath and should be valid (i.e., there should be a direct road between ans[i] and ans[i + 1]). If there are multiple answers return any one of them.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1568 Minimum Number Of Days To Disconnect Island

**HARD**

You are given an m x n binary grid grid where 1 represents land and 0 represents water. An island is a maximal 4-directionally (horizontal or vertical) connected group of 1's.

The grid is said to be connected if we have exactly one island, otherwise is said disconnected.

In one day, we are allowed to change any single land cell (1) into a water cell (0).

Return the minimum number of days to disconnect the grid.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1579 Remove Max Number Of Edges To Keep Graph Fully Traversable

**HARD**

Alice and Bob have an undirected graph of n nodes and three types of edges:
- Type 1: Can be traversed by Alice only.

- Type 2: Can be traversed by Bob only.
- Type 3: Can be traversed by both Alice and Bob.

Given an array edges where edges[i] = [typei, ui, vi] represents a bidirectional edge of type typei between nodes ui and vi, find the maximum number of edges you can remove so that after removing the edges, the graph can still be fully traversed by both Alice and Bob. The graph is fully traversed by Alice and Bob if starting from any node, they can reach all other nodes.

Return the maximum number of edges you can remove, or return -1 if Alice and Bob cannot fully traverse the graph.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1595 Minimum Cost To Connect Two Groups Of Points

**HARD**

You are given two groups of points where the first group has size1 points, the second group has size2 points, and size1 >= size2.

The cost of the connection between any two points are given in an size1 x size2 matrix where cost[i][j] is the cost of connecting point i of the first group and point j of the second group. The groups are connected if each point in both groups is connected to one or more points in the opposite group. In other words, each point in the first group must be connected to at least one point in the second group, and each point in the second group must be connected to at least one point in the first group.

Return the minimum cost it takes to connect the two groups.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1625 Lexicographically Smallest String After Applying Operations

**MEDIUM**

You are given a string s of even length consisting of digits from 0 to 9, and two integers a and b.

You can apply either of the following two operations any number of times and in any order on s:
- Add a to all odd indices of s (0-indexed). Digits post 9 are cycled back to 0. For example, if s = "3456" and a = 5, s becomes "3951".
- Rotate s to the right by b positions. For example, if s = "3456" and b = 1, s becomes "6345".

Return the lexicographically smallest string you can obtain by applying the above operations any number of times on s.

A string a is lexicographically smaller than a string b (of the same length) if in the first position where a and b differ, string a has a letter that appears earlier in the alphabet than the corresponding letter in b. For example, "0158" is lexicographically smaller than "0190" because the first position they differ is at the third letter, and '5' comes before '9'.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1627 Graph Connectivity With Threshold

**HARD**

We have n cities labeled from 1 to n. Two different cities with labels x and y are directly connected by a bidirectional road if and only if x and y share a common divisor strictly greater than some threshold. More formally, cities with labels x and y have a road between them if there exists an integer z such that all of the following are true:
- x % z == 0,
- y % z == 0, and
- z > threshold.

Given the two integers, n and threshold, and an array of queries, you must determine for each queries[i] = [ai, bi] if cities ai and bi are connected directly or indirectly. (i.e. there is some path between them).

Return an array answer, where answer.length == queries.length and answer[i] is true if for the ith query, there is a path between ai and bi, or answer[i] is false if there is no path.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1718 Construct The Lexicographically Largest Valid Sequence

**MEDIUM**

Given an integer n, find a sequence that satisfies all of the following:
- The integer 1 occurs once in the sequence.
- Each integer between 2 and n occurs twice in the sequence.
- For every integer i between 2 and n, the distance between the two occurrences of i is exactly i.

The distance between two numbers on the sequence, a[i] and a[j], is the absolute difference of their indices, |j - i|.

Return the lexicographically largest sequence. It is guaranteed that under the given constraints, there is always a solution.

A sequence a is lexicographically larger than a sequence b (of the same length) if in the first position where a and b differ, sequence a has a number greater than the corresponding number in b. For example, [0,1,9,0] is lexicographically larger than [0,1,5,6] because the first position they differ is at the third number, and 9 is greater than 5.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1730 Shortest Path To Get Food

**MEDIUM**

You are starving and you want to eat food as quickly as possible. You want to find the shortest path to arrive at any food cell.

You are given an m x n character matrix, grid, of these different types of cells:
- '*' is your location. There is exactly one '*' cell.
- '#' is a food cell. There may be multiple food cells.
- 'O' is free space, and you can travel through these cells.
- 'X' is an obstacle, and you cannot travel through these cells.

You can travel to any adjacent cell north, east, south, or west of your current location if there is not an obstacle.

Return the length of the shortest path for you to reach any food cell. If there is no path for you to reach food, return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1761 Minimum Degree Of A Connected Trio In A Graph

**HARD**

You are given an undirected graph. You are given an integer n which is the number of nodes in the graph and an array edges, where each edges[i] = [ui, vi] indicates that there is an undirected edge between ui and vi.

A connected trio is a set of three nodes where there is an edge between every pair of them.

The degree of a connected trio is the number of edges where one endpoint is in the trio, and the other is not.

Return the minimum degree of a connected trio in the graph, or -1 if the graph has no connected trios.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1778 Shortest Path In A Hidden Grid

**MEDIUM**

This is an interactive problem.

There is a robot in a hidden grid, and you are trying to get it from its starting cell to the target cell in this grid. The grid is of size m x n, and each cell in the grid is either empty or blocked. It is guaranteed that the starting cell and the target cell are different, and neither of them is blocked.

You want to find the minimum distance to the target cell. However, you do not know the grid's dimensions, the starting cell, nor the target cell. You are only allowed to ask queries to the GridMaster object.

The GridMaster class has the following functions:
- boolean canMove(char direction) Returns true if the robot can move in that direction. Otherwise, it returns false.
- void move(char direction) Moves the robot in that direction. If this move would move the robot to a blocked cell or off the grid, the move will be ignored, and the robot will remain in the same position.
- boolean isTarget() Returns true if the robot is currently on the target cell. Otherwise, it returns false.

Note that direction in the above functions should be a character from {'U','D','L','R'}, representing the directions up, down, left, and right, respectively.

Return the minimum distance between the robot's initial starting cell and the target cell. If there is no valid path between the cells, return -1.

Custom testing:

The test input is read as a 2D matrix grid of size m x n where:
- grid[i][j] == -1 indicates that the robot is in cell (i, j) (the starting cell).
- grid[i][j] == 0 indicates that the cell (i, j) is blocked.
- grid[i][j] == 1 indicates that the cell (i, j) is empty.
- grid[i][j] == 2 indicates that the cell (i, j) is the target cell.

There is exactly one -1 and 2 in grid. Remember that you will not have this information in your code.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1791 Find Center Of Star Graph

**EASY**

There is an undirected star graph consisting of n nodes labeled from 1 to n. A star graph is a graph where there is one center node and exactly n - 1 edges that connect the center node with every other node.

You are given a 2D integer array edges where each edges[i] = [ui, vi] indicates that there is an edge between the nodes ui and vi. Return the center of the given star graph.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1857 Largest Color Value In A Directed Graph

HARD

There is a directed graph of n colored nodes and m edges. The nodes are numbered from 0 to n - 1.

You are given a string colors where colors[i] is a lowercase English letter representing the color of the ith node in this graph (0-indexed). You are also given a 2D array edges where edges[j] = [aj, bj] indicates that there is a directed edge from node aj to node bj.

A valid path in the graph is a sequence of nodes x1 -> x2 -> x3 -> ... -> xk such that there is a directed edge from xi to xi+1 for every 1 <= i < k. The color value of the path is the number of nodes that are colored the most frequently occurring color along that path.

Return the largest color value of any valid path in the given graph, or -1 if the graph contains a cycle.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1900 The Earliest And Latest Rounds Where Players Compete

HARD

There is a tournament where n players are participating. The players are standing in a single row and are numbered from 1 to n based on their initial standing position (player 1 is the first player in the row, player 2 is the second player in the row, etc.).

The tournament consists of multiple rounds (starting from round number 1). In each round, the ith player from the front of the row competes against the ith player from the end of the row, and the winner advances to the next round. When the number of players is odd for the current round, the player in the middle automatically advances to the next round.
- For example, if the row consists of players 1, 2, 4, 6, 7
- Player 1 competes against player 7.
- Player 2 competes against player 6.
- Player 4 automatically advances to the next round.

After each round is over, the winners are lined back up in the row based on the original ordering assigned to them initially (ascending order).

The players numbered firstPlayer and secondPlayer are the best in the tournament. They can win against any other player before they compete against each other. If any two other players compete against each other, either of them might win, and thus you may choose the outcome of this round.

Given the integers n, firstPlayer, and secondPlayer, return an integer array containing two values, the earliest possible round number and the latest possible round number in which these two players will compete against each other, respectively.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1905 Count Sub Islands

MEDIUM

You are given two m x n binary matrices grid1 and grid2 containing only 0's (representing water) and 1's (representing land). An island is a group of 1's connected 4-directionally (horizontal or vertical). Any cells outside of the grid are considered water cells.

An island in grid2 is considered a sub-island if there is an island in grid1 that contains all the cells that make up this island in grid2.

Return the number of islands in grid2 that are considered sub-islands.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1971 Find If Path Exists In Graph

EASY

There is a bi-directional graph with n vertices, where each vertex is labeled from 0 to n - 1 (inclusive). The edges in the graph are represented as a 2D integer array edges, where each edges[i] = [ui, vi] denotes a bi-directional edge between vertex ui and vertex vi. Every vertex pair is connected by at most one edge, and no vertex has an edge to itself.

You want to determine if there is a valid path that exists from vertex source to vertex destination.

Given edges and the integers n, source, and destination, return true if there is a valid path from source to destination, or false otherwise.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2065 Maximum Path Quality Of A Graph

HARD

There is an undirected graph with n nodes numbered from 0 to n - 1 (inclusive). You are given a 0-indexed integer array values where values[i] is the value of the ith node. You are also given a 0-indexed 2D integer array edges, where each edges[j] = [uj, vj, timej] indicates that there is an undirected edge between the nodes uj and vj, and it takes timej seconds to travel between the two nodes. Finally, you are given an integer maxTime.

A valid path in the graph is any path that starts at node 0, ends at node 0, and takes at most maxTime seconds to complete. You may visit the same node multiple times. The quality of a valid path is the sum of the values of the unique nodes visited in the path (each node's value is added at most once to the sum).

Return the maximum quality of a valid path.

Note: There are at most four edges connected to each node.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2101 Detonate The Maximum Bombs

MEDIUM

You are given a list of bombs. The range of a bomb is defined as the area where its effect can be felt. This area is in the shape of a circle with the center as the location of the bomb.

The bombs are represented by a 0-indexed 2D integer array bombs where bombs[i] = [xi, yi, ri]. xi and yi denote the X-coordinate and Y-coordinate of the location of the ith bomb, whereas ri denotes the radius of its range.

You may choose to detonate a single bomb. When a bomb is detonated, it will detonate all bombs that lie in its range. These bombs will further detonate the bombs that lie in their ranges.

Given the list of bombs, return the maximum number of bombs that can be detonated if you are allowed to detonate only one bomb.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2127 Maximum Employees To Be Invited To A Meeting

HARD

A company is organizing a meeting and has a list of n employees, waiting to be invited. They have arranged for a large circular table, capable of seating any number of employees.

The employees are numbered from 0 to n - 1. Each employee has a favorite person and they will attend the meeting only if they can sit next to their favorite person at the table. The favorite person of an employee is not themself.

Given a 0-indexed integer array favorite, where favorite[i] denotes the favorite person of the ith employee, return the maximum number of employees that can be invited to the meeting.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2192 All Ancestors Of A Node In A Directed Acyclic Graph

**MEDIUM**

You are given a positive integer n representing the number of nodes of a Directed Acyclic Graph (DAG). The nodes are numbered from 0 to n - 1 (inclusive).

You are also given a 2D integer array edges, where edges[i] = [fromi, toi] denotes that there is a unidirectional edge from fromi to toi in the graph.

Return a list answer, where answer[i] is the list of ancestors of the ith node, sorted in ascending order.

A node u is an ancestor of another node v if u can reach v via a set of edges.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2203 Minimum Weighted Subgraph With The Required Paths

**HARD**

You are given an integer n denoting the number of nodes of a weighted directed graph. The nodes are numbered from 0 to n - 1.

You are also given a 2D integer array edges where edges[i] = [fromi, toi, weighti] denotes that there exists a directed edge from fromi to toi with weight weighti.

Lastly, you are given three distinct integers src1, src2, and dest denoting three distinct nodes of the graph.

Return the minimum weight of a subgraph of the graph such that it is possible to reach dest from both src1 and src2 via a set of edges of this subgraph. In case such a subgraph does not exist, return -1.

A subgraph is a graph whose vertices and edges are subsets of the original graph. The weight of a subgraph is the sum of weights of its constituent edges.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2204 Distance To A Cycle In Undirected Graph

**HARD**

You are given a positive integer n representing the number of nodes in a connected undirected graph containing exactly one cycle. The nodes are numbered from 0 to n - 1 (inclusive).

You are also given a 2D integer array edges, where edges[i] = [node1i, node2i] denotes that there is a bidirectional edge connecting node1i and node2i in the graph.

The distance between two nodes a and b is defined to be the minimum number of edges that are needed to go from a to b.

Return an integer array answer of size n, where answer[i] is the minimum distance between the ith node and any node in the cycle.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2307 Check For Contradictions In Equations

HARD

You are given a 2D array of strings equations and an array of real numbers values, where equations[i] = [Ai, Bi] and values[i] means that Ai / Bi = values[i].

Determine if there exists a contradiction in the equations. Return true if there is a contradiction, or false otherwise.

Note:
- When checking if two numbers are equal, check that their absolute difference is less than 10-5.
- The testcases are generated such that there are no cases targeting precision, i.e. using double is enough to solve the problem.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2316 Count Unreachable Pairs Of Nodes In An Undirected Graph

MEDIUM

You are given an integer n. There is an undirected graph with n nodes, numbered from 0 to n - 1. You are given a 2D integer array edges where edges[i] = [ai, bi] denotes that there exists an undirected edge connecting nodes ai and bi.

Return the number of pairs of different nodes that are unreachable from each other.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2360 Longest Cycle In A Graph

HARD

You are given a directed graph of n nodes numbered from 0 to n - 1, where each node has at most one outgoing edge.

The graph is represented with a given 0-indexed array edges of size n, indicating that there is a directed edge from node i to node edges[i]. If there is no outgoing edge from node i, then edges[i] == -1.

Return the length of the longest cycle in the graph. If no cycle exists, return -1.

A cycle is a path that starts and ends at the same node.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2434 Using A Robot To Print The Lexicographically Smallest String

MEDIUM

You are given a string s and a robot that currently holds an empty string t. Apply one of the following operations until s and t are both empty:
- Remove the first character of a string s and give it to the robot. The robot will append this character to the string t.
- Remove the last character of a string t and give it to the robot. The robot will write this character on paper.

Return the lexicographically smallest string that can be written on the paper.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2477 Minimum Fuel Cost To Report To The Capital

MEDIUM

There is a tree (i.e., a connected, undirected graph with no cycles) structure country network consisting of n cities numbered from 0 to n - 1 and exactly n - 1 roads. The capital city is city 0. You are given a 2D integer array roads where roads[i] = [ai, bi] denotes that there exists a bidirectional road connecting cities ai and bi.

There is a meeting for the representatives of each city. The meeting is in the capital city.

There is a car in each city. You are given an integer seats that indicates the number of seats in each car.

A representative can use the car in their city to travel or change the car and ride with another representative. The cost of traveling between two cities is one liter of fuel.

Return the minimum number of liters of fuel to reach the capital city.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2697 Lexicographically Smallest Palindrome

EASY

You are given a string s consisting of lowercase English letters, and you are allowed to perform operations on it. In one operation, you can replace a character in s with another lowercase English letter.

Your task is to make s a palindrome with the minimum number of operations possible. If there are multiple palindromes that can be made using the minimum number of operations, make the lexicographically smallest one.

A string a is lexicographically smaller than a string b (of the same length) if in the first position where a and b differ, string a has a letter that appears earlier in the alphabet than the corresponding letter in b.

Return the resulting palindrome string.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2714 Find Shortest Path With K Hops

HARD

You are given a positive integer n which is the number of nodes of a 0-indexed undirected weighted connected graph and a 0-indexed 2D array edges where edges[i] = [ui, vi, wi] indicates that there is an edge between nodes ui and vi with weight wi.

You are also given two nodes s and d, and a positive integer k, your task is to find the shortest path from s to d, but you can hop over at most k edges. In other words, make the weight of at most k edges 0 and then find the shortest path from s to d.

Return the length of the shortest path from s to d with the given condition.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2858 Minimum Edge Reversals So Every Node Is Reachable

HARD

There is a simple directed graph with n nodes labeled from 0 to n - 1. The graph would form a tree if its edges were bi-directional.

You are given an integer n and a 2D integer array edges, where edges[i] = [ui, vi] represents a directed edge going from node ui to node vi.

An edge reversal changes the direction of an edge, i.e., a directed edge going from node ui to node vi becomes a directed edge going from node vi to node ui.

For every node i in the range [0, n - 1], your task is to independently calculate the minimum number of edge reversals required so it is possible to reach any other node starting from node i through a sequence of directed edges.

Return an integer array answer, where answer[i] is the minimum number of edge reversals required so it is possible to reach any other node starting from node i through a sequence of directed edges.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2904 Shortest And Lexicographically Smallest Beautiful String

**MEDIUM**

You are given a binary string s and a positive integer k.

A substring of s is beautiful if the number of 1's in it is exactly k.

Let len be the length of the shortest beautiful substring.

Return the lexicographically smallest beautiful substring of string s with length equal to len. If s doesn't contain a beautiful substring, return an empty string.

A string a is lexicographically larger than a string b (of the same length) if in the first position where a and b differ, a has a character strictly larger than the corresponding character in b.
- For example, "abcd" is lexicographically larger than "abcc" because the first position they differ is at the fourth character, and d is greater than c.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2948 Make Lexicographically Smallest Array By Swapping Elements

**MEDIUM**

You are given a 0-indexed array of positive integers nums and a positive integer limit.

In one operation, you can choose any two indices i and j and swap nums[i] and nums[j] if |nums[i] - nums[j]| <= limit.

Return the lexicographically smallest array that can be obtained by performing the operation any number of times.

An array a is lexicographically smaller than an array b if in the first position where a and b differ, array a has an element that is less than the corresponding element in b. For example, the array [2,10,3] is lexicographically smaller than the array [10,2,3] because they differ at index 0 and 2 < 10.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3108 Minimum Cost Walk In Weighted Graph

**HARD**

There is an undirected weighted graph with n vertices labeled from 0 to n - 1.

You are given the integer n and an array edges, where edges[i] = [ui, vi, wi] indicates that there is an edge between vertices ui and vi with a weight of wi.

A walk on a graph is a sequence of vertices and edges. The walk starts and ends with a vertex, and each edge connects the vertex that comes before it and the vertex that comes after it. It's important to note that a walk may visit the same edge or vertex more than once.

The cost of a walk starting at node u and ending at node v is defined as the bitwise AND of the

weights of the edges traversed during the walk. In other words, if the sequence of edge weights encountered during the walk is w0, w1, w2, ..., wk, then the cost is calculated as w0 & w1 & w2 & ... & wk, where & denotes the bitwise AND operator.

You are also given a 2D array query, where query[i] = [si, ti]. For each query, you need to find the minimum cost of the walk starting at vertex si and ending at vertex ti. If there exists no such walk, the answer is -1.

Return the array answer, where answer[i] denotes the minimum cost of a walk for query i.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3170 Lexicographically Minimum String After Removing Stars

**MEDIUM**

You are given a string s. It may contain any number of '*' characters. Your task is to remove all '*' characters.

While there is a '*', do the following operation:
- Delete the leftmost '*' and the smallest non-'*' character to its left. If there are several smallest characters, you can delete any of them.

Return the lexicographically smallest resulting string after removing all '*' characters.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3403 Find The Lexicographically Largest String From The Box I

**MEDIUM**

You are given a string word, and an integer numFriends.

Alice is organizing a game for her numFriends friends. There are multiple rounds in the game, where in each round:
- word is split into numFriends non-empty strings, such that no previous round has had the exact same split.
- All the split words are put into a box.

Find the lexicographically largest string from the box after all the rounds are finished.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3406 Find The Lexicographically Largest String From The Box Ii

**HARD**

You are given a string word, and an integer numFriends.

Alice is organizing a game for her numFriends friends. There are multiple rounds in the game, where in each round:
- word is split into numFriends non-empty strings, such that no previous round has had the exact same split.
- All the split words are put into a box.

Find the lexicographically largest string from the box after all the rounds are finished.

A string a is lexicographically smaller than a string b if in the first position where a and b differ, string a has a letter that appears earlier in the alphabet than the corresponding letter in b.

If the first min(a.length, b.length) characters do not differ, then the shorter string is the lexicographically smaller one.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

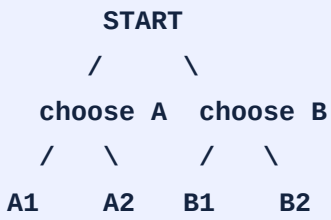
CHAPTER 09 / 53 EXERCISES

# Recursion & Backtracking

## 01 THE IDEA

Recursion solves a task through smaller instances. Backtracking explores choices, undoes them, and tries alternatives.

## 02 SEE IT




## 03 JAVASCRIPT PATTERN

```
function choose(path) {
 if (complete(path)) return;
 for (const option of options) {
  // choose, recurse, undo
 }
}
```

## 04 HOW TO THINK

Always identify the base case, the state passed to the next call, and what must be undone.

## 05 PERFORMANCE

Often exponential search; recursion space depends on depth.

## EXERCISE COLLECTION

Try each question before opening the matching problem number in the separate Solutions PDF. The topic classification is a study aid; some exercises combine multiple patterns.

### 0017 Letter Combinations Of A Phone Number

MEDIUM

Given a string containing digits from 2-9 inclusive, return all possible letter combinations that the number could represent. Return the answer in any order.

A mapping of digit to letters (just like on the telephone buttons) is

given below. Note that 1 does not map to any letters.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0031 Next Permutation

**MEDIUM**

Implement next permutation, which rearranges numbers into the lexicographically next greater permutation of numbers.

If such an arrangement is not possible, it must rearrange it as the lowest possible order (i.e., sorted in ascending order).

The replacement must be in place and use only constant extra memory.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0036 Valid Sudoku

**MEDIUM**

Determine if a 9x9 Sudoku board is valid.
Only the filled cells need to be validated according to the following rules:

- Each row must contain the digits 1-9 without repetition.
- Each column must contain the digits 1-9 without repetition.
- Each of the 9 3x3 sub-boxes of the grid must contain the digits 1-9 without repetition.

The Sudoku board could be partially filled, where empty cells are filled with the character '.'.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0037 Sudoku Solver

**HARD**

Write a program to solve a Sudoku puzzle by filling the empty cells.

A sudoku solution must satisfy all of the following rules:
- Each of the digits 1-9 must occur exactly once in each row.
- Each of the digits 1-9 must occur exactly once in each column.
- Each of the digits 1-9 must occur exactly once in each of the 9 3x3 sub-boxes of the grid.
- The '.' character indicates empty cells.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0039 Combination Sum

**MEDIUM**

Given an array of distinct integers candidates and a target integer target, return a list of all unique combinations of candidates where the chosen numbers sum to target. You may return the combinations in any order.

The same number may be chosen from candidates an unlimited number of times. Two combinations are unique if the frequency of at least one of the chosen numbers is different.

It is guaranteed that the number of unique combinations that sum up to target is less than 150 combinations for the given input.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0040 Combination Sum Ii

MEDIUM

Given a collection of candidate numbers (candidates) and a target number (target),
find all unique combinations in candidates where the candidate numbers sum to target.

Each number in candidates may only be used once in the combination.

Note: The solution set must not contain duplicate combinations.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0046 Permutations

MEDIUM

Given an array nums of distinct integers, return all the possible permutations.
You can return the answer in any order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0047 Permutations Ii

MEDIUM

Given a collection of numbers, nums, that might contain duplicates,
return all possible unique permutations in any order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0060 Permutation Sequence

HARD

The set [1, 2, 3, ..., n] contains a total of n! unique permutations.

By listing and labeling all of the permutations in order, we get the
following sequence for n = 3:
- "123"
- "132"
- "213"
- "231"
- "312"
- "321"

Given n and k, return the kth permutation sequence.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0077 Combinations

MEDIUM

Given two integers n and k, return all possible combinations of k numbers
out of the range [1, n].

You may return the answer in any order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0078 Subsets

MEDIUM

Given an integer array nums of unique elements, return all possible subsets (the power set).

The solution set must not contain duplicate subsets. Return the solution in any order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0079 Word Search

MEDIUM

Given an m x n grid of characters board and a string word, return true if word exists in the grid.

The word can be constructed from letters of sequentially adjacent cells, where adjacent cells are horizontally or vertically neighboring. The same letter cell may not be used more than once.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0090 Subsets Ii

MEDIUM

Given an integer array nums that may contain duplicates, return all possible subsets (the power set).

The solution set must not contain duplicate subsets. Return the solution in any order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0212 Word Search Ii

HARD

Given an m x n board of characters and a list of strings words, return all words on the board.

Each word must be constructed from letters of sequentially adjacent cells, where adjacent cells are horizontally or vertically neighboring. The same letter cell may not be used more than once in a word.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0216 Combination Sum Iii

MEDIUM

Find all valid combinations of k numbers that sum up to n such that the following conditions are true:
- Only numbers 1 through 9 are used.
- Each number is used at most once.

Return a list of all possible valid combinations. The list must not contain the same combination twice, and the combinations may be returned in any order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0254 Factor Combinations

MEDIUM

Numbers can be regarded as the product of their factors.

For example, 8 = 2 x 2 x 2 = 2 x 4.

Given an integer n, return all possible combinations of its factors. You may return the answer in any order.

Note that the factors should be in the range [2, n - 1].

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0266 Palindrome Permutation

**EASY**

Given a string s, return true if a permutation of the string could form a palindrome and false otherwise.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0267 Palindrome Permutation Ii

**MEDIUM**

Given a string s, return all the palindromic permutations (without duplicates) of it.

You may return the answer in any order. If s has no palindromic permutation, return an empty list.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0368 Largest Divisible Subset

**MEDIUM**

Given a set of distinct positive integers nums, return the largest subset answer such that every pair (answer[i], answer[j]) of elements in this subset satisfies:
- answer[i] % answer[j] == 0, or
- answer[j] % answer[i] == 0

If there are multiple solutions, return any of them.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0377 Combination Sum Iv

**MEDIUM**

Given an array of distinct integers nums and a target integer target, return the number of possible combinations that add up to target.

The test cases are generated so that the answer can fit in a 32-bit integer.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0416 Partition Equal Subset Sum

**MEDIUM**

Given an integer array nums, return true if you can partition the array into two subsets such that the sum of the elements in both subsets is equal or false otherwise.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0484 Find Permutation

MEDIUM

A permutation perm of n integers of all the integers in the range [1, n] can be represented as a string s of length n - 1 where:
- s[i] == 'I' if perm[i] < perm[i + 1], and
- s[i] == 'D' if perm[i] > perm[i + 1].

Given a string s, reconstruct the lexicographically smallest permutation perm and return it.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0567 Permutation In String

MEDIUM

Given two strings s1 and s2, return true if s2 contains a permutation of s1, or false otherwise.

In other words, return true if one of s1's permutations is the substring of s2.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0698 Partition To K Equal Sum Subsets

MEDIUM

Given an integer array nums and an integer k, return true if it is possible to divide this array into k non-empty subsets whose sums are all equal.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0784 Letter Case Permutation

MEDIUM

Given a string s, you can transform every letter individually to be lowercase or uppercase to create another string.

Return a list of all possible strings we could create. Return the output in any order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0903 Valid Permutations For Di Sequence

HARD

You are given a string s of length n where s[i] is either:
- 'D' means decreasing, or
- 'I' means increasing.

A permutation perm of n + 1 integers of all the integers in the range [0, n] is called a valid permutation if for all valid i:
- If s[i] == 'D', then perm[i] > perm[i + 1], and
- If s[i] == 'I', then perm[i] < perm[i + 1].

Return the number of valid permutations perm. Since the answer may be large, return it modulo 109 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0916 Word Subsets

MEDIUM

You are given two string arrays words1 and words2.

A string b is a subset of string a if every letter in b occurs in a including multiplicity.

For example, "wrr" is a subset of "warrior" but is not a subset of "world".

A string a from words1 is universal if for every string b in words2, b is a subset of a.

Return an array of all the universal strings in words1. You may return the answer in any order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1053 Previous Permutation With One Swap

MEDIUM

Given an array of positive integers arr (not necessarily distinct), return the lexicographically largest permutation that is smaller than arr, that can be made with exactly one swap. If it cannot be done, then return the same array.

Note that a swap exchanges the positions of two numbers arr[i] and arr[j]

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1220 Count Vowels Permutation

HARD

Given an integer n, your task is to count how many strings of length n can be formed under the following rules:
- Each character is a lower case vowel ('a', 'e', 'i', 'o', 'u')
- Each vowel 'a' may only be followed by an 'e'.
- Each vowel 'e' may only be followed by an 'a' or an 'i'.
- Each vowel 'i' may not be followed by another 'i'.
- Each vowel 'o' may only be followed by an 'i' or a 'u'.
- Each vowel 'u' may only be followed by an 'a'.

Since the answer may be too large, return it modulo 10^9 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1238 Circular Permutation In Binary Representation

MEDIUM

Given 2 integers n and start. Your task is return any permutation p of (0,1,2.....,2^n -1) such that:
- p[0] = start
- p[i] and p[i+1] differ by only one bit in their binary representation.
- p[0] and p[2^n -1] must also differ by only one bit in their binary representation.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1286 Iterator For Combination

MEDIUM

Design the CombinationIterator class:
- CombinationIterator(string characters, int combinationLength) Initializes the object with a string characters of sorted distinct lowercase English letters and a number combinationLength as arguments.

- next() Returns the next combination of length combinationLength in lexicographical order.
- hasNext() Returns true if and only if there exists a next combination.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1409 Queries On A Permutation With Key

**MEDIUM**

Given the array queries of positive integers between 1 and m, you have to process all queries[i] (from i=0 to i=queries.length-1) according to the following rules:
- In the beginning, you have the permutation P=[1,2,3,...,m].
- For the current i, find the position of queries[i] in the permutation P (indexing from 0) and then move this at the beginning of the permutation P. Notice that the position of queries[i] in P is the result for queries[i].

Return an array containing the result for the given queries.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1452 People Whose List Of Favorite Companies Is Not A Subset Of Another

**MEDIUM**

Given the array favoriteCompanies where favoriteCompanies[i] is the list of favorites companies for the ith person (indexed from 0).

Return the indices of people whose list of favorite companies is not a subset of any other list of favorites companies. You must return the indices in increasing order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1589 Maximum Sum Obtained Of Any Permutation

**MEDIUM**

We have an array of integers, nums, and an array of requests where requests[i] = [starti, endi]. The ith request asks for the sum of nums[starti] + nums[starti + 1] + ... + nums[endi - 1] + nums[endi]. Both starti and endi are 0-indexed.

Return the maximum total sum of all requests among all permutations of nums.

Since the answer may be too large, return it modulo 109 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1734 Decode Xored Permutation

**MEDIUM**

There is an integer array perm that is a permutation of the first n positive integers, where n is always odd.

It was encoded into another integer array encoded of length n - 1, such that encoded[i] = perm[i] XOR perm[i + 1]. For example, if perm = [1,3,2], then encoded = [2,1].

Given the encoded array, return the original array perm. It is guaranteed that the answer exists and is unique.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1806 Minimum Number Of Operations To Reinitialize A Permutation

MEDIUM

You are given an even integer n. You initially have a permutation perm of size n where perm[i] == i (0-indexed).

In one operation, you will create a new array arr, and for each i:
- If i % 2 == 0, then arr[i] = perm[i / 2].
- If i % 2 == 1, then arr[i] = perm[n / 2 + (i - 1) / 2].

You will then assign arr to perm.

Return the minimum non-zero number of operations you need to perform on perm to return the permutation to its initial value.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1863 Sum Of All Subset Xor Totals

EASY

The XOR total of an array is defined as the bitwise XOR of all its elements, or 0 if the array is empty.

For example, the XOR total of the array [2,5,6] is 2 XOR 5 XOR 6 = 1.
Given an array nums, return the sum of all XOR totals for every subset of nums.
Note: Subsets with the same elements should be counted multiple times.

An array a is a subset of an array b if a can be obtained from b by deleting some (possibly zero) elements of b.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1920 Build Array From Permutation

EASY

Given a zero-based permutation nums (0-indexed), build an array ans of the same length where ans[i] = nums[nums[i]] for each 0 <= i < nums.length and return it.

A zero-based permutation nums is an array of distinct integers from 0 to nums.length - 1 (inclusive).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1982 Find Array Given Subset Sums

HARD

You are given an integer n representing the length of an unknown array that you are trying to recover. You are also given an array sums containing the values of all 2n subset sums of the unknown array (in no particular order).

Return the array ans of length n representing the unknown array. If multiple answers exist, return any of them.

An array sub is a subset of an array arr if sub can be obtained from arr by deleting some (possibly zero or all) elements of arr. The sum of the elements in sub is one possible subset sum of arr. The sum of an empty array is considered to be 0.

Note: Test cases are generated such that there will always be at least one correct answer.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1994 The Number Of Good Subsets

HARD

You are given an integer array nums. We call a subset of nums good if its product can be represented as a product of one or more distinct prime numbers.

- For example, if nums = [1, 2, 3, 4]:
- [2, 3], [1, 2, 3], and [1, 3] are good subsets with products 6 = 2*3, 6 = 2*3, and 3 = 3 respectively.
- [1, 4] and [4] are not good subsets with products 4 = 2*2 and 4 = 2*2 respectively.

Return the number of different good subsets in nums modulo 109 + 7.

A subset of nums is any array that can be obtained by deleting some (possibly none or all) elements from nums. Two subsets are different if and only if the chosen indices to delete are different.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2044 Count Number Of Maximum Bitwise Or Subsets

MEDIUM

Given an integer array nums, find the maximum possible bitwise OR of a subset of nums and return the number of different non-empty subsets with the maximum bitwise OR.

An array a is a subset of an array b if a can be obtained from b by deleting some (possibly zero) elements of b. Two subsets are considered different if the indices of the elements chosen are different.

The bitwise OR of an array a is equal to a[0] OR a[1] OR ... OR a[a.length - 1] (0-indexed).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2056 Number Of Valid Move Combinations On Chessboard

HARD

There is an 8 x 8 chessboard containing n pieces (rooks, queens, or bishops). You are given a string array pieces of length n, where pieces[i] describes the type (rook, queen, or bishop) of the ith piece. In addition, you are given a 2D integer array positions also of length n, where positions[i] = [ri, ci] indicates that the ith piece is currently at the 1-based coordinate (ri, ci) on the chessboard.

When making a move for a piece, you choose a destination square that the piece will travel toward and stop on.
- A rook can only travel horizontally or vertically from (r, c) to the direction of (r+1, c), (r-1, c), (r, c+1), or (r, c-1).
- A queen can only travel horizontally, vertically, or diagonally from (r, c) to the direction of (r+1, c), (r-1, c), (r, c+1), (r, c-1), (r+1, c+1), (r+1, c-1), (r-1, c+1), (r-1, c-1).
- A bishop can only travel diagonally from (r, c) to the direction of (r+1, c+1), (r+1, c-1), (r-1, c+1), (r-1, c-1).

You must make a move for every piece on the board simultaneously. A move combination consists of all the moves performed on all the given pieces. Every second, each piece will instantaneously travel one square towards their destination if they are not already at it. All pieces start traveling at the 0th second. A move combination is invalid if, at a given time, two or more pieces occupy the same square.

Return the number of valid move combinations.

Notes:
- No two pieces will start in the same square.

- You may choose the square a piece is already on as its destination.
- If two pieces are directly adjacent to each other, it is valid for them to move past each other and swap positions in one second.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2275 Largest Combination With Bitwise And Greater Than Zero

**MEDIUM**

The bitwise AND of an array nums is the bitwise AND of all integers in nums.
- For example, for nums = [1, 5, 3], the bitwise AND is equal to 1 & 5 & 3 = 1.
- Also, for nums = [7], the bitwise AND is 7.

You are given an array of positive integers candidates. Compute the bitwise AND for all possible combinations of elements in the candidates array.

Return the size of the largest combination of candidates with a bitwise AND greater than 0.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2597 The Number Of Beautiful Subsets

**MEDIUM**

You are given an array nums of positive integers and a positive integer k.

A subset of nums is beautiful if it does not contain two integers with an absolute difference equal to k.

Return the number of non-empty beautiful subsets of the array nums.

A subset of nums is an array that can be obtained by deleting some (possibly none) elements from nums. Two subsets are different if and only if the chosen indices to delete are different.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2638 Count The Number Of K Free Subsets

**MEDIUM**

You are given an integer array nums, which contains distinct elements and an integer k.

A subset is called a k-Free subset if it contains no two elements with an absolute difference equal to k. Notice that the empty set is a k-Free subset.

Return the number of k-Free subsets of nums.

A subset of an array is a selection of elements (possibly none) of the array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2717 Semi Ordered Permutation

**EASY**

You are given a 0-indexed permutation of n integers nums.

A permutation is called semi-ordered if the first number equals 1 and the last number equals n. You can perform the below operation as many times as you want until you make nums a semi-ordered permutation:
- Pick two adjacent elements in nums, then swap them.

Return the minimum number of operations to make nums a semi-ordered permutation.

A permutation is a sequence of integers from 1 to n of length n containing each number exactly once.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 2732 Find A Good Subset Of The Matrix

HARD

You are given a 0-indexed m x n binary matrix grid.

Let us call a non-empty subset of rows good if the sum of each column of the subset is at most half of the length of the subset.

More formally, if the length of the chosen subset of rows is k, then the sum of each column should be at most floor(k / 2).

Return an integer array that contains row indices of a good subset sorted in ascending order.

If there are multiple good subsets, you can return any of them. If there are no good subsets, return an empty array.

A subset of rows of the matrix grid is any matrix that can be obtained by deleting some (possibly none or all) rows from grid.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 2862 Maximum Element Sum Of A Complete Subset Of Indices

HARD

You are given a 1-indexed array nums. Your task is to select a complete subset from nums where every pair of selected indices multiplied is a perfect square,. i. e. if you select ai and aj, i * j must be a perfect square.

Return the sum of the complete subset with the maximum sum.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 2992 Number Of Self Divisible Permutations

MEDIUM

Given an integer n, return the number of permutations of the 1-indexed array nums = [1, 2, ..., n], such that it's self-divisible.

A 1-indexed array a of length n is self-divisible if for every 1 <= i <= n, gcd(a[i], i) == 1.

A permutation of an array is a rearrangement of the elements of that array, for example here are all of the permutations of the array [1, 2, 3]:
- [1, 2, 3]
- [1, 3, 2]
- [2, 1, 3]
- [2, 3, 1]
- [3, 1, 2]
- [3, 2, 1]

TRY IT / Write examples, edge cases, and your own algorithm first.

## 3109 Find The Index Of Permutation

MEDIUM

Given an array perm of length n which is a permutation of [1, 2, ..., n], return the index of perm in the lexicographically sorted array of all of the permutations of [1, 2, ..., n].

Since the answer may be very large, return it modulo 109 + 7.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 3343 Count Number Of Balanced Permutations

HARD

You are given a string num. A string of digits is called balanced if the sum of the digits at even indices is equal to the sum of the digits at odd indices.

Return the number of distinct permutations of num that are balanced.

Since the answer may be very large, return it modulo 109 + 7.

A permutation is a rearrangement of all the characters of a string.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3437 Permutations Iii

MEDIUM

Given an integer n, an alternating permutation is a permutation of the first n positive integers such that no two adjacent elements are both odd or both even.

Return all such alternating permutations sorted in lexicographical order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3577 Count The Number Of Computer Unlocking Permutations

MEDIUM

You are given an array complexity of length n.

There are n locked computers in a room with labels from 0 to n - 1, each with its own unique password. The password of the computer i has a complexity complexity[i].

The password for the computer labeled 0 is already decrypted and serves as the root. All other computers must be unlocked using it or another previously unlocked computer, following this information:
- You can decrypt the password for the computer i using the password for computer j, where j is any integer less than i with a lower complexity. (i.e. j < i and complexity[j] < complexity[i])
- To decrypt the password for computer i, you must have already unlocked a computer j such that j < i and complexity[j] < complexity[i].

Find the number of permutations of [0, 1, 2, ..., (n - 1)] that represent a valid order in which the computers can be unlocked, starting from computer 0 as the only initially unlocked one.

Since the answer may be large, return it modulo 109 + 7.

Note that the password for the computer with label 0 is decrypted, and not the computer with the first position in the permutation.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

CHAPTER 10 / 20 EXERCISES

# Dynamic Programming

## 01 THE IDEA

Dynamic programming stores solutions to overlapping subproblems. Define a state, a transition and base cases.

## 02 SEE IT

| STATE | dp[0] | dp[1] | dp[2] | dp[3] | dp[4] |
|---|---|---|---|---|---|
| VALUE | 0 | 1 | 1 | 2 | 3 |

## 03 JAVASCRIPT PATTERN

```
dp[0] = 0; dp[1] = 1;
for (let i = 2; i <= n; i++)
 dp[i] = dp[i-1] + dp[i-2];
```

## 04 HOW TO THINK

Before coding, write: dp[state] means ___. Then derive a recurrence and determine evaluation order.

## 05 PERFORMANCE

State count × transition cost; memory may be optimized.

## EXERCISE COLLECTION

Try each question before opening the matching problem number in the separate Solutions PDF. The topic classification is a study aid; some exercises combine multiple patterns.

### 0070 Climbing Stairs

EASY

You are climbing a staircase. It takes n steps to reach the top.

Each time you can either climb 1 or 2 steps. In how many distinct ways can you climb to the top?

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0072 Edit Distance

MEDIUM

Given two strings word1 and word2, return the minimum number of operations required to convert word1 to word2.

You have the following three operations permitted on a word:
- Insert a character
- Delete a character
- Replace a character

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0091 Decode Ways

MEDIUM

You have intercepted a secret message encoded as a string of numbers. The message is decoded via the following mapping:
"1" -> 'A'
"2" -> 'B'
...
"25" -> 'Y'
"26" -> 'Z'

However, while decoding the message, you realize that there are many different ways you can decode the message because some codes are contained in other codes ("2" and "5" vs "25").

For example, "11106" can be decoded into:
- "AAJF" with the grouping (1, 1, 10, 6)
- "KJF" with the grouping (11, 10, 6)
- The grouping (1, 11, 06) is invalid because "06" is not a valid code (only "6" is valid).

Note: there may be strings that are impossible to decode.

Given a string s containing only digits, return the number of ways to decode it. If the entire string cannot be decoded in any valid way, return 0.

The test cases are generated so that the answer fits in a 32-bit integer.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0115 Distinct Subsequences

HARD

Given two strings s and t, return the number of distinct subsequences of s which equals t.

The test cases are generated so that the answer fits on a 32-bit signed integer.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0161 One Edit Distance

MEDIUM

Given two strings s and t, return true if they are both one edit distance apart, otherwise return false.

A string s is said to be one distance apart from a string t if you can:
- Insert exactly one character into s to get t.
- Delete exactly one character from s to get t.

- Replace exactly one character of s with a different character to get t.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0198 House Robber

**MEDIUM**

You are a professional robber planning to rob houses along a street. Each house has a certain amount of money stashed, the only constraint stopping you from robbing each of them is that adjacent houses have security systems connected and it will automatically contact the police if two adjacent houses were broken into on the same night.

Given an integer array nums representing the amount of money of each house, return the maximum amount of money you can rob tonight without alerting the police.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0213 House Robber Ii

**MEDIUM**

You are a professional robber planning to rob houses along a street. Each house has a certain amount of money stashed. All houses at this place are arranged in a circle. That means the first house is the neighbor of the last one. Meanwhile, adjacent houses have a security system connected, and it will automatically contact the police if two adjacent houses were broken into on the same night.

Given an integer array nums representing the amount of money of each house, return the maximum amount of money you can rob tonight without alerting the police.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0322 Coin Change

**MEDIUM**

You are given an integer array coins representing coins of different denominations and an integer amount representing a total amount of money.

Return the fewest number of coins that you need to make up that amount. If that amount of money cannot be made up by any combination of the coins, return -1.

You may assume that you have an infinite number of each kind of coin.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0337 House Robber Iii

**MEDIUM**

The thief has found himself a new place for his thievery again. There is only one entrance to this area, called root.

Besides the root, each house has one and only one parent house. After a tour, the smart thief realized that all houses in this place form a binary tree. It will automatically contact the police if two directly-linked houses were broken into on the same night.

Given the root of the binary tree, return the maximum amount of money the thief can rob without alerting the police.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0509 Fibonacci Number

EASY

The Fibonacci numbers, commonly denoted F(n) form a sequence, called the Fibonacci sequence, such that each number is the sum of the two preceding ones, starting from 0 and 1. That is:
- F(0) = 0, F(1) = 1
- F(n) = F(n - 1) + F(n - 2), for n > 1.
Given n, calculate F(n).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0518 Coin Change Ii

MEDIUM

You are given an integer array coins representing coins of different denominations and an integer amount representing a total amount of money.

Return the number of combinations that make up that amount. If that amount of money cannot be made up by any combination of the coins, return 0.

You may assume that you have an infinite number of each kind of coin.

The answer is guaranteed to fit into a signed 32-bit integer.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0639 Decode Ways Ii

HARD

A message containing letters from A-Z can be encoded into numbers using the following mapping:
- 'A' -> "1"
- 'B' -> "2"
- 'Z' -> "26"

To decode an encoded message, all the digits must be grouped then mapped back into letters using the reverse of the mapping above (there may be multiple ways). For example, "11106" can be mapped into:
- "AAJF" with the grouping (1 1 10 6)
- "KJF" with the grouping (11 10 6)

Note that the grouping (1 11 06) is invalid because "06" cannot be mapped into 'F' since "6" is different from "06".

In addition to the mapping above, an encoded message may contain the '*' character, which can represent any digit from '1' to '9' ('0' is excluded). For example, the encoded message "1*" may represent any of the encoded messages "11", "12", "13", "14", "15", "16", "17", "18", or "19". Decoding "1*" is equivalent to decoding any of the encoded messages it can represent.

Given a string s consisting of digits and '*' characters, return the number of ways to decode it.

Since the answer may be very large, return it modulo 109 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0746 Min Cost Climbing Stairs

EASY

You are given an integer array cost where cost[i] is the cost of ith step on a staircase. Once you pay the cost, you can either climb one or two steps.

You can either start from the step with index 0, or the step with index 1.

Return the minimum cost to reach the top of the floor.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0842 Split Array Into Fibonacci Sequence

**MEDIUM**

You are given a string of digits num, such as "123456579". We can split it into a Fibonacci-like sequence [123, 456, 579].

Formally, a Fibonacci-like sequence is a list f of non-negative integers such that:
- 0 <= f[i] < 231, (that is, each integer fits in a 32-bit signed integer type),
- f.length >= 3, and
- f[i] + f[i + 1] == f[i + 2] for all 0 <= i < f.length - 2.

Note that when splitting the string into pieces, each piece must not have extra leading zeroes, except if the piece is the number 0 itself.

Return any Fibonacci-like sequence split from num, or return [] if it cannot be done.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0873 Length Of Longest Fibonacci Subsequence

**MEDIUM**

A sequence x1, x2, ..., xn is Fibonacci-like if:
- n >= 3
- xi + xi+1 == xi+2 for all i + 2 <= n

Given a strictly increasing array arr of positive integers forming a sequence, return the length of the longest Fibonacci-like subsequence of arr. If one does not exist, return 0.

A subsequence is derived from another sequence arr by deleting any number of elements (including none) from arr, without changing the order of the remaining elements. For example, [3, 5, 8] is a subsequence of [3, 4, 5, 6, 7, 8].

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0940 Distinct Subsequences Ii

**HARD**

Given a string s, return the number of distinct non-empty subsequences of s. Since the answer may be very large, return it modulo 109 + 7.

A subsequence of a string is a new string that is formed from the original string by deleting some (can be none) of the characters without disturbing the relative positions of the remaining characters. (i.e., "ace" is a subsequence of "abcde" while "aec" is not.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1143 Longest Common Subsequence

**MEDIUM**

Given two strings text1 and text2, return the length of their longest common subsequence. If there is no common subsequence, return 0.

A subsequence of a string is a new string generated from the original string with some characters (can be none) deleted without changing the relative order of the remaining characters.

- For example, "ace" is a subsequence of "abcde".

A common subsequence of two strings is a subsequence that is common to both strings.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1414 Find The Minimum Number Of Fibonacci Numbers Whose Sum Is K

**MEDIUM**

Given an integer k, return the minimum number of Fibonacci numbers whose sum is equal to k. The same Fibonacci number can be used multiple times.

The Fibonacci numbers are defined as:
- F1 = 1
- F2 = 1
- Fn = Fn-1 + Fn-2 for n > 2.

It is guaranteed that for the given constraints we can always find such Fibonacci numbers that sum up to k.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2560 House Robber Iv

**MEDIUM**

There are several consecutive houses along a street, each of which has some money inside. There is also a robber, who wants to steal money from the homes, but he refuses to steal from adjacent homes.

The capability of the robber is the maximum amount of money he steals from one house of all the houses he robbed.

You are given an integer array nums representing how much money is stashed in each house. More formally, the ith house from the left has nums[i] dollars.

You are also given an integer k, representing the minimum number of houses the robber will steal from. It is always possible to steal at least k houses.

Return the minimum capability of the robber out of all the possible ways to steal at least k houses.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2648 Generate Fibonacci Sequence

**EASY**

Write a generator function that returns a generator object which yields the fibonacci sequence.

The fibonacci sequence is defined by the relation Xn = Xn-1 + Xn-2.

The first few numbers of the series are 0, 1, 1, 2, 3, 5, 8, 13.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

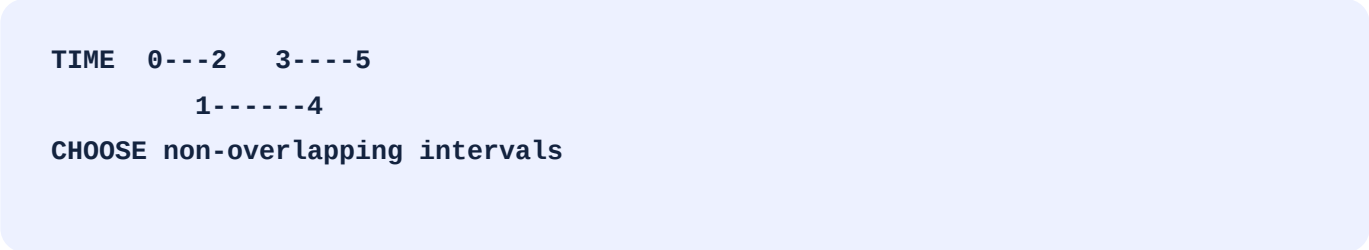
CHAPTER 11 / 37 EXERCISES

# Greedy, Intervals & Heaps

## 01 THE IDEA

Greedy algorithms make locally optimal choices; heaps repeatedly retrieve an extreme item efficiently.

## 02 SEE IT




## 03 JAVASCRIPT PATTERN

```
intervals.sort((a,b) => a[1]-b[1]);
// consider earliest finishing interval
```

## 04 HOW TO THINK

Explain why the local choice is safe, not just intuitive. Sort intervals by start or end depending on the invariant.

## 05 PERFORMANCE

Sorting O(n log n); heap insert/remove O(log n).

## EXERCISE COLLECTION

Try each question before opening the matching problem number in the separate Solutions PDF. The topic classification is a study aid; some exercises combine multiple patterns.

### 0045 Jump Game Ii

MEDIUM

You are given a 0-indexed array of integers nums of length n.
You are initially positioned at nums[0].

Each element nums[i] represents the maximum length of a forward jump from index i. In other words, if you are at nums[i], you can jump to any nums[i + j] where:
- 0 <= j <= nums[i] and
- i + j < n

Return the minimum number of jumps to reach nums[n - 1].

The test cases are generated such that you can reach nums[n - 1].

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0055 Jump Game

**MEDIUM**

You are given an integer array nums. You are initially positioned at the array's first index, and each element in the array represents your maximum jump length at that position.

Return true if you can reach the last index, or false otherwise.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0057 Insert Interval

**MEDIUM**

You are given an array of non-overlapping intervals intervals where intervals[i] = [starti, endi] represent the start and the end of the ith interval and intervals is sorted in ascending order by starti. You are also given an interval newInterval = [start, end] that represents the start and end of another interval.

Insert newInterval into intervals such that intervals is still sorted in ascending order by starti and intervals still does not have any overlapping intervals (merge overlapping intervals if necessary).

Return intervals after the insertion.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0134 Gas Station

**MEDIUM**

There are n gas stations along a circular route, where the amount of gas at the ith station is gas[i].

You have a car with an unlimited gas tank and it costs cost[i] of gas to travel from the ith station to its next (i + 1)th station. You begin the journey with an empty tank at one of the gas stations.

Given two integer arrays gas and cost, return the starting gas station's index if you can travel around the circuit once in the clockwise direction, otherwise return -1. If there exists a solution, it is guaranteed to be unique

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0215 Kth Largest Element In An Array

**MEDIUM**

Given an integer array nums and an integer k, return the kth largest element in the array.

Note that it is the kth largest element in the sorted order, not the kth distinct element.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0252 Meeting Rooms

**EASY**

Given an array of meeting time intervals where intervals[i] = [starti, endi], determine if a person could attend all meetings.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0253 Meeting Rooms Ii

MEDIUM

Given an array of meeting time intervals intervals where intervals[i] = [starti, endi], return the minimum number of conference rooms required.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0352 Data Stream As Disjoint Intervals

HARD

Given a data stream input of non-negative integers a1, a2, ..., an, summarize the numbers seen so far as a list of disjoint intervals.

Implement the SummaryRanges class:
- SummaryRanges() Initializes the object with an empty stream.
- void addNum(int value) Adds the integer value to the stream.
- int[][] getIntervals() Returns a summary of the integers in the stream currently as a list of disjoint intervals [starti, endi]. The answer should be sorted by starti.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0435 Non Overlapping Intervals

MEDIUM

Given an array of intervals intervals where intervals[i] = [starti, endi], return the minimum number of intervals you need to remove to make the rest of the intervals non-overlapping.

Note that intervals which only touch at a point are non-overlapping. For example, [1, 2] and [2, 3] are non-overlapping.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0436 Find Right Interval

MEDIUM

You are given an array of intervals, where intervals[i] = [starti, endi] and each starti is unique.

The right interval for an interval i is an interval j such that startj >= endi and startj is minimized. Note that i may equal j.

Return an array of right interval indices for each interval i. If no right interval exists for interval i, then put -1 at index i.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0452 Minimum Number Of Arrows To Burst Balloons

MEDIUM

There are some spherical balloons taped onto a flat wall that represents the XY-plane. The balloons are represented as a 2D integer array points where points[i] = [xstart, xend] denotes a balloon whose horizontal diameter stretches between xstart and xend. You do not know the exact y-coordinates of the balloons.

Arrows can be shot up directly vertically (in the positive y-direction) from different points along the x-axis. A balloon with xstart and xend is burst by an arrow shot at x if x(start) <= x <= x(end). There is no limit to the number of arrows that can be shot. A shot

arrow keeps traveling up infinitely, bursting any balloons in its path.

Given the array points, return the minimum number of arrows that must be shot to burst all balloons.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0621 Task Scheduler

**MEDIUM**

You are given an array of CPU tasks, each labeled with a letter from A to Z, and a number n. Each CPU interval can be idle or allow the completion of one task. Tasks can be completed in any order, but there's a constraint: there has to be a gap of at least n intervals between two tasks with the same label.

Return the minimum number of CPU intervals required to complete all tasks.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0703 Kth Largest Element In A Stream

**EASY**

You are part of a university admissions office and need to keep track of the kth highest test score from applicants in real-time. This helps to determine cut-off marks for interviews and admissions dynamically as new applicants submit their scores.

You are tasked to implement a class which, for a given integer k, maintains a stream of test scores and continuously returns the kth highest test score after a new score has been submitted. More specifically, we are looking for the kth highest score in the sorted list of all scores.

Implement the KthLargest class:
- KthLargest(int k, int[] nums) Initializes the object with the integer k and the stream of test scores nums.
- int add(int val) Adds a new test score val to the stream and returns the element representing the kth largest element in the pool of test scores so far.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0774 Minimize Max Distance To Gas Station

**HARD**

You are given an integer array stations that represents the positions of the gas stations on the x-axis. You are also given an integer k.

You should add k new gas stations. You can add the stations anywhere on the x-axis, and not necessarily on an integer position.

Let penalty() be the maximum distance between adjacent gas stations after adding the k new stations.

Return the smallest possible value of penalty(). Answers within 10-6 of the actual answer will be accepted.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0915 Partition Array Into Disjoint Intervals

**MEDIUM**

Given an integer array nums, partition it into two (contiguous) subarrays left and right so that:
- Every element in left is less than or equal to every element in right.
- left and right are non-empty.

- left has the smallest possible size.

Return the length of left after such a partitioning.

Test cases are generated such that partitioning exists.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0986 Interval List Intersections

**MEDIUM**

You are given two lists of closed intervals, firstList and secondList, where firstList[i] = [starti, endi] and secondList[j] = [startj, endj]. Each list of intervals is pairwise disjoint and in sorted order.

Return the intersection of these two interval lists.

A closed interval [a, b] (with a <= b) denotes the set of real numbers x with a <= x <= b.

The intersection of two closed intervals is a set of real numbers that are either empty or represented as a closed interval. For example, the intersection of [1, 3] and [2, 4] is [2, 3].

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1124 Longest Well Performing Interval

**MEDIUM**

We are given hours, a list of the number of hours worked per day for a given employee.

A day is considered to be a tiring day if and only if the number of hours worked is (strictly) greater than 8.

A well-performing interval is an interval of days for which the number of tiring days is strictly larger than the number of non-tiring days.

Return the length of the longest well-performing interval.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1272 Remove Interval

**MEDIUM**

A set of real numbers can be represented as the union of several disjoint intervals, where each interval is in the form [a, b). A real number x is in the set if one of its intervals [a, b) contains x (i.e. a <= x < b).

You are given a sorted list of disjoint intervals intervals representing a set of real numbers as described above, where intervals[i] = [ai, bi] represents the interval [ai, bi). You are also given another interval toBeRemoved.

Return the set of real numbers with the interval toBeRemoved removed from intervals. In other words, return the set of real numbers such that every x in the set is in intervals but not in toBeRemoved. Your answer should be a sorted list of disjoint intervals as described above.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1288 Remove Covered Intervals

**MEDIUM**

Given an array intervals where intervals[i] = [li, ri] represent the interval [li, ri), remove all intervals that are covered by another interval in the list.

The interval [a, b) is covered by the interval [c, d) if and only if c <= a and b <= d.

Return the number of remaining intervals.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1306 Jump Game Iii

MEDIUM

Given an array of non-negative integers arr, you are initially positioned at start index of the array. When you are at index i, you can jump to i + arr[i] or i - arr[i], check if you can reach any index with value 0.

Notice that you can not jump outside of the array at any time.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1340 Jump Game V

HARD

Given an array of integers arr and an integer d. In one step you can jump from index i to index:
- i + x where: i + x < arr.length and 0 < x <= d.
- i - x where: i - x >= 0 and 0 < x <= d.

In addition, you can only jump from index i to index j if arr[i] > arr[j] and arr[i] > arr[k] for all indices k between i and j (More formally min(i, j) < k < max(i, j)).

You can choose any index of the array and start jumping. Return the maximum number of indices you can visit.

Notice that you can not jump outside of the array at any time.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1345 Jump Game Iv

HARD

Given an array of integers arr, you are initially positioned at the first index of the array.

In one step you can jump from index i to index:
- i + 1 where: i + 1 < arr.length.
- i - 1 where: i - 1 >= 0.
- j where: arr[i] == arr[j] and i != j.

Return the minimum number of steps to reach the last index of the array.

Notice that you can not jump outside of the array at any time.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1523 Count Odd Numbers In An Interval Range

EASY

Given two non-negative integers low and high. Return the count of odd numbers between low and high (inclusive).
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1696 Jump Game Vi

MEDIUM

You are given a 0-indexed integer array nums and an integer k.

You are initially standing at index 0. In one move, you can jump at most k steps forward without going outside the boundaries of the array. That is, you can jump from index i to any index in the range [i + 1, min(n - 1, i + k)] inclusive.

You want to reach the last index of the array (index n - 1). Your score is the sum of all nums[j] for each index j you visited in the array.

Return the maximum score you can get.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1738 Find Kth Largest Xor Coordinate Value

MEDIUM

You are given a 2D matrix of size m x n, consisting of non-negative integers. You are also given an integer k.

The value of coordinate (a, b) of the matrix is the XOR of all matrix[i][j] where 0 <= i <= a < m and 0 <= j <= b < n (0-indexed).

Find the kth largest value (1-indexed) of all the coordinates of matrix.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1851 Minimum Interval To Include Each Query

HARD

You are given a 2D integer array intervals, where intervals[i] = [lefti, righti] describes the ith interval starting at lefti and ending at righti (inclusive). The size of an interval is defined as the number of integers it contains, or more formally righti - lefti + 1.

You are also given an integer array queries. The answer to the jth query is the size of the smallest interval i such that lefti <= queries[j] <= righti. If no such interval exists, the answer is -1.

Return an array containing the answers to the queries.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1871 Jump Game Vii

MEDIUM

You are given a 0-indexed binary string s and two integers minJump and maxJump. In the beginning, you are standing at index 0, which is equal to '0'. You can move from index i to index j if the following conditions are fulfilled:
- i + minJump <= j <= min(i + maxJump, s.length - 1), and
- s[j] == '0'.

Return true if you can reach index s.length - 1 in s, or false otherwise.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1985 Find The Kth Largest Integer In The Array

MEDIUM

You are given an array of strings nums and an integer k. Each string in nums represents an integer without leading zeros.

Return the string that represents the kth largest integer in nums.

Note: Duplicate numbers should be counted distinctly. For example, if nums is ["1","2","2"], "2" is the first largest integer, "2" is the second-largest integer, and "1" is the third-largest integer.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2121 Intervals Between Identical Elements

MEDIUM

You are given a 0-indexed array of n integers arr.

The interval between two elements in arr is defined as the absolute difference between their indices. More formally, the interval between arr[i] and arr[j] is |i - j|.

Return an array intervals of length n where intervals[i] is the sum of intervals between arr[i] and each element in arr with the same value as arr[i].

Note: |x| is the absolute value of x.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2276 Count Integers In Intervals

HARD

Given an empty set of intervals, implement a data structure that can:
- Add an interval to the set of intervals.
- Count the number of integers that are present in at least one interval.

Implement the CountIntervals class:
- CountIntervals() Initializes the object with an empty set of intervals.
- void add(int left, int right) Adds the interval [left, right] to the set of intervals.
- int count() Returns the number of integers that are present in at least one interval.

Note that an interval [left, right] denotes all the integers x where left <= x <= right.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2297 Jump Game Viii

MEDIUM

You are given a 0-indexed integer array nums of length n. You are initially standing at index 0. You can jump from index i to index j where i < j if:
- nums[i] <= nums[j] and nums[k] < nums[i] for all indexes k in the range i < k < j, or
- nums[i] > nums[j] and nums[k] >= nums[i] for all indexes k in the range i < k < j.

You are also given an integer array costs of length n where costs[i] denotes the cost of jumping to index i.

Return the minimum cost to jump to the index n - 1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2365 Task Scheduler Ii

MEDIUM

You are given a 0-indexed array of positive integers tasks, representing tasks that need to be completed in order, where tasks[i] represents the type of the ith task.

You are also given a positive integer space, which represents the minimum number of days that must pass after the completion of a task before another task of the same type can be performed.

Each day, until all tasks have been completed, you must either:
- Complete the next task from tasks, or
- Take a break.

Return the minimum number of days needed to complete all tasks.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2402 Meeting Rooms Iii

HARD

You are given an integer n. There are n rooms numbered from 0 to n - 1.

You are given a 2D integer array meetings where meetings[i] = [starti, endi] means that a meeting will be held during the half-closed time interval [starti, endi). All the values of starti are unique.

Meetings are allocated to rooms in the following manner:
The delayed meeting should have the same duration as the original meeting.
given the room.

Return the number of the room that held the most meetings. If there are multiple rooms, return the room with the lowest number.

A half-closed interval [a, b) is the interval between a and b including a and not including b.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2406 Divide Intervals Into Minimum Number Of Groups

MEDIUM

You are given a 2D integer array intervals where intervals[i] = [lefti, righti] represents the inclusive interval [lefti, righti].

You have to divide the intervals into one or more groups such that each interval is in exactly one group, and no two intervals that are in the same group intersect each other.

Return the minimum number of groups you need to make.

Two intervals intersect if there is at least one common number between them. For example, the intervals [1, 5] and [5, 8] intersect.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2725 Interval Cancellation

EASY

Given a function fn, an array of arguments args, and an interval time t, return a cancel function cancelFn.

After a delay of cancelTimeMs, the returned cancel function cancelFn will be invoked.
> setTimeout(cancelFn, cancelTimeMs)

The function fn should be called with args immediately and then called again every t milliseconds until cancelFn is called at cancelTimeMs ms.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2805 Custom Interval

**MEDIUM**

Function customInterval

Given a function fn, a number delay and a number period, return a number id.

customInterval is a function that should execute the provided function fn at intervals based on a linear pattern defined by the formula delay + period * count.

The count in the formula represents the number of times the interval has been executed starting from an initial value of 0.

Function customClearInterval

Given the id. id is the returned value from the function customInterval.

customClearInterval should stop executing provided function fn at intervals.

Note: The setTimeout and setInterval functions in Node.js return an object, not a number.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3323 Minimize Connected Groups By Inserting Interval

**MEDIUM**

You are given a 2D array intervals, where intervals[i] = [starti, endi] represents the start and the end of interval i. You are also given an integer k.

You must add exactly one new interval [startnew, endnew] to the array such that:
- The length of the new interval, endnew - startnew, is at most k.
- After adding, the number of connected groups in intervals is minimized.

A connected group of intervals is a maximal collection of intervals that, when considered together, cover a continuous range from the smallest point to the largest point with no gaps between them. Here are some examples:
- A group of intervals [[1, 2], [2, 5], [3, 3]] is connected because together they cover the range from 1 to 5 without any gaps.
- However, a group of intervals [[1, 2], [3, 4]] is not connected because the segment (2, 3) is not covered.

Return the minimum number of connected groups after adding exactly one new interval to the array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

CHAPTER 12 / 688 EXERCISES

# Math, Bits & Miscellaneous

## 01 THE IDEA

Some problems depend on number properties, modular arithmetic, bit operations, or custom simulation.

## 02 SEE IT

```
DECIMAL   13
BINARY  1101
BITS    8+4+0+1
```

## 03 JAVASCRIPT PATTERN

```
const odd = (x & 1) === 1;
const remainder = x % 10;
```

## 04 HOW TO THINK

Work through a small example by hand. Watch integer bounds, rounding, negative values and zero.

## 05 PERFORMANCE

Depends on number of digits, bits, or simulated steps.

## EXERCISE COLLECTION

Try each question before opening the matching problem number in the separate Solutions PDF. The topic classification is a study aid; some exercises combine multiple patterns.

### 0007 Reverse Integer

EASY

Given a signed 32-bit integer x, return x with its digits reversed.
If reversing x causes the value to go outside the signed 32-bit integer range [-2^31, 2^31 - 1], then return 0.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

### 0010 Regular Expression Matching

HARD

Given an input string s and a pattern p, implement regular expression

matching with support for '.' and '*' where:

'.' Matches any single character.
'*' Matches zero or more of the preceding element.

The matching should cover the entire input string (not partial).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0028 Implement Strstr

**EASY**

Return the index of the first occurrence of needle in haystack, or -1 if needle is not part of haystack.

Clarification:
- What should we return when needle is an empty string? This is a great question to ask during an interview.

- For the purpose of this problem, we will return 0 when needle is an empty string. This is consistent to C's strstr() and Java's indexOf().

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0029 Divide Two Integers

**MEDIUM**

Given two integers dividend and divisor, divide two integers without using multiplication, division, and mod operator.

The integer division should truncate toward zero, which means losing its fractional part. For example, 8.345 would be truncated to 8, and -2.7335 would be truncated to -2.

Return the quotient after dividing dividend by divisor.

Note: Assume we are dealing with an environment that could only store integers within the 32-bit signed integer range: [–231, 231 – 1]. For this problem, if the quotient is strictly greater than 231 - 1, then return 231 - 1, and if the quotient is strictly less than -231, then return -231.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0041 First Missing Positive

**HARD**

Given an unsorted integer array nums, return the smallest missing positive integer.

You must implement an algorithm that runs in O(n) time and uses constant extra space.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0044 Wildcard Matching

**HARD**

Given an input string (s) and a pattern (p), implement wildcard pattern matching with support for '?' and '*' where:

- '?' Matches any single character.
- '*' Matches any sequence of characters (including the empty sequence).

The matching should cover the entire input string (not partial).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0048 Rotate Image

MEDIUM

You are given an n x n 2D matrix representing an image, rotate the image by 90 degrees (clockwise).

You have to rotate the image in-place, which means you have to modify the input 2D matrix directly. DO NOT allocate another 2D matrix and do the rotation.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0050 Powx N

MEDIUM

Implement `pow(x, n)`, which calculates `x` raised to the power `n` (i.e., `x^n`).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0051 N Queens

HARD

The n-queens puzzle is the problem of placing n queens on an n x n chessboard such that no two queens attack each other.

Given an integer n, return all distinct solutions to the n-queens puzzle. You may return the answer in any order.

Each solution contains a distinct board configuration of the n-queens' placement, where 'Q' and '.' both indicate a queen and an empty space, respectively.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0052 N Queens Ii

HARD

The n-queens puzzle is the problem of placing n queens on an n x n chessboard such that no two queens attack each other.

Given an integer n, return the number of distinct solutions to the n-queens puzzle.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0062 Unique Paths

MEDIUM

There is a robot on an m x n grid. The robot is initially located at the top-left corner (i.e., grid[0][0]). The robot tries to move to the bottom-right corner (i.e., grid[m - 1][n - 1]). The robot can only move either down or right at any point in time.

Given the two integers m and n, return the number of possible unique paths that the robot can take to reach the bottom-right corner.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0063 Unique Paths Ii

MEDIUM

You are given an m x n integer array grid. There is a robot initially located at the top-left corner (i.e., grid[0][0]). The robot tries to move to the bottom-right corner (i.e., grid[m - 1][n - 1]). The robot can only move either down or right at any point in time.

An obstacle and space are marked as 1 or 0 respectively in grid. A path that the robot takes cannot include any square that is an obstacle.

Return the number of possible unique paths that the robot can take to reach the bottom-right corner.

The testcases are generated so that the answer will be less than or equal to 2 * 109.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0066 Plus One

EASY

Given a non-empty array of digits representing a non-negative integer, plus one to the integer.

The digits are stored such that the most significant digit is at the head of the list, and each element in the array contain a single digit.

You may assume the integer does not contain any leading zero, except the number 0 itself.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0067 Add Binary

EASY

Given two binary strings, return their sum (also a binary string).

The input strings are both non-empty and contains only characters 1 or 0.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0069 Sqrtx

MEDIUM

Given a non-negative integer x, return the square root of x rounded down to the nearest integer. The returned integer should be non-negative as well.

You must not use any built-in exponent function or operator.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0071 Simplify Path

MEDIUM

You are given an absolute path for a Unix-style file system, which always begins with a slash '/'. Your task is to transform this absolute path into its simplified canonical path.

The rules of a Unix-style file system are as follows:
- A single period '.' represents the current directory.
- A double period '..' represents the previous/parent directory.

- Multiple consecutive slashes such as '//' and '///' are treated as a single slash '/'.
- Any sequence of periods that does not match the rules above should be treated as a valid directory or file name. For example, '...' and '....' are valid directory or file names.

The simplified canonical path should follow these rules:
- The path must start with a single slash '/'.
- Directories within the path must be separated by exactly one slash '/'.
- The path must not end with a slash '/', unless it is the root directory.
- The path must not have any single or double periods ('.' and '..') used to denote current or parent directories.

Return the simplified canonical path.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0085 Maximal Rectangle

**HARD**

Given a rows x cols binary matrix filled with 0's and 1's, find the largest rectangle containing only 1's and return its area.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0089 Gray Code

**MEDIUM**

An n-bit gray code sequence is a sequence of 2n integers where:
- Every integer is in the inclusive range [0, 2n - 1],
- The first integer is 0,
- An integer appears no more than once in the sequence,
- The binary representation of every pair of adjacent integers differs by exactly one bit, and
- The binary representation of the first and last integers differs by exactly one bit.

Given an integer n, return any valid n-bit gray code sequence.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0093 Restore Ip Addresses

**MEDIUM**

A valid IP address consists of exactly four integers separated by single dots. Each integer is between 0 and 255 (inclusive) and cannot have leading zeros.

For example, "0.1.2.201" and "192.168.1.1" are valid IP addresses, but "0.011.255.245", "192.168.1.312" and "192.168@1.1" are invalid IP addresses.

Given a string s containing only digits, return all possible valid IP addresses that can be formed by inserting dots into s. You are not allowed to reorder or remove any digits in s. You may return the valid IP addresses in any order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0116 Populating Next Right Pointers In Each Node

MEDIUM

You are given a perfect binary tree where all leaves are on the same level, and every parent has two children. The binary tree has the following definition:

Populate each next pointer to point to its next right node. If there is no next right node, the next pointer should be set to NULL.

Initially, all next pointers are set to NULL.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0117 Populating Next Right Pointers In Each Node Ii

MEDIUM

Given a binary tree, populate each next pointer to point to its next right node. If there is no next right node, the next pointer should be set to NULL.

Initially, all next pointers are set to NULL.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0118 Pascals Triangle

EASY

Given an integer numRows, return the first numRows of Pascal's triangle.

In Pascal's triangle, each number is the sum of the two numbers directly above it as shown:

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0119 Pascals Triangle Ii

EASY

Given an integer rowIndex, return the rowIndexth (0-indexed) row of the Pascal's triangle.

In Pascal's triangle, each number is the sum of the two numbers directly above it as shown:

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0120 Triangle

MEDIUM

Given a triangle array, return the minimum path sum from top to bottom.

For each step, you may move to an adjacent number of the row below. More formally, if you are on index i on the current row, you may move to either index i or index i + 1 on the next row.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0121 Best Time To Buy And Sell Stock

EASY

You are given an array prices where prices[i] is the price of a given stock

on the ith day.

You want to maximize your profit by choosing a single day to buy one stock and choosing a different day in the future to sell that stock.

Return the maximum profit you can achieve from this transaction. If you cannot achieve any profit, return 0.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0122 Best Time To Buy And Sell Stock Ii

**MEDIUM**

You are given an integer array prices where prices[i] is the price of a given stock on the ith day.

On each day, you may decide to buy and/or sell the stock. You can only hold at most one share of the stock at any time. However, you can buy it then immediately sell it on the same day.

Find and return the maximum profit you can achieve.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0123 Best Time To Buy And Sell Stock Iii

**HARD**

You are given an array prices where prices[i] is the price of a given stock on the ith day.

Find the maximum profit you can achieve. You may complete at most two transactions.

Note: You may not engage in multiple transactions simultaneously (i.e., you must sell the stock before you buy again).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0125 Valid Palindrome

**EASY**

A phrase is a palindrome if, after converting all uppercase letters into lowercase letters and removing all non-alphanumeric characters, it reads the same forward and backward. Alphanumeric characters include letters and numbers.

Given a string s, return true if it is a palindrome, or false otherwise.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0131 Palindrome Partitioning

**MEDIUM**

Given a string s, partition s such that every substring of the partition is a palindrome. Return all possible palindrome partitioning of s.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0132 Palindrome Partitioning Ii

**HARD**

Given a string s, partition s such that every substring of the partition is a palindrome.

Return the minimum cuts needed for a palindrome partitioning of s.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0135 Candy

HARD

There are n children standing in a line. Each child is assigned a rating value given in the integer array ratings.

You are giving candies to these children subjected to the following requirements:
- Each child must have at least one candy.
- Children with a higher rating get more candies than their neighbors.

Return the minimum number of candies you need to have to distribute the candies to the children.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0150 Evaluate Reverse Polish Notation

MEDIUM

You are given an array of strings tokens that represents an arithmetic expression in a Reverse Polish Notation.

Evaluate the expression. Return an integer that represents the value of the expression.

Note that:
- The valid operators are '+', '-', '*', and '/'.
- Each operand may be an integer or another expression.
- The division between two integers always truncates toward zero.
- There will not be any division by zero.
- The input represents a valid arithmetic expression in a reverse polish notation.
- The answer and all the intermediate calculations can be represented in a 32-bit integer.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0163 Missing Ranges

EASY

You are given an inclusive range [lower, upper] and a sorted unique integer array nums, where all elements are within the inclusive range.

A number x is considered missing if x is in the range [lower, upper] and x is not in nums.

Return the shortest sorted list of ranges that exactly covers all the missing numbers. That is, no element of nums is included in any of the ranges, and each missing number is covered by one of the ranges.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0164 Maximum Gap

MEDIUM

Given an integer array nums, return the maximum difference between two successive elements in its sorted form. If the array contains less than two elements, return 0.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0168 Excel Sheet Column Title

EASY

Given an integer columnNumber, return its corresponding column title as it appears in an Excel sheet.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 0172 Factorial Trailing Zeroes

MEDIUM

Given an integer n, return the number of trailing zeroes in n!.

Note that n! = n * (n - 1) * (n - 2) * ... * 3 * 2 * 1.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 0174 Dungeon Game

HARD

The demons had captured the princess and imprisoned her in the bottom-right corner of a dungeon. The dungeon consists of m x n rooms laid out in a 2D grid. Our valiant knight was initially positioned in the top-left room and must fight his way through dungeon to rescue the princess.

The knight has an initial health point represented by a positive integer. If at any point his health point drops to 0 or below, he dies immediately.

Some of the rooms are guarded by demons (represented by negative integers), so the knight loses health upon entering these rooms; other rooms are either empty (represented as 0) or contain magic orbs that increase the knight's health (represented by positive integers).

To reach the princess as quickly as possible, the knight decides to move only rightward or downward in each step.

Return the knight's minimum initial health so that he can rescue the princess.

Note that any room can contain threats or power-ups, even the first room the knight enters and the bottom-right room where the princess is imprisoned.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 0188 Best Time To Buy And Sell Stock Iv

HARD

You are given an integer array prices where prices[i] is the price of a given stock on the ith day, and an integer k.

Find the maximum profit you can achieve. You may complete at most k transactions: i.e. you may buy at most k times and sell at most k times.

Note: You may not engage in multiple transactions simultaneously (i.e., you must sell the stock before you buy again).

TRY IT / Write examples, edge cases, and your own algorithm first.

## 0190 Reverse Bits

EASY

Reverse bits of a given 32 bits unsigned integer.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 0214 Shortest Palindrome

HARD

You are given a string s. You can convert s to a palindrome by adding characters in front of it.

Return the shortest palindrome you can find by performing this transformation.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0218 The Skyline Problem

**HARD**

A city's skyline is the outer contour of the silhouette formed by all the buildings in that city when viewed from a distance. Given the locations and heights of all the buildings, return the skyline formed by these buildings collectively.

The geometric information of each building is given in the array buildings where buildings[i] = [lefti, righti, heighti]:
- lefti is the x coordinate of the left edge of the ith building.
- righti is the x coordinate of the right edge of the ith building.
- heighti is the height of the ith building.

You may assume all buildings are perfect rectangles grounded on an absolutely flat surface at height 0.

The skyline should be represented as a list of "key points" sorted by their x-coordinate in the form [[x1,y1],[x2,y2],...]. Each key point is the left endpoint of some horizontal segment in the skyline except the last point in the list, which always has a y-coordinate 0 and is used to mark the skyline's termination where the rightmost building ends. Any ground between the leftmost and rightmost buildings should be part of the skyline's contour.

Note: There must be no consecutive horizontal lines of equal height in the output skyline. For instance, [...,[2 3],[4 5],[7 5],[11 5],[12 7],...] is not acceptable; the three lines of height 5 should be merged into one in the final output as such: [...,[2 3],[4 5],[12 7],...]

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0221 Maximal Square

**MEDIUM**

Given an m x n binary matrix filled with 0's and 1's, find the largest square containing only 1's and return its area.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0223 Rectangle Area

**MEDIUM**

Given the coordinates of two rectilinear rectangles in a 2D plane, return the total area covered by the two rectangles.

The first rectangle is defined by its bottom-left corner (ax1, ay1) and its top-right corner (ax2, ay2).

The second rectangle is defined by its bottom-left corner (bx1, by1) and its top-right corner (bx2, by2).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0231 Power Of Two

**EASY**

Given an integer n, return true if it is a power of two. Otherwise, return false.

An integer n is a power of two, if there exists an integer x such that n == 2x.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0251 Flatten 2D Vector

MEDIUM

Design an iterator to flatten a 2D vector. It should support the next and hasNext operations.

Implement the Vector2D class:
- Vector2D(int[][] vec) initializes the object with the 2D vector vec.
- next() returns the next element from the 2D vector and moves the pointer one step forward. You may assume that all the calls to next are valid.
- hasNext() returns true if there are still some elements in the vector, and false otherwise.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0256 Paint House

MEDIUM

There is a row of n houses, where each house can be painted one of three colors: red, blue, or green. The cost of painting each house with a certain color is different. You have to paint all the houses such that no two adjacent houses have the same color.

The cost of painting each house with a certain color is represented by an n x 3 cost matrix costs.
- For example, costs[0][0] is the cost of painting house 0 with the color red; costs[1][2] is the cost of painting house 1 with color green, and so on...

Return the minimum cost to paint all houses.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0265 Paint House Ii

HARD

There are a row of n houses, each house can be painted with one of the k colors. The cost of painting each house with a certain color is different. You have to paint all the houses such that no two adjacent houses have the same color.

The cost of painting each house with a certain color is represented by an n x k cost matrix costs.
- For example, costs[0][0] is the cost of painting house 0 with color 0; costs[1][2] is the cost of painting house 1 with color 2, and so on...

Return the minimum cost to paint all houses.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0274 H Index

MEDIUM

Given an array of integers citations where citations[i] is the number of citations a researcher received for their ith paper, return the researcher's h-index.

According to the definition of h-index on Wikipedia: The h-index is defined as the maximum value of h such that the given researcher has published at least h papers that have each been cited at least h times.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0275 H Index Ii

MEDIUM

Given an array of integers citations where citations[i] is the number of citations a researcher received for their ith paper and citations is sorted in ascending order, return the researcher's h-index.

According to the definition of h-index on Wikipedia: The h-index is defined as the maximum value of h such that the given researcher has published at least h papers that have each been cited at least h times.

You must write an algorithm that runs in logarithmic time.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0276 Paint Fence

MEDIUM

You are painting a fence of n posts with k different colors. You must paint the posts following these rules:
- Every post must be painted exactly one color.
- There cannot be three or more consecutive posts with the same color.

Given the two integers n and k, return the number of ways you can paint the fence.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0277 Find The Celebrity

MEDIUM

Suppose you are at a party with n people labeled from 0 to n - 1 and among them, there may exist one celebrity. The definition of a celebrity is that all the other n - 1 people know the celebrity, but the celebrity does not know any of them.

Now you want to find out who the celebrity is or verify that there is not one. You are only allowed to ask questions like: "Hi, A. Do you know B?" to get information about whether A knows B. You need to find out the celebrity (or verify there is not one) by asking as few questions as possible (in the asymptotic sense).

You are given an integer n and a helper function bool knows(a, b) that tells you whether a knows b. Implement a function int findCelebrity(n). There will be exactly one celebrity if they are at the party.

Return the celebrity's label if there is a celebrity at the party. If there is no celebrity, return -1.

Note that the n x n 2D array graph given as input is not directly available to you, and instead only accessible through the helper function knows. graph[i][j] == 1 represents person i knows person j, wherease graph[i][j] == 0 represents person j does not know person i.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0278 First Bad Version

MEDIUM

You are a product manager and currently leading a team to develop a new product. Unfortunately, the latest version of your product fails the quality check. Since each version is developed based on the previous version, all the versions after a bad version are also bad.

Suppose you have n versions [1, 2, ..., n] and you want to find out the first bad

one, which causes all the following ones to be bad.

You are given an API bool isBadVersion(version) which returns whether version is bad. Implement a function to find the first bad version. You should minimize the number of calls to the API.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0279 Perfect Squares

**MEDIUM**

Given an integer n, return the least number of perfect square numbers that sum to n.

A perfect square is an integer that is the square of an integer; in other words, it is the product of some integer with itself. For example, 1, 4, 9, and 16 are perfect squares while 3 and 11 are not.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0282 Expression Add Operators

**HARD**

Given a string num that contains only digits and an integer target, return all possibilities to insert the binary operators '+', '-', and/or '*' between the digits of num so that the resultant expression evaluates to the target value.

Note that operands in the returned expressions should not contain leading zeros.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0283 Move Zeroes

**EASY**

Given an integer array nums, move all 0's to the end of it while maintaining the relative order of the non-zero elements.

Note that you must do this in-place without making a copy of the array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0284 Peeking Iterator

**MEDIUM**

Design an iterator that supports the peek operation on an existing iterator in addition to the hasNext and the next operations.

Implement the PeekingIterator class:
- PeekingIterator(Iterator<int> nums) Initializes the object with the given integer iterator iterator.
- int next() Returns the next element in the array and moves the pointer to the next element.
- boolean hasNext() Returns true if there are still elements in the array.
- int peek() Returns the next element in the array without moving the pointer.

Note: Each language may have a different implementation of the constructor and Iterator, but they all support the int next() and boolean hasNext() functions.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0286 Walls And Gates

MEDIUM

You are given an m x n grid rooms initialized with these three possible values.
- -1 A wall or an obstacle.
- 0 A gate.
- INF Infinity means an empty room. We use the value 231 - 1 = 2147483647 to represent INF as you may assume that the distance to a gate is less than 2147483647.

Fill each empty room with the distance to its nearest gate. If it is impossible to reach a gate, it should be filled with INF.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0289 Game Of Life

MEDIUM

According to Wikipedia's article: "The Game of Life, also known simply as Life, is a cellular automaton devised by the British mathematician John Horton Conway in 1970."

The board is made up of an m x n grid of cells, where each cell has an initial state: live (represented by a 1) or dead (represented by a 0). Each cell interacts with its eight neighbors (horizontal, vertical, diagonal) using the following four rules (taken from the above Wikipedia article):

The next state of the board is determined by applying the above rules simultaneously to every cell in the current state of the m x n grid board. In this process, births and deaths occur simultaneously.

Given the current state of the board, update the board to reflect its next state.

Note that you do not need to return anything.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0292 Nim Game

EASY

You are playing the following Nim Game with your friend:
- Initially, there is a heap of stones on the table.
- You and your friend will alternate taking turns, and you go first.
- On each turn, the person whose turn it is will remove 1 to 3 stones from the heap.
- The one who removes the last stone is the winner.

Given n, the number of stones in the heap, return true if you can win the game assuming both you and your friend play optimally, otherwise return false.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0293 Flip Game

EASY

You are playing a Flip Game with your friend.

You are given a string currentState that contains only '+' and '-'. You and your friend take turns to flip two consecutive "++" into "--". The game ends when a person can no longer make a move, and therefore the other person will be the winner.

Return all possible states of the string currentState after one valid move. You may return the answer in any order. If there is no valid move, return an empty list [].

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0295 Find Median From Data Stream

HARD

The median is the middle value in an ordered integer list. If the size of the list is even, there is no middle value, and the median is the mean of the two middle values.

- For example, for arr = [2,3,4], the median is 3.
- For example, for arr = [2,3], the median is (2 + 3) / 2 = 2.5.

Implement the MedianFinder class:
- MedianFinder() initializes the MedianFinder object.
- void addNum(int num) adds the integer num from the data stream to the data structure.
- double findMedian() returns the median of all elements so far. Answers within 10-5 of the actual answer will be accepted.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0296 Best Meeting Point

HARD

Given an m x n binary grid grid where each 1 marks the home of one friend, return the minimal total travel distance.

The total travel distance is the sum of the distances between the houses of the friends and the meeting point.

The distance is calculated using Manhattan Distance, where distance(p1, p2) = |p2.x - p1.x| + |p2.y - p1.y|.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0299 Bulls And Cows

MEDIUM

You are playing the Bulls and Cows game with your friend.

You write down a secret number and ask your friend to guess what the number is. When your friend makes a guess, you provide a hint with the following info:
- The number of "bulls", which are digits in the guess that are in the correct position.
- The number of "cows", which are digits in the guess that are in your secret number but are located in the wrong position. Specifically, the non-bull digits in the guess that could be rearranged such that they become bulls.

Given the secret number secret and your friend's guess guess, return the hint for your friend's guess.

The hint should be formatted as "xAyB", where x is the number of bulls and y is the number of cows. Note that both secret and guess may contain duplicate digits.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0300 Longest Increasing Subsequence

MEDIUM

Given an integer array nums, return the length of the longest strictly increasing subsequence.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0302 Smallest Rectangle Enclosing Black Pixels

HARD

You are given an m x n binary matrix image where 0 represents a white pixel and 1 represents a black pixel.

The black pixels are connected (i.e., there is only one black region). Pixels are connected horizontally and vertically.

Given two integers x and y that represents the location of one of the black pixels, return the area of the smallest (axis-aligned) rectangle that encloses all black pixels.

You must write an algorithm with less than O(mn) runtime complexity

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0309 Best Time To Buy And Sell Stock With Cooldown

MEDIUM

You are given an array prices where prices[i] is the price of a given stock on the ith day.

Find the maximum profit you can achieve. You may complete as many transactions as you like (i.e., buy one and sell one share of the stock multiple times) with the following restrictions:
- After you sell your stock, you cannot buy stock on the next day (i.e., cooldown one day).

Note: You may not engage in multiple transactions simultaneously (i.e., you must sell the stock before you buy again).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0312 Burst Balloons

HARD

You are given n balloons, indexed from 0 to n - 1. Each balloon is painted with a number on it represented by an array nums. You are asked to burst all the balloons.

If you burst the ith balloon, you will get nums[i - 1] * nums[i] * nums[i + 1] coins. If i - 1 or i + 1 goes out of bounds of the array, then treat it as if there is a balloon with a 1 painted on it.

Return the maximum coins you can collect by bursting the balloons wisely.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0319 Bulb Switcher

MEDIUM

There are n bulbs that are initially off. You first turn on all the bulbs, then you turn off every second bulb.

On the third round, you toggle every third bulb (turning on if it's off or turning off if it's on). For the ith round, you toggle every i bulb. For the nth round, you only toggle the last bulb.

Return the number of bulbs that are on after n rounds.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0320 Generalized Abbreviation

MEDIUM

A word's generalized abbreviation can be constructed by taking any number of non-overlapping and non-adjacent substrings and replacing them with their respective lengths.
- For example, "abcde" can be abbreviated into:
- "a3e" ("bcd" turned into "3")
- "1bcd1" ("a" and "e" both turned into "1")
- "5" ("abcde" turned into "5")
- "abcde" (no substrings replaced)
- However, these abbreviations are invalid:
- "23" ("ab" turned into "2" and "cde" turned into "3") is invalid as the substrings chosen are adjacent.
- "22de" ("ab" turned into "2" and "bc" turned into "2") is invalid as the substring chosen overlap.

Given a string word, return a list of all the possible generalized abbreviations of word. Return the answer in any order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0326 Power Of Three

EASY

Given an integer n, return true if it is a power of three. Otherwise, return false.

An integer n is a power of three, if there exists an integer x such that n == 3x.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0335 Self Crossing

HARD

You are given an array of integers distance.

You start at the point (0, 0) on an X-Y plane, and you move distance[0] meters to the north, then distance[1] meters to the west, distance[2] meters to the south, distance[3] meters to the east, and so on. In other words, after each move, your direction changes counter-clockwise.

Return true if your path crosses itself or false if it does not.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0341 Flatten Nested List Iterator

MEDIUM

You are given a nested list of integers nestedList. Each element is either an integer or a list whose elements may also be integers or other lists. Implement an iterator to flatten it.

Implement the NestedIterator class:
- NestedIterator(List<NestedInteger> nestedList) Initializes the iterator with the nested list nestedList.
- int next() Returns the next integer in the nested list.
- boolean hasNext() Returns true if there are still some integers in the nested list and false otherwise.

Your code will be tested with the following pseudocode:
initialize iterator with nestedList
res = []
while iterator.hasNext()

append iterator.next() to the end of res
return res

If res matches the expected flattened list, then your code will be judged as correct.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0342 Power Of Four

**EASY**

Given an integer n, return true if it is a power of four. Otherwise, return false.

An integer n is a power of four, if there exists an integer x such that n == 4x.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0343 Integer Break

**MEDIUM**

Given an integer n, break it into the sum of k positive integers, where k >= 2,
and maximize the product of those integers.

Return the maximum product you can get.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0346 Moving Average From Data Stream

**EASY**

Given a stream of integers and a window size, calculate the moving average of all integers
in the sliding window.

Implement the MovingAverage class:
- MovingAverage(int size) Initializes the object with the size of the window size.
- double next(int val) Returns the moving average of the last size values of the stream.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0348 Design Tic Tac Toe

**MEDIUM**

Assume the following rules are for the tic-tac-toe game on an n x n board between two players:
row wins the game.

Implement the TicTacToe class:
- TicTacToe(int n) Initializes the object the size of the board n.
- int move(int row, int col, int player) Indicates that the player with id player plays at the
cell (row, col) of the board. The move is guaranteed to be a valid move, and the two players
alternate in making moves. Return
- 0 if there is no winner after the move,
- 1 if player 1 is the winner after the move, or
- 2 if player 2 is the winner after the move.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0354 Russian Doll Envelopes

**HARD**

You are given a 2D array of integers envelopes where envelopes[i] = [wi, hi] represents
the width and the height of an envelope.

One envelope can fit into another if and only if both the width and height of one envelope are greater than the other envelope's width and height.

Return the maximum number of envelopes you can Russian doll (i.e., put one inside the other).

Note: You cannot rotate an envelope.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0361 Bomb Enemy

**MEDIUM**

Given an m x n matrix grid where each cell is either a wall 'W', an enemy 'E' or empty '0', return the maximum enemies you can kill using one bomb. You can only place the bomb in an empty cell.

The bomb kills all the enemies in the same row and column from the planted point until it hits the wall since it is too strong to be destroyed.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0365 Water And Jug Problem

**MEDIUM**

You are given two jugs with capacities x liters and y liters. You have an infinite water supply. Return whether the total amount of water in both jugs may reach target using the following operations:
- Fill either jug completely with water.
- Completely empty either jug.
- Pour water from one jug into another until the receiving jug is full, or the transferring jug is empty.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0367 Valid Perfect Square

**EASY**

Given a positive integer num, return true if num is a perfect square or false otherwise.

A perfect square is an integer that is the square of an integer. In other words, it is the product of some integer with itself.

You must not use any built-in library function, such as sqrt.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0370 Range Addition

**MEDIUM**

You are given an integer length and an array updates where updates[i] = [startIdxi, endIdxi, inci].

You have an array arr of length length with all zeros, and you have some operation to apply on arr. In the ith operation, you should increment all the elements arr[startIdxi], arr[startIdxi + 1], ..., arr[endIdxi] by inci.

Return arr after applying all the updates.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0372 Super Pow

MEDIUM

Your task is to calculate ab mod 1337 where a is a positive integer and b is an extremely large positive integer given in the form of an array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0376 Wiggle Subsequence

MEDIUM

A wiggle sequence is a sequence where the differences between successive numbers strictly alternate between positive and negative. The first difference (if one exists) may be either positive or negative. A sequence with one element and a sequence with two non-equal elements are trivially wiggle sequences.
- For example, [1, 7, 4, 9, 2, 5] is a wiggle sequence because the differences (6, -3, 5, -7, 3) alternate between positive and negative.
- In contrast, [1, 4, 7, 2, 5] and [1, 7, 4, 5, 5] are not wiggle sequences. The first is not because its first two differences are positive, and the second is not because its last difference is zero.

A subsequence is obtained by deleting some elements (possibly zero) from the original sequence, leaving the remaining elements in their original order.

Given an integer array nums, return the length of the longest wiggle subsequence of nums.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0385 Mini Parser

MEDIUM

Given a string s represents the serialization of a nested list, implement a parser to deserialize it and return the deserialized NestedInteger.

Each element is either an integer or a list whose elements may also be integers or other lists.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0388 Longest Absolute File Path

MEDIUM

Suppose we have a file system that stores both files and directories. An example of one system is represented in the following picture:

Here, we have dir as the only directory in the root. dir contains two subdirectories, subdir1 and subdir2. subdir1 contains a file file1.ext and subdirectory subsubdir1. subdir2 contains a subdirectory subsubdir2, which contains a file file2.ext.
In text form, it looks like this (with ■ representing the tab character):

dir
■ subdir1
■ ■ file1.ext
■ ■ subsubdir1
■ subdir2
■ ■ subsubdir2
■ ■ ■ file2.ext

If we were to write this representation in code, it will look like this:
"dir\n\tsubdir1\n\t\tfile1.ext\n\t\tsubsubdir1\n\tsubdir2\n\t\tsubsubdir2\n\t\t\tfile2.ext".
Note that the '\n' and '\t' are the new-line and tab characters.

Every file and directory has a unique absolute path in the file system, which is the order of directories that must be opened to reach the file/directory itself, all concatenated by '/'s. Using the above example, the absolute path to file2.ext is "dir/subdir2/subsubdir2/file2.ext". Each directory name consists of letters, digits, and/or spaces. Each file name is of the form name.extension, where name and extension consist of letters, digits, and/or spaces.

Given a string input representing the file system in the explained format, return the length of the longest absolute path to a file in the abstracted file system. If there is no file in the system, return 0.

Note that the testcases are generated such that the file system is valid and no file or directory name has length 0.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0390 Elimination Game

**MEDIUM**

You have a list arr of all integers in the range [1, n] sorted in a strictly increasing order. Apply the following algorithm on arr:
- Starting from left to right, remove the first number and every other number afterward until you reach the end of the list.
- Repeat the previous step again, but this time from right to left, remove the rightmost number and every other number from the remaining numbers.
- Keep repeating the steps again, alternating left to right and right to left, until a single number remains.

Given the integer n, return the last number that remains in arr.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0392 Is Subsequence

**EASY**

Given two strings s and t, return true if s is a subsequence of t, or false otherwise.

A subsequence of a string is a new string that is formed from the original string by deleting some (can be none) of the characters without disturbing the relative positions of the remaining characters. (i.e., "ace" is a subsequence of "abcde" while "aec" is not).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0393 Utf 8 Validation

**MEDIUM**

Given an integer array data representing the data, return whether it is a valid UTF-8 encoding (i.e. it translates to a sequence of valid UTF-8 encoded characters).

A character in UTF8 can be from 1 to 4 bytes long, subjected to the following rules:
n - 1 bytes with the most significant 2 bits being 10.

This is how the UTF-8 encoding would work:

```
Number of Bytes   |        UTF-8 Octet Sequence
|             (binary)
-------------------+-----------------------------------------
1         |  0xxxxxxx
2         |  110xxxxx 10xxxxxx
3         |  1110xxxx 10xxxxxx 10xxxxxx
4         |  11110xxx 10xxxxxx 10xxxxxx 10xxxxxx
```

x denotes a bit in the binary form of a byte that may be either 0 or 1.

Note: The input is an array of integers. Only the least significant 8 bits of each integer is used to store the data. This means each integer represents only 1 byte of data.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0396 Rotate Function

**MEDIUM**

You are given an integer array nums of length n.

Assume arrk to be an array obtained by rotating nums by k positions clock-wise.
We define the rotation function F on nums as follow:
- F(k) = 0 * arrk[0] + 1 * arrk[1] + ... + (n - 1) * arrk[n - 1].

Return the maximum value of F(0), F(1), ..., F(n-1).

The test cases are generated so that the answer fits in a 32-bit integer.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0397 Integer Replacement

**MEDIUM**

Given a positive integer n, you can apply one of the following operations:

Return the minimum number of operations needed for n to become 1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0401 Binary Watch

**EASY**

A binary watch has 4 LEDs on the top to represent the hours (0-11), and 6 LEDs on the bottom to represent the minutes (0-59). Each LED represents a zero or one, with the least significant bit on the right.
- For example, the below binary watch reads "4:51".
Given an integer turnedOn which represents the number of LEDs that are currently on (ignoring the PM), return all possible times the watch could represent. You may return the answer in any order.
The hour must not contain a leading zero.
- For example, "01:00" is not valid. It should be "1:00".
The minute must consist of two digits and may contain a leading zero.
- For example, "10:2" is not valid. It should be "10:02".

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0412 Fizz Buzz

**EASY**

Given an integer n, return a string array answer (1-indexed) where:
- answer[i] == "FizzBuzz" if i is divisible by 3 and 5.
- answer[i] == "Fizz" if i is divisible by 3.
- answer[i] == "Buzz" if i is divisible by 5.
- answer[i] == i (as a string) if none of the above conditions are true.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0413 Arithmetic Slices

MEDIUM

An integer array is called arithmetic if it consists of at least three elements and if the difference between any two consecutive elements is the same.

For example, [1,3,5,7,9], [7,7,7,7], and [3,-1,-5,-9] are arithmetic sequences.

Given an integer array nums, return the number of arithmetic subarrays of nums.

A subarray is a contiguous subsequence of the array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0419 Battleships In A Board

MEDIUM

Given an m x n matrix board where each cell is a battleship 'X' or empty '.', return the number of the battleships on board.

Battleships can only be placed horizontally or vertically on board. In other words, they can only be made of the shape 1 x k (1 row, k columns) or k x 1 (k rows, 1 column), where k can be of any size. At least one horizontal or vertical cell separates between two battleships (i.e., there are no adjacent battleships).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0439 Ternary Expression Parser

MEDIUM

Given a string expression representing arbitrarily nested ternary expressions, evaluate the expression, and return the result of it.

You can always assume that the given expression is valid and only contains digits, '?', ':', 'T', and 'F' where 'T' is true and 'F' is false. All the numbers in the expression are one-digit numbers (i.e., in the range [0, 9]).

The conditional expressions group right-to-left (as usual in most languages), and the result of the expression will always evaluate to either a digit, 'T' or 'F'.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0441 Arranging Coins

EASY

You have n coins and you want to build a staircase with these coins. The staircase consists of k rows where the ith row has exactly i coins. The last row of the staircase may be incomplete.

Given the integer n, return the number of complete rows of the staircase you will build.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0455 Assign Cookies

EASY

Assume you are an awesome parent and want to give your children some cookies. But, you should give each child at most one cookie.

Each child i has a greed factor g[i], which is the minimum size of a cookie that the child will be content with; and each cookie j has a size s[j]. If s[j] >= g[i], we can assign the cookie j to the child i, and the child i will be content. Your goal is to

maximize the number of your content children and output the maximum number.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0456 132 Pattern

**MEDIUM**

Given an array of n integers nums, a 132 pattern is a subsequence of three integers nums[i], nums[j] and nums[k] such that i < j < k and nums[i] < nums[k] < nums[j].

Return true if there is a 132 pattern in nums, otherwise, return false.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0458 Poor Pigs

**HARD**

There are buckets buckets of liquid, where exactly one of the buckets is poisonous. To figure out which one is poisonous, you feed some number of (poor) pigs the liquid to see whether they will die or not. Unfortunately, you only have minutesToTest minutes to determine which bucket is poisonous.

You can feed the pigs according to these steps:
simultaneously and will take no time. Each pig can feed from any number of buckets, and each bucket can be fed from by any number of pigs.
will die, and all others will survive.

Given buckets, minutesToDie, and minutesToTest, return the minimum number of pigs needed to figure out which bucket is poisonous within the allotted time.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0461 Hamming Distance

**EASY**

The Hamming distance between two integers is the number of positions at which the corresponding bits are different.

Given two integers x and y, return the Hamming distance between them.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0468 Validate Ip Address

**MEDIUM**

Given a string queryIP, return "IPv4" if IP is a valid IPv4 address, "IPv6" if IP is a valid IPv6 address or "Neither" if IP is not a correct IP of any type.

A valid IPv4 address is an IP in the form "x1.x2.x3.x4" where 0 <= xi <= 255 and xi cannot contain leading zeros. For example, "192.168.1.1" and "192.168.1.0" are valid IPv4 addresses while "192.168.01.1", "192.168.1.00", and "192.168@1.1" are invalid IPv4 addresses.

A valid IPv6 address is an IP in the form "x1:x2:x3:x4:x5:x6:x7:x8" where:
- 1 <= xi.length <= 4
- xi is a hexadecimal string which may contain digits, lowercase English letter ('a' to 'f') and upper-case English letters ('A' to 'F').
- Leading zeros are allowed in xi.

For example, "2001:0db8:85a3:0000:0000:8a2e:0370:7334" and "2001:db8:85a3:0:0:8A2E:0370:7334" are valid IPv6 addresses, while "2001:0db8:85a3::8A2E:037j:7334" and "02001:0db8:85a3:0000:0000:8a2e:0370:7334" are invalid IPv6 addresses.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 0469 Convex Polygon

MEDIUM

You are given an array of points on the X-Y plane points where points[i] = [xi, yi]. The points form a polygon when joined sequentially.

Return true if this polygon is convex and false otherwise.

You may assume the polygon formed by given points is always a simple polygon. In other words, we ensure that exactly two edges intersect at each vertex and that edges otherwise don't intersect each other.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 0470 Implement Rand10 Using Rand7

MEDIUM

Given the API rand7() that generates a uniform random integer in the range [1, 7], write a function rand10() that generates a uniform random integer in the range [1, 10]. You can only call the API rand7(), and you shouldn't call any other API. Please do not use a language's built-in random API.

Each test case will have one internal argument n, the number of times that your implemented function rand10() will be called while testing. Note that this is not an argument passed to rand10().

TRY IT / Write examples, edge cases, and your own algorithm first.

## 0473 Matchsticks To Square

MEDIUM

You are given an integer array matchsticks where matchsticks[i] is the length of the ith matchstick. You want to use all the matchsticks to make one square. You should not break any stick, but you can link them up, and each matchstick must be used exactly one time.

Return true if you can make this square and false otherwise.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 0474 Ones And Zeroes

MEDIUM

You are given an array of binary strings strs and two integers m and n.

Return the size of the largest subset of strs such that there are at most m 0's and n 1's in the subset.

A set x is a subset of a set y if all elements of x are also elements of y.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 0475 Heaters

MEDIUM

Winter is coming! During the contest, your first job is to design a standard heater with a fixed warm radius to warm all the houses.

Every house can be warmed, as long as the house is within the heater's warm radius range.

Given the positions of houses and heaters on a horizontal line, return the minimum radius standard of heaters so that those heaters could cover all houses.

Notice that all the heaters follow your radius standard, and the warm radius will the same.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0477 Total Hamming Distance

**MEDIUM**

The Hamming distance between two integers is the number of positions at which the corresponding bits are different.

Given an integer array nums, return the sum of Hamming distances between all the pairs of the integers in nums.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0478 Generate Random Point In A Circle

**MEDIUM**

Given the radius and the position of the center of a circle, implement the function randPoint which generates a uniform random point inside the circle.

Implement the Solution class:
- Solution(double radius, double x_center, double y_center) initializes the object with the radius of the circle radius and the position of the center (x_center, y_center).
- randPoint() returns a random point inside the circle. A point on the circumference of the circle is considered to be in the circle. The answer is returned as an array [x, y].

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0482 License Key Formatting

**EASY**

You are given a license key represented as a string s that consists of only alphanumeric characters and dashes. The string is separated into n + 1 groups by n dashes. You are also given an integer k.

We want to reformat the string s such that each group contains exactly k characters, except for the first group, which could be shorter than k but still must contain at least one character. Furthermore, there must be a dash inserted between two groups, and you should convert all lowercase letters to uppercase.

Return the reformatted license key.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0483 Smallest Good Base

**HARD**

Given an integer n represented as a string, return the smallest good base of n.

We call k >= 2 a good base of n, if all digits of n base k are 1's.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0485 Max Consecutive Ones

EASY

Given a binary array nums, return the maximum number of consecutive 1's in the array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0486 Predict The Winner

MEDIUM

You are given an integer array nums. Two players are playing a game with this array: player 1 and player 2.

Player 1 and player 2 take turns, with player 1 starting first. Both players start the game with a score of 0. At each turn, the player takes one of the numbers from either end of the array (i.e., nums[0] or nums[nums.length - 1]) which reduces the size of the array by 1. The player adds the chosen number to their score. The game ends when there are no more elements in the array.

Return true if Player 1 can win the game. If the scores of both players are equal, then player 1 is still the winner, and you should also return true. You may assume that both players are playing optimally.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0487 Max Consecutive Ones Ii

MEDIUM

Given a binary array nums, return the maximum number of consecutive 1's in the array if you can flip at most one 0.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0492 Construct The Rectangle

EASY

A web developer needs to know how to design a web page's size. So, given a specific rectangular web page’s area, your job by now is to design a rectangular web page, whose length L and width W satisfy the following requirements:

The area of the rectangular web page you designed must equal to the given target area. The width W should not be larger than the length L, which means L >= W. The difference between length L and width W should be as small as possible.

Return an array [L, W] where L and W are the length and width of the web page you designed in sequence.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0495 Teemo Attacking

EASY

Our hero Teemo is attacking an enemy Ashe with poison attacks! When Teemo attacks Ashe, Ashe gets poisoned for a exactly duration seconds. More formally, an attack at second t will mean Ashe is poisoned during the inclusive time interval [t, t + duration - 1]. If Teemo attacks again before the poison effect ends, the timer for it is reset, and the poison effect will end duration seconds after the new attack.

You are given a non-decreasing integer array timeSeries, where timeSeries[i] denotes that Teemo attacks Ashe at second timeSeries[i], and an integer duration.

Return the total number of seconds that Ashe is poisoned.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0497 Random Point In Non Overlapping Rectangles

**MEDIUM**

You are given an array of non-overlapping axis-aligned rectangles rects where rects[i] = [ai, bi, xi, yi] indicates that (ai, bi) is the bottom-left corner point of the ith rectangle and (xi, yi) is the top-right corner point of the ith rectangle. Design an algorithm to pick a random integer point inside the space covered by one of the given rectangles. A point on the perimeter of a rectangle is included in the space covered by the rectangle.

Any integer point inside the space covered by one of the given rectangles should be equally likely to be returned.

Note that an integer point is a point that has integer coordinates.

Implement the Solution class:
- Solution(int[][] rects) Initializes the object with the given rectangles rects.
- int[] pick() Returns a random integer point [u, v] inside the space covered by one of the given rectangles.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0498 Diagonal Traverse

**MEDIUM**

Given an m x n matrix mat, return an array of all the elements of the array in a diagonal order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0499 The Maze Iii

**HARD**

There is a ball in a maze with empty spaces (represented as 0) and walls (represented as 1). The ball can go through the empty spaces by rolling up, down, left or right, but it won't stop rolling until hitting a wall. When the ball stops, it could choose the next direction. There is also a hole in this maze. The ball will drop into the hole if it rolls onto the hole.

Given the m x n maze, the ball's position ball and the hole's position hole, where ball = [ballrow, ballcol] and hole = [holerow, holecol], return a string instructions of all the instructions that the ball should follow to drop in the hole with the shortest distance possible. If there are multiple valid instructions, return the lexicographically minimum one. If the ball can't drop in the hole, return "impossible".

If there is a way for the ball to drop in the hole, the answer instructions should contain the characters 'u' (i.e., up), 'd' (i.e., down), 'l' (i.e., left), and 'r' (i.e., right).

The distance is the number of empty spaces traveled by the ball from the start position (excluded) to the destination (included).

You may assume that the borders of the maze are all walls (see examples).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0500 Keyboard Row

EASY

Given an array of strings words, return the words that can be typed using letters of the alphabet on only one row of American keyboard like the image below.

In the American keyboard:
- the first row consists of the characters "qwertyuiop",
- the second row consists of the characters "asdfghjkl", and
- the third row consists of the characters "zxcvbnm".

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0502 Ipo

HARD

Suppose LeetCode will start its IPO soon. In order to sell a good price of its shares to Venture Capital, LeetCode would like to work on some projects to increase its capital before the IPO. Since it has limited resources, it can only finish at most k distinct projects before the IPO. Help LeetCode design the best way to maximize its total capital after finishing at most k distinct projects.

You are given n projects where the ith project has a pure profit profits[i] and a minimum capital of capital[i] is needed to start it.

Initially, you have w capital. When you finish a project, you will obtain its pure profit and the profit will be added to your total capital.

Pick a list of at most k distinct projects from given projects to maximize your final capital, and return the final maximized capital.

The answer is guaranteed to fit in a 32-bit signed integer.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0504 Base 7

EASY

Given an integer num, return a string of its base 7 representation.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0505 The Maze Ii

MEDIUM

There is a ball in a maze with empty spaces (represented as 0) and walls (represented as 1). The ball can go through the empty spaces by rolling up, down, left or right, but it won't stop rolling until hitting a wall. When the ball stops, it could choose the next direction.

Given the m x n maze, the ball's start position and the destination, where start = [startrow, startcol] and destination = [destinationrow, destinationcol], return the shortest distance for the ball to stop at the destination. If the ball cannot stop at destination, return -1.

The distance is the number of empty spaces traveled by the ball from the start position (excluded) to the destination (included).

You may assume that the borders of the maze are all walls (see examples).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0516 Longest Palindromic Subsequence

MEDIUM

Given a string s, find the longest palindromic subsequence's length in s.

A subsequence is a sequence that can be derived from another sequence by deleting some or no elements without changing the order of the remaining elements.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0517 Super Washing Machines

HARD

You have n super washing machines on a line. Initially, each washing machine has some dresses or is empty.

For each move, you could choose any m (1 <= m <= n) washing machines, and pass one dress of each washing machine to one of its adjacent washing machines at the same time.

Given an integer array machines representing the number of dresses in each washing machine from left to right on the line, return the minimum number of moves to make all the washing machines have the same number of dresses. If it is not possible to do it, return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0520 Detect Capital

EASY

We define the usage of capitals in a word to be right when one of the following cases holds:
- All letters in this word are capitals, like "USA".
- All letters in this word are not capitals, like "leetcode".
- Only the first letter in this word is capital, like "Google".

Given a string word, return true if the usage of capitals in it is right.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0521 Longest Uncommon Subsequence I

EASY

Given two strings a and b, return the length of the longest uncommon subsequence between a and b. If no such uncommon subsequence exists, return -1.

An uncommon subsequence between two strings is a string that is a subsequence of exactly one of them.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0522 Longest Uncommon Subsequence Ii

MEDIUM

Given an array of strings strs, return the length of the longest uncommon subsequence between them. If the longest uncommon subsequence does not exist, return -1.

An uncommon subsequence between an array of strings is a string that is a subsequence of one string but not the others.

A subsequence of a string s is a string that can be obtained after deleting any number of characters from s.

For example, "abc" is a subsequence of "aebdc" because you can delete the underlined characters in "aebdc" to get "abc". Other subsequences of "aebdc" include "aebdc", "aeb", and "" (empty string).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0526 Beautiful Arrangement

**MEDIUM**

Suppose you have n integers labeled 1 through n. A permutation of those n integers perm (1-indexed) is considered a beautiful arrangement if for every i (1 <= i <= n), either of the following is true:
- perm[i] is divisible by i.
- i is divisible by perm[i].

Given an integer n, return the number of the beautiful arrangements that you can construct.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0528 Random Pick With Weight

**MEDIUM**

You are given a 0-indexed array of positive integers w where w[i] describes the weight of the ith index.

You need to implement the function pickIndex(), which randomly picks an index in the range [0, w.length - 1] (inclusive) and returns it. The probability of picking an index i is w[i] / sum(w).

For example, if w = [1, 3], the probability of picking index 0 is 1 / (1 + 3) = 0.25 (i.e., 25%), and the probability of picking index 1 is 3 / (1 + 3) = 0.75 (i.e., 75%).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0531 Lonely Pixel I

**MEDIUM**

Given an m x n picture consisting of black 'B' and white 'W' pixels, return the number of black lonely pixels.

A black lonely pixel is a character 'B' that located at a specific position where the same row and same column don't have any other black pixels.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0544 Output Contest Matches

**MEDIUM**

During the NBA playoffs, we always set the rather strong team to play with the rather weak team, like making the rank 1 team play with the rank nth team, which is a good strategy to make the contest more interesting.

Given n teams, return their final contest matches in the form of a string.

The n teams are labeled from 1 to n, which represents their initial rank (i.e., Rank 1 is the strongest team and Rank n is the weakest team).

We will use parentheses '(', and ')' and commas ',' to represent the contest team pairing. We use the parentheses for pairing and the commas for partition. During the pairing process in each round, you always need to follow the strategy of making the rather strong one pair with the rather weak one.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 0551 Student Attendance Record I

EASY

You are given a string s representing an attendance record for a student where each character signifies whether the student was absent, late, or present on that day. The record only contains the following three characters:
- 'A': Absent.
- 'L': Late.
- 'P': Present.

The student is eligible for an attendance award if they meet both of the following criteria:
- The student was absent ('A') for strictly fewer than 2 days total.
- The student was never late ('L') for 3 or more consecutive days.

Return true if the student is eligible for an attendance award, or false otherwise.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 0552 Student Attendance Record Ii

HARD

An attendance record for a student can be represented as a string where each character signifies whether the student was absent, late, or present on that day. The record only contains the following three characters:
- 'A': Absent.
- 'L': Late.
- 'P': Present.

Any student is eligible for an attendance award if they meet both of the following criteria:
- The student was absent ('A') for strictly fewer than 2 days total.
- The student was never late ('L') for 3 or more consecutive days.

Given an integer n, return the number of possible attendance records of length n that make a student eligible for an attendance award. The answer may be very large, so return it modulo 109 + 7.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 0553 Optimal Division

MEDIUM

You are given an integer array nums. The adjacent integers in nums will perform the float division.
- For example, for nums = [2,3,4], we will evaluate the expression "2/3/4".

However, you can add any number of parenthesis at any position to change the priority of operations. You want to add these parentheses such the value of the expression after the evaluation is maximum.

Return the corresponding expression that has the maximum value in string format.

Note: your expression should not contain redundant parenthesis.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 0564 Find The Closest Palindrome

HARD

Given a string n representing an integer, return the closest integer (not including itself), which is a palindrome. If there is a tie, return the smaller one.

The closest is defined as the absolute difference minimized between two integers.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0568 Maximum Vacation Days

HARD

LeetCode wants to give one of its best employees the option to travel among n cities to collect algorithm problems. But all work and no play makes Jack a dull boy, you could take vacations in some particular cities and weeks. Your job is to schedule the traveling to maximize the number of vacation days you could take, but there are certain rules and restrictions you need to follow.

Rules and restrictions:
are in the city indexed 0 on Monday.
necessarily symmetrical), called flights representing the airline status from the city i to the city j. If there is no flight from the city i to the city j, flights[i][j] == 0;
Otherwise, flights[i][j] == 1. Also, flights[i][i] == 0 for all i.
most once per day and can only take flights on each week's Monday morning. Since flight time is so short, we do not consider the impact of flight time.
matrix called days representing this relationship. For the value of days[i][j], it represents the maximum days you could take a vacation in the city i in the week j.
days, which will not be counted as vacation days.
vacation days will count towards the vacation days of city B in that week.

Given the two matrices flights and days, return the maximum vacation days you could take during k weeks.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0573 Squirrel Simulation

MEDIUM

You are given two integers height and width representing a garden of size height x width.
You are also given:
- an array tree where tree = [treer, treec] is the position of the tree in the garden,
- an array squirrel where squirrel = [squirrelr, squirrelc] is the position of the squirrel in the garden,
- and an array nuts where nuts[i] = [nutir, nutic] is the position of the ith nut in the garden.

The squirrel can only take at most one nut at one time and can move in four directions: up, down, left, and right, to the adjacent cell.

Return the minimal distance for the squirrel to collect all the nuts and put them under the tree one by one.

The distance is the number of moves.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0576 Out Of Boundary Paths

MEDIUM

There is an m x n grid with a ball. The ball is initially at the position [startRow, startColumn]. You are allowed to move the ball to one of the four adjacent cells in

the grid (possibly out of the grid crossing the grid boundary). You can apply at most maxMove moves to the ball.

Given the five integers m, n, maxMove, startRow, startColumn, return the number of paths to move the ball out of the grid boundary. Since the answer can be very large, return it modulo 109 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0591 Tag Validator

**HARD**

Given a string representing a code snippet, implement a tag validator to parse the code and return whether it is valid.

A code snippet is valid if all the following rules hold:
<TAG_NAME>TAG_CONTENT</TAG_NAME>. Among them, <TAG_NAME> is the start tag, and </TAG_NAME> is the end tag. The TAG_NAME in start and end tags should be the same. A closed tag is valid if and only if the TAG_NAME and TAG_CONTENT are valid.
the TAG_NAME is invalid.
note1) EXCEPT unmatched <, unmatched start and end tag, and unmatched or closed tags with invalid TAG_NAME. Otherwise, the TAG_CONTENT is invalid.
However, you also need to consider the issue of unbalanced when tags are nested.
the subsequent characters until the next > should be parsed as TAG_NAME (not necessarily valid).
is defined as the characters between <![CDATA[ and the first subsequent ]]>.
to parse CDATA_CONTENT, so even it has some characters that can be parsed as tag (no matter valid or invalid), you should treat it as regular characters.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0592 Fraction Addition And Subtraction

**MEDIUM**

Given a string expression representing an expression of fraction addition and subtraction, return the calculation result in string format.

The final result should be an irreducible fraction. If your final result is an integer, change it to the format of a fraction that has a denominator 1. So in this case, 2 should be converted to 2/1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0598 Range Addition Ii

**EASY**

You are given an m x n matrix M initialized with all 0's and an array of operations ops, where ops[i] = [ai, bi] means M[x][y] should be incremented by one for all 0 <= x < ai and 0 <= y < bi.

Count and return the number of maximum integers in the matrix after performing all the operations.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0600 Non Negative Integers Without Consecutive Ones

**HARD**

Given a positive integer n, return the number of the integers in the range [0, n] whose binary representations do not contain consecutive ones.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 0605 Can Place Flowers

EASY

You have a long flowerbed in which some of the plots are planted, and some are not. However, flowers cannot be planted in adjacent plots.

Given an integer array flowerbed containing 0's and 1's, where 0 means empty and 1 means not empty, and an integer n, return true if n new flowers can be planted in the flowerbed without violating the no-adjacent-flowers rule and false otherwise.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 0625 Minimum Factorization

MEDIUM

Given a positive integer num, return the smallest positive integer x whose multiplication of each digit equals num. If there is no answer or the answer is not fit in 32-bit signed integer, return 0.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 0635 Design Log Storage System

MEDIUM

You are given several logs, where each log contains a unique ID and timestamp. Timestamp is a string that has the following format: Year:Month:Day:Hour:Minute:Second, for example, 2017:01:01:23:59:59. All domains are zero-padded decimal numbers.

Implement the LogSystem class:
- LogSystem() Initializes the LogSystem object.
- void put(int id, string timestamp) Stores the given log (id, timestamp) in your storage system.
- int[] retrieve(string start, string end, string granularity) Returns the IDs of the logs whose timestamps are within the range from start to end inclusive. start and end all have the same format as timestamp, and granularity means how precise the range should be (i.e. to the exact Day, Minute, etc.). For example, start = "2017:01:01:23:59:59", end = "2017:01:02:23:59:59", and granularity = "Day" means that we need to find the logs within the inclusive range from Jan. 1st 2017 to Jan. 2nd 2017, and the Hour, Minute, and Second for each log entry can be ignored.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 0636 Exclusive Time Of Functions

MEDIUM

On a single-threaded CPU, we execute a program containing n functions. Each function has a unique ID between 0 and n-1.

Function calls are stored in a call stack: when a function call starts, its ID is pushed onto the stack, and when a function call ends, its ID is popped off the stack. The function whose ID is at the top of the stack is the current function being executed. Each time a function starts or ends, we write a log with the ID, whether it started or ended, and the timestamp.

You are given a list logs, where logs[i] represents the ith log message formatted as a string "{function_id}:{"start" | "end"}:{timestamp}". For example, "0:start:3" means a function call with function ID 0 started at the beginning of timestamp 3, and "1:end:2" means a function call with function ID 1 ended at the end of timestamp 2. Note that a function can be called

multiple times, possibly recursively.

A function's exclusive time is the sum of execution times for all function calls in the program. For example, if a function is called twice, one call executing for 2 time units and another call executing for 1 time unit, the exclusive time is 2 + 1 = 3.

Return the exclusive time of each function in an array, where the value at the ith index represents the exclusive time for the function with ID i.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0640 Solve The Equation

**MEDIUM**

Solve a given equation and return the value of 'x' in the form of a string "x=#value". The equation contains only '+', '-' operation, the variable 'x' and its coefficient. You should return "No solution" if there is no solution for the equation, or "Infinite solutions" if there are infinite solutions for the equation.

If there is exactly one solution for the equation, we ensure that the value of 'x' is an integer.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0641 Design Circular Deque

**MEDIUM**

Design your implementation of the circular double-ended queue (deque).

Implement the MyCircularDeque class:
- MyCircularDeque(int k) Initializes the deque with a maximum size of k.
- boolean insertFront() Adds an item at the front of Deque. Returns true if the operation is successful, or false otherwise.
- boolean insertLast() Adds an item at the rear of Deque. Returns true if the operation is successful, or false otherwise.
- boolean deleteFront() Deletes an item from the front of Deque. Returns true if the operation is successful, or false otherwise.
- boolean deleteLast() Deletes an item from the rear of Deque. Returns true if the operation is successful, or false otherwise.
- int getFront() Returns the front item from the Deque. Returns -1 if the deque is empty.
- int getRear() Returns the last item from Deque. Returns -1 if the deque is empty.
- boolean isEmpty() Returns true if the deque is empty, or false otherwise.
- boolean isFull() Returns true if the deque is full, or false otherwise.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0642 Design Search Autocomplete System

**HARD**

Design a search autocomplete system for a search engine. Users may input a sentence (at least one word and end with a special character '#').

You are given a string array sentences and an integer array times both of length n where sentences[i] is a previously typed sentence and times[i] is the corresponding number of times the sentence was typed. For each input character except '#', return the top 3 historical hot sentences that have the same prefix as the part of the sentence already typed.

Here are the specific rules:
- The hot degree for a sentence is defined as the number of times a user typed the exactly same sentence before.
- The returned top 3 hot sentences should be sorted by hot degree (The first is the hottest

one). If several sentences have the same hot degree, use ASCII-code order (smaller one appears first).
- If less than 3 hot sentences exist, return as many as you can.
- When the input is a special character, it means the sentence ends, and in this case, you need to return an empty list.

Implement the AutocompleteSystem class:
- AutocompleteSystem(String[] sentences, int[] times) Initializes the object with the sentences and times arrays.
- List<String> input(char c) This indicates that the user typed the character c.
- Returns an empty array [] if c == '#' and stores the inputted sentence in the system.
- Returns the top 3 historical hot sentences that have the same prefix as the part of the sentence already typed. If there are fewer than 3 matches, return them all.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0649 Dota2 Senate

**MEDIUM**

In the world of Dota2, there are two parties: the Radiant and the Dire.

The Dota2 senate consists of senators coming from two parties. Now the Senate wants to decide on a change in the Dota2 game. The voting for this change is a round-based procedure. In each round, each senator can exercise one of the two rights:
- Ban one senator's right: A senator can make another senator lose all his rights in this and all the following rounds.
- Announce the victory: If this senator found the senators who still have rights to vote are all from the same party, he can announce the victory and decide on the change in the game.

Given a string senate representing each senator's party belonging. The character 'R' and 'D' represent the Radiant party and the Dire party. Then if there are n senators, the size of the given string will be n.

The round-based procedure starts from the first senator to the last senator in the given order. This procedure will last until the end of voting. All the senators who have lost their rights will be skipped during the procedure.

Suppose every senator is smart enough and will play the best strategy for his own party. Predict which party will finally announce the victory and change the Dota2 game. The output should be "Radiant" or "Dire".

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0650 2 Keys Keyboard

**MEDIUM**

There is only one character 'A' on the screen of a notepad. You can perform one of two operations on this notepad for each step:
- Copy All: You can copy all the characters present on the screen (a partial copy is not allowed).
- Paste: You can paste the characters which are copied last time.

Given an integer n, return the minimum number of operations to get the character 'A' exactly n times on the screen.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0651 4 Keys Keyboard

MEDIUM

Imagine you have a special keyboard with the following keys:
- A: Print one 'A' on the screen.
- Ctrl-A: Select the whole screen.
- Ctrl-C: Copy selection to buffer.
- Ctrl-V: Print buffer on screen appending it after what has already been printed.

Given an integer n, return the maximum number of 'A' you can print on the screen with at most n presses on the keys.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0656 Coin Path

HARD

You are given an integer array coins (1-indexed) of length n and an integer maxJump. You can jump to any index i of the array coins if coins[i] != -1 and you have to pay coins[i] when you visit index i. In addition to that, if you are currently at index i, you can only jump to any index i + k where i + k <= n and k is a value in the range [1, maxJump].

You are initially positioned at index 1 (coins[1] is not -1). You want to find the path that reaches index n with the minimum cost.

Return an integer array of the indices that you will visit in order so that you can reach index n with the minimum cost. If there are multiple paths with the same cost, return the lexicographically smallest such path. If it is not possible to reach index n, return an empty array.

A path p1 = [Pa1, Pa2, ..., Pax] of length x is lexicographically smaller than p2 = [Pb1, Pb2, ..., Pbx] of length y, if and only if at the first j where Paj and Pbj differ, Paj < Pbj; when no such j exists, then x < y.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0657 Robot Return To Origin

EASY

There is a robot starting at the position (0, 0), the origin, on a 2D plane. Given a sequence of its moves, judge if this robot ends up at (0, 0) after it completes its moves.

You are given a string moves that represents the move sequence of the robot where moves[i] represents its ith move. Valid moves are 'R' (right), 'L' (left), 'U' (up), and 'D' (down).

Return true if the robot returns to the origin after it finishes all of its moves, or false otherwise.

Note: The way that the robot is "facing" is irrelevant. 'R' will always make the robot move to the right once, 'L' will always make it move left, etc. Also, assume that the magnitude of the robot's movement is the same for each move.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0660 Remove 9

HARD

Start from integer 1, remove any integer that contains 9 such as 9, 19, 29...

Now, you will have a new integer sequence [1, 2, 3, 4, 5, 6, 7, 8, 10, 11, ...].

Given an integer n, return the nth (1-indexed) integer in the new sequence.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0661 Image Smoother

**EASY**

An image smoother is a filter of the size 3 x 3 that can be applied to each cell of an image by rounding down the average of the cell and the eight surrounding cells (i.e., the average of the nine cells in the blue smoother). If one or more of the surrounding cells of a cell is not present, we do not consider it in the average (i.e., the average of the four cells in the red smoother).

Given an m x n integer matrix img representing the grayscale of an image, return the image after applying the smoother on each cell of it.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0664 Strange Printer

**HARD**

There is a strange printer with the following two special properties:
- The printer can only print a sequence of the same character each time.
- At each turn, the printer can print new characters starting from and ending at any place and will cover the original existing characters.

Given a string s, return the minimum number of turns the printer needed to print it.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0667 Beautiful Arrangement Ii

**MEDIUM**

Given two integers n and k, construct a list answer that contains n different positive integers ranging from 1 to n and obeys the following requirement:
- Suppose this list is answer = [a1, a2, a3, ... , an], then the list [|a1 - a2|, |a2 - a3|, |a3 - a4|, ... , |an-1 - an|] has exactly k distinct integers.

Return the list answer. If there multiple valid answers, return any of them.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0670 Maximum Swap

**MEDIUM**

You are given an integer num. You can swap two digits at most once to get the maximum valued number.

Return the maximum valued number you can get.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0672 Bulb Switcher Ii

**MEDIUM**

There is a room with n bulbs labeled from 1 to n that all are turned on initially, and four buttons on the wall. Each of the four buttons has a different functionality where:
- Button 1: Flips the status of all the bulbs.
- Button 2: Flips the status of all the bulbs with even labels (i.e., 2, 4, ...).
- Button 3: Flips the status of all the bulbs with odd labels (i.e., 1, 3, ...).

- Button 4: Flips the status of all the bulbs with a label j = 3k + 1 where k = 0, 1, 2, ... (i.e., 1, 4, 7, 10, ...).
- You must make exactly presses button presses in total. For each press, you may pick any of the four buttons to press.

Given the two integers n and presses, return the number of different possible statuses after performing all presses button presses.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0674 Longest Continuous Increasing Subsequence

**EASY**

Given an unsorted array of integers nums, return the length of the longest continuous increasing subsequence (i.e. subarray). The subsequence must be strictly increasing.

A continuous increasing subsequence is defined by two indices l and r (l < r) such that it is [nums[l], nums[l + 1], ..., nums[r - 1], nums[r]] and for each l <= i < r, nums[i] < nums[i + 1].

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0679 24 Game

**HARD**

You are given an integer array cards of length 4. You have four cards, each containing a number in the range [1, 9]. You should arrange the numbers on these cards in a mathematical expression using the operators ['+', '-', '*', '/'] and the parentheses '(' and ')' to get the value 24.

You are restricted with the following rules:
- The division operator '/' represents real division, not integer division.
- For example, 4 / (1 - 2 / 3) = 4 / (1 / 3) = 12.
- Every operation done is between two numbers. In particular, we cannot use '-' as a unary operator.
- For example, if cards = [1, 1, 1, 1], the expression "-1 - 1 - 1 - 1" is not allowed.
- You cannot concatenate numbers together
- For example, if cards = [1, 2, 1, 2], the expression "12 + 12" is not valid.

Return true if you can get such expression that evaluates to 24, and false otherwise.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0680 Valid Palindrome Ii

**EASY**

Given a string s, return true if the s can be palindrome after deleting at most one character from it.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0682 Baseball Game

**EASY**

You are keeping the scores for a baseball game with strange rules. At the beginning of the game, you start with an empty record.

You are given a list of strings operations, where operations[i] is the ith operation you must apply to the record and is one of the following:
- An integer x.
- Record a new score of x.
- '+'.

- Record a new score that is the sum of the previous two scores.
- 'D'.
- Record a new score that is the double of the previous score.
- 'C'.
- Invalidate the previous score, removing it from the record.

Return the sum of all the scores on the record after applying all the operations.

The test cases are generated such that the answer and all intermediate calculations fit in a 32-bit integer and that all operations are valid.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0683 K Empty Slots

**HARD**

You have n bulbs in a row numbered from 1 to n. Initially, all the bulbs are turned off. We turn on exactly one bulb every day until all bulbs are on after n days.

You are given an array bulbs of length n where bulbs[i] = x means that on the (i+1)th day, we will turn on the bulb at position x where i is 0-indexed and x is 1-indexed.

Given an integer k, return the minimum day number such that there exists two turned on bulbs that have exactly k bulbs between them that are all turned off. If there isn't such day, return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0685 Redundant Connection Ii

**HARD**

In this problem, a rooted tree is a directed graph such that, there is exactly one node (the root) for which all other nodes are descendants of this node, plus every node has exactly one parent, except for the root node which has no parents.

The given input is a directed graph that started as a rooted tree with n nodes (with distinct values from 1 to n), with one additional directed edge added. The added edge has two different vertices chosen from 1 to n, and was not an edge that already existed.

The resulting graph is given as a 2D-array of edges. Each element of edges is a pair [ui, vi] that represents a directed edge connecting nodes ui and vi, where ui is a parent of child vi.

Return an edge that can be removed so that the resulting graph is a rooted tree of n nodes. If there are multiple answers, return the answer that occurs last in the given 2D-array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0709 To Lower Case

**EASY**

Given a string s, return the string after replacing every uppercase letter with the same lowercase letter.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0714 Best Time To Buy And Sell Stock With Transaction Fee

**MEDIUM**

You are given an array prices where prices[i] is the price of a given stock on the ith day, and an integer fee representing a transaction fee.

Find the maximum profit you can achieve. You may complete as many transactions as you like, but you need to pay the transaction fee for each transaction.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0715 Range Module

**HARD**

A Range Module is a module that tracks ranges of numbers. Design a data structure to track the ranges represented as half-open intervals and query about them.

A half-open interval [left, right) denotes all the real numbers x where left <= x < right.

Implement the RangeModule class:
- RangeModule() Initializes the object of the data structure.
- void addRange(int left, int right) Adds the half-open interval [left, right), tracking every real number in that interval. Adding an interval that partially overlaps with currently tracked numbers should add any numbers in the interval [left, right) that are not already tracked.
- boolean queryRange(int left, int right) Returns true if every real number in the interval [left, right) is currently being tracked, and false otherwise.
- void removeRange(int left, int right) Stops tracking every real number currently being tracked in the half-open interval [left, right).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0722 Remove Comments

**MEDIUM**

Given a C++ program, remove comments from it. The program source is an array where source[i] is the i-th line of the source code. This represents the result of splitting the original source code string by the newline character \n.
In C++, there are two types of comments, line comments, and block comments.
The string // denotes a line comment, which represents that it and rest of the characters to the right of it in the same line should be ignored.
The string /* denotes a block comment, which represents that all characters until the next (non-overlapping) occurrence of should be ignored.
It is guaranteed that every open block comment will eventually be closed, so /* outside of a line or block comment always starts a new comment.
Finally, implicit newline characters can be deleted by block comments.
After removing the comments from the source code, return the source code in the same format.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0723 Candy Crush

**MEDIUM**

This question is about implementing a basic elimination algorithm for Candy Crush.

Given an m x n integer array board representing the grid of candy where board[i][j] represents the type of candy. A value of board[i][j] == 0 represents that the cell is empty.

The given board represents the state of the game following the player's move. Now, you need to restore the board to a stable state by crushing candies according to the following rules:
- If three or more candies of the same type are adjacent vertically or horizontally, crush them all at the same time - these positions become empty.
- After crushing all candies simultaneously, if an empty space on the board has candies on top of itself, then these candies will drop until they hit a candy or bottom at the same time. No new candies will drop outside the top boundary.
- After the above steps, there may exist more candies that can be crushed. If so, you need to repeat the above steps.

- If there does not exist more candies that can be crushed (i.e., the board is stable), then return the current board.

You need to perform the above rules until the board becomes stable, then return the stable board.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0724 Find Pivot Index

**EASY**

Given an array of integers nums, calculate the pivot index of this array.

The pivot index is the index where the sum of all the numbers strictly to the left of the index is equal to the sum of all the numbers strictly to the index's right.

If the index is on the left edge of the array, then the left sum is 0 because there are no elements to the left. This also applies to the right edge of the array.

Return the leftmost pivot index. If no such index exists, return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0729 My Calendar I

**MEDIUM**

You are implementing a program to use as your calendar. We can add a new event if adding the event will not cause a double booking.

A double booking happens when two events have some non-empty intersection (i.e., some moment is common to both events.).

The event can be represented as a pair of integers startTime and endTime that represents a booking on the half-open interval [startTime, endTime), the range of real numbers x such that startTime <= x < endTime.

Implement the MyCalendar class:
- MyCalendar() Initializes the calendar object.
- boolean book(int startTime, int endTime) Returns true if the event can be added to the calendar successfully without causing a double booking. Otherwise, return false and do not add the event to the calendar.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0731 My Calendar Ii

**MEDIUM**

You are implementing a program to use as your calendar. We can add a new event if adding the event will not cause a triple booking.

A triple booking happens when three events have some non-empty intersection (i.e., some moment is common to all the three events.).

The event can be represented as a pair of integers startTime and endTime that represents a booking on the half-open interval [startTime, endTime), the range of real numbers x such that startTime <= x < endTime.

Implement the MyCalendarTwo class:
- MyCalendarTwo() Initializes the calendar object.
- boolean book(int startTime, int endTime) Returns true if the event can be added to the calendar successfully without causing a triple booking. Otherwise, return false and do not add the event to the calendar.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0732 My Calendar Iii

HARD

A k-booking happens when k events have some non-empty intersection (i.e., there is some time that is common to all k events.)

You are given some events [startTime, endTime), after each given event, return an integer k representing the maximum k-booking between all the previous events.

Implement the MyCalendarThree class:
- MyCalendarThree() Initializes the object.
- int book(int startTime, int endTime) Returns an integer k representing the largest integer such that there exists a k-booking in the calendar.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0735 Asteroid Collision

MEDIUM

We are given an array asteroids of integers representing asteroids in a row. The indices of the asteriod in the array represent their relative position in space.

For each asteroid, the absolute value represents its size, and the sign represents its direction (positive meaning right, negative meaning left). Each asteroid moves at the same speed.

Find out the state of the asteroids after all collisions. If two asteroids meet, the smaller one will explode. If both are the same size, both will explode. Two asteroids moving in the same direction will never meet.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0740 Delete And Earn

MEDIUM

You are given an integer array nums. You want to maximize the number of points you get by performing the following operation any number of times:
- Pick any nums[i] and delete it to earn nums[i] points. Afterwards, you must delete every element equal to nums[i] - 1 and every element equal to nums[i] + 1.

Return the maximum number of points you can earn by applying the above operation some number of times.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0741 Cherry Pickup

HARD

You are given an n x n grid representing a field of cherries, each cell is one of three possible integers.
- 0 means the cell is empty, so you can pass through,
- 1 means the cell contains a cherry that you can pick up and pass through, or
- -1 means the cell contains a thorn that blocks your way.

Return the maximum number of cherries you can collect by following the rules below:
- Starting at the position (0, 0) and reaching (n - 1, n - 1) by moving right or down through valid path cells (cells with value 0 or 1).
- After reaching (n - 1, n - 1), returning to (0, 0) by moving left or up through valid path cells.
- When passing through a path cell containing a cherry, you pick it up, and the cell becomes

an empty cell 0.
- If there is no valid path between (0, 0) and (n - 1, n - 1), then no cherries can be collected.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0751 Ip To Cidr

**MEDIUM**

An IP address is a formatted 32-bit unsigned integer where each group of 8 bits is printed as a decimal number and the dot character '.' splits the groups.
- For example, the binary number 00001111 10001000 11111111 01101011 (spaces added for clarity) formatted as an IP address would be "15.136.255.107".

A CIDR block is a format used to denote a specific set of IP addresses. It is a string consisting of a base IP address, followed by a slash, followed by a prefix length k. The addresses it covers are all the IPs whose first k bits are the same as the base IP address.
- For example, "123.45.67.89/20" is a CIDR block with a prefix length of 20. Any IP address whose binary representation matches 01111011 00101101 0100xxxx xxxxxxxx, where x can be either 0 or 1, is in the set covered by the CIDR block.

You are given a start IP address ip and the number of IP addresses we need to cover n. Your goal is to use as few CIDR blocks as possible to cover all the IP addresses in the inclusive range [ip, ip + n - 1] exactly. No other IP addresses outside of the range should be covered.

Return the shortest list of CIDR blocks that covers the range of IP addresses. If there are multiple answers, return any of them.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0755 Pour Water

**MEDIUM**

You are given an elevation map represents as an integer array heights where heights[i] representing the height of the terrain at index i. The width at each index is 1. You are also given two integers volume and k. volume units of water will fall at index k.

Water first drops at the index k and rests on top of the highest terrain or water at that index. Then, it flows according to the following rules:
- If the droplet would eventually fall by moving left, then move left.
- Otherwise, if the droplet would eventually fall by moving right, then move right.
- Otherwise, rise to its current position.

Here, "eventually fall" means that the droplet will eventually be at a lower level if it moves in that direction. Also, level means the height of the terrain plus any water in that column.

We can assume there is infinitely high terrain on the two sides out of bounds of the array. Also, there could not be partial water being spread out evenly on more than one grid block, and each unit of water has to be in exactly one block.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0759 Employee Free Time

**HARD**

We are given a list schedule of employees, which represents the working time for each employee.

Each employee has a list of non-overlapping Intervals, and these intervals are in sorted order.

Return the list of finite intervals representing common, positive-length free time for all employees, also in sorted order.

(Even though we are representing Intervals in the form [x, y], the objects inside are Intervals,

not lists or arrays. For example, schedule[0][0].start = 1, schedule[0][0].end = 2, and schedule[0][0][0] is not defined). Also, we wouldn't include intervals like [5, 5] in our answer, as they have zero length.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0764 Largest Plus Sign

MEDIUM

You are given an integer n. You have an n x n binary grid grid with all values initially 1's except for some indices given in the array mines. The ith element of the array mines is defined as mines[i] = [xi, yi] where grid[xi][yi] == 0.

Return the order of the largest axis-aligned plus sign of 1's contained in grid. If there is none, return 0.

An axis-aligned plus sign of 1's of order k has some center grid[r][c] == 1 along with four arms of length k - 1 going up, down, left, and right, and made of 1's. Note that there could be 0's or 1's beyond the arms of the plus sign, only the relevant area of the plus sign is checked for 1's.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0765 Couples Holding Hands

HARD

There are n couples sitting in 2n seats arranged in a row and want to hold hands.

The people and seats are represented by an integer array row where row[i] is the ID of the person sitting in the ith seat. The couples are numbered in order, the first couple being (0, 1), the second couple being (2, 3), and so on with the last couple being (2n - 2, 2n - 1).

Return the minimum number of swaps so that every couple is sitting side by side. A swap consists of choosing any two people, then they stand up and switch seats.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0775 Global And Local Inversions

MEDIUM

You are given an integer array nums of length n which represents a permutation of all the integers in the range [0, n - 1].

The number of global inversions is the number of the different pairs (i, j) where:
- 0 <= i < j < n
- nums[i] > nums[j]

The number of local inversions is the number of indices i where:
- 0 <= i < n - 1
- nums[i] > nums[i + 1]

Return true if the number of global inversions is equal to the number of local inversions.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0778 Swim In Rising Water

HARD

You are given an n x n integer matrix grid where each value grid[i][j] represents the elevation at that point (i, j).

The rain starts to fall. At time t, the depth of the water everywhere is t. You can swim from

a square to another 4-directionally adjacent square if and only if the elevation of both squares individually are at most t. You can swim infinite distances in zero time. Of course, you must stay within the boundaries of the grid during your swim.

Return the least time until you can reach the bottom right square (n - 1, n - 1) if you start at the top left square (0, 0).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0779 K Th Symbol In Grammar

**MEDIUM**

We build a table of n rows (1-indexed). We start by writing 0 in the 1st row. Now in every subsequent row, we look at the previous row and replace each occurrence of 0 with 01, and each occurrence of 1 with 10.

- For example, for n = 3, the 1st row is 0, the 2nd row is 01, and the 3rd row is 0110.

Given two integer n and k, return the kth (1-indexed) symbol in the nth row of a table of n rows.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0780 Reaching Points

**HARD**

Given four integers sx, sy, tx, and ty, return true if it is possible to convert the point (sx, sy) to the point (tx, ty) through some operations, or false otherwise.

The allowed operation on some point (x, y) is to convert it to either (x, x + y) or (x + y, y).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0781 Rabbits In Forest

**MEDIUM**

There is a forest with an unknown number of rabbits. We asked n rabbits "How many rabbits have the same color as you?" and collected the answers in an integer array answers where answers[i] is the answer of the ith rabbit.

Given the array answers, return the minimum number of rabbits that could be in the forest.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0782 Transform To Chessboard

**HARD**

You are given an n x n binary grid board. In each move, you can swap any two rows with each other, or any two columns with each other.

Return the minimum number of moves to transform the board into a chessboard board. If the task is impossible, return -1.

A chessboard board is a board where no 0's and no 1's are 4-directionally adjacent.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0786 K Th Smallest Prime Fraction

**MEDIUM**

You are given a sorted integer array arr containing 1 and prime numbers, where all the integers of arr are unique. You are also given an integer k.

For every i and j where 0 <= i < j < arr.length, we consider the fraction arr[i] / arr[j].

Return the kth smallest fraction considered. Return your answer as an array of integers of size 2, where answer[0] == arr[i] and answer[1] == arr[j].

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0789 Escape The Ghosts

**MEDIUM**

You are playing a simplified PAC-MAN game on an infinite 2-D grid. You start at the point [0, 0], and you are given a destination point target = [xtarget, ytarget] that you are trying to get to. There are several ghosts on the map with their starting positions given as a 2D array ghosts, where ghosts[i] = [xi, yi] represents the starting position of the ith ghost. All inputs are integral coordinates.

Each turn, you and all the ghosts may independently choose to either move 1 unit in any of the four cardinal directions: north, east, south, or west, or stay still. All actions happen simultaneously.

You escape if and only if you can reach the target before any ghost reaches you. If you reach any square (including the target) at the same time as a ghost, it does not count as an escape.

Return true if it is possible to escape regardless of how the ghosts move, otherwise return false.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0790 Domino And Tromino Tiling

**MEDIUM**

You have two types of tiles: a 2 x 1 domino shape and a tromino shape. You may rotate these shapes.

Given an integer n, return the number of ways to tile an 2 x n board. Since the answer may be very large, return it modulo 109 + 7.

In a tiling, every square must be covered by a tile. Two tilings are different if and only if there are two 4-directionally adjacent cells on the board such that exactly one of the tilings has both squares occupied by a tile.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0793 Preimage Size Of Factorial Zeroes Function

**HARD**

Let f(x) be the number of zeroes at the end of x!. Recall that x! = 1 * 2 * 3 * ... * x and by convention, 0! = 1.

For example, f(3) = 0 because 3! = 6 has no zeroes at the end, while f(11) = 2 because 11! = 39916800 has two zeroes at the end.

Given an integer k, return the number of non-negative integers x have the property that f(x) = k.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0794 Valid Tic Tac Toe State

**MEDIUM**

Given a Tic-Tac-Toe board as a string array board, return true if and only if it is possible to reach this board position during the course of a valid tic-tac-toe game.

The board is a 3 x 3 array that consists of characters ' ', 'X', and 'O'. The ' ' character represents an empty square.

Here are the rules of Tic-Tac-Toe:
- Players take turns placing characters into empty squares ' '.
- The first player always places 'X' characters, while the second player always places 'O' characters.
- 'X' and 'O' characters are always placed into empty squares, never filled ones.
- The game ends when there are three of the same (non-empty) character filling any row, column, or diagonal.
- The game also ends if all squares are non-empty.
- No more moves can be played if the game is over.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0798 Smallest Rotation With Highest Score

**HARD**

You are given an array nums. You can rotate it by a non-negative integer k so that the array becomes [nums[k], nums[k + 1], ... nums[nums.length - 1], nums[0], nums[1], ..., nums[k-1]]. Afterward, any entries that are less than or equal to their index are worth one point.

For example, if we have nums = [2,4,1,3,0], and we rotate by k = 2, it becomes [1,3,0,2,4]. This is worth 3 points because 1 > 0 [no points], 3 > 1 [no points], 0 <= 2 [one point], 2 <= 3 [one point], 4 <= 4 [one point].

Return the rotation index k that corresponds to the highest score we can achieve if we rotated nums by it. If there are multiple answers, return the smallest such index k.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0799 Champagne Tower

**MEDIUM**

We stack glasses in a pyramid, where the first row has 1 glass, the second row has 2 glasses, and so on until the 100th row. Each glass holds one cup of champagne.

Then, some champagne is poured into the first glass at the top. When the topmost glass is full, any excess liquid poured will fall equally to the glass immediately to the left and right of it. When those glasses become full, any excess champagne will fall equally to the left and right of those glasses, and so on. (A glass at the bottom row has its excess champagne fall on the floor.)

For example, after one cup of champagne is poured, the top most glass is full. After two cups of champagne are poured, the two glasses on the second row are half full. After three cups of champagne are poured, those two cups become full - there are 3 full glasses total now. After four cups of champagne are poured, the third row has the middle glass half full, and the two outside glasses are a quarter full, as pictured below.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0800 Similar Rgb Color

**EASY**

The red-green-blue color "#AABBCC" can be written as "#ABC" in shorthand.
- For example, "#15c" is shorthand for the color "#1155cc".

The similarity between the two colors "#ABCDEF" and "#UVWXYZ" is -(AB - UV)2 - (CD - WX)2 - (EF - YZ)2.

Given a string color that follows the format "#ABCDEF", return a string represents the color that is most similar to the given color and has a shorthand (i.e., it can

be represented as some "#XYZ").

Any answer which has the same highest similarity as the best answer will be accepted.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0801 Minimum Swaps To Make Sequences Increasing

**HARD**

You are given two integer arrays of the same length nums1 and nums2. In one operation, you are allowed to swap nums1[i] with nums2[i].

For example, if nums1 = [1,2,3,8], and nums2 = [5,6,7,4], you can swap the element at i = 3 to obtain nums1 = [1,2,3,4] and nums2 = [5,6,7,8].

Return the minimum number of needed operations to make nums1 and nums2 strictly increasing. The test cases are generated so that the given input always makes it possible.

An array arr is strictly increasing if and only if
arr[0] < arr[1] < arr[2] < ... < arr[arr.length - 1].

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0802 Find Eventual Safe States

**MEDIUM**

There is a directed graph of n nodes with each node labeled from 0 to n - 1. The graph is represented by a 0-indexed 2D integer array graph where graph[i] is an integer array of nodes adjacent to node i, meaning there is an edge from node i to each node in graph[i].

A node is a terminal node if there are no outgoing edges. A node is a safe node if every possible path starting from that node leads to a terminal node (or another safe node).

Return an array containing all the safe nodes of the graph. The answer should be sorted in ascending order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0803 Bricks Falling When Hit

**HARD**

You are given an m x n binary grid, where each 1 represents a brick and 0 represents an empty space. A brick is stable if:
- It is directly connected to the top of the grid, or
- At least one other brick in its four adjacent cells is stable.

You are also given an array hits, which is a sequence of erasures we want to apply. Each time we want to erase the brick at the location hits[i] = (rowi, coli). The brick on that location (if it exists) will disappear. Some other bricks may no longer be stable because of that erasure and will fall. Once a brick falls, it is immediately erased from the grid (i.e., it does not land on other stable bricks).

Return an array result, where each result[i] is the number of bricks that will fall after the ith erasure is applied.

Note that an erasure may refer to a location with no brick, and if it does, no bricks drop.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0807 Max Increase To Keep City Skyline

MEDIUM

There is a city composed of n x n blocks, where each block contains a single building shaped like a vertical square prism. You are given a 0-indexed n x n integer matrix grid where grid[r][c] represents the height of the building located in the block at row r and column c.

A city's skyline is the outer contour formed by all the building when viewing the side of the city from a distance. The skyline from each cardinal direction north, east, south, and west may be different.

We are allowed to increase the height of any number of buildings by any amount (the amount can be different per building). The height of a 0-height building can also be increased. However, increasing the height of a building should not affect the city's skyline from any cardinal direction.

Return the maximum total sum that the height of the buildings can be increased by without changing the city's skyline from any cardinal direction.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0810 Chalkboard Xor Game

HARD

You are given an array of integers nums represents the numbers written on a chalkboard.

Alice and Bob take turns erasing exactly one number from the chalkboard, with Alice starting first. If erasing a number causes the bitwise XOR of all the elements of the chalkboard to become 0, then that player loses. The bitwise XOR of one element is that element itself, and the bitwise XOR of no elements is 0.

Also, if any player starts their turn with the bitwise XOR of all the elements of the chalkboard equal to 0, then that player wins.

Return true if and only if Alice wins the game, assuming both players play optimally.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0812 Largest Triangle Area

EASY

Given an array of points on the X-Y plane points where points[i] = [xi, yi], return the area of the largest triangle that can be formed by any three different points. Answers within 10-5 of the actual answer will be accepted.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0816 Ambiguous Coordinates

MEDIUM

We had some 2-dimensional coordinates, like "(1, 3)" or "(2, 0.5)". Then, we removed all commas, decimal points, and spaces and ended up with the string s.
- For example, "(1, 3)" becomes s = "(13)" and "(2, 0.5)" becomes s = "(205)".

Return a list of strings representing all possibilities for what our original coordinates could have been.

Our original representation never had extraneous zeroes, so we never started with numbers like "00", "0.0", "0.00", "1.0", "001", "00.01", or any other number that can be represented with fewer digits. Also, a decimal point within a number never occurs without at least one digit occurring before it, so we never started with numbers like ".1".

The final answer list can be returned in any order. All coordinates in the final answer have exactly one space between them (occurring after the comma.)

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0818 Race Car

**HARD**

Your car starts at position 0 and speed +1 on an infinite number line. Your car can go into negative positions. Your car drives automatically according to a sequence of instructions 'A' (accelerate) and 'R' (reverse):
- When you get an instruction 'A', your car does the following:
- position += speed
- speed *= 2
- When you get an instruction 'R', your car does the following:
- If your speed is positive then speed = -1
- otherwise speed = 1

Your position stays the same.

For example, after commands "AAR", your car goes to positions 0 --> 1 --> 3 --> 3, and your speed goes to 1 --> 2 --> 4 --> -1.

Given a target position target, return the length of the shortest sequence of instructions to get there.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0824 Goat Latin

**EASY**

A sentence S is given, composed of words separated by spaces.
Each word consists of lowercase and uppercase letters only.

We would like to convert the sentence to "Goat Latin"
(a made-up language similar to Pig Latin.)

The rules of Goat Latin are as follows:
- If a word begins with a vowel (a, e, i, o, or u), append "ma" to the end of the word. For example, the word 'apple' becomes 'applema'.
- If a word begins with a consonant (i.e. not a vowel), remove the first letter and append it to the end, then add "ma".
For example, the word "goat" becomes "oatgma".
- Add one letter 'a' to the end of each word per its word index in the sentence, starting with 1.
For example, the first word gets "a" added to the end, the second word gets "aa" added to the end and so on.

Return the final sentence representing the conversion from S to Goat Latin.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0825 Friends Of Appropriate Ages

**MEDIUM**

There are n persons on a social media website. You are given an integer array ages where ages[i] is the age of the ith person.

A Person x will not send a friend request to a person y (x != y) if any of the following conditions is true:
- age[y] <= 0.5 * age[x] + 7
- age[y] > age[x]

- age[y] > 100 && age[x] < 100

Otherwise, x will send a friend request to y.

Note that if x sends a request to y, y will not necessarily send a request to x. Also, a person will not send a friend request to themself.

Return the total number of friend requests made.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0826 Most Profit Assigning Work

**MEDIUM**

You have n jobs and m workers. You are given three arrays: difficulty, profit, and worker where:
- difficulty[i] and profit[i] are the difficulty and the profit of the ith job, and
- worker[j] is the ability of jth worker (i.e., the jth worker can only complete a job with difficulty at most worker[j]).

Every worker can be assigned at most one job, but one job can be completed multiple times.
- For example, if three workers attempt the same job that pays $1, then the total profit will be $3. If a worker cannot complete any job, their profit is $0.

Return the maximum profit we can achieve after assigning the workers to the jobs.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0830 Positions Of Large Groups

**EASY**

In a string s of lowercase letters, these letters form consecutive groups of the same character.

For example, a string like s = "abbxxxxzyy" has the groups "a", "bb", "xxxx", "z", and "yy".

A group is identified by an interval [start, end], where start and end denote the start and end indices (inclusive) of the group. In the above example, "xxxx" has the interval [3,6].

A group is considered large if it has 3 or more characters.

Return the intervals of every large group sorted in increasing order by start index.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0831 Masking Personal Information

**MEDIUM**

We are given a personal information string S, which may represent either an email address or a phone number.
We would like to mask this personal information according to the following rules:
We define a name to be a string of length ≥ 2 consisting of only lowercase letters a-z or uppercase letters A-Z.
An email address starts with a name, followed by the symbol '@', followed by a name, followed by the dot '.' and followed by a name. All email addresses are guaranteed to be valid and in the format of "name1@name2.name3".
To mask an email, all names must be converted to lowercase and all letters between the first and last letter of the first name must be replaced by 5 asterisks '*'.

A phone number is a string consisting of only the digits 0-9 or the characters from the set {'+', '-', '(', ')', ' '}. You may assume a phone number contains 10 to 13 digits. The last 10 digits make up the local number, while the digits before those make up the country code. Note that the country code is optional. We want to expose only the last 4 digits and mask all other digits. The local number should be formatted and masked as "***-***-1111", where 1 represents the exposed digits. To mask a phone number with country code like "+111 111 111 1111", we write it in the form "+***-***-***-1111". The '+' sign and the first '-' sign before the local number should only exist if there is a country code. For example, a 12 digit phone number mask should start with "+**-". Note that extraneous characters like "(", ")", " ", as well as extra dashes or plus signs not part of the above formatting scheme should be removed.

Return the correct "mask" of the information provided.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0832 Flipping An Image

**EASY**

Given an n x n binary matrix image, flip the image horizontally, then invert it, and return the resulting image.

To flip an image horizontally means that each row of the image is reversed.
- For example, flipping [1,1,0] horizontally results in [0,1,1].

To invert an image means that each 0 is replaced by 1, and each 1 is replaced by 0.
- For example, inverting [0,1,1] results in [1,0,0].

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0836 Rectangle Overlap

**EASY**

An axis-aligned rectangle is represented as a list [x1, y1, x2, y2], where (x1, y1) is the coordinate of its bottom-left corner, and (x2, y2) is the coordinate of its top-right corner. Its top and bottom edges are parallel to the X-axis, and its left and right edges are parallel to the Y-axis.

Two rectangles overlap if the area of their intersection is positive. To be clear, two rectangles that only touch at the corner or edges do not overlap.

Given two axis-aligned rectangles rec1 and rec2, return true if they overlap, otherwise return false.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0837 New 21 Game

**MEDIUM**

Alice plays the following game, loosely based on the card game "21".

Alice starts with 0 points and draws numbers while she has less than k points. During each draw, she gains an integer number of points randomly from the range [1, maxPts], where maxPts is an integer. Each draw is independent and the outcomes have equal probabilities.

Alice stops drawing numbers when she gets k or more points.

Return the probability that Alice has n or fewer points.

Answers within 10-5 of the actual answer are considered accepted.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0838 Push Dominoes

**MEDIUM**

There are n dominoes in a line, and we place each domino vertically upright. In the beginning, we simultaneously push some of the dominoes either to the left or to the right.

After each second, each domino that is falling to the left pushes the adjacent domino on the left. Similarly, the dominoes falling to the right push their adjacent dominoes standing on the right.

When a vertical domino has dominoes falling on it from both sides, it stays still due to the balance of the forces.

For the purposes of this question, we will consider that a falling domino expends no additional force to a falling or already fallen domino.

You are given a string dominoes representing the initial state where:
- dominoes[i] = 'L', if the ith domino has been pushed to the left,
- dominoes[i] = 'R', if the ith domino has been pushed to the right, and
- dominoes[i] = '.', if the ith domino has not been pushed.

Return a string representing the final state.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0849 Maximize Distance To Closest Person

**MEDIUM**

You are given an array representing a row of seats where seats[i] = 1 represents a person sitting in the ith seat, and seats[i] = 0 represents that the ith seat is empty (0-indexed).

There is at least one empty seat, and at least one person sitting.

Alex wants to sit in the seat such that the distance between him and the closest person to him is maximized.

Return that maximum distance to the closest person.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0853 Car Fleet

**MEDIUM**

There are n cars at given miles away from the starting mile 0, traveling to reach the mile target.

You are given two integer array position and speed, both of length n, where position[i] is the starting mile of the ith car and speed[i] is the speed of the ith car in miles per hour.

A car cannot pass another car, but it can catch up and then travel next to it at the speed of the slower car.

A car fleet is a car or cars driving next to each other. The speed of the car fleet is the minimum speed of any car in the fleet.

If a car catches up to a car fleet at the mile target, it will still be considered as part of the car fleet.

Return the number of car fleets that will arrive at the destination.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0855 Exam Room

**MEDIUM**

There is an exam room with n seats in a single row labeled from 0 to n - 1.

When a student enters the room, they must sit in the seat that maximizes the distance to the closest person. If there are multiple such seats, they sit in the seat with the lowest number. If no one is in the room, then the student sits at seat number 0.

Design a class that simulates the mentioned exam room.

Implement the ExamRoom class:
- ExamRoom(int n) Initializes the object of the exam room with the number of the seats n.
- int seat() Returns the label of the seat at which the next student will set.
- void leave(int p) Indicates that the student sitting at seat p will leave the room. It is guaranteed that there will be a student sitting at seat p.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 0857 Minimum Cost To Hire K Workers

HARD

There are n workers. You are given two integer arrays quality and wage where quality[i] is the quality of the ith worker and wage[i] is the minimum wage expectation for the ith worker.

We want to hire exactly k workers to form a paid group. To hire a group of k workers, we must pay them according to the following rules:
if a worker's quality is double that of another worker in the group, then they must be paid twice as much as the other worker.

Given the integer k, return the least amount of money needed to form a paid group satisfying the above conditions. Answers within 10-5 of the actual answer will be accepted.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 0858 Mirror Reflection

MEDIUM

There is a special square room with mirrors on each of the four walls. Except for the southwest corner, there are receptors on each of the remaining corners, numbered 0, 1, and 2.

The square room has walls of length p and a laser ray from the southwest corner first meets the east wall at a distance q from the 0th receptor.

Given the two integers p and q, return the number of the receptor that the ray meets first.

The test cases are guaranteed so that the ray will meet a receptor eventually.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 0860 Lemonade Change

EASY

At a lemonade stand, each lemonade costs $5. Customers are standing in a queue to buy from you and order one at a time (in the order specified by bills). Each customer will only buy one lemonade and pay with either a $5, $10, or $20 bill. You must provide the correct change to each customer so that the net transaction is that the customer pays $5.

Note that you do not have any change in hand at first.

Given an integer array bills where bills[i] is the bill the ith customer pays, return true if you can provide every customer with the correct change, or false otherwise.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 0866 Prime Palindrome

MEDIUM

Given an integer n, return the smallest prime palindrome greater than or equal to n.

An integer is prime if it has exactly two divisors: 1 and itself. Note that 1 is not a prime number.
- For example, 2, 3, 5, 7, 11, and 13 are all primes.

An integer is a palindrome if it reads the same from left to right as it does from right to left.
- For example, 101 and 12321 are palindromes.

The test cases are generated so that the answer always exists and is in the range [2, 2 * 108].

TRY IT / Write examples, edge cases, and your own algorithm first.

## 0868 Binary Gap

EASY

Given a positive integer n, find and return the longest distance between any two adjacent 1's in the binary representation of n. If there are no two adjacent 1's, return 0.

Two 1's are adjacent if there are only 0's separating them (possibly no 0's). The distance between two 1's is the absolute difference between their bit positions. For example, the two 1's in "1001" have a distance of 3.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0869 Reordered Power Of 2

MEDIUM

You are given an integer n. We reorder the digits in any order (including the original order) such that the leading digit is not zero.

Return true if and only if we can do this so that the resulting number is a power of two.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0870 Advantage Shuffle

MEDIUM

You are given two integer arrays nums1 and nums2 both of the same length. The advantage of nums1 with respect to nums2 is the number of indices i for which nums1[i] > nums2[i].

Return any permutation of nums1 that maximizes its advantage with respect to nums2.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0875 Koko Eating Bananas

MEDIUM

Koko loves to eat bananas. There are n piles of bananas, the ith pile has piles[i] bananas. The guards have gone and will come back in h hours.

Koko can decide her bananas-per-hour eating speed of k. Each hour, she chooses some pile of bananas and eats k bananas from that pile. If the pile has less than k bananas, she eats all of them instead and will not eat any more bananas during this hour.

Koko likes to eat slowly but still wants to finish eating all the bananas before the guards return.

Return the minimum integer k such that she can eat all the bananas within h hours.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0877 Stone Game

MEDIUM

Alice and Bob play a game with piles of stones. There are an even number of piles arranged in a row, and each pile has a positive integer number of stones piles[i].

The objective of the game is to end with the most stones. The total number of stones across all the piles is odd, so there are no ties.

Alice and Bob take turns, with Alice starting first. Each turn, a player takes the entire pile of stones either from the beginning or from the end of the row. This continues until there are no more piles left, at which point the person with the most stones wins.

Assuming Alice and Bob play optimally, return true if Alice wins the game, or false if Bob wins.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0879 Profitable Schemes

**HARD**

There is a group of n members, and a list of various crimes they could commit. The ith crime generates a profit[i] and requires group[i] members to participate in it. If a member participates in one crime, that member can't participate in another crime.

Let's call a profitable scheme any subset of these crimes that generates at least minProfit profit, and the total number of members participating in that subset of crimes is at most n.

Return the number of schemes that can be chosen. Since the answer may be very large, return it modulo 109 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0881 Boats To Save People

**MEDIUM**

You are given an array people where people[i] is the weight of the ith person, and an infinite number of boats where each boat can carry a maximum weight of limit. Each boat carries at most two people at the same time, provided the sum of the weight of those people is at most limit.

Return the minimum number of boats to carry every given person.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0883 Projection Area Of 3D Shapes

**EASY**

You are given an n x n grid where we place some 1 x 1 x 1 cubes that are axis-aligned with the x, y, and z axes.

Each value v = grid[i][j] represents a tower of v cubes placed on top of the cell (i, j).

We view the projection of these cubes onto the xy, yz, and zx planes.

A projection is like a shadow, that maps our 3-dimensional figure to a 2-dimensional plane. We are viewing the "shadow" when looking at the cubes from the top, the front, and the side.

Return the total area of all three projections.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0886 Possible Bipartition

**MEDIUM**

We want to split a group of n people (labeled from 1 to n) into two groups of any size. Each person may dislike some other people, and they should not go into the same group.

Given the integer n and the array dislikes where dislikes[i] = [ai, bi] indicates that the person labeled ai does not like the person labeled bi, return true if it is possible to split everyone into two groups in this way.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0887 Super Egg Drop

HARD

You are given k identical eggs and you have access to a building with n floors labeled from 1 to n.

You know that there exists a floor f where 0 <= f <= n such that any egg dropped at a floor higher than f will break, and any egg dropped at or below floor f will not break.

Each move, you may take an unbroken egg and drop it from any floor x (where 1 <= x <= n). If the egg breaks, you can no longer use it. However, if the egg does not break, you may reuse it in future moves.

Return the minimum number of moves that you need to determine with certainty what the value of f is.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0890 Find And Replace Pattern

MEDIUM

You have a list of words and a pattern, and you want to know which words in words matches the pattern.

A word matches the pattern if there exists a permutation of letters p so that after replacing every letter x in the pattern with p(x), we get the desired word.

(Recall that a permutation of letters is a bijection from letters to letters: every letter maps to another letter, and no two letters map to the same letter.)

Return a list of the words in words that match the given pattern.

You may return the answer in any order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0892 Surface Area Of 3D Shapes

EASY

You are given an n x n grid where you have placed some 1 x 1 x 1 cubes. Each value v = grid[i][j] represents a tower of v cubes placed on top of cell (i, j).

After placing these cubes, you have decided to glue any directly adjacent cubes to each other, forming several irregular 3D shapes.

Return the total surface area of the resulting shapes.

Note: The bottom face of each shape counts toward its surface area.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0900 Rle Iterator

MEDIUM

We can use run-length encoding (i.e., RLE) to encode a sequence of integers. In a run-length encoded array of even length encoding (0-indexed), for all even i, encoding[i] tells us the number of times that the non-negative integer value encoding[i + 1] is repeated in the sequence.
- For example, the sequence arr = [8,8,8,5,5] can be encoded to be encoding = [3,8,2,5]. encoding = [3,8,0,9,2,5] and encoding = [2,8,1,8,2,5] are also valid RLE of arr.

Given a run-length encoded array, design an iterator that iterates through it.

Implement the RLEIterator class:
- RLEIterator(int[] encoded) Initializes the object with the encoded array encoded.

- int next(int n) Exhausts the next n elements and returns the last element exhausted in this way. If there is no element left to exhaust, return -1 instead.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0901 Online Stock Span

**MEDIUM**

Design an algorithm that collects daily price quotes for some stock and returns the span of that stock's price for the current day.

The span of the stock's price in one day is the maximum number of consecutive days (starting from that day and going backward) for which the stock price was less than or equal to the price of that day.

- For example, if the prices of the stock in the last four days is [7,2,1,2] and the price of the stock today is 2, then the span of today is 4 because starting from today, the price of the stock was less than or equal 2 for 4 consecutive days.

- Also, if the prices of the stock in the last four days is [7,34,1,2] and the price of the stock today is 8, then the span of today is 3 because starting from today, the price of the stock was less than or equal 8 for 3 consecutive days.

Implement the StockSpanner class:
- StockSpanner() Initializes the object of the class.
- int next(int price) Returns the span of the stock's price given that today's price is price.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0906 Super Palindromes

**HARD**

Let's say a positive integer is a super-palindrome if it is a palindrome, and it is also the square of a palindrome.

Given two positive integers left and right represented as strings, return the number of super-palindromes integers in the inclusive range [left, right].

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0908 Smallest Range I

**EASY**

You are given an integer array nums and an integer k.

In one operation, you can choose any index i where 0 <= i < nums.length and change nums[i] to nums[i] + x where x is an integer from the range [-k, k]. You can apply this operation at most once for each index i.

The score of nums is the difference between the maximum and minimum elements in nums.

Return the minimum score of nums after applying the mentioned operation at most once for each index in it.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0910 Smallest Range Ii

**MEDIUM**

You are given an integer array nums and an integer k.

For each index i where 0 <= i < nums.length, change nums[i] to be either nums[i] + k or nums[i] - k.

The score of nums is the difference between the maximum and minimum elements in nums.

Return the minimum score of nums after changing the values at each index.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0913 Cat And Mouse

HARD

A game on an undirected graph is played by two players, Mouse and Cat, who alternate turns.

The graph is given as follows: graph[a] is a list of all nodes b such that ab is an edge of the graph.

The mouse starts at node 1 and goes first, the cat starts at node 2 and goes second, and there is a hole at node 0.

During each player's turn, they must travel along one edge of the graph that meets where they are. For example, if the Mouse is at node 1, it must travel to any node in graph[1].

Additionally, it is not allowed for the Cat to travel to the Hole (node 0).

Then, the game can end in three ways:
- If ever the Cat occupies the same node as the Mouse, the Cat wins.
- If ever the Mouse reaches the Hole, the Mouse wins.
- If ever a position is repeated (i.e., the players are in the same position as a previous turn, and it is the same player's turn to move), the game is a draw.

Given a graph, and assuming both players play optimally, return
- 1 if the mouse wins the game,
- 2 if the cat wins the game, or
- 0 if the game is a draw.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0924 Minimize Malware Spread

HARD

You are given a network of n nodes represented as an n x n adjacency matrix graph, where the ith node is directly connected to the jth node if graph[i][j] == 1.

Some nodes initial are initially infected by malware. Whenever two nodes are directly connected, and at least one of those two nodes is infected by malware, both nodes will be infected b malware. This spread of malware will continue until no more nodes can be infected in this manner.

Suppose M(initial) is the final number of nodes infected with malware in the entire network after the spread of malware stops. We will remove exactly one node from initial.

Return the node that, if removed, would minimize M(initial). If multiple nodes could be removed to minimize M(initial), return such a node with the smallest index.

Note that if a node was removed from the initial list of infected nodes, it might still be infected later due to the malware spread.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0925 Long Pressed Name

EASY

Your friend is typing his name into a keyboard. Sometimes, when typing a character c,

the key might get long pressed, and the character will be typed 1 or more times.

You examine the typed characters of the keyboard. Return True if it is possible that it was your friends name, with some characters (possibly none) being long pressed.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0927 Three Equal Parts

**HARD**

You are given an array arr which consists of only zeros and ones, divide the array into three non-empty parts such that all of these parts represent the same binary value.

If it is possible, return any [i, j] with i + 1 < j, such that:
- arr[0], arr[1], ..., arr[i] is the first part,
- arr[i + 1], arr[i + 2], ..., arr[j - 1] is the second part, and
- arr[j], arr[j + 1], ..., arr[arr.length - 1] is the third part.
- All three parts have equal binary values.

If it is not possible, return [-1, -1].

Note that the entire part is used when considering what binary value it represents. For example, [1,1,0] represents 6 in decimal, not 3. Also, leading zeros are allowed, so [0,1,1] and [1,1] represent the same value.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0934 Shortest Bridge

**MEDIUM**

You are given an n x n binary matrix grid where 1 represents land and 0 represents water.

An island is a 4-directionally connected group of 1's not connected to any other 1's. There are exactly two islands in grid.

You may change 0's to 1's to connect the two islands to form one island.

Return the smallest number of 0's you must flip to connect the two islands.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0935 Knight Dialer

**MEDIUM**

The chess knight has a unique movement, it may move two squares vertically and one square horizontally, or two squares horizontally and one square vertically (with both forming the shape of an L). The possible movements of chess knight are shown in this diagram:

A chess knight can move as indicated in the chess diagram below.

We have a chess knight and a phone pad as shown below, the knight can only stand on a numeric cell (i.e. blue cell).

Given an integer n, return how many distinct phone numbers of length n we can dial.

You are allowed to place the knight on any numeric cell initially and then you should perform n - 1 jumps to dial a number of length n. All jumps should be valid knight jumps.

As the answer may be very large, return the answer modulo 109 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0936 Stamping The Sequence

HARD

You are given two strings stamp and target. Initially, there is a string s of length target.length with all s[i] == '?'.

In one turn, you can place stamp over s and replace every letter in the s with the corresponding letter from stamp.

For example, if stamp = "abc" and target = "abcba", then s is "?????" initially.
In one turn you can:
- place stamp at index 0 of s to obtain "abc??",
- place stamp at index 1 of s to obtain "?abc?", or
- place stamp at index 2 of s to obtain "??abc".

Note that stamp must be fully contained in the boundaries of s in order to stamp (i.e., you cannot place stamp at index 3 of s).

We want to convert s to target using at most 10 * target.length turns.

Return an array of the index of the left-most letter being stamped at each turn. If we cannot obtain target from s within 10 * target.length turns, return an empty array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0937 Reorder Data In Log Files

MEDIUM

You are given an array of logs. Each log is a space-delimited string of words, where the first word is the identifier.

There are two types of logs:
- Letter-logs: All words (except the identifier) consist of lowercase English letters.
- Digit-logs: All words (except the identifier) consist of digits.

Reorder these logs so that:
contents are the same, then sort them lexicographically by their identifiers.

Return the final order of the logs.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0948 Bag Of Tokens

MEDIUM

You start with an initial power of power, an initial score of 0, and a bag of tokens given as an integer array tokens, where each tokens[i] denotes the value of tokeni.

Your goal is to maximize the total score by strategically playing these tokens. In one move, you can play an unplayed token in one of the two ways (but not both for the same token):
- Face-up: If your current power is at least tokens[i], you may play tokeni, losing tokens[i] power and gaining 1 score.
- Face-down: If your current score is at least 1, you may play tokeni, gaining tokens[i] power and losing 1 score.

Return the maximum possible score you can achieve after playing any number of tokens.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0950 Reveal Cards In Increasing Order

MEDIUM

You are given an integer array deck. There is a deck of cards where every card has a unique integer. The integer on the ith card is deck[i].

You can order the deck in any order you want. Initially, all the cards start face down (unrevealed) in one deck.

You will do the following steps repeatedly until all cards are revealed:
bottom of the deck.

Return an ordering of the deck that would reveal the cards in increasing order.

Note that the first entry in the answer is considered to be the top of the deck.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0959 Regions Cut By Slashes

MEDIUM

An n x n grid is composed of 1 x 1 squares where each 1 x 1 square consists of a '/', '\', or blank space ' '. These characters divide the square into contiguous regions.

Given the grid grid represented as a string array, return the number of regions.

Note that backslash characters are escaped, so a '\' is represented as '\\'.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0962 Maximum Width Ramp

MEDIUM

A ramp in an integer array nums is a pair (i, j) for which i < j and nums[i] <= nums[j]. The width of such a ramp is j - i.

Given an integer array nums, return the maximum width of a ramp in nums. If there is no ramp in nums, return 0.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0973 K Closest Points To Origin

MEDIUM

Given an array of points where points[i] = [xi, yi] represents a point on the X-Y plane and an integer k, return the k closest points to the origin (0, 0).

The distance between two points on the X-Y plane is the Euclidean distance (i.e., √(x1 - x2)2 + (y1 - y2)2).

You may return the answer in any order. The answer is guaranteed to be unique (except for the order that it is in).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0975 Odd Even Jump

HARD

You are given an integer array arr. From some starting index, you can make a series of jumps. The (1st, 3rd, 5th, ...) jumps in the series are called odd-numbered jumps, and the (2nd,

4th, 6th, ...) jumps in the series are called even-numbered jumps. Note that the jumps are numbered, not the indices.

You may jump forward from index i to index j (with i < j) in the following way:
- During odd-numbered jumps (i.e., jumps 1, 3, 5, ...), you jump to the index j such that arr[i] <= arr[j] and arr[j] is the smallest possible value. If there are multiple such indices j, you can only jump to the smallest such index j.
- During even-numbered jumps (i.e., jumps 2, 4, 6, ...), you jump to the index j such that arr[i] >= arr[j] and arr[j] is the largest possible value. If there are multiple such indices j, you can only jump to the smallest such index j.
- It may be the case that for some index i, there are no legal jumps.

A starting index is good if, starting from that index, you can reach the end of the array (index arr.length - 1) by jumping some number of times (possibly 0 or more than once).

Return the number of good starting indices.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0976 Largest Perimeter Triangle

**EASY**

Given an array A of positive lengths, return the largest perimeter of a triangle with non-zero area, formed from 3 of these lengths.

If it is impossible to form any triangle of non-zero area, return 0.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0980 Unique Paths Iii

**HARD**

You are given an m x n integer array grid where grid[i][j] could be:
- 1 representing the starting square. There is exactly one starting square.
- 2 representing the ending square. There is exactly one ending square.
- 0 representing empty squares we can walk over.
- -1 representing obstacles that we cannot walk over.

Return the number of 4-directional walks from the starting square to the ending square, that walk over every non-obstacle square exactly once.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0990 Satisfiability Of Equality Equations

**MEDIUM**

You are given an array of strings equations that represent relationships between variables where each string equations[i] is of length 4 and takes one of two different forms: "xi==yi" or "xi!=yi".Here, xi and yi are lowercase letters (not necessarily different) that represent one-letter variable names.

Return true if it is possible to assign integers to variable names so as to satisfy all the given equations, or false otherwise.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0994 Rotting Oranges

MEDIUM

You are given an m x n grid where each cell can have one of three values:
- 0 representing an empty cell,
- 1 representing a fresh orange, or
- 2 representing a rotten orange.

Every minute, any fresh orange that is 4-directionally adjacent to a rotten orange becomes rotten.

Return the minimum number of minutes that must elapse until no cell has a fresh orange. If this is impossible, return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 0999 Available Captures For Rook

EASY

You are given an 8 x 8 matrix representing a chessboard. There is exactly one white rook represented by 'R', some number of white bishops 'B', and some number of black pawns 'p'. Empty squares are represented by '.'.

A rook can move any number of squares horizontally or vertically (up, down, left, right) until it reaches another piece or the edge of the board. A rook is attacking a pawn if it can move to the pawn's square in one move.

Note: A rook cannot move through other pieces, such as bishops or pawns. This means a rook cannot attack a pawn if there is another piece blocking the path.

Return the number of pawns the white rook is attacking.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1000 Minimum Cost To Merge Stones

HARD

There are n piles of stones arranged in a row. The ith pile has stones[i] stones.

A move consists of merging exactly k consecutive piles into one pile, and the cost of this move is equal to the total number of stones in these k piles.

Return the minimum cost to merge all piles of stones into one pile. If it is impossible, return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1004 Max Consecutive Ones Iii

MEDIUM

Given a binary array nums and an integer k, return the maximum number of consecutive 1's in the array if you can flip at most k 0's.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1006 Clumsy Factorial

MEDIUM

The factorial of a positive integer n is the product of all positive integers less than or equal to n.

- For example, factorial(10) = 10 * 9 * 8 * 7 * 6 * 5 * 4 * 3 * 2 * 1.

We make a clumsy factorial using the integers in decreasing order by swapping out the multiply operations for a fixed rotation of operations with multiply '*', divide '/', add '+', and subtract '-' in this order.
- For example, clumsy(10) = 10 * 9 / 8 + 7 - 6 * 5 / 4 + 3 - 2 * 1.

However, these operations are still applied using the usual order of operations of arithmetic. We do all multiplication and division steps before any addition or subtraction steps, and multiplication and division steps are processed left to right.

Additionally, the division that we use is floor division such that 10 * 9 / 8 = 90 / 8 = 11.

Given an integer n, return the clumsy factorial of n.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1007 Minimum Domino Rotations For Equal Row

**MEDIUM**

In a row of dominoes, tops[i] and bottoms[i] represent the top and bottom halves of the ith domino. (A domino is a tile with two numbers from 1 to 6 - one on each half of the tile.)

We may rotate the ith domino, so that tops[i] and bottoms[i] swap values.

Return the minimum number of rotations so that all the values in tops are the same, or all the values in bottoms are the same.

If it cannot be done, return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1009 Complement Of Base 10 Integer

**EASY**

Every non-negative integer N has a binary representation.
For example, 5 can be represented as "101" in binary,
11 as "1011" in binary, and so on. Note that except for
N = 0, there are no leading zeroes in any binary representation.

The complement of a binary representation is the number in binary you get when changing every 1 to a 0 and 0 to a 1. For example, the complement of "101" in binary is "010" in binary.

For a given number N in base-10, return the complement of it's binary representation as a base-10 integer.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1011 Capacity To Ship Packages Within D Days

**MEDIUM**

A conveyor belt has packages that must be shipped from one port to another within days days.

The ith package on the conveyor belt has a weight of weights[i]. Each day, we load the ship with packages on the conveyor belt (in the order given by weights). We may not load more weight than the maximum weight capacity of the ship.

Return the least weight capacity of the ship that will result in all the packages on the conveyor belt being shipped within days days.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1015 Smallest Integer Divisible By K

MEDIUM

Given a positive integer k, you need to find the length of the smallest positive integer n such that n is divisible by k, and n only contains the digit 1.

Return the length of n. If there is no such n, return -1.

Note: n may not fit in a 64-bit signed integer.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1017 Convert To Base 2

MEDIUM

Given an integer n, return a binary string representing its representation in base -2.

Note that the returned string should not have leading zeros unless the string is "0".

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1023 Camelcase Matching

MEDIUM

Given an array of strings queries and a string pattern, return a boolean array answer where answer[i] is true if queries[i] matches pattern, and false otherwise.

A query word queries[i] matches pattern if you can insert lowercase English letters pattern so that it equals the query. You may insert each character at any position and you may not insert any characters.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1024 Video Stitching

MEDIUM

You are given a series of video clips from a sporting event that lasted time seconds. These video clips can be overlapping with each other and have varying lengths.

Each video clip is described by an array clips where clips[i] = [starti, endi] indicates that the ith clip started at starti and ended at endi.

We can cut these clips into segments freely.
- For example, a clip [0, 7] can be cut into segments [0, 1] + [1, 3] + [3, 7].

Return the minimum number of clips needed so that we can cut the clips into segments that cover the entire sporting event [0, time]. If the task is impossible, return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1025 Divisor Game

EASY

Alice and Bob take turns playing a game, with Alice starting first.

Initially, there is a number n on the chalkboard. On each player's turn, that player makes a move consisting of:
- Choosing any x with 0 < x < n and n % x == 0.
- Replacing the number n on the chalkboard with n - x.

Also, if a player cannot make a move, they lose the game.

Return true if and only if Alice wins the game, assuming both players play optimally.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1029 Two City Scheduling

**MEDIUM**

A company is planning to interview 2n people. Given the array costs where costs[i] = [aCosti, bCosti], the cost of flying the ith person to city a is aCosti, and the cost of flying the ith person to city b is bCosti.

Return the minimum cost to fly every person to a city such that exactly n people arrive in each city.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1033 Moving Stones Until Consecutive

**MEDIUM**

There are three stones in different positions on the X-axis. You are given three integers a, b, and c, the positions of the stones.

In one move, you pick up a stone at an endpoint (i.e., either the lowest or highest position stone), and move it to an unoccupied position between those endpoints. Formally, let's say the stones are currently at positions x, y, and z with x < y < z. You pick up the stone at either position x or position z, and move that stone to an integer position k, with x < k < z and k != y.

The game ends when you cannot make any more moves (i.e., the stones are in three consecutive positions).

Return an integer array answer of length 2 where:
- answer[0] is the minimum number of moves you can play, and
- answer[1] is the maximum number of moves you can play.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1035 Uncrossed Lines

**MEDIUM**

You are given two integer arrays nums1 and nums2. We write the integers of nums1 and nums2 (in the order they are given) on two separate horizontal lines.

We may draw connecting lines: a straight line connecting two numbers nums1[i] and nums2[j] such that:
- nums1[i] == nums2[j], and
- the line we draw does not intersect any other connecting (non-horizontal) line.

Note that a connecting line cannot intersect even at the endpoints (i.e., each number can only belong to one connecting line).

Return the maximum number of connecting lines we can draw in this way.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1037 Valid Boomerang

**EASY**

A boomerang is a set of 3 points that are all distinct and not in a straight line.

Given a list of three points in the plane, return whether these points are a boomerang.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1039 Minimum Score Triangulation Of Polygon

MEDIUM

You have a convex n-sided polygon where each vertex has an integer value. You are given an integer array values where values[i] is the value of the ith vertex in clockwise order.

Polygon triangulation is a process where you divide a polygon into a set of triangles and the vertices of each triangle must also be vertices of the original polygon. Note that no other shapes other than triangles are allowed in the division. This process will result in n - 2 triangles.

You will triangulate the polygon. For each triangle, the weight of that triangle is the product of the values at its vertices. The total score of the triangulation is the sum of these weights over all n - 2 triangles.

Return the minimum possible score that you can achieve with some triangulation of the polygon.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1040 Moving Stones Until Consecutive Ii

MEDIUM

There are some stones in different positions on the X-axis. You are given an integer array stones, the positions of the stones.

Call a stone an endpoint stone if it has the smallest or largest position. In one move, you pick up an endpoint stone and move it to an unoccupied position so that it is no longer an endpoint stone.
- In particular, if the stones are at say, stones = [1,2,5], you cannot move the endpoint stone at position 5, since moving it to any position (such as 0, or 3) will still keep that stone as an endpoint stone.

The game ends when you cannot make any more moves (i.e., the stones are in three consecutive positions).

Return an integer array answer of length 2 where:
- answer[0] is the minimum number of moves you can play, and
- answer[1] is the maximum number of moves you can play.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1041 Robot Bounded In Circle

MEDIUM

On an infinite plane, a robot initially stands at (0, 0) and faces north.
The robot can receive one of three instructions:
- "G": go straight 1 unit;
- "L": turn 90 degrees to the left;
- "R": turn 90 degrees to the right.

The robot performs the instructions given in order, and repeats them forever.
Return true if and only if there exists a circle in the plane such that the robot never leaves the circle.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1046 Last Stone Weight

EASY

You are given an array of integers stones where stones[i] is the weight of the ith stone.

We are playing a game with the stones. On each turn, we choose the heaviest two stones and smash them together. Suppose the heaviest two stones have weights x and y with x <= y. The result of this smash is:
- If x == y, both stones are destroyed, and
- If x != y, the stone of weight x is destroyed, and the stone of weight y has new weight y - x.

At the end of the game, there is at most one stone left.

Return the weight of the last remaining stone. If there are no stones left, return 0.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1051 Height Checker

**EASY**

A school is trying to take an annual photo of all the students. The students are asked to stand in a single file line in non-decreasing order by height. Let this ordering be represented by the integer array expected where expected[i] is the expected height of the ith student in line.

You are given an integer array heights representing the current order that the students are standing in. Each heights[i] is the height of the ith student in line (0-indexed).

Return the number of indices where heights[i] != expected[i].

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1052 Grumpy Bookstore Owner

**MEDIUM**

There is a bookstore owner that has a store open for n minutes. You are given an integer array customers of length n where customers[i] is the number of the customers that enter the store at the start of the ith minute and all those customers leave after the end of that minute.

During certain minutes, the bookstore owner is grumpy. You are given a binary array grumpy where grumpy[i] is 1 if the bookstore owner is grumpy during the ith minute, and is 0 otherwise.

When the bookstore owner is grumpy, the customers entering during that minute are not satisfied. Otherwise, they are satisfied.

The bookstore owner knows a secret technique to remain not grumpy for minutes consecutive minutes, but this technique can only be used once.

Return the maximum number of customers that can be satisfied throughout the day.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1058 Minimize Rounding Error To Meet Target

**MEDIUM**

Given an array of prices [p1,p2...,pn] and a target, round each price pi to Roundi(pi) so that the rounded array [Round1(p1),Round2(p2)...,Roundn(pn)] sums to the given target. Each operation Roundi(pi) could be either Floor(pi) or Ceil(pi).

Return the string "-1" if the rounded array is impossible to sum to target. Otherwise, return the smallest rounding error, which is defined as Σ |Roundi(pi) - (pi)| for i from 1 to n, as a string with three places after the decimal.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1064 Fixed Point

EASY

Given an array of distinct integers arr, where arr is sorted in ascending order, return the smallest index i that satisfies arr[i] == i. If there is no such index, return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1078 Occurrences After Bigram

EASY

Given two strings first and second, consider occurrences in some text of the form "first second third", where second comes immediately after first, and third comes immediately after second.

Return an array of all the words third for each occurrence of "first second third".

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1087 Brace Expansion

MEDIUM

You are given a string s representing a list of words. Each letter in the word has one or more options.
- If there is one option, the letter is represented as is.
- If there is more than one option, then curly braces delimit the options.
For example, "{a,b,c}" represents options ["a", "b", "c"].

For example, if s = "a{b,c}", the first character is always 'a', but the second character can be 'b' or 'c'. The original list is ["ab", "ac"].

Return all words that can be formed in this manner, sorted in lexicographical order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1092 Shortest Common Supersequence

HARD

Given two strings str1 and str2, return the shortest string that has both str1 and str2 as subsequences. If there are multiple valid strings, return any of them.

A string s is a subsequence of string t if deleting some number of characters from t (possibly 0) results in the string s.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1093 Statistics From A Large Sample

MEDIUM

You are given a large sample of integers in the range [0, 255]. Since the sample is so large, it is represented by an array count where count[k] is the number of times that k appears in the sample.

Calculate the following statistics:
- minimum: The minimum element in the sample.
- maximum: The maximum element in the sample.
- mean: The average of the sample, calculated as the total sum of all elements divided by the total number of elements.
- median:
- If the sample has an odd number of elements, then the median is the middle element once

the sample is sorted.
- If the sample has an even number of elements, then the median is the average of the two middle elements once the sample is sorted.
- mode: The number that appears the most in the sample. It is guaranteed to be unique.

Return the statistics of the sample as an array of floating-point numbers [minimum, maximum, mean, median, mode]. Answers within 10-5 of the actual answer will be accepted.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1094 Car Pooling

**MEDIUM**

There is a car with capacity empty seats. The vehicle only drives east (i.e., it cannot turn around and drive west).

You are given the integer capacity and an array trips where trips[i] = [numPassengersi, fromi, toi] indicates that the ith trip has numPassengersi passengers and the locations to pick them up and drop them off are fromi and toi respectively. The locations are given as the number of kilometers due east from the car's initial location.

Return true if it is possible to pick up and drop off all passengers for all the given trips, or false otherwise.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1101 The Earliest Moment When Everyone Become Friends

**MEDIUM**

There are n people in a social group labeled from 0 to n - 1. You are given an array logs where logs[i] = [timestampi, xi, yi] indicates that xi and yi will be friends at the time timestampi.

Friendship is symmetric. That means if a is friends with b, then b is friends with a. Also, person a is acquainted with a person b if a is friends with b, or a is a friend of someone acquainted with b.

Return the earliest time for which every person became acquainted with every other person. If there is no such earliest time, return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1102 Path With Maximum Minimum Value

**MEDIUM**

Given an m x n integer matrix grid, return the maximum score of a path starting at (0, 0) and ending at (m - 1, n - 1) moving in the 4 cardinal directions.

The score of a path is the minimum value in that path.
- For example, the score of the path 8 → 4 → 5 → 9 is 4.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1103 Distribute Candies To People

**EASY**

We distribute some number of `candies`, to a row of `n = num_people` people in the following way:

We then give 1 candy to the first person, 2 candies to the second person, and so on until we give `n` candies to the last person.

Then, we go back to the start of the row, giving `n + 1` candies to the first person, `n + 2` candies to the second person, and so on until we give `2 * n` candies to the last person.

This process repeats (with us giving one more candy each time, and moving to the start of the row after we reach the end) until we run out of candies. The last person will receive all of our remaining candies (not necessarily one more than the previous gift).

Return an array (of length `num_people` and sum `candies`) that represents the final distribution of candies.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1105 Filling Bookcase Shelves

**MEDIUM**

You are given an array books where books[i] = [thicknessi, heighti] indicates the thickness and height of the ith book. You are also given an integer shelfWidth.

We want to place these books in order onto bookcase shelves that have a total width shelfWidth.

We choose some of the books to place on this shelf such that the sum of their thickness is less than or equal to shelfWidth, then build another level of the shelf of the bookcase so that the total height of the bookcase has increased by the maximum height of the books we just put down. We repeat this process until there are no more books to place.

Note that at each step of the above process, the order of the books we place is the same order as the given sequence of books.

For example, if we have an ordered list of 5 books, we might place the first and second book onto the first shelf, the third book on the second shelf, and the fourth and fifth book on the last shelf.

Return the minimum possible height that the total bookshelf can be after placing shelves in this manner.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1106 Parsing A Boolean Expression

**HARD**

A boolean expression is an expression that evaluates to either true or false. It can be in one of the following shapes:
- 't' that evaluates to true.
- 'f' that evaluates to false.
- '!(subExpr)' that evaluates to the logical NOT of the inner expression subExpr.
- '&(subExpr1, subExpr2, ..., subExprn)' that evaluates to the logical AND of the inner expressions subExpr1, subExpr2, ..., subExprn where n >= 1.
- '|(subExpr1, subExpr2, ..., subExprn)' that evaluates to the logical OR of the inner expressions subExpr1, subExpr2, ..., subExprn where n >= 1.

Given a string expression that represents a boolean expression, return the evaluation of that expression.

It is guaranteed that the given expression is valid and follows the given rules.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1108 Defanging An Ip Address

EASY

Given a valid (IPv4) IP address, return a defanged version of that IP address.

A defanged IP address replaces every period "." with "[.]".

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1131 Maximum Of Absolute Value Expression

MEDIUM

Given two arrays of integers with equal lengths, return the maximum value of:
- |arr1[i] - arr1[j]| + |arr2[i] - arr2[j]| + |i - j|

where the maximum is taken over all 0 <= i, j < arr1.length.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1135 Connecting Cities With Minimum Cost

MEDIUM

There are n cities labeled from 1 to n. You are given the integer n and an array connections where connections[i] = [xi, yi, costi] indicates that the cost of connecting city xi and city yi (bidirectional connection) is costi.

Return the minimum cost to connect all the n cities such that there is at least one path between each pair of cities. If it is impossible to connect all the n cities, return -1,

The cost is the sum of the connections' costs used.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1138 Alphabet Board Path

MEDIUM

On an alphabet board, we start at position (0, 0), corresponding to character board[0][0].

Here, board = ["abcde", "fghij", "klmno", "pqrst", "uvwxy", "z"], as shown in the diagram below.

We may make the following moves:
- 'U' moves our position up one row, if the position exists on the board;
- 'D' moves our position down one row, if the position exists on the board;
- 'L' moves our position left one column, if the position exists on the board;
- 'R' moves our position right one column, if the position exists on the board;
- '!' adds the character board[r][c] at our current position (r, c) to the answer.

(Here, the only positions that exist on the board are positions with letters on them.)

Return a sequence of moves that makes our answer equal to target in the minimum number of moves. You may return any path that does so.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1139 Largest 1 Bordered Square

MEDIUM

Given a 2D grid of 0s and 1s, return the number of elements in the largest square subgrid that has all 1s on its border, or 0 if such a subgrid doesn't exist in the grid.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1147 Longest Chunked Palindrome Decomposition

HARD

You are given a string text. You should split it to k substrings (subtext1, subtext2, ..., subtextk) such that:
- subtexti is a non-empty string.
- The concatenation of all the substrings is equal to text (i.e., subtext1 + subtext2 + ... + subtextk == text).
- subtexti == subtextk - i + 1 for all valid values of i (i.e., 1 <= i <= k).

Return the largest possible value of k.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1151 Minimum Swaps To Group All 1S Together

MEDIUM

Given a binary array data, return the minimum number of swaps required to group all 1's present in the array together in any place in the array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1154 Day Of The Year

EASY

Given a string date representing a Gregorian calendar date formatted as YYYY-MM-DD, return the day number of the year.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1162 As Far From Land As Possible

MEDIUM

Given an n x n grid containing only values 0 and 1, where 0 represents water and 1 represents land, find a water cell such that its distance to the nearest land cell is maximized, and return the distance. If no land or water exists in the grid, return -1.

The distance used in this problem is the Manhattan distance: the distance between two cells (x0, y0) and (x1, y1) is |x0 - x1| + |y0 - y1|.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1167 Minimum Cost To Connect Sticks

MEDIUM

You have some number of sticks with positive integer lengths. These lengths are given as an array sticks, where sticks[i] is the length of the ith stick.

You can connect any two sticks of lengths x and y into one stick by paying a cost of x + y. You must connect all the sticks until there is only one stick remaining.

Return the minimum cost of connecting all the given sticks into one stick in this way.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1168 Optimize Water Distribution In A Village

HARD

There are n houses in a village. We want to supply water for all the houses by building wells and laying pipes.

For each house i, we can either build a well inside it directly with cost wells[i - 1] (note the -1 due to 0-indexing), or pipe in water from another well to it. The costs to lay pipes between houses are given by the array pipes where each pipes[j] = [house1j, house2j, costj] represents the cost to connect house1j and house2j together using a pipe. Connections are bidirectional, and there could be multiple valid connections between the same two houses with different costs.

Return the minimum total cost to supply water to all houses.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1175 Prime Arrangements

EASY

Return the number of permutations of 1 to n so that prime numbers are at prime indices (1-indexed.)

(Recall that an integer is prime if and only if it is greater than 1, and cannot be written as a product of two positive integers both smaller than it.)

Since the answer may be large, return the answer modulo 10^9 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1176 Diet Plan Performance

EASY

A dieter consumes calories[i] calories on the i-th day.

Given an integer k, for every consecutive sequence of k days (calories[i], calories[i+1], ..., calories[i+k-1] for all 0 <= i <= n-k), they look at T, the total calories consumed during that sequence of k days (calories[i] + calories[i+1] + ... + calories[i+k-1]):
- If T < lower, they performed poorly on their diet and lose 1 point;
- If T > upper, they performed well on their diet and gain 1 point;
- Otherwise, they performed normally and there is no change in points.

Initially, the dieter has zero points. Return the total number of points the dieter has after dieting for calories.length days.

Note that the total points can be negative.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1182 Shortest Distance To Target Color

MEDIUM

You are given an array colors, in which there are three colors: 1, 2 and 3.

You are also given some queries. Each query consists of two integers i and c, return the shortest distance between the given index i and the target color c. If there is no solution return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1184 Distance Between Bus Stops

EASY

A bus has n stops numbered from 0 to n - 1 that form a circle. We know the distance between all pairs of neighboring stops where distance[i] is the distance between the stops number i and (i + 1) % n.

The bus goes along both directions i.e. clockwise and counterclockwise.

Return the shortest distance between the given start and destination stops.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1185 Day Of The Week

EASY

Given a date, return the corresponding day of the week for that date.

The input is given as three integers representing the day, month and year respectively.

Return the answer as one of the following values {"Sunday", "Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday"}.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1192 Critical Connections In A Network

HARD

There are n servers numbered from 0 to n - 1 connected by undirected server-to-server connections forming a network where connections[i] = [ai, bi] represents a connection between servers ai and bi. Any server can reach other servers directly or indirectly through the network.

A critical connection is a connection that, if removed, will make some servers unable to reach some other server.

Return all critical connections in the network in any order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1196 How Many Apples Can You Put Into The Basket

EASY

You have some apples and a basket that can carry up to 5000 units of weight.

Given an integer array weight where weight[i] is the weight of the ith apple, return the maximum number of apples you can put in the basket.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1199 Minimum Time To Build Blocks

HARD

You are given a list of blocks, where blocks[i] = t means that the i-th block needs t units of time to be built. A block can only be built by exactly one worker.

A worker can either split into two workers (number of workers increases by one) or build a block then go home. Both decisions cost some time.

The time cost of spliting one worker into two workers is given as an integer split. Note that if two workers split at the same time, they split in parallel so the cost would be split.

Output the minimum time needed to build all blocks.

Initially, there is only one worker.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1206 Design Skiplist

HARD

Design a Skiplist without using any built-in libraries.

A skiplist is a data structure that takes O(log(n)) time to add, erase and search. Comparing with treap and red-black tree which has the same function and performance, the code length of Skiplist can be comparatively short and the idea behind Skiplists is just simple linked lists.

For example, we have a Skiplist containing [30,40,50,60,70,90] and we want to add 80 and 45 into it. The Skiplist works this way:

You can see there are many layers in the Skiplist. Each layer is a sorted linked list. With the help of the top layers, add, erase and search can be faster than O(n). It can be proven that the average time complexity for each operation is O(log(n)) and space complexity is O(n).

See more about Skiplist: https://en.wikipedia.org/wiki/Skip_list
Implement the Skiplist class:
- Skiplist() Initializes the object of the skiplist.
- bool search(int target) Returns true if the integer target exists in the Skiplist or false otherwise.
- void add(int num) Inserts the value num into the SkipList.
- bool erase(int num) Removes the value num from the Skiplist and returns true. If num does not exist in the Skiplist, do nothing and return false. If there exist multiple num values, removing any one of them is fine.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1216 Valid Palindrome Iii

HARD

Given a string s and an integer k, return true if s is a k-palindrome.

A string is k-palindrome if it can be transformed into a palindrome by removing at most k characters from it.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1217 Minimum Cost To Move Chips To The Same Position

EASY

We have n chips, where the position of the ith chip is position[i].

We need to move all the chips to the same position. In one step, we can change the position of the ith chip from position[i] to:
- position[i] + 2 or position[i] - 2 with cost = 0.
- position[i] + 1 or position[i] - 1 with cost = 1.

Return the minimum cost needed to move all the chips to the same position.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1219 Path With Maximum Gold

MEDIUM

In a gold mine grid of size m x n, each cell in this mine has an integer representing the

amount of gold in that cell, 0 if it is empty.

Return the maximum amount of gold you can collect under the conditions:
- Every time you are located in a cell you will collect all the gold in that cell.
- From your position, you can walk one step to the left, right, up, or down.
- You can't visit the same cell more than once.
- Never visit a cell with 0 gold.
- You can start and stop collecting gold from any position in the grid that has some gold.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1223 Dice Roll Simulation

**HARD**

A die simulator generates a random number from 1 to 6 for each roll. You introduced a constraint to the generator such that it cannot roll the number i more than rollMax[i] (1-indexed) consecutive times.

Given an array of integers rollMax and an integer n, return the number of distinct sequences that can be obtained with exact n rolls. Since the answer may be too large, return it modulo 1e9 + 7.

Two sequences are considered different if at least one element differs from each other.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1227 Airplane Seat Assignment Probability

**MEDIUM**

n passengers board an airplane with exactly n seats. The first passenger has lost the ticket and picks a seat randomly. But after that, the rest of the passengers will:
- Take their own seat if it is still available, and
- Pick other seats randomly when they find their seat occupied

Return the probability that the nth person gets his own seat.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1229 Meeting Scheduler

**MEDIUM**

Given the availability time slots arrays slots1 and slots2 of two people and a meeting duration duration, return the earliest time slot that works for both of them and is of duration duration.

If there is no common time slot that satisfies the requirements, return an empty array.

The format of a time slot is an array of two elements [start, end] representing an inclusive time range from start to end.

It is guaranteed that no two availability slots of the same person intersect with each other. That is, for any two time slots [start1, end1] and [start2, end2] of the same person, either start1 > end2 or start2 > end1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1230 Toss Strange Coins

**MEDIUM**

You have some coins. The i-th coin has a probability prob[i] of facing heads when tossed.

Return the probability that the number of coins facing heads equals target if you toss every

coin exactly once.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1231 Divide Chocolate

HARD

You have one chocolate bar that consists of some chunks. Each chunk has its own sweetness given by the array sweetness.

You want to share the chocolate with your k friends so you start cutting the chocolate bar into k + 1 pieces using k cuts, each piece consists of some consecutive chunks.

Being generous, you will eat the piece with the minimum total sweetness and give the other pieces to your friends.

Find the maximum total sweetness of the piece you can get by cutting the chocolate bar optimally.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1232 Check If It Is A Straight Line

EASY

You are given an array `coordinates`, `coordinates[i] = [x, y]`, where `[x, y] represents the coordinate of a point.

Check if these points make a straight line in the XY plane.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1235 Maximum Profit In Job Scheduling

HARD

We have n jobs, where every job is scheduled to be done from startTime[i] to endTime[i], obtaining a profit of profit[i].

You're given the startTime, endTime and profit arrays, return the maximum profit you can take such that there are no two jobs in the subset with overlapping time range.

If you choose a job that ends at time X you will be able to start another job that starts at time X.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1237 Find Positive Integer Solution For A Given Equation

MEDIUM

Given a callable function f(x, y) with a hidden formula and a value z, reverse engineer the formula and return all positive integer pairs x and y where f(x,y) == z. You may return the pairs in any order.

While the exact formula is hidden, the function is monotonically increasing, i.e.:
- f(x, y) < f(x + 1, y)
- f(x, y) < f(x, y + 1)

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1240 Tiling A Rectangle With The Fewest Squares

HARD

Given a rectangle of size n x m, return the minimum number of integer-sided squares that

tile the rectangle.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1246 Palindrome Removal

**HARD**

You are given an integer array arr.

In one move, you can select a palindromic subarray arr[i], arr[i + 1], ..., arr[j] where i <= j, and remove that subarray from the given array. Note that after removing a subarray, the elements on the left and on the right of that subarray move to fill the gap left by the removal.

Return the minimum number of moves needed to remove all numbers from the array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1259 Handshakes That Dont Cross

**HARD**

You are given an even number of people numPeople that stand around a circle and each person shakes hands with someone else so that there are numPeople / 2 handshakes total.

Return the number of ways these handshakes could occur such that none of the handshakes cross.

Since the answer could be very large, return it modulo 109 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1266 Minimum Time Visiting All Points

**EASY**

On a 2D plane, there are n points with integer coordinates points[i] = [xi, yi]. Return the minimum time in seconds to visit all the points in the order given by points.

You can move according to these rules:
- In 1 second, you can either:
- move vertically by one unit,
- move horizontally by one unit, or
- move diagonally sqrt(2) units (in other words, move one unit vertically then one unit horizontally in 1 second).
- You have to visit the points in the same order as they appear in the array.
- You are allowed to pass through points that appear later in the order, but these do not count as visits.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1268 Search Suggestions System

**MEDIUM**

You are given an array of strings products and a string searchWord.

Design a system that suggests at most three product names from products after each character of searchWord is typed. Suggested products should have common prefix with searchWord. If there are more than three products with a common prefix return the three lexicographically minimums products.

Return a list of lists of the suggested products after each character of searchWord is typed.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1275 Find Winner On A Tic Tac Toe Game

EASY

Tic-tac-toe is played by two players A and B on a 3 x 3 grid. The rules of Tic-Tac-Toe are:
- Players take turns placing characters into empty squares ' '.
- The first player A always places 'X' characters, while the second player B always places 'O' characters.
- 'X' and 'O' characters are always placed into empty squares, never on filled ones.
- The game ends when there are three of the same (non-empty) character filling any row, column, or diagonal.
- The game also ends if all squares are non-empty.
- No more moves can be played if the game is over.

Given a 2D integer array moves where moves[i] = [rowi, coli] indicates that the ith move will be played on grid[rowi][coli]. return the winner of the game if it exists (A or B).
In case the game ends in a draw return "Draw". If there are still movements to play return "Pending".

You can assume that moves is valid (i.e., it follows the rules of Tic-Tac-Toe), the grid is initially empty, and A will play first.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1278 Palindrome Partitioning Iii

HARD

You are given a string s containing lowercase letters and an integer k. You need to:
- First, change some characters of s to other lowercase English letters.
- Then divide s into k non-empty disjoint substrings such that each substring is a palindrome.

Return the minimal number of characters that you need to change to divide the string.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1283 Find The Smallest Divisor Given A Threshold

MEDIUM

Given an array of integers nums and an integer threshold, we will choose a positive integer divisor, divide all the array by it, and sum the division's result. Find the smallest divisor such that the result mentioned above is less than or equal to threshold.

Each result of the division is rounded to the nearest integer greater than or equal to that element. (For example: 7/3 = 3 and 10/2 = 5).

The test cases are generated so that there will be an answer.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1313 Decompress Run Length Encoded List

EASY

We are given a list nums of integers representing a list compressed with run-length encoding.

Consider each adjacent pair of elements [a, b] = [nums[2*i], nums[2*i+1]] (with i >= 0).
For each such pair, there are a elements with value b in the decompressed list.

Return the decompressed list.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1318 Minimum Flips To Make A Or B Equal To C

MEDIUM

Given 3 positives numbers a, b and c.
Return the minimum flips required in some bits of a and b
to make ( a OR b == c ). (bitwise OR operation).

Flip operation consists of change any single bit 1 to 0
or change the bit 0 to 1 in their binary representation.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1328 Break A Palindrome

MEDIUM

Given a palindromic string of lowercase English letters palindrome, replace exactly one character with any lowercase English letter so that the resulting string is not a palindrome and that it is the lexicographically smallest one possible.

Return the resulting string. If there is no way to replace a character to make it not a palindrome, return an empty string.

A string a is lexicographically smaller than a string b (of the same length) if in the first position where a and b differ, a has a character strictly smaller than the corresponding character in b. For example, "abcc" is lexicographically smaller than "abcd" because the first position they differ is at the fourth character, and 'c' is smaller than 'd'.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1332 Remove Palindromic Subsequences

EASY

Given a string s consisting only of letters 'a' and 'b'.
In a single step you can remove one palindromic subsequence from s.

Return the minimum number of steps to make the given string empty.
A string is a subsequence of a given string, if it is generated by
deleting some characters of a given string without changing its order.

A string is called palindrome if is one that reads the same backward
as well as forward.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1333 Filter Restaurants By Vegan Friendly Price And Distance

MEDIUM

Given the array restaurants where  restaurants[i] = [idi, ratingi, veganFriendlyi, pricei, distancei].
You have to filter the restaurants using three filters.

The veganFriendly filter will be either true (meaning you should only include restaurants with veganFriendlyi set to true) or false (meaning you can include any restaurant). In addition, you have the filters maxPrice and maxDistance which are the maximum value for price and distance of restaurants you should consider respectively.

Return the array of restaurant IDs after filtering, ordered by rating from highest to lowest.
For restaurants with the same rating, order them by id from highest to lowest. For simplicity
veganFriendlyi and veganFriendly take value 1 when it is true, and 0 when it is false.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1335 Minimum Difficulty Of A Job Schedule

HARD

You want to schedule a list of jobs in d days. Jobs are dependent (i.e To work on the ith job, you have to finish all the jobs j where 0 <= j < i).

You have to finish at least one task every day. The difficulty of a job schedule is the sum of difficulties of each day of the d days. The difficulty of a day is the maximum difficulty of a job done on that day.

You are given an integer array jobDifficulty and an integer d. The difficulty of the ith job is jobDifficulty[i].

Return the minimum difficulty of a job schedule. If you cannot find a schedule for the jobs return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1344 Angle Between Hands Of A Clock

MEDIUM

Given two numbers, hour and minutes, return the smaller angle (in degrees) formed between the hour and the minute hand.

Answers within 10-5 of the actual value will be accepted as correct.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1362 Closest Divisors

MEDIUM

Given an integer num, find the closest two integers in absolute difference whose product equals num + 1 or num + 2.

Return the two integers in any order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1363 Largest Multiple Of Three

HARD

Given an array of digits digits, return the largest multiple of three that can be formed by concatenating some of the given digits in any order. If there is no answer return an empty string.

Since the answer may not fit in an integer data type, return the answer as a string. Note that the returning answer must not contain unnecessary leading zeros.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1376 Time Needed To Inform All Employees

MEDIUM

A company has n employees with a unique ID for each employee from 0 to n - 1. The head of the company is the one with headID.

Each employee has one direct manager given in the manager array where manager[i] is the direct manager of the i-th employee, manager[headID] = -1. Also, it is guaranteed that the subordination relationships have a tree structure.

The head of the company wants to inform all the company employees of an urgent piece of

news. He will inform his direct subordinates, and they will inform their subordinates, and so on until all employees know about the urgent news.

The i-th employee needs informTime[i] minutes to inform all of his direct subordinates (i.e., After informTime[i] minutes, all his direct subordinates can start spreading the news).

Return the number of minutes needed to inform all the employees about the urgent news.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1383 Maximum Performance Of A Team

**HARD**

You are given two integers n and k and two integer arrays speed and efficiency both of length n. There are n engineers numbered from 1 to n. speed[i] and efficiency[i] represent the speed and efficiency of the ith engineer respectively.

Choose at most k different engineers out of the n engineers to form a team with the maximum performance.

The performance of a team is the sum of its engineers' speeds multiplied by the minimum efficiency among its engineers.

Return the maximum performance of this team. Since the answer can be a huge number, return it modulo 109 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1401 Circle And Rectangle Overlapping

**MEDIUM**

You are given a circle represented as (radius, xCenter, yCenter) and an axis-aligned rectangle represented as (x1, y1, x2, y2), where (x1, y1) are the coordinates of the bottom-left corner, and (x2, y2) are the coordinates of the top-right corner of the rectangle.

Return true if the circle and rectangle are overlapped otherwise return false. In other words, check if there is any point (xi, yi) that belongs to the circle and the rectangle at the same time.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1402 Reducing Dishes

**HARD**

A chef has collected data on the satisfaction level of his n dishes. Chef can cook any dish in 1 unit of time.

Like-time coefficient of a dish is defined as the time taken to cook that dish including previous dishes multiplied by its satisfaction level i.e. time[i] * satisfaction[i].

Return the maximum sum of like-time coefficient that the chef can obtain after dishes preparation.

Dishes can be prepared in any order and the chef can discard some dishes to get this maximum value.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1403 Minimum Subsequence In Non Increasing Order

EASY

Given the array nums, obtain a subsequence of the array whose sum of elements is strictly greater than the sum of the non included elements in such subsequence.

If there are multiple solutions, return the subsequence with minimum size and if there still exist multiple solutions, return the subsequence with the maximum total sum of all its elements. A subsequence of an array can be obtained by erasing some (possibly zero) elements from the array.

Note that the solution with the given constraints is guaranteed to be unique. Also return the answer sorted in non-increasing order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1406 Stone Game Iii

HARD

Alice and Bob continue their games with piles of stones. There are several stones arranged in a row, and each stone has an associated value which is an integer given in the array stoneValue.

Alice and Bob take turns, with Alice starting first. On each player's turn, that player can take 1, 2, or 3 stones from the first remaining stones in the row.

The score of each player is the sum of the values of the stones taken. The score of each player is 0 initially.

The objective of the game is to end with the highest score, and the winner is the player with the highest score and there could be a tie. The game continues until all the stones have been taken.

Assume Alice and Bob play optimally.

Return "Alice" if Alice will win, "Bob" if Bob will win, or "Tie" if they will end the game with the same score.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1410 Html Entity Parser

MEDIUM

HTML entity parser is the parser that takes HTML code as input and replace all the entities of the special characters by the characters itself.

The special characters and their entities for HTML are:
- Quotation Mark: the entity is " and symbol character is ".
- Single Quote Mark: the entity is ' and symbol character is '.
- Ampersand: the entity is & and symbol character is &.
- Greater Than Sign: the entity is > and symbol character is >.
- Less Than Sign: the entity is < and symbol character is <.
- Slash: the entity is ⁄ and symbol character is /.

Given the input text string to the HTML parser, you have to implement the entity parser.

Return the text after replacing the entities by the special characters.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1423 Maximum Points You Can Obtain From Cards

MEDIUM

There are several cards arranged in a row, and each card has an associated number of points. The points are given in the integer array cardPoints.

In one step, you can take one card from the beginning or from the end of the row. You have to take exactly k cards.

Your score is the sum of the points of the cards you have taken.

Given the integer array cardPoints and the integer k, return the maximum score you can obtain.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1424 Diagonal Traverse Ii

MEDIUM

Given a 2D integer array nums, return all elements of nums in diagonal order as shown in the below images.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1428 Leftmost Column With At Least A One

MEDIUM

A row-sorted binary matrix means that all elements are 0 or 1 and each row of the matrix is sorted in non-decreasing order.

Given a row-sorted binary matrix binaryMatrix, return the index (0-indexed) of the leftmost column with a 1 in it. If such an index does not exist, return -1.

You can't access the Binary Matrix directly. You may only access the matrix using a BinaryMatrix interface:
- BinaryMatrix.get(row, col) returns the element of the matrix at index (row, col) (0-indexed).
- BinaryMatrix.dimensions() returns the dimensions of the matrix as a list of 2 elements [rows, cols], which means the matrix is rows x cols.

Submissions making more than 1000 calls to BinaryMatrix.get will be judged Wrong Answer. Also, any solutions that attempt to circumvent the judge will result in disqualification.

For custom testing purposes, the input will be the entire binary matrix mat. You will not have access to the binary matrix directly.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1437 Check If All 1S Are At Least Length K Places Away

EASY

Given an array nums of 0s and 1s and an integer k, return True if all 1's are at least k places away from each other, otherwise return False.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1447 Simplified Fractions

MEDIUM

Given an integer n, return a list of all simplified fractions between 0 and 1 (exclusive) such that the denominator is less-than-or-equal-to n. The fractions can be in any order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1463 Cherry Pickup Ii

**HARD**

You are given a rows x cols matrix grid representing a field of cherries where grid[i][j] represents the number of cherries that you can collect from the (i, j) cell.

You have two robots that can collect cherries for you:
- Robot #1 is located at the top-left corner (0, 0), and
- Robot #2 is located at the top-right corner (0, cols - 1).

Return the maximum number of cherries collection using both robots by following the rules below:
- From a cell (i, j), robots can move to cell (i + 1, j - 1), (i + 1, j), or (i + 1, j + 1).
- When any robot passes through a cell, It picks up all cherries, and the cell becomes an empty cell.
- When both robots stay in the same cell, only one takes the cherries.
- Both robots cannot move outside of the grid at any moment.
- Both robots should reach the bottom row in grid.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1465 Maximum Area Of A Piece Of Cake After Horizontal And Vertical Cuts

**MEDIUM**

You are given a rectangular cake of size h x w and two arrays of integers horizontalCuts and verticalCuts where:
- horizontalCuts[i] is the distance from the top of the rectangular cake to the ith horizontal cut and similarly, and
- verticalCuts[j] is the distance from the left of the rectangular cake to the jth vertical cut.

Return the maximum area of a piece of cake after you cut at each horizontal and vertical position provided in the arrays horizontalCuts and verticalCuts. Since the answer can be a large number, return this modulo 109 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1472 Design Browser History

**MEDIUM**

You have a browser of one tab where you start on the homepage and you can visit another url, get back in the history number of steps or move forward in the history number of steps.

Implement the BrowserHistory class:

- `BrowserHistory(string homepage)` Initializes the object with the homepage of the browser.
- `void visit(string url)` Visits url from the current page. It clears up all the forward history.
- `string back(int steps)` Move steps back in history. If you can only return x steps in the history and steps > x, you will return only x steps. Return the current url after moving back in history at most steps.
- `string forward(int steps)` Move steps forward in history. If you can only forward x steps in the history and steps > x, you will forward only x steps. Return the current url after forwarding in history at most steps.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1476 Subrectangle Queries

MEDIUM

Implement the class SubrectangleQueries which receives a rows x cols rectangle as a matrix of integers in the constructor and supports two methods:
- Updates all values with newValue in the subrectangle whose upper left coordinate is (row1,col1) and bottom right coordinate is (row2,col2).

Returns the current value of the coordinate (row,col) from the rectangle.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1491 Average Salary Excluding The Minimum And Maximum Salary

EASY

Given an array of unique integers salary where salary[i] is the salary of the employee i.

Return the average salary of employees excluding the minimum and maximum salary.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1492 The Kth Factor Of N

MEDIUM

Given two positive integers n and k.

A factor of an integer n is defined as an integer i where n % i == 0.

Consider a list of all factors of n sorted in ascending order, return the kth factor in this list or return -1 if n has less than k factors.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1494 Parallel Courses Ii

HARD

You are given an integer n, which indicates that there are n courses labeled from 1 to n. You are also given an array relations where relations[i] = [prevCoursei, nextCoursei], representing a prerequisite relationship between course prevCoursei and course nextCoursei: course prevCoursei has to be taken before course nextCoursei. Also, you are given the integer k.

In one semester, you can take at most k courses as long as you have taken all the prerequisites in the previous semesters for the courses you are taking.

Return the minimum number of semesters needed to take all courses. The testcases will be generated such that it is possible to take every course.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1499 Max Value Of Equation

HARD

You are given an array points containing the coordinates of points on a 2D plane, sorted by the x-values, where points[i] = [xi, yi] such that xi < xj for all 1 <= i < j <= points.length. You are also given an integer k.

Return the maximum value of the equation yi + yj + |xi - xj| where |xi - xj| <= k and 1 <= i < j <= points.length.

It is guaranteed that there exists at least one pair of points that satisfy the constraint |xi - xj| <= k.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 1502 Can Make Arithmetic Progression From Sequence

EASY

A sequence of numbers is called an arithmetic progression if the difference between any two consecutive elements is the same.

Given an array of numbers arr, return true if the array can be rearranged to form an arithmetic progression. Otherwise, return false.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 1503 Last Moment Before All Ants Fall Out Of A Plank

MEDIUM

We have a wooden plank of the length n units. Some ants are walking on the plank, each ant moves with a speed of 1 unit per second. Some of the ants move to the left, the other move to the right.

When two ants moving in two different directions meet at some point, they change their directions and continue moving again. Assume changing directions does not take any additional time.

When an ant reaches one end of the plank at a time t, it falls out of the plank immediately.

Given an integer n and two integer arrays left and right, the positions of the ants moving to the left and the right, return the moment when the last ant(s) fall out of the plank.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 1507 Reformat Date

EASY

Given a date string in the form Day Month Year, where:
- Day is in the set {"1st", "2nd", "3rd", "4th", ..., "30th", "31st"}.
- Month is in the set {"Jan", "Feb", "Mar", "Apr", "May", "Jun", "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"}.
- Year is in the range [1900, 2100].

Convert the date string to the format YYYY-MM-DD, where:
- YYYY denotes the 4 digit year.
- MM denotes the 2 digit month.
- DD denotes the 2 digit day.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 1510 Stone Game Iv

HARD

Alice and Bob take turns playing a game, with Alice starting first.

Initially, there are n stones in a pile. On each player's turn, that player makes a move consisting of removing any non-zero square number of stones in the pile.

Also, if a player cannot make a move, he/she loses the game.

Given a positive integer n, return true if and only if Alice wins the game otherwise return false, assuming both players play optimally.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 1514 Path With Maximum Probability

MEDIUM

You are given an undirected weighted graph of n nodes (0-indexed), represented by an edge list where edges[i] = [a, b] is an undirected edge connecting the nodes a and b with a probability of success of traversing that edge succProb[i].

Given two nodes start and end, find the path with the maximum probability of success to go from start to end and return its success probability.

If there is no path from start to end, return 0. Your answer will be accepted if it differs from the correct answer by at most 1e-5.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1515 Best Position For A Service Centre

HARD

A delivery company wants to build a new service center in a new city. The company knows the positions of all the customers in this city on a 2D-Map and wants to build the new center in a position such that the sum of the euclidean distances to all customers is minimum.

Given an array positions where positions[i] = [xi, yi] is the position of the ith customer on the map, return the minimum sum of the euclidean distances to all customers.

In other words, you need to choose the position of the service center [xcentre, ycentre] such that the following formula is minimized:
- Answers within 10-5 of the actual value will be accepted.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1518 Water Bottles

EASY

There are numBottles water bottles that are initially full of water. You can exchange numExchange empty water bottles from the market with one full water bottle.

The operation of drinking a full water bottle turns it into an empty bottle.

Given the two integers numBottles and numExchange, return the maximum number of water bottles you can drink.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1533 Find The Index Of The Large Integer

MEDIUM

We have an integer array arr, where all the integers in arr are equal except for one integer which is larger than the rest of the integers. You will not be given direct access to the array, instead, you will have an API ArrayReader which have the following functions:
- int compareSub(int l, int r, int x, int y): where 0 <= l, r, x, y < ArrayReader.length(), l <= r and x <= y. The function compares the sum of sub-array arr[l..r] with the sum of the sub-array arr[x..y] and returns:
- 1 if arr[l]+arr[l+1]+...+arr[r] > arr[x]+arr[x+1]+...+arr[y].
- 0 if arr[l]+arr[l+1]+...+arr[r] == arr[x]+arr[x+1]+...+arr[y].
- -1 if arr[l]+arr[l+1]+...+arr[r] < arr[x]+arr[x+1]+...+arr[y].
- int length(): Returns the size of the array.

You are allowed to call compareSub() 20 times at most. You can assume both functions work in O(1) time.

Return the index of the array arr which has the largest integer.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1537 Get The Maximum Score

**HARD**

You are given two sorted arrays of distinct integers nums1 and nums2.

A valid path is defined as follows:
- Choose array nums1 or nums2 to traverse (from index-0).
- Traverse the current array from left to right.
- If you are reading any value that is present in nums1 and nums2 you are allowed to change your path to the other array. (Only one repeated value is considered in the valid path).

The score is defined as the sum of unique values in a valid path.

Return the maximum score you can obtain of all possible valid paths. Since the answer may be too large, return it modulo 109 + 7.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1550 Three Consecutive Odds

**EASY**

Given an integer array arr, return true if there are three consecutive odd numbers in the array. Otherwise, return false.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1552 Magnetic Force Between Two Balls

**MEDIUM**

In the universe Earth C-137, Rick discovered a special form of magnetic force between two balls if they are put in his new invented basket. Rick has n empty baskets, the ith basket is at position[i], Morty has m balls and needs to distribute the balls into the baskets such that the minimum magnetic force between any two balls is maximum.

Rick stated that magnetic force between two different balls at positions x and y is |x - y|.

Given the integer array position and the integer m. Return the required force.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1556 Thousand Separator

**EASY**

Given an integer n, add a dot (".") as the thousands separator and return it in string format.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1560 Most Visited Sector In A Circular Track

**EASY**

Given an integer n and an integer array rounds. We have a circular track which consists of n sectors labeled from 1 to n. A marathon will be held on this track, the marathon consists of m rounds. The ith round starts at sector rounds[i - 1] and ends at sector rounds[i]. For example, round 1 starts at sector rounds[0] and ends at sector rounds[1]

Return an array of the most visited sectors sorted in ascending order.

Notice that you circulate the track in ascending order of sector numbers in the counter-clockwise direction (See the first example).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1563 Stone Game V

**HARD**

There are several stones arranged in a row, and each stone has an associated value which is an integer given in the array stoneValue.

In each round of the game, Alice divides the row into two non-empty rows (i.e. left row and right row), then Bob calculates the value of each row which is the sum of the values of all the stones in this row. Bob throws away the row which has the maximum value, and Alice's score increases by the value of the remaining row. If the value of the two rows are equal, Bob lets Alice decide which row will be thrown away. The next round starts with the remaining row.

The game ends when there is only one stone remaining. Alice's is initially zero.

Return the maximum score that Alice can obtain.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1564 Put Boxes Into The Warehouse I

**MEDIUM**

You are given two arrays of positive integers, boxes and warehouse, representing the heights of some boxes of unit width and the heights of n rooms in a warehouse respectively. The warehouse's rooms are labelled from 0 to n - 1 from left to right where warehouse[i] (0-indexed) is the height of the ith room.

Boxes are put into the warehouse by the following rules:
- Boxes cannot be stacked.
- You can rearrange the insertion order of the boxes.
- Boxes can only be pushed into the warehouse from left to right only.
- If the height of some room in the warehouse is less than the height of a box, then that box and all other boxes behind it will be stopped before that room.

Return the maximum number of boxes you can put into the warehouse.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1566 Detect Pattern Of Length M Repeated K Or More Times

**EASY**

Given an array of positive integers arr, find a pattern of length m that is repeated k or more times.

A pattern is a subarray (consecutive sub-sequence) that consists of one or more values, repeated multiple times consecutively without overlapping. A pattern is defined by its length and the number of repetitions.

Return true if there exists a pattern of length m that is repeated k or more times, otherwise return false.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1578 Minimum Time To Make Rope Colorful

**MEDIUM**

Alice has n balloons arranged on a rope. You are given a 0-indexed string colors where colors[i] is the color of the ith balloon.

Alice wants the rope to be colorful. She does not want two consecutive balloons to be of the same

color, so she asks Bob for help. Bob can remove some balloons from the rope to make it colorful. You are given a 0-indexed integer array neededTime where neededTime[i] is the time (in seconds) that Bob needs to remove the ith balloon from the rope.

Return the minimum time Bob needs to make the rope colorful.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1580 Put Boxes Into The Warehouse Ii

**MEDIUM**

You are given two arrays of positive integers, boxes and warehouse, representing the heights of some boxes of unit width and the heights of n rooms in a warehouse respectively. The warehouse's rooms are labeled from 0 to n - 1 from left to right where warehouse[i] (0-indexed) is the height of the ith room.

Boxes are put into the warehouse by the following rules:
- Boxes cannot be stacked.
- You can rearrange the insertion order of the boxes.
- Boxes can be pushed into the warehouse from either side (left or right)
- If the height of some room in the warehouse is less than the height of a box, then that box and all other boxes behind it will be stopped before that room.

Return the maximum number of boxes you can put into the warehouse.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1598 Crawler Log Folder

**EASY**

The Leetcode file system keeps a log each time some user performs a change folder operation.

The operations are described below:

- `"../"` : Move to the parent folder of the current folder. (If you are already in the main folder, remain in the same folder).
- `"./"` : Remain in the same folder.
- `"x/"` : Move to the child folder named x (This folder is guaranteed to always exist).

You are given a list of strings logs where logs[i] is the operation performed by the user at the ith step.

The file system starts in the main folder, then the operations in logs are performed.

Return the minimum number of operations needed to go back to the main folder after the change folder operations.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1599 Maximum Profit Of Operating A Centennial Wheel

**MEDIUM**

You are the operator of a Centennial Wheel that has four gondolas, and each gondola has room for up to four people. You have the ability to rotate the gondolas counterclockwise, which costs you runningCost dollars.

You are given an array customers of length n where customers[i] is the number of new customers arriving just before the ith rotation (0-indexed). This means you must rotate the wheel i times before the customers[i] customers arrive. You cannot make customers wait if there is room in the gondola. Each customer pays boardingCost dollars when they board on the gondola closest to the

ground and will exit once that gondola reaches the ground again.

You can stop the wheel at any time, including before serving all customers. If you decide to stop serving customers, all subsequent rotations are free in order to get all the customers down safely. Note that if there are currently more than four customers waiting at the wheel, only four will board the gondola, and the rest will wait for the next rotation.

Return the minimum number of rotations you need to perform to maximize your profit. If there is no scenario where the profit is positive, return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1603 Design Parking System

**EASY**

Design a parking system for a parking lot. The parking lot has three kinds of parking spaces: big, medium, and small, with a fixed number of slots for each size.

Implement the ParkingSystem class:
- ParkingSystem(int big, int medium, int small) Initializes object of the ParkingSystem class. The number of slots for each parking space are given as part of the constructor.
- bool addCar(int carType) Checks whether there is a parking space of carType for the car that wants to get into the parking lot. carType can be of three kinds: big, medium, or small, which are represented by 1, 2, and 3 respectively. A car can only park in a parking space of its carType. If there is no space available, return false, else park the car in that size space and return true.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1611 Minimum One Bit Operations To Make Integers Zero

**HARD**

Given an integer n, you must transform it into 0 using the following operations any number of times:
- Change the rightmost (0th) bit in the binary representation of n.
- Change the ith bit in the binary representation of n if the (i-1)th bit is set to 1 and the (i-2)th through 0th bits are set to 0.

Return the minimum number of operations to transform n into 0.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1620 Coordinate With Maximum Network Quality

**MEDIUM**

You are given an array of network towers towers, where towers[i] = [xi, yi, qi] denotes the ith network tower with location (xi, yi) and quality factor qi. All the coordinates are integral coordinates on the X-Y plane, and the distance between the two coordinates is the Euclidean distance.

You are also given an integer radius where a tower is reachable if the distance is less than or equal to radius. Outside that distance, the signal becomes garbled, and the tower is not reachable.

The signal quality of the ith tower at a coordinate (x, y) is calculated with the formula ■qi / (1 + d)■, where d is the distance between the tower and the coordinate. The network quality at a coordinate is the sum of the signal qualities from all the reachable towers.

Return the array [cx, cy] representing the integral coordinate (cx, cy) where the network quality is maximum. If there are multiple coordinates with the same network quality, return the lexicographically minimum non-negative coordinate.

Note:
- A coordinate (x1, y1) is lexicographically smaller than (x2, y2) if either:
- x1 < x2, or
- x1 == x2 and y1 < y2.
- ■val■ is the greatest integer less than or equal to val (the floor function).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1622 Fancy Sequence

**HARD**

Write an API that generates fancy sequences using the append, addAll, and multAll operations.

Implement the Fancy class:
- Fancy() Initializes the object with an empty sequence.
- void append(val) Appends an integer val to the end of the sequence.
- void addAll(inc) Increments all existing values in the sequence by an integer inc.
- void multAll(m) Multiplies all existing values in the sequence by an integer m.
- int getIndex(idx) Gets the current value at index idx (0-indexed) of the sequence modulo 109 + 7. If the index is greater or equal than the length of the sequence, return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1626 Best Team With No Conflicts

**MEDIUM**

You are the manager of a basketball team. For the upcoming tournament, you want to choose the team with the highest overall score. The score of the team is the sum of scores of all the players in the team.

However, the basketball team is not allowed to have conflicts. A conflict exists if a younger player has a strictly higher score than an older player. A conflict does not occur between players of the same age.

Given two lists, scores and ages, where each scores[i] and ages[i] represents the score and age of the ith player, respectively, return the highest overall score of all possible basketball teams.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1629 Slowest Key

**EASY**

A newly designed keypad was tested, where a tester pressed a sequence of n keys, one at a time.

You are given a string keysPressed of length n, where keysPressed[i] was the ith key pressed in the testing sequence, and a sorted list releaseTimes, where releaseTimes[i] was the time the ith key was released. Both arrays are 0-indexed. The 0th key was pressed at the time 0, and every subsequent key was pressed at the exact time the previous key was released.

The tester wants to know the key of the keypress that had the longest duration. The ith keypress had a duration of releaseTimes[i] - releaseTimes[i - 1], and the 0th keypress had a duration of releaseTimes[0].

Note that the same key could have been pressed multiple times during the test, and these multiple presses of the same key may not have had the same duration.

Return the key of the keypress that had the longest duration. If there are multiple such keypresses, return the lexicographically largest key of the keypresses.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1631 Path With Minimum Effort

MEDIUM

You are a hiker preparing for an upcoming hike. You are given heights, a 2D array of size rows x columns, where heights[row][col] represents the height of cell (row, col). You are situated in the top-left cell, (0, 0), and you hope to travel to the bottom-right cell, (rows-1, columns-1) (i.e., 0-indexed). You can move up, down, left, or right, and you wish to find a route that requires the minimum effort.

A route's effort is the maximum absolute difference in heights between two consecutive cells of the route.

Return the minimum effort required to travel from the top-left cell to the bottom-right cell.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1637 Widest Vertical Area Between Two Points Containing No Points

EASY

Given n points on a 2D plane where points[i] = [xi, yi], Return the widest vertical area between two points such that no points are inside the area.

A vertical area is an area of fixed-width extending infinitely along the y-axis (i.e., infinite height). The widest vertical area is the one with the maximum width.

Note that points on the edge of a vertical area are not considered included in the area.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1642 Furthest Building You Can Reach

MEDIUM

You are given an integer array heights representing the heights of buildings, some bricks, and some ladders.

You start your journey from building 0 and move to the next building by possibly using bricks or ladders.

While moving from building i to building i+1 (0-indexed):
- If the current building's height is greater than or equal to the next building's height, you do not need a ladder or bricks.
- If the current building's height is less than the next building's height, you can either use one ladder or (h[i+1] - h[i]) bricks.

Return the furthest building index (0-indexed) you can reach if you use the given ladders and bricks optimally.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1643 Kth Smallest Instructions

HARD

Bob is standing at cell (0, 0), and he wants to reach destination: (row, column). He can only travel right and down. You are going to help Bob by providing instructions for him to reach destination.

The instructions are represented as a string, where each character is either:
- 'H', meaning move horizontally (go right), or
- 'V', meaning move vertically (go down).

Multiple instructions will lead Bob to destination. For example, if destination is (2, 3),

both "HHHVV" and "HVHVH" are valid instructions.

However, Bob is very picky. Bob has a lucky number k, and he wants the kth lexicographically smallest instructions that will lead him to destination. k is 1-indexed.

Given an integer array destination and an integer k, return the kth lexicographically smallest instructions that will take Bob to destination.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1648 Sell Diminishing Valued Colored Balls

**MEDIUM**

You have an inventory of different colored balls, and there is a customer that wants orders balls of any color.

The customer weirdly values the colored balls. Each colored ball's value is the number of balls of that color you currently have in your inventory. For example, if you own 6 yellow balls, the customer would pay 6 for the first yellow ball. After the transaction, there are only 5 yellow balls left, so the next yellow ball is then valued at 5 (i.e., the value of the balls decreases as you sell more to the customer).

You are given an integer array, inventory, where inventory[i] represents the number of balls of the ith color that you initially own. You are also given an integer orders, which represents the total number of balls that the customer wants. You can sell the balls in any order.

Return the maximum total value that you can attain after selling orders colored balls. As the answer may be too large, return it modulo 109 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1652 Defuse The Bomb

**EASY**

You have a bomb to defuse, and your time is running out! Your informer will provide you with a circular array code of length of n and a key k.

To decrypt the code, you must replace every number. All the numbers are replaced simultaneously.
- If k > 0, replace the ith number with the sum of the next k numbers.
- If k < 0, replace the ith number with the sum of the previous k numbers.
- If k == 0, replace the ith number with 0.

As code is circular, the next element of code[n-1] is code[0], and the previous element of code[0] is code[n-1].

Given the circular array code and an integer key k, return the decrypted code to defuse the bomb!

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1656 Design An Ordered Stream

**EASY**

There is a stream of n (idKey, value) pairs arriving in an arbitrary order, where idKey is an integer between 1 and n and value is a string. No two pairs have the same id.

Design a stream that returns the values in increasing order of their IDs by returning a chunk (list) of values after each insertion. The concatenation of all the chunks should result in a list of the sorted values.

Implement the OrderedStream class:
- OrderedStream(int n) Constructs the stream to take n values.
- String[] insert(int idKey, String value) Inserts the pair (idKey, value) into the stream, then returns the largest possible chunk of currently inserted values that appear next in

the order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1658 Minimum Operations To Reduce X To Zero

**MEDIUM**

You are given an integer array nums and an integer x. In one operation, you can either remove the leftmost or the rightmost element from the array nums and subtract its value from x. Note that this modifies the array for future operations.

Return the minimum number of operations to reduce x to exactly 0 if it is possible, otherwise, return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1665 Minimum Initial Energy To Finish Tasks

**HARD**

You are given an array tasks where tasks[i] = [actuali, minimumi]:
- actuali is the actual amount of energy you spend to finish the ith task.
- minimumi is the minimum amount of energy you require to begin the ith task.

For example, if the task is [10, 12] and your current energy is 11, you cannot start this task. However, if your current energy is 13, you can complete this task, and your energy will be 3 after finishing it.

You can finish the tasks in any order you like.

Return the minimum initial amount of energy you will need to finish all the tasks.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1672 Richest Customer Wealth

**EASY**

You are given an m x n integer grid accounts where accounts[i][j] is the amount of money the i[redacted]th[redacted] customer has in the j[redacted]th[redacted] bank. Return the wealth that the richest customer has.

A customer's wealth is the amount of money they have in all their bank accounts. The richest customer is the customer that has the maximum wealth.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1673 Find The Most Competitive Subsequence

**MEDIUM**

Given an integer array nums and a positive integer k, return the most competitive subsequence of nums of size k.

An array's subsequence is a resulting sequence obtained by erasing some (possibly zero) elements from the array.

We define that a subsequence a is more competitive than a subsequence b (of the same length) if in the first position where a and b differ, subsequence a has a number less than the corresponding number in b. For example, [1,3,4] is more competitive than [1,3,5] because the first position they differ is at the final number, and 4 is less than 5.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1678 Goal Parser Interpretation

EASY

You own a Goal Parser that can interpret a string command. The command consists of an alphabet of "G", "()" and/or "(al)" in some order. The Goal Parser will interpret "G" as the string "G", "()" as the string "o", and "(al)" as the string "al". The interpreted strings are then concatenated in the original order.

Given the string command, return the Goal Parser's interpretation of command.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1682 Longest Palindromic Subsequence Ii

MEDIUM

A subsequence of a string s is considered a good palindromic subsequence if:
- It is a subsequence of s.
- It is a palindrome (has the same value if reversed).
- It has an even length.
- No two consecutive characters are equal, except the two middle ones.

For example, if s = "abcabcabb", then "abba" is considered a good palindromic subsequence, while "bcb" (not even length) and "bbbb" (has equal consecutive characters) are not.

Given a string s, return the length of the longest good palindromic subsequence in s.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1686 Stone Game Vi

MEDIUM

Alice and Bob take turns playing a game, with Alice starting first.

There are n stones in a pile. On each player's turn, they can remove a stone from the pile and receive points based on the stone's value. Alice and Bob may value the stones differently.

You are given two integer arrays of length n, aliceValues and bobValues. Each aliceValues[i] and bobValues[i] represents how Alice and Bob, respectively, value the ith stone.

The winner is the person with the most points after all the stones are chosen. If both players have the same amount of points, the game results in a draw. Both players will play optimally. Both players know the other's values.

Determine the result of the game, and:
- If Alice wins, return 1.
- If Bob wins, return -1.
- If the game results in a draw, return 0.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1687 Delivering Boxes From Storage To Ports

HARD

You have the task of delivering some boxes from storage to their ports using only one ship. However, this ship has a limit on the number of boxes and the total weight that it can carry.

You are given an array boxes, where boxes[i] = [portsi, weighti], and three integers portsCount, maxBoxes, and maxWeight.
- portsi is the port where you need to deliver the ith box and weightsi is the weight of the ith box.
- portsCount is the number of ports.

- maxBoxes and maxWeight are the respective box and weight limits of the ship.

The boxes need to be delivered in the order they are given. The ship will follow these steps:
- The ship will take some number of boxes from the boxes queue, not violating the maxBoxes and maxWeight constraints.
- For each loaded box in order, the ship will make a trip to the port the box needs to be delivered to and deliver it. If the ship is already at the correct port, no trip is needed, and the box can immediately be delivered.
- The ship then makes a return trip to storage to take more boxes from the queue.

The ship must end at storage after all the boxes have been delivered.

Return the minimum number of trips the ship needs to make to deliver all boxes to their respective ports.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1690 Stone Game Vii

**MEDIUM**

Alice and Bob take turns playing a game, with Alice starting first.

There are n stones arranged in a row. On each player's turn, they can remove either the leftmost stone or the rightmost stone from the row and receive points equal to the sum of the remaining stones' values in the row. The winner is the one with the higher score when there are no stones left to remove.

Bob found that he will always lose this game (poor Bob, he always loses), so he decided to minimize the score's difference. Alice's goal is to maximize the difference in the score.

Given an array of integers stones where stones[i] represents the value of the ith stone from the left, return the difference in Alice and Bob's score if they both play optimally.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1697 Checking Existence Of Edge Length Limited Paths

**HARD**

An undirected graph of n nodes is defined by edgeList, where edgeList[i] = [ui, vi, disi] denotes an edge between nodes ui and vi with distance disi. Note that there may be multiple edges between two nodes.

Given an array queries, where queries[j] = [pj, qj, limitj], your task is to determine for each queries[j] whether there is a path between pj and qj such that each edge on the path has a distance strictly less than limitj.

Return a boolean array answer, where answer.length == queries.length and the jth value of answer is true if there is a path for queries[j] is true, and false otherwise.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1701 Average Waiting Time

**MEDIUM**

There is a restaurant with a single chef. You are given an array customers, where customers[i] = [arrivali, timei]:
- arrivali is the arrival time of the ith customer. The arrival times are sorted in non-decreasing order.
- timei is the time needed to prepare the order of the ith customer.

When a customer arrives, he gives the chef his order, and the chef starts preparing it once he is idle. The customer waits till the chef finishes preparing his order. The chef does not prepare food for more than one customer at a time. The chef prepares food for customers in

the order they were given in the input.

Return the average waiting time of all customers. Solutions within 10-5 from the actual answer are considered accepted.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1703 Minimum Adjacent Swaps For K Consecutive Ones

**HARD**

You are given an integer array, nums, and an integer k. nums comprises of only 0's and 1's. In one move, you can choose two adjacent indices and swap their values.

Return the minimum number of moves required so that nums has k consecutive 1's.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1706 Where Will The Ball Fall

**MEDIUM**

You have a 2-D grid of size m x n representing a box, and you have n balls. The box is open on the top and bottom sides.

Each cell in the box has a diagonal board spanning two corners of the cell that can redirect a ball to the right or to the left.
- A board that redirects the ball to the right spans the top-left corner to the bottom-right corner and is represented in the grid as 1.
- A board that redirects the ball to the left spans the top-right corner to the bottom-left corner and is represented in the grid as -1.

We drop one ball at the top of each column of the box. Each ball can get stuck in the box or fall out of the bottom. A ball gets stuck if it hits a "V" shaped pattern between two boards or if a board redirects the ball into either wall of the box.

Return an array answer of size n where answer[i] is the column that the ball falls out of at the bottom after dropping the ball from the ith column at the top, or -1 if the ball gets stuck in the box.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1710 Maximum Units On A Truck

**EASY**

You are assigned to put some amount of boxes onto one truck. You are given a 2D array boxTypes, where boxTypes[i] = [numberOfBoxesi, numberOfUnitsPerBoxi]:
- numberOfBoxesi is the number of boxes of type i.
- numberOfUnitsPerBoxi is the number of units in each box of the type i.

You are also given an integer truckSize, which is the maximum number of boxes that can be put on the truck. You can choose any boxes to put on the truck as long as the number of boxes does not exceed truckSize.

Return the maximum total number of units that can be put on the truck.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1716 Calculate Money In Leetcode Bank

**EASY**

Hercy wants to save money for his first car. He puts money in the Leetcode bank every day.

He starts by putting in $1 on Monday, the first day. Every day from Tuesday to Sunday, he will

put in $1 more than the day before. On every subsequent Monday, he will put in $1 more than the previous Monday. Given n, return the total amount of money he will have in the Leetcode bank at the end of the nth day.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1723 Find Minimum Time To Finish All Jobs

**HARD**

You are given an integer array jobs, where jobs[i] is the amount of time it takes to complete the ith job.

There are k workers that you can assign jobs to. Each job should be assigned to exactly one worker. The working time of a worker is the sum of the time it takes to complete all jobs assigned to them. Your goal is to devise an optimal assignment such that the maximum working time of any worker is minimized.

Return the minimum possible maximum working time of any assignment.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1732 Find The Highest Altitude

**EASY**

There is a biker going on a road trip. The road trip consists of n + 1 points at different altitudes. The biker starts his trip on point 0 with altitude equal 0.

You are given an integer array gain of length n where gain[i] is the net gain in altitude between points i and i + 1 for all (0 <= i < n). Return the highest altitude of a point.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1739 Building Boxes

**HARD**

You have a cubic storeroom where the width, length, and height of the room are all equal to n units. You are asked to place n boxes in this room where each box is a cube of unit side length. There are however some rules to placing the boxes:
- You can place the boxes anywhere on the floor.
- If box x is placed on top of the box y, then each side of the four vertical sides of the box y must either be adjacent to another box or to a wall.

Given an integer n, return the minimum possible number of boxes touching the floor.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1744 Can You Eat Your Favorite Candy On Your Favorite Day

**MEDIUM**

You are given a (0-indexed) array of positive integers candiesCount where candiesCount[i] represents the number of candies of the ith type you have. You are also given a 2D array queries where queries[i] = [favoriteTypei, favoriteDayi, dailyCapi].

You play a game with the following rules:
- You start eating candies on day 0.
- You cannot eat any candy of type i unless you have eaten all candies of type i - 1.
- You must eat at least one candy per day until you have eaten all the candies.

Construct a boolean array answer such that answer.length == queries.length and answer[i] is true if you can eat a candy of type favoriteTypei on day favoriteDayi without eating more than dailyCapi candies on any day, and false otherwise. Note that you can eat different

types of candy on the same day, provided that you follow rule 2.

Return the constructed array answer.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1745 Palindrome Partitioning Iv

HARD

Given a string s, return true if it is possible to split the string s into three non-empty palindromic substrings. Otherwise, return false.

A string is said to be palindrome if it the same string when reversed.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1753 Maximum Score From Removing Stones

MEDIUM

You are playing a solitaire game with three piles of stones of sizes a, b, and c respectively. Each turn you choose two different non-empty piles, take one stone from each, and add 1 point to your score. The game stops when there are fewer than two non-empty piles (meaning there are no more available moves).

Given three integers a, b, and c return the maximum score you can get.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1760 Minimum Limit Of Balls In A Bag

MEDIUM

You are given an integer array nums where the ith bag contains nums[i] balls. You are also given an integer maxOperations.

You can perform the following operation at most maxOperations times:
- Take any bag of balls and divide it into two new bags with a positive number of balls.
- For example, a bag of 5 balls can become two new bags of 1 and 4 balls, or two new bags of 2 and 3 balls.

Your penalty is the maximum number of balls in a bag. You want to minimize your penalty after the operations.

Return the minimum possible penalty after performing the operations.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1762 Buildings With An Ocean View

MEDIUM

There are n buildings in a line. You are given an integer array heights of size n that represents the heights of the buildings in the line.

The ocean is to the right of the buildings. A building has an ocean view if the building can see the ocean without obstructions. Formally, a building has an ocean view if all the buildings to its right have a smaller height.

Return a list of indices (0-indexed) of buildings that have an ocean view, sorted in increasing order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1765 Map Of Highest Peak

MEDIUM

You are given an integer matrix isWater of size m x n that represents a map of land and water cells:
- If isWater[i][j] == 0, cell (i, j) is a land cell.
- If isWater[i][j] == 1, cell (i, j) is a water cell.

You must assign each cell a height in a way that follows these rules:
- The height of each cell must be non-negative.
- If the cell is a water cell, its height must be 0.
- Any two adjacent cells must have an absolute height difference of at most 1. A cell is adjacent to another cell if the former is directly north, east, south, or west of the latter (i.e., their sides are touching).

Find an assignment of heights such that the maximum height in the matrix is maximized.

Return an integer matrix height of size m x n where height[i][j] is cell (i, j)'s height.
If there are multiple solutions, return any of them.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1770 Maximum Score From Performing Multiplication Operations

HARD

You are given two 0-indexed integer arrays nums and multipliers of size n and m respectively, where n >= m.

You begin with a score of 0. You want to perform exactly m operations. On the ith operation (0-indexed) you will:
- Choose one integer x from either the start or the end of the array nums.
- Add multipliers[i] * x to your score.
- Note that multipliers[0] corresponds to the first operation, multipliers[1] to the second operation, and so on.
- Remove x from nums.

Return the maximum score after performing m operations.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1771 Maximize Palindrome Length From Subsequences

HARD

You are given two strings, word1 and word2. You want to construct a string in the following manner:
- Choose some non-empty subsequence subsequence1 from word1.
- Choose some non-empty subsequence subsequence2 from word2.
- Concatenate the subsequences: subsequence1 + subsequence2, to make the string.

Return the length of the longest palindrome that can be constructed in the described manner.
If no palindromes can be constructed, return 0.

A subsequence of a string s is a string that can be made by deleting some (possibly none) characters from s without changing the order of the remaining characters.

A palindrome is a string that reads the same forward as well as backward.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1774 Closest Dessert Cost

MEDIUM

You would like to make dessert and are preparing to buy the ingredients. You have n ice cream base flavors and m types of toppings to choose from. You must follow these rules when making your dessert:
- There must be exactly one ice cream base.
- You can add one or more types of topping or have no toppings at all.
- There are at most two of each type of topping.

You are given three inputs:
- baseCosts, an integer array of length n, where each baseCosts[i] represents the price of the ith ice cream base flavor.
- toppingCosts, an integer array of length m, where each toppingCosts[i] is the price of one of the ith topping.
- target, an integer representing your target price for dessert.

You want to make a dessert with a total cost as close to target as possible.

Return the closest possible cost of the dessert to target. If there are multiple, return the lower one.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1776 Car Fleet Ii

HARD

There are n cars traveling at different speeds in the same direction along a one-lane road. You are given an array cars of length n, where cars[i] = [positioni, speedi] represents:
- positioni is the distance between the ith car and the beginning of the road in meters. It is guaranteed that positioni < positioni+1.
- speedi is the initial speed of the ith car in meters per second.

For simplicity, cars can be considered as points moving along the number line. Two cars collide when they occupy the same position. Once a car collides with another car, they unite and form a single car fleet. The cars in the formed fleet will have the same position and the same speed, which is the initial speed of the slowest car in the fleet.

Return an array answer, where answer[i] is the time, in seconds, at which the ith car collides with the next car, or -1 if the car does not collide with the next car. Answers within 10-5 of the actual answers are accepted.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1779 Find Nearest Point That Has The Same X Or Y Coordinate

EASY

You are given two integers, x and y, which represent your current location on a Cartesian grid: (x, y). You are also given an array points where each points[i] = [ai, bi] represents that a point exists at (ai, bi). A point is valid if it shares the same x-coordinate or the same y-coordinate as your location.

Return the index (0-indexed) of the valid point with the smallest Manhattan distance from your current location. If there are multiple, return the valid point with the smallest index. If there are no valid points, return -1.

The Manhattan distance between two points (x1, y1) and (x2, y2) is abs(x1 - x2) + abs(y1 - y2).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1792 Maximum Average Pass Ratio

MEDIUM

There is a school that has classes of students and each class will be having a final exam. You are given a 2D integer array classes, where classes[i] = [passi, totali]. You know beforehand that in the ith class, there are totali total students, but only passi number of students will pass the exam.

You are also given an integer extraStudents. There are another extraStudents brilliant students that are guaranteed to pass the exam of any class they are assigned to. You want to assign each of the extraStudents students to a class in a way that maximizes the average pass ratio across all the classes.

The pass ratio of a class is equal to the number of students of the class that will pass the exam divided by the total number of students of the class. The average pass ratio is the sum of pass ratios of all the classes divided by the number of the classes.

Return the maximum possible average pass ratio after assigning the extraStudents students. Answers within 10-5 of the actual answer will be accepted.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1799 Maximize Score After N Operations

HARD

You are given nums, an array of positive integers of size 2 * n. You must perform n operations on this array.

In the ith operation (1-indexed), you will:
- Choose two elements, x and y.
- Receive a score of i * gcd(x, y).
- Remove x and y from nums.

Return the maximum score you can receive after performing n operations.

The function gcd(x, y) is the greatest common divisor of x and y.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1812 Determine Color Of A Chessboard Square

EASY

You are given coordinates, a string that represents the coordinates of a square of the chessboard. Below is a chessboard for your reference.

Return true if the square is white, and false if the square is black.

The coordinate will always represent a valid chessboard square. The coordinate will always have the letter first, and the number second.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1823 Find The Winner Of The Circular Game

MEDIUM

There are n friends that are playing a game. The friends are sitting in a circle and are numbered from 1 to n in clockwise order. More formally, moving clockwise from the ith friend brings you to the (i+1)th friend for 1 <= i < n, and moving clockwise from the nth friend brings you to the 1st friend.

The rules of the game are as follows:

The counting wraps around the circle and may count some friends more than once.
friend immediately clockwise of the friend who just lost and repeat.

Given the number of friends, n, and an integer k, return the winner of the game.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1824 Minimum Sideway Jumps

**MEDIUM**

There is a 3 lane road of length n that consists of n + 1 points labeled from 0 to n. A frog starts at point 0 in the second lane and wants to jump to point n. However, there could be obstacles along the way.

You are given an array obstacles of length n + 1 where each obstacles[i] (ranging from 0 to 3) describes an obstacle on the lane obstacles[i] at point i. If obstacles[i] == 0, there are no obstacles at point i. There will be at most one obstacle in the 3 lanes at each point.
- For example, if obstacles[2] == 1, then there is an obstacle on lane 1 at point 2.

The frog can only travel from point i to point i + 1 on the same lane if there is not an obstacle on the lane at point i + 1. To avoid obstacles, the frog can also perform a side jump to jump to another lane (even if they are not adjacent) at the same point if there is no obstacle on the new lane.
- For example, the frog can jump from lane 3 at point 3 to lane 1 at point 3.

Return the minimum number of side jumps the frog needs to reach any lane at point n starting from lane 2 at point 0.

Note: There will be no obstacles on points 0 and n.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1825 Finding Mk Average

**HARD**

You are given two integers, m and k, and a stream of integers. You are tasked to implement a data structure that calculates the MKAverage for the stream.

The MKAverage can be calculated using these steps:
- If the number of the elements in the stream is less than m you should consider the MKAverage to be -1. Otherwise, copy the last m elements of the stream to a separate container.
- Remove the smallest k elements and the largest k elements from the container.
- Calculate the average value for the rest of the elements rounded down to the nearest integer.

Implement the MKAverage class:
- MKAverage(int m, int k) Initializes the MKAverage object with an empty stream and the two integers m and k.
- void addElement(int num) Inserts a new element num into the stream.
- int calculateMKAverage() Calculates and returns the MKAverage for the current stream rounded down to the nearest integer.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1826 Faulty Sensor

**EASY**

An experiment is being conducted in a lab. To ensure accuracy, there are two sensors collecting data simultaneously. You are given two arrays sensor1 and sensor2, where sensor1[i] and sensor2[i] are the ith data points collected by the two sensors.

However, this type of sensor has a chance of being defective, which causes exactly one data point to be dropped. After the data is dropped, all the data points to the right of the

dropped data are shifted one place to the left, and the last data point is replaced with some random value. It is guaranteed that this random value will not be equal to the dropped value.
- For example, if the correct data is [1,2,3,4,5] and 3 is dropped, the sensor could return [1,2,4,5,7] (the last position can be any value, not just 7).

We know that there is a defect in at most one of the sensors. Return the sensor number (1 or 2) with the defect. If there is no defect in either sensor or if it is impossible to determine the defective sensor, return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1829 Maximum Xor For Each Query

**MEDIUM**

You are given a sorted array nums of n non-negative integers and an integer maximumBit. You want to perform the following query n times:
- Find a non-negative integer k < 2maximumBit such that nums[0] XOR nums[1] XOR ... XOR nums[nums.length-1] XOR k is maximized. k is the answer to the ith query.
- Remove the last element from the current array nums.

Return an array answer, where answer[i] is the answer to the ith query.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1833 Maximum Ice Cream Bars

**MEDIUM**

It is a sweltering summer day, and a boy wants to buy some ice cream bars.

At the store, there are n ice cream bars. You are given an array costs of length n, where costs[i] is the price of the ith ice cream bar in coins. The boy initially has coins coins to spend, and he wants to buy as many ice cream bars as possible.

Note: The boy can buy the ice cream bars in any order.

Return the maximum number of ice cream bars the boy can buy with coins.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1834 Single Threaded Cpu

**MEDIUM**

You are given n tasks labeled from 0 to n - 1 represented by a 2D integer array tasks, where tasks[i] = [enqueueTimei, processingTimei] means that the ith task will be available to process at enqueueTimei and will take processingTimei to finish processing.

You have a single-threaded CPU that can process at most one task at a time and will act in the following way:
- If the CPU is idle and there are no available tasks to process, the CPU remains idle.
- If the CPU is idle and there are available tasks, the CPU will choose the one with the shortest processing time. If multiple tasks have the same shortest processing time, it will choose the task with the smallest index.
- Once a task is started, the CPU will process the entire task without stopping.
- The CPU can finish a task then start a new one instantly.

Return the order in which the CPU will process the tasks.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1840 Maximum Building Height

HARD

You want to build n new buildings in a city. The new buildings will be built in a line and are labeled from 1 to n.

However, there are city restrictions on the heights of the new buildings:
- The height of each building must be a non-negative integer.
- The height of the first building must be 0.
- The height difference between any two adjacent buildings cannot exceed 1.

Additionally, there are city restrictions on the maximum height of specific buildings. These restrictions are given as a 2D integer array restrictions where restrictions[i] = [idi, maxHeighti] indicates that building idi must have a height less than or equal to maxHeighti.

It is guaranteed that each building will appear at most once in restrictions, and building 1 will not be in restrictions.

Return the maximum possible height of the tallest building.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1847 Closest Room

HARD

There is a hotel with n rooms. The rooms are represented by a 2D integer array rooms where rooms[i] = [roomIdi, sizei] denotes that there is a room with room number roomIdi and size equal to sizei. Each roomIdi is guaranteed to be unique.

You are also given k queries in a 2D array queries where queries[j] = [preferredj, minSizej]. The answer to the jth query is the room number id of a room such that:
- The room has a size of at least minSizej, and
- abs(id - preferredj) is minimized, where abs(x) is the absolute value of x.

If there is a tie in the absolute difference, then use the room with the smallest such id. If there is no such room, the answer is -1.

Return an array answer of length k where answer[j] contains the answer to the jth query.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1854 Maximum Population Year

EASY

You are given a 2D integer array logs where each logs[i] = [birthi, deathi] indicates the birth and death years of the ith person.

The population of some year x is the number of people alive during that year. The ith person is counted in year x's population if x is in the inclusive range [birthi, deathi - 1]. Note that the person is not counted in the year that they die.

Return the earliest year with the maximum population.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1860 Incremental Memory Leak

MEDIUM

You are given two integers memory1 and memory2 representing the available memory in bits on two memory sticks. There is currently a faulty program running that consumes an increasing

amount of memory every second.

At the ith second (starting from 1), i bits of memory are allocated to the stick with more available memory (or from the first memory stick if both have the same available memory). If neither stick has at least i bits of available memory, the program crashes.

Return an array containing [crashTime, memory1crash, memory2crash], where crashTime is the time (in seconds) when the program crashed and memory1crash and memory2crash are the available bits of memory in the first and second sticks respectively.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1861 Rotating The Box

**MEDIUM**

You are given an m x n matrix of characters boxGrid representing a side-view of a box. Each cell of the box is one of the following:
- A stone '#'
- A stationary obstacle '*'
- Empty '.'

The box is rotated 90 degrees clockwise, causing some of the stones to fall due to gravity. Each stone falls down until it lands on an obstacle, another stone, or the bottom of the box. Gravity does not affect the obstacles' positions, and the inertia from the box's rotation does not affect the stones' horizontal positions.

It is guaranteed that each stone in boxGrid rests on an obstacle, another stone, or the bottom of the box.

Return an n x m matrix representing the box after the rotation described above.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1869 Longer Contiguous Segments Of Ones Than Zeros

**EASY**

Given a binary string s, return true if the longest contiguous segment of 1's is strictly longer than the longest contiguous segment of 0's in s, or return false otherwise.
- For example, in s = "110100010" the longest continuous segment of 1s has length 2, and the longest continuous segment of 0s has length 3.

Note that if there are no 0's, then the longest continuous segment of 0's is considered to have a length 0. The same applies if there is no 1's.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1870 Minimum Speed To Arrive On Time

**MEDIUM**

You are given a floating-point number hour, representing the amount of time you have to reach the office. To commute to the office, you must take n trains in sequential order. You are also given an integer array dist of length n, where dist[i] describes the distance (in kilometers) of the ith train ride.

Each train can only depart at an integer hour, so you may need to wait in between each train ride.
- For example, if the 1st train ride takes 1.5 hours, you must wait for an additional 0.5 hours before you can depart on the 2nd train ride at the 2 hour mark.

Return the minimum positive integer speed (in kilometers per hour) that all the trains must travel at for you to reach the office on time, or -1 if it is impossible to be on time.

Tests are generated such that the answer will not exceed 107 and hour will have at most two

digits after the decimal point.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1872 Stone Game Viii

HARD

Alice and Bob take turns playing a game, with Alice starting first.

There are n stones arranged in a row. On each player's turn, while the number of stones is more than one, they will do the following:

The game stops when only one stone is left in the row.

The score difference between Alice and Bob is (Alice's score - Bob's score). Alice's goal is to maximize the score difference, and Bob's goal is the minimize the score difference.

Given an integer array stones of length n where stones[i] represents the value of the ith stone from the left, return the score difference between Alice and Bob if they both play optimally.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1881 Maximum Value After Insertion

MEDIUM

You are given a very large integer n, represented as a string, and an integer digit x. The digits in n and the digit x are in the inclusive range [1, 9], and n may represent a negative number.

You want to maximize n's numerical value by inserting x anywhere in the decimal representation of n. You cannot insert x to the left of the negative sign.
- For example, if n = 73 and x = 6, it would be best to insert it between 7 and 3, making n = 763.
- If n = -55 and x = 2, it would be best to insert it before the first 5, making n = -255.

Return a string representing the maximum value of n after the insertion.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1882 Process Tasks Using Servers

MEDIUM

You are given two 0-indexed integer arrays servers and tasks of lengths n and m respectively. servers[i] is the weight of the ith server, and tasks[j] is the time needed to process the jth task in seconds.

Tasks are assigned to the servers using a task queue. Initially, all servers are free, and the queue is empty.

At second j, the jth task is inserted into the queue (starting with the 0th task being inserted at second 0). As long as there are free servers and the queue is not empty, the task in the front of the queue will be assigned to a free server with the smallest weight, and in case of a tie, it is assigned to a free server with the smallest index.

If there are no free servers and the queue is not empty, we wait until a server becomes free and immediately assign the next task. If multiple servers become free at the same time, then multiple tasks from the queue will be assigned in order of insertion following the weight and index priorities above.

A server that is assigned task j at second t will be free again at second t + tasks[j].

Build an array ans of length m, where ans[j] is the index of the server the jth task will be assigned to.

Return the array ans.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1883 Minimum Skips To Arrive At Meeting On Time

HARD

You are given an integer hoursBefore, the number of hours you have to travel to your meeting. To arrive at your meeting, you have to travel through n roads. The road lengths are given as an integer array dist of length n, where dist[i] describes the length of the ith road in kilometers. In addition, you are given an integer speed, which is the speed (in km/h) you will travel at.

After you travel road i, you must rest and wait for the next integer hour before you can begin traveling on the next road. Note that you do not have to rest after traveling the last road because you are already at the meeting.
- For example, if traveling a road takes 1.4 hours, you must wait until the 2 hour mark before traveling the next road. If traveling a road takes exactly 2 hours, you do not need to wait.

However, you are allowed to skip some rests to be able to arrive on time, meaning you do not need to wait for the next integer hour. Note that this means you may finish traveling future roads at different hour marks.
- For example, suppose traveling the first road takes 1.4 hours and traveling the second road takes 0.6 hours. Skipping the rest after the first road will mean you finish traveling the second road right at the 2 hour mark, letting you start traveling the third road immediately.

Return the minimum number of skips required to arrive at the meeting on time, or -1 if it is impossible.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1884 Egg Drop With 2 Eggs And N Floors

MEDIUM

You are given two identical eggs and you have access to a building with n floors labeled from 1 to n.

You know that there exists a floor f where 0 <= f <= n such that any egg dropped at a floor higher than f will break, and any egg dropped at or below floor f will not break.

In each move, you may take an unbroken egg and drop it from any floor x (where 1 <= x <= n). If the egg breaks, you can no longer use it. However, if the egg does not break, you may reuse it in future moves.

Return the minimum number of moves that you need to determine with certainty what the value of f is.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1889 Minimum Space Wasted From Packaging

HARD

You have n packages that you are trying to place in boxes, one package in each box. There are m suppliers that each produce boxes of different sizes (with infinite supply). A package can be placed in a box if the size of the package is less than or equal to the size of the box.

The package sizes are given as an integer array packages, where packages[i] is the size of the ith package. The suppliers are given as a 2D integer array boxes, where boxes[j] is an array of box sizes that the jth supplier produces.

You want to choose a single supplier and use boxes from them such that the total wasted space is minimized. For each package in a box, we define the space wasted to be size of the box - size

of the package. The total wasted space is the sum of the space wasted in all the boxes.
- For example, if you have to fit packages with sizes [2,3,5] and the supplier offers boxes of sizes [4,8], you can fit the packages of size-2 and size-3 into two boxes of size-4 and the package with size-5 into a box of size-8. This would result in a waste of (4-2) + (4-3) + (8-5) = 6.

Return the minimum total wasted space by choosing the box supplier optimally, or -1 if it is impossible to fit all the packages inside boxes. Since the answer may be large, return it modulo 109 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1891 Cutting Ribbons

**MEDIUM**

You are given an integer array ribbons, where ribbons[i] represents the length of the ith ribbon, and an integer k. You may cut any of the ribbons into any number of segments of positive integer lengths, or perform no cuts at all.

- For example, if you have a ribbon of length 4, you can:
- Keep the ribbon of length 4,
- Cut it into one ribbon of length 3 and one ribbon of length 1,
- Cut it into two ribbons of length 2,
- Cut it into one ribbon of length 2 and two ribbons of length 1, or
- Cut it into four ribbons of length 1.

Your task is to determine the maximum length of ribbon, x, that allows you to cut at least k ribbons, each of length x. You can discard any leftover ribbon from the cuts. If it is impossible to cut k ribbons of the same length, return 0.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1893 Check If All The Integers In A Range Are Covered

**EASY**

You are given a 2D integer array ranges and two integers left and right. Each ranges[i] = [starti, endi] represents an inclusive interval between starti and endi.

Return true if each integer in the inclusive range [left, right] is covered by at least one interval in ranges. Return false otherwise.

An integer x is covered by an interval ranges[i] = [starti, endi] if starti <= x <= endi.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1894 Find The Student That Will Replace The Chalk

**MEDIUM**

There are n students in a class numbered from 0 to n - 1. The teacher will give each student a problem starting with the student number 0, then the student number 1, and so on until the teacher reaches the student number n - 1. After that, the teacher will restart the process, starting with the student number 0 again.

You are given a 0-indexed integer array chalk and an integer k. There are initially k pieces of chalk. When the student number i is given a problem to solve, they will use chalk[i] pieces of chalk to solve that problem. However, if the current number of chalk pieces is strictly less than chalk[i], then the student number i will be asked to replace the chalk.

Return the index of the student that will replace the chalk pieces.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1895 Largest Magic Square

MEDIUM

A k x k magic square is a k x k grid filled with integers such that every row sum, every column sum, and both diagonal sums are all equal. The integers in the magic square do not have to be distinct. Every 1 x 1 grid is trivially a magic square.

Given an m x n integer grid, return the size (i.e., the side length k) of the largest magic square that can be found within this grid.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1896 Minimum Cost To Change The Final Value Of Expression

HARD

You are given a valid boolean expression as a string expression consisting of the characters '1','0','&' (bitwise AND operator),'|' (bitwise OR operator),'(', and ')'.
- For example, "()1|1" and "(1)&()" are not valid while "1", "(((1))|(0))", and "1|(0&(1))" are valid expressions.

Return the minimum cost to change the final value of the expression.
- For example, if expression = "1|1|(0&0)&1", its value is 1|1|(0&0)&1 = 1|1|0&1 = 1|0&1 = 1&1 = 1. We want to apply operations so that the new expression evaluates to 0.

The cost of changing the final value of an expression is the number of operations performed on the expression. The types of operations are described as follows:
- Turn a '1' into a '0'.
- Turn a '0' into a '1'.
- Turn a '&' into a '|'.
- Turn a '|' into a '&'.

Note: '&' does not take precedence over '|' in the order of calculation. Evaluate parentheses first, then in left-to-right order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1924 Erect The Fence Ii

HARD

You are given a 2D integer array trees where trees[i] = [xi, yi] represents the location of the ith tree in the garden.

You are asked to fence the entire garden using the minimum length of rope possible. The garden is well-fenced only if all the trees are enclosed and the rope used forms a perfect circle. A tree is considered enclosed if it is inside or on the border of the circle.

More formally, you must form a circle using the rope with a center (x, y) and radius r where all trees lie inside or on the circle and r is minimum.

Return the center and radius of the circle as a length 3 array [x, y, r]. Answers within 10-5 of the actual answer will be accepted.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1926 Nearest Exit From Entrance In Maze

MEDIUM

You are given an m x n matrix maze (0-indexed) with empty cells (represented as '.') and walls (represented as '+'). You are also given the entrance of the maze, where entrance

= [entrancerow, entrancecol] denotes the row and column of the cell you are initially standing at.

In one step, you can move one cell up, down, left, or right. You cannot step into a cell with a wall, and you cannot step outside the maze. Your goal is to find the nearest exit from the entrance. An exit is defined as an empty cell that is at the border of the maze. The entrance does not count as an exit.

Return the number of steps in the shortest path from the entrance to the nearest exit, or -1 if no such path exists.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1928 Minimum Cost To Reach Destination In Time

**HARD**

There is a country of n cities numbered from 0 to n - 1 where all the cities are connected by bi-directional roads. The roads are represented as a 2D integer array edges where edges[i] = [xi, yi, timei] denotes a road between cities xi and yi that takes timei minutes to travel. There may be multiple roads of differing travel times connecting the same two cities, but no road connects a city to itself.

Each time you pass through a city, you must pay a passing fee. This is represented as a 0-indexed integer array passingFees of length n where passingFees[j] is the amount of dollars you must pay when you pass through city j.

In the beginning, you are at city 0 and want to reach city n - 1 in maxTime minutes or less. The cost of your journey is the summation of passing fees for each city that you passed through at some moment of your journey (including the source and destination cities).

Given maxTime, edges, and passingFees, return the minimum cost to complete your journey, or -1 if you cannot complete it within maxTime minutes.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1943 Describe The Painting

**MEDIUM**

There is a long and thin painting that can be represented by a number line. The painting was painted with multiple overlapping segments where each segment was painted with a unique color. You are given a 2D integer array segments, where segments[i] = [starti, endi, colori] represents the half-closed segment [starti, endi) with colori as the color.

The colors in the overlapping segments of the painting were mixed when it was painted. When two or more colors mix, they form a new color that can be represented as a set of mixed colors.
- For example, if colors 2, 4, and 6 are mixed, then the resulting mixed color is {2,4,6}.

For the sake of simplicity, you should only output the sum of the elements in the set rather than the full set.

You want to describe the painting with the minimum number of non-overlapping half-closed segments of these mixed colors. These segments can be represented by the 2D array painting where painting[j] = [leftj, rightj, mixj] describes a half-closed segment [leftj, rightj) with the mixed color sum of mixj.
- For example, the painting created with segments = [[1,4,5],[1,7,7]] can be described by painting = [[1,4,12],[4,7,7]] because:
- [1,4) is colored {5,7} (with a sum of 12) from both the first and second segments.
- [4,7) is colored {7} from only the second segment.

Return the 2D array painting describing the finished painting (excluding any parts that are not painted). You may return the segments in any order.

A half-closed segment [a, b) is the section of the number line between points a and b including

point a and not including point b.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1952 Three Divisors

**EASY**

Given an integer n, return true if n has exactly three positive divisors. Otherwise, return false.

An integer m is a divisor of n if there exists an integer k such that n = k * m.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1954 Minimum Garden Perimeter To Collect Enough Apples

**MEDIUM**

In a garden represented as an infinite 2D grid, there is an apple tree planted at every integer coordinate. The apple tree planted at an integer coordinate (i, j) has |i| + |j| apples growing on it.

You will buy an axis-aligned square plot of land that is centered at (0, 0).

Given an integer neededApples, return the minimum perimeter of a plot such that at least neededApples apples are inside or on the perimeter of that plot.

The value of |x| is defined as:
- x if x >= 0
- -x if x < 0

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1956 Minimum Time For K Virus Variants To Spread

**HARD**

There are n unique virus variants in an infinite 2D grid. You are given a 2D array points, where points[i] = [xi, yi] represents a virus originating at (xi, yi) on day 0. Note that it is possible for multiple virus variants to originate at the same point.

Every day, each cell infected with a virus variant will spread the virus to all neighboring points in the four cardinal directions (i.e. up, down, left, and right). If a cell has multiple variants, all the variants will spread without interfering with each other.

Given an integer k, return the minimum integer number of days for any point to contain at least k of the unique virus variants.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1958 Check If Move Is Legal

**MEDIUM**

You are given a 0-indexed 8 x 8 grid board, where board[r][c] represents the cell (r, c) on a game board. On the board, free cells are represented by '.', white cells are represented by 'W', and black cells are represented by 'B'.

Each move in this game consists of choosing a free cell and changing it to the color you are playing as (either white or black). However, a move is only legal if, after changing it, the cell becomes the endpoint of a good line (horizontal, vertical, or diagonal).

A good line is a line of three or more cells (including the endpoints) where the endpoints of the line are one color, and the remaining cells in the middle are the opposite color (no cells in the line are free). You can find examples for good lines in the figure below.

Given two integers rMove and cMove and a character color representing the color you are playing as (white or black), return true if changing cell (rMove, cMove) to color color is a legal move, or false if it is not legal.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1959 Minimum Total Space Wasted With K Resizing Operations

**MEDIUM**

You are currently designing a dynamic array. You are given a 0-indexed integer array nums, where nums[i] is the number of elements that will be in the array at time i. In addition, you are given an integer k, the maximum number of times you can resize the array (to any size).

The size of the array at time t, sizet, must be at least nums[t] because there needs to be enough space in the array to hold all the elements. The space wasted at time t is defined as sizet - nums[t], and the total space wasted is the sum of the space wasted across every time t where 0 <= t < nums.length.

Return the minimum total space wasted if you can resize the array at most k times.

Note: The array can have any size at the start and does not count towards the number of resizing operations.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1962 Remove Stones To Minimize The Total

**MEDIUM**

You are given a 0-indexed integer array piles, where piles[i] represents the number of stones in the ith pile, and an integer k. You should apply the following operation exactly k times:
- Choose any piles[i] and remove ceil(piles[i] / 2) stones from it.

Notice that you can apply the operation on the same pile more than once.

Return the minimum possible total number of stones remaining after applying the k operations.

ceil(x) is the smallest integer that is greater than or equal to x (i.e., rounds x up).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1970 Last Day Where You Can Still Cross

**HARD**

There is a 1-based binary matrix where 0 represents land and 1 represents water. You are given integers row and col representing the number of rows and columns in the matrix, respectively.

Initially on day 0, the entire matrix is land. However, each day a new cell becomes flooded with water. You are given a 1-based 2D array cells, where cells[i] = [ri, ci] represents that on the ith day, the cell on the rith row and cith column (1-based coordinates) will be covered with water (i.e., changed to 1).

You want to find the last day that it is possible to walk from the top to the bottom by only walking on land cells. You can start from any cell in the top row and end at any cell in the bottom row. You can only travel in the four cardinal directions (left, right, up, and down).

Return the last day where it is possible to walk from the top to the bottom by only walking on land cells.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1992 Find All Groups Of Farmland

MEDIUM

You are given a 0-indexed m x n binary matrix land where a 0 represents a hectare of forested land and a 1 represents a hectare of farmland.

To keep the land organized, there are designated rectangular areas of hectares that consist entirely of farmland. These rectangular areas are called groups. No two groups are adjacent, meaning farmland in one group is not four-directionally adjacent to another farmland in a different group.

land can be represented by a coordinate system where the top left corner of land is (0, 0) and the bottom right corner of land is (m-1, n-1). Find the coordinates of the top left and bottom right corner of each group of farmland. A group of farmland with a top left corner at (r1, c1) and a bottom right corner at (r2, c2) is represented by the 4-length array [r1, c1, r2, c2].

Return a 2D array containing the 4-length arrays described above for each group of farmland in land. If there are no groups of farmland, return an empty array. You may return the answer in any order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 1997 First Day Where You Have Been In All The Rooms

MEDIUM

There are n rooms you need to visit, labeled from 0 to n - 1. Each day is labeled, starting from 0. You will go in and visit one room a day.

Initially on day 0, you visit room 0. The order you visit the rooms for the coming days is determined by the following rules and a given 0-indexed array nextVisit of length n:
- Assuming that on a day, you visit room i,
- if you have been in room i an odd number of times (including the current visit), on the next day you will visit a room with a lower or equal room number specified by nextVisit[i] where 0 <= nextVisit[i] <= i;
- if you have been in room i an even number of times (including the current visit), on the next day you will visit room (i + 1) mod n.

Return the label of the first day where you have been in all the rooms. It can be shown that such a day exists. Since the answer may be very large, return it modulo 109 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2008 Maximum Earnings From Taxi

MEDIUM

There are n points on a road you are driving your taxi on. The n points on the road are labeled from 1 to n in the direction you are going, and you want to drive from point 1 to point n to make money by picking up passengers. You cannot change the direction of the taxi.

The passengers are represented by a 0-indexed 2D integer array rides, where rides[i] = [starti, endi, tipi] denotes the ith passenger requesting a ride from point starti to point endi who is willing to give a tipi dollar tip.

For each passenger i you pick up, you earn endi - starti + tipi dollars. You may only drive at most one passenger at a time.

Given n and rides, return the maximum number of dollars you can earn by picking up the passengers optimally.

Note: You may drop off a passenger and pick up a different passenger at the same point.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2011 Final Value Of Variable After Performing Operations

EASY

There is a programming language with only four operations and one variable X:
- ++X and X++ increments the value of the variable X by 1.
- --X and X-- decrements the value of the variable X by 1.

Initially, the value of X is 0.

Given an array of strings operations containing a list of operations, return the final value of X after performing all the operations.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2014 Longest Subsequence Repeated K Times

HARD

You are given a string s of length n, and an integer k. You are tasked to find the longest subsequence repeated k times in string s.

A subsequence is a string that can be derived from another string by deleting some or no characters without changing the order of the remaining characters.

A subsequence seq is repeated k times in the string s if seq * k is a subsequence of s, where seq * k represents a string constructed by concatenating seq k times.
- For example, "bba" is repeated 2 times in the string "bababcba", because the string "bbabba", constructed by concatenating "bba" 2 times, is a subsequence of the string "bababcba".

Return the longest subsequence repeated k times in string s. If multiple such subsequences are found, return the lexicographically largest one. If there is no such subsequence, return an empty string.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2015 Average Height Of Buildings In Each Segment

MEDIUM

A perfectly straight street is represented by a number line. The street has building(s) on it and is represented by a 2D integer array buildings, where buildings[i] = [starti, endi, heighti]. This means that there is a building with heighti in the half-closed segment [starti, endi).

You want to describe the heights of the buildings on the street with the minimum number of non-overlapping segments. The street can be represented by the 2D integer array street where street[j] = [leftj, rightj, averagej] describes a half-closed segment [leftj, rightj) of the road where the average heights of the buildings in the segment is averagej.

- For example, if buildings = [[1,5,2],[3,10,4]], the street could be represented by street = [[1,3,2],[3,5,3],[5,10,4]] because:
- From 1 to 3, there is only the first building with an average height of 2 / 1 = 2.
- From 3 to 5, both the first and the second building are there with an average height of (2+4) / 2 = 3.
- From 5 to 10, there is only the second building with an average height of 4 / 1 = 4.

Given buildings, return the 2D integer array street as described above (excluding any areas of the street where there are no buldings). You may return the array in any order.

The average of n elements is the sum of the n elements divided (integer division) by n.

A half-closed segment [a, b) is the section of the number line between points a and b including point a and not including point b.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 2024 Maximize The Confusion Of An Exam

MEDIUM

A teacher is writing a test with n true/false questions, with 'T' denoting true and 'F' denoting false. He wants to confuse the students by maximizing the number of consecutive questions with the same answer (multiple trues or multiple falses in a row).

You are given a string answerKey, where answerKey[i] is the original answer to the ith question. In addition, you are given an integer k, the maximum number of times you may perform the following operation:
- Change the answer key for any question to 'T' or 'F' (i.e., set answerKey[i] to 'T' or 'F').

Return the maximum number of consecutive 'T's or 'F's in the answer key after performing the operation at most k times.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 2028 Find Missing Observations

MEDIUM

You have observations of n + m 6-sided dice rolls with each face numbered from 1 to 6. n of the observations went missing, and you only have the observations of m rolls. Fortunately, you have also calculated the average value of the n + m rolls.

You are given an integer array rolls of length m where rolls[i] is the value of the ith observation. You are also given the two integers mean and n.

Return an array of length n containing the missing observations such that the average value of the n + m rolls is exactly mean. If there are multiple valid answers, return any of them. If no such array exists, return an empty array.

The average value of a set of k numbers is the sum of the numbers divided by k.

Note that mean is an integer, so the sum of the n + m rolls should be divisible by n + m.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 2029 Stone Game Ix

MEDIUM

Alice and Bob continue their games with stones. There is a row of n stones, and each stone has an associated value. You are given an integer array stones, where stones[i] is the value of the ith stone.

Alice and Bob take turns, with Alice starting first. On each turn, the player may remove any stone from stones. The player who removes a stone loses if the sum of the values of all removed stones is divisible by 3. Bob will win automatically if there are no remaining stones (even if it is Alice's turn).

Assuming both players play optimally, return true if Alice wins and false if Bob wins.

TRY IT / Write examples, edge cases, and your own algorithm first.

## 2038 Remove Colored Pieces If Both Neighbors Are The Same Color

MEDIUM

There are n pieces arranged in a line, and each piece is colored either by 'A' or by 'B'. You are given a string colors of length n where colors[i] is the color of the ith piece.

Alice and Bob are playing a game where they take alternating turns removing pieces from the line. In this game, Alice moves first.
- Alice is only allowed to remove a piece colored 'A' if both its neighbors are also colored 'A'. She is not allowed to remove pieces that are colored 'B'.
- Bob is only allowed to remove a piece colored 'B' if both its neighbors are also colored 'B'. He is not allowed to remove pieces that are colored 'A'.
- Alice and Bob cannot remove pieces from the edge of the line.
- If a player cannot make a move on their turn, that player loses and the other player wins.

Assuming Alice and Bob play optimally, return true if Alice wins, or return false if Bob wins.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2039 The Time When The Network Becomes Idle

**MEDIUM**

There is a network of n servers, labeled from 0 to n - 1. You are given a 2D integer array edges, where edges[i] = [ui, vi] indicates there is a message channel between servers ui and vi, and they can pass any number of messages to each other directly in one second. You are also given a 0-indexed integer array patience of length n.

All servers are connected, i.e., a message can be passed from one server to any other server(s) directly or indirectly through the message channels.

The server labeled 0 is the master server. The rest are data servers. Each data server needs to send its message to the master server for processing and wait for a reply. Messages move between servers optimally, so every message takes the least amount of time to arrive at the master server. The master server will process all newly arrived messages instantly and send a reply to the originating server via the reversed path the message had gone through.

At the beginning of second 0, each data server sends its message to be processed. Starting from second 1, at the beginning of every second, each data server will check if it has received a reply to the message it sent (including any newly arrived replies) from the master server:
- If it has not, it will resend the message periodically. The data server i will resend the message every patience[i] second(s), i.e., the data server i will resend the message if patience[i] second(s) have elapsed since the last time the message was sent from this server.
- Otherwise, no more resending will occur from this server.

The network becomes idle when there are no messages passing between servers or arriving at servers.

Return the earliest second starting from which the network becomes idle.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2043 Simple Bank System

**MEDIUM**

You have been tasked with writing a program for a popular bank that will automate all its incoming transactions (transfer, deposit, and withdraw). The bank has n accounts numbered from 1 to n. The initial balance of each account is stored in a 0-indexed integer array balance, with the (i + 1)th account having an initial balance of balance[i].

Execute all the valid transactions. A transaction is valid if:
- The given account number(s) are between 1 and n, and
- The amount of money withdrawn or transferred from is less than or equal to the balance of the account.

Implement the Bank class:

- Bank(long[] balance) Initializes the object with the 0-indexed integer array balance.
- boolean transfer(int account1, int account2, long money) Transfers money dollars from the account numbered account1 to the account numbered account2. Return true if the transaction was successful, false otherwise.
- boolean deposit(int account, long money) Deposit money dollars into the account numbered account. Return true if the transaction was successful, false otherwise.
- boolean withdraw(int account, long money) Withdraw money dollars from the account numbered account. Return true if the transaction was successful, false otherwise.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2045 Second Minimum Time To Reach Destination

**HARD**

A city is represented as a bi-directional connected graph with n vertices where each vertex is labeled from 1 to n (inclusive). The edges in the graph are represented as a 2D integer array edges, where each edges[i] = [ui, vi] denotes a bi-directional edge between vertex ui and vertex vi. Every vertex pair is connected by at most one edge, and no vertex has an edge to itself. The time taken to traverse any edge is time minutes.

Each vertex has a traffic signal which changes its color from green to red and vice versa every change minutes. All signals change at the same time. You can enter a vertex at any time, but can leave a vertex only when the signal is green. You cannot wait at a vertex if the signal is green.

The second minimum value is defined as the smallest value strictly larger than the minimum value.
- For example the second minimum value of [2, 3, 4] is 3, and the second minimum value of [2, 2, 4] is 4.

Given n, edges, time, and change, return the second minimum time it will take to go from vertex 1 to vertex n.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2050 Parallel Courses Iii

**HARD**

You are given an integer n, which indicates that there are n courses labeled from 1 to n. You are also given a 2D integer array relations where relations[j] = [prevCoursej, nextCoursej] denotes that course prevCoursej has to be completed before course nextCoursej (prerequisite relationship). Furthermore, you are given a 0-indexed integer array time where time[i] denotes how many months it takes to complete the (i+1)th course.

You must find the minimum number of months needed to complete all the courses following these rules:
- You may start taking a course at any time if the prerequisites are met.
- Any number of courses can be taken at the same time.

Return the minimum number of months needed to complete all the courses.

Note: The test cases are generated such that it is possible to complete every course (i.e., the graph is a directed acyclic graph).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2054 Two Best Non Overlapping Events

**MEDIUM**

You are given a 0-indexed 2D integer array of events where events[i] = [startTimei, endTimei, valuei]. The ith event starts at startTimei and ends at endTimei, and if you attend this event, you will

receive a value of valuei. You can choose at most two non-overlapping events to attend such that the sum of their values is maximized.

Return this maximum sum.

Note that the start time and end time is inclusive: that is, you cannot attend two events where one of them starts and the other ends at the same time. More specifically, if you attend an event with end time t, the next event must start at or after t + 1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2055 Plates Between Candles

**MEDIUM**

There is a long table with a line of plates and candles arranged on top of it. You are given a 0-indexed string s consisting of characters '*' and '|' only, where a '*' represents a plate and a '|' represents a candle.

You are also given a 0-indexed 2D integer array queries where queries[i] = [lefti, righti] denotes the substring s[lefti...righti] (inclusive). For each query, you need to find the number of plates between candles that are in the substring. A plate is considered between candles if there is at least one candle to its left and at least one candle to its right in the substring.
- For example, s = "||**||**|*", and a query [3, 8] denotes the substring "*||**|". The number of plates between candles in this substring is 2, as each of the two plates has at least one candle in the substring to its left and right.

Return an integer array answer where answer[i] is the answer to the ith query.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2057 Smallest Index With Equal Value

**EASY**

Given a 0-indexed integer array nums, return the smallest index i of nums such that i mod 10 == nums[i], or -1 if such index does not exist.

x mod y denotes the remainder when x is divided by y.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2069 Walking Robot Simulation Ii

**MEDIUM**

A width x height grid is on an XY-plane with the bottom-left cell at (0, 0) and the top-right cell at (width - 1, height - 1). The grid is aligned with the four cardinal directions ("North", "East", "South", and "West"). A robot is initially at cell (0, 0) facing direction "East".

The robot can be instructed to move for a specific number of steps. For each step, it does the following.
counterclockwise and retries the step.

After the robot finishes moving the number of steps required, it stops and awaits the next instruction.

Implement the Robot class:
- Robot(int width, int height) Initializes the width x height grid with the robot at (0, 0) facing "East".
- void step(int num) Instructs the robot to move forward num steps.
- int[] getPos() Returns the current cell the robot is at, as an array of length 2, [x, y].

- String getDir() Returns the current direction of the robot, "North", "East", "South", or "West".

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2070 Most Beautiful Item For Each Query

**MEDIUM**

You are given a 2D integer array items where items[i] = [pricei, beautyi] denotes the price and beauty of an item respectively.

You are also given a 0-indexed integer array queries. For each queries[j], you want to determine the maximum beauty of an item whose price is less than or equal to queries[j]. If no such item exists, then the answer to this query is 0.

Return an array answer of the same length as queries where answer[j] is the answer to the jth query.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2073 Time Needed To Buy Tickets

**EASY**

There are n people in a line queuing to buy tickets, where the 0th person is at the front of the line and the (n - 1)th person is at the back of the line.

You are given a 0-indexed integer array tickets of length n where the number of tickets that the ith person would like to buy is tickets[i].

Each person takes exactly 1 second to buy a ticket. A person can only buy 1 ticket at a time and has to go back to the end of the line (which happens instantaneously) in order to buy more tickets. If a person does not have any tickets left to buy, the person will leave the line.

Return the time taken for the person initially at position k (0-indexed) to finish buying tickets.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2076 Process Restricted Friend Requests

**HARD**

You are given an integer n indicating the number of people in a network. Each person is labeled from 0 to n - 1.

You are also given a 0-indexed 2D integer array restrictions, where restrictions[i] = [xi, yi] means that person xi and person yi cannot become friends, either directly or indirectly through other people.

Initially, no one is friends with each other. You are given a list of friend requests as a 0-indexed 2D integer array requests, where requests[j] = [uj, vj] is a friend request between person uj and person vj.

A friend request is successful if uj and vj can be friends. Each friend request is processed in the given order (i.e., requests[j] occurs before requests[j + 1]), and upon a successful request, uj and vj become direct friends for all future friend requests.

Return a boolean array result, where each result[j] is true if the jth friend request is successful or false if it is not.

Note: If uj and vj are already direct friends, the request is still successful.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2078 Two Furthest Houses With Different Colors

EASY

There are n houses evenly lined up on the street, and each house is beautifully painted. You are given a 0-indexed integer array colors of length n, where colors[i] represents the color of the ith house.

Return the maximum distance between two houses with different colors.

The distance between the ith and jth houses is abs(i - j), where abs(x) is the absolute value of x.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2079 Watering Plants

MEDIUM

You want to water n plants in your garden with a watering can. The plants are arranged in a row and are labeled from 0 to n - 1 from left to right where the ith plant is located at x = i. There is a river at x = -1 that you can refill your watering can at.

Each plant needs a specific amount of water. You will water the plants in the following way:
- Water the plants in order from left to right.
- After watering the current plant, if you do not have enough water to completely water the next plant, return to the river to fully refill the watering can.
- You cannot refill the watering can early.

You are initially at the river (i.e., x = -1). It takes one step to move one unit on the x-axis.

Given a 0-indexed integer array plants of n integers, where plants[i] is the amount of water the ith plant needs, and an integer capacity representing the watering can capacity, return the number of steps needed to water all the plants.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2092 Find All People With Secret

HARD

You are given an integer n indicating there are n people numbered from 0 to n - 1. You are also given a 0-indexed 2D integer array meetings where meetings[i] = [xi, yi, timei] indicates that person xi and person yi have a meeting at timei. A person may attend multiple meetings at the same time. Finally, you are given an integer firstPerson.

Person 0 has a secret and initially shares the secret with a person firstPerson at time 0. This secret is then shared every time a meeting takes place with a person that has the secret. More formally, for every meeting, if a person xi has the secret at timei, then they will share the secret with person yi, and vice versa.

The secrets are shared instantaneously. That is, a person may receive the secret and share it with people in other meetings within the same time frame.

Return a list of all the people that have the secret after all the meetings have taken place. You may return the answer in any order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2100 Find Good Days To Rob The Bank

MEDIUM

You and a gang of thieves are planning on robbing a bank. You are given a 0-indexed integer array security, where security[i] is the number of guards on duty on the ith day. The days

are numbered starting from 0. You are also given an integer time.

The ith day is a good day to rob the bank if:
- There are at least time days before and after the ith day,
- The number of guards at the bank for the time days before i are non-increasing, and
- The number of guards at the bank for the time days after i are non-decreasing.

More formally, this means day i is a good day to rob the bank if and only if security[i - time] >= security[i - time + 1] >= ... >= security[i] <= ... <= security[i + time - 1] <= security[i + time].

Return a list of all days (0-indexed) that are good days to rob the bank. The order that the days are returned in does not matter.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2102 Sequentially Ordinal Rank Tracker

**HARD**

A scenic location is represented by its name and attractiveness score, where name is a unique string among all locations and score is an integer. Locations can be ranked from the best to the worst. The higher the score, the better the location. If the scores of two locations are equal, then the location with the lexicographically smaller name is better.

You are building a system that tracks the ranking of locations with the system initially starting with no locations. It supports:
- Adding scenic locations, one at a time.
- Querying the ith best location of all locations already added, where i is the number of times the system has been queried (including the current query).
- For example, when the system is queried for the 4th time, it returns the 4th best location of all locations already added.

Note that the test data are generated so that at any time, the number of queries does not exceed the number of locations added to the system.

Implement the SORTracker class:
- SORTracker() Initializes the tracker system.
- void add(string name, int score) Adds a scenic location with name and score to the system.
- string get() Queries and returns the ith best location, where i is the number of times this method has been invoked (including this invocation).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2105 Watering Plants Ii

**MEDIUM**

Alice and Bob want to water n plants in their garden. The plants are arranged in a row and are labeled from 0 to n - 1 from left to right where the ith plant is located at x = i.

Each plant needs a specific amount of water. Alice and Bob have a watering can each, initially full. They water the plants in the following way:
- Alice waters the plants in order from left to right, starting from the 0th plant. Bob waters the plants in order from right to left, starting from the (n - 1)th plant. They begin watering the plants simultaneously.
- It takes the same amount of time to water each plant regardless of how much water it needs.
- Alice/Bob must water the plant if they have enough in their can to fully water it. Otherwise, they first refill their can (instantaneously) then water the plant.
- In case both Alice and Bob reach the same plant, the one with more water currently in his/her watering can should water this plant. If they have the same amount of water, then Alice should water this plant.

Given a 0-indexed integer array plants of n integers, where plants[i] is the amount of water the

ith plant needs, and two integers capacityA and capacityB representing the capacities of Alice's and Bob's watering cans respectively, return the number of times they have to refill to water all the plants.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2106 Maximum Fruits Harvested After At Most K Steps

**HARD**

Fruits are available at some positions on an infinite x-axis. You are given a 2D integer array fruits where fruits[i] = [positioni, amounti] depicts amounti fruits at the position positioni. fruits is already sorted by positioni in ascending order, and each positioni is unique.

You are also given an integer startPos and an integer k. Initially, you are at the position startPos. From any position, you can either walk to the left or right. It takes one step to move one unit on the x-axis, and you can walk at most k steps in total. For every position you reach, you harvest all the fruits at that position, and the fruits will disappear from that position.

Return the maximum total number of fruits you can harvest.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2124 Check If All As Appears Before All Bs

**EASY**

Given a string s consisting of only the characters 'a' and 'b', return true if every 'a' appears before every 'b' in the string. Otherwise, return false.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2126 Destroying Asteroids

**MEDIUM**

You are given an integer mass, which represents the original mass of a planet. You are further given an integer array asteroids, where asteroids[i] is the mass of the ith asteroid.

You can arrange for the planet to collide with the asteroids in any arbitrary order. If the mass of the planet is greater than or equal to the mass of the asteroid, the asteroid is destroyed and the planet gains the mass of the asteroid. Otherwise, the planet is destroyed.

Return true if all asteroids can be destroyed. Otherwise, return false.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2128 Remove All Ones With Row And Column Flips

**MEDIUM**

You are given an m x n binary matrix grid.

In one operation, you can choose any row or column and flip each value in that row or column (i.e., changing all 0's to 1's, and all 1's to 0's).

Return true if it is possible to remove all 1's from grid using any number of operations or false otherwise.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2129 Capitalize The Title

EASY

You are given a string title consisting of one or more words separated by a single space, where each word consists of English letters. Capitalize the string by changing the capitalization of each word such that:
- If the length of the word is 1 or 2 letters, change all letters to lowercase.
- Otherwise, change the first letter to uppercase and the remaining letters to lowercase.
Return the capitalized title.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2134 Minimum Swaps To Group All 1S Together Ii

MEDIUM

A swap is defined as taking two distinct positions in an array and swapping the values in them.

A circular array is defined as an array where we consider the first element and the last element to be adjacent.

Given a binary circular array nums, return the minimum number of swaps required to group all 1's present in the array together at any location.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2136 Earliest Possible Day Of Full Bloom

HARD

You have n flower seeds. Every seed must be planted first before it can begin to grow, then bloom. Planting a seed takes time and so does the growth of a seed. You are given two 0-indexed integer arrays plantTime and growTime, of length n each:
- plantTime[i] is the number of full days it takes you to plant the ith seed. Every day, you can work on planting exactly one seed. You do not have to work on planting the same seed on consecutive days, but the planting of a seed is not complete until you have worked plantTime[i] days on planting it in total.
- growTime[i] is the number of full days it takes the ith seed to grow after being completely planted. After the last day of its growth, the flower blooms and stays bloomed forever.

From the beginning of day 0, you can plant the seeds in any order.

Return the earliest possible day where all seeds are blooming.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2137 Pour Water Between Buckets To Make Water Levels Equal

MEDIUM

You have n buckets each containing some gallons of water in it, represented by a 0-indexed integer array buckets, where the ith bucket contains buckets[i] gallons of water. You are also given an integer loss.

You want to make the amount of water in each bucket equal. You can pour any amount of water from one bucket to another bucket (not necessarily an integer). However, every time you pour k gallons of water, you spill loss percent of k.

Return the maximum amount of water in each bucket after making the amount of water equal. Answers within 10-5 of the actual answer will be accepted.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2139 Minimum Moves To Reach Target Score

MEDIUM

You are playing a game with integers. You start with the integer 1 and you want to reach the integer target.

In one move, you can either:
- Increment the current integer by one (i.e., x = x + 1).
- Double the current integer (i.e., x = 2 * x).

You can use the increment operation any number of times, however, you can only use the double operation at most maxDoubles times.

Given the two integers target and maxDoubles, return the minimum number of moves needed to reach target starting with 1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2140 Solving Questions With Brainpower

MEDIUM

You are given a 0-indexed 2D integer array questions where questions[i] = [pointsi, brainpoweri].

The array describes the questions of an exam, where you have to process the questions in order (i.e., starting from question 0) and make a decision whether to solve or skip each question. Solving question i will earn you pointsi points but you will be unable to solve each of the next brainpoweri questions. If you skip question i, you get to make the decision on the next question.

For example, given questions = [[3, 2], [4, 3], [4, 4], [2, 5]]:
- If question 0 is solved, you will earn 3 points but you will be unable to solve questions 1 and 2.
- If instead, question 0 is skipped and question 1 is solved, you will earn 4 points but you will be unable to solve questions 2 and 3.

Return the maximum points you can earn for the exam.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2141 Maximum Running Time Of N Computers

HARD

You have n computers. You are given the integer n and a 0-indexed integer array batteries where the ith battery can run a computer for batteries[i] minutes. You are interested in running all n computers simultaneously using the given batteries.

Initially, you can insert at most one battery into each computer. After that and at any integer time moment, you can remove a battery from a computer and insert another battery any number of times. The inserted battery can be a totally new battery or a battery from another computer. You may assume that the removing and inserting processes take no time.

Note that the batteries cannot be recharged.

Return the maximum number of minutes you can run all the n computers simultaneously.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2151 Maximum Good People Based On Statements

HARD

There are two types of persons:
- The good person: The person who always tells the truth.

- The bad person: The person who might tell the truth and might lie.

You are given a 0-indexed 2D integer array statements of size n x n that represents the statements made by n people about each other. More specifically, statements[i][j] could be one of the following:
- 0 which represents a statement made by person i that person j is a bad person.
- 1 which represents a statement made by person i that person j is a good person.
- 2 represents that no statement is made by person i about person j.

Additionally, no person ever makes a statement about themselves. Formally, we have that statements[i][i] = 2 for all 0 <= i < n.

Return the maximum number of people who can be good based on the statements made by the n people.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2154 Keep Multiplying Found Values By Two

**EASY**

You are given an array of integers nums. You are also given an integer original which is the first number that needs to be searched for in nums.

You then do the following steps:
If original is found in nums, multiply it by two (i.e., set original = 2 * original).
Otherwise, stop the process.
Repeat this process with the new number as long as you keep finding the number.
Return the final value of original.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2167 Minimum Time To Remove All Cars Containing Illegal Goods

**HARD**

You are given a 0-indexed binary string s which represents a sequence of train cars. s[i] = '0' denotes that the ith car does not contain illegal goods and s[i] = '1' denotes that the ith car does contain illegal goods.

As the train conductor, you would like to get rid of all the cars containing illegal goods. You can do any of the following three operations any number of times:
of time.

Return the minimum time to remove all the cars containing illegal goods.

Note that an empty sequence of cars is considered to have no cars containing illegal goods.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2178 Maximum Split Of Positive Even Integers

**MEDIUM**

You are given an integer finalSum. Split it into a sum of a maximum number of unique positive even integers.
- For example, given finalSum = 12, the following splits are valid (unique positive even integers summing up to finalSum): (12), (2 + 10), (2 + 4 + 6), and (4 + 8). Among them, (2 + 4 + 6) contains the maximum number of integers. Note that finalSum cannot be split into (2 + 2 + 4 + 4) as all the numbers should be unique.

Return a list of integers that represent a valid split containing a maximum number of integers. If no valid split exists for finalSum, return an empty list. You may return the integers in any order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2187 Minimum Time To Complete Trips

MEDIUM

You are given an array time where time[i] denotes the time taken by the ith bus to complete one trip.

Each bus can make multiple trips successively; that is, the next trip can start immediately after completing the current trip. Also, each bus operates independently; that is, the trips of one bus do not influence the trips of any other bus.

You are also given an integer totalTrips, which denotes the number of trips all buses should make in total. Return the minimum time required for all buses to complete at least totalTrips trips.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2188 Minimum Time To Finish The Race

HARD

You are given a 0-indexed 2D integer array tires where tires[i] = [fi, ri] indicates that the ith tire can finish its xth successive lap in fi * ri(x-1) seconds.
- For example, if fi = 3 and ri = 2, then the tire would finish its 1st lap in 3 seconds, its 2nd lap in 3 * 2 = 6 seconds, its 3rd lap in 3 * 22 = 12 seconds, etc.

You are also given an integer changeTime and an integer numLaps.

The race consists of numLaps laps and you may start the race with any tire. You have an unlimited supply of each tire and after every lap, you may change to any given tire (including the current tire type) if you wait changeTime seconds.

Return the minimum time to finish the race.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2194 Cells In A Range On An Excel Sheet

EASY

A cell (r, c) of an excel sheet is represented as a string "<col><row>" where:
- <col> denotes the column number c of the cell. It is represented by alphabetical letters.
- For example, the 1st column is denoted by 'A', the 2nd by 'B', the 3rd by 'C', and so on.
- <row> is the row number r of the cell. The rth row is represented by the integer r.
- You are given a string s in the format "<col1><row1>:<col2><row2>", where <col1> represents the column c1, <row1> represents the row r1, <col2> represents the column c2, and <row2> represents the row r2, such that r1 <= r2 and c1 <= c2.

Return the list of cells (x, y) such that r1 <= x <= r2 and c1 <= y <= c2. The cells should be represented as strings in the format mentioned above and be sorted in non-decreasing order first by columns and then by rows.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2209 Minimum White Tiles After Covering With Carpets

HARD

You are given a 0-indexed binary string floor, which represents the colors of tiles on a floor:
- floor[i] = '0' denotes that the ith tile of the floor is colored black.
- On the other hand, floor[i] = '1' denotes that the ith tile of the floor is colored white.

You are also given numCarpets and carpetLen. You have numCarpets black carpets, each of length carpetLen tiles. Cover the tiles with the given carpets such that the number of white tiles

still visible is minimum. Carpets may overlap one another.

Return the minimum number of white tiles still visible.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2212 Maximum Points In An Archery Competition

**MEDIUM**

Alice and Bob are opponents in an archery competition. The competition has set the following rules:
shot ak and bk arrows on that section respectively. If ak >= bk, then Alice takes k points.
If ak < bk, then Bob takes k points.
- For example, if Alice and Bob both shot 2 arrows on the section with score 11, then Alice takes 11 points. On the other hand, if Alice shot 0 arrows on the section with score 11 and Bob shot 2 arrows on that same section, then Bob takes 11 points.

You are given the integer numArrows and an integer array aliceArrows of size 12, which represents the number of arrows Alice shot on each scoring section from 0 to 11. Now, Bob wants to maximize the total number of points he can obtain.

Return the array bobArrows which represents the number of arrows Bob shot on each scoring section from 0 to 11. The sum of the values in bobArrows should equal numArrows.

If there are multiple ways for Bob to earn the maximum total points, return any one of them.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2214 Minimum Health To Beat Game

**MEDIUM**

You are playing a game that has n levels numbered from 0 to n - 1. You are given a 0-indexed integer array damage where damage[i] is the amount of health you will lose to complete the ith level.

You are also given an integer armor. You may use your armor ability at most once during the game on any level which will protect you from at most armor damage.

You must complete the levels in order and your health must be greater than 0 at all times to beat the game.

Return the minimum health you need to start with to beat the game.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2217 Find Palindrome With Fixed Length

**MEDIUM**

Given an integer array queries and a positive integer intLength, return an array answer where answer[i] is either the queries[i]th smallest positive palindrome of length intLength or -1 if no such palindrome exists.

A palindrome is a number that reads the same backwards and forwards. Palindromes cannot have leading zeros.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2218 Maximum Value Of K Coins From Piles

**HARD**

There are n piles of coins on a table. Each pile consists of a positive number of coins of assorted denominations.

In one move, you can choose any coin on top of any pile, remove it, and add it to your wallet.

Given a list piles, where piles[i] is a list of integers denoting the composition of the ith pile from top to bottom, and a positive integer k, return the maximum total value of coins you can have in your wallet if you choose exactly k coins optimally.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2226 Maximum Candies Allocated To K Children

**MEDIUM**

You are given a 0-indexed integer array candies. Each element in the array denotes a pile of candies of size candies[i]. You can divide each pile into any number of sub piles, but you cannot merge two piles together.

You are also given an integer k. You should allocate piles of candies to k children such that each child gets the same number of candies. Each child can be allocated candies from only one pile of candies and some piles of candies may go unused.

Return the maximum number of candies each child can get.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2234 Maximum Total Beauty Of The Gardens

**HARD**

Alice is a caretaker of n gardens and she wants to plant flowers to maximize the total beauty of all her gardens.

You are given a 0-indexed integer array flowers of size n, where flowers[i] is the number of flowers already planted in the ith garden. Flowers that are already planted cannot be removed. You are then given another integer newFlowers, which is the maximum number of flowers that Alice can additionally plant. You are also given the integers target, full, and partial.

A garden is considered complete if it has at least target flowers. The total beauty of the gardens is then determined as the sum of the following:
- The number of complete gardens multiplied by full.
- The minimum number of flowers in any of the incomplete gardens multiplied by partial.
If there are no incomplete gardens, then this value will be 0.

Return the maximum total beauty that Alice can obtain after planting at most newFlowers flowers.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2235 Add Two Integers

**EASY**

Given two integers num1 and num2, return the sum of the two integers.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2241 Design An Atm Machine

**MEDIUM**

There is an ATM machine that stores banknotes of 5 denominations: 20, 50, 100, 200, and 500 dollars. Initially the ATM is empty. The user can use the machine to deposit or withdraw any amount of money.

When withdrawing, the machine prioritizes using banknotes of larger values.
- For example, if you want to withdraw $300 and there are 2 $50 banknotes, 1 $100 banknote,

and 1 $200 banknote, then the machine will use the $100 and $200 banknotes.
- However, if you try to withdraw $600 and there are 3 $200 banknotes and 1 $500 banknote, then the withdraw request will be rejected because the machine will first try to use the $500 banknote and then be unable to use banknotes to complete the remaining $100. Note that the machine is not allowed to use the $200 banknotes instead of the $500 banknote.

Implement the ATM class:
- ATM() Initializes the ATM object.
- void deposit(int[] banknotesCount) Deposits new banknotes in the order $20, $50, $100, $200, and $500.
- int[] withdraw(int amount) Returns an array of length 5 of the number of banknotes that will be handed to the user in the order $20, $50, $100, $200, and $500, and update the number of banknotes in the ATM after withdrawing. Returns [-1] if it is not possible (do not withdraw any banknotes in this case).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2242 Maximum Score Of A Node Sequence

**HARD**

There is an undirected graph with n nodes, numbered from 0 to n - 1.

You are given a 0-indexed integer array scores of length n where scores[i] denotes the score of node i. You are also given a 2D integer array edges where edges[i] = [ai, bi] denotes that there exists an undirected edge connecting nodes ai and bi.

A node sequence is valid if it meets the following conditions:
- There is an edge connecting every pair of adjacent nodes in the sequence.
- No node appears more than once in the sequence.

The score of a node sequence is defined as the sum of the scores of the nodes in the sequence.

Return the maximum score of a valid node sequence with a length of 4. If no such sequence exists, return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2245 Maximum Trailing Zeros In A Cornered Path

**MEDIUM**

You are given a 2D integer array grid of size m x n, where each cell contains a positive integer.

A cornered path is defined as a set of adjacent cells with at most one turn. More specifically, the path should exclusively move either horizontally or vertically up to the turn (if there is one), without returning to a previously visited cell. After the turn, the path will then move exclusively in the alternate direction: move vertically if it moved horizontally, and vice versa, also without returning to a previously visited cell.

The product of a path is defined as the product of all the values in the path.

Return the maximum number of trailing zeros in the product of a cornered path found in grid.

Note:
- Horizontal movement means moving in either the left or right direction.
- Vertical movement means moving in either the up or down direction.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2258 Escape The Spreading Fire

**HARD**

You are given a 0-indexed 2D integer array grid of size m x n which represents a field.
Each cell has one of three values:

- 0 represents grass,
- 1 represents fire,
- 2 represents a wall that you and fire cannot pass through.

You are situated in the top-left cell, (0, 0), and you want to travel to the safehouse at the bottom-right cell, (m - 1, n - 1). Every minute, you may move to an adjacent grass cell. After your move, every fire cell will spread to all adjacent cells that are not walls.

Return the maximum number of minutes that you can stay in your initial position before moving while still safely reaching the safehouse. If this is impossible, return -1. If you can always reach the safehouse regardless of the minutes stayed, return 109.

Note that even if the fire spreads to the safehouse immediately after you have reached it, it will be counted as safely reaching the safehouse.

A cell is adjacent to another cell if the former is directly north, east, south, or west of the latter (i.e., their sides are touching).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2271 Maximum White Tiles Covered By A Carpet

**MEDIUM**

You are given a 2D integer array tiles where tiles[i] = [li, ri] represents that every tile j in the range li <= j <= ri is colored white.

You are also given an integer carpetLen, the length of a single carpet that can be placed anywhere.

Return the maximum number of white tiles that can be covered by the carpet.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2274 Maximum Consecutive Floors Without Special Floors

**MEDIUM**

Alice manages a company and has rented some floors of a building as office space. Alice has decided some of these floors should be special floors, used for relaxation only.

You are given two integers bottom and top, which denote that Alice has rented all the floors from bottom to top (inclusive). You are also given the integer array special, where special[i] denotes a special floor that Alice has designated for relaxation.

Return the maximum number of consecutive floors without a special floor.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2279 Maximum Bags With Full Capacity Of Rocks

**MEDIUM**

You have n bags numbered from 0 to n - 1. You are given two 0-indexed integer arrays capacity and rocks. The ith bag can hold a maximum of capacity[i] rocks and currently contains rocks[i] rocks. You are also given an integer additionalRocks, the number of additional rocks you can place in any of the bags.

Return the maximum number of bags that could have full capacity after placing the additional rocks in some bags.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2280 Minimum Lines To Represent A Line Chart

MEDIUM

You are given a 2D integer array stockPrices where stockPrices[i] = [dayi, pricei] indicates the price of the stock on day dayi is pricei. A line chart is created from the array by plotting the points on an XY plane with the X-axis representing the day and the Y-axis representing the price and connecting adjacent points. One such example is shown below.

Return the minimum number of lines needed to represent the line chart.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2285 Maximum Total Importance Of Roads

MEDIUM

You are given an integer n denoting the number of cities in a country. The cities are numbered from 0 to n - 1.

You are also given a 2D integer array roads where roads[i] = [ai, bi] denotes that there exists a bidirectional road connecting cities ai and bi.

You need to assign each city with an integer value from 1 to n, where each value can only be used once. The importance of a road is then defined as the sum of the values of the two cities it connects.

Return the maximum total importance of all roads possible after assigning the values optimally.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2286 Booking Concert Tickets In Groups

HARD

A concert hall has n rows numbered from 0 to n - 1, each with m seats, numbered from 0 to m - 1. You need to design a ticketing system that can allocate seats in the following cases:
- If a group of k spectators can sit together in a row.
- If every member of a group of k spectators can get a seat. They may or may not sit together.

Note that the spectators are very picky. Hence:
- They will book seats only if each member of their group can get a seat with row number less than or equal to maxRow. maxRow can vary from group to group.
- In case there are multiple rows to choose from, the row with the smallest number is chosen. If there are multiple seats to choose in the same row, the seat with the smallest number is chosen.

Implement the BookMyShow class:
- BookMyShow(int n, int m) Initializes the object with n as number of rows and m as number of seats per row.
- int[] gather(int k, int maxRow) Returns an array of length 2 denoting the row and seat number (respectively) of the first seat being allocated to the k members of the group, who must sit together. In other words, it returns the smallest possible r and c such that all [c, c + k - 1] seats are valid and empty in row r, and r <= maxRow. Returns [] in case it is not possible to allocate seats to the group.
- boolean scatter(int k, int maxRow) Returns true if all k members of the group can be allocated seats in rows 0 to maxRow, who may or may not sit together. If the seats can be allocated, it allocates k seats to the group with the smallest row numbers, and the smallest possible seat numbers in each row. Otherwise, returns false.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2291 Maximum Profit From Trading Stocks

MEDIUM

You are given two 0-indexed integer arrays of the same length present and future where present[i] is the current price of the ith stock and future[i] is the price of the ith stock a year in the future. You may buy each stock at most once. You are also given an integer budget representing the amount of money you currently have.

Return the maximum amount of profit you can make.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2293 Min Max Game

EASY

You are given a 0-indexed integer array nums whose length is a power of 2.

Apply the following algorithm on nums:
integer array newNums of length n / 2.
min(nums[2 * i], nums[2 * i + 1]).
max(nums[2 * i], nums[2 * i + 1]).

Return the last number that remains in nums after applying the algorithm.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2303 Calculate Amount Paid In Taxes

EASY

You are given a 0-indexed 2D integer array brackets where brackets[i] = [upperi, percenti] means that the ith tax bracket has an upper bound of upperi and is taxed at a rate of percenti. The brackets are sorted by upper bound (i.e. upperi-1 < upperi for 0 < i < brackets.length).

Tax is calculated as follows:
- The first upper0 dollars earned are taxed at a rate of percent0.
- The next upper1 - upper0 dollars earned are taxed at a rate of percent1.
- The next upper2 - upper1 dollars earned are taxed at a rate of percent2.
- And so on.

You are given an integer income representing the amount of money you earned. Return the amount of money that you have to pay in taxes. Answers within 10-5 of the actual answer will be accepted.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2305 Fair Distribution Of Cookies

MEDIUM

You are given an integer array cookies, where cookies[i] denotes the number of cookies in the ith bag. You are also given an integer k that denotes the number of children to distribute all the bags of cookies to. All the cookies in the same bag must go to the same child and cannot be split up.

The unfairness of a distribution is defined as the maximum total cookies obtained by a single child in the distribution.

Return the minimum unfairness of all distributions.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2311 Longest Binary Subsequence Less Than Or Equal To K

MEDIUM

You are given a binary string s and a positive integer k.

Return the length of the longest subsequence of s that makes up a binary number less than or equal to k.

Note:
- The subsequence can contain leading zeroes.
- The empty string is considered to be equal to 0.
- A subsequence is a string that can be derived from another string by deleting some or no characters without changing the order of the remaining characters.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2312 Selling Pieces Of Wood

HARD

You are given two integers m and n that represent the height and width of a rectangular piece of wood. You are also given a 2D integer array prices, where prices[i] = [hi, wi, pricei] indicates you can sell a rectangular piece of wood of height hi and width wi for pricei dollars.

To cut a piece of wood, you must make a vertical or horizontal cut across the entire height or width of the piece to split it into two smaller pieces. After cutting a piece of wood into some number of smaller pieces, you can sell pieces according to prices. You may sell multiple pieces of the same shape, and you do not have to sell all the shapes. The grain of the wood makes a difference, so you cannot rotate a piece to swap its height and width.

Return the maximum money you can earn after cutting an m x n piece of wood.

Note that you can cut the piece of wood as many times as you want.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2317 Maximum Xor After Operations

MEDIUM

You are given a 0-indexed integer array nums. In one operation, select any non-negative integer x and an index i, then update nums[i] to be equal to nums[i] AND (nums[i] XOR x).

Note that AND is the bitwise AND operation and XOR is the bitwise XOR operation.

Return the maximum possible bitwise XOR of all elements of nums after applying the operation any number of times.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2323 Find Minimum Time To Finish All Jobs Ii

MEDIUM

You are given two 0-indexed integer arrays jobs and workers of equal length, where jobs[i] is the amount of time needed to complete the ith job, and workers[j] is the amount of time the jth worker can work each day.

Each job should be assigned to exactly one worker, such that each worker completes exactly one job.

Return the minimum number of days needed to complete all the jobs after assignment.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2330 Valid Palindrome Iv

MEDIUM

You are given a 0-indexed string s consisting of only lowercase English letters. In one operation, you can change any character of s to any other character.

Return true if you can make s a palindrome after performing exactly one or two operations, or return false otherwise.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2335 Minimum Amount Of Time To Fill Cups

EASY

You have a water dispenser that can dispense cold, warm, and hot water. Every second, you can either fill up 2 cups with different types of water, or 1 cup of any type of water.

You are given a 0-indexed integer array amount of length 3 where amount[0], amount[1], and amount[2] denote the number of cold, warm, and hot water cups you need to fill respectively. Return the minimum number of seconds needed to fill up all the cups.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2361 Minimum Costs Using The Train Line

HARD

A train line going through a city has two routes, the regular route and the express route. Both routes go through the same n + 1 stops labeled from 0 to n. Initially, you start on the regular route at stop 0.

You are given two 1-indexed integer arrays regular and express, both of length n. regular[i] describes the cost it takes to go from stop i - 1 to stop i using the regular route, and express[i] describes the cost it takes to go from stop i - 1 to stop i using the express route.

You are also given an integer expressCost which represents the cost to transfer from the regular route to the express route.

Note that:
- There is no cost to transfer from the express route back to the regular route.
- You pay expressCost every time you transfer from the regular route to the express route.
- There is no extra cost to stay on the express route.

Return a 1-indexed array costs of length n, where costs[i] is the minimum cost to reach stop i from stop 0.

Note that a stop can be counted as reached from either route.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2370 Longest Ideal Subsequence

MEDIUM

You are given a string s consisting of lowercase letters and an integer k. We call a string t ideal if the following conditions are satisfied:
- t is a subsequence of the string s.
- The absolute difference in the alphabet order of every two adjacent letters in t is less than or equal to k.

Return the length of the longest ideal string.

A subsequence is a string that can be derived from another string by deleting some or no characters without changing the order of the remaining characters.

Note that the alphabet order is not cyclic. For example, the absolute difference in the alphabet order of 'a' and 'z' is 25, not 1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2374 Node With Highest Edge Score

**MEDIUM**

You are given a directed graph with n nodes labeled from 0 to n - 1, where each node has exactly one outgoing edge.

The graph is represented by a given 0-indexed integer array edges of length n, where edges[i] indicates that there is a directed edge from node i to node edges[i].

The edge score of a node i is defined as the sum of the labels of all the nodes that have an edge pointing to i.

Return the node with the highest edge score. If multiple nodes have the same edge score, return the node with the smallest index.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2379 Minimum Recolors To Get K Consecutive Black Blocks

**EASY**

You are given a 0-indexed string blocks of length n, where blocks[i] is either 'W' or 'B', representing the color of the ith block. The characters 'W' and 'B' denote the colors white and black, respectively.

You are also given an integer k, which is the desired number of consecutive black blocks.

In one operation, you can recolor a white block such that it becomes a black block.

Return the minimum number of operations needed such that there is at least one occurrence of k consecutive black blocks.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2383 Minimum Hours Of Training To Win A Competition

**EASY**

You are entering a competition, and are given two positive integers initialEnergy and initialExperience denoting your initial energy and initial experience respectively.

You are also given two 0-indexed integer arrays energy and experience, both of length n.

You will face n opponents in order. The energy and experience of the ith opponent is denoted by energy[i] and experience[i] respectively. When you face an opponent, you need to have both strictly greater experience and energy to defeat them and move to the next opponent if available.

Defeating the ith opponent increases your experience by experience[i], but decreases your energy by energy[i].

Before starting the competition, you can train for some number of hours. After each hour of training, you can either choose to increase your initial experience by one, or increase your initial energy by one.

Return the minimum number of training hours required to defeat all n opponents.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2391 Minimum Amount Of Time To Collect Garbage

MEDIUM

You are given a 0-indexed array of strings garbage where garbage[i] represents the assortment of garbage at the ith house. garbage[i] consists only of the characters 'M', 'P' and 'G' representing one unit of metal, paper and glass garbage respectively. Picking up one unit of any type of garbage takes 1 minute.

You are also given a 0-indexed integer array travel where travel[i] is the number of minutes needed to go from house i to house i + 1.

There are three garbage trucks in the city, each responsible for picking up one type of garbage. Each garbage truck starts at house 0 and must visit each house in order; however, they do not need to visit every house.

Only one garbage truck may be used at any given moment. While one truck is driving or picking up garbage, the other two trucks cannot do anything.

Return the minimum number of minutes needed to pick up all the garbage.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2397 Maximum Rows Covered By Columns

MEDIUM

You are given an m x n binary matrix matrix and an integer numSelect.

Your goal is to select exactly numSelect distinct columns from matrix such that you cover as many rows as possible.

A row is considered covered if all the 1's in that row are also part of a column that you have selected. If a row does not have any 1s, it is also considered covered.

More formally, let us consider selected = {c1, c2, ...., cnumSelect} as the set of columns selected by you. A row i is covered by selected if:
- For each cell where matrix[i][j] == 1, the column j is in selected.
- Or, no cell in row i has a value of 1.

Return the maximum number of rows that can be covered by a set of numSelect columns.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2407 Longest Increasing Subsequence Ii

HARD

You are given an integer array nums and an integer k.

Find the longest subsequence of nums that meets the following requirements:
- The subsequence is strictly increasing and
- The difference between adjacent elements in the subsequence is at most k.

Return the length of the longest subsequence that meets the requirements.

A subsequence is an array that can be derived from another array by deleting some or no elements without changing the order of the remaining elements.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2410 Maximum Matching Of Players With Trainers

MEDIUM

You are given a 0-indexed integer array players, where players[i] represents the ability of

the ith player. You are also given a 0-indexed integer array trainers, where trainers[j] represents the training capacity of the jth trainer.

The ith player can match with the jth trainer if the player's ability is less than or equal to the trainer's training capacity. Additionally, the ith player can be matched with at most one trainer, and the jth trainer can be matched with at most one player.

Return the maximum number of matchings between players and trainers that satisfy these conditions.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2412 Minimum Money Required Before Transactions

**HARD**

You are given a 0-indexed 2D integer array transactions, where transactions[i] = [costi, cashbacki].

The array describes transactions, where each transaction must be completed exactly once in some order. At any given moment, you have a certain amount of money. In order to complete transaction i, money >= costi must hold true. After performing a transaction, money becomes money - costi + cashbacki.

Return the minimum amount of money required before any transaction so that all of the transactions can be completed regardless of the order of the transactions.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2413 Smallest Even Multiple

**EASY**

Given a positive integer n, return the smallest positive integer that is a multiple of both 2 and n.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2417 Closest Fair Integer

**MEDIUM**

You are given a positive integer n.

We call an integer k fair if the number of even digits in k is equal to the number of odd digits in it.

Return the smallest fair integer that is greater than or equal to n.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2420 Find All Good Indices

**MEDIUM**

You are given a 0-indexed integer array nums of size n and a positive integer k.

We call an index i in the range k <= i < n - k good if the following conditions are satisfied:
- The k elements that are just before the index i are in non-increasing order.
- The k elements that are just after the index i are in non-decreasing order.

Return an array of all good indices sorted in increasing order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2429 Minimize Xor

MEDIUM

Given two positive integers num1 and num2, find the positive integer x such that:
- x has the same number of set bits as num2, and
- The value x XOR num1 is minimal.

Note that XOR is the bitwise XOR operation.

Return the integer x. The test cases are generated such that x is uniquely determined.

The number of set bits of an integer is the number of 1's in its binary representation.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2432 The Employee That Worked On The Longest Task

EASY

There are n employees, each with a unique id from 0 to n - 1.

You are given a 2D integer array logs where logs[i] = [idi, leaveTimei] where:
- idi is the id of the employee that worked on the ith task, and
- leaveTimei is the time at which the employee finished the ith task. All the values leaveTimei are unique.

Note that the ith task starts the moment right after the (i - 1)th task ends, and the 0th task starts at time 0.

Return the id of the employee that worked the task with the longest time. If there is a tie between two or more employees, return the smallest id among them.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2440 Create Components With Same Value

HARD

There is an undirected tree with n nodes labeled from 0 to n - 1.

You are given a 0-indexed integer array nums of length n where nums[i] represents the value of the ith node. You are also given a 2D integer array edges of length n - 1 where edges[i] = [ai, bi] indicates that there is an edge between nodes ai and bi in the tree.

You are allowed to delete some edges, splitting the tree into multiple connected components. Let the value of a component be the sum of all nums[i] for which node i is in the component.

Return the maximum number of edges you can delete, such that every connected component in the tree has the same value.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2446 Determine If Two Events Have Conflict

EASY

You are given two arrays of strings that represent two inclusive events that happened on the same day, event1 and event2, where:
- event1 = [startTime1, endTime1] and
- event2 = [startTime2, endTime2].

Event times are valid 24 hours format in the form of HH:MM.

A conflict happens when two events have some non-empty intersection (i.e., some moment is common to both events).

Return true if there is a conflict between two events. Otherwise, return false.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2462 Total Cost To Hire K Workers

**MEDIUM**

You are given a 0-indexed integer array costs where costs[i] is the cost of hiring the ith worker.

You are also given two integers k and candidates. We want to hire exactly k workers according to the following rules:
- You will run k sessions and hire exactly one worker in each session.
- In each hiring session, choose the worker with the lowest cost from either the first candidates workers or the last candidates workers. Break the tie by the smallest index.
- For example, if costs = [3,2,7,7,1,2] and candidates = 2, then in the first hiring session, we will choose the 4th worker because they have the lowest cost [3,2,7,7,1,2].
- In the second hiring session, we will choose 1st worker because they have the same lowest cost as 4th worker but they have the smallest index [3,2,7,7,2]. Please note that the indexing may be changed in the process.
- If there are fewer than candidates workers remaining, choose the worker with the lowest cost among them. Break the tie by the smallest index.
- A worker can only be chosen once.

Return the total cost to hire exactly k workers.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2463 Minimum Total Distance Traveled

**HARD**

There are some robots and factories on the X-axis. You are given an integer array robot where robot[i] is the position of the ith robot. You are also given a 2D integer array factory where factory[j] = [positionj, limitj] indicates that positionj is the position of the jth factory and that the jth factory can repair at most limitj robots.

The positions of each robot are unique. The positions of each factory are also unique. Note that a robot can be in the same position as a factory initially.

All the robots are initially broken; they keep moving in one direction. The direction could be the negative or the positive direction of the X-axis. When a robot reaches a factory that did not reach its limit, the factory repairs the robot, and it stops moving.

At any moment, you can set the initial direction of moving for some robot. Your target is to minimize the total distance traveled by all the robots.

Return the minimum total distance traveled by all the robots. The test cases are generated such that all the robots can be repaired.

Note that:
- All robots move at the same speed.
- If two robots move in the same direction, they will never collide.
- If two robots move in opposite directions and they meet at some point, they do not collide. They cross each other.
- If a robot passes by a factory that reached its limits, it crosses it as if it does not exist.
- If the robot moved from a position x to a position y, the distance it moved is |y - x|.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2468 Split Message Based On Limit

HARD

You are given a string, message, and a positive integer, limit.

You must split message into one or more parts based on limit. Each resulting part should have the suffix "<a/b>", where "b" is to be replaced with the total number of parts and "a" is to be replaced with the index of the part, starting from 1 and going up to b. Additionally, the length of each resulting part (including its suffix) should be equal to limit, except for the last part whose length can be at most limit.

The resulting parts should be formed such that when their suffixes are removed and they are all concatenated in order, they should be equal to message. Also, the result should contain as few parts as possible.

Return the parts message would be split into as an array of strings. If it is impossible to split message as required, return an empty array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2469 Convert The Temperature

EASY

You are given a non-negative floating point number rounded to two decimal places celsius, that denotes the temperature in Celsius.

You should convert Celsius into Kelvin and Fahrenheit and return it as an array ans = [kelvin, fahrenheit].

Return the array ans. Answers within 10-5 of the actual answer will be accepted.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2473 Minimum Cost To Buy Apples

MEDIUM

You are given a positive integer n representing n cities numbered from 1 to n. You are also given a 2D array roads, where roads[i] = [ai, bi, costi] indicates that there is a bidirectional road between cities ai and bi with a cost of traveling equal to costi.

You can buy apples in any city you want, but some cities have different costs to buy apples. You are given the 1-based array appleCost where appleCost[i] is the cost of buying one apple from city i.

You start at some city, traverse through various roads, and eventually buy exactly one apple from any city. After you buy that apple, you have to return back to the city you started at, but now the cost of all the roads will be multiplied by a given factor k.

Given the integer k, return a 1-based array answer of size n where answer[i] is the minimum total cost to buy an apple if you start at city i.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2481 Minimum Cuts To Divide A Circle

EASY

A valid cut in a circle can be:
- A cut that is represented by a straight line that touches two points on the edge of the circle and passes through its center, or
- A cut that is represented by a straight line that touches one point on the edge of the circle and its center.

Some valid and invalid cuts are shown in the figures below.

Given the integer n, return the minimum number of cuts needed to divide a circle into n equal slices.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2483 Minimum Penalty For A Shop

**MEDIUM**

You are given the customer visit log of a shop represented by a 0-indexed string customers consisting only of characters 'N' and 'Y':
- if the ith character is 'Y', it means that customers come at the ith hour
- whereas 'N' indicates that no customers come at the ith hour.

If the shop closes at the jth hour (0 <= j <= n), the penalty is calculated as follows:
- For every hour when the shop is open and no customers come, the penalty increases by 1.
- For every hour when the shop is closed and customers come, the penalty increases by 1.

Return the earliest hour at which the shop must be closed to incur a minimum penalty.

Note that if a shop closes at the jth hour, it means the shop is closed at the hour j.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2485 Find The Pivot Integer

**EASY**

Given a positive integer n, find the pivot integer x such that:
- The sum of all elements between 1 and x inclusively equals the sum of all elements between x and n inclusively.

Return the pivot integer x. If no such integer exists, return -1. It is guaranteed that there will be at most one pivot index for the given input.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2491 Divide Players Into Teams Of Equal Skill

**MEDIUM**

You are given a positive integer array skill of even length n where skill[i] denotes the skill of the ith player. Divide the players into n / 2 teams of size 2 such that the total skill of each team is equal.

The chemistry of a team is equal to the product of the skills of the players on that team.

Return the sum of the chemistry of all the teams, or return -1 if there is no way to divide the players into teams such that the total skill of each team is equal.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2498 Frog Jump Ii

**MEDIUM**

You are given a 0-indexed integer array stones sorted in strictly increasing order representing the positions of stones in a river.

A frog, initially on the first stone, wants to travel to the last stone and then return to the first stone. However, it can jump to any stone at most once.

The length of a jump is the absolute difference between the position of the stone the frog is currently on and the position of the stone to which the frog jumps.

- More formally, if the frog is at stones[i] and is jumping to stones[j], the length of the jump is |stones[i] - stones[j]|.

The cost of a path is the maximum length of a jump among all jumps in the path.

Return the minimum cost of a path for the frog.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2500 Delete Greatest Value In Each Row

**EASY**

You are given an m x n matrix grid consisting of positive integers.

Perform the following operation until grid becomes empty:
- Delete the element with the greatest value from each row. If multiple such elements exist, delete any of them.
- Add the maximum of deleted elements to the answer.

Note that the number of columns decreases by one after each operation.

Return the answer after performing the operations described above.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2502 Design Memory Allocator

**MEDIUM**

You are given an integer n representing the size of a 0-indexed memory array. All memory units are initially free.

You have a memory allocator with the following functionalities:

Note that:
- Multiple blocks can be allocated to the same mID.
- You should free all the memory units with mID, even if they were allocated in different blocks.

Implement the Allocator class:
- Allocator(int n) Initializes an Allocator object with a memory array of size n.
- int allocate(int size, int mID) Find the leftmost block of size consecutive free memory units and allocate it with the id mID. Return the block's first index. If such a block does not exist, return -1.
- int freeMemory(int mID) Free all memory units with the id mID. Return the number of memory units you have freed.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2511 Maximum Enemy Forts That Can Be Captured

**EASY**

You are given a 0-indexed integer array forts of length n representing the positions of several forts. forts[i] can be -1, 0, or 1 where:
- -1 represents there is no fort at the ith position.
- 0 indicates there is an enemy fort at the ith position.
- 1 indicates the fort at the ith the position is under your command.

Now you have decided to move your army from one of your forts at position i to an empty position j such that:
- 0 <= i, j <= n - 1
- The army travels over enemy forts only. Formally, for all k where min(i,j) < k < max(i,j), forts[k] == 0.

While moving the army, all the enemy forts that come in the way are captured.

Return the maximum number of enemy forts that can be captured. In case it is impossible to move your army, or you do not have any fort under your command, return 0.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2517 Maximum Tastiness Of Candy Basket

**MEDIUM**

You are given an array of positive integers price where price[i] denotes the price of the ith candy and a positive integer k.

The store sells baskets of k distinct candies. The tastiness of a candy basket is the smallest absolute difference of the prices of any two candies in the basket.

Return the maximum tastiness of a candy basket.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2534 Time Taken To Cross The Door

**HARD**

There are n persons numbered from 0 to n - 1 and a door. Each person can enter or exit through the door once, taking one second.

You are given a non-decreasing integer array arrival of size n, where arrival[i] is the arrival time of the ith person at the door. You are also given an array state of size n, where state[i] is 0 if person i wants to enter through the door or 1 if they want to exit through the door.

If two or more persons want to use the door at the same time, they follow the following rules:
- If the door was not used in the previous second, then the person who wants to exit goes first.
- If the door was used in the previous second for entering, the person who wants to enter goes first.
- If the door was used in the previous second for exiting, the person who wants to exit goes first.
- If multiple persons want to go in the same direction, the person with the smallest index goes first.

Return an array answer of size n where answer[i] is the second at which the ith person crosses the door.

Note that:
- Only one person can cross the door at each second.
- A person may arrive at the door and wait without entering or exiting to follow the mentioned rules.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2536 Increment Submatrices By One

**MEDIUM**

You are given a positive integer n, indicating that we initially have an n x n 0-indexed integer matrix mat filled with zeroes.

You are also given a 2D integer array query. For each query[i] = [row1i, col1i, row2i, col2i], you should do the following operation:
- Add 1 to every element in the submatrix with the top left corner (row1i, col1i) and the bottom right corner (row2i, col2i). That is, add 1 to mat[x][y] for all row1i <= x <= row2i and col1i <= y <= col2i.

Return the matrix mat after performing every query.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2540 Minimum Common Value

**EASY**

Given two integer arrays nums1 and nums2, sorted in non-decreasing order, return the minimum integer common to both arrays. If there is no common integer amongst nums1 and nums2, return -1.

Note that an integer is said to be common to nums1 and nums2 if both arrays have at least one occurrence of that integer.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2542 Maximum Subsequence Score

**MEDIUM**

You are given two 0-indexed integer arrays nums1 and nums2 of equal length n and a positive integer k. You must choose a subsequence of indices from nums1 of length k.

For chosen indices i0, i1, ..., ik - 1, your score is defined as:
- The sum of the selected elements from nums1 multiplied with the minimum of the selected elements from nums2.
- It can defined simply as: (nums1[i0] + nums1[i1] +...+ nums1[ik - 1]) * min(nums2[i0], nums2[i1], ... ,nums2[ik - 1]).

Return the maximum possible score.

A subsequence of indices of an array is a set that can be derived from the set {0, 1, ..., n-1} by deleting some or no elements.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2543 Check If Point Is Reachable

**HARD**

There exists an infinitely large grid. You are currently at point (1, 1), and you need to reach the point (targetX, targetY) using a finite number of steps.

In one step, you can move from point (x, y) to any one of the following points:
- (x, y - x)
- (x - y, y)
- (2 * x, y)
- (x, 2 * y)

Given two integers targetX and targetY representing the X-coordinate and Y-coordinate of your final position, return true if you can reach the point from (1, 1) using some number of steps, and false otherwise.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2548 Maximum Price To Fill A Bag

**MEDIUM**

You are given a 2D integer array items where items[i] = [pricei, weighti] denotes the price and weight of the ith item, respectively.

You are also given a positive integer capacity.

Each item can be divided into two items with ratios part1 and part2, where part1 + part2 == 1.
- The weight of the first item is weighti * part1 and the price of the first item is

pricei * part1.
- Similarly, the weight of the second item is weighti * part2 and the price of the second item is pricei * part2.

Return the maximum total price to fill a bag of capacity capacity with given items. If it is impossible to fill a bag return -1. Answers within 10-5 of the actual answer will be considered accepted.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2551 Put Marbles In Bags

**HARD**

You have k bags. You are given a 0-indexed integer array weights where weights[i] is the weight of the ith marble. You are also given the integer k.

Divide the marbles into the k bags according to the following rules:
- No bag is empty.
- If the ith marble and jth marble are in a bag, then all marbles with an index between the ith and jth indices should also be in that same bag.
- If a bag consists of all the marbles with an index from i to j inclusively, then the cost of the bag is weights[i] + weights[j].

The score after distributing the marbles is the sum of the costs of all the k bags.

Return the difference between the maximum and minimum scores among marble distributions.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2571 Minimum Operations To Reduce An Integer To 0

**MEDIUM**

You are given a positive integer n, you can do the following operation any number of times:
- Add or subtract a power of 2 from n.

Return the minimum number of operations to make n equal to 0.

A number x is power of 2 if x == 2i where i >= 0.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2582 Pass The Pillow

**EASY**

There are n people standing in a line labeled from 1 to n. The first person in the line is holding a pillow initially. Every second, the person holding the pillow passes it to the next person standing in the line. Once the pillow reaches the end of the line, the direction changes, and people continue passing the pillow in the opposite direction.
- For example, once the pillow reaches the nth person they pass it to the n - 1th person, then to the n - 2th person and so on.

Given the two positive integers n and time, return the index of the person holding the pillow after time seconds.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2596 Check Knight Tour Configuration

**MEDIUM**

There is a knight on an n x n chessboard. In a valid configuration, the knight starts at the top-left cell of the board and visits every cell on the board exactly once.

You are given an n x n integer matrix grid consisting of distinct integers from the range [0, n * n - 1] where grid[row][col] indicates that the cell (row, col) is the grid[row][col]th cell that the knight visited. The moves are 0-indexed.

Return true if grid represents a valid configuration of the knight's movements or false otherwise.

Note that a valid knight move consists of moving two squares vertically and one square horizontally, or two squares horizontally and one square vertically. The figure below illustrates all the possible eight moves of a knight from some cell.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2601 Prime Subtraction Operation

**MEDIUM**

You are given a 0-indexed integer array nums of length n.

You can perform the following operation as many times as you want:
- Pick an index i that you haven’t picked before, and pick a prime p strictly less than nums[i], then subtract p from nums[i].

Return true if you can make nums a strictly increasing array using the above operation and false otherwise.

A strictly increasing array is an array whose each element is strictly greater than its preceding element.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2604 Minimum Time To Eat All Grains

**HARD**

There are n hens and m grains on a line. You are given the initial positions of the hens and the grains in two integer arrays hens and grains of size n and m respectively.

Any hen can eat a grain if they are on the same position. The time taken for this is negligible. One hen can also eat multiple grains.

In 1 second, a hen can move right or left by 1 unit. The hens can move simultaneously and independently of each other.

Return the minimum time to eat all grains if the hens act optimally.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2618 Check If Object Instance Of Class

**MEDIUM**

Write a function that checks if a given value is an instance of a given class or superclass. For this problem, an object is considered an instance of a given class if that object has access to that class's methods.

There are no constraints on the data types that can be passed to the function. For example, the value or the class could be undefined.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2621 Sleep

**EASY**

Given a positive integer `ms`, write an asynchronous function that

sleeps for `ms` milliseconds. It can resolve any value.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2624 Snail Traversal

**MEDIUM**

Write code that enhances all arrays such that you can call the snail(rowsCount, colsCount) method that transforms the 1D array into a 2D array organised in the pattern known as snail traversal order. Invalid input values should output an empty array. If rowsCount * colsCount !== nums.length, the input is considered invalid.

Snail traversal order starts at the top left cell with the first value of the current array. It then moves through the entire first column from top to bottom, followed by moving to the next column on the right and traversing it from bottom to top. This pattern continues, alternating the direction of traversal with each column, until the entire current array is covered. For example, when given the input array [19, 10, 3, 7, 9, 8, 5, 2, 1, 17, 16, 14, 12, 18, 6, 13, 11, 20, 4, 15] with rowsCount = 5 and colsCount = 4, the desired output matrix is shown below. Note that iterating the matrix following the arrows corresponds to the order of numbers in the original array.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2627 Debounce

**MEDIUM**

Given a function fn and a time in milliseconds t, return a debounced version of that function.

A debounced function is a function whose execution is delayed by t milliseconds and whose execution is cancelled if it is called again within that window of time. The debounced function should also receive the passed parameters.

For example, let's say t = 50ms, and the function was called at 30ms, 60ms, and 100ms.

The first 2 function calls would be cancelled, and the 3rd function call would be executed at 150ms.

If instead t = 35ms, The 1st call would be cancelled, the 2nd would be executed at 95ms, and the 3rd would be executed at 135ms.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2628 Json Deep Equal

**MEDIUM**

Given two values o1 and o2, return a boolean value indicating whether two values, o1 and o2, are deeply equal.

For two values to be deeply equal, the following conditions must be met:
- If both values are primitive types, they are deeply equal if they pass the === equality check.
- If both values are arrays, they are deeply equal if they have the same elements in the same order, and each element is also deeply equal according to these conditions.
- If both values are objects, they are deeply equal if they have the same keys, and the associated values for each key are also deeply equal according to these conditions.

You may assume both values are the output of JSON.parse. In other words, they are valid JSON.

Please solve it without using lodash's _.isEqual() function

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2629 Function Composition

EASY

Given an array of functions [f1, f2, f3, ..., fn], return a new function fn that is the function composition of the array of functions.

The function composition of [f(x), g(x), h(x)] is fn(x) = f(g(h(x))).

The function composition of an empty list of functions is the identity function f(x) = x.

You may assume each function in the array accepts one integer as input and returns one integer as output.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2631 Group By

MEDIUM

Write code that enhances all arrays such that you can call the array.groupBy(fn) method on any array and it will return a grouped version of the array.

A grouped array is an object where each key is the output of fn(arr[i]) and each value is an array containing all items in the original array which generate that key.

The provided callback fn will accept an item in the array and return a string key.

The order of each value list should be the order the items appear in the array. Any order of keys is acceptable.

Please solve it without lodash's _.groupBy function.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2632 Curry

MEDIUM

Given a function fn, return a curried version of that function.

A curried function is a function that accepts fewer or an equal number of parameters as the original function and returns either another curried function or the same value the original function would have returned.

In practical terms, if you called the original function like sum(1,2,3), you would call the curried version like csum(1)(2)(3), csum(1)(2,3), csum(1,2)(3), or csum(1,2,3). All these methods of calling the curried function should return the same value as the original.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2636 Promise Pool

MEDIUM

Given an array of asynchronous functions functions and a pool limit n, return an asynchronous function promisePool. It should return a promise that resolves when all the input functions resolve.

Pool limit is defined as the maximum number promises that can be pending at once. promisePool should begin execution of as many functions as possible and continue executing new functions when old promises resolve. promisePool should execute functions[i] then functions[i + 1] then functions[i + 2], etc. When the last promise resolves, promisePool should also resolve.

For example, if n = 1, promisePool will execute one function at a time in series. However, if n = 2, it first executes two functions. When either of the two functions resolve, a 3rd

function should be executed (if available), and so on until there are no functions left to execute.

You can assume all functions never reject. It is acceptable for promisePool to return a promise that resolves any value.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2637 Promise Time Limit

**MEDIUM**

Given an asynchronous function fn and a time t in milliseconds, return a new time limited version of the input function. fn takes arguments provided to the time limited function.

The time limited function should follow these rules:
- If the fn completes within the time limit of t milliseconds, the time limited function should resolve with the result.
- If the execution of the fn exceeds the time limit, the time limited function should reject with the string "Time Limit Exceeded".

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2643 Row With Maximum Ones

**EASY**

Given a m x n binary matrix mat, find the 0-indexed position of the row that contains the maximum count of ones, and the number of ones in that row.

In case there are multiple rows that have the maximum count of ones, the row with the smallest row number should be selected.

Return an array containing the index of the row, and the number of ones in it.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2644 Find The Maximum Divisibility Score

**EASY**

You are given two integer arrays nums and divisors.

The divisibility score of divisors[i] is the number of indices j such that nums[j] is divisible by divisors[i].

Return the integer divisors[i] with the maximum divisibility score. If multiple integers have the maximum score, return the smallest one.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2646 Minimize The Total Price Of The Trips

**HARD**

There exists an undirected and unrooted tree with n nodes indexed from 0 to n - 1. You are given the integer n and a 2D integer array edges of length n - 1, where edges[i] = [ai, bi] indicates that there is an edge between nodes ai and bi in the tree.

Each node has an associated price. You are given an integer array price, where price[i] is the price of the ith node.

The price sum of a given path is the sum of the prices of all nodes lying on that path.

Additionally, you are given a 2D integer array trips, where trips[i] = [starti, endi] indicates that you start the ith trip from the node starti and travel to the node endi

by any path you like.

Before performing your first trip, you can choose some non-adjacent nodes and halve the prices.

Return the minimum total price sum to perform all the given trips.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2647 Color The Triangle Red

**HARD**

You are given an integer n. Consider an equilateral triangle of side length n, broken up into n2 unit equilateral triangles. The triangle has n 1-indexed rows where the ith row has 2i - 1 unit equilateral triangles.

The triangles in the ith row are also 1-indexed with coordinates from (i, 1) to (i, 2i - 1). The following image shows a triangle of side length 4 with the indexing of its triangle.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2650 Design Cancellable Function

**HARD**

Sometimes you have a long running task, and you may wish to cancel it before it completes. To help with this goal, write a function cancellable that accepts a generator object and returns an array of two values: a cancel function and a promise.

You may assume the generator function will only yield promises. It is your function's responsibility to pass the values resolved by the promise back to the generator. If the promise rejects, your function should throw that error back to the generator.

If the cancel callback is called before the generator is done, your function should throw an error back to the generator. That error should be the string "Cancelled" (Not an Error object). If the error was caught, the returned promise should resolve with the next value that was yielded or returned. Otherwise, the promise should reject with the thrown error. No more code should be executed.

When the generator is done, the promise your function returned should resolve the value the generator returned. If, however, the generator throws an error, the returned promise should reject with the error.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2651 Calculate Delayed Arrival Time

**EASY**

You are given a positive integer arrivalTime denoting the arrival time of a train in hours, and another positive integer delayedTime denoting the amount of delay in hours.

Return the time when the train will arrive at the station.

Note that the time in this problem is in 24-hours format.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2655 Find Maximal Uncovered Ranges

**MEDIUM**

You are given an integer n which is the length of a 0-indexed array nums, and a 0-indexed 2D-array ranges, which is a list of sub-ranges of nums (sub-ranges may overlap).

Each row ranges[i] has exactly 2 cells:
- ranges[i][0], which shows the start of the ith range (inclusive)
- ranges[i][1], which shows the end of the ith range (inclusive)

These ranges cover some cells of nums and leave some cells uncovered. Your task is to find all of the uncovered ranges with maximal length.

Return a 2D-array answer of the uncovered ranges, sorted by the starting point in ascending order.

By all of the uncovered ranges with maximal length, we mean satisfying two conditions:
- Each uncovered cell should belong to exactly one sub-range
- There should not exist two ranges (l1, r1) and (l2, r2) such that r1 + 1 = l2

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2664 The Knights Tour

**MEDIUM**

Given two positive integers m and n which are the height and width of a 0-indexed 2D-array board, a pair of positive integers (r, c) which is the starting position of the knight on the board.

Your task is to find an order of movements for the knight, in a manner that every cell of the board gets visited exactly once (the starting cell is considered visited and you shouldn't visit it again).

Return the array board in which the cells' values show the order of visiting the cell starting from 0 (the initial place of the knight).

Note that a knight can move from cell (r1, c1) to cell (r2, c2) if 0 <= r2 <= m - 1 and 0 <= c2 <= n - 1 and min(abs(r1 - r2), abs(c1 - c2)) = 1 and max(abs(r1 - r2), abs(c1 - c2)) = 2.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2666 Allow One Function Call

**EASY**

Given a function fn, return a new function that is identical to the original function except that it ensures fn is called at most once.

- The first time the returned function is called, it should return the same result as fn.
- Every subsequent time it is called, it should return undefined.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2667 Create Hello World Function

**EASY**

Write a function createHelloWorld. It should return a new function that always returns "Hello World".

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2676 Throttle

**MEDIUM**

Given a function fn and a time in milliseconds t, return a throttled version of that function.

A throttled function is first called without delay and then, for a time interval of t milliseconds, can't be executed but should store the latest function arguments provided

to call fn with them after the end of the delay.

For instance, t = 50ms, and the function was called at 30ms, 40ms, and 60ms.

At 30ms, without delay, the throttled function fn should be called with the arguments, and calling the throttled function fn should be blocked for the following t milliseconds.

At 40ms, the function should just save arguments.

At 60ms, arguments should overwrite currently stored arguments from the second call because the second and third calls are made before 80ms. Once the delay has passed, the throttled function fn should be called with the latest arguments provided during the delay period, and it should also create another delay period of 80ms + t.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2680 Maximum Or

**MEDIUM**

You are given a 0-indexed integer array nums of length n and an integer k. In an operation, you can choose an element and multiply it by 2.

Return the maximum possible value of nums[0] | nums[1] | ... | nums[n - 1] that can be obtained after applying the operation on nums at most k times.

Note that a | b denotes the bitwise or between two integers a and b.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2683 Neighboring Bitwise Xor

**MEDIUM**

A 0-indexed array derived with length n is derived by computing the bitwise XOR (⊕) of adjacent values in a binary array original of length n.

Specifically, for each index i in the range [0, n - 1]:
- If i = n - 1, then derived[i] = original[i] ⊕ original[0].
- Otherwise, derived[i] = original[i] ⊕ original[i + 1].

Given an array derived, your task is to determine whether there exists a valid binary array original that could have formed derived.

Return true if such an array exists or false otherwise.
- A binary array is an array containing only 0's and 1's

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2690 Infinite Method Object

**EASY**

Write a function that returns an infinite-method object.

An infinite-method object is defined as an object that allows you to call any method and it will always return the name of the method.

For example, if you execute obj.abc123(), it will return "abc123".

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2691 Immutability Helper

**HARD**

Creating clones of immutable objects with minor alterations can be a tedious process.

Write a class ImmutableHelper that serves as a tool to help with this requirement.
The constructor accepts an immutable object obj which will be a JSON object or array.

The class has a single method produce which accepts a function mutator. The function returns a new object which is similar to the original except it has those mutations applied.

mutator accepts a proxied version of obj. A user of this function can (appear to) mutate this object, but the original object obj should not actually be effected.

For example, a user could write code like this:

```
const originalObj = {"x": 5};
const helper = new ImmutableHelper(originalObj);
const newObj = helper.produce((proxy) => {
proxy.x = proxy.x + 1;
});
console.log(originalObj); // {"x": 5}
console.log(newObj); // {"x": 6}
```

Properties of the mutator function:
- It will always return undefined.
- It will never access keys that don't exist.
- It will never delete keys (delete obj.key)
- It will never call methods on a proxied object (push, shift, etc).
- It will never set keys to objects (proxy.x = {})

Note on how the solution will be tested: the solution validator will only analyze differences between what was returned and the original obj. Doing a full comparison would be too computationally expensive. Also, any mutations to the original object will result in a wrong answer.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2703 Return Length Of Arguments Passed

**EASY**

Write a function argumentsLength that returns the count of arguments passed to it.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2704 To Be Or Not To Be

**EASY**

Write a function expect that helps developers test their code. It should take in any value val and return an object with the following two functions.

- toBe(val) accepts another value and returns true if the two values === each other. If they are not equal, it should throw an error "Not Equal".
- notToBe(val) accepts another value and returns true if the two values !== each other. If they are equal, it should throw an error "Equal".

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2705 Compact Object

**MEDIUM**

Given an object or array obj, return a compact object.

A compact object is the same as the original object, except with keys containing falsy values removed. This operation applies to the object and any nested objects. Arrays are considered objects where the indices are keys. A value is considered falsy when Boolean(value) returns false.

You may assume the obj is the output of JSON.parse. In other words, it is valid JSON.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2706 Buy Two Chocolates

**EASY**

You are given an integer array prices representing the prices of various chocolates in a store. You are also given a single integer money, which represents your initial amount of money.

You must buy exactly two chocolates in such a way that you still have some non-negative leftover money. You would like to minimize the sum of the prices of the two chocolates you buy.

Return the amount of money you will have leftover after buying the two chocolates. If there is no way for you to buy two chocolates without ending up in debt, return money. Note that the leftover must be non-negative.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2715 Timeout Cancellation

**EASY**

Given a function fn, an array of arguments args, and a timeout t in milliseconds, return a cancel function cancelFn.

After a delay of cancelTimeMs, the returned cancel function cancelFn will be invoked.
> setTimeout(cancelFn, cancelTimeMs)

Initially, the execution of the function fn should be delayed by t milliseconds.

If, before the delay of t milliseconds, the function cancelFn is invoked, it should cancel the delayed execution of fn. Otherwise, if cancelFn is not invoked within the specified delay t, fn should be executed with the provided args as arguments.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2721 Execute Asynchronous Functions In Parallel

**MEDIUM**

Given an array of asynchronous functions functions, return a new promise promise. Each function in the array accepts no arguments and returns a promise. All the promises should be executed in parallel.

promise resolves:
- When all the promises returned from functions were resolved successfully in parallel. The resolved value of promise should be an array of all the resolved values of promises in the same order as they were in the functions. The promise should resolve when all the asynchronous functions in the array have completed execution in parallel.

promise rejects:
When any of the promises returned from functions were rejected. promise should also reject with the reason of the first rejection.

Please solve it without using the built-in Promise.all function.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2723 Add Two Promises

EASY

Given two promises promise1 and promise2, return a new promise. promise1 and promise2 will both resolve with a number. The returned promise should resolve with the sum of the two numbers.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2727 Is Object Empty

EASY

Given an object or an array, return if it is empty.

- An empty object contains no key-value pairs.
- An empty array contains no elements.

You may assume the object or array is the output of JSON.parse.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2733 Neither Minimum Nor Maximum

EASY

Given an integer array nums containing distinct positive integers, find and return any number from the array that is neither the minimum nor the maximum value in the array, or -1 if there is no such number.

Return the selected integer.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2739 Total Distance Traveled

EASY

A truck has two fuel tanks. You are given two integers, mainTank representing the fuel present in the main tank in liters and additionalTank representing the fuel present in the additional tank in liters.

The truck has a mileage of 10 km per liter. Whenever 5 liters of fuel get used up in the main tank, if the additional tank has at least 1 liters of fuel, 1 liters of fuel will be transferred from the additional tank to the main tank.

Return the maximum distance which can be traveled.

Note: Injection from the additional tank is not continuous. It happens suddenly and immediately for every 5 liters consumed.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2740 Find The Value Of The Partition

MEDIUM

You are given a positive integer array nums.

Partition nums into two arrays, nums1 and nums2, such that:
- Each element of the array nums belongs to either the array nums1 or the array nums2.
- Both arrays are non-empty.
- The value of the partition is minimized.

The value of the partition is |max(nums1) - min(nums2)|.

Here, max(nums1) denotes the maximum element of the array nums1, and min(nums2) denotes the minimum element of the array nums2.

Return the integer denoting the value of such partition.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2742 Painting The Walls

**HARD**

You are given two 0-indexed integer arrays, cost and time, of size n representing the costs and the time taken to paint n different walls respectively. There are two painters available:
- A paid painter that paints the ith wall in time[i] units of time and takes cost[i] units of money.
- A free painter that paints any wall in 1 unit of time at a cost of 0. But the free painter can only be used if the paid painter is already occupied.

Return the minimum amount of money required to paint the n walls.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2749 Minimum Operations To Make The Integer Zero

**MEDIUM**

You are given two integers num1 and num2.

In one operation, you can choose integer i in the range [0, 60] and subtract 2i + num2 from num1.

Return the integer denoting the minimum number of operations needed to make num1 equal to 0.

If it is impossible to make num1 equal to 0, return -1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2751 Robot Collisions

**HARD**

There are n 1-indexed robots, each having a position on a line, health, and movement direction.

You are given 0-indexed integer arrays positions, healths, and a string directions (directions[i] is either 'L' for left or 'R' for right). All integers in positions are unique.

All robots start moving on the line simultaneously at the same speed in their given directions. If two robots ever share the same position while moving, they will collide.

If two robots collide, the robot with lower health is removed from the line, and the health of the other robot decreases by one. The surviving robot continues in the same direction it was going. If both robots have the same health, they are both removed from the line.

Your task is to determine the health of the robots that survive the collisions, in the same order that the robots were given, i.e. final health of robot 1 (if survived), final health of robot 2 (if survived), and so on. If there are no survivors, return an empty array.

Return an array containing the health of the remaining robots (in the order they were given in the input), after no further collisions can occur.

Note: The positions may be unsorted.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2755 Deep Merge Of Two Objects

MEDIUM

Given two values obj1 and obj2, return a deepmerged value.

Values should be deepmerged according to these rules:
- If the two values are objects, the resulting object should have all the keys that exist on either object. If a key belongs to both objects, deepmerge the two associated values. Otherwise, add the key-value pair to the resulting object.
- If the two values are arrays, the resulting array should be the same length as the longer array. Apply the same logic as you would with objects, but treat the indices as keys.
- Otherwise the resulting value is obj2.

You can assume obj1 and obj2 are the output of JSON.parse().

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2756 Query Batching

HARD

Batching multiple small queries into a single large query can be a useful optimization. Write a class QueryBatcher that implements this functionality.

The constructor should accept two parameters:
- An asynchronous function queryMultiple which accepts an array of string keys input. It will resolve with an array of values that is the same length as the input array. Each index corresponds to the value associated with input[i]. You can assume the promise will never reject.
- A throttle time in milliseconds t.

The class has a single method.
- async getValue(key). Accepts a single string key and resolves with a single string value.

The keys passed to this function should eventually get passed to the queryMultiple function. queryMultiple should never be called consecutively within t milliseconds. The first time getValue is called, queryMultiple should immediately be called with that single key. If after t milliseconds, getValue had been called again, all the passed keys should be passed to queryMultiple and ultimately returned. You can assume every key passed to this method is unique.

The following diagram illustrates how the throttling algorithm works. Each rectangle represents 100ms. The throttle time is 400ms.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2758 Next Day

EASY

Write code that enhances all date objects such that you can call the date.nextDay() method on any date object and it will return the next day in the format YYYY-MM-DD as a string.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2775 Undefined To Null

MEDIUM

Given a deeply nested object or array obj, return the object obj with any undefined values replaced by null.

undefined values are handled differently than null values when objects are converted to a JSON string using JSON.stringify(). This function helps ensure serialized data

is free of unexpected errors.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2776 Convert Callback Based Function To Promise Based Function

**MEDIUM**

Write a function that accepts another function fn and converts the callback-based function into a promise-based function.

The function fn takes a callback as its first argument, along with any additional arguments args passed as separate inputs.

The promisify function returns a new function that should return a promise. The promise should resolve with the argument passed as the first parameter of the callback when the callback is invoked without error, and reject with the error when the callback is called with an error as the second argument.

The following is an example of a function that could be passed into promisify.

```
function sum(callback, a, b) {
if (a < 0 || b < 0) {
const err = Error('a and b must be positive');
callback(undefined, err);
} else {
callback(a + b);
}
}
```

This is the equivalent code based on promises:

```
async function sum(a, b) {
if (a < 0 || b < 0) {
throw Error('a and b must be positive');
} else {
return a + b;
}
}
```

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2777 Date Range Generator

**MEDIUM**

Given a start date start, an end date end, and a positive integer step, return a generator object that yields dates in the range from start to end inclusive.

The value of step indicates the number of days between consecutive yielded values.

All yielded dates must be in the string format YYYY-MM-DD.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2797 Partial Function With Placeholders

**EASY**

Given a function fn and an array args, return a function partialFn.

Placeholders "_" in the args should be replaced with values from restArgs starting from index 0. Any remaining values in the restArgs should be added at the end of the args.

partialFn should return a result of fn. fn should be called with the elements of the modified args passed as separate arguments.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2803 Factorial Generator

EASY

Write a generator function that takes an integer n as an argument and returns a generator object which yields the factorial sequence.

The factorial sequence is defined by the relation n! = n * (n-1) * (n-2) * ... * 2 * 1.

The factorial of 0 is defined as 1.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2810 Faulty Keyboard

EASY

Your laptop keyboard is faulty, and whenever you type a character 'i' on it, it reverses the string that you have written. Typing other characters works as expected.

You are given a 0-indexed string s, and you type each character of s using your faulty keyboard.

Return the final string that will be present on your laptop screen.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2814 Minimum Time Takes To Reach Destination Without Drowning

HARD

You are given an n * m 0-indexed grid of string land. Right now, you are standing at the cell that contains "S", and you want to get to the cell containing "D". There are three other types of cells in this land:
- ".": These cells are empty.
- "X": These cells are stone.
- "*": These cells are flooded.

At each second, you can move to a cell that shares a side with your current cell (if it exists). Also, at each second, every empty cell that shares a side with a flooded cell becomes flooded as well.

There are two problems ahead of your journey:
- You can't step on stone cells.
- You can't step on flooded cells since you will drown (also, you can't step on a cell that will be flooded at the same time as you step on it).

Return the minimum time it takes you to reach the destination in seconds, or -1 if it is impossible.

Note that the destination will never be flooded.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2818 Apply Operations To Maximize Score

HARD

You are given an array nums of n positive integers and an integer k.

Initially, you start with a score of 1. You have to maximize your score by applying the following operation at most k times:
- Choose any non-empty subarray nums[l, ..., r] that you haven't chosen previously.
- Choose an element x of nums[l, ..., r] with the highest prime score. If multiple such elements exist, choose the one with the smallest index.
- Multiply your score by x.

Here, nums[l, ..., r] denotes the subarray of nums starting at index l and ending at the index r, both ends being inclusive.

The prime score of an integer x is equal to the number of distinct prime factors of x. For example, the prime score of 300 is 3 since 300 = 2 * 2 * 3 * 5 * 5.

Return the maximum possible score after applying at most k operations.

Since the answer may be large, return it modulo 109 + 7.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2819 Minimum Relative Loss After Buying Chocolates

**HARD**

You are given an integer array prices, which shows the chocolate prices and a 2D integer array queries, where queries[i] = [ki, mi].

Alice and Bob went to buy some chocolates, and Alice suggested a way to pay for them, and Bob agreed.

The terms for each query are as follows:
- If the price of a chocolate is less than or equal to ki, Bob pays for it.
- Otherwise, Bob pays ki of it, and Alice pays the rest.

Bob wants to select exactly mi chocolates such that his relative loss is minimized, more formally, if, in total, Alice has paid ai and Bob has paid bi, Bob wants to minimize bi - ai.

Return an integer array ans where ans[i] is Bob's minimum relative loss possible for queries[i].

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2821 Delay The Resolution Of Each Promise

**MEDIUM**

Given an array functions and a number ms, return a new array of functions.
- functions is an array of functions that return promises.
- ms represents the delay duration in milliseconds. It determines the amount of time to wait before resolving or rejecting each promise in the new array.

Each function in the new array should return a promise that resolves or rejects after an additional delay of ms milliseconds, preserving the order of the original functions array.

The delayAll function should ensure that each promise from functions is executed with a delay, forming the new array of functions returning delayed promises.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2822 Inversion Of Object

**EASY**

Given an object or an array obj, return an inverted object or array invertedObj.

The invertedObj should have the keys of obj as values and the values of obj as keys. The indices of array should be treated as keys.

The function should handle duplicates, meaning that if there are multiple keys in obj with the same value, the invertedObj should map the value to an array containing all corresponding keys.

It is guaranteed that the values in obj are only strings.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2823 Deep Object Filter

MEDIUM

Given an object or an array obj and a function fn, return a filtered object or array filteredObject.

Function deepFilter should perform a deep filter operation on the obj. The deep filter operation should remove properties for which the output of the filter function fn is false, as well as any empty objects or arrays that remain after the keys have been removed.

If the deep filter operation results in an empty object or array, with no remaining properties, deepFilter should return undefined to indicate that there is no valid data left in the filteredObject.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2833 Furthest Point From Origin

EASY

You are given a string moves of length n consisting only of characters 'L', 'R', and '_'. The string represents your movement on a number line starting from the origin 0.

In the ith move, you can choose one of the following directions:
- move to the left if moves[i] = 'L' or moves[i] = '_'
- move to the right if moves[i] = 'R' or moves[i] = '_'

Return the distance from the origin of the furthest point you can get to after n moves.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2838 Maximum Coins Heroes Can Collect

MEDIUM

There is a battle and n heroes are trying to defeat m monsters. You are given two 1-indexed arrays of positive integers heroes and monsters of length n and m, respectively. heroes[i] is the power of ith hero, and monsters[i] is the power of ith monster.

The ith hero can defeat the jth monster if monsters[j] <= heroes[i].

You are also given a 1-indexed array coins of length m consisting of positive integers. coins[i] is the number of coins that each hero earns after defeating the ith monster.

Return an array ans of length n where ans[i] is the maximum number of coins that the ith hero can collect from this battle.

Notes
- The health of a hero doesn't get reduced after defeating a monster.
- Multiple heroes can defeat a monster, but each monster can be defeated by a given hero only once.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2860 Happy Students

MEDIUM

You are given a 0-indexed integer array nums of length n where n is the total number of students in the class. The class teacher tries to select a group of students so that all the students remain happy.

The ith student will become happy if one of these two conditions is met:
- The student is selected and the total number of selected students is strictly greater

than nums[i].
- The student is not selected and the total number of selected students is strictly less than nums[i].

Return the number of ways to select a group of students so that everyone remains happy.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2865 Beautiful Towers I

**MEDIUM**

You are given an array heights of n integers representing the number of bricks in n consecutive towers. Your task is to remove some bricks to form a mountain-shaped tower arrangement. In this arrangement, the tower heights are non-decreasing, reaching a maximum peak value with one or multiple consecutive towers and then non-increasing.

Return the maximum possible sum of heights of a mountain-shaped tower arrangement.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2895 Minimum Processing Time

**MEDIUM**

You have a certain number of processors, each having 4 cores. The number of tasks to be executed is four times the number of processors. Each task must be assigned to a unique core, and each core can only be used once.

You are given an array processorTime representing the time each processor becomes available and an array tasks representing how long each task takes to complete. Return the minimum time needed to complete all tasks.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2899 Last Visited Integers

**EASY**

Given an integer array nums where nums[i] is either a positive integer or -1. We need to find for each -1 the respective positive integer, which we call the last visited integer.

To achieve this goal, let's define two empty arrays: seen and ans.

Start iterating from the beginning of the array nums.
- If a positive integer is encountered, prepend it to the front of seen.
- If -1 is encountered, let k be the number of consecutive -1s seen so far (including the current -1),
- If k is less than or equal to the length of seen, append the k-th element of seen to ans.
- If k is strictly greater than the length of seen, append -1 to ans.

Return the array ans.
**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2900 Longest Unequal Adjacent Groups Subsequence I

**EASY**

You are given a string array words and a binary array groups both of length n, where words[i] is associated with groups[i].

Your task is to select the longest alternating subsequence from words. A subsequence of words is alternating if for any two consecutive strings in the sequence, their corresponding elements in the binary array groups differ. Essentially, you are to choose strings such that adjacent

elements have non-matching corresponding bits in the groups array.

Formally, you need to find the longest subsequence of an array of indices [0, 1, ..., n - 1] denoted as [i0, i1, ..., ik-1], such that groups[ij] != groups[ij+1] for each 0 <= j < k - 1 and then find the words corresponding to these indices.

Return the selected subsequence. If there are multiple answers, return any of them.

Note: The elements in words are distinct.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2901 Longest Unequal Adjacent Groups Subsequence Ii

**MEDIUM**

You are given a string array words, and an array groups, both arrays having length n.

The hamming distance between two strings of equal length is the number of positions at which the corresponding characters are different.

You need to select the longest subsequence from an array of indices [0, 1, ..., n - 1], such that for the subsequence denoted as [i0, i1, ..., ik-1] having length k, the following holds:
- For adjacent indices in the subsequence, their corresponding groups are unequal, i.e., groups[ij] != groups[ij+1], for each j where 0 < j + 1 < k.
- words[ij] and words[ij+1] are equal in length, and the hamming distance between them is 1, where 0 < j + 1 < k, for all indices in the subsequence.

Return a string array containing the words corresponding to the indices (in order) in the selected subsequence. If there are multiple answers, return any of them.

Note: strings in words may be unequal in length.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2923 Find Champion I

**EASY**

There are n teams numbered from 0 to n - 1 in a tournament.

Given a 0-indexed 2D boolean matrix grid of size n * n. For all i, j that 0 <= i, j <= n - 1 and i != j team i is stronger than team j if grid[i][j] == 1, otherwise, team j is stronger than team i.

Team a will be the champion of the tournament if there is no team b that is stronger than team a.

Return the team that will be the champion of the tournament.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2924 Find Champion Ii

**MEDIUM**

There are n teams numbered from 0 to n - 1 in a tournament; each team is also a node in a DAG.

You are given the integer n and a 0-indexed 2D integer array edges of length m representing the DAG, where edges[i] = [ui, vi] indicates that there is a directed edge from team ui to team vi in the graph.

A directed edge from a to b in the graph means that team a is stronger than team b and team b is weaker than team a.

Team a will be the champion of the tournament if there is no team b that is stronger than team a.

Return the team that will be the champion of the tournament if there is a unique champion,

otherwise, return -1.

Notes:
- A cycle is a series of nodes a1, a2, ..., an, an+1 such that node a1 is the same node as node an+1, the nodes a1, a2, ..., an are distinct, and there is a directed edge from the node ai to node ai+1 for every i in the range [1, n].
- A DAG is a directed graph that does not have any cycle.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2927 Distribute Candies Among Children Iii

**HARD**

You are given two positive integers n and limit.

Return the total number of ways to distribute n candies among 3 children such that no child gets more than limit candies.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2928 Distribute Candies Among Children I

**EASY**

You are given two positive integers n and limit.

Return the total number of ways to distribute n candies among 3 children such that no child gets more than limit candies.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2929 Distribute Candies Among Children Ii

**MEDIUM**

You are given two positive integers n and limit.

Return the total number of ways to distribute n candies among 3 children such that no child gets more than limit candies.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2938 Separate Black And White Balls

**MEDIUM**

There are n balls on a table, each ball has a color black or white.

You are given a 0-indexed binary string s of length n, where 1 and 0 represent black and white balls, respectively.

In each step, you can choose two adjacent balls and swap them.

Return the minimum number of steps to group all the black balls to the right and all the white balls to the left.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2951 Find The Peaks

**EASY**

You are given a 0-indexed array mountain. Your task is to find all the peaks in the mountain array.

Return an array that consists of indices of peaks in the given array in any order.

Notes:
- A peak is defined as an element that is strictly greater than its neighboring elements.
- The first and last elements of the array are not a peak.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2961 Double Modular Exponentiation

**MEDIUM**

You are given a 0-indexed 2D array variables where variables[i] = [ai, bi, ci, mi], and an integer target.

An index i is good if the following formula holds:
- 0 <= i < variables.length
- ((aibi % 10)ci) % mi == target

Return an array consisting of good indices in any order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2971 Find Polygon With The Largest Perimeter

**MEDIUM**

You are given an array of positive integers nums of length n.

A polygon is a closed plane figure that has at least 3 sides. The longest side of a polygon is smaller than the sum of its other sides.

Conversely, if you have k (k >= 3) positive real numbers a1, a2, a3, ..., ak where a1 <= a2 <= a3 <= ... <= ak and a1 + a2 + a3 + ... + ak-1 > ak, then there always exists a polygon with k sides whose lengths are a1, a2, a3, ..., ak.

The perimeter of a polygon is the sum of lengths of its sides.

Return the largest possible perimeter of a polygon whose sides can be formed from nums, or -1 if it is not possible to create a polygon.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2979 Most Expensive Item That Can Not Be Bought

**MEDIUM**

You are given two distinct prime numbers primeOne and primeTwo.

Alice and Bob are visiting a market. The market has an infinite number of items, for any positive integer x there exists an item whose price is x. Alice wants to buy some items from the market to gift to Bob. She has an infinite number of coins in the denomination primeOne and primeTwo. She wants to know the most expensive item she can not buy to gift to Bob.

Return the price of the most expensive item which Alice can not gift to Bob.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 2980 Check If Bitwise Or Has Trailing Zeros

**EASY**

You are given an array of positive integers nums.

You have to check if it is possible to select two or more elements in the array such that the bitwise OR of the selected elements has at least one trailing zero in its binary representation.

For example, the binary representation of 5, which is "101", does not have any trailing zeros, whereas the binary representation of 4, which is "100", has two trailing zeros.

Return true if it is possible to select two or more elements whose bitwise OR has trailing zeros, return false otherwise.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3000 Maximum Area Of Longest Diagonal Rectangle

**EASY**

You are given a 2D 0-indexed integer array dimensions.

For all indices i, 0 <= i < dimensions.length, dimensions[i][0] represents the length and dimensions[i][1] represents the width of the rectangle i.

Return the area of the rectangle having the longest diagonal. If there are multiple rectangles with the longest diagonal, return the area of the rectangle having the maximum area.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3021 Alice And Bob Playing Flower Game

**MEDIUM**

Alice and Bob are playing a turn-based game on a circular field surrounded by flowers. The circle represents the field, and there are x flowers in the clockwise direction between Alice and Bob, and y flowers in the anti-clockwise direction between them.

The game proceeds as follows:
and pick one flower from that side.
captures their opponent and wins the game.

Given two integers, n and m, the task is to compute the number of possible pairs (x, y) that satisfy the conditions:
- Alice must win the game according to the described rules.
- The number of flowers x in the clockwise direction must be in the range [1,n].
- The number of flowers y in the anti-clockwise direction must be in the range [1,m].

Return the number of possible pairs (x, y) that satisfy the conditions mentioned in the statement.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3023 Find Pattern In Infinite Stream I

**MEDIUM**

You are given a binary array pattern and an object stream of class InfiniteStream representing a 0-indexed infinite stream of bits.

The class InfiniteStream contains the following function:
- int next(): Reads a single bit (which is either 0 or 1) from the stream and returns it.

Return the first starting index where the pattern matches the bits read from the stream. For example, if the pattern is [1, 0], the first match is the highlighted part in the stream [0, 1, 0, 1, ...].

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3024 Type Of Triangle

**EASY**

You are given a 0-indexed integer array nums of size 3 which can form the sides of a triangle.

- A triangle is called equilateral if it has all sides of equal length.
- A triangle is called isosceles if it has exactly two sides of equal length.
- A triangle is called scalene if all its sides are of different lengths.

Return a string representing the type of triangle that can be formed or "none" if it cannot form a triangle.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3028 Ant On The Boundary

**EASY**

An ant is on a boundary. It sometimes goes left and sometimes right.

You are given an array of non-zero integers nums. The ant starts reading nums from the first element of it to its end. At each step, it moves according to the value of the current element:
- If nums[i] < 0, it moves left by -nums[i] units.
- If nums[i] > 0, it moves right by nums[i] units.

Return the number of times the ant returns to the boundary.

Notes:
- There is an infinite space on both sides of the boundary.
- We check whether the ant is on the boundary only after it has moved |nums[i]| units. In other words, if the ant crosses the boundary during its movement, it does not count.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3035 Maximum Palindromes After Operations

**MEDIUM**

You are given a 0-indexed string array words having length n and containing 0-indexed strings.

You are allowed to perform the following operation any number of times (including zero):
- Choose integers i, j, x, and y such that 0 <= i, j < n, 0 <= x < words[i].length, 0 <= y < words[j].length, and swap the characters words[i][x] and words[j][y].

Return an integer denoting the maximum number of palindromes words can contain, after performing some operations.

Note: i and j may be equal during an operation.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3037 Find Pattern In Infinite Stream Ii

**HARD**

You are given a binary array pattern and an object stream of class InfiniteStream representing a 0-indexed infinite stream of bits.

The class InfiniteStream contains the following function:
- int next(): Reads a single bit (which is either 0 or 1) from the stream and returns it.

Return the first starting index where the pattern matches the bits read from the stream. For example, if the pattern is [1, 0], the first match is the highlighted part in the stream [0, 1, 0, 1, ...].

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3047 Find The Largest Area Of Square Inside Two Rectangles

MEDIUM

There exist n rectangles in a 2D plane with edges parallel to the x and y axis. You are given two 2D integer arrays bottomLeft and topRight where bottomLeft[i] = [a_i, b_i] and topRight[i] = [c_i, d_i] represent the bottom-left and top-right coordinates of the ith rectangle, respectively.

You need to find the maximum area of a square that can fit inside the intersecting region of at least two rectangles. Return 0 if such a square does not exist.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3065 Minimum Operations To Exceed Threshold Value I

EASY

You are given a 0-indexed integer array nums, and an integer k.

In one operation, you can remove one occurrence of the smallest element of nums.

Return the minimum number of operations needed so that all elements of the array are greater than or equal to k.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3066 Minimum Operations To Exceed Threshold Value Ii

MEDIUM

You are given a 0-indexed integer array nums, and an integer k.

You are allowed to perform some operations on nums, where in a single operation, you can:
- Select the two smallest integers x and y from nums.
- Remove x and y from nums.
- Insert (min(x, y) * 2 + max(x, y)) at any position in the array.

Note that you can only apply the described operation if nums contains at least two elements.

Return the minimum number of operations needed so that all elements of the array are greater than or equal to k.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3074 Apple Redistribution Into Boxes

EASY

You are given an array apple of size n and an array capacity of size m.

There are n packs where the ith pack contains apple[i] apples. There are m boxes as well, and the ith box has a capacity of capacity[i] apples.

Return the minimum number of boxes you need to select to redistribute these n packs of apples into boxes.

Note that, apples from the same pack can be distributed into different boxes.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3075 Maximize Happiness Of Selected Children

MEDIUM

You are given an array happiness of length n, and a positive integer k.

There are n children standing in a queue, where the ith child has happiness value happiness[i]. You want to select k children from these n children in k turns.

In each turn, when you select a child, the happiness value of all the children that have not been selected till now decreases by 1. Note that the happiness value cannot become negative and gets decremented only if it is positive.

Return the maximum sum of the happiness values of the selected children you can achieve by selecting k children.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3100 Water Bottles Ii

**MEDIUM**

You are given two integers numBottles and numExchange.

numBottles represents the number of full water bottles that you initially have. In one operation, you can perform one of the following operations:
- Drink any number of full water bottles turning them into empty bottles.
- Exchange numExchange empty bottles with one full water bottle. Then, increase numExchange by one.

Note that you cannot exchange multiple batches of empty bottles for the same value of numExchange. For example, if numBottles == 3 and numExchange == 1, you cannot exchange 3 empty water bottles for 3 full bottles.

Return the maximum number of water bottles you can drink.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3141 Maximum Hamming Distances

**HARD**

Given an array nums and an integer m, with each element nums[i] satisfying 0 <= nums[i] < 2m, return an array answer. The answer array should be of the same length as nums, where each element answer[i] represents the maximum Hamming distance between nums[i] and any other element nums[j] in the array.

The Hamming distance between two binary integers is defined as the number of positions at which the corresponding bits differ (add leading zeroes if needed).

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3147 Taking Maximum Energy From The Mystic Dungeon

**MEDIUM**

In a mystic dungeon, n magicians are standing in a line. Each magician has an attribute that gives you energy. Some magicians can give you negative energy, which means taking energy from you.

You have been cursed in such a way that after absorbing energy from magician i, you will be instantly transported to magician (i + k). This process will be repeated until you reach the magician where (i + k) does not exist.

In other words, you will choose a starting point and then teleport with k jumps until you reach the end of the magicians' sequence, absorbing all the energy during the journey.

You are given an array energy and an integer k. Return the maximum possible energy you can gain.

Note that when you are reach a magician, you must take energy from them, whether it is negative or positive energy.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3178 Find The Child Who Has The Ball After K Seconds

EASY

You are given two positive integers n and k. There are n children numbered from 0 to n - 1 standing in a queue in order from left to right.

Initially, child 0 holds a ball and the direction of passing the ball is towards the right direction. After each second, the child holding the ball passes it to the child next to them. Once the ball reaches either end of the line, i.e. child 0 or child n - 1, the direction of passing is reversed.

Return the number of the child who receives the ball after k seconds.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3189 Minimum Moves To Get A Peaceful Board

MEDIUM

Given a 2D array rooks of length n, where rooks[i] = [xi, yi] indicates the position of a rook on an n x n chess board. Your task is to move the rooks 1 cell at a time vertically or horizontally (to an adjacent cell) such that the board becomes peaceful.

A board is peaceful if there is exactly one rook in each row and each column.

Return the minimum number of moves required to get a peaceful board.

Note that at no point can there be two rooks in the same cell.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3195 Find The Minimum Area To Cover All Ones I

MEDIUM

You are given a 2D binary array grid. Find a rectangle with horizontal and vertical sides with the smallest area, such that all the 1's in grid lie inside this rectangle.

Return the minimum possible area of the rectangle.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3197 Find The Minimum Area To Cover All Ones Ii

HARD

You are given a 2D binary array grid. You need to find 3 non-overlapping rectangles having non-zero areas with horizontal and vertical sides such that all the 1's in grid lie inside these rectangles.

Return the minimum possible sum of the area of these rectangles.

Note that the rectangles are allowed to touch.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3201 Find The Maximum Length Of Valid Subsequence I

MEDIUM

You are given an integer array nums.

A subsequence sub of nums with length x is called valid if it satisfies:
- (sub[0] + sub[1]) % 2 == (sub[1] + sub[2]) % 2 == ... == (sub[x - 2] + sub[x - 1]) % 2.

Return the length of the longest valid subsequence of nums.

A subsequence is an array that can be derived from another array by deleting some or no elements without changing the order of the remaining elements.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3202 Find The Maximum Length Of Valid Subsequence Ii

**MEDIUM**

You are given an integer array nums and a positive integer k.

A subsequence sub of nums with length x is called valid if it satisfies:
- (sub[0] + sub[1]) % k == (sub[1] + sub[2]) % k == ... == (sub[x - 2] + sub[x - 1]) % k.

Return the length of the longest valid subsequence of nums.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3208 Alternating Groups Ii

**MEDIUM**

There is a circle of red and blue tiles. You are given an array of integers colors and an integer k. The color of tile i is represented by colors[i]:
- colors[i] == 0 means that tile i is red.
-colors[i] == 1 means that tile i is blue.

An alternating group is every k contiguous tiles in the circle with alternating colors (each tile in the group except the first and last one has a different color from its left and right tiles).

Return the number of alternating groups.

Note that since colors represents a circle, the first and the last tiles are considered to be next to each other.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3341 Find Minimum Time To Reach Last Room I

**MEDIUM**

There is a dungeon with n x m rooms arranged as a grid.

You are given a 2D array moveTime of size n x m, where moveTime[i][j] represents the minimum time in seconds when you can start moving to that room. You start from the room (0, 0) at time t = 0 and can move to an adjacent room. Moving between adjacent rooms takes exactly one second.

Return the minimum time to reach the room (n - 1, m - 1).

Two rooms are adjacent if they share a common wall, either horizontally or vertically.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3342 Find Minimum Time To Reach Last Room Ii

**MEDIUM**

There is a dungeon with n x m rooms arranged as a grid.

You are given a 2D array moveTime of size n x m, where moveTime[i][j] represents the minimum time in seconds when you can start moving to that room. You start from the room (0, 0) at time t = 0 and can move to an adjacent room. Moving between adjacent rooms takes one second for one move and two seconds for the next, alternating between the two.

Return the minimum time to reach the room (n - 1, m - 1).

Two rooms are adjacent if they share a common wall, either horizontally or vertically.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3353 Minimum Total Operations

**EASY**

Given an array of integers nums, you can perform any number of operations on this array.

In each operation, you can:
- Choose a prefix of the array.
- Choose an integer k (which can be negative) and add k to each element in the chosen prefix.

A prefix of an array is a subarray that starts from the beginning of the array and extends to any point within it.

Return the minimum number of operations required to make all elements in arr equal.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3383 Minimum Runes To Add To Cast Spell

**HARD**

Alice has just graduated from wizard school, and wishes to cast a magic spell to celebrate. The magic spell contains certain focus points where magic needs to be concentrated, and some of these focus points contain magic crystals which serve as the spell's energy source. Focus points can be linked through directed runes, which channel magic flow from one focus point to another.

You are given a integer n denoting the number of focus points and an array of integers crystals where crystals[i] indicates a focus point which holds a magic crystal. You are also given two integer arrays flowFrom and flowTo, which represent the existing directed runes. The ith rune allows magic to freely flow from focus point flowFrom[i] to focus point flowTo[i].

You need to find the number of directed runes Alice must add to her spell, such that each focus point either:
- Contains a magic crystal.
- Receives magic flow from another focus point.

Return the minimum number of directed runes that she should add.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3385 Minimum Time To Break Locks Ii

**HARD**

Bob is stuck in a dungeon and must break n locks, each requiring some amount of energy to break. The required energy for each lock is stored in an array called strength where strength[i] indicates the energy needed to break the ith lock.

To break a lock, Bob uses a sword with the following characteristics:
- The initial energy of the sword is 0.
- The initial factor X by which the energy of the sword increases is 1.
- Every minute, the energy of the sword increases by the current factor X.
- To break the ith lock, the energy of the sword must reach at least strength[i].
- After breaking a lock, the energy of the sword resets to 0, and the factor X increases by 1.

Your task is to determine the minimum time in minutes required for Bob to break all n locks and escape the dungeon.

Return the minimum time required for Bob to break all n locks.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3402 Minimum Operations To Make Columns Strictly Increasing

EASY

You are given a m x n matrix grid consisting of non-negative integers.

In one operation, you can increment the value of any grid[i][j] by 1.

Return the minimum number of operations needed to make all columns of grid strictly increasing.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3408 Design Task Manager

MEDIUM

There is a task management system that allows users to manage their tasks, each associated with a priority. The system should efficiently handle adding, modifying, executing, and removing tasks.

Implement the TaskManager class:
- TaskManager(vector<vector<int>>& tasks) initializes the task manager with a list of user-task-priority triples. Each element in the input list is of the form [userId, taskId, priority], which adds a task to the specified user with the given priority.
- void add(int userId, int taskId, int priority) adds a task with the specified taskId and priority to the user with userId. It is guaranteed that taskId does not exist in the system.
- void edit(int taskId, int newPriority) updates the priority of the existing taskId to newPriority. It is guaranteed that taskId exists in the system.
- void rmv(int taskId) removes the task identified by taskId from the system. It is guaranteed that taskId exists in the system.
- int execTop() executes the task with the highest priority across all users. If there are multiple tasks with the same highest priority, execute the one with the highest taskId. After executing, the taskId is removed from the system. Return the userId associated with the executed task. If no tasks are available, return -1.

Note that a user may be assigned multiple tasks.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3439 Reschedule Meetings For Maximum Free Time I

MEDIUM

You are given an integer eventTime denoting the duration of an event, where the event occurs from time t = 0 to time t = eventTime.

You are also given two integer arrays startTime and endTime, each of length n. These represent the start and end time of n non-overlapping meetings, where the ith meeting occurs during the time [startTime[i], endTime[i]].

You can reschedule at most k meetings by moving their start time while maintaining the same duration, to maximize the longest continuous period of free time during the event.

The relative order of all the meetings should stay the same and they should remain non-overlapping.

Return the maximum amount of free time possible after rearranging the meetings.

Note that the meetings can not be rescheduled to a time outside the event.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3440 Reschedule Meetings For Maximum Free Time Ii

MEDIUM

You are given an integer eventTime denoting the duration of an event. You are also given two integer arrays startTime and endTime, each of length n.

These represent the start and end times of n non-overlapping meetings that occur during the event between time t = 0 and time t = eventTime, where the ith meeting occurs during the time [startTime[i], endTime[i]].

You can reschedule at most one meeting by moving its start time while maintaining the same duration, such that the meetings remain non-overlapping, to maximize the longest continuous period of free time during the event.

Return the maximum amount of free time possible after rearranging the meetings.

Note that the meetings can not be rescheduled to a time outside the event and they should remain non-overlapping.

Note: In this version, it is valid for the relative ordering of the meetings to change after rescheduling one meeting.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3443 Maximum Manhattan Distance After K Changes

MEDIUM

You are given a string s consisting of the characters 'N', 'S', 'E', and 'W', where s[i] indicates movements in an infinite grid:
- 'N' : Move north by 1 unit.
- 'S' : Move south by 1 unit.
- 'E' : Move east by 1 unit.
- 'W' : Move west by 1 unit.

Initially, you are at the origin (0, 0). You can change at most k characters to any of the four directions.

Find the maximum Manhattan distance from the origin that can be achieved at any time while performing the movements in order.

The Manhattan Distance between two cells (xi, yi) and (xj, yj) is |xi - xj| + |yi - yj|.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3464 Maximize The Distance Between Points On A Square

HARD

You are given an integer side, representing the edge length of a square with corners at (0, 0), (0, side), (side, 0), and (side, side) on a Cartesian plane.

You are also given a positive integer k and a 2D integer array points, where points[i] = [xi, yi] represents the coordinate of a point lying on the boundary of the square.

You need to select k elements among points such that the minimum Manhattan distance between any two points is maximized.

Return the maximum possible minimum Manhattan distance between the selected k points.

The Manhattan Distance between two cells (xi, yi) and (xj, yj) is |xi - xj| + |yi - yj|.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3466 Maximum Coin Collection

MEDIUM

Mario drives on a two-lane freeway with coins every mile. You are given two integer arrays, lane1 and lane2, where the value at the ith index represents the number of coins he gains or loses in the ith mile in that lane.
- If Mario is in lane 1 at mile i and lane1[i] > 0, Mario gains lane1[i] coins.
- If Mario is in lane 1 at mile i and lane1[i] < 0, Mario pays a toll and loses abs(lane1[i]) coins.
- The same rules apply for lane2.

Mario can enter the freeway anywhere and exit anytime after traveling at least one mile. Mario always enters the freeway on lane 1 but can switch lanes at most 2 times.

A lane switch is when Mario goes from lane 1 to lane 2 or vice versa.

Return the maximum number of coins Mario can earn after performing at most 2 lane switches.

Note: Mario can switch lanes immediately upon entering or just before exiting the freeway.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3477 Fruits Into Baskets Ii

EASY

You are given two arrays of integers, fruits and baskets, each of length n, where fruits[i] represents the quantity of the ith type of fruit, and baskets[j] represents the capacity of the jth basket.

From left to right, place the fruits according to these rules:
- Each fruit type must be placed in the leftmost available basket with a capacity greater than or equal to the quantity of that fruit type.
- Each basket can hold only one type of fruit.
- If a fruit type cannot be placed in any basket, it remains unplaced.

Return the number of fruit types that remain unplaced after all possible allocations are made.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3479 Fruits Into Baskets Iii

MEDIUM

You are given two arrays of integers, fruits and baskets, each of length n, where fruits[i] represents the quantity of the ith type of fruit, and baskets[j] represents the capacity of the jth basket.

From left to right, place the fruits according to these rules:
- Each fruit type must be placed in the leftmost available basket with a capacity greater than or equal to the quantity of that fruit type.
- Each basket can hold only one type of fruit.
- If a fruit type cannot be placed in any basket, it remains unplaced.

Return the number of fruit types that remain unplaced after all possible allocations are made.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3494 Find The Minimum Amount Of Time To Brew Potions

MEDIUM

You are given two integer arrays, skill and mana, of length n and m, respectively.

In a laboratory, n wizards must brew m potions in order. Each potion has a mana capacity mana[j] and must pass through all the wizards sequentially to be brewed properly. The time taken by the ith wizard on the jth potion is timeij = skill[i] * mana[j].

Since the brewing process is delicate, a potion must be passed to the next wizard immediately after the current wizard completes their work. This means the timing must be synchronized so that each wizard begins working on a potion exactly when it arrives.

Return the minimum amount of time required for the potions to be brewed properly.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3506 Find Time Required To Eliminate Bacterial Strains

**HARD**

You are given an integer array timeReq and an integer splitTime.

In the microscopic world of the human body, the immune system faces an extraordinary challenge: combatting a rapidly multiplying bacterial colony that threatens the body's survival.

Initially, only one white blood cell (WBC) is deployed to eliminate the bacteria. However, the lone WBC quickly realizes it cannot keep up with the bacterial growth rate.

The WBC devises a clever strategy to fight the bacteria:
- The ith bacterial strain takes timeReq[i] units of time to be eliminated.
- A single WBC can eliminate only one bacterial strain. Afterwards, the WBC is exhausted and cannot perform any other tasks.
- A WBC can split itself into two WBCs, but this requires splitTime units of time. Once split, the two WBCs can work in parallel on eliminating the bacteria.
- Only one WBC can work on a single bacterial strain. Multiple WBCs cannot attack one strain in parallel.

You must determine the minimum time required to eliminate all the bacterial strains.

Note that the bacterial strains can be eliminated in any order.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3516 Find Closest Person

**EASY**

You are given three integers x, y, and z, representing the positions of three people on a number line:
- x is the position of Person 1.
- y is the position of Person 2.
- z is the position of Person 3, who does not move.

Both Person 1 and Person 2 move toward Person 3 at the same speed.

Determine which person reaches Person 3 first:
- Return 1 if Person 1 arrives first.
- Return 2 if Person 2 arrives first.
- Return 0 if both arrive at the same time.

Return the result accordingly.

**TRY IT / Write examples, edge cases, and your own algorithm first.**

## 3606 Coupon Code Validator

**EASY**

You are given three arrays of length n that describe the properties of n coupons: code, businessLine, and isActive. The ith coupon has:

- code[i]: a string representing the coupon identifier.
- businessLine[i]: a string denoting the business category of the coupon.
- isActive[i]: a boolean indicating whether the coupon is currently active.

A coupon is considered valid if all of the following conditions hold:
and underscores (_).
"pharmacy", "restaurant".

Return an array of the codes of all valid coupons, sorted first by their businessLine in the order: "electronics", "grocery", "pharmacy", "restaurant", and then by code in lexicographical (ascending) order within each category.

**TRY IT / Write examples, edge cases, and your own algorithm first.**